Registers for aic8820_hard

August 29, 2022

1 INT_ROM

1.1 boot rom				0x00000000
S.NO.	Count	0	1	98303
0	boot rom	0x00000000	0x00000004	0x0005FFFC

2 INT_BTROM

2.1 bt rom				0x00080000
S.NO.	Count	0	1	131071
0	bt rom	0x00080000	0x00080004	0x000FFFFC

3 INT_SRAM0

3.1 int sram0				0x00100000
S.NO.	Count	0	1	32767
0	int sram0	0x00100000	0x00100004	0x0011FFFC

4 INT_SRAM1

4.1 int sram1				0x00120000
S.NO.	Count	0	1	32767
0	int sram1	0x00120000	0x00120004	0x0013FFFC

5 INT_SRAM2

5.1 int sram2				0x00140000
S.NO.	Count	0	1	32767
0	int sram2	0x00140000	0x00140004	0x0015FFFC

6 INT_SRAM3

6.1 int sram3				0x00160000
S.NO.	Count	0	1	32767
0	int sram3	0x00160000	0x00160004	0x0017FFFC

7 INT_SRAM4

7.1 int sram4				0x00180000
S.NO.	Count	0	1	32767
0	int sram4	0x00180000	0x00180004	0x0019FFFC

8 INT_SRAM5

8.1 int sram5				0x001A0000
S.NO.	Count	0	1	32767
0	int sram5	0x001A0000	0x001A0004	0x001BFFFC

9 INT_SRAM6

9.1 int sram6				0x001C0000
S.NO.	Count	0	1	32767
0	int sram6	0x001C0000	0x001C0004	0x001DFFFC

10 INT_SHARED_CACHE_SRAM

10.1 int shared_cache_sram				0x001E0000
S.NO.	Count	0	1	8191
0	int shared_cache_sram	0x001E0000	0x001E0004	0x001E7FFC

11 INT_BTRAM

11.1 bt ram				0x00200000
S.NO.	Count	0	1	16383
0	bt ram	0x00200000	0x00200004	0x0020FFFC

12 INT_AON_SRAM

12.1 aon_sram				0x00300000
S.NO.	Count	0	1	8191
0	aon_sram	0x00300000	0x00300004	0x00307FFC

13 FLASH_MEM

13.1 flash_mem				0x04000000
S.NO.	Count	0	1	16777215
0	flash_mem	0x04000000	0x04000004	0x07FFFFFC

14 FLASH_CACHE_MEM

14.1 flash_cache_mem				0x08000000
S.NO.	Count	0	1	16777215
0	flash_cache_mem	0x08000000	0x08000004	0x0BFFFFFC

15 cache

15.1 BUS_CFG0				0x40000000
Bits	Field Name	Acc	Default	Description
0	rf_blk_cache_en0	rw	0x0	0: block 0 cache disabled 1: block 0 cache enabled
1	rf_blk_cache_en1	rw	0x0	0: block 1 cache disabled 1: block 1 cache enabled
2	rf_blk_cache_en2	rw	0x0	0: block 2 cache disabled 1: block 2 cache enabled
3	rf_blk_cache_en3	rw	0x0	0: block 3 cache disabled 1: block 3 cache enabled
4	rf_blk_cache_en4	rw	0x1	0: block 4 cache disabled 1: block 4 cache enabled
5	rf_blk_cache_en5	rw	0x1	0: block 5 cache disabled 1: block 5 cache enabled
6	rf_blk_cache_en6	rw	0x1	0: block 6 cache disabled 1: block 6 cache enabled
7	rf_blk_cache_en7	rw	0x1	0: block 7 cache disabled 1: block 7 cache enabled
8	rf_blk_remap_en0	rw	0x0	0: block 0 cache remap disabled 1: block 0 cache remap enabled
9	rf_blk_remap_en1	rw	0x0	0: block 1 cache remap disabled 1: block 1 cache remap enabled
10	rf_blk_remap_en2	rw	0x0	0: block 2 cache remap disabled 1: block 2 cache remap enabled
11	rf_blk_remap_en3	rw	0x0	0: block 3 cache remap disabled 1: block 3 cache remap enabled
12	rf_blk_remap_en4	rw	0x1	0: block 4 cache remap disabled 1: block 4 cache remap enabled
13	rf_blk_remap_en5	rw	0x1	0: block 5 cache remap disabled 1: block 5 cache remap enabled
14	rf_blk_remap_en6	rw	0x1	0: block 6 cache remap disabled 1: block 6 cache remap enabled
15	rf_blk_remap_en7	rw	0x1	0: block 7 cache remap disabled 1: block 7 cache remap enabled
16	rf_blk_prot_en0	rw	0x0	0: block 0 protect disabled or access permitted 1: block 0 protect enabled or access forbidden
17	rf_blk_prot_en1	rw	0x0	0: block 1 protect disabled or access permitted 1: block 1 protect enabled or access forbidden
18	rf_blk_prot_en2	rw	0x0	0: block 2 protect disabled or access permitted 1: block 2 protect enabled or access forbidden
19	rf_blk_prot_en3	rw	0x0	0: block 3 protect disabled or access permitted 1: block 3 protect enabled or access forbidden
20	rf_blk_prot_en4	rw	0x0	0: block 4 protect disabled or access permitted 1: block 4 protect enabled or access forbidden
21	rf_blk_prot_en5	rw	0x0	0: block 5 protect disabled or access permitted 1: block 5 protect enabled or access forbidden
22	rf_blk_prot_en6	rw	0x0	0: block 6 protect disabled or access permitted 1: block 6 protect enabled or access forbidden
23	rf_blk_prot_en7	rw	0x0	0: block 7 protect disabled or access permitted 1: block 7 protect enabled or access forbidden

15.2 BUS_CFG1				0x40000004
Bits	Field Name	Acc	Default	Description
28 : 13	rf_blk_base_addr1	rw	0x0	block 1 start address

15.3 BUS_CFG2				0x40000008
Bits	Field Name	Acc	Default	Description
28 : 13	rf_blk_base_addr2	rw	0x0	block 2 start address

15.4 BUS_CFG3	0x4000000C

Bits	Field Name	Acc	Default	Description
28 : 13	rf_blk_base_addr3	rw	0x0	block 3 start address

15.5 BUS_CFG4				0x40000010
Bits	Field Name	Acc	Default	Description
28 : 13	rf_blk_base_addr4	rw	0x4000	block 4 start address

15.6 BUS_CFG5				0x40000014
Bits	Field Name	Acc	Default	Description
28 : 13	rf_blk_base_addr5	rw	0x4000	block 5 start address

15.7 BUS_CFG6				0x40000018
Bits	Field Name	Acc	Default	Description
28 : 13	rf_blk_base_addr6	rw	0x4000	block 6 start address

15.8 BUS_CFG7				0x4000001C
Bits	Field Name	Acc	Default	Description
28 : 13	rf_blk_base_addr7	rw	0x4000	block 7 start address

15.9 BUS_REMAP0				0x40000020
Bits	Field Name	Acc	Default	Description
31 : 12	rf_blk0_remap_offset	rw	0x0	block 0 remap offset when the block 0 remap function is enabled. The address after remap will be original addr + rf_blk0_remap_offset

15.10 BUS_REMAP1				0x40000024
Bits	Field Name	Acc	Default	Description
31 : 12	rf_blk1_remap_offset	rw	0x0	block 1 remap offset when the block 1 remap function is enabled. The address after remap will be original addr + rf_blk1_remap_offset

15.11 BUS_REMAP2				0x40000028
Bits	Field Name	Acc	Default	Description
31 : 12	rf_blk2_remap_offset	rw	0x0	block 2 remap offset when the block 2 remap function is enabled. The address after remap will be original addr + rf_blk2_remap_offset

15.12 BUS_REMAP3				0x4000002C
Bits	Field Name	Acc	Default	Description
31 : 12	rf_blk3_remap_offset	rw	0x0	block 3 remap offset when the block 3 remap function is enabled. The address after remap will be original addr + rf_blk3_remap_offset

15.13 BUS_REMAP4				0x40000030
Bits	Field Name	Acc	Default	Description

31 : 12	rf_blk4_remap_offset	rw	0xFC000	block 4 remap offset when the block 4 remap function is enabled. The address after remap will be original addr + rf_blk4_remap_offset

15.14 BUS_REMAP5				0x40000034
Bits	Field Name	Acc	Default	Description
31 : 12	rf_blk5_remap_offset	rw	0xFC000	block 5 remap offset when the block 5 remap function is enabled. The address after remap will be original addr + rf_blk5_remap_offset

15.15 BUS_REMAP6				0x40000038
Bits	Field Name	Acc	Default	Description
31 : 12	rf_blk6_remap_offset	rw	0xFC000	block 6 remap offset when the block 6 remap function is enabled. The address after remap will be original addr + rf_blk6_remap_offset

15.16 BUS_REMAP7				0x4000003C
Bits	Field Name	Acc	Default	Description
31 : 12	rf_blk7_remap_offset	rw	0xFC000	block 7 remap offset when the block 7 remap function is enabled. The address after remap will be original addr + rf_blk7_remap_offset

15.17 CACHE_CFG0				0x40000040
Bits	Field Name	Acc	Default	Description
1 : 0	rf_write_mode	rw	0x0	cache write operation mode: 2b00: write through 2b01: write back, no write allocate 2b10: write back, write allocate
5 : 2	rf_hprot_byp	rw	0x0	Bit [5]: 1b1: hprot[3] from CM4 go through cache controller without modification 1b0: hprot[3] is provided by cache controller register Bit [4]: 1b1: hprot[2] from CM4 go through cache controller without modification 1b0: hprot[2] is provided by cache controller register Bit [3]: 1b1: hprot[1] from CM4 go through cache controller without modification 1b0: hprot[1] is provided by cache controller register Bit [2]: 1b1: hprot[0] from CM4 go through cache controller without modification 1b0: hprot[0] is provided by cache controller register
9 : 6	rf_hprot_bus_reg	rw	0x3	hprot control register which provide 4bit hprot for cache bus AHB
13 : 10	rf_hprot_cache_reg	rw	0xB	hprot control register which provide 4bit hprot for cache ctrl AHB
14	ahb_ram_en	rw	0x0	Cache Data Ram Can be Write/Read Throught rf AHB Bus
29 : 28	rf_cache_size_sel	rw	0x3	cache size selection: 0: 4K Byte 1: 8K Byte 2: 16K Byte 3: 32K Byte
30	rf_cmd_exec_mode	rw	0x0	0: (recommended) software can run in cacheable region during software command processing 1: software cannot run in cacheable region during software command processing

31	rf_debug_en	rw	0x0	cache debug mode enable: 0: normal mode 1: debug mode This bit MUST always be cleared during cache operating

15.18 BUS_STS0				0x40000044
Bits	Field Name	Acc	Default	Description
27 : 0	rf_prot_detect_addr	ro	0x0	trigging address for protect block

15.19 CACHE_STS0				0x40000048
Bits	Field Name	Acc	Default	Description
3 : 0	rf_cache_st	ro	0x0	rf_cache_st
6 : 4	rf_cmd_st	ro	0x0	rf_cmd_st
7	rf_write_ongoing	ro	0x0	rf_write_ongoing, this is a status register for write through mode to avoid potential coherence issue. 1b1: the cache is still doing AHB write transaction to the main memory and the data is not written into the main memory. 1b0: the cache has finished AHB write transaction and the data is written into the main memory.

15.20 CMD_CFG0				0x40000050
Bits	Field Name	Acc	Default	Description
27 : 0	rf_cmd_str_addr	rw	0x0	cmd_all : not used cmd_range : start address cmd_entry : entry address

15.21 CMD_CFG1				0x40000054
Bits	Field Name	Acc	Default	Description
27 : 0	rf_cmd_end_addr	rw	0x0	cmd_all : not used cmd_range : end address cmd_entry : not used

15.22 CMD_CFG2				0x40000058
Bits	Field Name	Acc	Default	Description
5 : 0	rf_cmd_type	rw	0x0	software command type: 6h0 : clean all 6h1 : clean range 6h2 : clean entry 6h3 : reserved 6h4 : invalid all 6h5 : invalid range 6h6 : invalid entry 6h7 : reserved 6h8 : clean and invalid all 6h9 : clean and invalid range 6hA : clean and invalid entry 6hB : reserved
31	rf_cmd_str	wo	0x0	software command start: write 1 to this bit to issue one command

15.23 INT_EN				0x40000060
Bits	Field Name	Acc	Default	Description
0	rf_cmd_irq_en	rw	0x0	interrupt enable for software command done
1	rf_prot_irq_en	rw	0x0	interrupt enable for protect block trigging

15.24 INT_RAW				0x40000064
Bits	Field Name	Acc	Default	Description
0	rf_cmd_irq_raw	ro	0x0	interrupt raw status for software command done
1	rf_prot_irq_raw	ro	0x0	interrupt raw status for protect block trigging

15.25 INT_MASK				0x40000068
Bits	Field Name	Acc	Default	Description
0	rf_cmd_irq_mask	ro	0x0	interrupt masked status for software command done
1	rf_prot_irq_mask	ro	0x0	interrupt masked status for protect block trigging

15.26 INT_CLR				0x4000006C
Bits	Field Name	Acc	Default	Description
0	rf_cmd_irq_clr	wo	0x0	interrupt clear for software command done
1	rf_prot_irq_clr	wo	0x0	interrupt clear for protect block trigging

15.27 CACHE_WRITE_HIT_CNT				0x40000070
Bits	Field Name	Acc	Default	Description
31 : 0	rf_write_hit_cnt	ro	0x0	Cache write hit times. When cache write hit, the counter value increment by 1

15.28 CACHE_WRITE_MISS_CNT				0x40000074
Bits	Field Name	Acc	Default	Description
29 : 0	rf_write_miss_cnt	ro	0x0	Cache write miss times. When cache write miss, the counter value increment by 1
30	rf_write_cnt_run	rw	0x0	1: write hit/miss counter will run 0: write hit/miss counter will stop
31	rf_write_cnt_clr	wo	0x0	clear write counter values to zero

15.29 CACHE_READ_HIT_CNT				0x40000078
Bits	Field Name	Acc	Default	Description
31 : 0	rf_read_hit_cnt	ro	0x0	Cache read hit times. When cache read hit, the counter value increment by 1

15.30 CACHE_READ_MISS_CNT				0x4000007C
Bits	Field Name	Acc	Default	Description
29 : 0	rf_read_miss_cnt	ro	0x0	Cache read miss times. When cache read miss, the counter value increment by 1
30	rf_read_cnt_run	rw	0x0	1: read hit/miss counter will run 0: read hit/miss counter will stop
31	rf_read_cnt_clr	wo	0x0	clear read counter values to zero

15.31 CACHE_HACT_CNT				0x40000080
Bits	Field Name	Acc	Default	Description
31 : 0	rf_hact_cnt	ro	0x0	Cache master AHB active cycles in total. When master AHB is active (hsel and htrans[1]), the counter value increment by 1

15.32 CACHE_HRDY_CNT				0x40000084
Bits	Field Name	Acc	Default	Description

29 : 0	rf_hrdy_cnt	ro	0x0	The HRDY counter counts the valid cycles of HREADY signal from the cache controller to the master when the master AHB is active. When HREADY signal to the master is high and the master AHB is active, the counter value increment by 1
30	rf_hact_hrdy_run	rw	0x0	1: HACT and HRDY counters will run 0: HACT and HRDY counters will stop
31	rf_hact_hrdy_clr	wo	0x0	clear HACT and HRDY counter values to zero

16 dma_cx

16.1 DMA_PAUSE				0x40020000
Discription:				
Bits	Field Name	Acc	Default	Description
0	DMA_PAUSE	rw	0x0	Active High, pause all the channels. Note: If "DMA_PAUSE" asserted active when the DMA is busy, DMA would finish the current fragment before enter Pause State.
16	DMA_PAUSE_STATUS	ro	0x0	DMA Pause status. 0 : In active state. 1 : In Pause state.

16.2 DMA_FRAG_WAIT				0x40020004
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	DMA_FRAG_WAIT	rw	0x0	Wait Time between two successive Fragment. With unit of AHB clock.
28	O_ENDIAN	rw	0x0	DMA output endian, 0- little endian 1- big endian

16.3 REQ_PEND0_EN				0x40020008
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	REQ_PEND_EN_31to0	rw	0x0	Requests Pending Enable. Acitve high, enable DMA_REQ[31:0] to wakeup DMA, when MCU deep sleep;

16.4 REQ_PEND1_EN				0x4002000C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	REQ_PEND_EN_63to32	rw	0x0	Requests Pending Enable. Acitve high, enable DMA_REQ[63:32] to wakeup DMA, when MCU deep sleep;

16.5 DMA_INT_RAW_STATUS				0x40020010
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_INT_RAW_STATUS	ro	0x0	DMA 32 Channels Interrupt Raw Status. 0 : No Interrupt need to be served. 1 : Interrupt need to be served.

16.6 DMA_INT_MASK_STATUS				0x40020014
Discription:				
Bits	Field Name	Acc	Default	Description

31 : 0	DMA_INT_MASK_STATUS	ro	0x0	DMA 32 Channels Interrupt Mask Status. 0 : No Interrupt need to be served. 1 : Interrupt need to be served.

16.7 DMA_REQ_STATUS				0x40020018
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_REQ_STATUS	ro	0x0	DMA 32 Channels Request Status. 0 : No Request need to be served. 1 : Request need to be served.

16.8 DMA_EN_STATUS				0x4002001C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_EN_STATUS	ro	0x0	DMA 32 Channels Enable Status. 0 : Channel Not Enabled. 1 : Channel Enabled.

16.9 DMA_DEBUG_STATUS				0x40020020
Discription:				
Bits	Field Name	Acc	Default	Description
3 : 0	DEST_FSM_STATUS	ro	0x0	Destination Side FSM status. 4'd0: IDLE 4'd1: calculate burst parameter 4'd2: calculate data number of burst 4'd3: wait enough data buffer space for write burst 4'd4: start write burst 4'd5: write burst execute 4'd6: wait fragment done or limit outstanding number 4'd7: fragment write done
7 : 4	SRC_FSM_STATUS	ro	0x0	Source Side FSM status. 4'd0: IDLE 4'd1: calculate burst parameter 4'd2: calculate data number of burst 4'd3: wait enough data buffer space for read burst 4'd4: start read burst 4'd5: read burst execute 4'd6: wait fragment done or limit outstanding number 4'd7: fragment read done
12 : 8	MAIN_FSM_STATUS	ro	0x0	MAIN FSM status. 5'd0: IDLE 5'd1: load link list pointer 5'd2: load link list source and dest address 5'd3: load link list fragment and block length 5'd4: load link list transaction length and src/dest transfer step 5'd5: load link list wrap_ptr and wrap_to 5'd6: load full mode source and dest address 5'd7: load full mode fragment and block length 5'd8: load full mode transaction length and src/dest transfer step 5'd9: load full mode wrap_ptr and wrap_to 5'd10: update block length 5'd11: update transaction length 5'd12: fragment execute 5'd13: load list pointer and use src/dest fragment step 5'd14: load src/dest block step 5'd15: load link list high 4bits pointer and src/dest fragment step 5'd16: load src/dest block step 5'd17: transaction or link list done 5'd18: update wrap info.
16	HREADY_DMAR	ro	0x1	dma read hready
17	HREADY_DMAW	ro	0x1	dma write hready
20	DMA_BUSY	ro	0x0	DMA Busy Status.

16.10 DMA_ARB_SEL_STATUS				0x40020024
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	CHN_ARB_SEL	ro	0x0	Channel Arbitrator Selection Status;

16.11 CRC_CFG				0x40020030
Discription:				
Bits	Field Name	Acc	Default	Description
3 : 0	DATA_MODE	rw	0xE	CRC data mode bit0: In data byte receive bit1: In data bit receive bit2: Out data byte receive bit3: Out data bit receive
5 : 4	CRC_GEN_SIZE	rw	0x3	CRC generator size 2'b0:crc8 2'b1:crc16 2'b10:crc24 2'b11:crc32
9 : 8	CRC_INIT_MODE	rw	0x0	CRC initial mode 2'b0:frag done 2'b1:blk done 2'b10:trsc done
20 : 16	CRC_CH_NUM	rw	0x1	DMA channel number for CRC check
24	CRC_HOLD	rw	0x0	CRC hold
28	CRC_EN	rw	0x0	CRC enable
29	CRC_CLK_GATE_EN	rw	0x1	CRC clock gate enable

16.12 CRC_GEN				0x40020034
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CRC_GEN	rw	0x4C11DB7	CRC32 generator

16.13 CRC_OUT_XOR				0x40020038
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CRC_OUT_XOR	rw	0xFFFFFFFF	CRC out data xor

16.14 CRC_INIT				0x4002003C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CRC_INIT	rw	0xFFFFFFFF	CRC init val

16.15 I_CRC				0x40020040
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	I_CRC	ro	0x0	

16.16 AES_KEY_0_ADDR				0x40020044
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_0_ADDR	rw	0x0	

16.17 AES_KEY_1_ADDR				0x40020048
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_1_ADDR	rw	0x0	

16.18 AES_KEY_2_ADDR				0x4002004C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_2_ADDR	rw	0x0	

16.19 AES_KEY_3_ADDR				0x40020050
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_3_ADDR	rw	0x0	

16.20 AES_KEY_4_ADDR				0x40020054
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_4_ADDR	rw	0x0	

16.21 AES_KEY_5_ADDR				0x40020058
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_5_ADDR	rw	0x0	

16.22 AES_KEY_6_ADDR				0x4002005C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_6_ADDR	rw	0x0	

16.23 AES_KEY_7_ADDR				0x40020060
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_KEY_7_ADDR	rw	0x0	

16.24 AES_IV_0_ADDR				0x40020064
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_IV_0_ADDR	rw	0x0	

16.25 AES_IV_1_ADDR				0x40020068
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_IV_1_ADDR	rw	0x0	

16.26 AES_IV_2_ADDR				0x4002006C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_IV_2_ADDR	rw	0x0	

16.27 AES_IV_3_ADDR				0x40020070
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_IV_3_ADDR	rw	0x0	

16.28 AES_PLAIN_LEN_H_ADDR				0x40020074
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_PLAIN_LEN_H_ADDR	rw	0x0	

16.29 AES_PLAIN_LEN_L_ADDR				0x40020078
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_PLAIN_LEN_L_ADDR	rw	0x0	

16.30 AES_A_LEN_H_ADDR				0x4002007C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_A_LEN_H_ADDR	rw	0x0	

16.31 AES_A_LEN_L_ADDR				0x40020080
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	AES_A_LEN_L_ADDR	rw	0x0	

16.32 CE_CTRL_ADDR				0x40020084
Discription:				
Bits	Field Name	Acc	Default	Description
0	rsa_en	rw	0x0	
1	ecc_en	rw	0x0	
2	rsa_mode	rw	0x0	
5 : 3	ecc_mode	rw	0x0	
6	kpEccAEqP3En	rw	0x0	
12 : 7	macKeyIndex	rw	0x0	
13	aes_en	rw	0x0	
14	aes_enc_dec_sel	rw	0x0	
17 : 15	aes_mode	rw	0x0	
18	key192	rw	0x0	
19	key256	rw	0x0	
20	hash_en	rw	0x0	
26 : 21	ce_chn_sel	rw	0x0	
27	dma_clr	rw	0x0	
28	hash_endian_sel	rw	0x1	
29	aes_endian_sel	rw	0x1	
30	hash_done	ro	0x1	
31	aes_done	ro	0x1	

16.33 PKA_REG0_ADDR				0x40020088
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	PKA_REG0_ADDR	rw	0x0	

16.34 CE_CLOCK_GATE_CTRL				0x4002008C
Discription:				
Bits	Field Name	Acc	Default	Description
0	PKA_CLOCK_GATE_EN	rw	0x1	
1	AES_CLOCK_GATE_EN	rw	0x1	

2	HASH_CLOCK_GATE_EN	rw	0x1	
3	reserved0	rw	0x1	

16.35 TRANS_CTRL0_ADDR				0x40020090
Discription:				
Bits	Field Name	Acc	Default	Description
2 : 0	fft_len	rw	0x0	
5 : 3	fft_scale	rw	0x0	
6	fft_dir	rw	0x0	
7	fft_en	rw	0x0	
13 : 8	trans_chn_sel	rw	0x0	
14	fft_in_format_sel	rw	0x0	
15	fft_out_format_sel	rw	0x0	
31	trans_done	ro	0x1	

16.36 TRANS_CTRL1_ADDR				0x40020094
Discription:				
Bits	Field Name	Acc	Default	Description
0	fltr_en	rw	0x0	
1	fltr_cofshw_upd	rw	0x0	
2	sbc_mode_en	rw	0x0	
3	fltr_cofcos_upd	rw	0x0	
4	fltr_cof_len	rw	0x0	
18 : 5	fltr_len	rw	0x0	
23 : 19	fltr_order	rw	0x0	
24	fltr_cof_init_sel	rw	0x0	

16.37 TRANS_CLOCK_GATE_CTRL				0x40020098
Discription:				
Bits	Field Name	Acc	Default	Description
0	fft_clk_gate_en	rw	0x1	
1	flt_clk_gate_en	rw	0x1	
2	reserved0	rw	0x0	

16.38 DMA_MON_SEL				0x4002009C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	dma_mon_sel	rw	0x0	

16.39 dma_reserved0				0x400200A0
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	dma_reserved0	rw	0x0	

16.40 dma_reserved1				0x400200A4
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	dma_reserved1	rw	0x0	

16.41 dma_reserved2	0x400200A8
Discription:	

Bits	Field Name	Acc	Default	Description
31 : 0	dma_reserved2	rw	0x0	

16.42 CHKSUM_CTRL_ADDR				0x400200B8
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	chksum_len	rw	0x0	
16	chksum_en	rw	0x0	
22 : 17	chksum_chn_sel	rw	0x0	
23	chksum_endian_sel	rw	0x0	0: big endian 1: little endian
24	chksum_dest_bypass	rw	0x0	0: dest trans enable 1: dest trans bypass
25	chksum_clk_gate_en	rw	0x1	

16.43 CHKSUM_RES_ADDR				0x400200BC
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	chksum_res	ro	0x0	
31	chksum_done	ro	0x0	

16.44 HASH_LEN_BIT_ADDR				0x400200C0
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	hash_len_bit	rw	0x0	hash padding length

16.45 CE_CTRL_2_ADDR				0x400200C4
Discription:				
Bits	Field Name	Acc	Default	Description
0	hash_hw_pad_en	rw	0x0	
2 : 1	hash_mode_sel	rw	0x0	0: sha-224 1: sha-256 2: sha-1

16.46 FULL_DMA_CHS				0x40021000
S.NO.	Count	0	1	13
0	FULL_CHN_PAUSE	0x40021000	0x40021040	0x04136340
1	FULL_CHN_REQ	0x40021004	0x40021044	0x04136344
2	FULL_CHN_CFG	0x40021008	0x40021048	0x04136348
3	FULL_CHN_INT	0x4002100C	0x4002104C	0x0413634C
4	FULL_CHN_SRC_ADDR	0x40021010	0x40021050	0x04136350
5	FULL_CHN_DEST_ADDR	0x40021014	0x40021054	0x04136354
6	FULL_CHN_FRAG_LEN	0x40021018	0x40021058	0x04136358
7	FULL_CHN_BLK_LEN	0x4002101C	0x4002105C	0x0413635C
8	FULL_CHN_TRSC_LEN	0x40021020	0x40021060	0x04136360
9	FULL_CHN_TRSF_STEP	0x40021024	0x40021064	0x04136364
10	FULL_CHN_WRAP_PTR	0x40021028	0x40021068	0x04136368
11	FULL_CHN_WRAP_TO	0x4002102C	0x4002106C	0x0413636C
12	FULL_CHN_LLIST_PTR	0x40021030	0x40021070	0x04136370
13	FULL_CHN_FRAG_STEP	0x40021034	0x40021074	0x04136374
14	FULL_CHN_SRC_-BLK_STEP	0x40021038	0x40021078	0x04136378
15	FULL_CHN_DEST-_BLK_STEP	0x4002103C	0x4002107C	0x0413637C

16.46.1 FULL_CHN_PAUSE				0x40021000
Discription :				
Bits	Field Name	Acc	Default	Description
0	CHN_PAUSE	rw	0x0	Channel Pause Enable. 0 : Channel in active mode. 1 : Channel pause. Active high, only pause the channel. If the CHN_PAUSE asserted when the channel is busy, it should be paused after the current fragment finished.
16	CHN_PAUSE_STATUS	ro	0x0	Channel Pause status.

16.46.2 FULL_CHN_REQ				0x40021004
Discription :				
Bits	Field Name	Acc	Default	Description
0	CHN_REQ	rw	0x0	Channel Software DMA Request.
1	CHN_EXREQ_MODE	rw	0x0	Channel External DMA Request Mode: 1- level, 0- edge
2	CHN_EXREQ_LEVEL	rw	0x1	Channel External DMA Valid Level: 1- high, 0- low

16.46.3 FULL_CHN_CFG				0x40021008
Discription :				
Bits	Field Name	Acc	Default	Description
0	CHN_EN	rw	0x0	Channel Enable: 0 : Disable the channel. 1 : Enable the channel.
4	LLIST_EN	rw	0x0	Link list Enable, only available for full DMA channel. 0 : Disable the link list. 1 : Enable the link list.
8	LLIST_CFG_VALID	rw	0x0	If "LLIST_EN" asserted, active high indicates Link list configure valid. 0 : Link list configure in-valid. DMA controller need obtain link list configure information by link list pointer. 1 : link list configure information valid. DMA controller would continue to complete the node and clear this bit. Note: Strongly recommend to observe only. Carefully writing to this bit, which may cause DMA crashed. If had to do so, pause this channel before changing.
13 : 12	CHN_PRIORITY	rw	0x0	Channel Request Priority. Four level of Priority supported as list : 2'b00 : Lowest priority. 2'b01 : Secondary lowest priority. 2'b10 : Tertiary lowest priority. 2'b11 : Highest priority.
17 : 16	AHB_BURST_SEL	rw	0x0	AMBA Burst Select. 2'b00 : INCR8; 2'b01 : INCR4; 2'b10 : INCR16; 2'b11 : INCR16;

23 : 20	ERR_TYPE_STATUS	ro	0x0	Error Configure , which would be cleared when "ERR_INT_STATUS" cleared:
				4'b0001 : Addr Wrap Error Configure;
				4'b0010 : Request Mode Error Configure;
				4'b0100 : Length Error Configure;
				4'b1000 :Unalign Error Configure;
				1 gt; Addr wrap error configure
				Set wrap enable in standard DMA channel.
				2 gt; Request mode error configure
				Req_mode==2 or 3 in in standard DMA channel.
				3 gt; Length Error Configure
				d gt; Frag_len==0;
				e gt; Blk_len==0;
				f gt; FULL DMA channel,trsc_len==0;
				4 gt; Unalign Error Configure
				c gt; Src unsigned error
				a1 gt; address unalign in halfword width
				error=(src_data_size[1:0]==1) amp; amp;(src_addr[0] =0)
				a2 gt; address unalign and trsf_step unalign in word width
				error=(src_data_size[1:0]==2) amp; amp;(src_addr[1:0] =0)
				amp; amp;(src_trsf_step =4)
				d gt; dest unsigned error
				b1 gt; address unalign in halfword width
				error=(dst_data_size[1:0]==1) amp; amp;(dst_addr[0] =0)
				b2 gt; address unalign and trsf_step unalign in word width
				error=(dst_data_size[1:0]==2) amp; amp;(dst_addr[1:0] =0)
				amp; amp;(dst_trsf_step =4)

16.46.4 FULL_CHN_INT 0x4002100C

Discription :

Bits	Field Name	Acc	Default	Description
0	CHN_FRAGMENT_INT_EN	rw	0x0	Fragment done interrupt enable.
1	CHN_BLK_INT_EN	rw	0x0	Block done interrupt enable.
2	CHN_TRSC_INT_EN	rw	0x0	Transaction done interrupt enable
3	CHN_LLIST_INT_EN	rw	0x0	Link list done interrupt enable
4	CHN_CFG_ERR_INT_EN	rw	0x0	Configuration error interrupt enable.
8	CHN_FRAGMENT_-INT_RAW_STATUS	ro	0x0	Fragment done Raw interrupt status.
9	CHN_BLK_INT_-RAW_STATUS	ro	0x0	Block done interrupt status.
10	CHN_TRSC_INT_-RAW_STATUS	ro	0x0	Channel Transaction interrupts status.
11	CHN_LLIST_INT-_RAW_STATUS	ro	0x0	Channel Link List interrupts status.
12	CHN_CFG_ERR_-INT_RAW_STATUS	ro	0x0	Configuration error raw interrupts status.
16	CHN_FRAGMENT_-INT_MASK_STATUS	ro	0x0	Channel Fragment Masked interrupts status.
17	CHN_BLK_INT_-MASK_STATUS	ro	0x0	Channel Block Masked interrupts status.
18	CHN_TRSC_INT_-MASK_STATUS	ro	0x0	Channel Transaction Masked interrupts status.
19	CHN_LLIST_INT-_MASK_STATUS	ro	0x0	Channel Link list Masked interrupt status.
20	CHN_CFG_ERR_-INT_MASK_STATUS	ro	0x0	Configuration error masked interrupt status.
24	CHN_FRAGMENT_INT_CLR	wo	0x0	Fragment done interrupt clear.
25	CHN_BLK_INT_CLR	wo	0x0	Block done interrupt clear.

26	CHN_TRSC_INT_CLR	wo	0x0	Transaction done interrupt clear
27	CHN_LLIST_INT_CLR	wo	0x0	Link list done interrupt clear
28	CHN_CFG_ERR_INT_CLR	wo	0x0	Configuration error interrupt clear.

16.46.5 FULL_CHN_SRC_ADDR 0x40021010

Discription :

Bits	Field Name	Acc	Default	Description
31 : 0	CHN_SRC_ADDR	rw	0x0	Channel Source Address

16.46.6 FULL_CHN_DEST_ADDR 0x40021014

Discription :

Bits	Field Name	Acc	Default	Description
31 : 0	CHN_DEST_ADDR	rw	0x0	Channel Destination Address

16.46.7 FULL_CHN_FRAG_LEN 0x40021018

Discription :

Bits	Field Name	Acc	Default	Description
16 : 0	FRG_LEN	rw	0x0	Fragment Length, with byte unit. Max 128K-1 bytes
18 : 17	BLK_LEN_REC_H	rw	0x0	Block counter[16:15], update after each fragment done; Note: Normally configured to 2'b0. Only to be configured other value during state retention after wake up
19	LLIST_END	rw	0x0	Link list end flag. Active only when "LLIST_EN" asserted.
20	SRC_ADDR_FIX_EN	rw	0x0	Source Address Fix Enable: 0 : Disable Source Address Fix mode. 1 : Enable Source Address Fix mode. Active High would make source address fixed within the fragment, while "FRAG_STEP" and "BLK_STEP" would be applied to address after fragment done;
21	DEST_ADDR_FIX_EN	rw	0x0	Selection of the side using Address Fix mode. 0 : Disable Destination Address Fix mode. 1 : Enable Destination Address Fix mode. Active High would make destination address fixed within the fragment, while "FRAG_STEP" and "BLK_STEP" would be applied to address after fragment done;
22	ADDR_WRAP_EN	rw	0x0	Active High enable Address-wrapping for ring buffer. The next address of DMA jumps to "WRAP_START_ADDR" address when the current address matches "WRAP_END_ADDR" address: 0 : Disable address wrap mode. 1 : Enable address wrap mode. "WRAP_START_ADDR" and "WRAP_END_ADDR" would be used to limit the address space of either Source or Destination.
23	ADDR_WRAP_SEL	rw	0x0	Selection of the side using Address wrap mode. 0 : Address wrap on Source. 1 : Address wrap on Destination. Only one side of a DMA channel can activate address-wrapping mode at a time.
25 : 24	REQ_MODE	rw	0x0	Request mode, For Full DMA channel: 2'b00 : Fragment; 2'b01 : Block; 2'b10 : Transaction; 2'b11 : Link List;

27 : 26	SWT_MODE	rw	0x0	Data Switch mode selection. 2'b00 : ABCD = gt; ABCD; 2'b01 : ABCD = gt; DCBA 2'b10 : ABCD = gt; BADC; 2'b11 : ABCD = gt; CDAB
29 : 28	DEST_SIZE	rw	0x0	Destination AHB Size. 2'b00 : Byte; 2'b01 : Half Word 2'b10 : Word 2'b11 : DWord.
31 : 30	SRC_SIZE	rw	0x0	Source AHB Size. 2'b00 : Byte; 2'b01 : Half Word 2'b10 : Word 2'b11 : DWord.

16.46.8 FULL_CHN_BLK_LEN 0x4002101C

Discription :

Bits	Field Name	Acc	Default	Description
16 : 0	BLK_LEN	rw	0x0	Block Length, with byte unit. Max 128K-1 bytes
31 : 17	BLK_LEN_REC_L	rw	0x0	Block counter[14:0], update after each fragment done; Note: Normally configured to 15'b0. Only to be configured other value during state retention after wake up

16.46.9 FULL_CHN_TRSC_LEN 0x40021020

Discription :

Bits	Field Name	Acc	Default	Description
27 : 0	TRSC_LEN	rw	0x0	Transaction Length, with byte unit. (Max 256M-1 bytes)
28	LLIST_DEDICATED_INT_EN	rw	0x0	Interrupt global enable dedicated for a certain linklist, only valid when linklist enable
29	LLIST_DEDICATED_COUNT_EN	rw	0x0	Counter enable dedicated for a certain linklist, only valid when linklist enable

16.46.10 FULL_CHN_TRSF_STEP 0x40021024

Discription :

Bits	Field Name	Acc	Default	Description
15 : 0	SRC_TRSF_STEP	rw	0x0	Channel destination side transfer step, with byte unit.
31 : 16	DEST_TRSF_STEP	rw	0x0	Channel Source side transfer step, with byte unit.

16.46.11 FULL_CHN_WRAP_PTR 0x40021028

Discription :

Bits	Field Name	Acc	Default	Description
27 : 0	WRAP_PTR	rw	0x0	Wrap jump pointer address. Once the transfer address reaches WPPT, the next address jumps to the WPTO address after completing the WPPT data transfer.
31 : 28	CHN_SRC_ADDR_35to32	rw	0x0	Chn source address high 4bits

16.46.12 FULL_CHN_WRAP_TO				0x4002102C
Discription :				
Bits	Field Name	Acc	Default	Description
27 : 0	WRAP_TO	rw	0x0	Wrap jump-to address. Once the transfer address reaches WPPT, the next address jumps to the WPTO address after completing the WPPT data transfer.
31 : 28	CHN_DEST_ADDR_35_32	rw	0x0	Chn Destination address high 4bits

16.46.13 FULL_CHN_LLIST_PTR				0x40021030
Discription :				
Bits	Field Name	Acc	Default	Description
31 : 0	LLIST_PTR	rw	0x0	Link List pointer. Must DWord align.

16.46.14 FULL_CHN_FRAG_STEP				0x40021034
Discription :				
Bits	Field Name	Acc	Default	Description
15 : 0	SRC_FRAG_STEP	rw	0x0	Channel Source side transfer step. With byte unit.
31 : 16	DEST_FRAG_STEP	rw	0x0	Channel Destination side transfer step. With byte unit.

16.46.15 FULL_CHN_SRC_BLK_STEP				0x40021038
Discription :				
Bits	Field Name	Acc	Default	Description
27 : 0	CHN_SRC_BLK_STEP	rw	0x0	Channel Source block step. With byte unit.
31 : 28	LLIST_PTR_35_32	rw	0x0	Link List pointer high 4bits

16.46.16 FULL_CHN_DEST_BLK_STEP				0x4002103C
Discription :				
Bits	Field Name	Acc	Default	Description
27 : 0	CHN_DEST_BLK_STEP	rw	0x0	Channel Destination block step. With byte unit.

16.47 REQ_CID_0				0x40022000
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.48 REQ_CID_1				0x40022004
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.49 REQ_CID_2				0x40022008
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.50 REQ_CID_3				0x4002200C
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.51 REQ_CID_4				0x40022010
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.52 REQ_CID_5				0x40022014
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.53 REQ_CID_6				0x40022018
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.54 REQ_CID_7				0x4002201C
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.55 REQ_CID_8				0x40022020
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.56 REQ_CID_9				0x40022024
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.57 REQ_CID_10				0x40022028
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.58 REQ_CID_11				0x4002202C
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.59 REQ_CID_12				0x40022030
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.60 REQ_CID_13				0x40022034
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.61 REQ_CID_14				0x40022038
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.62 REQ_CID_15				0x4002203C
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.63 REQ_CID_16				0x40022040
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.64 REQ_CID_17				0x40022044
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.65 REQ_CID_18				0x40022048
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.66 REQ_CID_19				0x4002204C
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.67 REQ_CID_20				0x40022050
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.68 REQ_CID_21				0x40022054
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.69 REQ_CID_22				0x40022058
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.70 REQ_CID_23				0x4002205C
Discription:				
Bits	Field Name	Acc	Default	Description
5 : 0	REQ_CID	rw	0x0	Req channel identify number

16.71 CHN_1_DMA_ADDR_RD				0x40022100
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_1_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 1 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.72 CHN_1_DMA_ADDR_WR				0x40022104
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_1_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 1 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.73 CHN_2_DMA_ADDR_RD				0x40022108
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_2_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 2 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.74 CHN_2_DMA_ADDR_WR				0x4002210C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_2_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 2 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.75 CHN_3_DMA_ADDR_RD				0x40022110
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_3_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 3 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.76 CHN_3_DMA_ADDR_WR				0x40022114
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_3_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 3 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.77 CHN_4_DMA_ADDR_RD	0x40022118

Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_4_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 4 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.78 CHN_4_DMA_ADDR_WR				0x4002211C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_4_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 4 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.79 CHN_5_DMA_ADDR_RD				0x40022120
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_5_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 5 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.80 CHN_5_DMA_ADDR_WR				0x40022124
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_5_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 5 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.81 CHN_6_DMA_ADDR_RD				0x40022128
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_6_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 6 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.82 CHN_6_DMA_ADDR_WR				0x4002212C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_6_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 6 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.83 CHN_7_DMA_ADDR_RD				0x40022130
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_7_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 7 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.84 CHN_7_DMA_ADDR_WR				0x40022134
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_7_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 7 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffffff

16.85 CHN_8_DMA_ADDR_RD				0x40022138
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_8_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 8 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.86 CHN_8_DMA_ADDR_WR				0x4002213C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_8_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 8 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.87 CHN_9_DMA_ADDR_RD				0x40022140
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_9_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 9 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.88 CHN_9_DMA_ADDR_WR				0x40022144
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_9_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 9 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.89 CHN_10_DMA_ADDR_RD				0x40022148
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_10_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 10 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.90 CHN_10_DMA_ADDR_WR				0x4002214C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_10_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 10 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.91 CHN_11_DMA_ADDR_RD				0x40022150
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_11_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 11 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.92 CHN_11_DMA_ADDR_WR				0x40022154
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_11_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 11 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.93 CHN_12_DMA_ADDR_RD				0x40022158
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_12_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 12 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.94 CHN_12_DMA_ADDR_WR				0x4002215C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_12_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 12 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.95 CHN_13_DMA_ADDR_RD				0x40022160
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_13_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 13 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.96 CHN_13_DMA_ADDR_WR				0x40022164
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_13_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 13 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.97 CHN_14_DMA_ADDR_RD				0x40022168
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_14_DMA_ADDR_RD	ro	0xFFFFFFFF	channel 14 dma addr read monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.98 CHN_14_DMA_ADDR_WR				0x4002216C
Discription:				
Bits	Field Name	Acc	Default	Description
31 : 0	CHN_14_DMA_ADDR_WR	ro	0xFFFFFFFF	channel 14 dma addr write monitor write bit0 to 1'b1, dma addr monitor value reset to 0xffffff

16.99 CHN_LLI_NODE_COUNTER_0				0x40022200
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.100 CHN_LLI_NODE_COUNTER_1				0x40022204
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.101 CHN_LLI_NODE_COUNTER_2	0x40022208

Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.102 CHN_LLI_NODE_COUNTER_3				0x4002220C
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.103 CHN_LLI_NODE_COUNTER_4				0x40022210
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.104 CHN_LLI_NODE_COUNTER_5				0x40022214
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.105 CHN_LLI_NODE_COUNTER_6				0x40022218
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.106 CHN_LLI_NODE_COUNTER_7				0x4002221C
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.107 CHN_LLI_NODE_COUNTER_8				0x40022220
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.108 CHN_LLI_NODE_COUNTER_9				0x40022224
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.109 CHN_LLI_NODE_COUNTER_10				0x40022228
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.110 CHN_LLI_NODE_COUNTER_11				0x4002222C
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.111 CHN_LLI_NODE_COUNTER_12				0x40022230
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

16.112 CHN_LLI_NODE_COUNTER_13				0x40022234
Discription:				
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_NODE_COUNTER	rw	0x0	Linklist Node Counter of Dma Channel n

17 cpu_p_trap

17.1 addr_in_0				0x40030000
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.2 addr_in_1				0x40030004
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.3 addr_in_2				0x40030008
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.4 addr_in_3				0x4003000C
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.5 addr_in_4				0x40030010
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.6 addr_in_5				0x40030014
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.7 addr_in_6				0x40030018
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.8 addr_in_7				0x4003001C
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

17.9 en				0x40030080
Bits	Field Name	Acc	Default	Description
7 : 0	en	rw	0x0	trap enable for 8 channels
31 : 8	en_reserved_0	ro	0x0	

17.10 trap_out_base				0x40030084
Bits	Field Name	Acc	Default	Description
6 : 0	base_out_reserved	ro	0x0	
31 : 7	base_out	rw	0x0	trap out base address, only high 25 bits are valid.

18 cpu_s0_trap

18.1 addr_in_0				0x40031000
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.2 addr_in_1				0x40031004
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.3 addr_in_2				0x40031008
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.4 addr_in_3				0x4003100C
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.5 addr_in_4				0x40031010
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.6 addr_in_5				0x40031014
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.7 addr_in_6				0x40031018
Bits	Field Name	Acc	Default	Description

19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.8 addr_in_7				0x4003101C
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

18.9 en				0x40031080
Bits	Field Name	Acc	Default	Description
7 : 0	en	rw	0x0	trap enable for 8 channels
31 : 8	en_reserved_0	ro	0x0	

18.10 trap_out_base				0x40031084
Bits	Field Name	Acc	Default	Description
6 : 0	base_out_reserved	ro	0x0	
31 : 7	base_out	rw	0x0	trap out base address, only high 25 bits are valid.

19 uart0

19.1 txrx_fifo_divl				0x40032000
Bits	Field Name	Acc	Default	Description
7 : 0	txrx_data_divl	rw	0x0	Tx data(when write data), Rx data(when read data), Divisor latch(low) config when divisor latch is enable.

19.2 int_en_divh				0x40032004
Discription: Interrupt enable register; divisor latch high when divisor latch is enable.				
Bits	Field Name	Acc	Default	Description
0	rx_int_en_divh_0	rw	0x0	Enable Received Data Available Interrupt; Divisor latch high bit 0 when divisor latch is enable.
1	tx_int_en_divh_1	rw	0x0	Enable Transmit Holding Register Empty Interrupt; Divisor latch high bit 1 when divisor latch is enable.
2	lstatus_int_en_divh_2	rw	0x0	Enable Receiver Line Status Interrupt; Divisor latch high bit 2 when divisor latch is enable.
3	mstatus_int_en_divh_3	rw	0x0	Enable Modem Status Interrupt; Divisor latch high bit 3 when divisor latch is enable.
4	divh_4	rw	0x0	Divisor latch high bit 4 when divisor latch is enable.
5	timeout_int_en_divh_5	rw	0x0	Enable Timeout Interrupt; Divisor latch high bit 5 when divisor latch is enable.
6	divh_6	rw	0x0	Divisor latch high bit 6 when divisor latch is enable.
7	ptime_divh_7	rw	0x0	Programmable THRE Interrupt Mode Enable; Divisor latch high bit 7 when divisor latch is enable.

19.3 int_id_fifo_ctrl				0x40032008
Discription: Interrupt identity register or fifo control register				
Bits	Field Name	Acc	Default	Description
0	int_id_0_fifo_en	rw	0x1	Read: Interrupt identity bit 0, reset value is '0'; Write: Fifo enable, reset value is '1'
1	int_id_1_rx_reset	rw	0x0	Read: Interrupt identity bit 1; Write: Receiver reset
2	int_id_2_tx_reset	rw	0x0	Read: Interrupt identity bit 2; Write: Transmit reset
3	int_id_3_dma_mode	rw	0x0	Read: Interrupt identity bit 3; Write: Dma mode enable
7 : 6	fifo_enabled	ro	0x3	

19.4 line_ctrl				0x4003200C
Bits	Field Name	Acc	Default	Description
1 : 0	data_length_sel	rw	0x3	0: 5 bit 1: 6 bit 2: 7 bit 3: 8 bit
2	stop	rw	0x0	Number of stop bits
3	parity_en	rw	0x0	Parity enable
4	even_parity_sel	rw	0x0	'1' select even parity
6	break_irq	rw	0x0	
7	div_latch_acc	rw	0x0	Divisor Latch access enable
8	div_mode	rw	0x0	
9	break_en	rw	0x0	

19.5 modem_ctrl				0x40032010
Bits	Field Name	Acc	Default	Description
0	dtr	rw	0x0	Data terminal ready
1	rts	rw	0x0	Request to send
2	out1	rw	0x0	
3	out2	rw	0x0	
4	loopback	rw	0x0	
5	auto_flow_ctrl_enable	rw	0x0	Auto flow control enable
6	serial_infrared_radiation_en	rw	0x0	Serial infrared radiation enable
7	auto_det	rw	0x0	
8	clk_p	rw	0x0	1: 48M 0: 24M

19.6 line_status				0x40032014
Bits	Field Name	Acc	Default	Description
0	rx_data_ready	ro	0x0	Rx Data ready(rxfifo not empty)
1	rx_overflow	ro	0x0	Overflow error
5	tx_fifo_empty	ro	0x0	Tx fifo empty or almost empty
6	tx_empty	ro	0x0	Tx fifo empty and shift register empty
7	rx_fifo_error	ro	0x0	Rx fifo error
8	trigger_ready	ro	0x0	Trigger ready(rxfifo not empty)

19.7 modem_status				0x40032018
Bits	Field Name	Acc	Default	Description
0	delayed_cts	ro	0x0	
4	cts	ro	0x0	

19.8 scratchpad				0x4003201C
Bits	Field Name	Acc	Default	Description
6 : 0	scratchpad	rw	0x0	not used

19.9 fifo_status				0x40032020
Bits	Field Name	Acc	Default	Description
7 : 0	tx_count	ro	0x0	TX fifo count
16 : 9	rx_count	ro	0x0	RX fifo count
18	tx_fifo_empty	ro	0x1	TX fifo empty
19	tx_fifo_full	ro	0x0	TX fifo full
20	rx_fifo_empty	ro	0x1	RX fifo empty
21	rx_fifo_full	ro	0x0	RX fifo full

19.10 fifo_trigger				0x40032024
Bits	Field Name	Acc	Default	Description
7 : 0	rx_trigger_th	rw	0x10	
16 : 9	tx_trigger_th	rw	0x10	
18	rx_tout_trig_en	rw	0x0	Enable rx legacy dma req triggered when timeout

19.11 divisor_latch2				0x40032028
Bits	Field Name	Acc	Default	Description
7 : 0	divisor_adjust_coef	rw	0x0	accuray adjust baudrate

19.12 counter_real				0x4003203C
Bits	Field Name	Acc	Default	Description
19 : 0	counter_real	ro	0x0	

19.13 div_real_slow				0x40032040
Bits	Field Name	Acc	Default	Description
31 : 0	div_real_slow	rw	0x1000E	when clk_p is 0: slow div_real = div_real_slow[15:1]; slow div_adjust_real = div_real_slow[31:24]; when clk_p is 1: slow div_real = div_real_slow[15:0]; slow div_adjust_real = div_real_slow[23:16];

19.14 div_real_fast				0x40032044
Bits	Field Name	Acc	Default	Description
31 : 0	div_real_fast	rw	0x660001	when clk_p is 0: fast div_real = div_real_fast[15:1]; fast div_adjust_real = div_real_fast[31:24]; when clk_p is 1: fast div_real = div_real_fast[15:0]; fast div_adjust_real = div_real_fast[23:16];

19.15 tports_sel				0x40032048
Bits	Field Name	Acc	Default	Description
7 : 0	tports_sel	rw	0x0	Tports Select

20 uart1

20.1 txrx_fifo_divl				0x40033000
Bits	Field Name	Acc	Default	Description
7 : 0	txrx_data_divl	rw	0x0	Tx data(when write data), Rx data(when read data), Divisor latch(low) config when divisor latch is enable.

20.2 int_en_divh				0x40033004
Discription: Interrupt enable register; divisor latch high when divisor latch is enable.				
Bits	Field Name	Acc	Default	Description
0	rx_int_en_divh_0	rw	0x0	Enable Received Data Available Interrupt; Divisor latch high bit 0 when divisor latch is enable.
1	tx_int_en_divh_1	rw	0x0	Enable Transmit Holding Register Empty Interrupt; Divisor latch high bit 1 when divisor latch is enable.
2	lstatus_int_en_divh_2	rw	0x0	Enable Receiver Line Status Interrupt; Divisor latch high bit 2 when divisor latch is enable.
3	mstatus_int_en_divh_3	rw	0x0	Enable Modem Status Interrupt; Divisor latch high bit 3 when divisor latch is enable.
4	divh_4	rw	0x0	Divisor latch high bit 4 when divisor latch is enable.
5	timeout_int_en_divh_5	rw	0x0	Enable Timeout Interrupt; Divisor latch high bit 5 when divisor latch is enable.

6	divh_6	rw	0x0	Divisor latch high bit 6 when divisor latch is enable.
7	ptime_divh_7	rw	0x0	Programmable THRE Interrupt Mode Enable; Divisor latch high bit 7 when divisor latch is enable.

20.3 int_id_fifo_ctrl				0x40033008
Discription: Interrupt identity register or fifo control register				
Bits	Field Name	Acc	Default	Description
0	int_id_0_fifo_en	rw	0x1	Read: Interrupt identity bit 0, reset value is '0'; Write: Fifo enable, reset value is '1'
1	int_id_1_rx_reset	rw	0x0	Read: Interrupt identity bit 1; Write: Receiver reset
2	int_id_2_tx_reset	rw	0x0	Read: Interrupt identity bit 2; Write: Transmit reset
3	int_id_3_dma_mode	rw	0x0	Read: Interrupt identity bit 3; Write: Dma mode enable
7 : 6	fifo_enabled	ro	0x3	

20.4 line_ctrl				0x4003300C
Bits	Field Name	Acc	Default	Description
1 : 0	data_length_sel	rw	0x3	0: 5 bit 1: 6 bit 2: 7 bit 3: 8 bit
2	stop	rw	0x0	Number of stop bits
3	parity_en	rw	0x0	Parity enable
4	even_parity_sel	rw	0x0	'1' select even parity
6	break_irq	rw	0x0	
7	div_latch_acc	rw	0x0	Divisor Latch access enable
8	div_mode	rw	0x0	
9	break_en	rw	0x0	

20.5 modem_ctrl				0x40033010
Bits	Field Name	Acc	Default	Description
0	dtr	rw	0x0	Data terminal ready
1	rts	rw	0x0	Request to send
2	out1	rw	0x0	
3	out2	rw	0x0	
4	loopback	rw	0x0	
5	auto_flow_ctrl_enable	rw	0x0	Auto flow control enable
6	serial_infrared_radiation_en	rw	0x0	Serial infrared radiation enable
7	auto_det	rw	0x0	
8	clk_p	rw	0x0	1: 48M 0: 24M

20.6 line_status				0x40033014
Bits	Field Name	Acc	Default	Description
0	rx_data_ready	ro	0x0	Rx Data ready(rxfifo not empty)
1	rx_overflow	ro	0x0	Overflow error
5	tx_fifo_empty	ro	0x0	Tx fifo empty or almost empty
6	tx_empty	ro	0x0	Tx fifo empty and shift register empty
7	rx_fifo_error	ro	0x0	Rx fifo error
8	trigger_ready	ro	0x0	Trigger ready(rxfifo not empty)

20.7 modem_status				0x40033018
Bits	Field Name	Acc	Default	Description
0	delayed_cts	ro	0x0	
4	cts	ro	0x0	

20.8 scratchpad				0x4003301C
Bits	Field Name	Acc	Default	Description
6 : 0	scratchpad	rw	0x0	not used

20.9 fifo_status				0x40033020
Bits	Field Name	Acc	Default	Description
7 : 0	tx_count	ro	0x0	TX fifo count
16 : 9	rx_count	ro	0x0	RX fifo count
18	tx_fifo_empty	ro	0x1	TX fifo empty
19	tx_fifo_full	ro	0x0	TX fifo full
20	rx_fifo_empty	ro	0x1	RX fifo empty
21	rx_fifo_full	ro	0x0	RX fifo full

20.10 fifo_trigger				0x40033024
Bits	Field Name	Acc	Default	Description
7 : 0	rx_trigger_th	rw	0x10	
16 : 9	tx_trigger_th	rw	0x10	
18	rx_tout_trig_en	rw	0x0	Enable rx legacy dma req triggered when timeout

20.11 divisor_latch2				0x40033028
Bits	Field Name	Acc	Default	Description
7 : 0	divisor_adjust_coef	rw	0x0	accuray adjust baudrate

20.12 counter_real				0x4003303C
Bits	Field Name	Acc	Default	Description
19 : 0	counter_real	ro	0x0	

20.13 div_real_slow				0x40033040
Bits	Field Name	Acc	Default	Description

31 : 0	div_real_slow	rw	0x1000E	when clk_p is 0: slow div_real = div_real_slow[15:1];
				slow div_adjust_real = div_real_slow[31:24];
				when clk_p is 1: slow div_real = div_real_slow[15:0];
				slow div_adjust_real = div_real_slow[23:16];

20.14 div_real_fast				0x40033044
Bits	Field Name	Acc	Default	Description

31 : 0	div_real_fast	rw	0x660001	when clk_p is 0: fast div_real = div_real_fast[15:1];
				fast div_adjust_real = div_real_fast[31:24];
				when clk_p is 1: fast div_real = div_real_fast[15:0];
				fast div_adjust_real = div_real_fast[23:16];

20.15 tports_sel				0x40033048
Bits	Field Name	Acc	Default	Description
7 : 0	tports_sel	rw	0x0	Tports Select

21 uart2

21.1 txrx_fifo_divl				0x40034000
Bits	Field Name	Acc	Default	Description
7 : 0	txrx_data_divl	rw	0x0	Tx data(when write data), Rx data(when read data), Divisor latch(low) config when divisor latch is enable.

21.2 int_en_divh				0x40034004
Discription: Interrupt enable register; divisor latch high when divisor latch is enable.				
Bits	Field Name	Acc	Default	Description
0	rx_int_en_divh_0	rw	0x0	Enable Received Data Available Interrupt; Divisor latch high bit 0 when divisor latch is enable.
1	tx_int_en_divh_1	rw	0x0	Enable Transmit Holding Register Empty Interrupt; Divisor latch high bit 1 when divisor latch is enable.
2	lstatus_int_en_divh_2	rw	0x0	Enable Receiver Line Status Interrupt; Divisor latch high bit 2 when divisor latch is enable.
3	mstatus_int_en_divh_3	rw	0x0	Enable Modem Status Interrupt; Divisor latch high bit 3 when divisor latch is enable.
4	divh_4	rw	0x0	Divisor latch high bit 4 when divisor latch is enable.
5	timeout_int_en_divh_5	rw	0x0	Enable Timeout Interrupt; Divisor latch high bit 5 when divisor latch is enable.
6	divh_6	rw	0x0	Divisor latch high bit 6 when divisor latch is enable.
7	ptime_divh_7	rw	0x0	Programmable THRE Interrupt Mode Enable; Divisor latch high bit 7 when divisor latch is enable.

21.3 int_id_fifo_ctrl				0x40034008
Discription: Interrupt identity register or fifo control register				
Bits	Field Name	Acc	Default	Description
0	int_id_0_fifo_en	rw	0x1	Read: Interrupt identity bit 0, reset value is '0'; Write: Fifo enable, reset value is '1'
1	int_id_1_rx_reset	rw	0x0	Read: Interrupt identity bit 1; Write: Receiver reset
2	int_id_2_tx_reset	rw	0x0	Read: Interrupt identity bit 2; Write: Transmit reset
3	int_id_3_dma_mode	rw	0x0	Read: Interrupt identity bit 3; Write: Dma mode enable
7 : 6	fifo_enabled	ro	0x3	

21.4 line_ctrl				0x4003400C
Bits	Field Name	Acc	Default	Description
1 : 0	data_length_sel	rw	0x3	0: 5 bit 1: 6 bit 2: 7 bit 3: 8 bit
2	stop	rw	0x0	Number of stop bits
3	parity_en	rw	0x0	Parity enable
4	even_parity_sel	rw	0x0	'1' select even parity
6	break_irq	rw	0x0	
7	div_latch_acc	rw	0x0	Divisor Latch access enable
8	div_mode	rw	0x0	
9	break_en	rw	0x0	

21.5 modem_ctrl				0x40034010
Bits	Field Name	Acc	Default	Description
0	dtr	rw	0x0	Data terminal ready
1	rts	rw	0x0	Request to send
2	out1	rw	0x0	
3	out2	rw	0x0	
4	loopback	rw	0x0	
5	auto_flow_ctrl_enable	rw	0x0	Auto flow control enable
6	serial_infrared_radiation_en	rw	0x0	Serial infrared radiation enable
7	auto_det	rw	0x0	
8	clk_p	rw	0x0	1: 48M 0: 24M

21.6 line_status	0x40034014

Bits	Field Name	Acc	Default	Description
0	rx_data_ready	ro	0x0	Rx Data ready(rxfifo not empty)
1	rx_overflow	ro	0x0	Overflow error
5	tx_fifo_empty	ro	0x0	Tx fifo empty or almost empty
6	tx_empty	ro	0x0	Tx fifo empty and shift register empty
7	rx_fifo_error	ro	0x0	Rx fifo error
8	trigger_ready	ro	0x0	Trigger ready(rxfifo not empty)

21.7 modem_status				0x40034018
Bits	Field Name	Acc	Default	Description
0	delayed_cts	ro	0x0	
4	cts	ro	0x0	

21.8 scratchpad				0x4003401C
Bits	Field Name	Acc	Default	Description
6 : 0	scratchpad	rw	0x0	not used

21.9 fifo_status				0x40034020
Bits	Field Name	Acc	Default	Description
7 : 0	tx_count	ro	0x0	TX fifo count
16 : 9	rx_count	ro	0x0	RX fifo count
18	tx_fifo_empty	ro	0x1	TX fifo empty
19	tx_fifo_full	ro	0x0	TX fifo full
20	rx_fifo_empty	ro	0x1	RX fifo empty
21	rx_fifo_full	ro	0x0	RX fifo full

21.10 fifo_trigger				0x40034024
Bits	Field Name	Acc	Default	Description
7 : 0	rx_trigger_th	rw	0x10	
16 : 9	tx_trigger_th	rw	0x10	
18	rx_tout_trig_en	rw	0x0	Enable rx legacy dma req triggered when timeout

21.11 divisor_latch2				0x40034028
Bits	Field Name	Acc	Default	Description
7 : 0	divisor_adjust_coef	rw	0x0	accuray adjust baudrate

21.12 counter_real				0x4003403C
Bits	Field Name	Acc	Default	Description
19 : 0	counter_real	ro	0x0	

21.13 div_real_slow				0x40034040
Bits	Field Name	Acc	Default	Description
31 : 0	div_real_slow	rw	0x1000E	when clk_p is 0: slow div_real = div_real_slow[15:1]; slow div_adjust_real = div_real_slow[31:24]; when clk_p is 1: slow div_real = div_real_slow[15:0]; slow div_adjust_real = div_real_slow[23:16];

21.14 div_real_fast				0x40034044
Bits	Field Name	Acc	Default	Description

31 : 0	div_real_fast	rw	0x660001	when clk_p is 0: fast div_real = div_real_fast[15:1];
				fast div_adjust_real = div_real_fast[31:24];
				when clk_p is 1: fast div_real = div_real_fast[15:0];
				fast div_adjust_real = div_real_fast[23:16];

21.15 tports_sel				0x40034048
Bits	Field Name	Acc	Default	Description
7 : 0	tports_sel	rw	0x0	Tports Select

22 mailbox

22.1 app2emb_trigger				0x40035000
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_trigger	rw	0x0	write to set

22.2 emb2app_rawstatus				0x40035004
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_rawstatus	ro	0x0	

22.3 emb2app_ack				0x40035008
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_ack	wo	0x0	write to clear

22.4 emb2app_unmaskset				0x4003500C
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_unmaskset	rw	0x0	write to set

22.5 emb2app_unmaskclear				0x40035010
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_unmaskclear	rw	0x0	write to clear

22.6 emb2app_linesel_low				0x40035014
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_linesel_low	rw	0x0	

22.7 emb2app_linesel_high				0x40035018
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_linesel_high	rw	0x0	

22.8 emb2app_status				0x4003501C
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_status	ro	0x0	

22.9 mutex0				0x40035040
Bits	Field Name	Acc	Default	Description
0	mutex0	rw	0x1	read to clear

22.10 mutex1				0x40035044
Bits	Field Name	Acc	Default	Description
0	mutex1	rw	0x1	read to clear

22.11 mutex2				0x40035048
Bits	Field Name	Acc	Default	Description
0	mutex2	rw	0x1	read to clear

22.12 mutex3				0x4003504C
Bits	Field Name	Acc	Default	Description
0	mutex3	rw	0x1	read to clear

22.13 mutex4				0x40035050
Bits	Field Name	Acc	Default	Description
0	mutex4	rw	0x1	read to clear

22.14 mutex5				0x40035054
Bits	Field Name	Acc	Default	Description
0	mutex5	rw	0x1	read to clear

22.15 mutex6				0x40035058
Bits	Field Name	Acc	Default	Description
0	mutex6	rw	0x1	read to clear

22.16 mutex7				0x4003505C
Bits	Field Name	Acc	Default	Description
0	mutex7	rw	0x1	read to clear

22.17 comreg0				0x40035080
Bits	Field Name	Acc	Default	Description
31 : 0	comreg0	rw	0x0	common register

22.18 comreg1				0x40035084
Bits	Field Name	Acc	Default	Description
31 : 0	comreg1	rw	0x0	common register

22.19 comreg2				0x40035088
Bits	Field Name	Acc	Default	Description
31 : 0	comreg2	rw	0x0	common register

22.20 comreg3				0x4003508C
Bits	Field Name	Acc	Default	Description
31 : 0	comreg3	rw	0x0	common register

22.21 comreg4				0x40035090
Bits	Field Name	Acc	Default	Description
31 : 0	comreg4	rw	0x0	common register

22.22 comreg5				0x40035094
Bits	Field Name	Acc	Default	Description
31 : 0	comreg5	rw	0x0	common register

22.23 comreg6				0x40035098
Bits	Field Name	Acc	Default	Description
31 : 0	comreg6	rw	0x0	common register

22.24 comreg7				0x4003509C
Bits	Field Name	Acc	Default	Description
31 : 0	comreg7	rw	0x0	common register

22.25 emb2app_trigger				0x40035100
Bits	Field Name	Acc	Default	Description
31 : 0	emb2app_trigger	rw	0x0	write to set

22.26 app2emb_rawstatus				0x40035104
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_rawstatus	ro	0x0	

22.27 app2emb_ack				0x40035108
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_ack	wo	0x0	write to clear

22.28 app2emb_unmaskset				0x4003510C
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_unmaskset	rw	0x0	write to set

22.29 app2emb_unmaskclear				0x40035110
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_unmaskclear	rw	0x0	write to clear

22.30 app2emb_linesel_low				0x40035114
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_linesel_low	rw	0x0	

22.31 app2emb_linesel_high				0x40035118
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_linesel_high	rw	0x0	

22.32 app2emb_status				0x4003511C
Bits	Field Name	Acc	Default	Description
31 : 0	app2emb_status	ro	0x0	

22.33 g1_app2emb_trigger				0x40035200
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_trigger	rw	0x0	write to set

22.34 g1_emb2app_rawstatus				0x40035204
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_rawstatus	ro	0x0	

22.35 g1_emb2app_ack				0x40035208
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_ack	wo	0x0	write to clear

22.36 g1_emb2app_unmaskset				0x4003520C
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_unmaskset	rw	0x0	write to set

22.37 g1_emb2app_unmaskclear				0x40035210
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_unmaskclear	rw	0x0	write to clear

22.38 g1_emb2app_linesel_low				0x40035214
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_linesel_low	rw	0x0	

22.39 g1_emb2app_linesel_high				0x40035218
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_linesel_high	rw	0x0	

22.40 g1_emb2app_status				0x4003521C
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_status	ro	0x0	

22.41 g1_mutex0				0x40035240
Bits	Field Name	Acc	Default	Description
0	g1_mutex0	rw	0x1	read to clear

22.42 g1_mutex1				0x40035244
Bits	Field Name	Acc	Default	Description
0	g1_mutex1	rw	0x1	read to clear

22.43 g1_mutex2				0x40035248
Bits	Field Name	Acc	Default	Description
0	g1_mutex2	rw	0x1	read to clear

22.44 g1_mutex3				0x4003524C
Bits	Field Name	Acc	Default	Description
0	g1_mutex3	rw	0x1	read to clear

22.45 g1_mutex4				0x40035250
Bits	Field Name	Acc	Default	Description
0	g1_mutex4	rw	0x1	read to clear

22.46 g1_mutex5				0x40035254
Bits	Field Name	Acc	Default	Description
0	g1_mutex5	rw	0x1	read to clear

22.47 g1_mutex6				0x40035258
Bits	Field Name	Acc	Default	Description
0	g1_mutex6	rw	0x1	read to clear

22.48 g1_mutex7				0x4003525C
Bits	Field Name	Acc	Default	Description
0	g1_mutex7	rw	0x1	read to clear

22.49 g1_comreg0				0x40035280
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg0	rw	0x0	common register

22.50 g1_comreg1				0x40035284
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg1	rw	0x0	common register

22.51 g1_comreg2				0x40035288
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg2	rw	0x0	common register

22.52 g1_comreg3				0x4003528C
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg3	rw	0x0	common register

22.53 g1_comreg4				0x40035290
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg4	rw	0x0	common register

22.54 g1_comreg5				0x40035294
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg5	rw	0x0	common register

22.55 g1_comreg6				0x40035298
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg6	rw	0x0	common register

22.56 g1_comreg7				0x4003529C
Bits	Field Name	Acc	Default	Description
31 : 0	g1_comreg7	rw	0x0	common register

22.57 g1_emb2app_trigger				0x40035300
Bits	Field Name	Acc	Default	Description
31 : 0	g1_emb2app_trigger	rw	0x0	write to set

22.58 g1_app2emb_rawstatus				0x40035304
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_rawstatus	ro	0x0	

22.59 g1_app2emb_ack				0x40035308
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_ack	wo	0x0	write to clear

22.60 g1_app2emb_unmaskset				0x4003530C
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_unmaskset	rw	0x0	write to set

22.61 g1_app2emb_unmaskclear				0x40035310
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_unmaskclear	rw	0x0	write to clear

22.62 g1_app2emb_linesel_low				0x40035314
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_linesel_low	rw	0x0	

22.63 g1_app2emb_linesel_high				0x40035318
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_linesel_high	rw	0x0	

22.64 g1_app2emb_status				0x4003531C
Bits	Field Name	Acc	Default	Description
31 : 0	g1_app2emb_status	ro	0x0	

23 sd_host

23.1 ctrl				0x40036000
Bits	Field Name	Acc	Default	Description
2 : 0	endian	rw	0x0	Endian Type SD TX FIFO Endian 3'b000: {pwdata_byte[3],pwdata_byte[2],pwdata_byte[1],pwdata_byte 3'b001: {pwdata_byte[0],pwdata_byte[1],pwdata_byte[2],pwdata_byte 3'b010: {pwdata_byte[1],pwdata_byte[0],pwdata_byte[3],pwdata_byte 3'b011: pwdata_bit[0-31] 3'b100: {pwdata_byte[2],pwdata_byte[3],pwdata_byte[0],pwdata_byte default: 32'h0
3	reset_n	rw	0x1	Software reset, active Low Software reset for TX and RX FIFOs of Sd host, active low
4	dma_rx_en	rw	0x0	Enable rx dma req triggered when rxfifo level arrived dma_rx_th
5	dma_tx_en	rw	0x0	Enable tx dma req triggered when txfifo level arrived dma_tx_th
11 : 8	dma_rx_th	rw	0x0	Rx dma req triggered threshold
19 : 16	dma_tx_th	rw	0x0	Tx dma req triggered threshold

23.2 txfifo_entry				0x40036008
Bits	Field Name	Acc	Default	Description

31 : 0	txfifo_entry	wo	0x0	Tx FIFO Write Entry Write data for 8x32 bit TxFIFO

23.3 rxfifo_entry				0x4003600C
Bits	Field Name	Acc	Default	Description
31 : 0	rxfifo_entry	ro	0x0	Rx FIFO Read Entry Read data from 8x32 bit RxFIFO

23.4 cfg				0x40036800
Bits	Field Name	Acc	Default	Description
0	cmd_start	rw	0x0	Configuration begin
4	response_en	rw	0x0	Response enable
6 : 5	response_type	rw	0x0	Response select register 3:Reserved 2:R2 1:R3 0:Other
8	read_write_en	rw	0x0	Read/write enable
9	write_mode	rw	0x0	'1' means write operation,'0' means read operation. 0:READ 1:WRITE
10	multi_block_mode	rw	0x0	'1' means multiple block data operation. 0:SIMPLE 1:MULTIPLE
16	auto_cmd12_en	rw	0x1	Auto cmd12 enable

23.5 status				0x40036804
Bits	Field Name	Acc	Default	Description
0	sd_not_over	ro	0x0	'1' means the sd/mmc operation is not over.
1	busy	ro	0x0	'1' means sd modem is busy.
2	data_line_busy	ro	0x0	'1' means the data line is busy.
8	response_error	ro	0x0	Response CRC checks error register, '1' means response CRC check error.
9	no_response_error	ro	0x0	'1' means the card has no response to command.

23.6 cmd_index				0x40036808
Bits	Field Name	Acc	Default	Description
5 : 0	command	rw	0x0	Sd command register.

23.7 cmd_arg				0x4003680C
Bits	Field Name	Acc	Default	Description
31 : 0	argument	rw	0x0	Sd command argument register, write data to the Sd card.

23.8 rsp_index				0x40036810
Bits	Field Name	Acc	Default	Description
5 : 0	resp	ro	0x0	Sd response index register.

23.9 rsp_arg3				0x40036814
Bits	Field Name	Acc	Default	Description
31 : 0	arg3	ro	0x0	Response argument of R1, R3 and R6, or 127 to 96 bit response argument of R2.

23.10 rsp_arg2				0x40036818
Bits	Field Name	Acc	Default	Description
31 : 0	arg2	ro	0x0	95 to 64 bit response argument of R2.

23.11 rsp_arg1				0x4003681C
Bits	Field Name	Acc	Default	Description
31 : 0	arg1	ro	0x0	63 to 32 bit response argument of R2.

23.12 rsp_arg0				0x40036820
Bits	Field Name	Acc	Default	Description
31 : 0	arg0	ro	0x0	31 to 0 bit response argument of R2.

23.13 dline_width				0x40036824
Bits	Field Name	Acc	Default	Description
3 : 0	dline_width	rw	0x0	Sd data width: 0x1: 1 data line 0x2: 2 reserved 0x4: 4 data lines 0x8: 8 data lines

23.14 block_size				0x40036828
Bits	Field Name	Acc	Default	Description
3 : 0	block_size	rw	0x0	Sd size of one block: 0-1:reserved 2: 1 word 3: 2 words 4: 4 words 5: 8 words 6: 16 words 11: 512 words 12-15 reserved

23.15 block_cnt				0x4003682C
Bits	Field Name	Acc	Default	Description
15 : 0	block_cnt	rw	0x0	Block number that wants to transfer.

23.16 int_status				0x40036830
Bits	Field Name	Acc	Default	Description
4	data_done_int	ro	0x0	Read write correct int raw status.
5	txdma_done_int	ro	0x0	'1' means rx dma done.
6	rxdma_done_int	ro	0x0	'1' means tx dma done.
14	data_done_cause	ro	0x0	Read write correct int status after masking
15	txdma_done_cause	ro	0x0	'1' means rx dma done is the source of interrupt.
16	rx_dma_done_cause	ro	0x0	'1' means tx dma done is the source of interrupt.

23.17 int_enable				0x40036834
Bits	Field Name	Acc	Default	Description
4	data_done_en	rw	0x0	When data transmission is over, '0' means INT is disable.
5	txdma_done_en	rw	0x0	When tx dma done, '0' means INT is disabled.
6	rxdma_done_en	rw	0x0	When rx dma done, '0' means INT is disabled.

23.18 int_clear	0x40036838

Bits	Field Name	Acc	Default	Description
4	data_done_clear	wo	0x0	Write a '1' to this bit to clear the data_over interrupt.
5	txdma_done_clear	wo	0x0	Write a '1' to this bit to clear the txdma_done interrupt.
6	rxdma_done_clear	wo	0x0	Write a '1' to this bit to clear the rxdma_done interrupt.

23.19 trans_speed				0x4003683C
Bits	Field Name	Acc	Default	Description
11 : 0	trans_speed	rw	0x9	Mclk = Pclk/(2*(trans_speed +1)).

23.20 mclk_ctrl				0x40036840
Bits	Field Name	Acc	Default	Description
3 : 0	mclk_delay	rw	0x0	This register may delay the mclk output. When MCLK_DELAY = n, Mclk is outputted with n Pclk.
4	mclk_inv_en	rw	0x0	Mclk invert enable
8	mclk_disable	rw	0x1	Mclk disable

23.21 fifo_status				0x40036844
Bits	Field Name	Acc	Default	Description
11 : 8	rxfifo_level	ro	0x0	Rx fifo level monitor
19 : 16	txfifo_level	ro	0x8	Tx fifo level monitor

24 aud_proc

24.1 aud_proc_function_id				0x40037000
Bits	Field Name	Acc	Default	Description
15 : 0	rev_id	ro	0x0	revision id
31 : 16	func_id	ro	0x302	function id

24.2 aud_proc_cfg				0x40037004
Bits	Field Name	Acc	Default	Description
0	aud_proc_enable	rw	0x0	enable aud_proc
1	aud_proc_adc_path_reset	rw	0x0	reset aud_proc adc path
2	aud_proc_dac_path_reset	rw	0x0	reset aud_proc dac path
6	loopback_mode	rw	0x0	aud_proc loopback test mode, from rx path to tx path
7	high_speed_mode	rw	0x0	aud_proc high speed mode enable
8	sine_tone_en	rw	0x0	1:select sine tone output
9	auto_clock_gating_en	rw	0x1	enable auto clock gating to save power
10	rx2tx_loop_en	rw	0x0	1:enable rx to tx path, rx path sample rate must be same as tx path when this bit is enable
15 : 14	tx_lr_fifo_share_en	rw	0x0	1: tx L/R share one buffer. 0: tx L/R two buffer
17 : 16	rx_lr_fifo_share_en	rw	0x0	1: rx L/R share one buffer. 0: rx L/R two buffer
19 : 18	rxtx_lr_fifo_share_en	rw	0x0	1: rxtx L/R share one buffer. 0: rxtx L/R two buffer
20	txrx_lr_fifo_share_en	rw	0x0	1:txrx channel L/R share mode. 0: txrx channel
21	txrx_burst_mode	rw	0x0	1: txrx channel burst incr 4 mode, 0: txrx channel single mode
22	rxtx_burst_mode	rw	0x0	1: rxtx channel burst incr 4 mode, 0: rxtx channel single mode
23	tx_burst_mode	rw	0x0	1: tx channel burst incr 4 mode, 0: tx channel single mode
24	rx_burst_mode	rw	0x0	1: rx channel burst incr 4 mode, 0: rx channel single mode
31 : 25	monitor_sel	rw	0x0	

24.3 aud_proc_tx_cfg_ch0				0x40037008
Bits	Field Name	Acc	Default	Description
0	tx_en_ch_0	rw	0x0	channel enable
1	tx_fifo_flush_ch_0	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
3 : 2	tx_mode_ch_01	rw	0x0	channel 0and1 bit mode 2'b00: 16-bit mode 2'b01: 24-bit mode 2'b1x: 32-bit mode
5 : 4	tx_fifo_stat_ch_0	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full

24.4 aud_proc_tx_cfg_ch1				0x4003700C
Bits	Field Name	Acc	Default	Description
0	tx_en_ch_1	rw	0x0	channel enable
1	tx_fifo_flush_ch_1	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
4 : 3	tx_fifo_stat_ch_1	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full

24.5 aud_proc_tx_cfg_ch2				0x40037010
Bits	Field Name	Acc	Default	Description
0	tx_en_ch_2	rw	0x0	channel enable
1	tx_fifo_flush_ch_2	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
3 : 2	tx_mode_ch_23	rw	0x0	channel 2and3 bit mode 2'b00: 16-bit mode 2'b01: 24-bit mode 2'b1x: 32-bit mode
5 : 4	tx_fifo_stat_ch_2	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterhalf 2'b10: halff̃ull 2'b11: full

24.6 aud_proc_tx_cfg_ch3				0x40037014
Bits	Field Name	Acc	Default	Description
0	tx_en_ch_3	rw	0x0	channel enable
1	tx_fifo_flush_ch_3	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
4 : 3	tx_fifo_stat_ch_3	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full

24.7 aud_proc_rx_cfg_ch01				0x40037018
Bits	Field Name	Acc	Default	Description
0	rx_en_ch_0	rw	0x0	channel enable

1	rx_fifo_flush_ch_01	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
3 : 2	rx_mode_ch_01	rw	0x0	channel mode 1'b0: 16-bit mode 1'b1: 24-bit mode
5 : 4	rx_fifo_stat_ch_01	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full
6	rx_en_ch_1	rw	0x0	channel enable

24.8 aud_proc_rx_cfg_ch23				0x4003701C
Bits	Field Name	Acc	Default	Description
0	rx_en_ch_2	rw	0x0	channel enable
1	rx_fifo_flush_ch_23	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
3 : 2	rx_mode_ch_23	rw	0x0	channel mode 1'b0: 16-bit mode 1'b1: 24-bit mode
5 : 4	rx_fifo_stat_ch_23	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full
6	rx_en_ch_3	rw	0x0	channel enable

24.9 aud_proc_dac_cfg1				0x40037020
Bits	Field Name	Acc	Default	Description
0	src_in_sel	rw	0x0	0:select tx0andtx1 into src 1:select rx2tx loop data into src
4 : 1	src_mode	rw	0x8	0:mode_8k; 1:mode_11p025k; 2:mode_12k; 3: mode_16k ; 4: mode_22p05k; 5:mode_24k; 6:mode_32k; 7:mode_44p1k; 8: mode_48k;
5	src_freq_sel	rw	0x0	0: output 48k mode 1: output 26m/512 mode
7 : 6	src_ch_en	rw	0x0	[0]: l channel enable [1]: r channel enable
9 : 8	rx_mux_mode_l	rw	0x3	dac mux left channel add rx mux mode 2'b00: add rx ch0 data 2'b01: add rx ch1 data other: disable rx mux
11 : 10	tx_mux_mode_l	rw	0x3	dac mux left channel add tx mux mode 2'b00: add tx ch2 data 2'b01: add tx ch3 data other: disable tx23 mux
13 : 12	rx_mux_mode_r	rw	0x3	dac mux right channel add rx mux mode 2'b00: add rx ch0 data 2'b01: add rx ch1 data other: disable rx mux
15 : 14	tx_mux_mode_r	rw	0x3	dac mux right channel add tx mux mode 2'b00: add tx ch0 data 2'b01: add tx ch1 data other: disable tx01 mux
23 : 16	dac_volume_l	rw	0xA0	dac left channel volume
31 : 24	dac_volume_r	rw	0xA0	dac right channel volume

24.10 aud_proc_dac_cfg2				0x40037024
Bits	Field Name	Acc	Default	Description
2 : 0	dac_mixer_mode_l	rw	0x0	dac left channel mixer mode 3'h0:left channel data 3'h1:mute 3'h2:right channel data 3'h3:(left channel + right channel)/2 3'h4:(left channel - right channel)/2 other: left channel data
5 : 3	dac_mixer_mode_r	rw	0x2	dac right channel mixer mode 3'h0:left channel data 3'h1:mute 3'h2:right channel data 3'h3:(left channel + right channel)/2 3'h4:(left channel - right channel)/2 other: right channel data
7 : 6	eq_ch_en	rw	0x0	EQ channel enable 2'b11: enable both channel 2'b10: enable right channel only 2'b01: enable left channel only 2'b00: Bypass EQ
12 : 8	eq_stage	rw	0xA	set eq number of stage
13	eq_clr_done	ro	0x0	EQ clear is done
14	eq_clr	w1c	0x0	clear EQ module
15	src_clr	rw	0x0	clear src path
17 : 16	tx_serial_data_stat	ro	0x0	
19 : 18	tx_parallel_data_stat	ro	0x0	
23 : 20	src_flt_bypass	rw	0x0	[0]: force half band filter 1 bypass [1]: force half band filter 2 bypass [2]:force upsample filter bypass [3]:force src filter bypass
24	aud_proc_dac_trig_en	rw	0x0	1:enable dac trigger
25	aud_proc_tx_i2s_trig_en	rw	0x0	1:enable i2s trigger

24.11 aud_proc_adc_cfg1				0x40037028
Bits	Field Name	Acc	Default	Description
7 : 0	adc01_volume_l	rw	0xA0	adc left channel volume
15 : 8	adc01_volume_r	rw	0xA0	adc right channel volume
23 : 16	adc23_volume_l	rw	0xA0	adc left channel volume
31 : 24	adc23_volume_r	rw	0xA0	adc right channel volume

24.12 aud_proc_adc_cfg2				0x4003702C
Bits	Field Name	Acc	Default	Description
0	side_tone_l_input_sel	rw	0x0	1:side_tone select right channel; 0:side_tone select left channel
1	side_tone_r_input_sel	rw	0x0	1:side_tone select left channel; 0:side_tone select right channel

6 : 2	side_tone_gain	rw	0x0	0 : 0dB 1:- 2dB 2:- 4dB 3:- 6dB 4:- 8dB 5:-10dB 6:-12dB 7:-14dB 8:-16dB 9:-18dB 10-20dB 11:-22dB 12:-24dB 13:-26dB 14:-28dB 15:-30dB 16:-32dB 17:-34dB 18:-36dB 19:-38dB 20:-40dB 21:-42dB 22:-44dB 23:-46dB 24:-48dB 25:-50dB 26:-52dB 27:-54dB 28: 56dB 29:-58dB 30-60dB
9 : 7	side_tone_mixer_mode_l	rw	0x0	dac left channel mixer mode 3'h0:left channel data 3'h1:mute 3'h2:right channel data 3'h3:(left channel + right channel)/2 3'h4:(left channel - right channel)/2 other: left channel data
12 : 10	side_tone_mixer_mode_r	rw	0x2	dac right channel mixer mode 3'h0:left channel data 3'h1:mute 3'h2:right channel data 3'h3:(left channel + right channel)/2 3'h4:(left channel - right channel)/2 other: right channel data
13	side_tone_mode	rw	0x0	side tone mode 1'h0: add side tone after eq 1'h1: add side tone before eq
14	side_tone_en	rw	0x0	side tone enable
15	aud_proc_rx01_i2s_trig_en	rw	0x0	1:enable i2s trigger
16	aud_proc_rx23_i2s_trig_en	rw	0x0	1:enable i2s trigger
18 : 17	eq_adc_ch_en	rw	0x0	EQ channel enable [1]: enable chanl1 adc EQ [0]: enable chanl0 adc EQ
22 : 19	eq_adc_stage	rw	0xA	set eq number of stage
23	eq_adc_clr_done	ro	0x0	EQ clear is done
24	eq_adc_clr	w1c	0x0	clear EQ module
25	adc1_source_sel	rw	0x0	adc source select 1'h1: select Ext Interface 1'h0: select codec
26	adc2_source_sel	rw	0x0	adc source select 1'h1: select Ext Interface 1'h0: select codec
27	adc_sync_mode	rw	0x0	1:adc01 and adc23 are sync; 0:adc01 and adc23 are async
31 : 28	loop_volume	rw	0x7	rx to tx loop path volume

24.13 aud_proc_txrx_cfg_ch0 0x40037030

Bits	Field Name	Acc	Default	Description
0	txrx_en_ch_0	rw	0x0	channel enable
1	txrx_fifo_flush_ch_0	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
2	txrx_mode_ch_0	rw	0x0	channel mode 1'b0: 16-bit mode 1'b1: 24-bit mode
4 : 3	txrx_fifo_stat_ch_0	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full
14 : 8	txrx_fifo_ae_th_ch_0	rw	0x3	
15	txrx_fifo_full_sel_ch_0	rw	0x0	1:select almost full, thus can use af_th_ch_0 to control fifo level
22 : 16	txrx_fifo_af_th_ch_0	rw	0x1E	
24	txrx_fifo_clr_ch_0	rw	0x0	1:clear txrx fifo

24.14 aud_proc_txrx_cfg_ch1				0x40037034
Bits	Field Name	Acc	Default	Description
0	txrx_en_ch_1	rw	0x0	channel enable
1	txrx_fifo_flush_ch_1	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
2	txrx_mode_ch_1	rw	0x0	channel mode 1'b0: 16-bit mode 1'b1: 24-bit mode
4 : 3	txrx_fifo_stat_ch_1	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full

24.15 aud_proc_rxtx_cfg_ch0				0x40037038
Bits	Field Name	Acc	Default	Description
0	rxtx_en_ch_0	rw	0x0	channel enable
1	rxtx_fifo_flush_ch_0	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
2	rxtx_mode_ch_0	rw	0x0	channel mode 1'b0: 16-bit mode 1'b1: 24-bit mode
4 : 3	rxtx_fifo_stat_ch_0	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full
5	rxtx_fifo_clr_ch_0	rw	0x0	rxtx fifo clr
6	rxtx_ch01_4mac_in_en	rw	0x0	1: enable 4 mac sync mode

24.16 aud_proc_rxtx_cfg_ch1				0x4003703C
Bits	Field Name	Acc	Default	Description
0	rxtx_en_ch_1	rw	0x0	channel enable
1	rxtx_fifo_flush_ch_1	rw	0x0	flush fifo 1'b1: flush status 1'b0: normal status
2	rxtx_mode_ch_1	rw	0x0	channel mode 1'b0: 16-bit mode 1'b1: 24-bit mode

4 : 3	rxtx_fifo_stat_ch_1	ro	0x0	fifo_stat 2'b00: emptyq̃uarter 2'b01: quarterh̃alf 2'b10: halff̃ull 2'b11: full
5	rxtx_fifo_clr_ch_1	rw	0x0	rxtx fifo clr

24.17 aud_proc_frac_src				0x40037040
Bits	Field Name	Acc	Default	Description
0	frac_src_en	rw	0x0	1: fractional src enable
1	frac_rate_adj_soft_en	rw	0x0	1:set software configure src rate
2	frac_rate_soft_set	rw	0x0	1: directly set frac src value
27 : 4	frac_rate_adj_sw	rw	0x0	96k input : src adj rate = frac_rate_adj_sw*(96000/240/2Î5); 88.2k input : src adj rate = frac_rate_adj_sw*(88200/240/2Î5); 64k input: src adj rate = frac_rate_adj_sw*(64000/240/2Î5);

24.18 aud_proc_dac_trig				0x40037044
Bits	Field Name	Acc	Default	Description
0	aud_dac_trig_cfg_update	w1c	0x0	1:update trig configure, such as *hw_en,*_sw, trig_cnt_th
2	aud_proc_dac_strb_lrck_sel	rw	0x0	0: select lrck0, 1: select lrck1
27 : 4	dac_src_trig_cnt_th	rw	0x0	hw trigger counter threshold,
28	aud_proc_dac_strb_mode	rw	0x0	
29	aud_dac_trig_sw_oe	rw	0x0	1:enable software to start aud oe
30	aud_dac_trig_sw	rw	0x0	1:enable software to start aud
31	aud_dac_trig_hw_en	rw	0x0	1:enable hw trigger to start aud

24.19 aud_proc_dac_strb				0x40037048
Bits	Field Name	Acc	Default	Description
23 : 0	dac_src_strb_lo_cnt_th	rw	0x0	strb cnt low 24 bits threshold
29 : 28	aud_proc_dac_stb_sel	rw	0x0	0: aud codec strb, 1: aud intf strb l, 2: aud intf strb r
30	bt_clk_strb_inv	rw	0x0	1:sel invert strb edge to cnt
31	dac_src_strb_cnt_en	rw	0x0	1:enable strb cnt

24.20 aud_proc_irq				0x4003704C
Bits	Field Name	Acc	Default	Description
3 : 0	irq_aud_proc_masked	ro	0x0	
7 : 4	irq_aud_proc_raw	ro	0x0	
19 : 16	irq_aud_proc_mask	rw	0x0	
23 : 20	irq_aud_proc_clr	w1c	0x0	

24.21 bt_clk_lo_cnt				0x40037050
Bits	Field Name	Acc	Default	Description
23 : 0	bt_clk_lo_cnt	ro	0x0	

24.22 bt_clk_hi_cnt				0x40037054
Bits	Field Name	Acc	Default	Description
31 : 0	bt_clk_hi_cnt	ro	0x0	

24.23 bt_clk_lo_cnt_latch	0x40037058

Bits	Field Name	Acc	Default	Description
23 : 0	bt_clk_lo_cnt_latch	ro	0x0	
29 : 24	bt_clk_lo_frac_cnt_latch	ro	0x0	

24.24 bt_clk_hi_cnt_latch				0x4003705C
Bits	Field Name	Acc	Default	Description
31 : 0	bt_clk_hi_cnt_latch	ro	0x0	

24.25 dac_src_strb_lo_cnt				0x40037060
Bits	Field Name	Acc	Default	Description
23 : 0	dac_src_strb_lo_cnt	ro	0x0	
31 : 24	src_frac_strb_latch	ro	0x0	

24.26 dac_src_strb_hi_cnt				0x40037064
Bits	Field Name	Acc	Default	Description
31 : 0	dac_src_strb_hi_cnt	ro	0x0	

24.27 dpll_alfa1_cfg				0x40037068
Bits	Field Name	Acc	Default	Description
13 : 0	dpll_1_sub_alfa1	rw	0x0	
29 : 16	dpll_alfa1	rw	0x0	

24.28 dpll_k1_k2_cfg				0x4003706C
Bits	Field Name	Acc	Default	Description
13 : 0	dpll_k2	rw	0x0	
29 : 16	dpll_k1	rw	0x0	

24.29 dpll_alfa2_cfg				0x40037070
Bits	Field Name	Acc	Default	Description
13 : 0	dpll_1_sub_alfa2	rw	0x0	
29 : 16	dpll_alfa2	rw	0x0	

24.30 dpll_other_cfg1				0x40037074
Bits	Field Name	Acc	Default	Description
13 : 0	dpll_gain	rw	0x0	

24.31 dpll_trig_cfg				0x4003707C
Bits	Field Name	Acc	Default	Description
0	dpll_step_trig	w1c	0x0	
1	dpll_hw_en	rw	0x0	
18 : 2	dpll_din	rw	0x0	
19	dpll_cal_finish	ro	0x0	

24.32 frac_rate_adj_up_th				0x40037080
Bits	Field Name	Acc	Default	Description
23 : 0	frac_rate_adj_up_th	rw	0x7FFFFF	
28 : 24	frac_48k_cnt_th_h	rw	0x0	

24.33 frac_rate_adj_dn_th				0x40037084
Bits	Field Name	Acc	Default	Description
23 : 0	frac_rate_adj_dn_th	rw	0x800000	
31 : 24	frac_48k_cnt_th_l	rw	0x87	

24.34 frac_rate_adj_hw				0x40037088
Bits	Field Name	Acc	Default	Description
23 : 0	frac_rate_adj_hw	ro	0x0	

24.35 aud_intf_i2s_cfg0				0x4003708C
Bits	Field Name	Acc	Default	Description
1 : 0	swap_chlr_0	rw	0x0	Swap Channel L and Channel R.[0] [0] for output channel swap, 0: left, 1:right [1] for input channel swap, 0: left, 1: right
2	tx_data_half_cycle_dly_0	rw	0x0	delayed Audio data out by half cycle. 0:: NO_DLY 1:: DLY
3	lsb_0	rw	0x0	When high, the output data format is with the least significant bit first. 0:: MSB 1:: LSB
4	right_justified_0	rw	0x0	right justified mode.
9 : 5	bck_lrck_0	rw	0x0	configure extend cycles of bck after data transfer is finished. Voice_Mode:: each sample takes wlen + tx_bck_lrck BCLK cycle. Audio_Mode:: each sample takes 2*(wlen + tx_bck_lrck) BCLK cycle. 2 times than Voice
11 : 10	wlen_0	rw	0x0	i2s data length 00: 16-bit 01: 20-bit 10: 24-bit 11: 32-bit
12	lrck_pol_0	rw	0x0	configure LRCK polarity. 0:: LEFT_H_RIGHT_L: high level on LRCK means left channel, low level on LRCK means right channel. 1:: LEFT_L_RIGHT_H: high level on LRCK means right channel, low level on LRCK means left channel. Note: this bit should be set to '0' (LEFT_H_RIGHT_L) in voice mode.
13	bck_pol_0	rw	0x0	if Master Mode, invert BCLK out. if slave Mode, invert BCLK in. 0:: NORMAL 1:: INVERT
14	bckout_gate_0	rw	0x0	Sets the BckOut gating. This bit decide if aud intf continue to output BCK clock after data has been sent. 0:: NO_GATE 1:: GATED
15	rx_half_cycle_dly_0	rw	0x0	delayed Audio input data by half cycle. 0:: NO_DLY 1:: DLY
16	tx_lrck_half_cycle_dly_0	rw	0x0	delayed Audio LRCK by half cycle. 0:: NO_DLY 1:: DLY
18 : 17	rx_dly_0	rw	0x0	Indicates the delay between serial data in MSB and LRCK edge. 2'b00:: ALIGN: Digital audio in MSB is aligned with LRCLK edge. 2'b01:: DLY_1: Digital audio in MSB is 1 cycle delayed to LRCLK edge. 2'b10:: DLY_2: Digital audio in MSB is 2 cycle delayed to LRCLK edge. 2'b11:: DLY_3: Digital audio in MSB is 3 cycle delayed to LRCLK edge.

19	tx_dly_s_0	rw	0x0	ONLY for slave mode:: configure 1 cycle supplementary Tx delay. 0:: NO_DLY: No supplementary Tx delay. 1:: DLY: One Cycle supplementary Tx delay.
20	tx_dly_0	rw	0x0	configure the delay between serial data out MSB and LRCK edge. 0:: ALIGN: Digital audio out MSB is aligned with LRCLK edge. 1:: DLY_1: Digital audio out MSB is 1 cycle delayed to LRCLK edge
21	master_mode_0	rw	0x1	configure aud intf works in master mode (LRCLK and BCK timing signals are generated internally) or slave mode (LRCLK and BCK timing signals are generated externally). 0:: SLAVE 1:: MASTER
24 : 22	rx_serial_mode_0	rw	0x0	Configure serial aud intf mode. 0:: I2S_PCM 1:: VOICE 2:: DAI When mode is set DAI, the bit Master Mode should be set to '1'. Data should be sent out on falling edge, which requires either Bclk_Pol = '0' and Half_Cycle_DLY = '1' or Bclk_Pol = '1' and Half_Cycle_DLY = '0'. Bits Tx_DLY and BCK-Out_Gate must be configured to '0' and '1'. The DAI mode must NOT be modified after aud intf is enabled.
27 : 25	tx_serial_mode_0	rw	0x0	Configure serial aud intf mode. 0:: I2S_PCM 1:: VOICE 2:: DAI When mode is set DAI, the bit Master Mode should be set to '1'. Data should be sent out on falling edge, which requires either Bclk_Pol = '0' and Half_Cycle_DLY = '1' or Bclk_Pol = '1' and Half_Cycle_DLY = '0'. Bits Tx_DLY and BCK-Out_Gate must be configured to '0' and '1'. The DAI mode must NOT be modified after aud intf is enabled.
28	tx_off_0	rw	0x1	Disable aud intf Tx functions. Important:: if you want to do record only, you must set this bit otherwise aud intf state machine will not start. 0:: TX_ON: Both Tx Rx enabled. 1:: TX_OFF: Rx enabled only, Tx disabled.
30 : 29	tx_mode_0	rw	0x0	Configure mono or stereo format for Audio data out: 2'b00:: STEREO_STEREO: stereo input from DMA, stereo output to pin. 2'b01:: MONOL_STEREO_DUPLI: mono channel L input from DMA, stereo output duplicate in both channels to pin. 2'b10:: MONOR_STEREO_DUPLI: mono channel R input from DMA, stereo output duplicate in both channels to pin. 2'b11:: MONO_IN_MONO_OUT Funtion mono input from DMA, stereo output in left channel to pin, can be implemented by setting aud intf_tx_cfg_ch[1] or aud intf_tx_cfg_ch[0] disable. When works in DAI, or DSD mode, always select "00" mode STEREO_STEREO.
31	i2s_enable_0	rw	0x0	enable i2s interface

24.36 aud_intf_i2s_cfg1				0x40037090
Bits	Field Name	Acc	Default	Description
1 : 0	swap_chlr_1	rw	0x0	Swap Channel L and Channel R.[0] [0] for output channel swap, 0: left, 1:right [1] for input channel swap, 0: left, 1: right
2	tx_data_half_cycle_dly_1	rw	0x0	delayed Audio data out by half cycle. 0:: NO_DLY 1:: DLY

3	lsb_1	rw	0x0	When high, the output data format is with the least significant bit first. 0:: MSB 1:: LSB
4	right_justified_1	rw	0x0	right justified mode.
9 : 5	bck_lrck_1	rw	0x0	configure extend cycles of bck after data transfer is finished. Voice_Mode:: each sample takes wlen + tx_bck_lrck BCLK cycle. Audio_Mode:: each sample takes 2*(wlen + tx_bck_lrck) BCLK cycle. 2 times than Voice
11 : 10	wlen_1	rw	0x0	i2s data length 00: 16-bit 01: 20-bit 10: 24-bit 11: 32-bit
12	lrck_pol_1	rw	0x0	configure LRCK polarity. 0:: LEFT_H_RIGHT_L: high level on LRCK means left channel, low level on LRCK means right channel. 1:: LEFT_L_RIGHT_H: high level on LRCK means right channel, low level on LRCK means left channel. Note: this bit should be set to '0' (LEFT_H_RIGHT_L) in voice mode.
13	bck_pol_1	rw	0x0	if Master Mode, invert BCLK out. if slave Mode, invert BCLK in. 0:: NORMAL 1:: INVERT
14	bckout_gate_1	rw	0x0	Sets the BckOut gating. This bit decide if aud intf continue to output BCK clock after data has been sent. 0:: NO_GATE 1:: GATED
15	rx_half_cycle_dly_1	rw	0x0	delayed Audio input data by half cycle. 0:: NO_DLY 1:: DLY
16	tx_lrck_half_cycle_dly_1	rw	0x0	delayed Audio output LRCK by half cycle. 0:: NO_DLY 1:: DLY
18 : 17	rx_dly_1	rw	0x0	Indicates the delay between serial data in MSB and LRCK edge. 2'b00:: ALIGN: Digital audio in MSB is aligned with LRCLK edge. 2'b01:: DLY_1: Digital audio in MSB is 1 cycle delayed to LRCLK edge. 2'b10:: DLY_2: Digital audio in MSB is 2 cycle delayed to LRCLK edge. 2'b11:: DLY_3: Digital audio in MSB is 3 cycle delayed to LRCLK edge.
19	tx_dly_s_1	rw	0x0	ONLY for slave mode:: configure 1 cycle supplementary Tx delay. 0:: NO_DLY: No supplementary Tx delay. 1:: DLY: One Cycle supplementary Tx delay.
20	tx_dly_1	rw	0x0	configure the delay between serial data out MSB and LRCK edge. 0:: ALIGN: Digital audio out MSB is aligned with LRCLK edge. 1:: DLY_1: Digital audio out MSB is 1 cycle delayed to LRCLK edge
21	master_mode_1	rw	0x1	configure aud intf works in master mode (LRCLK and BCK timing signals are generated internally) or slave mode (LRCLK and BCK timing signals are generated externally). 0:: SLAVE 1:: MASTER

24 : 22	rx_serial_mode_1	rw	0x0	Configure serial aud intf mode. 0:: I2S_PCM 1:: VOICE 2:: DAI When mode is set DAI, the bit Master Mode should be set to '1'. Data should be sent out on falling edge, which requires either Bclk_Pol = '0' and Half_Cycle_DLY = '1' or Bclk_Pol = '1' and Half_Cycle_DLY = '0'. Bits Tx_DLY and BCK-Out_Gate must be configured to '0' and '1'. The DAI mode must NOT be modified after aud intf is enabled.
27 : 25	tx_serial_mode_1	rw	0x0	Configure serial aud intf mode. 0:: I2S_PCM 1:: VOICE 2:: DAI When mode is set DAI, the bit Master Mode should be set to '1'. Data should be sent out on falling edge, which requires either Bclk_Pol = '0' and Half_Cycle_DLY = '1' or Bclk_Pol = '1' and Half_Cycle_DLY = '0'. Bits Tx_DLY and BCK-Out_Gate must be configured to '0' and '1'. The DAI mode must NOT be modified after aud intf is enabled.
28	tx_off_1	rw	0x1	Disable aud intf Tx functions. Important:: if you want to do record only, you must set this bit otherwise aud intf state machine will not start. 0:: TX_ON: Both Tx Rx enabled. 1:: TX_OFF: Rx enabled only, Tx disabled.
30 : 29	tx_mode_1	rw	0x0	Configure mono or stereo format for Audio data out: 2'b00:: STEREO_STEREO: stereo input from DMA, stereo output to pin. 2'b01:: MONOL_STEREO_DUPLI: mono channel L input from DMA, stereo output duplicate in both channels to pin. 2'b10:: MONOR_STEREO_DUPLI: mono channel R input from DMA, stereo output duplicate in both channels to pin. 2'b11:: MONO_IN_MONO_OUT Funtion mono input from DMA, stereo output in left channel to pin, can be implemented by setting aud intf_tx_cfg_ch[1] or aud intf_tx_cfg_ch[0] disable. When works in DAI, or DSD mode, always select "00" mode STEREO_STEREO.
31	i2s_enable_1	rw	0x0	enable i2s interface

24.37 pcm_cfg_0				0x40037094
Bits	Field Name	Acc	Default	Description
7 : 0	slotnum_0	rw	0x0	slots number to be transferred in one frame
8	longsync_0	rw	0x0	longsync enable
9	tx_bit_mode_0	rw	0x0	0: low bit are inserted zero to 24bits 1:high bits are inserted zero to 24bits
10	rx_bit_mode_0	rw	0x0	0: low bit are inserted zero to 24bits 1:high bits are inserted zero to 24bits
11	lrck_m_dly_0	rw	0x0	1: output lrck delay one bck
12	tx_sync_mode_0	rw	0x0	1:output fast sync in master mode
13	loop_back_0	rw	0x0	Sets i2s in loop back mode. The feature is for debug only and can not work in DAI mode. 0:: NORMAL 1:: LOOPBACK
14	bckout_auto_gate_bypass_0	rw	0x0	1: bypss bck auto gate, keep bck output always ok
15	tx_ch01_iis_en_mux	rw	0x0	
16	rx_ch01_iis_en_mux	rw	0x0	
18 : 17	rx_mode_0	rw	0x0	
19	i2s_chn_out_sel	rw	0x0	1:i2s tx sel channel 1; 0:i2s tx sel channel 0
20	i2s_chn_in_sel	rw	0x0	1:i2s rx sel channel 1; 0:i2s rx sel channel 0

24.38 pcm_cfg_1				0x40037098
Bits	Field Name	Acc	Default	Description
7 : 0	slotnum_1	rw	0x0	slots number to be transferred in one frame
8	longsync_1	rw	0x0	longsync enable
9	tx_bit_mode_1	rw	0x0	0: low bit are inserted zero to 24bits 1:high bits are inserted zero to 24bits
10	rx_bit_mode_1	rw	0x0	0: low bit are inserted zero to 24bits 1:high bits are inserted zero to 24bits
11	lrck_m_dly_1	rw	0x0	1: output lrck delay one bck
12	tx_sync_mode_1	rw	0x0	1:output fast sync in master mode
13	loop_back_1	rw	0x0	Sets i2s in loop back mode. The feature is for debug only and can not work in DAI mode. 0:: NORMAL 1:: LOOPBACK
14	bckout_auto_gate_bypass_1	rw	0x0	1: bypss bck auto gate, keep bck output always ok
15	tx_ch23_iis_en_mux	rw	0x0	
16	rx_ch23_iis_en_mux	rw	0x0	
18 : 17	rx_mode_1	rw	0x0	

24.39 spdif_userdata_0				0x4003709C
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_userdata_0	rw	0x0	

24.40 spdif_userdata_1				0x400370A0
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_userdata_1	rw	0x0	

24.41 spdif_userdata_2				0x400370A4
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_userdata_2	rw	0x0	

24.42 spdif_userdata_3				0x400370A8
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_userdata_3	rw	0x0	

24.43 spdif_userdata_4				0x400370AC
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_userdata_4	rw	0x0	

24.44 spdif_userdata_5				0x400370B0
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_userdata_5	rw	0x0	

24.45 spdif_channel_stat_0				0x400370B4
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_channel_stat_0	rw	0x0	spdif channel status out

24.46 spdif_channel_stat_1				0x400370B8
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_channel_stat_1	rw	0x0	

24.47 spdif_channel_stat_2				0x400370BC
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_channel_stat_2	rw	0x0	

24.48 spdif_channel_stat_3				0x400370C0
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_channel_stat_3	rw	0x0	

24.49 spdif_channel_stat_4				0x400370C4
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_channel_stat_4	rw	0x0	

24.50 spdif_channel_stat_5				0x400370C8
Bits	Field Name	Acc	Default	Description
31 : 0	spdif_channel_stat_5	rw	0x0	

24.51 spdif_output_cfg				0x400370CC
Bits	Field Name	Acc	Default	Description
0	validity_flag	rw	0x0	
2 : 1	vucz_mode	rw	0x0	
7 : 3	req_point	rw	0x0	
8	start_zen	rw	0x0	

24.52 spdif_input_cfg				0x400370D0
Bits	Field Name	Acc	Default	Description
0	spdif_in_fsdet_en	rw	0x0	
13 : 1	spdif_in_cfg	rw	0x0	
24 : 14	iaau_spdif_in	ro	0x0	

24.53 spdif_input_stat1				0x400370D4
Bits	Field Name	Acc	Default	Description
7 : 0	spdif_in_bit_width	ro	0x0	
29 : 8	status_spdif_in	ro	0x0	

24.54 spdif_input_stat2				0x400370D8
Bits	Field Name	Acc	Default	Description
31 : 0	channel_spdif_in	ro	0x0	spdif input status2

24.55 aud_drc_cfg0				0x400370DC
Bits	Field Name	Acc	Default	Description
1 : 0	cfg_aud_drc_chnl_en	rw	0x0	
8	cfg_aud_drc_clr	w1c	0x0	
9	aud_drc_clr_done	ro	0x0	
12 : 10	cfg_drc_band3_mode	rw	0x0	
13	cfg_drc_band3_en	rw	0x0	
14	cfg_drc_eq_swap	rw	0x0	
15	cfg_drc_crossover_bypass	rw	0x0	
16	cfg_drc_en	rw	0x0	
18 : 17	cfg_drc_stage_cnt	rw	0x2	
23 : 20	cfg_drc_band2_mode	rw	0x0	

26 : 24	cfg_drc_band1_mode	rw	0x0	
29 : 27	cfg_drc_band0_mode	rw	0x0	

24.56 eq_stereo_coef_ram_cfg0				0x40037100
Bits	Field Name	Acc	Default	Description
29 : 0	eq_stereo_coef_wdata	rw	0x0	
31	eq_stereo_coef_wr	w1c	0x0	

24.57 eq_stereo_coef_ram_cfg1				0x40037104
Bits	Field Name	Acc	Default	Description
29 : 0	eq_stereo_coef_rdata	ro	0x0	
31	eq_stereo_coef_rd	w1c	0x0	

24.58 eq_mono_coef_ram_cfg0				0x40037108
Bits	Field Name	Acc	Default	Description
29 : 0	eq_mono_coef_wdata	rw	0x0	
31	eq_mono_coef_wr	w1c	0x0	

24.59 eq_mono_coef_ram_cfg1				0x4003710C
Bits	Field Name	Acc	Default	Description
29 : 0	eq_mono_coef_rdata	ro	0x0	
31	eq_mono_coef_rd	w1c	0x0	

24.60 aud_drc_ram_cfg0				0x40037110
Bits	Field Name	Acc	Default	Description
0	cfg_aud_drc_ram_wr	w1c	0x0	
1	cfg_aud_drc_ram_rd	w1c	0x0	
14 : 8	cfg_aud_drc_ram_addr	rw	0x0	
17 : 16	cfg_aud_drc_ram_sel	rw	0x0	

24.61 aud_drc_ram_cfg1				0x40037114
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_aud_drc_ram_wdata	rw	0x0	

24.62 aud_drc_ram_cfg2				0x40037118
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_aud_drc_ram_rdata	ro	0x0	

24.63 aud_src_ctrl0				0x4003711C
Bits	Field Name	Acc	Default	Description
0	aud_src_enable	rw	0x0	1: aud src function enable
1	aud_src_path_reset	rw	0x0	1: aud src path reset
2	aud_src_txrx_lr_fifo_share_en	rw	0x0	1: enable txrx channel share one buffer
3	aud_src_tx_lr_fifo_share_en	rw	0x0	1: enable tx channel data share one buffer
4	aud_src_txrx_en_ch_1	rw	0x0	1: enable src txrx channel r.
5	aud_src_txrx_en_ch_0	rw	0x0	1: enable src txrx channel l,
6	aud_src_txrx_fifo_flush_ch	rw	0x0	1: flush txrx src buffer
7	aud_src_txrx_mode_ch	rw	0x0	1: select src txrx data is 24 bits; 0; select src txrx data is 16 bits
8	aud_src_tx_en_ch_1	rw	0x0	1: enable src tx channel r.

9	aud_src_tx_en_ch_0	rw	0x0	1: enable src tx channel l,
10	aud_src_tx_fifo_flush_ch	rw	0x0	
11	aud_src_tx_mode_ch	rw	0x0	1: select src tx data is 24 bits; 0; select src tx data is 16 bits
12	aud_src_auto_clock_gating_en	rw	0x1	1: enable src audo clock gating
13	aud_src_adc_in_en	rw	0x0	1: select src tx data from adc channel, 0: select src tx data from src tx channel
14	aud_src_txrx_burst_mode	rw	0x0	1:src txrx channel burst inc4 mode; 0: src txrx channel single mode
15	aud_src_tx_burst_mode	rw	0x0	1:src tx channel burst inc4 mode; 0: src tx channel single mode
17 : 16	aud_src_txrx_ch_stat	ro	0x0	
19 : 18	aud_src_tx_ch_stat	ro	0x0	

24.64 aud_src_ctrl1				0x40037120
Bits	Field Name	Acc	Default	Description
23 : 0	aud_src_frac_rate_adj_sw	rw	0x0	
24	aud_src_frac_rate_soft_set	rw	0x0	

24.65 aud_src_ctrl2				0x40037124
Bits	Field Name	Acc	Default	Description
0	aud_src_frac_src_en	rw	0x0	1. enable aud rx src function mode
2 : 1	aud_src_ch_en	rw	0x0	[0]:1, enable aud src l channel [1]:1, enable aud src r channel
3	aud_src_freq_sel	rw	0x0	0: output 48k mode 1: output 26m/512 mode
7 : 4	aud_src_mode	rw	0x0	0:mode_8k; 1:mode_11p025k; 2:mode_12k; 3: mode_16k ; 4: mode_22p05k; 5:mode_24k; 6:mode_32k; 7:mode_44p1k; 8: mode_48k;
8	aud_src_clr	rw	0x0	1: clear aud src data path
12 : 9	aud_src_flt_bypass	rw	0x0	[0]:1, bypass 1th src half band filter [1]:1, bypass 2th src half band filter [2]:1, bypass 3th src half band filter [3]:1, bypass src filter
13	aud_src_decim24_sel	rw	0x0	1: select src decimation 2; 0: select src decimation 4;
14	aud_src_decim24_bypass	rw	0x0	1:bypass src decimation 2/4
15	aud_src_decim3_sel	rw	0x0	1:select src decimation 3
16	aud_src_decim3_bypass	rw	0x0	1: bypass src decimation 3
17	aud_src_iir_hpf_bypass	rw	0x0	
18	aud_src_iir_bank_en	rw	0x0	
21 : 19	aud_src_iir_flt_stage	rw	0x0	
22	aud_adc_iir_hpf_bypass	rw	0x0	
23	aud_adc_iir_bank_en	rw	0x0	
26 : 24	aud_adc_iir_flt_stage	rw	0x0	

24.66 aud_src_hpf_coef1_a1				0x40037128
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_hpf_coef1_a1_high32	rw	0x0	

24.67 aud_src_hpf_coef1_a2				0x4003712C
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_hpf_coef1_a2_high32	rw	0x0	

24.68 aud_src_hpf_coef1_b1				0x40037130
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_hpf_coef1_b1_high32	rw	0x0	

24.69 aud_src_hpf_coef1_g0				0x40037134
Bits	Field Name	Acc	Default	Description
19 : 0	aud_src_hpf_coef1_g0	rw	0x40000	
23 : 20	aud_src_hpf_coef1_b1_low4	rw	0x0	
27 : 24	aud_src_hpf_coef1_a2_low4	rw	0x0	
31 : 28	aud_src_hpf_coef1_a1_low4	rw	0x0	

24.70 aud_src_lpf_coef2_a1				0x40037138
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_lpf_coef2_a1_high32	rw	0x0	

24.71 aud_src_lpf_coef2_a2				0x4003713C
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_lpf_coef2_a2_high32	rw	0x0	

24.72 aud_src_lpf_coef2_b1				0x40037140
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_lpf_coef2_b1_high32	rw	0x0	

24.73 aud_src_lpf_coef2_g0				0x40037144
Bits	Field Name	Acc	Default	Description
19 : 0	aud_src_lpf_coef2_g0	rw	0x40000	
23 : 20	aud_src_lpf_coef2_b1_low4	rw	0x0	
27 : 24	aud_src_lpf_coef2_a2_low4	rw	0x0	
31 : 28	aud_src_lpf_coef2_a1_low4	rw	0x0	

24.74 aud_src_lpf_coef3_a1				0x40037148
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_lpf_coef3_a1_high32	rw	0x0	

24.75 aud_src_lpf_coef3_a2				0x4003714C
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_lpf_coef3_a2_high32	rw	0x0	

24.76 aud_src_lpf_coef3_b1				0x40037150
Bits	Field Name	Acc	Default	Description
31 : 0	aud_src_lpf_coef3_b1_high32	rw	0x0	

24.77 aud_src_lpf_coef3_g0				0x40037154
Bits	Field Name	Acc	Default	Description
19 : 0	aud_src_lpf_coef3_g0	rw	0x40000	
23 : 20	aud_src_lpf_coef3_b1_low4	rw	0x0	
27 : 24	aud_src_lpf_coef3_a2_low4	rw	0x0	
31 : 28	aud_src_lpf_coef3_a1_low4	rw	0x0	

24.78 aud_adc_hpf_coef1_a1				0x40037158
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_hpf_coef1_a1_high32	rw	0x0	

24.79 aud_adc_hpf_coef1_a2				0x4003715C
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_hpf_coef1_a2_high32	rw	0x0	

24.80 aud_adc_hpf_coef1_b1				0x40037160
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_hpf_coef1_b1_high32	rw	0x0	

24.81 aud_adc_hpf_coef1_g0				0x40037164
Bits	Field Name	Acc	Default	Description
19 : 0	aud_adc_hpf_coef1_g0	rw	0x40000	
23 : 20	aud_adc_hpf_coef1_b1_low4	rw	0x0	
27 : 24	aud_adc_hpf_coef1_a2_low4	rw	0x0	
31 : 28	aud_adc_hpf_coef1_a1_low4	rw	0x0	

24.82 aud_adc_lpf_coef2_a1				0x40037168
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_lpf_coef2_a1_high32	rw	0x0	

24.83 aud_adc_lpf_coef2_a2				0x4003716C
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_lpf_coef2_a2_high32	rw	0x0	

24.84 aud_adc_lpf_coef2_b1				0x40037170
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_lpf_coef2_b1_high32	rw	0x0	

24.85 aud_adc_lpf_coef2_g0				0x40037174
Bits	Field Name	Acc	Default	Description
19 : 0	aud_adc_lpf_coef2_g0	rw	0x40000	
23 : 20	aud_adc_lpf_coef2_b1_low4	rw	0x0	
27 : 24	aud_adc_lpf_coef2_a2_low4	rw	0x0	
31 : 28	aud_adc_lpf_coef2_a1_low4	rw	0x0	

24.86 aud_adc_lpf_coef3_a1				0x40037178
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_lpf_coef3_a1_high32	rw	0x0	

24.87 aud_adc_lpf_coef3_a2				0x4003717C
Bits	Field Name	Acc	Default	Description
31 : 0	aud_adc_lpf_coef3_a2_high32	rw	0x0	

24.88 aud_adc_lpf_coef3_b1				0x40037180
Bits	Field Name	Acc	Default	Description

31 : 0	aud_adc_lpf_coef3_b1_high32	rw	0x0	

24.89 aud_adc_lpf_coef3_g0				0x40037184
Bits	Field Name	Acc	Default	Description
19 : 0	aud_adc_lpf_coef3_g0	rw	0x40000	
23 : 20	aud_adc_lpf_coef3_b1_low4	rw	0x0	
27 : 24	aud_adc_lpf_coef3_a2_low4	rw	0x0	
31 : 28	aud_adc_lpf_coef3_a1_low4	rw	0x0	

24.90 eq_coef_ram_cfg3				0x40037190
Bits	Field Name	Acc	Default	Description
7 : 0	eq_stereo_coef_addr	rw	0x0	
21 : 16	eq_mono_coef_addr	rw	0x0	

24.91 aud_adc_drc_cfg0				0x40037194
Bits	Field Name	Acc	Default	Description
1 : 0	cfg_adc_drc_chnl_en	rw	0x0	
8	cfg_adc_drc_clr	w1c	0x0	
9	aud_adc_drc_clr_done	ro	0x0	
12 : 10	cfg_adc_drc_band_mode	rw	0x0	
13	cfg_adc_drc_en	rw	0x0	
14	cfg_adc_drc_auto_clk_gating	rw	0x0	

24.92 aud_adc_drc_ram_cfg0				0x40037198
Bits	Field Name	Acc	Default	Description
0	cfg_aud_adc_drc_ram_wr	w1c	0x0	
1	cfg_aud_adc_drc_ram_rd	w1c	0x0	
14 : 8	cfg_aud_adc_drc_ram_addr	rw	0x0	
17 : 16	cfg_aud_adc_drc_ram_sel	rw	0x0	

24.93 aud_adc_drc_ram_cfg1				0x4003719C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_aud_adc_drc_ram_wdata	rw	0x0	

24.94 aud_adc_drc_ram_cfg2				0x400371A0
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_aud_adc_drc_ram_rdata	ro	0x0	

25 spiflash

25.1 tx_cmd_addr				0x40038000
Bits	Field Name	Acc	Default	Description
7 : 0	tx_cmd	rw	0x0	Transmit command.
31 : 8	tx_addr	rw	0x0	Transmit address.

25.2 tx_block_size				0x40038004
Bits	Field Name	Acc	Default	Description
7 : 0	tx_mode	rw	0x0	Transmit mode bits.
16 : 8	tx_block_size	rw	0x1	Transmit block size.
24	tx_cont_mode	rw	0x0	Transmit continuous mode enable.

25.3 tx_fifo_data				0x40038008
Bits	Field Name	Acc	Default	Description
7 : 0	tx_data	wo	0x0	Transmit FIFO data.

25.4 status				0x4003800C
Bits	Field Name	Acc	Default	Description
0	busy_stat	ro	0x0	Busy status.
1	tx_fifo_empty	ro	0x1	Transmit FIFO empty.
2	tx_fifo_full	ro	0x0	Transmit FIFO full.
3	rx_fifo_empty	ro	0x1	Receive FIFO empty.
7 : 4	rx_fifo_cnt	ro	0x0	Receive FIFO count.
8	read_state_busy	ro	0x0	Read state busy.

25.5 rx_fifo_data				0x40038010
Bits	Field Name	Acc	Default	Description
31 : 0	rx_data	ro	0x0	

25.6 flash_config				0x40038014
Bits	Field Name	Acc	Default	Description
0	quad_mode	rw	0x0	Quad mode.
1	wproctect_pin	rw	0x0	Set wproctect pin active.
2	hold_pin	rw	0x0	Set hold pin active.
3	dual_mode	rw	0x0	Dual mode.
6 : 4	sample_delay	rw	0x2	Sample delay.
7	din_cap_edge	rw	0x0	sample clock edge selector, 0: posedge of clk spiflash 1: negedge of clk spiflash
15 : 8	clk_divider	rw	0x8	Clock divider.
16	cmd_quad	rw	0x0	Quad Command Enable. Hardware will regard current command as quad command if this bit is active.

25.7 fifo_ctrl				0x40038018
Bits	Field Name	Acc	Default	Description
0	rx_fifo_clr	rw	0x0	Write 1 to clear rx FIFO.
1	tx_fifo_clr	rw	0x0	Write 1 to clear tx FIFO.

25.8 dual_spi				0x4003801C
Bits	Field Name	Acc	Default	Description
0	spi_cs_num	rw	0x0	SPI cs number. 1'b0: One cs; 1'b1: Two cs.
2 : 1	spi_size	rw	0x0	SPI flash size. 2'b00: 32M bits flash; 2'b01: 64M bits flash; 2'b10: 16M bits flash; 2'b11: 8M bits flash.
3	spi_128m	rw	0x0	128M bits flash.
4	ahb_read_disable	rw	0x0	
5	spi_cs1_sel	rw	0x0	
6	spi_cs1_sel1	rw	0x0	
7	diff_128m_diff_cmd_en	rw	0x0	

25.9 read_cmd				0x40038020
Bits	Field Name	Acc	Default	Description
7 : 0	qread_cmd	rw	0xEB	Quad read command.
15 : 8	fread_cmd	rw	0xB	Fast read command.
23 : 16	read_cmd	rw	0x3	Read command.
31 : 24	dread_cmd	rw	0x3B	Dual read command.

25.10 nand_cfg1				0x40038024
Bits	Field Name	Acc	Default	Description
0	nand_sel	rw	0x0	Internal nand flash selection. Nand flash also can be selected from pin. 1'b0: Nor FLASH; 1'b1: Nand flash.
2 : 1	nand_addr_26_25	rw	0x0	Bit 26 to 25 of nand_address.
15 : 8	page_read_cmd	rw	0x13	Page read command.
23 : 16	get_sts_cmd	rw	0xF	Get status command.
31 : 24	ram_read_cmd	rw	0x3	RAM read command.

25.11 nand_cfg2				0x40038028
Bits	Field Name	Acc	Default	Description
7 : 0	get_sts_addr	rw	0xC0	Get status address.
15 : 8	program_exe_cmd	rw	0x10	Program execute command.
23 : 16	stst_qip	rw	0x1	

25.12 dummy_cycle_cfg				0x4003802C
Bits	Field Name	Acc	Default	Description
0	four_byte_addr	rw	0x0	Four bytes address. 1'b0: 3 bytes address; 1'b1: 4 bytes address.
1	dummy_cycle_en	rw	0x0	Dummy cycle enable.
11 : 8	dummy_cycle	rw	0x8	Dummy cycle count.

25.13 diff_128m_cmd				0x40038030
Bits	Field Name	Acc	Default	Description
7 : 0	first_128m_cmd	ro	0x8C	1st 128M flash command.
15 : 8	second_128m_cmd	ro	0x8D	2nd 128M flash command.
23 : 16	third_128m_cmd	ro	0x0	3rd 128M flash command.
31 : 24	fourth_128m_cmd	ro	0x0	4th 128M flash command.

25.14 aesKey0				0x40038034
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey0	rw	0x0	aesKey bit 255 to 224

25.15 aesKey1				0x40038038
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey1	rw	0x0	aesKey bit 223 to 192

25.16 aesKey2				0x4003803C
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey2	rw	0x0	aesKey bit 191 to 160

25.17 aesKey3				0x40038040
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey3	rw	0x0	aesKey bit 159 to 128

25.18 aesKey4				0x40038044
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey4	rw	0x0	aesKey bit 127 to 96

25.19 aesKey5				0x40038048
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey5	rw	0x0	aesKey bit 95 to 64

25.20 aesKey6				0x4003804C
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey6	rw	0x0	aesKey bit 63 to 32

25.21 aesKey7				0x40038050
Bits	Field Name	Acc	Default	Description
31 : 0	aesKey7	rw	0x0	aesKey bit 31 to 0

25.22 initCtrNonce0				0x40038054
Bits	Field Name	Acc	Default	Description
31 : 0	initCtrNonce0	rw	0x0	ctr bit 127 to 96 of 128 bit

25.23 initCtrNonce1				0x40038058
Bits	Field Name	Acc	Default	Description
31 : 0	initCtrNonce1	rw	0x0	ctr bit 95 to 64 of 128 bit

25.24 initCtrNonce2				0x4003805C
Bits	Field Name	Acc	Default	Description
31 : 0	initCtrNonce2	rw	0x0	ctr bit 63 to 32 of 128 bit

25.25 encStartAddr				0x40038060
Bits	Field Name	Acc	Default	Description
23 : 0	encStartAddr	rw	0x0	data encryption start addr

25.26 encEndAddr				0x40038064
Bits	Field Name	Acc	Default	Description
23 : 0	encEndAddr	rw	0xFFFFFF	data encryption end addr

25.27 flash_decrypt_ctrl				0x40038068
Bits	Field Name	Acc	Default	Description
0	flashDecrEn	rw	0x0	flash data decryption enable
1	aesKeyValid	rw	0x0	aes key valid set this bit when aes key write done
2	ce_clear	wo	0x0	decryptor rst and clear
31	debug_mode	wo	0x0	key and nonce can be read when this bit set

26 cpu_sysctrl

26.1 hclk_ctrl_mode				0x40100000
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_hclk_mode	rw	0xFFFFFFFF	1'b1: manual mode; 1'b0: auto mode

26.2 hclk1_ctrl_mode				0x40100004
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_hclk1_mode	rw	0xFFFFFFFF	1'b1: manual mode; 1'b0: auto mode

26.3 pclk_ctrl_mode				0x40100008
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_pclk_mode	rw	0xFFFFFFFF	1'b1: manual mode; 1'b0: auto mode

26.4 others_clk_ctrl_mode				0x4010000C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_others_clk_mode	rw	0xFFFFFFFF	1'b1: manual mode; 1'b0: auto mode

26.5 cfg_mem_rom				0x40100010
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_rom	rw	0x2006	memory configration

26.6 cfg_mem_1prf				0x40100014
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_1prf	rw	0x2488	memory configration

26.7 cfg_mem_2prf				0x40100018
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_2prf	rw	0x20D6	memory configration

26.8 trap_bypass				0x4010001C
Bits	Field Name	Acc	Default	Description
0	cfg_trap_bypass	rw	0x1	1'b1: bypass trap

26.9 cache_bypass				0x40100020
Bits	Field Name	Acc	Default	Description
0	cfg_cache_bypass	rw	0x1	1'b1: bypass cache, 32KB ram used for normal sram
1	cfg_burst_incr_only_en	rw	0x1	1'b1: trans ahb incr4/8/16 burst to incr only when noncache access

26.10 pclk_div				0x40100024
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_pclk_div_denom	rw	0x4	APB clock divider denominator
8	cfg_pclk_div_update	rw	0x0	APB clk divider update

26.11 codec_mclk_div				0x40100028
Bits	Field Name	Acc	Default	Description
5 : 0	cfg_mclk_div_denom	rw	0x4	mclk clock divider denominator
8	cfg_mclk_div_update	rw	0x0	mclk divider update

26.12 bclk_sel				0x4010002C
Bits	Field Name	Acc	Default	Description
0	cfg_bck_0_sel	rw	0x0	1'b0: pad_i2s_bck0; 1'b1: mclk_div
1	cfg_bck_0_inv	rw	0x0	bck0 invert enable
2	cfg_bck_1_sel	rw	0x0	1'b0: pad_i2s_bck1; 1'b1: mclk_div
3	cfg_bck_1_inv	rw	0x0	bck1 invert enable

26.13 cpu_sys_tports_sel	0x40100030

Bits	Field Name	Acc	Default	Description
7 : 0	cfg_cpu_sys_tports_sel	rw	0x0	test ports selector
15 : 8	cfg_dbg_clk_sel	rw	0x0	debug clock selector

26.14 phy_dump_sel				0x40100034
Bits	Field Name	Acc	Default	Description
0	cfg_phy_dump_sel	rw	0x0	1: BT; 0: WLAN

26.15 clk_msadc_div				0x40100038
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_clk_msadc_div_denom	rw	0x8	msadc clock divider denominator
8	cfg_clk_msadc_div_update	rw	0x0	msadc clock divider update
9	cfg_clk_msadc_and_inv	rw	0x0	msadc clock invert sel for analog ADC

26.16 cfg_cpu_stclk_div				0x4010003C
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_cpu_stclk_div	rw	0x27	CPU stclk divider, default div40; 40MHz/40=1MHz

26.17 cfg_usb_gpio				0x40100040
Bits	Field Name	Acc	Default	Description
0	cfg_usb_gpio_mode	rw	0x0	
2 : 1	cfg_usb_gpio_pullup	rw	0x0	
4 : 3	cfg_usb_gpio_pulldown	rw	0x0	

26.18 cfg_cpu_p_rxev				0x40100044
Bits	Field Name	Acc	Default	Description
0	cfg_cpu_p_rxev_bt_dma_en	rw	0x0	
1	cfg_cpu_p_rxev_bt_uart_en	rw	0x0	
2	cfg_cpu_p_rxev_pwrctrl_en	rw	0x0	
3	cfg_cpu_p_rxev_pmu_rtc0_en	rw	0x0	
4	cfg_cpu_p_rxev_pmu_rtc1_en	rw	0x0	
5	cfg_cpu_p_rxev_gpiob_en	rw	0x0	
6	cfg_cpu_p_rxev_aon_pwm_en	rw	0x0	
7	cfg_cpu_p_rxev_uart0_en	rw	0x0	
8	cfg_cpu_p_rxev_uart1_en	rw	0x0	
9	cfg_cpu_p_rxev_uart2_en	rw	0x0	
10	cfg_cpu_p_rxev_dma_en	rw	0x0	
11	cfg_cpu_p_rxev_usb_dma_en	rw	0x0	
12	cfg_cpu_p_rxev_pwm_en	rw	0x0	
13	cfg_cpu_p_rxev_gpioa_en	rw	0x0	
14	cfg_cpu_p_rxev_spi_en	rw	0x0	
15	cfg_cpu_p_rxev_pcie_en	rw	0x0	
16	cfg_cpu_p_rxev_trng_en	rw	0x0	
17	cfg_cpu_p_rxev_i2cm_en	rw	0x0	
18	cfg_cpu_p_rxev_sdmmc_en	rw	0x0	
19	cfg_cpu_p_rxev_psi_mst_en	rw	0x0	
20	cfg_cpu_p_rxev_usb_ready_en	rw	0x0	
21	cfg_cpu_p_rxev_pcie_ready_en	rw	0x0	

26.19 cfg_cpu_s0_rxev				0x40100048
Bits	Field Name	Acc	Default	Description
0	cfg_cpu_s0_rxev_bt_dma_en	rw	0x0	

1	cfg_cpu_s0_rxev_bt_uart_en	rw	0x0	
2	cfg_cpu_s0_rxev_pwrctrl_en	rw	0x0	
3	cfg_cpu_s0_rxev_pmu_rtc0_en	rw	0x0	
4	cfg_cpu_s0_rxev_pmu_rtc1_en	rw	0x0	
5	cfg_cpu_s0_rxev_gpiob_en	rw	0x0	
6	cfg_cpu_s0_rxev_aon_pwm_en	rw	0x0	
7	cfg_cpu_s0_rxev_uart0_en	rw	0x0	
8	cfg_cpu_s0_rxev_uart1_en	rw	0x0	
9	cfg_cpu_s0_rxev_uart2_en	rw	0x0	
10	cfg_cpu_s0_rxev_dma_en	rw	0x0	
11	cfg_cpu_s0_rxev_usb_dma_en	rw	0x0	
12	cfg_cpu_s0_rxev_pwm_en	rw	0x0	
13	cfg_cpu_s0_rxev_gpioa_en	rw	0x0	
14	cfg_cpu_s0_rxev_spi_en	rw	0x0	
15	cfg_cpu_s0_rxev_pcie_en	rw	0x0	
16	cfg_cpu_s0_rxev_trng_en	rw	0x0	
17	cfg_cpu_s0_rxev_i2cm_en	rw	0x0	
18	cfg_cpu_s0_rxev_sdmmc_en	rw	0x0	
19	cfg_cpu_s0_rxev_psi_mst_en	rw	0x0	
20	cfg_cpu_s0_rxev_usb_ready_en	rw	0x0	
21	cfg_cpu_s0_rxev_pcie_ready_en	rw	0x0	

26.20 cfg_cpu_nmi_en				0x4010004C
Bits	Field Name	Acc	Default	Description
0	cfg_cpu_p_nmi_en	rw	0x0	

26.21 ro_top_tports				0x40100050
Bits	Field Name	Acc	Default	Description
15 : 0	ro_top_tports	ro	0x0	

26.22 hclk_manual_enable				0x40100100
Bits	Field Name	Acc	Default	Description
0	hclk_cpu_p_manual_enable	rw	0x1	write 1'b1 enable manual mode
1	hclk_cpu_p_g-ated_manual_enable	rw	0x1	write 1'b1 enable manual mode
2	hclk_cpu_s0_manual_enable	rw	0x1	write 1'b1 enable manual mode
3	hclk_cpu_s0_-gated_manual_enable	rw	0x1	write 1'b1 enable manual mode
4	hclk_rom_manual_enable	rw	0x1	write 1'b1 enable manual mode
5	hclk_ram0_manual_enable	rw	0x1	write 1'b1 enable manual mode
6	hclk_ram1_manual_enable	rw	0x1	write 1'b1 enable manual mode
7	hclk_ram2_manual_enable	rw	0x1	write 1'b1 enable manual mode
8	hclk_ram3_manual_enable	rw	0x1	write 1'b1 enable manual mode
9	hclk_ram4_manual_enable	rw	0x1	write 1'b1 enable manual mode
10	hclk_ram5_manual_enable	rw	0x1	write 1'b1 enable manual mode
11	hclk_ram6_manual_enable	rw	0x1	write 1'b1 enable manual mode
12	hclk_sys_matr-ix_manual_enable	rw	0x1	write 1'b1 enable manual mode
13	hclk_ahb0_manual_enable	rw	0x1	write 1'b1 enable manual mode
14	hclk_apb0_manual_enable	rw	0x1	write 1'b1 enable manual mode
15	hclk_exmem_bu-s_manual_enable	rw	0x1	write 1'b1 enable manual mode
16	hclk_cache_manual_enable	rw	0x1	write 1'b1 enable manual mode
17	hclk_dma_manual_enable	rw	0x1	write 1'b1 enable manual mode
18	hclk_flash_manual_enable	rw	0x1	write 1'b1 enable manual mode
19	hclk_trap_p_manual_enable	rw	0x1	write 1'b1 enable manual mode

20	hclk_trap_s0_manual_enable	rw	0x1	write 1'b1 enable manual mode
21	hclk_uart0_manual_enable	rw	0x1	write 1'b1 enable manual mode
22	hclk_uart1_manual_enable	rw	0x1	write 1'b1 enable manual mode
23	hclk_uart2_manual_enable	rw	0x1	write 1'b1 enable manual mode
24	hclk_usbc_manual_enable	rw	0x1	write 1'b1 enable manual mode
25	hclk_aud_proc_manual_enable	rw	0x1	write 1'b1 enable manual mode
26	hclk_mailbox_manual_enable	rw	0x1	write 1'b1 enable manual mode
27	hclk_sdio_manual_enable	rw	0x1	write 1'b1 enable manual mode
28	hclk_mm_manual_enable	rw	0x1	write 1'b1 enable manual mode
29	hclk_pcie_manual_enable	rw	0x1	write 1'b1 enable manual mode

26.23 hclk_manual_disable				0x40100104
Bits	Field Name	Acc	Default	Description
0	hclk_cpu_p_manual_disable	rw	0x1	write 1'b1 disable manual mode
1	hclk_cpu_p_gated_manual_disable	rw	0x1	write 1'b1 disable manual mode
2	hclk_cpu_s0_manual_disable	rw	0x1	write 1'b1 disable manual mode
3	hclk_cpu_s0_gated_manual_disable	rw	0x1	write 1'b1 disable manual mode
4	hclk_rom_manual_disable	rw	0x1	write 1'b1 disable manual mode
5	hclk_ram0_manual_disable	rw	0x1	write 1'b1 disable manual mode
6	hclk_ram1_manual_disable	rw	0x1	write 1'b1 disable manual mode
7	hclk_ram2_manual_disable	rw	0x1	write 1'b1 disable manual mode
8	hclk_ram3_manual_disable	rw	0x1	write 1'b1 disable manual mode
9	hclk_ram4_manual_disable	rw	0x1	write 1'b1 disable manual mode
10	hclk_ram5_manual_disable	rw	0x1	write 1'b1 disable manual mode
11	hclk_ram6_manual_disable	rw	0x1	write 1'b1 disable manual mode
12	hclk_sys_matrix_manual_disable	rw	0x1	write 1'b1 disable manual mode
13	hclk_ahb0_manual_disable	rw	0x1	write 1'b1 disable manual mode
14	hclk_apb0_manual_disable	rw	0x1	write 1'b1 disable manual mode
15	hclk_exmem_bus_manual_disable	rw	0x1	write 1'b1 disable manual mode
16	hclk_cache_manual_disable	rw	0x1	write 1'b1 disable manual mode
17	hclk_dma_manual_disable	rw	0x1	write 1'b1 disable manual mode
18	hclk_flash_manual_disable	rw	0x1	write 1'b1 disable manual mode
19	hclk_trap_p_manual_disable	rw	0x1	write 1'b1 disable manual mode
20	hclk_trap_s0_manual_disable	rw	0x1	write 1'b1 disable manual mode
21	hclk_uart0_manual_disable	rw	0x1	write 1'b1 disable manual mode
22	hclk_uart1_manual_disable	rw	0x1	write 1'b1 disable manual mode
23	hclk_uart2_manual_disable	rw	0x1	write 1'b1 disable manual mode
24	hclk_usbc_manual_disable	rw	0x1	write 1'b1 disable manual mode
25	hclk_aud_proc_manual_disable	rw	0x1	write 1'b1 disable manual mode
26	hclk_mailbox_manual_disable	rw	0x1	write 1'b1 disable manual mode
27	hclk_sdio_manual_disable	rw	0x1	write 1'b1 disable manual mode
28	hclk_mm_manual_disable	rw	0x1	write 1'b1 disable manual mode
29	hclk_pcie_manual_disable	rw	0x1	write 1'b1 disable manual mode

26.24 hclk1_manual_enable				0x40100108
Bits	Field Name	Acc	Default	Description
0	hclk_sdmmc_always_manual_enable	rw	0x1	write 1'b1 enable manual mode
1	hclk_sdmmc_manual_enable	rw	0x1	write 1'b1 enable manual mode
2	hclk_sram0_low_manual_enable	rw	0x1	write 1'b1 enable manual mode
3	hclk_sram0_high_manual_enable	rw	0x1	write 1'b1 enable manual mode

4	hclk_sram1_low_manual_enable	rw	0x1	write 1'b1 enable manual mode
5	hclk_sram1_high_manual_enable	rw	0x1	write 1'b1 enable manual mode

26.25 hclk1_manual_disable				0x4010010C
Bits	Field Name	Acc	Default	Description
0	hclk_sdmmc_always_manual_disable	rw	0x1	write 1'b1 disable manual mode
1	hclk_sdmmc_manual_disable	rw	0x1	write 1'b1 disable manual mode
2	hclk_sram0_low_manual_disable	rw	0x1	write 1'b1 disable manual mode
3	hclk_sram0_high_manual_disable	rw	0x1	write 1'b1 disable manual mode
4	hclk_sram1_low_manual_disable	rw	0x1	write 1'b1 disable manual mode
5	hclk_sram1_high_manual_disable	rw	0x1	write 1'b1 disable manual mode

26.26 pclk_manual_enable				0x40100110
Bits	Field Name	Acc	Default	Description
0	pclk_cpu_sysctrl_manual_enable	rw	0x1	write 1'b1 enable manual mode
1	pclk_wdg0_manual_enable	rw	0x1	write 1'b1 enable manual mode
2	pclk_wdg1_manual_enable	rw	0x1	write 1'b1 enable manual mode
3	pclk_timer0_manual_enable	rw	0x1	write 1'b1 enable manual mode
4	pclk_timer1_manual_enable	rw	0x1	write 1'b1 enable manual mode
5	pclk_pwm_manual_enable	rw	0x1	write 1'b1 enable manual mode
6	pclk_i2cm_manual_enable	rw	0x1	write 1'b1 enable manual mode
7	pclk_mod_i2cm_manual_enable	rw	0x1	write 1'b1 enable manual mode
8	pclk_spi_manual_enable	rw	0x1	write 1'b1 enable manual mode
9	pclk_trng_manual_enable	rw	0x1	write 1'b1 enable manual mode
10	pclk_process_mon_manual_enable	rw	0x1	write 1'b1 enable manual mode
11	pclk_analog_reg0_manual_enable	rw	0x1	write 1'b1 enable manual mode
12	pclk_gpioa_manual_enable	rw	0x1	write 1'b1 enable manual mode
13	pclk_msadc_manual_enable	rw	0x1	write 1'b1 enable manual mode
14	pclk_efuse_manual_enable	rw	0x1	write 1'b1 enable manual mode

26.27 pclk_manual_disable				0x40100114
Bits	Field Name	Acc	Default	Description
0	pclk_cpu_sysctrl_manual_disable	rw	0x1	write 1'b1 disable manual mode
1	pclk_wdg0_manual_disable	rw	0x1	write 1'b1 disable manual mode
2	pclk_wdg1_manual_disable	rw	0x1	write 1'b1 disable manual mode
3	pclk_timer0_manual_disable	rw	0x1	write 1'b1 disable manual mode
4	pclk_timer1_manual_disable	rw	0x1	write 1'b1 disable manual mode
5	pclk_pwm_manual_disable	rw	0x1	write 1'b1 disable manual mode
6	pclk_i2cm_manual_disable	rw	0x1	write 1'b1 disable manual mode
7	pclk_mod_i2cm_manual_disable	rw	0x1	write 1'b1 disable manual mode
8	pclk_spi_manual_disable	rw	0x1	write 1'b1 disable manual mode
9	pclk_trng_manual_disable	rw	0x1	write 1'b1 disable manual mode
10	pclk_process_mon_manual_disable	rw	0x1	write 1'b1 disable manual mode

11	pclk_analog_reg0_manual_disable	rw	0x1	write 1'b1 disable manual mode
12	pclk_gpioa_manual_disable	rw	0x1	write 1'b1 disable manual mode
13	pclk_msadc_manual_disable	rw	0x1	write 1'b1 disable manual mode
14	pclk_efuse_manual_disable	rw	0x1	write 1'b1 disable manual mode

26.28 others_manual_enable				0x40100118
Bits	Field Name	Acc	Default	Description
0	others_aux_pcie_manual_enable	rw	0x1	write 1'b1 enable manual mode
1	others_sdmmc_manual_enable	rw	0x1	write 1'b1 enable manual mode
2	others_flash_manual_enable	rw	0x1	write 1'b1 enable manual mode
3	others_time0_manual_enable	rw	0x1	write 1'b1 enable manual mode
4	others_time1_manual_enable	rw	0x1	write 1'b1 enable manual mode
5	others_uart0_manual_enable	rw	0x1	write 1'b1 enable manual mode
6	others_uart1_manual_enable	rw	0x1	write 1'b1 enable manual mode
7	others_uart2_manual_enable	rw	0x1	write 1'b1 enable manual mode
8	others_sdio_manual_enable	rw	0x1	write 1'b1 enable manual mode
9	others_pwm_manual_enable	rw	0x1	write 1'b1 enable manual mode
10	others_rtc_always_manual_enable	rw	0x1	write 1'b1 enable manual mode
11	others_rtc_wdg0_manual_enable	rw	0x1	write 1'b1 enable manual mode
12	others_rtc_wdg1_manual_enable	rw	0x1	write 1'b1 enable manual mode
13	others_audio_proc_manual_enable	rw	0x1	write 1'b1 enable manual mode
14	others_process_mon_manual_enable	rw	0x1	write 1'b1 enable manual mode
15	others_bck_0_manual_enable	rw	0x1	write 1'b1 enable manual mode
16	others_bck_1_manual_enable	rw	0x1	write 1'b1 enable manual mode
17	others_spdif_manual_enable	rw	0x1	write 1'b1 enable manual mode
18	others_spi_manual_enable	rw	0x1	write 1'b1 enable manual mode
19	others_msadc_manual_enable	rw	0x1	write 1'b1 enable manual mode
20	others_efuse_manual_enable	rw	0x1	write 1'b1 enable manual mode
21	others_1m_pcie_manual_enable	rw	0x1	write 1'b1 enable manual mode
22	others_usbc_manual_enable	rw	0x1	write 1'b1 enable manual mode

26.29 others_manual_disable				0x4010011C
Bits	Field Name	Acc	Default	Description
0	others_aux_pcie_manual_disable	rw	0x1	write 1'b1 disable manual mode
1	others_sdmmc_manual_disable	rw	0x1	write 1'b1 disable manual mode
2	others_flash_manual_disable	rw	0x1	write 1'b1 disable manual mode
3	others_time0_manual_disable	rw	0x1	write 1'b1 disable manual mode
4	others_time1_manual_disable	rw	0x1	write 1'b1 disable manual mode
5	others_uart0_manual_disable	rw	0x1	write 1'b1 disable manual mode
6	others_uart1_manual_disable	rw	0x1	write 1'b1 disable manual mode
7	others_uart2_manual_disable	rw	0x1	write 1'b1 disable manual mode
8	others_sdio_manual_disable	rw	0x1	write 1'b1 disable manual mode
9	others_pwm_manual_disable	rw	0x1	write 1'b1 disable manual mode
10	others_rtc_always_manual_disable	rw	0x1	write 1'b1 disable manual mode
11	others_rtc_wdg0_manual_disable	rw	0x1	write 1'b1 disable manual mode
12	others_rtc_wdg1_manual_disable	rw	0x1	write 1'b1 disable manual mode

13	others_audio_-proc_manual_disable	rw	0x1	write 1'b1 disable manual mode
14	others_process-_mon_manual_disable	rw	0x1	write 1'b1 disable manual mode
15	others_bck_0_manual_disable	rw	0x1	write 1'b1 disable manual mode
16	others_bck_1_manual_disable	rw	0x1	write 1'b1 disable manual mode
17	others_spdif_manual_disable	rw	0x1	write 1'b1 disable manual mode
18	others_spi_manual_disable	rw	0x1	write 1'b1 disable manual mode
19	others_msadc_manual_disable	rw	0x1	write 1'b1 disable manual mode
20	others_efuse_manual_disable	rw	0x1	write 1'b1 disable manual mode
21	others_1m_pci-e_manual_disable	rw	0x1	write 1'b1 disable manual mode
22	others_usbc_manual_disable	rw	0x1	write 1'b1 disable manual mode

26.30 hclk_soft_resetn_set				0x40100120
Bits	Field Name	Acc	Default	Description
0	hresetn_rom_set	rw	0x1	write 1'b1 set soft reset
1	hresetn_sram0_set	rw	0x1	write 1'b1 set soft reset
2	hresetn_sram1_set	rw	0x1	write 1'b1 set soft reset
3	hresetn_sram2_set	rw	0x1	write 1'b1 set soft reset
4	hresetn_sram3_set	rw	0x1	write 1'b1 set soft reset
5	hresetn_sram4_set	rw	0x1	write 1'b1 set soft reset
6	hresetn_sram5_set	rw	0x1	write 1'b1 set soft reset
7	hresetn_sram6_set	rw	0x1	write 1'b1 set soft reset
8	hresetn_ahb0_dec_set	rw	0x1	write 1'b1 set soft reset
9	hresetn_apb0_set	rw	0x1	write 1'b1 set soft reset
10	hresetn_exmem_bus_set	rw	0x1	write 1'b1 set soft reset
11	hresetn_cache_set	rw	0x1	write 1'b1 set soft reset
12	hresetn_dma_set	rw	0x1	write 1'b1 set soft reset
13	hresetn_flash_set	rw	0x1	write 1'b1 set soft reset
14	hresetn_trap_p_set	rw	0x1	write 1'b1 set soft reset
15	hresetn_trap_s0_set	rw	0x1	write 1'b1 set soft reset
16	hresetn_uart0_set	rw	0x1	write 1'b1 set soft reset
17	hresetn_uart1_set	rw	0x1	write 1'b1 set soft reset
18	hresetn_uart2_set	rw	0x1	write 1'b1 set soft reset
19	hresetn_pcie_set	rw	0x1	write 1'b1 set soft reset
20	hresetn_aud_proc_set	rw	0x1	write 1'b1 set soft reset
21	hresetn_mailbox_set	rw	0x1	write 1'b1 set soft reset
22	hresetn_sdio_set	rw	0x1	write 1'b1 set soft reset
23	hresetn_mm_sys_set	rw	0x1	write 1'b1 set soft reset
24	hresetn_sdmmc_set	rw	0x1	write 1'b1 set soft reset

26.31 hclk_soft_resetn_clr				0x40100124
Bits	Field Name	Acc	Default	Description
0	hresetn_rom_clear	rw	0x1	write 1'b1 clear soft reset
1	hresetn_sram0_clear	rw	0x1	write 1'b1 clear soft reset
2	hresetn_sram1_clear	rw	0x1	write 1'b1 clear soft reset
3	hresetn_sram2_clear	rw	0x1	write 1'b1 clear soft reset
4	hresetn_sram3_clear	rw	0x1	write 1'b1 clear soft reset
5	hresetn_sram4_clear	rw	0x1	write 1'b1 clear soft reset
6	hresetn_sram5_clear	rw	0x1	write 1'b1 clear soft reset
7	hresetn_sram6_clear	rw	0x1	write 1'b1 clear soft reset
8	hresetn_ahb0_dec_clear	rw	0x1	write 1'b1 clear soft reset
9	hresetn_apb0_clear	rw	0x1	write 1'b1 clear soft reset
10	hresetn_exmem_bus_clear	rw	0x1	write 1'b1 clear soft reset
11	hresetn_cache_clear	rw	0x1	write 1'b1 clear soft reset

12	hresetn_dma_clear	rw	0x1	write 1'b1 clear soft reset
13	hresetn_flash_clear	rw	0x1	write 1'b1 clear soft reset
14	hresetn_trap_p_clear	rw	0x1	write 1'b1 clear soft reset
15	hresetn_trap_s0_clear	rw	0x1	write 1'b1 clear soft reset
16	hresetn_uart0_clear	rw	0x1	write 1'b1 clear soft reset
17	hresetn_uart1_clear	rw	0x1	write 1'b1 clear soft reset
18	hresetn_uart2_clear	rw	0x1	write 1'b1 clear soft reset
19	hresetn_pcie_clear	rw	0x1	write 1'b1 clear soft reset
20	hresetn_aud_proc_clear	rw	0x1	write 1'b1 clear soft reset
21	hresetn_mailbox_clear	rw	0x1	write 1'b1 clear soft reset
22	hresetn_sdio_clear	rw	0x1	write 1'b1 clear soft reset
23	hresetn_mm_sys_clear	rw	0x1	write 1'b1 clear soft reset
24	hresetn_sdmmc_clear	rw	0x1	write 1'b1 clear soft reset

26.32 hclk1_soft_resetn_set				0x40100128
Bits	Field Name	Acc	Default	Description

26.33 hclk1_soft_resetn_clr				0x4010012C
Bits	Field Name	Acc	Default	Description

26.34 pclk_soft_resetn_set				0x40100130
Bits	Field Name	Acc	Default	Description
0	presetn_cpu_sysctrl_set	rw	0x1	write 1'b1 set soft reset
1	presetn_wdg0_set	rw	0x1	write 1'b1 set soft reset
2	presetn_wdg1_set	rw	0x1	write 1'b1 set soft reset
3	presetn_timer0_set	rw	0x1	write 1'b1 set soft reset
4	presetn_timer1_set	rw	0x1	write 1'b1 set soft reset
5	presetn_pwm_set	rw	0x1	write 1'b1 set soft reset
6	presetn_i2cm_set	rw	0x1	write 1'b1 set soft reset
7	presetn_mod_i2cm_set	rw	0x1	write 1'b1 set soft reset
8	presetn_spi_set	rw	0x1	write 1'b1 set soft reset
9	presetn_trng_set	rw	0x1	write 1'b1 set soft reset
10	presetn_process_mon_set	rw	0x1	write 1'b1 set soft reset
11	presetn_analog_reg0_set	rw	0x1	write 1'b1 set soft reset
12	presetn_gpioa_set	rw	0x1	write 1'b1 set soft reset
13	presetn_msadc_set	rw	0x1	write 1'b1 set soft reset
14	presetn_efuse_set	rw	0x1	write 1'b1 set soft reset

26.35 pclk_soft_resetn_clr				0x40100134
Bits	Field Name	Acc	Default	Description
0	presetn_cpu_sysctrl_clear	rw	0x1	write 1'b1 clear soft reset
1	presetn_wdg0_clear	rw	0x1	write 1'b1 clear soft reset
2	presetn_wdg1_clear	rw	0x1	write 1'b1 clear soft reset
3	presetn_timer0_clear	rw	0x1	write 1'b1 clear soft reset
4	presetn_timer1_clear	rw	0x1	write 1'b1 clear soft reset
5	presetn_pwm_clear	rw	0x1	write 1'b1 clear soft reset
6	presetn_i2cm_clear	rw	0x1	write 1'b1 clear soft reset
7	presetn_mod_i2cm_clear	rw	0x1	write 1'b1 clear soft reset
8	presetn_spi_clear	rw	0x1	write 1'b1 clear soft reset
9	presetn_trng_clear	rw	0x1	write 1'b1 clear soft reset
10	presetn_process_mon_clear	rw	0x1	write 1'b1 clear soft reset
11	presetn_analog_reg0_clear	rw	0x1	write 1'b1 clear soft reset
12	presetn_gpioa_clear	rw	0x1	write 1'b1 clear soft reset
13	presetn_msadc_clear	rw	0x1	write 1'b1 clear soft reset

14	presetn_efuse_clear	rw	0x1	write 1'b1 clear soft reset

26.36 others_soft_resetn_set				0x40100138
Bits	Field Name	Acc	Default	Description
0	resetn_timer0_set	rw	0x1	write 1'b1 set soft reset
1	resetn_timer1_set	rw	0x1	write 1'b1 set soft reset
2	resetn_uart0_set	rw	0x1	write 1'b1 set soft reset
3	resetn_uart1_set	rw	0x1	write 1'b1 set soft reset
4	resetn_uart2_set	rw	0x1	write 1'b1 set soft reset
5	resetn_sdio_set	rw	0x1	write 1'b1 set soft reset
6	resetn_pwm_set	rw	0x1	write 1'b1 set soft reset
7	resetn_rtc_wdg0_set	rw	0x1	write 1'b1 set soft reset
8	resetn_rtc_wdg1_set	rw	0x1	write 1'b1 set soft reset
9	resetn_audio_proc_set	rw	0x1	write 1'b1 set soft reset
10	resetn_process_mon_set	rw	0x1	write 1'b1 set soft reset
11	resetn_spi_set	rw	0x1	write 1'b1 set soft reset
12	resetn_flash_set	rw	0x1	write 1'b1 set soft reset
13	resetn_msadc_set	rw	0x1	write 1'b1 set soft reset
14	resetn_efuse_set	rw	0x1	write 1'b1 set soft reset
15	resetn_1m_pcie_set	rw	0x1	write 1'b1 set soft reset
16	resetn_usbc_set	rw	0x1	write 1'b1 set soft reset

26.37 others_soft_resetn_clr				0x4010013C
Bits	Field Name	Acc	Default	Description
0	resetn_timer0_clear	rw	0x1	write 1'b1 clear soft reset
1	resetn_timer1_clear	rw	0x1	write 1'b1 clear soft reset
2	resetn_uart0_clear	rw	0x1	write 1'b1 clear soft reset
3	resetn_uart1_clear	rw	0x1	write 1'b1 clear soft reset
4	resetn_uart2_clear	rw	0x1	write 1'b1 clear soft reset
5	resetn_sdio_clear	rw	0x1	write 1'b1 clear soft reset
6	resetn_pwm_clear	rw	0x1	write 1'b1 clear soft reset
7	resetn_rtc_wdg0_clear	rw	0x1	write 1'b1 clear soft reset
8	resetn_rtc_wdg1_clear	rw	0x1	write 1'b1 clear soft reset
9	resetn_audio_proc_clear	rw	0x1	write 1'b1 clear soft reset
10	resetn_process_mon_clear	rw	0x1	write 1'b1 clear soft reset
11	resetn_spi_clear	rw	0x1	write 1'b1 clear soft reset
12	resetn_flash_clear	rw	0x1	write 1'b1 clear soft reset
13	resetn_msadc_clear	rw	0x1	write 1'b1 clear soft reset
14	resetn_efuse_clear	rw	0x1	write 1'b1 clear soft reset
15	resetn_1m_pcie_clear	rw	0x1	write 1'b1 clear soft reset
16	resetn_usbc_clear	rw	0x1	write 1'b1 clear soft reset

27 pwm

27.1 tmr0_ld_value				0x40101000
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_ld_value	rw	0x0	timer0 load value

27.2 tmr0_cnt_value				0x40101008
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_value	rw	0x0	timer0 counter value

27.3 tmr0_cfg				0x40101010
Bits	Field Name	Acc	Default	Description
0	tmr_mode	rw	0x0	0: free running 1: periodic
1	tmr_run	rw	0x0	timer0 enable

27.4 tmr0_int_ctrl				0x40101014
Bits	Field Name	Acc	Default	Description
0	tmr_int_en	rw	0x0	timer0 interrupt enable
1	tmr_int_raw	ro	0x0	timer0 interrupt raw
2	tmr_int_status	ro	0x0	timer0 interrupt status
3	tmr_int_clr	wo	0x0	timer0 interrupt clear

27.5 tmr0_cnt_read				0x40101018
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_read	rw	0x0	timer0 counter read value

27.6 tmr1_ld_value				0x40101020
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_ld_value	rw	0x0	timer1 load value

27.7 tmr1_cnt_value				0x40101028
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_value	rw	0x0	timer1 counter value

27.8 tmr1_cfg				0x40101030
Bits	Field Name	Acc	Default	Description
0	tmr_mode	rw	0x0	0: free running 1: periodic
1	tmr_run	rw	0x0	timer1 enable

27.9 tmr1_int_ctrl				0x40101034
Bits	Field Name	Acc	Default	Description
0	tmr_int_en	rw	0x0	timer1 interrupt enable
1	tmr_int_raw	ro	0x0	timer1 interrupt raw
2	tmr_int_status	ro	0x0	timer1 interrupt status
3	tmr_int_clr	wo	0x0	timer1 interrupt clear

27.10 tmr1_cnt_read				0x40101038
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_read	rw	0x0	timer1 counter read value

27.11 tmr2_ld_value				0x40101040
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_ld_value	rw	0x0	timer2 load value

27.12 tmr2_cnt_value				0x40101048
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_value	rw	0x0	timer2 counter value

27.13 tmr2_cfg				0x40101050
Bits	Field Name	Acc	Default	Description
0	tmr_mode	rw	0x0	0: free running 1: periodic
1	tmr_run	rw	0x0	timer2 enable

27.14 tmr2_int_ctrl				0x40101054
Bits	Field Name	Acc	Default	Description
0	tmr_int_en	rw	0x0	timer2 interrupt enable
1	tmr_int_raw	ro	0x0	timer2 interrupt raw
2	tmr_int_status	ro	0x0	timer2 interrupt status
3	tmr_int_clr	wo	0x0	timer2 interrupt clear

27.15 tmr2_cnt_read				0x40101058
Bits	Field Name	Acc	Default	Description
31 : 0	tmr2_cnt_read	rw	0x0	timer2 counter read value

27.16 pwm0_sta_val				0x40101100
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_sta_val	rw	0x0	

27.17 pwm0_end_val				0x40101104
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val	rw	0x0	

27.18 pwm0_step_cfg0				0x40101108
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc	rw	0x0	
31 : 16	pwm_step_num	rw	0x0	

27.19 pwm0_step_cfg1				0x4010110C
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_step_value	rw	0x0	

27.20 pwm0_cfg				0x40101110
Bits	Field Name	Acc	Default	Description
0	pwm_run	rw	0x0	
1	pwm_inc_mode	rw	0x0	
2	pwm_default_val	rw	0x0	
17 : 16	pwm_tmr_sel	rw	0x0	
31	pwm_update	rw	0x0	

27.21 pwm0_int_ctrl				0x40101114
Bits	Field Name	Acc	Default	Description
0	pwm_int_en	rw	0x0	
1	pwm_int_raw	ro	0x0	
2	pwm_int_status	ro	0x0	
3	pwm_int_clr	wo	0x0	

27.22 pwm0_end_val_use	0x40101118

Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val_use	ro	0x0	

27.23 pwm0_step_cnt				0x4010111C
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc_cnt	ro	0x0	
31 : 16	pwm_step_num_cnt	ro	0x0	

27.24 pwm1_sta_val				0x40101120
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_sta_val	rw	0x0	

27.25 pwm1_end_val				0x40101124
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val	rw	0x0	

27.26 pwm1_step_cfg0				0x40101128
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc	rw	0x0	
31 : 16	pwm_step_num	rw	0x0	

27.27 pwm1_step_cfg1				0x4010112C
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_step_value	rw	0x0	

27.28 pwm1_cfg				0x40101130
Bits	Field Name	Acc	Default	Description
0	pwm_run	rw	0x0	
1	pwm_inc_mode	rw	0x0	
2	pwm_default_val	rw	0x0	
17 : 16	pwm_tmr_sel	rw	0x0	
31	pwm_update	rw	0x0	

27.29 pwm1_int_ctrl				0x40101134
Bits	Field Name	Acc	Default	Description
0	pwm_int_en	rw	0x0	
1	pwm_int_raw	ro	0x0	
2	pwm_int_status	ro	0x0	
3	pwm_int_clr	wo	0x0	

27.30 pwm1_end_val_use				0x40101138
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val_use	ro	0x0	

27.31 pwm1_step_cnt				0x4010113C
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc_cnt	ro	0x0	
31 : 16	pwm_step_num_cnt	ro	0x0	

27.32 pwm2_sta_val				0x40101140
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_sta_val	rw	0x0	

27.33 pwm2_end_val				0x40101144
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val	rw	0x0	

27.34 pwm2_step_cfg0				0x40101148
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc	rw	0x0	
31 : 16	pwm_step_num	rw	0x0	

27.35 pwm2_step_cfg1				0x4010114C
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_step_value	rw	0x0	

27.36 pwm2_cfg				0x40101150
Bits	Field Name	Acc	Default	Description
0	pwm_run	rw	0x0	
1	pwm_inc_mode	rw	0x0	
2	pwm_default_val	rw	0x0	
17 : 16	pwm_tmr_sel	rw	0x0	
31	pwm_update	rw	0x0	

27.37 pwm2_int_ctrl				0x40101154
Bits	Field Name	Acc	Default	Description
0	pwm_int_en	rw	0x0	
1	pwm_int_raw	ro	0x0	
2	pwm_int_status	ro	0x0	
3	pwm_int_clr	wo	0x0	

27.38 pwm2_end_val_use				0x40101158
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val_use	ro	0x0	

27.39 pwm2_step_cnt				0x4010115C
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc_cnt	ro	0x0	
31 : 16	pwm_step_num_cnt	ro	0x0	

28 i2cm

28.1 IC_CON				0x40102000
Bits	Field Name	Acc	Default	Description
0	master_mode	rw	0x1	
2 : 1	speed	rw	0x3	
3	ic_10bitaddr_slave	rw	0x0	
5	ic_restart_en	rw	0x1	
6	ic_slave_disable	rw	0x1	

7	stop_det_ifaddressed	rw	0x0	
8	tx_empty_ctrl	rw	0x0	

28.2 IC_TAR				0x40102004
Bits	Field Name	Acc	Default	Description
9 : 0	ic_tar	rw	0x55	
10	gc_or_start	rw	0x0	
11	special	rw	0x0	
12	ic_10bitaddr_master	rw	0x0	

28.3 IC_SAR				0x40102008
Bits	Field Name	Acc	Default	Description
9 : 0	ic_sar	rw	0x55	

28.4 IC_HS_MADDR				0x4010200C
Bits	Field Name	Acc	Default	Description
2 : 0	ic_hs_mar	rw	0x1	

28.5 IC_DATA_CMD				0x40102010
Bits	Field Name	Acc	Default	Description
7 : 0	dat	rw	0x0	
8	cmd	wo	0x0	

28.6 IC_SS_SCL_HCNT				0x40102014
Bits	Field Name	Acc	Default	Description
15 : 0	ic_ss_scl_hcnt	rw	0x320	

28.7 IC_SS_SCL_LCNT				0x40102018
Bits	Field Name	Acc	Default	Description
15 : 0	ic_ss_scl_lcnt	rw	0x3AC	

28.8 IC_FS_SCL_HCNT				0x4010201C
Bits	Field Name	Acc	Default	Description
15 : 0	ic_fs_scl_hcnt	rw	0x78	

28.9 IC_FS_SCL_LCNT				0x40102020
Bits	Field Name	Acc	Default	Description
15 : 0	ic_fs_scl_lcnt	rw	0x104	

28.10 IC_HS_SCl_HCNT				0x40102024
Bits	Field Name	Acc	Default	Description
15 : 0	ic_hs_scl_hcnt	rw	0xC	

28.11 IC_HS_SCL_LCNT				0x40102028
Bits	Field Name	Acc	Default	Description
15 : 0	ic_hs_scl_lcnt	rw	0x20	

28.12 IC_INTR_STAT				0x4010202C
Bits	Field Name	Acc	Default	Description
0	r_rx_under	ro	0x0	
1	r_rx_over	ro	0x0	
2	r_rx_full	ro	0x0	
3	r_tx_over	ro	0x0	
4	r_tx_empty	ro	0x0	
5	r_rd_req	ro	0x0	
6	r_tx_abrt	ro	0x0	
7	r_rx_done	ro	0x0	
8	r_activity	ro	0x0	
9	r_stop_det	ro	0x0	
10	r_start_det	ro	0x0	
11	r_gen_call	ro	0x0	
12	r_restart_det	ro	0x0	
13	r_master_on_hold	ro	0x0	

28.13 IC_INTR_MASK				0x40102030
Bits	Field Name	Acc	Default	Description
0	m_rx_under	rw	0x1	
1	m_rx_over	rw	0x1	
2	m_rx_full	rw	0x1	
3	m_tx_over	rw	0x1	
4	m_tx_empty	rw	0x1	
5	m_rd_req	rw	0x1	
6	m_tx_abrt	rw	0x1	
7	m_rx_done	rw	0x0	
8	m_activity	rw	0x0	
9	m_stop_det	rw	0x0	
10	m_start_det	rw	0x0	
11	m_gen_call	rw	0x1	
12	m_restart_det	rw	0x0	
13	m_master_on_hold	rw	0x0	

28.14 IC_RAW_INTR_STAT				0x40102034
Bits	Field Name	Acc	Default	Description
0	rx_under	ro	0x0	
1	rx_over	ro	0x0	
2	rx_full	ro	0x0	
3	tx_over	ro	0x0	
4	tx_empty	ro	0x0	
5	rd_req	ro	0x0	
6	tx_abrt	ro	0x0	
7	rx_done	ro	0x0	
8	activity	ro	0x0	
9	stop_det	ro	0x0	
10	start_det	ro	0x0	
11	gen_call	ro	0x0	
12	restart_det	ro	0x0	
13	master_on_hold	ro	0x0	

28.15 IC_RX_TL				0x40102038
Bits	Field Name	Acc	Default	Description
7 : 0	rx_tl	rw	0x0	

28.16 IC_TX_TL				0x4010203C
Bits	Field Name	Acc	Default	Description
7 : 0	tx_tl	rw	0x0	

28.17 IC_CLR_INTR				0x40102040
Bits	Field Name	Acc	Default	Description
0	clr_intr	ro	0x0	

28.18 IC_CLR_RX_UNDER				0x40102044
Bits	Field Name	Acc	Default	Description
0	clr_rx_under	ro	0x0	

28.19 IC_CLR_RX_OVER				0x40102048
Bits	Field Name	Acc	Default	Description
0	clr_rx_over	ro	0x0	

28.20 IC_CLR_TX_OVER				0x4010204C
Bits	Field Name	Acc	Default	Description
0	clr_tx_over	ro	0x0	

28.21 IC_CLR_RD_REQ				0x40102050
Bits	Field Name	Acc	Default	Description
0	clr_rd_req	ro	0x0	

28.22 IC_CLR_TX_ABRT				0x40102054
Bits	Field Name	Acc	Default	Description
0	clr_tx_abrt	ro	0x0	

28.23 IC_CLR_RX_DONE				0x40102058
Bits	Field Name	Acc	Default	Description
0	clr_tx_done	ro	0x0	

28.24 IC_CLR_ACTIVITY				0x4010205C
Bits	Field Name	Acc	Default	Description
0	clr_activity	ro	0x0	

28.25 IC_CLR_STOP_DET				0x40102060
Bits	Field Name	Acc	Default	Description
0	clr_stop_det	ro	0x0	

28.26 IC_CLR_START_DET				0x40102064
Bits	Field Name	Acc	Default	Description
0	clr_start_det	ro	0x0	

28.27 IC_CLR_GEN_CALL				0x40102068
Bits	Field Name	Acc	Default	Description
0	clr_gen_call	ro	0x0	

28.28 IC_ENABLE				0x4010206C
Bits	Field Name	Acc	Default	Description
0	enable	rw	0x0	
1	abort	rw	0x0	
2	tx_cmd_block	rw	0x0	

28.29 IC_STATUS				0x40102070
Bits	Field Name	Acc	Default	Description
0	activity	ro	0x0	
1	tfnf	ro	0x1	
2	tfe	ro	0x1	
3	rfne	ro	0x0	
4	rff	ro	0x0	
5	mat_activity	ro	0x0	
6	slv_activity	ro	0x0	

28.30 IC_TXFLR				0x40102074
Bits	Field Name	Acc	Default	Description
4 : 0	txflr	ro	0x0	

28.31 IC_RXFLR				0x40102078
Bits	Field Name	Acc	Default	Description
4 : 0	rxflr	ro	0x0	

28.32 IC_SDA_HOLD				0x4010207C
Bits	Field Name	Acc	Default	Description
15 : 0	ic_sda_tx_hold	rw	0x1	
23 : 16	ic_sda_rx_hold	rw	0x0	

28.33 IC_TX_ABRT_SOURCE				0x40102080
Bits	Field Name	Acc	Default	Description
0	abrt_7b_addr_noack	ro	0x0	
1	abrt_10addr1_noack	ro	0x0	
2	abrt_10addr2_noack	ro	0x0	
3	abrt_txdata_noack	ro	0x0	
4	abrt_gcall_noack	ro	0x0	
5	abrt_gcall_read	ro	0x0	
6	abrt_hs_ackdet	ro	0x0	
7	abrt_sbyte_ackdet	ro	0x0	
8	abrt_hs_norstrt	ro	0x0	
9	abrt_sbyte_norstrt	ro	0x0	
10	abrt_10b_rd_norstrt	ro	0x0	
11	abrt_master_dis	ro	0x0	
12	arb_lost	ro	0x0	
13	abrt_slvflush_txfifo	ro	0x0	
14	abrt_slv_arblost	ro	0x0	
15	abrt_slvrd_intx	ro	0x0	
16	abrt_user_abrt	ro	0x0	
31 : 23	tx_flush_cnt	ro	0x0	

28.34 IC_DMA_CR				0x40102088
Bits	Field Name	Acc	Default	Description

0	rdmae	rw	0x0	
1	tdmae	rw	0x0	

28.35 IC_DMA_TDLR				0x4010208C
Bits	Field Name	Acc	Default	Description
3 : 0	dmatdl	rw	0x0	

28.36 IC_DMA_RDLR				0x40102090
Bits	Field Name	Acc	Default	Description
3 : 0	dmardl	rw	0x0	

28.37 IC_SDA_SETUP				0x40102094
Bits	Field Name	Acc	Default	Description
7 : 0	sda_setup	rw	0x64	

28.38 IC_ACK_GENERAL_CALL				0x40102098
Bits	Field Name	Acc	Default	Description
0	ack_gen_call	rw	0x0	

28.39 IC_ENABLE_STATUS				0x4010209C
Bits	Field Name	Acc	Default	Description
0	ic_en	ro	0x0	
1	slv_disabled_while_busy	ro	0x0	
2	slv_rx_data_lost	ro	0x0	

28.40 IC_FS_SPKLEN				0x401020A0
Bits	Field Name	Acc	Default	Description
7 : 0	ic_fs_spklen	rw	0x10	

28.41 IC_HS_SPKLEN				0x401020A4
Bits	Field Name	Acc	Default	Description
7 : 0	ic_hs_spklen	rw	0x8	

28.42 IC_CLR_RESTART_DET				0x401020A8
Bits	Field Name	Acc	Default	Description
0	clr_restart_det	ro	0x0	

28.43 IC_COMP_PARAM_1				0x401020F4
Bits	Field Name	Acc	Default	Description
1 : 0	apb_data_width	ro	0x2	
3 : 2	max_speed_mode	ro	0x3	
4	hc_count_values	ro	0x0	
5	intr_io	ro	0x1	
6	has_dma	ro	0x1	
7	add_encoded_params	ro	0x1	
15 : 8	rx_buffer_depth	ro	0x8	
23 : 16	tx_buffer_depth	ro	0x8	

28.44 IC_COMP_VERSION				0x401020F8
Bits	Field Name	Acc	Default	Description
31 : 0	ic_comp_version	ro	0x3230322A	

28.45 IC_COMP_TYPE				0x401020FC
Bits	Field Name	Acc	Default	Description
31 : 0	ic_comp_type	ro	0x44570140	

29 wdg0

29.1 WDG_LOAD_VAL				0x40103000
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_load_val	rw	0x3FFFF	watchdog timer load value. In reset mode, software should load new value before timer decrease to 0. In interrupt mode, this value is counting start number. The default value is about 8 seconds.

29.2 WDG_CNT				0x40103004
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_cnt	rw	0xFFFFFFFF	watchdog counter value

29.3 WDG_CTRL				0x40103008
Bits	Field Name	Acc	Default	Description
0	wdg_irq_en	rw	0x0	Watchdog interrupt enable bit 0: interrupt is disabled 1: interrupt is enabled For reset mode: wdg_rst_en =1, wdg_irq_en=0. For interrupt mode: wdg_rst_en =0, wdg_irq_en=1. For combined mode: wdg_rst_en =1, wdg_irq_en=1.
1	wdg_run	rw	0x0	Watchdog counter open: 0: counter stops. 1: counter runs.

29.4 WDG_INT_CLR				0x4010300C
Bits	Field Name	Acc	Default	Description
0	wdg_int_clr	wo	0x0	Watchdog interrupt clear Write 1 to this bit to clear interrupt Read this bit always get 0.

29.5 WDG_INT_RAW				0x40103010
Bits	Field Name	Acc	Default	Description
0	wdg_int_raw	ro	0x0	Watchdog interrupt raw status. Watch dog reset cannot clear this raw status. Write wdg_int_clr can clear this raw status.
4	wdg_ld_busy	ro	0x0	Watchdog load busy status 0: Watchdog is ready for new loading 1: Last loading is not completed Software must not load new value when this bit is busy, that is, this bit should be checked before any new loading. This bit is set after a new loading, and lasts two or three RTC clock cycles, about 60us - 92us.

29.6 WDG_INT_MASK	0x40103014

Bits	Field Name	Acc	Default	Description
0	wdg_int_mask	ro	0x0	Watchdog interrupt masked status

29.7 WDG_LOCK				0x40103C00
Bits	Field Name	Acc	Default	Description
15 : 0	wdg_lock	rw	0x0	Watchdog lock control Write 32'h1ACCE551 to this register to unlock watchdog. Write other value to this register to lock watchdog If reading this register, bit-0 is lock status, and other bits are reserved. If watchdog is locked, all control registers cannot be written by software.

30 wdg1

30.1 WDG_LOAD_VAL				0x40104000
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_load_val	rw	0x3FFFF	watchdog timer load value. In reset mode, software should load new value before timer decrease to 0. In interrupt mode, this value is counting start number. The default value is about 8 seconds.

30.2 WDG_CNT				0x40104004
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_cnt	rw	0xFFFFFFFF	watchdog counter value

30.3 WDG_CTRL				0x40104008
Bits	Field Name	Acc	Default	Description
0	wdg_irq_en	rw	0x0	Watchdog interrupt enable bit 0: interrupt is disabled 1: interrupt is enabled For reset mode: wdg_rst_en =1, wdg_irq_en=0. For interrupt mode: wdg_rst_en =0, wdg_irq_en=1. For combined mode: wdg_rst_en =1, wdg_irq_en=1.
1	wdg_run	rw	0x0	Watchdog counter open: 0: counter stops. 1: counter runs.

30.4 WDG_INT_CLR				0x4010400C
Bits	Field Name	Acc	Default	Description
0	wdg_int_clr	wo	0x0	Watchdog interrupt clear Write 1 to this bit to clear interrupt Read this bit always get 0.

30.5 WDG_INT_RAW				0x40104010
Bits	Field Name	Acc	Default	Description
0	wdg_int_raw	ro	0x0	Watchdog interrupt raw status. Watch dog reset cannot clear this raw status. Write wdg_int_clr can clear this raw status.

4	wdg_ld_busy	ro	0x0	Watchdog load busy status 0: Watchdog is ready for new loading 1: Last loading is not completed Software must not load new value when this bit is busy, that is, this bit should be checked before any new loading. This bit is set after a new loading, and lasts two or three RTC clock cycles, about 60us - 92us.

30.6 WDG_INT_MASK				0x40104014
Bits	Field Name	Acc	Default	Description
0	wdg_int_mask	ro	0x0	Watchdog interrupt masked status

30.7 WDG_LOCK				0x40104C00
Bits	Field Name	Acc	Default	Description
15 : 0	wdg_lock	rw	0x0	Watchdog lock control Write 32'h1ACCE551 to this register to unlock watchdog. Write other value to this register to lock watchdog If reading this register, bit-0 is lock status, and other bits are reserved. If watchdog is locked, all control registers cannot be written by software.

31 spi

31.1 SPI_TXD				0x40105000
Discription: Write data to tx fifo Read data from rx fifo				
Bits	Field Name	Acc	Default	Description
31 : 0	SPI_TXD_RXD	rw	0x0	Write data to this address initiates a character transmission through TX FIFO Read from this address retrieve data from RX FIFO

31.2 SPI_CLKD				0x40105004
Discription: Clock divider bit 15 to 0.divider is 2(n+1)				
Bits	Field Name	Acc	Default	Description
15 : 0	SPI_CLKD	rw	0x3	Specify the clock ratio between spi_sck and clk_spi spi_sck = clk_spi/2(SPI_CLKD + 1)

31.3 SPI_CTL0				0x40105008
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
0	NG_RX	rw	0x0	"1" enable RX data shift in at clock neg-edge
1	NG_TX	rw	0x1	"1" enable TX data shift out at clock neg-edge
6 : 2	CHNL_LEN	rw	0x0	"0" 32 bits per word "1" 1 bits per word "31" 31 bits per word
7	LSB	rw	0x0	"1": enable transmit data from LSB
11 : 8	SPI_CSN_PRE	rw	0xF	"1": cs is invalid "0": cs is valid
12	Reserved	rw	0x0	Reserved
13	SCK_REV	rw	0x0	SCK inverse enable
14	SYNC_MD	rw	0x0	"1" : sync mode
15	SYNC_POL	rw	0x0	Sync polarity, positive or negative pulse for SPI or 3-wire mode, read command polarity

31.4 SPI_CTL1				0x4010500C
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
4 : 0	S3W_POS	rw	0x0	3-wire mode, w/r control position or the sync pulse position
5	S3W_MD	rw	0x0	"1" enable 3-wire mode
6	CS_H_MD	rw	0x0	3-wire melody timing 1, csn high mode enable
7	S8_MD	rw	0x0	enable S8 mode
11 : 8	SYN_CSN_SEL	rw	0x0	S8 CD or SYNC signal maps to csn number 0x1 selects csn0 as cd signal 0x2 selects csn1 as cd signal in SPI_HS it must be 0x0
13 : 12	RTX_MD	rw	0x0	"0x00":idle mode "0x01":receive mode "0x10":transmit mode "0x11":transmit and receive

31.5 SPI_CTL2				0x40105010
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
4 : 0	S3W_RD_START	rw	0x0	Read date start bit, used for 3-wire mode and 3-wire 9bit RW mode
5	IS_SLVD	rw	0x0	"1" slave,only support microplus mode "0" master
6	DMA_EN	rw	0x0	"1" : enable DMA
7	RX_ONLY_HLD	rw	0x0	"0" : working on only receive mode, when rxf_realfull is high, SPI will be held until rxf realfull is low
8	RX_DMA_SEL	rw	0x0	"0" : rx_dma_req keep high until receiving the rx_dma_ack "1" : rx_dma_req is high when rx full is high
9	TX_DMA_SEL	rw	0x0	"0" : tx_dma_req keep high until receiving the tx_dma_ack "1" : tx_dma_req is high when tx full is high
10	DMA_REQ_SEL	rw	0x0	"1" enable 3-wire mode

31.6 SPI_CTL3				0x40105014
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
4 : 0	RXF_FULL_THLD	rw	0x10	Receive data empty threshold
12 : 8	RXF_EMPTY_THLD	rw	0x10	Receive data full threshold

31.7 SPI_CTL4				0x40105018
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
8 : 0	BLOCK_NUM	rw	0x0	Number of data words ready to receive in "receive only" mode. Only used for master mode
9	IS_RX_ONLY	rw	0x0	"1":receive data only.The bits should be written at last. Only used for master mode
10	SYNC_HALF	rw	0x0	Sync half,sync width is half spi_sck cycle
11	SYNC_CLKMASK	rw	0x0	"1" Mask out the first clock pulse in SPI mode
13 : 12	PHS_DLY	rw	0x0	Phase delay. Relate to fast mode When in normal mode,this bit is not used. Only used for slave mode
14	IS_FST	rw	0x0	"0": normal mode "1": fast mode
15	RX_ONLY_DO	rw	0x0	"0": SPI send all 0 to slave "1": SPI send all 1 to slave

31.8 SPI_CTL5				0x4010501C
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
15 : 0	ITVL_NUM	rw	0x0	For master, transmit data interval, programmabl n from 0 to 65535, delay is (4*n+3) clock cycle For slave, max receive data interval. If slave has not sampled data,in the interval (4*n+3) spi_clk,slave will stop the receive process and send timeout interrupt

31.9 SPI_INT_EN				0x40105020
Discription: SPI interrupt enable register				
Bits	Field Name	Acc	Default	Description
1	RXF_EMPTY_INT_EN	rw	0x0	Rx_fifo_empty interrupt enable
2	TXF_FUL_INT_EN	rw	0x0	Tx_fifo_full interrupt enable
4	RX_OVF_INT_EN	rw	0x0	Rx_overrun_reg interrupt enable
5	TIME_OUT_INT_EN	rw	0x0	Slave mode timeout interrupt enable
6	RXF_FULL_INT_EN	rw	0x0	Rx_fifo_full interrupt enable
7	TXF_EMPTY_INT_EN	rw	0x0	Tx_fifo_empty interrupt enable
8	TXF_END_INT_EN	rw	0x0	Tx end interrupt enable
9	RXF_END_INT_EN	rw	0x0	Rx end interrupt enable

31.10 SPI_INT_CLR				0x40105024
Discription: SPI interrupt clear register				
Bits	Field Name	Acc	Default	Description
0	RXF_FULL_INT_CLR	wo	0x0	Write "1" clear Rx_fifo_full interrupt
1	RXF_EMPTY_INT_CLR	wo	0x0	Write "1" clear Rx_fifo_empty interrupt
2	TXF_FULL_INT_CLR	wo	0x0	Write "1" clear Tx_fifo_full interrupt
3	TXF_EMPTY_INT_CLR	wo	0x0	Write "1" clear Tx_fifo_empty interrupt
4	RX_OVF_INT_CLR	wo	0x0	Write "1" clear Rx_overrun_reg interrupt
5	TIME_OUT_INT_CLR	wo	0x0	Write "1" clear slave mode timeout interrupt
8	TXF_END_INT_CLR	rw	0x0	Tx data end interrupt clear
9	RXF_END_INT_CLR	rw	0x0	Rx data end interrupt clear

31.11 SPI_INT_RAW_STS				0x40105028
Discription: SPI interrupt raw status				
Bits	Field Name	Acc	Default	Description
0	RX_FIFO_FULL_R	ro	0x0	Rx_fifo_full_r
1	RXF_EMPTY_RAW_STS	ro	0x1	Raw rx_fifo_empty interrupt
2	TXF_FULL_RAW_STS	ro	0x0	Raw Tx_fifo_full interrupt
3	TX_FIFO_EMPTY_W	ro	0x1	Raw txf_fifo_empty_empty interrupt
4	RX_OVF_RAW_STS	ro	0x0	Raw Rx_overrun_reg interrupt
5	TIME_OUT_RAW_STS	ro	0x0	Raw slave mode time out interrupt
6	RXF_FULL_RAW_STS	ro	0x0	Raw rxf_full interrupt. when the number of rx fifo data byte is larger than the the rx empty watermark value.Atuo clear
7	TXF_EMPTY_RAW_STS	ro	0x1	Raw txf_empty interrupt. when the number of tx fifo data byte is less than the the tx empty watermark value.Atuo clear
8	TX_END_IRQ	ro	0x0	Raw tx data end interrupt. bit is set when spi controller send TX_DATA_LEN data
9	RX_END_IRQ	ro	0x0	Raw rx data end interrupt. bit is set when spi controller receive RX_DATA_LEN data

31.12 SPI_INT_MASK_STS				0x4010502C
Discription: SPI interrupt mask status				
Bits	Field Name	Acc	Default	Description

1	RX_FIFO_EMPTY-_MASK_STS	ro	0x0	Rx_fifo_empty interrupt mask status
2	TX_FIFO_FULL_MASK_STS	ro	0x0	Tx_fifo_full interrupt mask status
4	RX_OVF_MASK_STS	ro	0x0	Rx_overrun_reg interrupt mask status
5	TIME_OUT_MASK_STS	ro	0x0	Slave mode time out interrupt mask status
6	RXF_FULL_MASK_STS	ro	0x0	Rxf_full interrupt mask status.
7	TXF_EMPTY_MASK_STS	ro	0x0	Txf_empty interrupt mask status.
8	TXF_END_IRQ_MASK_STS	ro	0x0	Raw tx data end interrupt. bit is set when spi controller send TX_DATA_LEN data
9	RXF_END_IRQ_MASK_STS	ro	0x0	Raw rx data end interrupt. bit is set when spi controller receive RX_DATA_LEN data

31.13 SPI_STS1				0x40105030
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description
4 : 0	RXF_RD_ADDR	ro	0x0	RX FIFO read address
12 : 8	RXF_WR_ADDR	ro	0x0	RX FIFO write address

31.14 SPI_STS2				0x40105034
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description
0	RXF_FULL	ro	0x0	This bit is set when the number of RX FIFO data byte is larger than the RX full watermark value. Auto cleared when the condition disappears.
1	RXF_EMPTY	ro	0x1	This bit is set when the number of RX FIFO data byte is less than the RX empty interrupt watermark value. Auto cleared when the condition disappears.
2	TXF_FULL	ro	0x0	This bit is set when the number of TX FIFO data byte is larger than the TX full watermark value. Auto cleared when the condition disappears.
3	TXF_EMPTY	ro	0x1	This bit is set when the number of TX FIFO data byte is less than the TX empty watermark value. Auto cleared when the condition disappears.
4	RXF_REALFULL	ro	0x0	RX FIFO is real full.(not relates to register RXF_FULL_THLD)
5	RXF_REALEMPTY	ro	0x1	RX FIFO is real empty. RX FIFO has no data
6	TXF_REALFULL	ro	0x0	TX FIFO is real full.(not relates to register TXF_FULL_THLD)
7	TXF_REALEMPTY	ro	0x1	TX FIFO is real empty. TX FIFO has no data
8	BUSY	ro	0x0	"1": busy, transmit process "0": idle state
9	SPI_RXD	ro	0x0	Spi_rxd(monitor)
10	SPI_TXD	ro	0x0	Spi_txd(monitor)
11	SPI_SCK	ro	0x0	Spi_sck(monitor)
12	SPI_CS	ro	0x0	Spi_cs(monitor)

31.15 DSP_WAIT				0x40105038
Discription: Use for DSP control				
Bits	Field Name	Acc	Default	Description
3 : 0	SPI_DSPWAIT	rw	0x1	This register is used for DSP control
5 : 4	RX_DATA_SWT	rw	0x0	Read data switch: 0: rdata[31:0] = pdata[31:0] 1: rdata = {pdata[7:0], pdata[15:8], pdata[23:16], pdata[31:24]} 2: rdata = {pdata[15:0],pdata[31:16]}
7 : 6	TX_DATA_SWT	rw	0x0	Write data switch: 0: wdata[31:0] = pdata[31:0] 1: wdata = {pdata[7:0], pdata[15:8], pdata[23:16], pdata[31:24]} 2: wdata = {pdata[15:0],pdata[31:16]}

31.16 SPI_STS3				0x4010503C
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description
8 : 0	RX_CNT	ro	0x0	Working in only receive mode as master

31.17 SPI_CTL6				0x40105040
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
4 : 0	TXF_FULL_THLD	rw	0x8	TX FIFO data full threshold
12 : 8	TXF_EMPTY_THLD	rw	0x8	TX FIFO data empty threshold

31.18 SPI_STS4				0x40105044
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description
4 : 0	TXF_RD_ADDR	ro	0x0	TX FIFO read address
12 : 8	TXF_WR_ADDR	ro	0x0	TX FIFO write address

31.19 SPI_FIFO_RST				0x40105048
Discription: SPI tx/rx fifo reset bit				
Bits	Field Name	Acc	Default	Description
0	SPI_FIFO_RST	rw	0x0	"1":reset all FIFOs

31.20 SPI_CTL7				0x4010504C
Discription: SPI control register				
Bits	Field Name	Acc	Default	Description
0	CSN_IE_CTL	rw	0x0	CSN IE output set 0: not support csn input 1: support csn input
2 : 1	CSN_I_SEL	rw	0x0	CSN select control: 0: CSN 0 1: CSN 1 2: CSN 2 3: CSN 3
5 : 3	SPI_MODE	rw	0x0	Used for master only 0: SPI_MODE disable 1: 3 wire 9 bit, cd bit, SDI/SDO share one IO 2: 3 wire 9 bit, cd bit, SDI,SDO 3: 4 wire 8 bit, cd pin, SDI/SDO 4: 4 wire 8 bit, cd pin, SDI,SDO
6	TX_CMD_SET	rw	0x0	1: select fmark as the dma request 0:select software dma request
7	SPI_TX_HLD_EN	rw	0x0	1: enable ahb2apb bridge write hold when tx fifo full 0: disable ahb2apb bridge write hold
8	SPI_RX_HLD_EN	rw	0x0	1: enable ahb2apb bridge read hold when rx fifo empty 0: disable ahb2apb bridge write hold
9	SPI_SLV_CLK_SEl	rw	0x0	
10	SPI_SLV_EN	rw	0x0	Used when SPI slave in High speed mode 1:enable spi slave rtx 0:disable spi slave rtx
11	SPI_SLV_SEL	rw	0x0	1: SPI slave in low speed mode 0: SPI slave in low speed mode
12	RGB_888_EN	rw	0x0	
13	RGB_666_EN	rw	0x0	
14	RGB_565_EN	rw	0x0	
15	DATA_LINE2_EN	rw	0x0	

31.21 SPI_STS5				0x40105050
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description

0	CSN_IN_SYNC2	ro	0x0	CSN for slave
4	CSN_IN_ERR_SYNC2	ro	0x0	1:indicates csn occurring a exception

31.22 SPI_CTL8				0x40105054
Discription: SPI tx data control				
Bits	Field Name	Acc	Default	Description
3 : 0	SPI_TX_DATA_LEN_H	rw	0x0	Indicates tx data length from tx fifo, High 4 bits of spi tx data length
9 : 4	SPI_TX_DUMY_LEN	rw	0x0	Spi tx dummy clock length
12	RGB_PIX_MODE	rw	0x0	
13	CD_DATA2_SEL	rw	0x0	
14	SPI_CD_BIT2	rw	0x0	
15	SPI_CD_BIT	rw	0x0	Spi tx cd bit 0:indicates command 1:indicates data

31.23 SPI_CTL9				0x40105058
Discription: SPI tx data control				
Bits	Field Name	Acc	Default	Description
15 : 0	SPI_TX_DATA_LEN_L	rw	0x0	Indicates: spi tx data length from tx fifo, Low 16bit of tx data length

31.24 SPI_CTL10				0x4010505C
Discription: SPI tx data control				
Bits	Field Name	Acc	Default	Description
3 : 0	SPI_RX_DATA_LEN_H	rw	0x0	Indicates rx data length from slave, High 4 bits of spi rx data length
9 : 4	SPI_RX_DUMY_LEN	rw	0x0	Spi rx dummy clock length
15 : 10	CHNL_RX_LEN	rw	0x0	spi rx data len, used in spi mode 0

31.25 SPI_CTL11				0x40105060
Discription: SPI tx data control				
Bits	Field Name	Acc	Default	Description
15 : 0	SPI_RX_DATA_LEN_L	rw	0x0	Indicates: spi rx data length from slave, Low 16bit of rx data length

31.26 SPI_CTL12				0x40105064
Discription: SPI tx data control				
Bits	Field Name	Acc	Default	Description
0	SW_RX_REQ	wo	0x0	Software RX data request, for read LCD
1	SW_TX_REQ	wo	0x0	Software TX data request, for write LCD

31.27 SPI_STS6				0x40105068
Discription: tx data counter				
Bits	Field Name	Acc	Default	Description
15 : 0	TX_DATA_CNT	ro	0x0	Tx data cnt[15:0]

31.28 SPI_STS7				0x4010506C
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description
3 : 0	TX_DATA_CNT_H	ro	0x0	Tx data cnt[19:16]
15 : 10	TX_DUMMY_CNT	ro	0x0	Tx dummy cnt

31.29 SPI_STS8				0x40105070
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description
15 : 0	RX_DATA_CNT	ro	0x0	Rx data cnt[15:0]

31.30 SPI_STS9				0x40105074
Discription: SPI status register				
Bits	Field Name	Acc	Default	Description
3 : 0	RX_DATA_CNT_H	ro	0x0	Rx data cnt[19:16]
15 : 10	RX_DUMMY_CNT	ro	0x0	Rx dummy cnt

32 timer0

32.1 tmr_cfg				0x40106000
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_run	rw	0x0	Timer run enable, high active.
1	cfg_tmr_reset	rw	0x0	Timer PC reset, write 1 auto clear.

32.2 tmr_pr				0x40106004
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_pr	rw	0xFFFFFFFF	Prescale register.

32.3 tmr_pc				0x40106008
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_pc	rw	0xFFFFFFFF	Current counter number.

32.4 tmr_mr0				0x4010600C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr0	rw	0xFFFFFFFF	Match point for CPU0.

32.5 tmr_mr1				0x40106010
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr1	rw	0xFFFFFFFF	Match point for CPU1.

32.6 tmr_mr2				0x40106014
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr2	rw	0xFFFFFFFF	Match point for CPU2.

32.7 tmr_delta0				0x40106018
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_delta0	rw	0xFFFFFFFF	Delta counter for CPU0.

32.8 tmr_delta1				0x4010601C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_delta1	rw	0xFFFFFFFF	Delta counter for CPU1.

32.9 tmr_delta2				0x40106020
Bits	Field Name	Acc	Default	Description

31 : 0	cfg_tmr_delta2	rw	0xFFFFFFFF	Delta counter for CPU2.

32.10 tmr_rtc_clk_div				0x40106024
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_tmr_rtc_clk_div	rw	0x19	Timer rtc clock dividier. Default is 26M/26, 1 bit means 1us for PC.
16	cfg_tmr_rtc_clk_div_update	rw	0x0	RTC clock divider update, write auto clear.

32.11 tmr_int_mask_enable				0x40106028
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match-0_int_mask_enable	rw	0x0	Match point0 interrupt mask enable. Write1 to enable.
1	cfg_tmr_match-1_int_mask_enable	rw	0x0	Match point1 interrupt mask enable. Write1 to enable.
2	cfg_tmr_match-2_int_mask_enable	rw	0x0	Match point2 interrupt mask enable. Write1 to enable.

32.12 tmr_int_mask_clear				0x4010602C
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match-0_int_mask_clear	rw	0x0	Match point0 interrupt mask clear. Write1 to clear.
1	cfg_tmr_match-1_int_mask_clear	rw	0x0	Match point1 interrupt mask clear. Write1 to clear.
2	cfg_tmr_match-2_int_mask_clear	rw	0x0	Match point2 interrupt mask clear. Write1 to clear.

32.13 tmr_int_raw				0x40106030
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match0_int_raw	rw	0x0	Match point0 raw interrupt. Write 1 to clear.
1	cfg_tmr_match1_int_raw	rw	0x0	Match point1 raw interrupt. Write 1 to clear.
2	cfg_tmr_match2_int_raw	rw	0x0	Match point2 raw interrupt. Write 1 to clear.

32.14 tmr_int_status				0x40106034
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match0_int_status	rw	0x0	Match point0 interrupt status. Write 1 to clear.
1	cfg_tmr_match1_int_status	rw	0x0	Match point1 interrupt status. Write 1 to clear.
2	cfg_tmr_match2_int_status	rw	0x0	Match point2 interrupt status. Write 1 to clear.

33 timer1

33.1 tmr_cfg				0x40107000
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_run	rw	0x0	Timer run enable, high active.
1	cfg_tmr_reset	rw	0x0	Timer PC reset, write 1 auto clear.

33.2 tmr_pr				0x40107004
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_pr	rw	0xFFFFFFFF	Prescale register.

33.3 tmr_pc				0x40107008
Bits	Field Name	Acc	Default	Description

31 : 0	cfg_tmr_pc	rw	0xFFFFFFFF	Current counter number.

33.4 tmr_mr0				0x4010700C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr0	rw	0xFFFFFFFF	Match point for CPU0.

33.5 tmr_mr1				0x40107010
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr1	rw	0xFFFFFFFF	Match point for CPU1.

33.6 tmr_mr2				0x40107014
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr2	rw	0xFFFFFFFF	Match point for CPU2.

33.7 tmr_delta0				0x40107018
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_delta0	rw	0xFFFFFFFF	Delta counter for CPU0.

33.8 tmr_delta1				0x4010701C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_delta1	rw	0xFFFFFFFF	Delta counter for CPU1.

33.9 tmr_delta2				0x40107020
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_delta2	rw	0xFFFFFFFF	Delta counter for CPU2.

33.10 tmr_rtc_clk_div				0x40107024
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_tmr_rtc_clk_div	rw	0x19	Timer rtc clock dividier. Default is 26M/26, 1 bit means 1us for PC.
16	cfg_tmr_rtc_clk_div_update	rw	0x0	RTC clock divider update, write auto clear.

33.11 tmr_int_mask_enable				0x40107028
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match-0_int_mask_enable	rw	0x0	Match point0 interrupt mask enable. Write1 to enable.
1	cfg_tmr_match-1_int_mask_enable	rw	0x0	Match point1 interrupt mask enable. Write1 to enable.
2	cfg_tmr_match-2_int_mask_enable	rw	0x0	Match point2 interrupt mask enable. Write1 to enable.

33.12 tmr_int_mask_clear				0x4010702C
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match-0_int_mask_clear	rw	0x0	Match point0 interrupt mask clear. Write1 to clear.
1	cfg_tmr_match-1_int_mask_clear	rw	0x0	Match point1 interrupt mask clear. Write1 to clear.
2	cfg_tmr_match-2_int_mask_clear	rw	0x0	Match point2 interrupt mask clear. Write1 to clear.

33.13 tmr_int_raw				0x40107030
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match0_int_raw	rw	0x0	Match point0 raw interrupt. Write 1 to clear.
1	cfg_tmr_match1_int_raw	rw	0x0	Match point1 raw interrupt. Write 1 to clear.
2	cfg_tmr_match2_int_raw	rw	0x0	Match point2 raw interrupt. Write 1 to clear.

33.14 tmr_int_status				0x40107034
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match0_int_status	rw	0x0	Match point0 interrupt status. Write 1 to clear.
1	cfg_tmr_match1_int_status	rw	0x0	Match point1 interrupt status. Write 1 to clear.
2	cfg_tmr_match2_int_status	rw	0x0	Match point2 interrupt status. Write 1 to clear.

34 trng

34.1 trng_en				0x40108000
Bits	Field Name	Acc	Default	Description
2	sTrngEnQ	rw	0x0	Block enable, high active.
3	sTrngIEQ	rw	0x0	Interrupte enable, high active.
4	sTrngEEQ	rw	0x0	

34.2 trng_status				0x40108004
Bits	Field Name	Acc	Default	Description
0	sDRDYQ	ro	0x0	Random data ready status. Read sDRQ(0x008) to clear status.
2	sSECSQ	ro	0x0	Clock error current status. Write bit[6] 1'b0 to clear status.
6	sSEISQ	ro	0x0	Seed error interrupt status. Write bit[6] 1'b0 to clear status.

34.3 trng_data				0x40108008
Bits	Field Name	Acc	Default	Description
31 : 0	sDRQ	ro	0x0	Random 32-bit data.

34.4 trng_div_dly				0x40108010
Bits	Field Name	Acc	Default	Description
7 : 0	sDivNumQ	rw	0x40	Enable period
15 : 8	SDivHighQ	rw	0x10	Enable high duration
23 : 16	sDlyNumQ	rw	0x8	Delay number

34.5 trng_sel				0x40108014
Bits	Field Name	Acc	Default	Description
7 : 0	sTrngSel0Q	rw	0xF	Trng Sel0
15 : 8	sTrngSel1Q	rw	0xF0	Trng Sel1
23 : 16	sLfsrGen0Q	rw	0x0	LFSR gen0
31 : 24	sLfsrGen1Q	rw	0x0	LFSR gen1

35 ana_reg0

35.1 flash_pad_pull				0x40109000
Bits	Field Name	Acc	Default	Description
3 : 0	cfg_ana_fls_sio_pull_dn	rw	0xC	FLASH SIO pull down

7 : 4	cfg_ana_fls_sio_pull_up	rw	0x3	FLASH SIO pull up
8	cfg_ana_fls_cs_pull_dn	rw	0x0	FLASH CS pull down
9	cfg_ana_fls_cs_pull_up	rw	0x1	FLASH CS pull up
10	cfg_ana_fls_clk_pull_dn	rw	0x0	FLASH clock pull down
11	cfg_ana_fls_clk_pull_up	rw	0x0	FLASH clock pull up

36 proc_mon

36.1 auto_mode				0x4010A000
Bits	Field Name	Acc	Default	Description
0	cfg_auto_mode	rw	0x0	1'b1: auto pvt monitor mode; 1'b0: single mode

36.2 cfg_start				0x4010A004
Bits	Field Name	Acc	Default	Description
0	cfg_start	rw	0x0	Write to generate start pulse

36.3 cfg_auto_stop				0x4010A008
Bits	Field Name	Acc	Default	Description
0	cfg_auto_stop	rw	0x0	Write to generate stop pulse when auto monitor mode

36.4 enable_duration				0x4010A00C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_enable_duration	rw	0x190	1'b1: manual mode; 1'b0: auto mode

36.5 cfg_auto_interval				0x4010A010
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_auto_interval	rw	0x2625A00	1'b1: manual mode; 1'b0: auto mode

36.6 ro_pvt_cnt_hvt				0x4010A014
Bits	Field Name	Acc	Default	Description
31 : 0	ro_pvt_cnt_hvt	ro	0x0	PVT counter for HVT cell

36.7 cfg_cnt_last				0x4010A018
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_cnt_last	rw	0xA	PVT counter last cycle.

37 gpioa

37.1 GPIODATA				0x4010B000
Bits	Field Name	Acc	Default	Description
15 : 0	GPIODATA	rw	0x0	GPIO bits data input

37.2 GPIODMSK				0x4010B004
Bits	Field Name	Acc	Default	Description
15 : 0	GPIODMSK	rw	0x0	GPIODATA register can be read/write if GPIO DMSK set "1"

37.3 GPIODIR	0x4010B008

Bits	Field Name	Acc	Default	Description
15 : 0	GPIODIR	rw	0x0	"1" configure gpio bits to be output "0" configure gpio bits to be input

37.4 GPIOIS				0x4010B00C
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIS	rw	0xFFFF	"1" detect signals level "0" detect signals edge

37.5 GPIOIBE				0x4010B010
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIBE	rw	0x0	"1" both edges trigger an interrupt, "0" interrupt generation event is controlled by GPIOIEV

37.6 GPIOIEV				0x4010B014
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIEV	rw	0xFFFF	GPIO bits interrupt event register: "1" high levels trigger interrupts, "0" low levels trigger interrupts.

37.7 GPIOIE				0x4010B018
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIE	rw	0x0	GPIO bits interrupt enable register: "1" corresponding bit interrupt is enabled. "0" corresponding bit interrupt isn't enabled

37.8 GPIORIS				0x4010B01C
Bits	Field Name	Acc	Default	Description
15 : 0	GPIORIS	ro	0x0	GPIO bits raw interrupt status register: "1" interrupt condition met "0" condition not met

37.9 GPIOMIS				0x4010B020
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOMIS	ro	0x0	GPIO bits masked interrupt status register: "1" Interrupt active "0" interrupt not active

37.10 GPIOIC				0x4010B024
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIC	wo	0x0	GPIO bits interrupt clear register: "1" clears detected interrupt. "0" has no effect.

37.11 GPIOINEN				0x4010B028
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOINEN	rw	0x0	GPIO input enable register: "1" input enable. "0" input disable.

37.12 GPIO_FILTER_NUM				0x4010B02C
Bits	Field Name	Acc	Default	Description

15 : 0	GPIO_FILTER_NUM	rw	0x3333	

37.13 GPIO_FILTER_DIV_L				0x4010B030
Bits	Field Name	Acc	Default	Description
15 : 0	GPIO_FILTER_DIV_L	rw	0x0	

37.14 GPIO_FILTER_DIV_H				0x4010B034
Bits	Field Name	Acc	Default	Description
15 : 0	GPIO_FILTER_DIV_H	rw	0x0	

38 efuse

38.1 EFUSE_ALL0_INDEX				0x4010C008
Bits	Field Name	Acc	Default	Description
15 : 0	EFUSE_ALL0_END_INDEX	rw	0x1F	All 0 check end index
31 : 16	EFUSE_ALL0_START_INDEX	rw	0x0	All 0 check start index

38.2 EFUSE_MODE_CTRL				0x4010C00C
Bits	Field Name	Acc	Default	Description
0	EFUSE_ALL0_CHECK_START	rw	0x0	write 1 to this bit will trigger all0 check to efuse, this bit is self-clear ,read this bit will always get 0.(use PREADY)
31 : 1	reserved_0	ro	0x0	RESERVED

38.3 EFUSE_IP_VER_reg				0x4010C014
Bits	Field Name	Acc	Default	Description
15 : 0	EFUSE_IP_VER	ro	0x100	IP version, now is r1p0
17 : 16	EFUSE_TYPE	ro	0x0	Efuse type: 00 TSMC
31 : 18	reserved_0	ro	0x0	RESERVED

38.4 EFUSE_CFG0				0x4010C018
Bits	Field Name	Acc	Default	Description
8 : 0	TPGM_TIME_CNT	rw	0x140	Program strobe high time. If set n, the Tpgm time will last for (n+1) clk_efuse cycle, only when PGM_EN=1 can write this field.
15 : 9	reserved_0	ro	0x0	RESERVED
23 : 16	EFUSE_STROBE_LOW_WIDTH	rw	0x1C	This counter is used to control STROBE signal low level width in PGM mode, for TSMC efuse memory, no extra requirement for this signal, For 26Mhz efuse controller clock, by default, this width will be: 38.4*28=1075ns ¿ 1us. If you want to speed up program speed, can configure the register to a smaller value.
31 : 24	CLK_EFS_DIV	rw	0x0	Clk_efs divider, if this value is n, the frequency of controller will be divided by (n+1) from clk_efs. In most case, this field not need to change.

38.5 EFUSE_SEC_EN				0x4010C040
Bits	Field Name	Acc	Default	Description
0	SEC_VDD_EN	rw	0x0	efuse vdd enable
1	SEC_AUTO_CHECK_ENABLE	rw	0x0	program read back auto-check enable
2	DOUBLE_BIT_EN_SEC	rw	0x0	efuse double bit enable

3	SEC_MARGIN_RD_ENABLE	rw	0x0	efuse margin read mode enable
4	SEC_LOCK_BIT_WR_EN	rw	0x0	If set the bit, lock bits will be written after PGM process.
31 : 5	reserved_0	ro	0x0	RESERVED

38.6 EFUSE_SEC_ERR_FLAG				0x4010C044
Bits	Field Name	Acc	Default	Description
0	block_auto_check_failed	ro	0x0	The bit indicates block auto check failed after programming. If PGM_AUTO_CHK_EN is set, and controller compared the value read after PGM from the same block, and found the two value not match, this bit will set as an error flag. But if this block is protected, the PGM command in-fact not really send, so this bit will not set.
1	shadow_block_-auto_check_failed	ro	0x0	The bit indicates shadow block auto check failed after programming if double_bit_en is set. If PGM_AUTO_CHK_EN is set, and controller compared the value read after PGM from the same block, and found the two value not match, this bit will set as an error flag. But if this block is protected, the PGM command in-fact not really send, so this bit will not set.
3 : 2	reserved_2	ro	0x0	RESERVED
4	SEC_WORD0_PROT_FLAG	ro	0x0	The bit indicates block auto check failed after programming. If PGM_AUTO_CHK_EN is set, and controller compared the value read after PGM from the same block, and found the two value not match, this bit will set as an error flag. But if this block is protected, the PGM command in-fact not really send, so this bit will not set.
5	SEC_WORD1_PROT_FLAG	ro	0x0	The bit indicates shadow block is protected and can not be programmed if double_bit_en is set. If SW send a PGM command to memory block[i], and the controller found this memory block is protected(which means the highest bit is 1), this bit will set to 1.
7 : 6	reserved_1	ro	0x0	RESERVED
8	SEC_PG_EN_WR_FLAG	ro	0x0	the bit indicates write process without setting pg_en to 1.
9	SEC_VDD_ON_RD_FLAG	ro	0x0	the bit indicates read process without setting vdd_on to 1.
10	SEC_BLOCK0_RD_FLAG	ro	0x0	the bit indicates arbiter read block0 which may indicates unexpected access.
11	SEC_MAGNUM_WR_FLAG	ro	0x0	the bit indicates write process without setting magic number.
12	SEC_ENK_ERR_FLAG	ro	0x0	the bit enk1 and enk2 is not switch correctly.
13	SEC_ALL0_CHECK_FLAG	ro	0x0	the bit indicates all 0 check fail.
31 : 14	reserved_0	ro	0x0	RESERVED

38.7 EFUSE_SEC_FLAG_CLR				0x4010C048
Bits	Field Name	Acc	Default	Description
0	SEC_WORD0_ERR_CLR	rw	0x0	write 1 will clear SEC_WORD0_ERR_FLAG. Write 0 will do nothing.
1	SEC_WORD1_ERR_CLR	rw	0x0	write 1 will clear SEC_WORD1_ERR_FLAG. Write 0 will do nothing.
3 : 2	reserved_2	ro	0x0	RESERVED
4	SEC_WORD0_PROT_CLR	rw	0x0	write 1 will clear SEC_WORD0_PROT_FLAG. Write 0 will do nothing.
5	SEC_WORD1_PROT_CLR	rw	0x0	write 1 will clear SEC_WORD1_PROT_FLAG. Write 0 will do nothing.
7 : 6	reserved_1	ro	0x0	RESERVED
8	SEC_PG_EN_WR_CLR	rw	0x0	write 1 will clear SEC_PG_EN_WR_FLAG
9	SEC_VDD_ON_RD_CLR	rw	0x0	write 1 will clear SEC_VDD_ON_RD_FLAG

10	SEC_BLOCK0_RD_CLR	rw	0x0	write 1 will clear SEC_BLOCK0_RD_FLAG
11	SEC_MAGNUM_WR_CLR	rw	0x0	write 1 will clear SEC_MAGNUM_WR_FLAG
12	SEC_ENK_ERR_CLR	rw	0x0	write 1 will clear SEC_ENK_ERR_FLAG
13	SEC_ALL0_CHECK_CLR	rw	0x0	write 1 will clear SEC_ALL0_CHECK_FLAG
31 : 14	reserved_0	ro	0x0	RESERVED

38.8 EFUSE_SEC_MAGIC_NUMBER 0x4010C04C

Bits	Field Name	Acc	Default	Description
15 : 0	SEC_EFUSE_MAGIC_NUMBER	rw	0x0	Magic number, only when this field is 0x8800, the efuse programming command can be handle. Set the magic number right will lock the power switch and PGM enable. So if SW want to program efuse memory, except open clocks and power, 2 other conditions must be met : (1) SEC_EFUSE_MAGIC_NUMBER =0x8800 (2) PG_EN=1; (3) Switch the power right;
31 : 16	reserved_0	ro	0x0	RESERVED

38.9 EFUSE_FW_CFG 0x4010C050

Bits	Field Name	Acc	Default	Description
0	CONF_PROT	rw	0x0	SEC/NON-SEC address configure. 1: indicates SEC can access EFUSE_CFG0; 0: indicates NON-SEC can access EFUSE_CFG0;
1	ACCESS_PROT	rw	0x0	0: SEC access; 1: NON SEC address enable switch
31 : 2	reserved_0	ro	0x0	RESERVED

38.10 EFUSE_PW_SWT 0x4010C054

Bits	Field Name	Acc	Default	Description
0	EFS_ENK1_ON	rw	0x0	VDDQ power switch K2, to safely control this power switch.
1	EFS_ENK2_ON	rw	0x1	VDDQ power switch K2, to safely control this power switch.
2	NS_S_PG_EN	rw	0x0	set this bit will open static power supply for efuse memory, before any operation towards to efuse memory this bit have to set to 1. once this bit is cleared, the efuse will go to power down mose.
3	ANLG_CFG_EFUSE_SAR_READ_MODE	rw	0x0	read mode, 1'b1: sar mode; 1'b0: normal mode
4	ANLG_CFG_EFUSE_SAR_LCH_BIAS_MODE	rw	0x0	SAR bias mode
8 : 5	ANLG_CFG_EFUSE_SAR_DLY_BIT	rw	0x0	SAR delay bits
11 : 9	ANLG_CFG_EFUSE_SAR_BIAS_BIT	rw	0x0	SAR bias bit control
16 : 12	SAR_READ_BITS	rw	0x0	Read bits index for SAR mode
31 : 17	reserved_0	ro	0x0	RESERVED

38.11 EFUSE_FUNC_EN_0 0x4010C058

Bits	Field Name	Acc	Default	Description
0	efuse_func_en_0	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.12 EFUSE_FUNC_EN_1 0x4010C05C

Bits	Field Name	Acc	Default	Description

0	efuse_func_en_1	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.13 EFUSE_FUNC_EN_2				0x4010C060
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_2	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.14 EFUSE_FUNC_EN_3				0x4010C064
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_3	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.15 EFUSE_FUNC_EN_4				0x4010C068
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_4	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.16 EFUSE_FUNC_EN_5				0x4010C06C
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_5	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.17 EFUSE_FUNC_EN_6				0x4010C070
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_6	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.18 EFUSE_FUNC_EN_7				0x4010C074
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_7	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.19 EFUSE_FUNC_EN_8				0x4010C078
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_8	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.20 EFUSE_FUNC_EN_9				0x4010C07C
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_9	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.21 EFUSE_FUNC_EN_10				0x4010C080
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_10	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.22 EFUSE_FUNC_EN_11				0x4010C084
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_11	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.23 EFUSE_FUNC_EN_12				0x4010C088
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_12	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.24 EFUSE_FUNC_EN_13				0x4010C08C
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_13	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.25 EFUSE_FUNC_EN_14				0x4010C090
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_14	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.26 EFUSE_FUNC_EN_15				0x4010C094
Bits	Field Name	Acc	Default	Description
0	efuse_func_en_15	rw	0x1	write once register, security function control bit: 0 = disabled 1 = enabled
31 : 1	reserved_0	ro	0x0	RESERVED

38.27 efuse_bist_en_reg				0x4010C098
Bits	Field Name	Acc	Default	Description
0	efuse_bist_en	rw	0x0	efuse bist enable, write once register
31 : 1	reserved_0	ro	0x0	RESERVED

38.28 efuse_block0_rw_ctrl_reg				0x4010C09C
Bits	Field Name	Acc	Default	Description
0	efuse_block0_rw_ctrl	rw	0x1	control efuse block0 read/write, write once register: 0 = not enable read/write efuse block0 1 = enable read/write efuse block0
31 : 1	reserved_0	ro	0x0	RESERVED

38.29 efuse_block1_rw_ctrl_reg				0x4010C0A0
Bits	Field Name	Acc	Default	Description
0	efuse_block1_rw_ctrl	rw	0x1	control efuse block1 read/write, write once register: 0 = not enable read/write efuse block1 1 = enable read/write efuse block1
31 : 1	reserved_0	ro	0x0	RESERVED

38.30 efuse_block2_rw_ctrl_reg				0x4010C0A4
Bits	Field Name	Acc	Default	Description
0	efuse_block2_rw_ctrl	rw	0x1	control efuse block2 read/write, write once register: 0 = not enable read/write efuse block2 1 = enable read/write efuse block2
31 : 1	reserved_0	ro	0x0	RESERVED

38.31 efuse_block3_rw_ctrl_reg				0x4010C0A8
Bits	Field Name	Acc	Default	Description
0	efuse_block3_rw_ctrl	rw	0x1	control efuse block3 read/write, write once register: 0 = not enable read/write efuse block3 1 = enable read/write efuse block3
31 : 1	reserved_0	ro	0x0	RESERVED

38.32 efuse_block4_rw_ctrl_reg				0x4010C0AC
Bits	Field Name	Acc	Default	Description
0	efuse_block4_rw_ctrl	rw	0x1	control efuse block4 read/write, write once register: 0 = not enable read/write efuse block4 1 = enable read/write efuse block4
31 : 1	reserved_0	ro	0x0	RESERVED

38.33 efuse_block5_rw_ctrl_reg				0x4010C0B0
Bits	Field Name	Acc	Default	Description
0	efuse_block5_rw_ctrl	rw	0x1	control efuse block5 read/write, write once register: 0 = not enable read/write efuse block5 1 = enable read/write efuse block5
31 : 1	reserved_0	ro	0x0	RESERVED

38.34 efuse_block6_rw_ctrl_reg				0x4010C0B4
Bits	Field Name	Acc	Default	Description
0	efuse_block6_rw_ctrl	rw	0x1	control efuse block6 read/write, write once register: 0 = not enable read/write efuse block6 1 = enable read/write efuse block6
31 : 1	reserved_0	ro	0x0	RESERVED

38.35 efuse_block7_rw_ctrl_reg				0x4010C0B8
Bits	Field Name	Acc	Default	Description
0	efuse_block7_rw_ctrl	rw	0x1	control efuse block7 read/write, write once register: 0 = not enable read/write efuse block7 1 = enable read/write efuse block7
31 : 1	reserved_0	ro	0x0	RESERVED

38.36 efuse_block8_rw_ctrl_reg				0x4010C0BC
Bits	Field Name	Acc	Default	Description
0	efuse_block8_rw_ctrl	rw	0x1	control efuse block8 read/write, write once register: 0 = not enable read/write efuse block8 1 = enable read/write efuse block8
31 : 1	reserved_0	ro	0x0	RESERVED

38.37 efuse_block9_rw_ctrl_reg				0x4010C0C0
Bits	Field Name	Acc	Default	Description
0	efuse_block9_rw_ctrl	rw	0x1	control efuse block9 read/write, write once register: 0 = not enable read/write efuse block9 1 = enable read/write efuse block9
31 : 1	reserved_0	ro	0x0	RESERVED

38.38 efuse_block10_rw_ctrl_reg				0x4010C0C4
Bits	Field Name	Acc	Default	Description
0	efuse_block10_rw_ctrl	rw	0x1	control efuse block10 read/write, write once register: 0 = not enable read/write efuse block10 1 = enable read/write efuse block10
31 : 1	reserved_0	ro	0x0	RESERVED

38.39 efuse_block11_rw_ctrl_reg				0x4010C0C8
Bits	Field Name	Acc	Default	Description
0	efuse_block11_rw_ctrl	rw	0x1	control efuse block11 read/write, write once register: 0 = not enable read/write efuse block11 1 = enable read/write efuse block11
31 : 1	reserved_0	ro	0x0	RESERVED

38.40 efuse_block12_rw_ctrl_reg				0x4010C0CC
Bits	Field Name	Acc	Default	Description
0	efuse_block12_rw_ctrl	rw	0x1	control efuse block12 read/write, write once register: 0 = not enable read/write efuse block12 1 = enable read/write efuse block12
31 : 1	reserved_0	ro	0x0	RESERVED

38.41 efuse_block13_rw_ctrl_reg				0x4010C0D0
Bits	Field Name	Acc	Default	Description
0	efuse_block13_rw_ctrl	rw	0x1	control efuse block13 read/write, write once register: 0 = not enable read/write efuse block13 1 = enable read/write efuse block13
31 : 1	reserved_0	ro	0x0	RESERVED

38.42 efuse_block14_rw_ctrl_reg				0x4010C0D4
Bits	Field Name	Acc	Default	Description
0	efuse_block14_rw_ctrl	rw	0x1	control efuse block14 read/write, write once register: 0 = not enable read/write efuse block14 1 = enable read/write efuse block14
31 : 1	reserved_0	ro	0x0	RESERVED

38.43 efuse_block15_rw_ctrl_reg				0x4010C0D8
Bits	Field Name	Acc	Default	Description
0	efuse_block15_rw_ctrl	rw	0x1	control efuse block15 read/write, write once register: 0 = not enable read/write efuse block15 1 = enable read/write efuse block15
31 : 1	reserved_0	ro	0x0	RESERVED

38.44 efuse_block16_rw_ctrl_reg				0x4010C0DC
Bits	Field Name	Acc	Default	Description
0	efuse_block16_rw_ctrl	rw	0x1	control efuse block16 read/write, write once register: 0 = not enable read/write efuse block16 1 = enable read/write efuse block16
31 : 1	reserved_0	ro	0x0	RESERVED

38.45 efuse_block17_rw_ctrl_reg				0x4010C0E0
Bits	Field Name	Acc	Default	Description
0	efuse_block17_rw_ctrl	rw	0x1	control efuse block17 read/write, write once register: 0 = not enable read/write efuse block17 1 = enable read/write efuse block17
31 : 1	reserved_0	ro	0x0	RESERVED

38.46 efuse_block18_rw_ctrl_reg				0x4010C0E4
Bits	Field Name	Acc	Default	Description
0	efuse_block18_rw_ctrl	rw	0x1	control efuse block18 read/write, write once register: 0 = not enable read/write efuse block18 1 = enable read/write efuse block18
31 : 1	reserved_0	ro	0x0	RESERVED

38.47 efuse_block19_rw_ctrl_reg				0x4010C0E8
Bits	Field Name	Acc	Default	Description
0	efuse_block19_rw_ctrl	rw	0x1	control efuse block19 read/write, write once register: 0 = not enable read/write efuse block19 1 = enable read/write efuse block19
31 : 1	reserved_0	ro	0x0	RESERVED

38.48 efuse_block20_rw_ctrl_reg				0x4010C0EC
Bits	Field Name	Acc	Default	Description
0	efuse_block20_rw_ctrl	rw	0x1	control efuse block20 read/write, write once register: 0 = not enable read/write efuse block20 1 = enable read/write efuse block20
31 : 1	reserved_0	ro	0x0	RESERVED

38.49 efuse_block21_rw_ctrl_reg				0x4010C0F0
Bits	Field Name	Acc	Default	Description
0	efuse_block21_rw_ctrl	rw	0x1	control efuse block21 read/write, write once register: 0 = not enable read/write efuse block21 1 = enable read/write efuse block21
31 : 1	reserved_0	ro	0x0	RESERVED

38.50 efuse_block22_rw_ctrl_reg				0x4010C0F4
Bits	Field Name	Acc	Default	Description
0	efuse_block22_rw_ctrl	rw	0x1	control efuse block22 read/write, write once register: 0 = not enable read/write efuse block22 1 = enable read/write efuse block22
31 : 1	reserved_0	ro	0x0	RESERVED

38.51 efuse_block23_rw_ctrl_reg				0x4010C0F8
Bits	Field Name	Acc	Default	Description
0	efuse_block23_rw_ctrl	rw	0x1	control efuse block23 read/write, write once register: 0 = not enable read/write efuse block23 1 = enable read/write efuse block23
31 : 1	reserved_0	ro	0x0	RESERVED

38.52 efuse_block24_rw_ctrl_reg				0x4010C0FC
Bits	Field Name	Acc	Default	Description
0	efuse_block24_rw_ctrl	rw	0x1	control efuse block24 read/write, write once register: 0 = not enable read/write efuse block24 1 = enable read/write efuse block24
31 : 1	reserved_0	ro	0x0	RESERVED

38.53 efuse_block25_rw_ctrl_reg				0x4010C100
Bits	Field Name	Acc	Default	Description
0	efuse_block25_rw_ctrl	rw	0x1	control efuse block25 read/write, write once register: 0 = not enable read/write efuse block25 1 = enable read/write efuse block25
31 : 1	reserved_0	ro	0x0	RESERVED

38.54 efuse_block26_rw_ctrl_reg				0x4010C104
Bits	Field Name	Acc	Default	Description
0	efuse_block26_rw_ctrl	rw	0x1	control efuse block26 read/write, write once register: 0 = not enable read/write efuse block26 1 = enable read/write efuse block26
31 : 1	reserved_0	ro	0x0	RESERVED

38.55 efuse_block27_rw_ctrl_reg				0x4010C108
Bits	Field Name	Acc	Default	Description
0	efuse_block27_rw_ctrl	rw	0x1	control efuse block27 read/write, write once register: 0 = not enable read/write efuse block27 1 = enable read/write efuse block27
31 : 1	reserved_0	ro	0x0	RESERVED

38.56 efuse_block28_rw_ctrl_reg				0x4010C10C
Bits	Field Name	Acc	Default	Description
0	efuse_block28_rw_ctrl	rw	0x1	control efuse block28 read/write, write once register: 0 = not enable read/write efuse block28 1 = enable read/write efuse block28
31 : 1	reserved_0	ro	0x0	RESERVED

38.57 efuse_block29_rw_ctrl_reg				0x4010C110
Bits	Field Name	Acc	Default	Description
0	efuse_block29_rw_ctrl	rw	0x1	control efuse block29 read/write, write once register: 0 = not enable read/write efuse block29 1 = enable read/write efuse block29
31 : 1	reserved_0	ro	0x0	RESERVED

38.58 efuse_block30_rw_ctrl_reg				0x4010C114
Bits	Field Name	Acc	Default	Description
0	efuse_block30_rw_ctrl	rw	0x1	control efuse block30 read/write, write once register: 0 = not enable read/write efuse block30 1 = enable read/write efuse block30
31 : 1	reserved_0	ro	0x0	RESERVED

38.59 efuse_block31_rw_ctrl_reg				0x4010C118
Bits	Field Name	Acc	Default	Description
0	efuse_block31_rw_ctrl	rw	0x1	control efuse block31 read/write, write once register: 0 = not enable read/write efuse block31 1 = enable read/write efuse block31
31 : 1	reserved_0	ro	0x0	RESERVED

38.60 por_read_done_reg				0x4010C128
Bits	Field Name	Acc	Default	Description
0	por_read_done	ro	0x0	read config bits done status from efuse macro after por, then this bit is set to 1.
31 : 1	reserved_0	ro	0x0	RESERVED

38.61 efuse_cfg_reg				0x4010C12C
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_efuse_func_en	ro	0xFFFF	efuse function enable control bit
16	cfg_efuse_bist_en	ro	0x0	efuse bist enable
31 : 17	reserved_0	ro	0x0	RESERVED

38.62 efuse_block_en_reg				0x4010C130
Bits	Field Name	Acc	Default	Description
31 : 0	efuse_block_en	ro	0xFFFFFFFF	every block(31$\tilde{0}$) read/write enable status

38.63 por_read_data0_reg				0x4010C134
Bits	Field Name	Acc	Default	Description
31 : 0	por_read_data0	ro	0x0	read block1 config bit

38.64 por_read_data1_reg				0x4010C138
Bits	Field Name	Acc	Default	Description
31 : 0	por_read_data1	ro	0x0	read block2 config bit

38.65 por_read_data2_reg				0x4010C13C
Bits	Field Name	Acc	Default	Description
31 : 0	por_read_data2	ro	0x0	read block3 config bit

38.66 por_read_data3_reg				0x4010C140
Bits	Field Name	Acc	Default	Description
31 : 0	por_read_data3	ro	0x0	read block4 config bit

38.67 EFUSE_MEM				0x4010C7F8
Bits	Field Name	Acc	Default	Description
31 : 0	EFUSE_DATA	rw	0x0	the registers are the mapping address to efuse macro. Write the 12'hxxx address means burn the data into block (12'hxxx¡¡2) of efuse. Read the 12'hxxx address will get the block (12'hxxx¡¡2) data of efuse.

39 msadc

39.1 cfg_msadc_mode				0x4010D000
Bits	Field Name	Acc	Default	Description
0	cfg_msadc_mode	rw	0x0	1'b1: hardware mode auto test; 1'b0: software mode;
1	cfg_msadc_hw_test_channel	rw	0x0	1'b1: 2 channels; 1'b0: 1 channel

39.2 cfg_msadc_sw_ctrl0				0x4010D004
Bits	Field Name	Acc	Default	Description
0	cfg_msadc_sw_start_pulse	rw	0x0	Start pulse, write 1'b1 to generate one cycle pulse

1	cfg_msadc_sw_stop_pulse	rw	0x0	Stop pulse, write 1'b1 to generate one cycle pulse
2	cfg_msadc_sw_clear_pulse	rw	0x0	Clear pulse, write 1'b1 to generate one cycle pulse

39.3 cfg_msadc_sw_ctrl1				0x4010D008
Bits	Field Name	Acc	Default	Description
3 : 0	cfg_msadc_sw_mux_bits	rw	0x0	msadc meter mux bits when software mode
7 : 4	cfg_msadc_test_0_mux_bits	rw	0x0	msadc meter mux bits when power off
11 : 8	cfg_msadc_test_1_mux_bits	rw	0x0	msadc meter mux bits when power off
15 : 12	cfg_msadc_wait_cnt	rw	0x3	msadc meter wait counter
27 : 16	cfg_msadc_window	rw	0x200	msadc meter window
28	cfg_msadc_sw_diff_mode	rw	0x0	msadc diff mode

39.4 cfg_msadc_ana_ctrl0				0x4010D00C
Bits	Field Name	Acc	Default	Description
0	cfg_ana_msadc_chnp_sel	rw	0x0	
1	cfg_ana_msadc_cmp_bit	rw	0x0	
9 : 2	cfg_ana_msadc_vbe_cal_bit	rw	0x80	
10	cfg_ana_msadc_sdm_mode	rw	0x1	
11	cfg_ana_msadc_test_mode	rw	0x0	
14 : 12	cfg_ana_msadc_ibias_bit	rw	0x1	
15	cfg_ana_msadc_chop_en	rw	0x1	
16	cfg_ana_msadc_sdm_gain_bit	rw	0x0	
18 : 17	cfg_ana_msadc_dem_en	rw	0x3	
19	cfg_ana_msadc_adc_ff_en	rw	0x1	
20	cfg_ana_msadc_ts_mode	rw	0x0	
24 : 21	cfg_ana_msadc_rsvd_bit	rw	0x0	
26 : 25	cfg_ana_msadc_vcom_bit	rw	0x1	
28 : 27	cfg_ana_msadc_dither_mode	rw	0x0	

39.5 cfg_msadc_ro_acc				0x4010D010
Bits	Field Name	Acc	Default	Description
21 : 0	ro_msadc_dat_acc	ro	0x0	read only, msadc acc vaule when software

39.6 cfg_msadc_ro_test_acc_0				0x4010D014
Bits	Field Name	Acc	Default	Description
21 : 0	ro_msadc_test_acc_0	ro	0x0	read only, msadc acc vaule when power off

39.7 cfg_msadc_ro_test_acc_1				0x4010D018
Bits	Field Name	Acc	Default	Description
21 : 0	ro_msadc_test_acc_1	ro	0x0	read only, msadc acc vaule when power off

39.8 cfg_msadc_int_raw				0x4010D01C
Bits	Field Name	Acc	Default	Description
0	masdc_int_raw	rw	0x0	masdc interrupt raw status, write 1'b1 to clear

39.9 cfg_msadc_int_status				0x4010D020
Bits	Field Name	Acc	Default	Description
0	msadc_int_status	rw	0x0	masdc interrupt status, write 1'b1 to clear

39.10 cfg_msadc_int_mask	0x4010D024

Bits	Field Name	Acc	Default	Description
0	cfg_msadc_int_mask	rw	0x0	masdc interrupt mask

40 usbc

40.1 GOTGCTL				0x40200000
Bits	Field Name	Acc	Default	Description
0	SesReqScs	ro	0x0	
1	SesReq	rw	0x0	
2	VbvalidOvEn	rw	0x0	
3	VbvalidOvVal	rw	0x0	
4	AvalidOvEn	rw	0x0	
5	AvalidOvVal	rw	0x0	
6	BvalidOvEn	rw	0x0	
7	BvalidOvVal	rw	0x0	
8	HstNegScs	ro	0x0	
9	HNPReq	rw	0x0	
10	HstSetHNPEn	rw	0x0	
11	DevHNPEn	rw	0x0	
12	EHEn	rw	0x0	
14 : 13	GOTGCTL_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	DbnceFltrBypass	rw	0x0	
16	ConIDSts	ro	0x1	
17	DbncTime	ro	0x0	
18	ASesVld	rw	0x0	bit type is changed from ru to rw.
19	BSesVld	ro	0x0	
20	OTGVer	rw	0x0	
21	CurMod	ro	0x0	
31 : 22	GOTGCTL_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.2 GOTGINT				0x40200004
Bits	Field Name	Acc	Default	Description
1 : 0	GOTGINT_reserved_0	rc	0x0	bit type is changed from dc to rc.
2	SesEndDet	rc	0x0	bit type is changed from w1c to rc.
7 : 3	GOTGINT_reserved_1	rc	0x0	bit type is changed from dc to rc.
8	SesReqSucStsChng	rc	0x0	bit type is changed from w1c to rc.
9	HstNegSucStsChng	rc	0x0	bit type is changed from w1c to rc.
16 : 10	GOTGINT_reserved_2	rc	0x0	bit type is changed from dc to rc.
17	HstNegDet	rc	0x0	bit type is changed from w1c to rc.
18	ADevTOUTChg	rc	0x0	bit type is changed from w1c to rc.
19	DbnceDone	rc	0x0	bit type is changed from w1c to rc.
31 : 20	GOTGINT_reserved_3	rc	0x0	bit type is changed from dc to rc.

40.3 GAHBCFG				0x40200008
Bits	Field Name	Acc	Default	Description
0	GlblIntrMsk	rw	0x0	
4 : 1	HBstLen	rw	0x0	
5	DMAEn	rw	0x0	
6	GAHBCFG_reserved_0	rc	0x0	bit type is changed from dc to rc.
7	NPTxFEmpLvl	rw	0x0	
8	PTxFEmpLvl	rw	0x0	
20 : 9	GAHBCFG_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	RemMemSupp	rw	0x0	
22	NotiAllDmaWrit	rw	0x0	

23	AHBSingle	rw	0x0	
24	InvDescEndianess	rw	0x0	
31 : 25	GAHBCFG_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.4 GUSBCFG				0x4020000C
Bits	Field Name	Acc	Default	Description
2 : 0	TOutCal	rw	0x0	
3	PHYIf	ro	0x0	
4	ULPI_UTMI_Sel	rw	0x0	bit type is changed from ru to rw.
5	FSIntf	rw	0x0	bit type is changed from othr to rw.
6	PHYSel	rw	0x0	
7	GUSBCFG_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	SRPCap	rw	0x0	
9	HNPCap	rw	0x0	
13 : 10	USBTrdTim	rw	0x5	
14	GUSBCFG_reserved_1	rc	0x0	bit type is changed from dc to rc.
15	PhyLPwrClkSel	rw	0x0	
21 : 16	GUSBCFG_reserved_2	rc	0x0	bit type is changed from dc to rc.
22	TermSelDLPulse	rw	0x0	
25 : 23	GUSBCFG_reserved_3	rc	0x0	bit type is changed from dc to rc.
26	IC_USBCap	ro	0x0	
27	GUSBCFG_reserved_4	rc	0x0	bit type is changed from dc to rc.
28	TxEndDelay	rw	0x0	
29	ForceHstMode	rw	0x0	
30	ForceDevMode	rw	0x0	
31	CorruptTxPkt	rw	0x0	bit type is changed from w1 to rw.

40.5 GRSTCTL				0x40200010
Bits	Field Name	Acc	Default	Description
0	CSftRst	rw	0x0	bit type is changed from othr to rw.
1	PIUFSSftRst	rw	0x0	bit type is changed from othr to rw.
2	FrmCntrRst	rw	0x0	bit type is changed from othr to rw.
3	GRSTCTL_reserved_0	rc	0x0	bit type is changed from dc to rc.
4	RxFFlsh	rw	0x0	bit type is changed from othr to rw.
5	TxFFlsh	rw	0x0	bit type is changed from othr to rw.
10 : 6	TxFNum	rw	0x0	
11	GRSTCTL_reserved_1	rc	0x0	bit type is changed from dc to rc.
15 : 12	StreamSwRstChannel	rw	0x0	
16	StreamSwRstReq	rw	0x0	
17	StreamStsFifoSwRst	rw	0x0	
29 : 18	GRSTCTL_reserved_2	rc	0x0	bit type is changed from dc to rc.
30	DMAReq	ro	0x0	
31	AHBIdle	ro	0x1	

40.6 GINTSTS				0x40200014
Bits	Field Name	Acc	Default	Description
0	CurMod	ro	0x0	
1	ModeMis	rc	0x0	bit type is changed from w1c to rc.
2	OTGInt	rw	0x0	bit type is changed from ru to rw.
3	Sof	rw	0x0	bit type is changed from othr to rw.
4	RxFLvl	ro	0x0	
5	NPTxFEmp	ro	0x1	
6	GINNakEff	ro	0x0	
7	GOUTNakEff	ro	0x0	
8	StreamStsFifoNotEmpt	ro	0x0	

9	StreamStsFifoOvf	ro	0x0	
10	ErlySusp	rc	0x0	bit type is changed from w1c to rc.
11	USBSusp	rc	0x0	bit type is changed from w1c to rc.
12	USBRst	rc	0x0	bit type is changed from w1c to rc.
13	EnumDone	rc	0x0	bit type is changed from w1c to rc.
14	ISOOutDrop	rc	0x0	bit type is changed from w1c to rc.
15	EOPF	rc	0x0	bit type is changed from w1c to rc.
16	GINTSTS_reserved_1	rc	0x0	bit type is changed from dc to rc.
17	EPMis	rc	0x0	bit type is changed from w1c to rc.
18	IEPInt	ro	0x0	
19	OEPInt	ro	0x0	
20	incompISOIN	rc	0x0	bit type is changed from w1c to rc.
21	incomplP	rc	0x0	bit type is changed from w1c to rc.
22	FetSusp	rc	0x0	bit type is changed from w1c to rc.
23	ResetDet	rc	0x0	bit type is changed from w1c to rc.
24	PrtInt	ro	0x0	
25	HChInt	ro	0x0	
26	PTxFEmp	rw	0x1	bit type is changed from ru to rw.
27	GINTSTS_reserved_2	rc	0x0	bit type is changed from dc to rc.
28	ConIDStsChng	rw	0x0	bit type is changed from othr to rw.
29	DisconnInt	rc	0x0	bit type is changed from w1c to rc.
30	SessReqInt	rc	0x0	bit type is changed from w1c to rc.
31	WkUpInt	rc	0x0	bit type is changed from w1c to rc.

40.7 GINTMSK				0x40200018
Bits	Field Name	Acc	Default	Description
0	GINTMSK_reserved_0	rc	0x0	bit type is changed from dc to rc.
1	ModeMisMsk	rw	0x0	
2	OTGIntMsk	rw	0x0	
3	SofMsk	rw	0x0	
4	RxFLvlMsk	rw	0x0	
5	NPTxFEmpMsk	rw	0x0	
6	GINNakEffMsk	rw	0x0	
7	GOUTNakEffMsk	rw	0x0	
8	StreamStsFifoNotEmptMsk	ro	0x0	
9	StreamStsFifoOvfMsk	ro	0x0	
10	ErlySuspMsk	rw	0x0	
11	USBSuspMsk	rw	0x0	
12	USBRstMsk	rw	0x0	
13	EnumDoneMsk	rw	0x0	
14	ISOOutDropMsk	rw	0x0	
15	EOPFMsk	rw	0x0	
16	GINTMSK_reserved_2	rc	0x0	bit type is changed from dc to rc.
17	EPMisMsk	rw	0x0	
18	IEPIntMsk	rw	0x0	
19	OEPIntMsk	rw	0x0	
20	GINTMSK_reserved_3	rc	0x0	bit type is changed from dc to rc.
21	incomplPMsK	rw	0x0	
22	FetSuspMsk	rw	0x0	
23	ResetDetMsk	rw	0x0	
24	PrtIntMsk	rw	0x0	
25	HChIntMsk	rw	0x0	
26	PTxFEmpMsk	rw	0x0	
27	GINTMSK_reserved_4	rc	0x0	bit type is changed from dc to rc.
28	ConIDStsChngMsk	rw	0x0	
29	DisconnIntMsk	rw	0x0	
30	SessReqIntMsk	rw	0x0	

31	WkUpIntMsk	rw	0x0	

40.8 GRXSTSR				0x4020001C
Bits	Field Name	Acc	Default	Description
3 : 0	ChNum	rw	0x0	bit type is changed from ru to rw.
14 : 4	BCnt	ro	0x0	
16 : 15	DPID	ro	0x0	
20 : 17	PktSts	rw	0x0	bit type is changed from ru to rw.
24 : 21	FN	ro	0x0	
31 : 25	GRXSTSR_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.9 GRXSTSP				0x40200020
Bits	Field Name	Acc	Default	Description
3 : 0	ChNum	rw	0x0	bit type is changed from ru to rw.
14 : 4	BCnt	ro	0x0	
16 : 15	DPID	ro	0x0	
20 : 17	PktSts	rw	0x0	bit type is changed from ru to rw.
24 : 21	FN	ro	0x0	
31 : 25	GRXSTSP_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.10 GRXFSIZ				0x40200024
Bits	Field Name	Acc	Default	Description
11 : 0	RxFDep	rw	0xEAF	
31 : 12	GRXFSIZ_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.11 GNPTXFSIZ				0x40200028
Bits	Field Name	Acc	Default	Description
11 : 0	NPTXFStAddr	rw	0xEAF	
15 : 12	GNPTXFSIZ_reserved_0	rc	0x0	bit type is changed from dc to rc.
27 : 16	NPTXFDep	rw	0xEAF	
31 : 28	GNPTXFSIZ_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.12 GNPTXSTS				0x4020002C
Bits	Field Name	Acc	Default	Description
15 : 0	NPTxFSpcAvail	ro	0xEAF	
23 : 16	NPTxQSpcAvail	ro	0x8	
30 : 24	NPTxQTop	ro	0x0	
31	GNPTXSTS_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.13 GPVNDCTL				0x40200034
Bits	Field Name	Acc	Default	Description
7 : 0	RegData	rw	0x0	Register Data (RegData) Contains the write data for register write. Read data for register read, valid when VStatus Done is set.
15 : 8	VCtrl	rw	0x0	UTMI+ Vendor Control Register Address (VCtrl) The 4-bit register address a vendor defined 4-bit parallel output bus. Bits 11:8 of this field are placed on utmi_vcontrol[3:0]. ULPI Extended Register Address (ExtRegAddr) The 6-bit PHY extended register address.

21 : 16	RegAddr	rw	0x0	Register Address (RegAddr) The 6-bit PHY register address for immediate PHY Register set access. Set to 6'h2F for Extended PHY Register set access.
22	RegWr	rw	0x0	Register Write (RegWr) Set this bit for register writes, and clear it for register reads.
25	NewRegReq	rw	0x0	New Register Request (NewRegReq) The application sets this bit for a new vendor control access. Values: 0x0 (INACTIVE): New Register Request not active 0x1 (ACTIVE): New Register Request active
26	VStsBsy	ro	0x0	VStatus Busy (VStsBsy) The core sets this bit when the vendor control access is in progress and clears this bit when done.
27	VStsDone	rw	0x0	VStatus Done (VStsDone) The core sets this bit when the vendor control access is done. This bit is cleared by the core when the application sets the New Register Request bit (bit 25). Values: 0x0 (INACTIVE): VStatus Done inactive 0x1 (ACTIVE): VStatus Done active

40.14 GGPIO				0x40200038
Bits	Field Name	Acc	Default	Description
15 : 0	GPI	ro	0x0	
31 : 16	GPO	rw	0x0	

40.15 GUID				0x4020003C
Bits	Field Name	Acc	Default	Description
31 : 0	GUID	rw	0x88000000	

40.16 GSNPSID				0x40200040
Bits	Field Name	Acc	Default	Description
31 : 0	SynopsysID	ro	0x4F54400A	

40.17 GHWCFG1				0x40200044
Bits	Field Name	Acc	Default	Description
31 : 0	EpDir	ro	0x0	

40.18 GHWCFG2				0x40200048
Bits	Field Name	Acc	Default	Description
2 : 0	OtgMode	ro	0x0	
4 : 3	OtgArch	ro	0x2	
5	SingPnt	ro	0x0	
7 : 6	HSPhyType	ro	0x1	
9 : 8	FSPhyType	ro	0x1	
13 : 10	NumDevEps	ro	0xC	
17 : 14	NumHstChnl	ro	0xF	
18	PerioSupport	ro	0x1	
19	DynFifoSizing	ro	0x1	
20	MultiProcIntrpt	ro	0x0	
21	GHWCFG2_reserved_0	rc	0x0	bit type is changed from dc to rc.

23 : 22	NPTxQDepth	ro	0x2	
25 : 24	PTxQDepth	ro	0x2	
30 : 26	TknQDepth	ro	0x8	
31	GHWCFG2_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.19 GHWCFG3				0x4020004C
Bits	Field Name	Acc	Default	Description
3 : 0	XferSizeWidth	ro	0x8	
6 : 4	PktSizeWidth	ro	0x6	
7	OtgEn	ro	0x1	
8	I2CIntSel	ro	0x0	
9	VndctlSupt	ro	0x0	
10	OptFeature	ro	0x0	
11	RstType	ro	0x0	
12	ADPSupport	ro	0x1	
13	HSICMode	ro	0x0	
14	BCSupport	ro	0x0	
15	LPMMode	ro	0x0	
31 : 16	DfifoDepth	ro	0xE47	

40.20 GHWCFG4				0x40200050
Bits	Field Name	Acc	Default	Description
3 : 0	NumDevPerioEps	ro	0x0	
4	PartialPwrDn	ro	0x0	
5	AhbFreq	ro	0x1	
6	Hibernation	ro	0x0	
7	ExtendedHibernation	ro	0x0	
11 : 8	GHWCFG4_reserved_0	rc	0x0	bit type is changed from dc to rc.
12	ACGSupt	ro	0x0	
13	EnhancedLPMSupt	ro	0x1	
15 : 14	PhyDataWidth	ro	0x0	
19 : 16	NumCtlEps	ro	0x0	
20	IddgFltr	ro	0x1	
21	VBusValidFltr	ro	0x1	
22	AValidFltr	ro	0x1	
23	BValidFltr	ro	0x1	
24	SessEndFltr	ro	0x1	
25	DedFifoMode	ro	0x1	
29 : 26	INEps	ro	0x8	
30	DescDMAEnabled	ro	0x1	
31	DescDMA	ro	0x1	

40.21 GPWRDN				0x40200058
Bits	Field Name	Acc	Default	Description
0	PMUIntSel	rw	0x0	
1	PMUActv	rw	0x0	
2	GPWRDN_reserved_0	rc	0x0	bit type is changed from dc to rc.
3	PwrDnClmp	rw	0x0	
4	PwrDnRst_n	rw	0x1	
5	PwrDnSwtch	rw	0x0	
6	DisableVBUS	rw	0x0	
14 : 7	GPWRDN_reserved_1	rc	0x0	bit type is changed from dc to rc.
15	SRPDetect	rc	0x0	bit type is changed from w1c to rc.
16	SRPDetectMsk	rw	0x0	
17	StsChngInt	rc	0x0	bit type is changed from w1c to rc.

18	StsChngIntMsk	rw	0x0	
20 : 19	LineState	ro	0x0	
21	IDDIG	ro	0x0	
22	BSessVld	ro	0x0	
23	ADPInt	rc	0x0	bit type is changed from w1c to rc.
31 : 24	GPWRDN_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.22 GDFIFOCFG				0x4020005C
Bits	Field Name	Acc	Default	Description
15 : 0	GDFIFOCfg	rw	0xEAF	
31 : 16	EPInfoBaseAddr	rw	0xE47	

40.23 GADPCTL				0x40200060
Bits	Field Name	Acc	Default	Description
1 : 0	PrbDschg	rw	0x0	
3 : 2	PrbDelta	rw	0x0	
5 : 4	PrbPer	rw	0x0	
16 : 6	RTIM	ro	0x0	
17	EnaPrb	rw	0x0	
18	EnaSns	rw	0x0	
19	ADPRes	rw	0x0	bit type is changed from othr to rw.
20	ADPEn	rw	0x0	
21	AdpPrbInt	rc	0x0	bit type is changed from w1c to rc.
22	AdpSnsInt	rc	0x0	bit type is changed from w1c to rc.
23	AdpToutInt	rc	0x0	bit type is changed from w1c to rc.
24	AdpPrbIntMsk	rw	0x0	
25	AdpSnsIntMsk	rw	0x0	
26	AdpToutMsk	rw	0x0	
28 : 27	AR	rw	0x0	
31 : 29	GADPCTL_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.24 STREAMSTSFIFO				0x40200080
Bits	Field Name	Acc	Default	Description
0	StreamValid	ro	0x0	
1	StreamSwRsted	ro	0x0	
3 : 2	STREAMSTSFIFO_-reserved_0	rc	0x0	
7 : 4	StreamEpNum	ro	0x0	
15 : 8	StreamFileNum	ro	0x0	
31 : 16	STREAMSTSFIFO_-reserved_1	rc	0x0	

40.25 STREAMFIRSTPTRFIFO				0x40200084
Bits	Field Name	Acc	Default	Description
31 : 0	StreamFirstPtr	ro	0x0	

40.26 STREAMLASTPTRFIFO				0x40200088
Bits	Field Name	Acc	Default	Description
31 : 0	StreamLastPtr	ro	0x0	

40.27 HPTXFSIZ				0x40200100
Bits	Field Name	Acc	Default	Description

12 : 0	PTxFStAddr	rw	0x1D5E	
15 : 13	HPTXFSIZ_reserved_0	rc	0x0	bit type is changed from dc to rc.
27 : 16	PTxFSize	rw	0xEAF	
31 : 28	HPTXFSIZ_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.28 DIEPTXF1				0x40200104
Bits	Field Name	Acc	Default	Description
11 : 0	INEPnTxFStAddr	rw	0xECF	
15 : 12	DIEPTXF1_reserved_0	rc	0x0	bit type is changed from dc to rc.
25 : 16	INEPnTxFDep	rw	0x300	
31 : 26	DIEPTXF1_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.29 DIEPTXF2				0x40200108
Bits	Field Name	Acc	Default	Description
12 : 0	INEPnTxFStAddr	rw	0x11CF	
15 : 13	DIEPTXF2_reserved_0	rc	0x0	bit type is changed from dc to rc.
25 : 16	INEPnTxFDep	rw	0x300	
31 : 26	DIEPTXF2_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.30 DIEPTXF3				0x4020010C
Bits	Field Name	Acc	Default	Description
12 : 0	INEPnTxFStAddr	rw	0x14CF	
15 : 13	DIEPTXF3_reserved_0	rc	0x0	bit type is changed from dc to rc.
24 : 16	INEPnTxFDep	rw	0x100	
31 : 25	DIEPTXF3_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.31 DIEPTXF4				0x40200110
Bits	Field Name	Acc	Default	Description
12 : 0	INEPnTxFStAddr	rw	0x15CF	
15 : 13	DIEPTXF4_reserved_0	rc	0x0	bit type is changed from dc to rc.
24 : 16	INEPnTxFDep	rw	0x100	
31 : 25	DIEPTXF4_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.32 DIEPTXF5				0x40200114
Bits	Field Name	Acc	Default	Description
12 : 0	INEPnTxFStAddr	rw	0x16CF	
15 : 13	DIEPTXF5_reserved_0	rc	0x0	bit type is changed from dc to rc.
24 : 16	INEPnTxFDep	rw	0x100	
31 : 25	DIEPTXF5_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.33 DIEPTXF6				0x40200118
Bits	Field Name	Acc	Default	Description
12 : 0	INEPnTxFStAddr	rw	0x17CF	
15 : 13	DIEPTXF6_reserved_0	rc	0x0	bit type is changed from dc to rc.
24 : 16	INEPnTxFDep	rw	0x100	
31 : 25	DIEPTXF6_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.34 DIEPTXF7				0x4020011C
Bits	Field Name	Acc	Default	Description
12 : 0	INEPnTxFStAddr	rw	0x18CF	
15 : 13	DIEPTXF7_reserved_0	rc	0x0	bit type is changed from dc to rc.

24 : 16	INEPnTxFDep	rw	0x100	
31 : 25	DIEPTXF7_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.35 DIEPTXF8				0x40200120
Bits	Field Name	Acc	Default	Description
12 : 0	INEPnTxFStAddr	rw	0x19CF	
15 : 13	DIEPTXF8_reserved_0	rc	0x0	bit type is changed from dc to rc.
24 : 16	INEPnTxFDep	rw	0x100	
31 : 25	DIEPTXF8_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.36 HCFG				0x40200400
Bits	Field Name	Acc	Default	Description
1 : 0	FSLSPclkSel	rw	0x0	
2	FSLSSupp	rw	0x0	
6 : 3	HCFG_reserved_0	rc	0x0	bit type is changed from dc to rc.
7	Ena32KHzS	rw	0x0	
15 : 8	ResValid	rw	0x2	
22 : 16	HCFG_reserved_1	rc	0x0	bit type is changed from dc to rc.
23	DescDMA	rw	0x0	
25 : 24	FrListEn	rw	0x0	
26	PerSchedEna	rw	0x0	
30 : 27	HCFG_reserved_2	rc	0x0	bit type is changed from dc to rc.
31	ModeChTimEn	rw	0x0	

40.37 HFIR				0x40200404
Bits	Field Name	Acc	Default	Description
15 : 0	FrInt	rw	0xEA60	
16	HFIRRldCtrl	rw	0x0	
31 : 17	HFIR_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.38 HFNUM				0x40200408
Bits	Field Name	Acc	Default	Description
15 : 0	FrNum	ro	0x3FFF	
31 : 16	FrRem	ro	0x0	

40.39 HPTXSTS				0x40200410
Bits	Field Name	Acc	Default	Description
15 : 0	PTxFSpcAvail	ro	0xEAF	
23 : 16	PTxQSpcAvail	ro	0x8	
31 : 24	PTxQTop	ro	0x0	

40.40 HAINT				0x40200414
Bits	Field Name	Acc	Default	Description
15 : 0	HAINT	ro	0x0	
31 : 16	HAINT_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.41 HAINTMSK				0x40200418
Bits	Field Name	Acc	Default	Description
15 : 0	HAINTMsk	rw	0x0	
31 : 16	HAINTMSK_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.42 HFLBAddr				0x4020041C
Bits	Field Name	Acc	Default	Description
31 : 0	HFLBAddr	rw	0x0	

40.43 HPRT				0x40200440
Bits	Field Name	Acc	Default	Description
0	PrtConnSts	ro	0x0	
1	PrtConnDet	rc	0x0	bit type is changed from w1c to rc.
2	PrtEna	rc	0x0	bit type is changed from w1c to rc.
3	PrtEnChng	rc	0x0	bit type is changed from w1c to rc.
4	PrtOvrCurrAct	ro	0x0	
5	PrtOvrCurrChng	rc	0x0	bit type is changed from w1c to rc.
6	PrtRes	rw	0x0	
7	PrtSusp	rw	0x0	bit type is changed from othr to rw.
8	PrtRst	rw	0x0	
9	HPRT_reserved_0	rc	0x0	bit type is changed from dc to rc.
11 : 10	PrtLnSts	ro	0x0	
12	PrtPwr	rw	0x0	
16 : 13	PrtTstCtl	rw	0x0	
18 : 17	PrtSpd	ro	0x0	
31 : 19	HPRT_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.44 HCCHAR0				0x40200500
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR0_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.45 HCSPLT0				0x40200504
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT0_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.46 HCINT0				0x40200508
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.

7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT0_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.47 HCINTMSK0				0x4020050C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK0_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK0_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK0_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.48 HCTSIZ0				0x40200510
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.49 HCDMA0				0x40200514
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.50 HCDMAB0				0x4020051C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.51 HCCHAR1				0x40200520
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR1_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.52 HCSPLT1				0x40200524
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	

13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT1_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.53 HCINT1				0x40200528
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT1_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.54 HCINTMSK1				0x4020052C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK1_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK1_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK1_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.55 HCTSIZ1				0x40200530
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.56 HCDMA1				0x40200534
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.57 HCDMAB1				0x4020053C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.58 HCCHAR2				0x40200540
Bits	Field Name	Acc	Default	Description

10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR2_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.59 HCSPLT2				0x40200544
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT2_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.60 HCINT2				0x40200548
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT2_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.61 HCINTMSK2				0x4020054C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK2_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK2_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK2_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.62 HCTSIZ2				0x40200550
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	

30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.63 HCDMA2				0x40200554
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.64 HCDMAB2				0x4020055C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.65 HCCHAR3				0x40200560
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR3_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.66 HCSPLT3				0x40200564
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT3_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.67 HCINT3				0x40200568
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT3_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.68 HCINTMSK3				0x4020056C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK3_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK3_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK3_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.69 HCTSIZ3				0x40200570
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.70 HCDMA3				0x40200574
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.71 HCDMAB3				0x4020057C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.72 HCCHAR4				0x40200580
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR4_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.73 HCSPLT4				0x40200584
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT4_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.74 HCINT4				0x40200588
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.

1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT4_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.75 HCINTMSK4				0x4020058C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK4_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK4_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK4_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.76 HCTSIZ4				0x40200590
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.77 HCDMA4				0x40200594
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.78 HCDMAB4				0x4020059C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.79 HCCHAR5				0x402005A0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR5_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.

31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.80 HCSPLT5				0x402005A4
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT5_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.81 HCINT5				0x402005A8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT5_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.82 HCINTMSK5				0x402005AC
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK5_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK5_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK5_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.83 HCTSIZ5				0x402005B0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.84 HCDMA5				0x402005B4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.85 HCDMAB5	0x402005BC

Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.86 HCCHAR6				0x402005C0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR6_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.87 HCSPLT6				0x402005C4
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT6_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.88 HCINT6				0x402005C8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT6_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.89 HCINTMSK6				0x402005CC
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK6_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK6_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK6_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.90 HCTSIZ6				0x402005D0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.91 HCDMA6				0x402005D4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.92 HCDMAB6				0x402005DC
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.93 HCCHAR7				0x402005E0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR7_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.94 HCSPLT7				0x402005E4
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT7_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.95 HCINT7				0x402005E8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.

13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT7_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.96 HCINTMSK7				0x402005EC
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK7_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK7_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK7_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.97 HCTSIZ7				0x402005F0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.98 HCDMA7				0x402005F4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.99 HCDMAB7				0x402005FC
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.100 HCCHAR8				0x40200600
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR8_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.101 HCSPLT8				0x40200604
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT8_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.102 HCINT8				0x40200608
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT8_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.103 HCINTMSK8				0x4020060C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK8_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK8_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK8_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.104 HCTSIZ8				0x40200610
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.105 HCDMA8				0x40200614
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.106 HCDMAB8				0x4020061C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.107 HCCHAR9				0x40200620
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR9_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	

28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.108 HCSPLT9				0x40200624
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT9_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.109 HCINT9				0x40200628
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT9_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.110 HCINTMSK9				0x4020062C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK9_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK9_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK9_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.111 HCTSIZ9				0x40200630
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.112 HCDMA9				0x40200634
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.113 HCDMAB9				0x4020063C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.114 HCCHAR10				0x40200640
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR10_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.115 HCSPLT10				0x40200644
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT10_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.116 HCINT10				0x40200648
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT10_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.117 HCINTMSK10				0x4020064C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK10_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK10_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	

31 : 14	HCINTMSK10_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.118 HCTSIZ10				0x40200650
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.119 HCDMA10				0x40200654
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.120 HCDMAB10				0x4020065C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.121 HCCHAR11				0x40200660
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR11_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.122 HCSPLT11				0x40200664
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT11_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.123 HCINT11				0x40200668
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.

10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT11_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.124 HCINTMSK11				0x4020066C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK11_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK11_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK11_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.125 HCTSIZ11				0x40200670
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.126 HCDMA11				0x40200674
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.127 HCDMAB11				0x4020067C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.128 HCCHAR12				0x40200680
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR12_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.129 HCSPLT12				0x40200684
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	

30 : 17	HCSPLT12_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.130 HCINT12				0x40200688
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT12_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.131 HCINTMSK12				0x4020068C
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK12_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK12_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK12_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.132 HCTSIZ12				0x40200690
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.133 HCDMA12				0x40200694
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.134 HCDMAB12				0x4020069C
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.135 HCCHAR13				0x402006A0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	

16	HCCHAR13_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.136 HCSPLT13				0x402006A4
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT13_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.137 HCINT13				0x402006A8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT13_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.138 HCINTMSK13				0x402006AC
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK13_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK13_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK13_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.139 HCTSIZ13				0x402006B0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.140 HCDMA13				0x402006B4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.141 HCDMAB13				0x402006BC
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.142 HCCHAR14				0x402006C0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR14_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.143 HCSPLT14				0x402006C4
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT14_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.144 HCINT14				0x402006C8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.
5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT14_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.145 HCINTMSK14				0x402006CC
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	

2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK14_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK14_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK14_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.146 HCTSIZ14				0x402006D0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.147 HCDMA14				0x402006D4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.148 HCDMAB14				0x402006DC
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.149 HCCHAR15				0x402006E0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	EPNum	rw	0x0	
15	EPDir	rw	0x0	
16	HCCHAR15_reserved_0	rc	0x0	bit type is changed from dc to rc.
17	LSpdDev	rw	0x0	
19 : 18	EPType	rw	0x0	
21 : 20	EC	rw	0x0	
28 : 22	DevAddr	rw	0x0	
29	OddFrm	rw	0x0	
30	ChDis	rw	0x0	bit type is changed from othr to rw.
31	ChEna	rw	0x0	bit type is changed from othr to rw.

40.150 HCSPLT15				0x402006E4
Bits	Field Name	Acc	Default	Description
6 : 0	PrtAddr	rw	0x0	
13 : 7	HubAddr	rw	0x0	
15 : 14	XactPos	rw	0x0	
16	CompSplt	rw	0x0	
30 : 17	HCSPLT15_reserved_0	rc	0x0	bit type is changed from dc to rc.
31	SpltEna	rw	0x0	

40.151 HCINT15				0x402006E8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	ChHltd	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	STALL	rc	0x0	bit type is changed from w1c to rc.
4	NAK	rc	0x0	bit type is changed from w1c to rc.

5	ACK	rc	0x0	bit type is changed from w1c to rc.
6	NYET	rc	0x0	bit type is changed from w1c to rc.
7	XactErr	rc	0x0	bit type is changed from w1c to rc.
8	BblErr	rc	0x0	bit type is changed from w1c to rc.
9	FrmOvrun	rc	0x0	bit type is changed from w1c to rc.
10	DataTglErr	rc	0x0	bit type is changed from w1c to rc.
11	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
12	XCS_XACT_ERR	rc	0x0	bit type is changed from w1c to rc.
13	DESC_LST_ROLLIntr	rc	0x0	bit type is changed from w1c to rc.
31 : 14	HCINT15_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.152 HCINTMSK15				0x402006EC
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	ChHltdMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
10 : 3	HCINTMSK15_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	BNAIntrMsk	rw	0x0	
12	HCINTMSK15_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	DESC_LST_ROLLIntrMsk	rw	0x0	
31 : 14	HCINTMSK15_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.153 HCTSIZ15				0x402006F0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	Pid	rw	0x0	
31	DoPng	rw	0x0	

40.154 HCDMA15				0x402006F4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.155 HCDMAB15				0x402006FC
Bits	Field Name	Acc	Default	Description
31 : 0	HCDMAB	ro	0x0	

40.156 DCFG				0x40200800
Bits	Field Name	Acc	Default	Description
1 : 0	DevSpd	rw	0x0	
2	NZStsOUTHShk	rw	0x0	
3	Ena32KHzSusp	rw	0x0	
10 : 4	DevAddr	rw	0x0	
12 : 11	PerFrInt	rw	0x0	
13	EnDevOutNak	rw	0x0	
14	XCVRDLY	rw	0x0	
15	ErraticIntMsk	rw	0x0	
22 : 16	DCFG_reserved_0	rc	0x0	bit type is changed from dc to rc.
23	DescDMA	rw	0x0	
25 : 24	PerSchIntvl	rw	0x0	
31 : 26	ResValid	rw	0x2	

40.157 DCTL				0x40200804
Bits	Field Name	Acc	Default	Description
0	RmtWkUpSig	rw	0x0	
1	SftDiscon	rw	0x1	
2	GNPINNakSts	ro	0x0	
3	GOUTNakSts	ro	0x0	
6 : 4	TstCtl	rw	0x0	
7	SGNPInNak	rw	0x0	bit type is changed from w1 to rw.
8	CGNPInNak	rw	0x0	bit type is changed from w1 to rw.
9	SGOUTNak	rw	0x0	bit type is changed from w1 to rw.
10	CGOUTNak	rw	0x0	bit type is changed from w1 to rw.
11	PWROnPrgDone	rw	0x0	
12	DCTL_reserved_0	rc	0x0	bit type is changed from dc to rc.
14 : 13	GMC	rw	0x0	
15	IgnrFrmNum	rw	0x0	
16	NakOnBble	rw	0x0	
17	EnContOnBNA	rw	0x0	
31 : 18	DCTL_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.158 DSTS				0x40200808
Bits	Field Name	Acc	Default	Description
0	SuspSts	ro	0x0	
2 : 1	EnumSpd	ro	0x1	
3	ErrticErr	ro	0x0	
7 : 4	DSTS_reserved_0	rc	0x0	bit type is changed from dc to rc.
21 : 8	SOFFN	rw	0x0	bit type is changed from ru to rw.
23 : 22	DevLnSts	ro	0x0	
28 : 24	StreamStsFifoUsedCnt	ro	0x0	
31 : 29	DSTS_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.159 DIEPMSK				0x40200810
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	EPDisbldMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
3	TimeOUTMsk	rw	0x0	
4	INTknTXFEmpMsk	rw	0x0	
5	INTknEPMisMsk	rw	0x0	
6	INEPNakEffMsk	rw	0x0	
7	DIEPMSK_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	TxfifoUndrnMsk	rw	0x0	
9	BNAInIntrMsk	rw	0x0	
12 : 10	DIEPMSK_reserved_1	rc	0x0	bit type is changed from dc to rc.
13	NAKMsk	rw	0x0	
31 : 14	DIEPMSK_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.160 DOEPMSK				0x40200814
Bits	Field Name	Acc	Default	Description
0	XferComplMsk	rw	0x0	
1	EPDisbldMsk	rw	0x0	
2	AHBErrMsk	rw	0x0	
3	SetUPMsk	rw	0x0	
4	OUTTknEPdisMsk	rw	0x0	
5	StsPhseRcvdMsk	rw	0x0	
6	Back2BackSETup	rw	0x0	

7	DOEPMSK_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErrMsk	rw	0x0	
9	BnaOutIntrMsk	rw	0x0	
11 : 10	DOEPMSK_reserved_1	rc	0x0	bit type is changed from dc to rc.
12	BbleErrMsk	rw	0x0	
13	NAKMsk	rw	0x0	
14	NYETMsk	rw	0x0	
15	DOEPMSK_reserved_3	rc	0x0	bit type is changed from dc to rc.
16	StreamTimeOutMsk	rw	0x0	
17	StreamHdrErrMsk	rw	0x0	
18	StreamEndErrMsk	rw	0x0	
19	StreamLenIllegalMsk	rw	0x0	
31 : 20	DOEPMSK_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.161 DAINT				0x40200818
Bits	Field Name	Acc	Default	Description
0	InEpInt0	ro	0x0	
1	InEpInt1	ro	0x0	
2	InEpInt2	ro	0x0	
3	InEpInt3	ro	0x0	
4	InEpInt4	ro	0x0	
5	InEpInt5	ro	0x0	
6	InEpInt6	ro	0x0	
7	InEpInt7	ro	0x0	
8	InEpInt8	ro	0x0	
9	InEpInt9	ro	0x0	
10	InEpInt10	ro	0x0	
11	InEpInt11	ro	0x0	
12	InEpInt12	ro	0x0	
15 : 13	DAINT_reserved_0	rc	0x0	bit type is changed from dc to rc.
16	OutEPInt0	ro	0x0	
17	OutEPInt1	ro	0x0	
18	OutEPInt2	ro	0x0	
19	OutEPInt3	ro	0x0	
20	OutEPInt4	ro	0x0	
21	OutEPInt5	ro	0x0	
22	OutEPInt6	ro	0x0	
23	OutEPInt7	ro	0x0	
24	OutEPInt8	ro	0x0	
25	OutEPInt9	ro	0x0	
26	OutEPInt10	ro	0x0	
27	OutEPInt11	ro	0x0	
28	OutEPInt12	ro	0x0	
31 : 29	DAINT_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.162 DAINTMSK				0x4020081C
Bits	Field Name	Acc	Default	Description
0	InEpMsk0	rw	0x0	
1	InEpMsk1	rw	0x0	
2	InEpMsk2	rw	0x0	
3	InEpMsk3	rw	0x0	
4	InEpMsk4	rw	0x0	
5	InEpMsk5	rw	0x0	
6	InEpMsk6	rw	0x0	
7	InEpMsk7	rw	0x0	
8	InEpMsk8	rw	0x0	

9	InEpMsk9	rw	0x0	
10	InEpMsk10	rw	0x0	
11	InEpMsk11	rw	0x0	
12	InEpMsk12	rw	0x0	
15 : 13	DAINTMSK_reserved_0	rc	0x0	bit type is changed from dc to rc.
16	OutEPMsk0	rw	0x0	
17	OutEPMsk1	rw	0x0	
18	OutEPMsk2	rw	0x0	
19	OutEPMsk3	rw	0x0	
20	OutEPMsk4	rw	0x0	
21	OutEPMsk5	rw	0x0	
22	OutEPMsk6	rw	0x0	
23	OutEPMsk7	rw	0x0	
24	OutEPMsk8	rw	0x0	
25	OutEPMsk9	rw	0x0	
26	OutEPMsk10	rw	0x0	
27	OutEPMsk11	rw	0x0	
28	OutEPMsk12	rw	0x0	
31 : 29	DAINTMSK_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.163 DVBUSDIS				0x40200828
Bits	Field Name	Acc	Default	Description
15 : 0	DVBUSDis	rw	0x17D7	
31 : 16	DVBUSDIS_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.164 DVBUSPULSE				0x4020082C
Bits	Field Name	Acc	Default	Description
11 : 0	DVBUSPulse	rw	0x5B8	
31 : 12	DVBUSPULSE_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.165 DTHRCTL				0x40200830
Bits	Field Name	Acc	Default	Description
0	NonISOThrEn	rw	0x0	
1	ISOThrEn	rw	0x0	
10 : 2	TxThrLen	rw	0x8	
12 : 11	AHBThrRatio	rw	0x0	
15 : 13	DTHRCTL_reserved_0	rc	0x0	bit type is changed from dc to rc.
16	RxThrEn	rw	0x0	
25 : 17	RxThrLen	rw	0x8	
26	DTHRCTL_reserved_1	rc	0x0	bit type is changed from dc to rc.
27	ArbPrkEn	rw	0x1	
31 : 28	DTHRCTL_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.166 DIEPEMPMSK				0x40200834
Bits	Field Name	Acc	Default	Description
15 : 0	InEpTxfEmpMsk	rw	0x0	
31 : 16	DIEPEMPMSK_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.167 DOEPSTREAMTIMEOUTTH				0x402008C0
Bits	Field Name	Acc	Default	Description
15 : 0	StreamTimeoutTh	rw	0xFFFF	
31 : 16	DOEPSTREAMTIMEO-UTTH_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.168 DOEPSTREAMDBGCTRL				0x402008C4
Bits	Field Name	Acc	Default	Description
2 : 0	StreamDbgEpNum	rw	0x0	
31 : 3	DOEPSTREAMDBGCT-RL_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.169 DOEPSTREAMDBGSTATUS1				0x402008C8
Bits	Field Name	Acc	Default	Description
31 : 0	LpCurDescAddr	ro	0x0	

40.170 DOEPSTREAMDBGSTATUS2				0x402008CC
Bits	Field Name	Acc	Default	Description
10 : 0	LpState	ro	0x0	
17 : 13	AiuEffChannel	ro	0x0	
31 : 18	TxRxBytes	ro	0x0	

40.171 DOEPSTREAMDBGSTATUS3				0x402008D0
Bits	Field Name	Acc	Default	Description
10 : 0	RxPktCnt	ro	0x0	
20 : 11	RxPktRem	ro	0x0	
31 : 21	RxFileCnt	ro	0x0	

40.172 DOEPSTREAMDBGSTATUS4				0x402008D4
Bits	Field Name	Acc	Default	Description
15 : 0	RxFileLen	ro	0x0	
31 : 16	RxFileRem	ro	0x0	

40.173 DIEPCTL0				0x40200900
Bits	Field Name	Acc	Default	Description
1 : 0	MPS	rw	0x0	
14 : 2	DIEPCTL0_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	ro	0x1	
16	DIEPCTL0_reserved_1	rc	0x0	bit type is changed from dc to rc.
17	NAKSts	ro	0x0	
19 : 18	EPType	ro	0x0	
20	DIEPCTL0_reserved_2	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	wo	0x0	
29 : 28	DIEPCTL0_reserved_3	rc	0x0	bit type is changed from dc to rc.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.174 DIEPINT0				0x40200908
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.

6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT0_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT0_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.175 DIEPTSIZ0				0x40200910
Bits	Field Name	Acc	Default	Description
6 : 0	XferSize	rw	0x0	
18 : 7	DIEPTSIZ0_reserved_0	rc	0x0	bit type is changed from dc to rc.
20 : 19	PktCnt	rw	0x0	
31 : 21	DIEPTSIZ0_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.176 DIEPDMA0				0x40200914
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.177 DTXFSTS0				0x40200918
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS0_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.178 DIEPDMAB0				0x4020091C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	ro	0x0	

40.179 DIEPCTL1				0x40200920
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL1_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.180 DIEPINT1				0x40200928
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.

1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT1_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT1_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.181 DIEPTSIZ1				0x40200930
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ1_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.182 DIEPDMA1				0x40200934
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.183 DTXFSTS1				0x40200938
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS1_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.184 DIEPDMAB1				0x4020093C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.185 DIEPCTL2				0x40200940
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL2_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL2_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.186 DIEPINT2				0x40200948
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT2_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT2_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.187 DIEPTSIZ2				0x40200950
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ2_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.188 DIEPDMA2				0x40200954
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.189 DTXFSTS2				0x40200958
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS2_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.190 DIEPDMAB2				0x4020095C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.191 DIEPCTL3				0x40200960
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL3_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL3_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.

29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.192 DIEPINT3				0x40200968
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT3_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT3_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.193 DIEPTSIZ3				0x40200970
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ3_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.194 DIEPDMA3				0x40200974
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.195 DTXFSTS3				0x40200978
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS3_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.196 DIEPDMAB3				0x4020097C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.197 DIEPCTL4				0x40200980
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL4_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL4_reserved_1	rc	0x0	bit type is changed from dc to rc.

21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.198 DIEPINT4				0x40200988
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT4_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT4_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.199 DIEPTSIZ4				0x40200990
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ4_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.200 DIEPDMA4				0x40200994
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.201 DTXFSTS4				0x40200998
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS4_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.202 DIEPDMAB4				0x4020099C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.203 DIEPCTL5				0x402009A0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL5_reserved_0	rc	0x0	bit type is changed from dc to rc.

15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL5_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.204 DIEPINT5				0x402009A8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT5_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT5_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.205 DIEPTSIZ5				0x402009B0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ5_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.206 DIEPDMA5				0x402009B4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.207 DTXFSTS5				0x402009B8
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS5_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.208 DIEPDMAB5				0x402009BC
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.209 DIEPCTL6				0x402009C0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL6_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL6_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.210 DIEPINT6				0x402009C8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT6_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT6_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.211 DIEPTSIZ6				0x402009D0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ6_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.212 DIEPDMA6				0x402009D4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.213 DTXFSTS6				0x402009D8
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS6_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.214 DIEPDMAB6				0x402009DC
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.215 DIEPCTL7				0x402009E0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL7_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL7_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.216 DIEPINT7				0x402009E8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT7_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT7_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.217 DIEPTSIZ7				0x402009F0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ7_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.218 DIEPDMA7				0x402009F4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.219 DTXFSTS7	0x402009F8

Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS7_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.220 DIEPDMAB7				0x402009FC
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.221 DIEPCTL8				0x40200A00
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL8_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL8_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.222 DIEPINT8				0x40200A08
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT8_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT8_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.223 DIEPTSIZ8				0x40200A10
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ8_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.224 DIEPDMA8	0x40200A14

Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.225 DTXFSTS8				0x40200A18
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS8_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.226 DIEPDMAB8				0x40200A1C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.227 DIEPCTL9				0x40200A20
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL9_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL9_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.228 DIEPINT9				0x40200A28
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT9_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT9_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.229 DIEPTSIZ9				0x40200A30
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	

30 : 29	MC	rw	0x0	
31	DIEPTSIZ9_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.230 DIEPDMA9				0x40200A34
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.231 DTXFSTS9				0x40200A38
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS9_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.232 DIEPDMAB9				0x40200A3C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.233 DIEPCTL10				0x40200A40
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL10_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL10_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.234 DIEPINT10				0x40200A48
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT10_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT10_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.235 DIEPTSIZ10				0x40200A50
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ10_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.236 DIEPDMA10				0x40200A54
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.237 DTXFSTS10				0x40200A58
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS10_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.238 DIEPDMAB10				0x40200A5C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.239 DIEPCTL11				0x40200A60
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL11_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL11_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.240 DIEPINT11				0x40200A68
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	
8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT11_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.

13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT11_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.241 DIEPTSIZ11				0x40200A70
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ11_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.242 DIEPDMA11				0x40200A74
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.243 DTXFSTS11				0x40200A78
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS11_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.244 DIEPDMAB11				0x40200A7C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.245 DIEPCTL12				0x40200A80
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
14 : 11	DIEPCTL12_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	DIEPCTL12_reserved_1	rc	0x0	bit type is changed from dc to rc.
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	TxFNum	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.246 DIEPINT12				0x40200A88
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	TimeOUT	rc	0x0	bit type is changed from w1c to rc.
4	INTknTXFEmp	rc	0x0	bit type is changed from w1c to rc.
5	INTknEPMis	rc	0x0	bit type is changed from w1c to rc.
6	INEPNakEff	rc	0x0	bit type is changed from w1c to rc.
7	TxFEmp	ro	0x1	

8	TxfifoUndrn	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DIEPINT12_reserved_0	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
31 : 15	DIEPINT12_reserved_1	rc	0x0	bit type is changed from dc to rc.

40.247 DIEPTSIZ12				0x40200A90
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	MC	rw	0x0	
31	DIEPTSIZ12_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.248 DIEPDMA12				0x40200A94
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.249 DTXFSTS12				0x40200A98
Bits	Field Name	Acc	Default	Description
15 : 0	INEPTxFSpcAvail	ro	0xEAF	
31 : 16	DTXFSTS12_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.250 DIEPDMAB12				0x40200A9C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.251 DOEPCTL0				0x40200B00
Bits	Field Name	Acc	Default	Description
1 : 0	MPS	ro	0x0	
14 : 2	DOEPCTL0_reserved_0	rc	0x0	bit type is changed from dc to rc.
15	USBActEP	ro	0x1	
16	DOEPCTL0_reserved_1	rc	0x0	bit type is changed from dc to rc.
17	NAKSts	ro	0x0	
19 : 18	EPType	ro	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
25 : 22	DOEPCTL0_reserved_2	rc	0x0	bit type is changed from dc to rc.
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
29 : 28	DOEPCTL0_reserved_3	rc	0x0	bit type is changed from dc to rc.
30	EPDis	ro	0x0	
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.252 DOEPINT0				0x40200B08
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.

4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT0_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT0_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT0_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.253 DOEPTSIZ0				0x40200B10
Bits	Field Name	Acc	Default	Description
6 : 0	XferSize	rw	0x0	
18 : 7	DOEPTSIZ0_reserved_0	rc	0x0	bit type is changed from dc to rc.
19	PktCnt	rw	0x0	
28 : 20	DOEPTSIZ0_reserved_1	rc	0x0	bit type is changed from dc to rc.
30 : 29	SUPCnt	rw	0x0	
31	DOEPTSIZ0_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.254 DOEPDMA0				0x40200B14
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.255 DOEPDMAB0				0x40200B1C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.256 DOEPCTL1				0x40200B20
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.

28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.257 DOEPCTL1_1				0x40200B24
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.258 DOEPINT1				0x40200B28
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT1_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT1_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT1_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.259 DOEPTSIZ1				0x40200B30
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	

30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ1_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.260 DOEPDMA1				0x40200B34
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.261 DOEPDMAB1				0x40200B3C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.262 DOEPCTL2				0x40200B40
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.263 DOEPCTL2_1				0x40200B44
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.

23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.264 DOEPINT2				0x40200B48
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT2_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT2_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT2_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.265 DOEPTSIZ2				0x40200B50
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ2_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.266 DOEPDMA2				0x40200B54
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.267 DOEPDMAB2				0x40200B5C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.268 DOEPCTL3				0x40200B60
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	

20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.269 DOEPCTL3_1				0x40200B64
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.270 DOEPINT3				0x40200B68
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT3_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT3_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.

19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT3_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.271 DOEPTSIZ3				0x40200B70
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ3_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.272 DOEPDMA3				0x40200B74
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.273 DOEPDMAB3				0x40200B7C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.274 DOEPCTL4				0x40200B80
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.275 DOEPCTL4_1				0x40200B84
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not

5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.276 DOEPINT4				0x40200B88
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT4_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT4_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT4_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.277 DOEPTSIZ4				0x40200B90
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ4_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.278 DOEPDMA4				0x40200B94
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.279 DOEPDMAB4				0x40200B9C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.280 DOEPCTL5				0x40200BA0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	

11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.281 DOEPCTL5_1				0x40200BA4
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.282 DOEPINT5				0x40200BA8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT5_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT5_reserved_1	rc	0x0	bit type is changed from dc to rc.

11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT5_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.283 DOEPTSIZ5				0x40200BB0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ5_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.284 DOEPDMA5				0x40200BB4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.285 DOEPDMAB5				0x40200BBC
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.286 DOEPCTL6				0x40200BC0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.287 DOEPCTL6_1				0x40200BC4
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.

1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.288 DOEPINT6				0x40200BC8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT6_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT6_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT6_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.289 DOEPTSIZ6				0x40200BD0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ6_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.290 DOEPDMA6				0x40200BD4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.291 DOEPDMAB6				0x40200BDC
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.292 DOEPCTL7				0x40200BE0
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.293 DOEPCTL7_1				0x40200BE4
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.294 DOEPINT7				0x40200BE8
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.

3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT7_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT7_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT7_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.295 DOEPTSIZ7				0x40200BF0
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ7_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.296 DOEPDMA7				0x40200BF4
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.297 DOEPDMAB7				0x40200BFC
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.298 DOEPCTL8				0x40200C00
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.

29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.299 DOEPCTL8_1				0x40200C04
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.300 DOEPINT8				0x40200C08
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT8_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT8_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT8_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.301 DOEPTSIZ8				0x40200C10
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	

31	DOEPTSIZ8_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.302 DOEPDMA8				0x40200C14
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.303 DOEPDMAB8				0x40200C1C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.304 DOEPCTL9				0x40200C20
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.305 DOEPCTL9_1				0x40200C24
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak

31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.306 DOEPINT9				0x40200C28
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT9_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT9_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT9_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.307 DOEPTSIZ9				0x40200C30
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ9_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.308 DOEPDMA9				0x40200C34
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.309 DOEPDMAB9				0x40200C3C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.310 DOEPCTL10				0x40200C40
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	

21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.311 DOEPCTL10_1				0x40200C44
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.312 DOEPINT10				0x40200C48
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT10_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT10_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.

31 : 20	DOEPINT10_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.313 DOEPTSIZ10				0x40200C50
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ10_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.314 DOEPDMA10				0x40200C54
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.315 DOEPDMAB10				0x40200C5C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.316 DOEPCTL11				0x40200C60
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	
12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.317 DOEPCTL11_1				0x40200C64
Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, Hardware buf counter will be increased by DmaBufCnt
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged

7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.318 DOEPINT11				0x40200C68
Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT11_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT11_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.
12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT11_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.319 DOEPTSIZ11				0x40200C70
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ11_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.320 DOEPDMA11				0x40200C74
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.321 DOEPDMAB11				0x40200C7C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.322 DOEPCTL12				0x40200C80
Bits	Field Name	Acc	Default	Description
10 : 0	MPS	rw	0x0	
11	StreamLenChkEn	rw	0x0	

12	StreamEndChkEn	rw	0x0	
13	StreamHdrChkEn	rw	0x0	
14	XferCmpltIntDis	rw	0x0	
15	USBActEP	rw	0x0	
16	DPID	ro	0x0	
17	NAKSts	ro	0x0	
19 : 18	EPType	rw	0x0	
20	Snp	rw	0x0	
21	Stall	rw	0x0	bit type is changed from othr to rw.
22	StreamEn	rw	0x0	
23	XferCmpltCtrl	rw	0x0	
24	EPDisCtrl	rw	0x0	
25	StreamTimeOutEn	rw	0x0	
26	CNAK	rw	0x0	bit type is changed from w1 to rw.
27	SNAK	rw	0x0	bit type is changed from w1 to rw.
28	SetD0PID	rw	0x0	bit type is changed from w1 to rw.
29	SetD1PID	rw	0x0	bit type is changed from w1 to rw.
30	EPDis	rw	0x0	bit type is changed from othr to rw.
31	EPEna	rw	0x0	bit type is changed from othr to rw.

40.323 DOEPCTL12_1 0x40200C84

Bits	Field Name	Acc	Default	Description
0	EnNakAfterShort	rw	0x0	Enable Set Nak after receiving a Short packet.
1	EnAvoidDmaOvf	rw	0x0	Enable function to avoid dma buffer overflow
2	AutoSetNakEn	rw	0x0	Enable Nak Auto Setting when dma buf is not enough
3	DmaBufCntInit	wo	0x0	When Set 1, Hardware buf counter will be initiated by DmaBufCnt When Set 0, not
4	DmaBufCntIncr	wo	0x0	When Set 1 and DmaBufCntInit is set 0, Hardware buf counter will be increased by DmaBufCnt When Set 0, not
5	AutoSetNakSts	ro	0x0	1, Auto Nak has been trigged
7 : 6	DOEPINT1_1_reserved_0	rc	0x0	bit type is changed from dc to rc.
15 : 8	DmaBufCnt	wo	0x0	When DmaBufCntInit Set 1, Hardware buf counter will be initiated by DmaBufCnt When DmaBufCntInit Set 0, Hardware buf counter will be increased by DmaBufCnt
16	DmaBufCntSign	ro	0x0	1, Hardware buf counter is minus 0, Hardware buf counter is positive
17	DOEPINT1_1_reserved_1	rc	0x0	bit type is changed from dc to rc.
23 : 18	DmaBufTh	rw	0x0	Dma Buf counter Threshold for trigger auto Nak
31 : 24	DmaBufIndex	rw	0x0	Only used for software to storing dma desc index

40.324 DOEPINT12 0x40200C88

Bits	Field Name	Acc	Default	Description
0	XferCompl	rc	0x0	bit type is changed from w1c to rc.
1	EPDisbld	rc	0x0	bit type is changed from w1c to rc.
2	AHBErr	rc	0x0	bit type is changed from w1c to rc.
3	SetUp	rc	0x0	bit type is changed from w1c to rc.
4	OUTTknEPdis	rc	0x0	bit type is changed from w1c to rc.
5	StsPhseRcvd	rc	0x0	bit type is changed from w1c to rc.
6	Back2BackSETup	rc	0x0	bit type is changed from w1c to rc.
7	DOEPINT12_reserved_0	rc	0x0	bit type is changed from dc to rc.
8	OutPktErr	rc	0x0	bit type is changed from w1c to rc.
9	BNAIntr	rc	0x0	bit type is changed from w1c to rc.
10	DOEPINT12_reserved_1	rc	0x0	bit type is changed from dc to rc.
11	PktDrpSts	rc	0x0	bit type is changed from w1c to rc.

12	BbleErr	rc	0x0	bit type is changed from w1c to rc.
13	NAKIntrpt	rc	0x0	bit type is changed from w1c to rc.
14	NYETIntrpt	rc	0x0	bit type is changed from w1c to rc.
15	StupPktRcvd	rc	0x0	bit type is changed from w1c to rc.
16	StreamTimeOut	rc	0x0	bit type is changed from w1c to rc.
17	StreamHdrErr	rc	0x0	bit type is changed from w1c to rc.
18	StreamEndErr	rc	0x0	bit type is changed from w1c to rc.
19	StreamLenIllegal	rc	0x0	bit type is changed from w1c to rc.
31 : 20	DOEPINT12_reserved_2	rc	0x0	bit type is changed from dc to rc.

40.325 DOEPTSIZ12				0x40200C90
Bits	Field Name	Acc	Default	Description
18 : 0	XferSize	rw	0x0	
28 : 19	PktCnt	rw	0x0	
30 : 29	RxDPID	ro	0x0	
31	DOEPTSIZ12_reserved_0	rc	0x0	bit type is changed from dc to rc.

40.326 DOEPDMA12				0x40200C94
Bits	Field Name	Acc	Default	Description
31 : 0	DMAAddr	rw	0x0	

40.327 DOEPDMAB12				0x40200C9C
Bits	Field Name	Acc	Default	Description
31 : 0	DMABufferAddr	rw	0x0	bit type is changed from ru to rw.

40.328 PCGCCTL				0x40200E00
Bits	Field Name	Acc	Default	Description
0	StopPclk	rw	0x0	
2 : 1	PCGCCTL_reserved_0	rc	0x0	bit type is changed from dc to rc.
3	RstPdwnModule	rw	0x0	
5 : 4	PCGCCTL_reserved_1	rc	0x0	bit type is changed from dc to rc.
6	PhySleep	ro	0x0	
7	L1Suspended	ro	0x0	
31 : 8	PCGCCTL_reserved_2	rc	0x0	bit type is changed from dc to rc.

41 mm_sys_ctrl

41.1 mem_cfg				0x40241000
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_1pram	rw	0x21C8	memory configration
31 : 16	cfg_mem_2prf	rw	0x20C6	memory configration

41.2 usb_cfg0				0x40241004
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_usb0	rw	0x220	

41.3 usb_cfg1				0x40241008
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_usb1	rw	0xFFFF03	

41.4 usb_cfg2				0x4024100C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_usb2	rw	0x0	

41.5 usb_tport_sel				0x40241010
Bits	Field Name	Acc	Default	Description
5 : 0	rg_usb_tport_sel	rw	0x3F	

41.6 usb_status				0x40241014
Bits	Field Name	Acc	Default	Description
0	ro_usb_ready	ro	0x0	
4	ro_usb_rxovf_int_raw_status	ro	0x0	
5	ro_usb_hseof_-err_int_raw_status	ro	0x0	
6	ro_usb_fseof_err_int_raw_status	ro	0x0	
7	ro_usb_busact-_err_int_raw_status	ro	0x0	
8	ro_usb_sync_-timeout_int_raw_status	ro	0x0	
12	ro_usb_rxovf_int_status	ro	0x0	
13	ro_usb_hseof_err_int_status	ro	0x0	
14	ro_usb_fseof_err_int_status	ro	0x0	
15	ro_usb_busact_err_int_status	ro	0x0	
16	ro_usb_sync_-timeout_int_status	ro	0x0	

41.7 usb_digphy_int_enable				0x40241018
Bits	Field Name	Acc	Default	Description
0	rg_usb_rxovf_int_enable	rw	0x0	
1	rg_usb_hseof_err_int_enable	rw	0x0	
2	rg_usb_fseof_err_int_enable	rw	0x0	
3	rg_usb_busact_err_int_enable	rw	0x0	
4	rg_usb_sync_-timeout_int_enable	rw	0x0	

41.8 usb_digphy_int_clr				0x4024101C
Bits	Field Name	Acc	Default	Description
0	rg_usb_rxovf_int_clr_tgl	rw	0x0	
1	rg_usb_hseof_err_int_clr_tgl	rw	0x0	
2	rg_usb_fseof_err_int_clr_tgl	rw	0x0	
3	rg_usb_busact_err_int_clr_tgl	rw	0x0	
4	rg_usb_sync_-timeout_int_clr_tgl	rw	0x0	

41.9 cfg_mm_sys_tports_sel				0x40241020
Bits	Field Name	Acc	Default	Description
1 : 0	cfg_mm_sys_tports_sel	rw	0x0	mm system test ports select
5 : 2	rg_host_dma_tports_sel	rw	0x0	host dma test ports select

41.10 mem_cfg1				0x40241024
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_1prf	rw	0x2488	

41.11 usb_prbs_cfg				0x40241028
Bits	Field Name	Acc	Default	Description
0	rg_usb_tx_prbs_mode	rw	0x0	
1	rg_usb_tx_prbs_load	rw	0x0	
8 : 2	rg_usb_tx_prbs_seed	rw	0x1	
9	rg_usb_rx_prbs_mode	rw	0x0	
13 : 10	rg_usb_rx_prbs_time	rw	0x0	
20 : 14	rg_usb_rx_err_tolerance	rw	0x0	

41.12 usb_prbs_status				0x4024102C
Bits	Field Name	Acc	Default	Description
0	ro_usb_rx_prbs_valid	ro	0x0	
1	ro_usb_rx_prbs_end	ro	0x0	
2	ro_usb_rx_prbs_pass	ro	0x0	
19 : 4	ro_usb_rx_prbs_errcnt	ro	0x0	

41.13 cfg_pu_usb				0x40241050
Bits	Field Name	Acc	Default	Description
0	rg_pu_usb_dr	rw	0x1	
1	rg_pu_usb	rw	0x0	
2	rg_pu_usb_ana_dr	rw	0x0	
3	rg_pu_usb_ana	rw	0x0	
4	rg_usb_dll_rst_dr	rw	0x0	
5	rg_usb_dll_rst	rw	0x0	
6	rg_usb_dll_clk_en_dr	rw	0x0	
7	rg_usb_dll_clk_en	rw	0x0	
8	rg_usb_dll_clk2dig_en_dr	rw	0x0	
9	rg_usb_dll_clk2dig_en	rw	0x0	
10	rg_usb_hs_bypass	rw	0x1	
15 : 11	rg_pu_usb_dll_us_x2	rw	0x10	
20 : 16	rg_usb_dll_rst_us_x2	rw	0x10	
26 : 21	rg_timer_0p5us_th	rw	0x27	
27	rg_sw_resetn_usbc_core	rw	0x1	
28	rg_sw_resetn_usbc_digphy	rw	0x1	
29	rg_usb_cdr_clk_dis	rw	0x0	
30	rg_bbpll_clk_usb_en_dr	rw	0x0	
31	rg_bbpll_clk_usb_en	rw	0x0	

41.14 usb_ana_cfg0				0x40241054
Bits	Field Name	Acc	Default	Description
0	rg_usb_JC_mode	rw	0x1	
1	rg_usb_zps_mode	rw	0x1	
2	rg_usb_rst_intp_mode	rw	0x0	
3	rg_usb_hold_last_mode	rw	0x0	
4	rg_usb_loop_back	rw	0x0	
5	rg_usb_sel_terr	rw	0x1	
6	rg_usb_sel_hs_gain	rw	0x0	
7	rg_usb_squach_det_mode	rw	0x0	
8	rg_usb_en_idcheck_sel	rw	0x0	
9	rg_usb_en_idcheck_reg	rw	0x0	
11 : 10	rg_usb_hs_clk_sel	rw	0x0	
14 : 12	rg_usb_iset_seq	rw	0x4	
18 : 16	rg_usb_iset_otg	rw	0x0	
22 : 20	rg_usb_iset_hs_disconnect	rw	0x4	
25 : 24	rg_usb_cdr_gn_	rw	0x1	

27 : 26	rg_usb_hs_idac_mode	rw	0x0	
31 : 28	rg_usb_hs_idac_bit	rw	0x7	

41.15 usb_ana_cfg1				0x40241058
Bits	Field Name	Acc	Default	Description
3 : 0	rg_usb_rterm_hs	rw	0x8	
8	rg_usb_dll_clk_sel_bbpll_pcie	rw	0x0	
17 : 12	rg_usb_en_otg	rw	0x0	
19 : 18	rg_usb_dll_pu_bit	rw	0x3	
21 : 20	rg_usb_fsls_drv_bit	rw	0x3	

41.16 host_dma_cfg				0x4024105C
Bits	Field Name	Acc	Default	Description
2 : 0	rg_host_dma_upstream_hburst	rw	0x1	
6 : 4	rg_host_dma_-downstream_hburst	rw	0x1	

42 host_dma

42.1 CH_LLI_ROOT0				0x40242000
Bits	Field Name	Acc	Default	Description
31 : 0	CH_LLI_ROOT0	rw	0x0	

42.2 CH_LLI_ROOT1				0x40242004
Bits	Field Name	Acc	Default	Description
31 : 0	CH_LLI_ROOT1	rw	0x0	

42.3 CH_LLI_ROOT2				0x40242008
Bits	Field Name	Acc	Default	Description
31 : 0	CH_LLI_ROOT2	rw	0x0	

42.4 CH_LLI_ROOT3				0x4024200C
Bits	Field Name	Acc	Default	Description
31 : 0	CH_LLI_ROOT3	rw	0x0	

42.5 DMA_STATUS				0x40242010
Bits	Field Name	Acc	Default	Description
15 : 0	OFT_FREE	ro	0x0	
16	CH0_STOPPED	ro	0x0	
17	CH1_STOPPED	ro	0x0	
18	CH2_STOPPED	ro	0x0	
19	CH3_STOPPED	ro	0x0	
22 : 20	REQUEST_STATE	ro	0x0	
23	ARB_A0_VALID	ro	0x0	
24	ARB_A1_VALID	ro	0x0	
25	ARB_A2_VALID	ro	0x0	
26	ARB_A3_VALID	ro	0x0	
27	ARB_A4_VALID	ro	0x0	
28	UP_STREAM_BSY	ro	0x0	
29	DOWN_STREAM_BSY	ro	0x0	
30	MID_STREAM_BSY	ro	0x0	

31	CH4_STOPPED	ro	0x0	

42.6 INT_RAWSTATUS				0x40242014
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_IRQ_RAWSTATUS	ro	0x0	
16	ERROR_RAWSTATUS	ro	0x0	
20	CH0_EOT_RAWSTATUS	ro	0x0	
21	CH1_EOT_RAWSTATUS	ro	0x0	
22	CH2_EOT_RAWSTATUS	ro	0x0	
23	CH3_EOT_RAWSTATUS	ro	0x0	
24	CH4_EOT_RAWSTATUS	ro	0x0	

42.7 INT_UNMASK_SET				0x40242018
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_IRQ_UNMASK_SET	wo	0x0	
16	ERROR_UNMASK_SET	wo	0x0	
20	CH0_EOT_UNMASK_SET	wo	0x0	
21	CH1_EOT_UNMASK_SET	wo	0x0	
22	CH2_EOT_UNMASK_SET	wo	0x0	
23	CH3_EOT_UNMASK_SET	wo	0x0	
24	CH4_EOT_UNMASK_SET	wo	0x0	

42.8 INT_UNMASK_CLEAR				0x4024201C
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_IRQ_UNMASK_CLEAR	wo	0x0	
16	ERROR_UNMASK_CLEAR	wo	0x0	
20	CH0_EOT_UNMAS-K_CLEAR	wo	0x0	
21	CH1_EOT_UNMAS-K_CLEAR	wo	0x0	
22	CH2_EOT_UNMAS-K_CLEAR	wo	0x0	
23	CH3_EOT_UNMAS-K_CLEAR	wo	0x0	
24	CH4_EOT_UNMAS-K_CLEAR	wo	0x0	

42.9 INT_ACK				0x40242020
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_IRQ_ACK	wo	0x0	
16	ERROR_ACK	wo	0x0	
20	CH0_EOT_ACK	wo	0x0	
21	CH1_EOT_ACK	wo	0x0	
22	CH2_EOT_ACK	wo	0x0	
23	CH3_EOT_ACK	wo	0x0	
24	CH4_EOT_ACK	wo	0x0	

42.10 INT_STATUS				0x40242024
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_IRQ_STATUS	wo	0x0	
16	ERROR_STATUS	wo	0x0	
20	CH0_EOT_STATUS	wo	0x0	
21	CH1_EOT_STATUS	wo	0x0	
22	CH2_EOT_STATUS	wo	0x0	

23	CH3_EOT_STATUS	wo	0x0	
24	CH4_EOT_STATUS	wo	0x0	

42.11 DMA_REVISION				0x40242030
Bits	Field Name	Acc	Default	Description
31 : 0	REVISION	ro	0x2	

42.12 ARBITRATION				0x40242034
Bits	Field Name	Acc	Default	Description
3 : 0	DOWNSTREAM_TAG-_USAGE	rw	0xC	

42.13 CHANNEL_MUTEX_SET				0x40242038
Bits	Field Name	Acc	Default	Description
0	CH0_MUTEX_SET	wo	0x0	
1	CH1_MUTEX_SET	wo	0x0	
2	CH2_MUTEX_SET	wo	0x0	
3	CH3_MUTEX_SET	wo	0x0	
4	CH4_MUTEX_SET	wo	0x0	

42.14 CHANNEL_MUTEX_CLEAR				0x4024203C
Bits	Field Name	Acc	Default	Description
0	CH0_MUTEX_CLEAR	wo	0x0	
1	CH1_MUTEX_CLEAR	wo	0x0	
2	CH2_MUTEX_CLEAR	wo	0x0	
3	CH3_MUTEX_CLEAR	wo	0x0	
4	CH4_MUTEX_CLEAR	wo	0x0	

42.15 CH_LLI_ROOT4				0x40242040
Bits	Field Name	Acc	Default	Description
31 : 0	CH_LLI_ROOT4	rw	0x0	

42.16 AXI_BURST_LENGTH_LIMIT				0x40242044
Bits	Field Name	Acc	Default	Description
3 : 0	AXI_BURST_LEN-GTH_LIMIT	rw	0x5	

42.17 LLI_COUNTER0				0x40242080
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER0	ro	0x0	

42.18 LLI_COUNTER1				0x40242084
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER1	ro	0x0	

42.19 LLI_COUNTER2				0x40242088
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER2	ro	0x0	

42.20 LLI_COUNTER3				0x4024208C
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER3	ro	0x0	

42.21 LLI_COUNTER4				0x40242090
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER4	ro	0x0	

42.22 LLI_COUNTER5				0x40242094
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER5	ro	0x0	

42.23 LLI_COUNTER6				0x40242098
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER6	ro	0x0	

42.24 LLI_COUNTER7				0x4024209C
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER7	ro	0x0	

42.25 LLI_COUNTER8				0x402420A0
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER8	ro	0x0	

42.26 LLI_COUNTER9				0x402420A4
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER9	ro	0x0	

42.27 LLI_COUNTER10				0x402420A8
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER10	ro	0x0	

42.28 LLI_COUNTER11				0x402420AC
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER11	ro	0x0	

42.29 LLI_COUNTER12				0x402420B0
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER12	ro	0x0	

42.30 LLI_COUNTER13				0x402420B4
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER13	ro	0x0	

42.31 LLI_COUNTER14				0x402420B8
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER14	ro	0x0	

42.32 LLI_COUNTER15				0x402420BC
Bits	Field Name	Acc	Default	Description
15 : 0	LLI_COUNTER15	ro	0x0	

42.33 CONVERSION_CONFIG0				0x402420E0
Bits	Field Name	Acc	Default	Description
10 : 0	CONV_OFFSET0	rw	0x0	
15	CONV_LLC_LEN_EN0	rw	0x0	
19 : 16	CONV_DATA_LENGTH0	rw	0x0	

42.34 CONVERSION_CONFIG1				0x402420E4
Bits	Field Name	Acc	Default	Description
10 : 0	CONV_OFFSET1	rw	0x0	
15	CONV_LLC_LEN_EN1	rw	0x0	
19 : 16	CONV_DATA_LENGTH1	rw	0x0	

42.35 CONVERSION_CONFIG2				0x402420E8
Bits	Field Name	Acc	Default	Description
10 : 0	CONV_OFFSET2	rw	0x0	
15	CONV_LLC_LEN_EN2	rw	0x0	
19 : 16	CONV_DATA_LENGTH2	rw	0x0	

42.36 CONVERSION_CONFIG3				0x402420EC
Bits	Field Name	Acc	Default	Description
10 : 0	CONV_OFFSET3	rw	0x0	
15	CONV_LLC_LEN_EN3	rw	0x0	
19 : 16	CONV_DATA_LENGTH3	rw	0x0	

42.37 CONVERSION_DATA00				0x40242100
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA00	rw	0x0	

42.38 CONVERSION_DATA01				0x40242104
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA01	rw	0x0	

42.39 CONVERSION_DATA10				0x40242110
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA10	rw	0x0	

42.40 CONVERSION_DATA11				0x40242114
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA11	rw	0x0	

42.41 CONVERSION_DATA20				0x40242120
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA20	rw	0x0	

42.42 CONVERSION_DATA21				0x40242124
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA21	rw	0x0	

42.43 CONVERSION_DATA30				0x40242130
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA30	rw	0x0	

42.44 CONVERSION_DATA31				0x40242134
Bits	Field Name	Acc	Default	Description
31 : 0	CONVERSION_DATA31	rw	0x0	

43 dfe_reg_bank

43.1 a0x000				0x40342000
Bits	Field Name	Acc	Default	Description
9 : 0	stop_delay_reg___	rw	0x199	
11 : 10	dump_size_____	rw	0x0	
15 : 12	dump_start_wr_addr	rw	0x0	
19 : 16	dump_data_sel	rw	0x1	
23 : 20	stop_trig_sel	rw	0x0	
27 : 24	mon_i_sel	rw	0x0	
31 : 28	mon_q_sel	rw	0x0	

43.2 a0x004				0x40342004
Bits	Field Name	Acc	Default	Description
0	dump_enable_____	rw	0x0	
3	manual_stop_reg___	rw	0x0	
4	dump_wrap_mode_n___	rw	0x0	
6 : 5	dump_sampl_rate_sel	rw	0x1	
7	restop_trig_mode__	rw	0x0	
11 : 8	dump_ram_en____	rw	0x0	
12	dump_lane_mode	rw	0x0	
13	dump_rst_reg	rw	0x0	
24	tx_dump_sel	rw	0x0	
31	dump_clk_sel_	rw	0x0	

43.3 a0x008				0x40342008
Bits	Field Name	Acc	Default	Description
0	imbala_cancel_bypass	rw	0x1	
1	imbala_calc_rels	rw	0x0	
2	imbala_cal_dr	rw	0x0	
3	imbala_cancel_hold_dr	rw	0x0	
7 : 4	imbala_cancel_bw_ct_wa	rw	0x3	
11 : 8	imbala_cancel_bw_ct_wq	rw	0x4	
15 : 12	rx_iqim_mem_addr_reg	rw	0x0	
16	rx_iqim_mem_addr_dr	rw	0x0	
17	rx_iqim_mem_wen0	rw	0x0	
18	rx_iqim_mem_cen0	rw	0x1	
19	dac_iq_80m_swap	rw	0x0	
20	adc_i_clkedge_sel	rw	0x0	
21	adc_q_clkedge_sel	rw	0x0	

22	adc_i_dly	rw	0x0	
23	adc_q_dly	rw	0x0	
24	adc_iq_format	rw	0x0	
25	adc_iq_swap	rw	0x0	
26	adc_i_negate	rw	0x0	
27	adc_q_negate	rw	0x0	
28	dac_iq_format	rw	0x0	
29	dac_iq_swap	rw	0x0	
30	dac_i_negate	rw	0x0	
31	dac_q_negate	rw	0x0	

43.4 a0x00c				0x4034200C
Bits	Field Name	Acc	Default	Description
30 : 0	rx_iqim_mem_din0	rw	0x0	

43.5 a0x010				0x40342010
Bits	Field Name	Acc	Default	Description
30 : 0	sdm_freq_reg	rw	0x4051EB8	

43.6 a0x014				0x40342014
Bits	Field Name	Acc	Default	Description
0	vco_cal_resetn_dr	rw	0x0	
1	vco_cal_resetn_reg	rw	0x0	
2	pll_cal_resetn_dr	rw	0x0	
3	pll_cal_resetn_reg	rw	0x0	
4	timer_sdm_resetn_dr	rw	0x0	
5	timer_sdm_resetn_reg	rw	0x0	
6	vco_aac_polarity	rw	0x1	
7	vco_aac_freq_chg	rw	0x0	
9 : 8	reset_pll_lock_delay	rw	0x0	
11 : 10	vco_band_mode	rw	0x0	
12	pll_cal_freq_chg	rw	0x0	
13	pll_cal_eachtime	rw	0x1	
14	pll_cal_det_en	rw	0x0	
15	pll_recal_en	rw	0x0	
16	pll_cal_opt	rw	0x0	
17	pll_cal_manual	rw	0x0	
18	pll_cal_cnt_en_sel	rw	0x0	
19	pll_cal_cnt_up	rw	0x0	
24 : 20	pll_cal_lock_delay	rw	0x0	
27 : 25	pll_cal_init_delay	rw	0x3	
30 : 28	pll_cal_wait_time	rw	0x0	
31	sdm_freq_dr	rw	0x0	

43.7 a0x018				0x40342018
Bits	Field Name	Acc	Default	Description
3 : 0	reset_pll_sdm_delay	rw	0x1	
4	pll_sdm_resetn_dr	rw	0x0	
5	pll_sdm_resetn_reg	rw	0x0	
7 : 6	pll_sdm_clk_sel_reg	rw	0x1	
17 : 8	pll_cal_each_cnt_time_reg	rw	0x28	
18	pll_cal_each_cnt_time_dr	rw	0x1	
20 : 19	pll_cal_eachbit_time	rw	0x1	
21	vco_aac_en_dr	rw	0x0	

22	vco_aac_en_reg	rw	0x0	
23	vco_reg_vbit_dr	rw	0x0	
27 : 24	vco_reg_vbit_reg	rw	0x0	
28	pll_cal_en_dr	rw	0x0	
29	pll_cal_en_reg	rw	0x0	
30	pll_vco_band_dr	rw	0x0	

43.8 a0x01c				0x4034201C
Bits	Field Name	Acc	Default	Description
12 : 0	pll_vco_band_reg	rw	0x0	
29 : 13	pll_cal_freq_reg	rw	0x3C5	
30	pll_cal_freq_dr	rw	0x1	
31	pll_freq_change_bt_en	rw	0x0	

43.9 a0x020				0x40342020
Bits	Field Name	Acc	Default	Description
11 : 0	idac_data_reg	rw	0x0	
23 : 12	idac_data_reg_bt	rw	0x0	
27	tx_dsp_ramp_dr	rw	0x0	
28	tx_dsp_ramp_reg	rw	0x0	
29	tx_dsp_ramp_sel	rw	0x0	
31 : 30	idac_data_dr	rw	0x1	

43.10 a0x024				0x40342024
Bits	Field Name	Acc	Default	Description
5 : 0	ramp_speed	rw	0x10	
9 : 6	ramp_dly	rw	0x0	
14 : 10	ramp_init	rw	0x0	
15	ramp_bypass	rw	0x1	
21 : 16	gain_ct_u	rw	0x0	
26 : 24	ampm_calib_lpf_bw	rw	0x5	
28 : 27	ampm_calib_en	rw	0x0	
29	ampm_calib_sel_pm	rw	0x0	
30	ampm_calib_out_lock	rw	0x0	

43.11 a0x028				0x40342028
Bits	Field Name	Acc	Default	Description
11 : 0	ref_amp	rw	0x0	
23 : 12	amp_offset	rw	0x0	
28 : 24	tx_iq_dly_sel	rw	0x0	
31 : 29	lpf_ct_u	rw	0x0	

43.12 a0x02c				0x4034202C
Bits	Field Name	Acc	Default	Description
7 : 0	dfe_clken_cfg1	rw	0x0	
8	aux_adc_i_clkedge_sel	rw	0x1	
9	aux_adc_q_clkedge_sel	rw	0x1	
10	aux_adc_iq_format	rw	0x0	
11	aux_adc_iq_swap	rw	0x0	
15 : 12	aux_fifo_din_sel___	rw	0x0	
16	tx_cal_en_____	rw	0x0	
17	rx_cal_en_____	rw	0x0	
18	gain_index_test_sel	rw	0x0	

19	rx_sel_aux_adc	rw	0x0	
20	aux_fifo_wr_sel	rw	0x0	
22	dac_iq45_swap	rw	0x0	
23	dac_iq45_no_rot	rw	0x0	
29 : 28	pre_gain	rw	0x0	
30	pre_gain_en	rw	0x0	
31	regb_fmetre_start	rw	0x0	

43.13 a0x030				0x40342030
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_clken_cfg	rw	0x1B7E7	

43.14 a0x034				0x40342034
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_clken_cfg2	rw	0x0	

43.15 a0x038				0x40342038
Bits	Field Name	Acc	Default	Description
3 : 0	plllock_mon_reset_delay	rw	0xF	
7 : 4	pll_sdm_clksel_delay	rw	0x4	
8	dc_scale_factor_bit3	rw	0x0	
10 : 9	dc_reg_wr_bb_ctrl	rw	0x3	
11	dc_rf_update_en	rw	0x0	
12	dcc_bypass	rw	0x0	

43.16 a0x03c				0x4034203C
Bits	Field Name	Acc	Default	Description
4 : 0	dc_send_period	rw	0xE	
14 : 5	dc_update_thd	rw	0x3	
27 : 15	dc_dir_thd	rw	0x14	
31 : 28	dc_rf_update_ctrl	rw	0x1	

43.17 a0x04c				0x4034204C
Bits	Field Name	Acc	Default	Description
9 : 0	rc_interval_offset_n	rw	0x3F0	
25 : 16	rc_interval_offset_p	rw	0x10	

43.18 a0x050				0x40342050
Bits	Field Name	Acc	Default	Description
0	rc_mode_clk_adc	rw	0x1	
1	rc_core_bypass	rw	0x0	
2	rc_iqm_bypass	rw	0x0	
3	rc_hbf_bypass	rw	0x0	
4	rc_bypass_rx	rw	0x0	
5	rc_bypass_tx	rw	0x0	
6	bypass_intf_for_agc	rw	0x0	
7	rc_fifo_wr_en_ctrl	rw	0x1	
11 : 8	rc_fifo_empty_threshold	rw	0x5	
15 : 12	rc_fifo_full_threshold	rw	0xB	
19 : 16	rc_fifo_rd_en_threshold	rw	0x4	
21 : 20	rc_interval_dr_tx	rw	0x2	
23 : 22	rc_interval_dr_rx	rw	0x2	

43.19 a0x054				0x40342054
Bits	Field Name	Acc	Default	Description
27 : 0	rc_interval_reg_tx	rw	0x21F6A8F	
28	interval_auto_sel	rw	0x1	

43.20 a0x058				0x40342058
Bits	Field Name	Acc	Default	Description
27 : 0	rc_interval_reg_rx	rw	0x20B4807	

43.21 a0x05c				0x4034205C
Bits	Field Name	Acc	Default	Description
0	dc_calib_en	rw	0x0	
1	dc_cancel_en	rw	0x0	
2	dc_reg_update_en	rw	0x1	
3	dc_est_mode	rw	0x0	
6 : 4	dc_reg_wr_ctrl	rw	0x3	
7	dc_reg_set_en	rw	0x0	
10 : 8	dc_alpha_rst_ctrl	rw	0x6	
14 : 12	dc_avg_len_coef	rw	0x3	
15	dcc_fil_bypass	rw	0x1	
23 : 16	dc_est_wait_time	rw	0x8	
26 : 24	dcc_mode_ct	rw	0x0	
29 : 27	dc_scale_factor	rw	0x3	
31 : 30	dcc_bw_ct2	rw	0x1	

43.22 a0x060				0x40342060
Bits	Field Name	Acc	Default	Description
11 : 0	aux_adc_dc_set_q	rw	0x0	
15 : 12	dc_reg_index	rw	0x0	
27 : 16	aux_adc_dc_set_i	rw	0x0	
29 : 28	dc_reg_rd_sel	rw	0x0	
30	aux_adc_dccancel_en	rw	0x0	
31	agc_dccancel_en	rw	0x0	

43.23 a0x064				0x40342064
Bits	Field Name	Acc	Default	Description
15 : 0	dcc_bw_ct	rw	0xA008	
23 : 16	dcc_tmr_ct	rw	0x30	
30 : 24	dcc_tmr_ct2	rw	0x10	
31	dc_reg_set_dr	rw	0x0	

43.24 a0x068				0x40342068
Bits	Field Name	Acc	Default	Description
13 : 1	dc_reg_set_q	rw	0x0	
29 : 17	dc_reg_set_i	rw	0x0	

43.25 a0x06c				0x4034206C
Bits	Field Name	Acc	Default	Description
11 : 0	amp_tone1	rw	0x500	
25 : 12	tone_freq_num1	rw	0xCCD	
27 : 26	tone_typ	rw	0x0	
28	tx_tone_sel	rw	0x0	

31 : 30	slope_mode	rw	0x0	

43.26 a0x070				0x40342070
Bits	Field Name	Acc	Default	Description
11 : 0	amp_tone2	rw	0xFFC	
25 : 12	tone_freq_num2	rw	0x99A	
26	repeat_mode	rw	0x1	
29 : 28	dac_i_dly_sel	rw	0x0	
31 : 30	dac_q_dly_sel	rw	0x0	

43.27 a0x074				0x40342074
Bits	Field Name	Acc	Default	Description
3 : 0	t_low_amp	rw	0xF	
7 : 4	t_high_amp	rw	0xF	
20	angle_init_set	rw	0x0	
21	tone_com_sel	rw	0x0	
22	tx_tone_150m_sel	rw	0x0	

43.28 a0x078				0x40342078
Bits	Field Name	Acc	Default	Description
11 : 0	coef0_imag_rx	rw	0x0	
23 : 12	coef0_real_rx	rw	0x0	
27 : 24	coef0_k_rx	rw	0x0	
28	notch_mux_reg	rw	0x0	
29	notch_mux_dr	rw	0x0	
30	bypass_rx_notch0	rw	0x1	
31	bypass_tx_notch0	rw	0x1	

43.29 a0x07c				0x4034207C
Bits	Field Name	Acc	Default	Description
11 : 0	coef1_imag_rx	rw	0x0	
23 : 12	coef1_real_rx	rw	0x0	
27 : 24	coef1_k_rx	rw	0x0	
30	bypass_rx_notch1	rw	0x1	
31	bypass_tx_notch1	rw	0x1	

43.30 a0x080				0x40342080
Bits	Field Name	Acc	Default	Description
3 : 2	am_shift	rw	0x0	
5 : 4	am_p_shift	rw	0x1	
10 : 8	pm_shift	rw	0x0	
11	ampm_interval_sel_select	rw	0x0	

43.31 a0x084				0x40342084
Bits	Field Name	Acc	Default	Description
11 : 0	coef0_imag_tx	rw	0x0	
23 : 12	coef0_real_tx	rw	0x0	
27 : 24	coef0_k_tx	rw	0x0	

43.32 a0x088				0x40342088
Bits	Field Name	Acc	Default	Description

11 : 0	coef1_imag_tx	rw	0x0	
23 : 12	coef1_real_tx	rw	0x0	
27 : 24	coef1_k_tx	rw	0x0	

43.33 a0x0c8 — 0x403420C8

Bits	Field Name	Acc	Default	Description
0	loft_calib_en	rw	0x0	
1	loft_cancel_bypass	rw	0x1	
2	lpf_bypass	rw	0x0	
3	bpf_bypass	rw	0x0	
6 : 4	pwr_lpf_ct	rw	0x5	
13 : 8	gain1_ct	rw	0x0	
14	gain1_ct_sel	rw	0x0	
15	gain2_ct_sel	rw	0x0	
21 : 16	gain2_ct	rw	0x0	
26 : 24	rfgain_setting_time_ctrl	rw	0x5	
30 : 28	fdb_ring_delay_ctrl	rw	0x2	

43.34 a0x0cc — 0x403420CC

Bits	Field Name	Acc	Default	Description
5 : 0	gain1_dyna_p1	rw	0x0	
13 : 8	gain1_dyna_p2	rw	0x0	
21 : 16	gain1_dyna_p3	rw	0x0	
29 : 24	gain1_dyna_p4	rw	0x0	

43.35 a0x0d0 — 0x403420D0

Bits	Field Name	Acc	Default	Description
5 : 0	gain2_dyna_p1	rw	0x0	
13 : 8	gain2_dyna_p2	rw	0x0	
21 : 16	gain2_dyna_p3	rw	0x0	
29 : 24	gain2_dyna_p4	rw	0x0	

43.36 a0x0d4 — 0x403420D4

Bits	Field Name	Acc	Default	Description
5 : 0	gain2_dyna_p5	rw	0x0	
6	din_loft_sel	rw	0x0	
7	bpf_din_sel	rw	0x0	
11 : 8	loft_bit_ctrl	rw	0x0	
14 : 12	phase_bit_ctrl	rw	0x0	
18 : 16	amp_bit_ctrl	rw	0x0	
19	pwr_sel_abs	rw	0x0	
20	phase_err_dr	rw	0x0	
21	amp_err_dr	rw	0x0	
22	loft_offset_dr	rw	0x0	
23	loft_offset_i_dr	rw	0x0	
24	loft_calib_set_dr	rw	0x0	
25	loft_calib_set_reg	rw	0x0	
26	pwr_out_lock	rw	0x0	
27	loft_calib_reg_clear	rw	0x0	
29 : 28	loft_calib_sel	rw	0x0	
30	loft_calib_singl_mode	rw	0x0	

43.37 a0x0d8 — 0x403420D8

Bits	Field Name	Acc	Default	Description
15 : 12	gain_index_ph	rw	0x0	
31 : 28	gain_index_amp	rw	0x0	

43.38 a0x0dc				0x403420DC
Bits	Field Name	Acc	Default	Description
15 : 12	gain_index_loft_i	rw	0x0	
31 : 28	gain_index_loft_q	rw	0x0	

43.39 a0x0e0				0x403420E0
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_iqim_coef_a11	rw	0x37A4	
29 : 16	bpf_iqim_coef_a12	rw	0xAB8	

43.40 a0x0e4				0x403420E4
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_iqim_coef_a21	rw	0x3180	
29 : 16	bpf_iqim_coef_a22	rw	0xE52	

43.41 a0x0e8				0x403420E8
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_iqim_coef_a31	rw	0x3C34	
29 : 16	bpf_iqim_coef_a32	rw	0xE1E	

43.42 a0x0ec				0x403420EC
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_iqim_coef_a41	rw	0x3456	
29 : 16	bpf_iqim_coef_a42	rw	0xBCA	

43.43 a0x0f0				0x403420F0
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_iqim_coef_a51	rw	0x3A93	
29 : 16	bpf_iqim_coef_a52	rw	0xB7F	

43.44 a0x0f4				0x403420F4
Bits	Field Name	Acc	Default	Description
3 : 0	bpf_iqim_coef_g1	rw	0x3	
7 : 4	bpf_iqim_coef_g2	rw	0x2	
11 : 8	bpf_iqim_coef_g3	rw	0x5	
15 : 12	bpf_iqim_coef_g4	rw	0x3	
19 : 16	bpf_iqim_coef_g5	rw	0x4	
23 : 20	bpf_loleakage_coef_g1	rw	0x2	
27 : 24	bpf_loleakage_coef_g2	rw	0x4	
31 : 28	bpf_loleakage_coef_g3	rw	0x3	

43.45 a0x104				0x40342104
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_loleakage_coef_a11	rw	0x2595	
29 : 16	bpf_loleakage_coef_a12	rw	0xE1B	

43.46 a0x108				0x40342108
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_loleakage_coef_a21	rw	0x2B09	
29 : 16	bpf_loleakage_coef_a22	rw	0xD50	

43.47 a0x10c				0x4034210C
Bits	Field Name	Acc	Default	Description
13 : 0	bpf_loleakage_coef_a31	rw	0x2991	
29 : 16	bpf_loleakage_coef_a32	rw	0xBA5	

43.48 a0x124				0x40342124
Bits	Field Name	Acc	Default	Description
11 : 0	dlpf_det_fine1_thresh	rw	0x1E0	
23 : 12	dlpf_det_fine2_thresh	rw	0x1E0	
30 : 24	gro_mode2_digital_offset	rw	0x0	

43.49 a0x128				0x40342128
Bits	Field Name	Acc	Default	Description
0	dlpf_restart	rw	0x0	
1	dlpf_freq_mode	rw	0x1	
3 : 2	dlpf_mode	rw	0x0	
4	dlpf_coarse_bypass_lpf1	rw	0x1	
5	dlpf_bypass_sdm	rw	0x0	
6	gro_mode_sel	rw	0x0	
7	gro_sample_clkedge_sel	rw	0x0	
9 : 8	dlpf_sinc_order	rw	0x3	
11 : 10	dlpf_sinc_order1	rw	0x3	
13 : 12	cors_avg_num	rw	0x1	
14	cors_avg_bps	rw	0x0	
15	dlpf_det_bypass	rw	0x0	
27 : 16	dlpf_det_coarse_thresh	rw	0x1E0	

43.50 a0x12c				0x4034212C
Bits	Field Name	Acc	Default	Description
15 : 0	dlpf_phase_offset	rw	0xC8	

43.51 a0x130				0x40342130
Bits	Field Name	Acc	Default	Description
11 : 0	dlpf_coarse_code_initial	rw	0x800	
28 : 16	dlpf_dco_code_initial	rw	0x1000	

43.52 a0x134				0x40342134
Bits	Field Name	Acc	Default	Description
1 : 0	dlpf_k1_idx	rw	0x2	
3 : 2	dlpf_k2_idx	rw	0x2	
5 : 4	dlpf_k5_idx	rw	0x2	
7 : 6	dlpf_k6_idx	rw	0x2	
10 : 8	dlpf_k3_idx_fine1	rw	0x2	
14 : 12	dlpf_k4_idx_fine1	rw	0x2	
18 : 16	dlpf_k3_idx_fine2	rw	0x2	
22 : 20	dlpf_k4_idx_fine2	rw	0x2	
26 : 24	dlpf_k3_idx_coarse	rw	0x2	

30 : 28	dlpf_k4_idx_coarse	rw	0x2	

43.53 a0x138				0x40342138
Bits	Field Name	Acc	Default	Description
9 : 0	dlpf_cnt_fine1	rw	0x1F4	
21 : 12	dlpf_cnt_fine2	rw	0x1F4	

43.54 a0x13c				0x4034213C
Bits	Field Name	Acc	Default	Description
9 : 0	dlpf_cycle_slip_fine_code	rw	0x0	
21 : 12	dlpf_cycle_slip_coarse_code	rw	0x3C	
25 : 24	dlpf_kp_idx_fine1	rw	0x1	
27 : 26	dlpf_kp_idx_fine2	rw	0x1	
29 : 28	dlpf_kp_idx_coarse	rw	0x1	
30	dlpf_cycle_slip_coarse_en	rw	0x1	
31	dlpf_cycle_slip_fine_en	rw	0x0	

43.55 a0x140				0x40342140
Bits	Field Name	Acc	Default	Description
3 : 0	mon_sel_dlpf	rw	0x0	
4	mon_clk_sel	rw	0x0	
5	mon_lock_en	rw	0x0	
7	dlpf_coarse_code_dr	rw	0x0	
19 : 8	dlpf_coarse_code_dr_reg	rw	0x800	
29 : 20	dlpf_cnt_coarse	rw	0x190	

43.56 a0x144				0x40342144
Bits	Field Name	Acc	Default	Description
15 : 0	txon_start	rw	0x20	
31 : 16	txon_end	rw	0x800	

43.57 a0x148				0x40342148
Bits	Field Name	Acc	Default	Description
1 : 0	feedback_pow_en_reg	rw	0x0	
2	is_maxpow_dr	rw	0x1	
3	is_maxpow_reg	rw	0x0	
7 : 4	maxpow_thr	rw	0xF	
11 : 8	txgain_cmp_reg	rw	0x8	
12	txgain_cmp_dr	rw	0x1	
19 : 16	sw_cfg_pow	rw	0x8	
23 : 20	notch_bypass_thr	rw	0xF	

43.58 a0x14c				0x4034214C
Bits	Field Name	Acc	Default	Description
31 : 0	pow_acc_upthr	rw	0x0	

43.59 a0x150				0x40342150
Bits	Field Name	Acc	Default	Description
31 : 0	pow_acc_lowthr	rw	0x0	

43.60 a0x15c				0x4034215C
Bits	Field Name	Acc	Default	Description
9 : 0	cfg_a	rw	0x1D8	
22 : 12	cfg_b	rw	0x7B3	
27 : 24	cfg_et_cmp	rw	0x8	
31 : 28	et_bypass_thr	rw	0xF	

43.61 a0x160				0x40342160
Bits	Field Name	Acc	Default	Description
8 : 0	cfg_v0	rw	0xE6	
14 : 12	cfg_v1	rw	0x3	
15	et_max_sel	rw	0x0	
24 : 16	cfg_c	rw	0xD7	

43.62 a0x164				0x40342164
Bits	Field Name	Acc	Default	Description
0	generator_mem_hw_access	rw	0x0	
1	generator_mem_rd_only_once	rw	0x0	
2	generator_mem_hw_rden	rw	0x0	
4	generator_mem_clk_sel	rw	0x0	
6 : 5	generator_mem_en	rw	0x0	
11 : 8	generator_addr_offset	rw	0x0	
30 : 16	generator_mem_end_addr	rw	0x0	

43.63 a0x168				0x40342168
Bits	Field Name	Acc	Default	Description
14 : 0	generator_mem_repeat_addr1	rw	0x0	
30 : 16	generator_mem_repeat_addr0	rw	0x0	

43.64 a0x16c				0x4034216C
Bits	Field Name	Acc	Default	Description
7 : 0	generator_mem_repeat_num	rw	0x0	
15 : 8	generator_mem_tx_ramp_time	rw	0x0	
23 : 16	generator_mem_interval_time	rw	0x0	
31 : 24	num_1us_in_tx_clk	rw	0x0	

43.65 a0x170				0x40342170
Bits	Field Name	Acc	Default	Description
2	papr_lpf_bypass	rw	0x0	
7 : 4	dly_ct_din	rw	0x0	
11 : 8	dly_ct_clipflg	rw	0x0	
12	papr_lpf_coef_sel	rw	0x0	
13	clip_thres_sel_11ax	rw	0x0	
15 : 14	is_software_ctrl	rw	0x0	
22	tx_data_select	rw	0x1	
31 : 24	generator_mem_rd_pkt_num	rw	0x0	

43.66 a0x180				0x40342180
Bits	Field Name	Acc	Default	Description
0	pll_freq_change_dr	rw	0x0	
1	dc_est_restart_dr	rw	0x0	
10 : 7	tpd_sel_dpll	rw	0x0	

15 : 11	mon_port_sel	rw	0x0	
23 : 16	tpm_dac	rw	0x0	
26 : 24	mon_to_dac_sel	rw	0x0	

43.67 a0x184				0x40342184
Bits	Field Name	Acc	Default	Description
0	pll_freq_change_reg	rw	0x0	WT Reg.
1	dc_est_restart_reg	rw	0x0	WT Reg. (normal regs, do not use this addr)

43.68 iqim_estg_ctrl				0x403421A0
Bits	Field Name	Acc	Default	Description
0	iqim_estg_en0	rw	0x0	
1	iqim_estg_en1	rw	0x0	
2	iqim_estg_ok0	ro	0x0	
3	iqim_estg_ok1	ro	0x0	
6 : 4	iqim_estg_mode	rw	0x7	
9 : 8	iqim_estg_amp_max	rw	0x0	
13 : 12	iqim_estg_amp_min	rw	0x2	
19 : 16	iqim_estg_len_phi	rw	0x4	
23 : 20	iqim_estg_len_amp	rw	0x4	
27 : 24	iqim_estg_len_dc	rw	0x4	
28	rstn_iqim_estg_en	rw	0x1	
29	clk_iqim_estg_en	rw	0x0	

43.69 iqim_estg_phi				0x403421A4
Bits	Field Name	Acc	Default	Description
13 : 0	oiqim_estg_phi0	ro	0x0	
29 : 16	oiqim_estg_phi1	ro	0x0	

43.70 iqim_estg_idc0				0x403421B4
Bits	Field Name	Acc	Default	Description
13 : 0	iqim_estg_dcq0	rw	0x0	
29 : 16	iqim_estg_dci0	rw	0x0	

43.71 iqim_estg_odc0				0x403421B8
Bits	Field Name	Acc	Default	Description
13 : 0	oiqim_estg_dcq0	ro	0x0	
29 : 16	oiqim_estg_dci0	ro	0x0	

43.72 iqim_estg_oam0				0x403421BC
Bits	Field Name	Acc	Default	Description
13 : 0	oiqim_estg_amq0	ro	0x0	
29 : 16	oiqim_estg_ami0	ro	0x0	

43.73 iqim_estg_idc1				0x403421C4
Bits	Field Name	Acc	Default	Description
13 : 0	iqim_estg_dcq1	rw	0x0	
29 : 16	iqim_estg_dci1	rw	0x0	

43.74 iqim_estg_odc1	0x403421C8

Bits	Field Name	Acc	Default	Description
13 : 0	oiqim_estg_dcq1	ro	0x0	
29 : 16	oiqim_estg_dci1	ro	0x0	

43.75 iqim_estg_oam1				0x403421CC
Bits	Field Name	Acc	Default	Description
13 : 0	oiqim_estg_amq1	ro	0x0	
29 : 16	oiqim_estg_ami1	ro	0x0	

43.76 wdfs_ctrl_1d0				0x403421D0
Bits	Field Name	Acc	Default	Description
23 : 0	RegParam	rw	0xA020E0	
31 : 24	RegCommand	rw	0xD0	

43.77 wdfs_ctrl_1d4				0x403421D4
Bits	Field Name	Acc	Default	Description
0	RadarFifoRdEn	w1c	0x0	
1	RegCmdValid	w1c	0x0	
2	RegPulseTimerClear	w1c	0x0	
3	RadarFifoEmpty	ro	0x0	
6 : 4	RegPulseIrqThr	rw	0x1	
15 : 8	wdfs_data_fs	rw	0x51	
20 : 16	pwr_gap	rw	0xA	
31 : 24	dbm_thd	rw	0x40	

43.78 wdfs_fifo_1d8				0x403421D8
Bits	Field Name	Acc	Default	Description
31 : 0	RadarFifoRdData	ro	0x0	

43.79 wdfs_stat_1dc				0x403421DC
Bits	Field Name	Acc	Default	Description
23 : 0	RegStatus	ro	0x0	
30	rstn_wlan_dfs_en	rw	0x0	
31	clk_wlan_dfs_en	rw	0x0	

43.80 wdfs_ctrl_1e0				0x403421E0
Bits	Field Name	Acc	Default	Description
7 : 0	RegFomSing	rw	0x0	
15 : 8	RegFreqSing	rw	0x0	
19 : 16	RegFomSingGap	rw	0x0	
23 : 20	RegFreqSingGap	rw	0x0	
31 : 24	RegPulseIrqTimeOut	rw	0x0	

43.81 a0x1e4				0x403421E4
Bits	Field Name	Acc	Default	Description
0	aux_dc_lut_sel	rw	0x0	
1	aux_dc_crtl_dr	rw	0x1	
2	aux_dc_crtl_reg	rw	0x1	

43.82 wdfs_ctrl_1e8	0x403421E8

Bits	Field Name	Acc	Default	Description
6 : 0	idx_chg_tim	rw	0x28	
8	wdfs_clk_sel	rw	0x0	
10 : 9	wdfs_dat_sel	rw	0x0	
17 : 12	dif_tim	rw	0x10	

43.83 a0x200				0x40342200
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg0	ro	0x0	

43.84 a0x204				0x40342204
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg1	ro	0x0	

43.85 a0x208				0x40342208
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg2	ro	0x0	

43.86 a0x20c				0x4034220C
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg3	ro	0x0	

43.87 a0x210				0x40342210
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg4	ro	0x0	

43.88 a0x214				0x40342214
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg5	ro	0x0	

43.89 a0x218				0x40342218
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg6	ro	0x0	

43.90 a0x21c				0x4034221C
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg7	ro	0x0	

43.91 a0x220				0x40342220
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg8	ro	0x0	

43.92 a0x224				0x40342224
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg9	ro	0x0	

43.93 a0x228	0x40342228

Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg10	ro	0x0	

43.94 a0x22c				0x4034222C
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg11	ro	0x0	

43.95 a0x230				0x40342230
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg12	ro	0x0	

43.96 a0x234				0x40342234
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_reg13	ro	0x0	

43.97 a0x238				0x40342238
Bits	Field Name	Acc	Default	Description
31 : 0	DumpDataIn0	ro	0x0	

43.98 a0x23c				0x4034223C
Bits	Field Name	Acc	Default	Description
31 : 0	DumpDataIn1	ro	0x0	

43.99 a0x240				0x40342240
Bits	Field Name	Acc	Default	Description
31	DumpDataInEn	ro	0x0	

43.100 a0x244				0x40342244
Bits	Field Name	Acc	Default	Description
31 : 0	mon_port	ro	0x0	

43.101 a0x248				0x40342248
Bits	Field Name	Acc	Default	Description
7 : 0	txpowerout_offset	rw	0xA	
15 : 8	dr_rc_tx1_power	rw	0x0	
16	tx_power_dr	rw	0x0	
24 : 17	txpowerout_offset_11b	rw	0xA	
26	cfg_txgain_ram_en	rw	0x0	
31	step_mode	rw	0x1	

43.102 a0x24c				0x4034224C
Bits	Field Name	Acc	Default	Description
7 : 0	txpowerout_lowrate_offset	rw	0xA	
8	cfg_am_ram_en_	rw	0x0	
9	scale_step_6db_dr	rw	0x1	
11 : 10	scale_step_6db_reg	rw	0x0	
14 : 12	tx_gain_map_qam_threshold1	rw	0x3	
17 : 15	tx_gain_map_qam_threshold0	rw	0x0	
18	dac_rnd_sw_reg	rw	0x0	

19	dac_rnd_sw_dr	rw	0x1	

43.103 a0x26c				0x4034226C
Bits	Field Name	Acc	Default	Description
1 : 0	delay_0p25_tlo_mode	rw	0x1	
2	delay_0p25_tlo_bypass	rw	0x0	
3	delay_0p5_tdac_bypass	rw	0x0	
7	pre_hb_dr	rw	0x1	
8	pre_hb_reg	rw	0x1	
9	mem_update_reg	rw	0x0	
11 : 10	ext_out_select	rw	0x3	
16 : 12	level_qam_other	rw	0xF	
21 : 17	level_qam64	rw	0xF	
26 : 22	level_qam256	rw	0xF	
31 : 27	level_qam1024	rw	0xF	

43.104 a0x270				0x40342270
Bits	Field Name	Acc	Default	Description
31 : 0	tx_gain_map_reg	rw	0xC0C080	

43.105 a0x274				0x40342274
Bits	Field Name	Acc	Default	Description
1 : 0	gain_map_select3	rw	0x2	
3 : 2	gain_map_select2	rw	0x1	
5 : 4	gain_map_select1	rw	0x0	
7	idac_interval_sel_select	rw	0x0	
9 : 8	idac_am_shift	rw	0x0	
11 : 10	idac_am_p_shift	rw	0x1	
12	idac_bypass_dr	rw	0x1	
13	idac_bypass_reg	rw	0x1	
14	loft_mem_dr	rw	0x0	
15	loft_mem_addr_dr	rw	0x0	
20 : 16	loft_mem_addr_reg	rw	0x0	
21	loft_mem_wen	rw	0x0	
22	loft_mem_cen	rw	0x1	
31	tx_gain_map_dr	rw	0x0	

43.106 a0x278				0x40342278
Bits	Field Name	Acc	Default	Description
11 : 0	loft_offset_reg	rw	0x0	
27 : 16	loft_offset_i_reg	rw	0x0	

43.107 a0x27c				0x4034227C
Bits	Field Name	Acc	Default	Description
11 : 0	amp_err_reg	rw	0x0	
27 : 16	phase_err_reg	rw	0x0	

43.108 a0x280				0x40342280
Bits	Field Name	Acc	Default	Description
31 : 0	loft_mem_din	rw	0x0	

43.109 a0x284				0x40342284
Bits	Field Name	Acc	Default	Description
31 : 0	loft_mem_rd	ro	0x0	

43.110 a0x2b4				0x403422B4
Bits	Field Name	Acc	Default	Description
0	bt_txon_rf	rw	0x0	
1	bt_rxon_rf	rw	0x0	
2	bt_tws_rf	rw	0x0	

43.111 a0x2c4				0x403422C4
Bits	Field Name	Acc	Default	Description
7 : 0	papr_g1	rw	0x1	
15 : 8	papr_g3	rw	0x1	
17 : 16	cfg_am_ram_offset	rw	0x0	

43.112 a0x2c8				0x403422C8
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg0	ro	0x0	

43.113 a0x2cc				0x403422CC
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg1	ro	0x0	

43.114 a0x2d0				0x403422D0
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg2	ro	0x0	

43.115 a0x2d4				0x403422D4
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg3	ro	0x0	

43.116 a0x2d8				0x403422D8
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg4	ro	0x0	

43.117 a0x2dc				0x403422DC
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg5	ro	0x0	

43.118 a0x2e0				0x403422E0
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg6	ro	0x0	

43.119 a0x2e4				0x403422E4
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg7	ro	0x0	

43.120 a0x2e8				0x403422E8
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg8	ro	0x0	

43.121 a0x2ec				0x403422EC
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg9	ro	0x0	

43.122 a0x2f0				0x403422F0
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg10	ro	0x0	

43.123 a0x2f4				0x403422F4
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg11	ro	0x0	

43.124 a0x2f8				0x403422F8
Bits	Field Name	Acc	Default	Description
31 : 0	dfe_rd_mem_reg12	ro	0x0	

43.125 a0x2fc				0x403422FC
Bits	Field Name	Acc	Default	Description
6 : 0	reg_aux_rad_gain0	rw	0x0	
14 : 8	reg_aux_rad_gain1	rw	0x6	
22 : 16	reg_aux_rad_gain2	rw	0xC	
30 : 24	reg_aux_rad_gain3	rw	0x12	

43.126 a0x300				0x40342300
Bits	Field Name	Acc	Default	Description
6 : 0	reg_aux_rad_gain4	rw	0x18	
14 : 8	reg_aux_rad_gain5	rw	0x1E	
22 : 16	reg_aux_rad_gain6	rw	0x24	
30 : 24	reg_aux_rad_gain7	rw	0x2A	

43.127 a0x304				0x40342304
Bits	Field Name	Acc	Default	Description
6 : 0	reg_aux_rad_gain8	rw	0x30	
14 : 8	reg_aux_rad_gain9	rw	0x36	
22 : 16	reg_aux_rad_gaina	rw	0x3C	
30 : 24	reg_aux_rad_gainb	rw	0x42	

43.128 a0x308				0x40342308
Bits	Field Name	Acc	Default	Description
6 : 0	reg_aux_rad_gainc	rw	0x48	
14 : 8	reg_aux_rad_gaind	rw	0x4E	
22 : 16	reg_aux_rad_gaine	rw	0x54	
30 : 24	reg_aux_rad_gainf	rw	0x5A	

44 wf_rf_interface

44.1 reg_00				0x40344000
Bits	Field Name	Acc	Default	Description
0	wf_pa_pkdet_rstn_reg	rw	0x0	
1	wf_pa_pkdet_rstn_dr	rw	0x0	
2	wf_trx24g_wf2bt_rx_en_reg	rw	0x0	
3	wf_trx24g_wf2bt_rx_en_dr	rw	0x0	
4	wf_trx24g_sw_sel_avdd13_reg	rw	0x0	
5	wf_trx24g_sw_sel_avdd13_dr	rw	0x0	
6	wf_pu_trx24g_capsw_reg	rw	0x0	
7	wf_pu_trx24g_capsw_dr	rw	0x0	
8	wf_trx5g_sw_sel_avdd13_reg	rw	0x0	
9	wf_trx5g_sw_sel_avdd13_dr	rw	0x0	
10	wf_pu_trx5g_capsw_reg	rw	0x0	
11	wf_pu_trx5g_capsw_dr	rw	0x0	
12	wf_pa24g_gain_en_reg	rw	0x1	
13	wf_pa24g_gain_en_dr	rw	0x0	
14	wf_pa5g_gain_en_reg	rw	0x1	
15	wf_pa5g_gain_en_dr	rw	0x0	
16	wf_pa24g_bt_mode_en_reg	rw	0x0	
17	wf_pa24g_bt_mode_en_dr	rw	0x0	
18	wf_pu_pa_reg	rw	0x0	
19	wf_pu_pa_dr	rw	0x0	
20	wf_tia_rstn_reg	rw	0x0	
21	wf_tia_rstn_dr	rw	0x0	
22	wf_pu_tia_reg	rw	0x0	
23	wf_pu_tia_dr	rw	0x0	
24	wf_rmx_logen_dll_rstn_reg	rw	0x0	
25	wf_rmx_logen_dll_rstn_dr	rw	0x0	
26	wf_pu_rmx_reg	rw	0x0	
27	wf_pu_rmx_dr	rw	0x0	
28	wf_pu_lna_reg	rw	0x0	
29	wf_pu_lna_dr	rw	0x0	
30	wf_pu_lna5g_preamp_reg	rw	0x0	
31	wf_pu_lna5g_preamp_dr	rw	0x0	

44.2 reg_04				0x40344004
Bits	Field Name	Acc	Default	Description
0	wf_vco_txlo_en_reg	rw	0x1	
1	wf_vco_txlo_en_dr	rw	0x0	
2	wf_pu_vco_reg	rw	0x0	
3	wf_pu_vco_dr	rw	0x0	
4	wf_logen_phase_mux_rstn_reg	rw	0x0	
5	wf_logen_phase_mux_rstn_dr	rw	0x0	
6	wf_rfpll_vctrl_det_en_reg	rw	0x0	
7	wf_rfpll_vctrl_det_en_dr	rw	0x0	
8	wf_rfpll_adc_clk_en_reg	rw	0x0	
9	wf_rfpll_adc_clk_en_dr	rw	0x0	
10	wf_pu_rfpll_reg	rw	0x0	
11	wf_pu_rfpll_dr	rw	0x0	
12	wf_pu_pdadc_reg	rw	0x0	
13	wf_pu_pdadc_dr	rw	0x0	
14	wf_pu_adc_vreg_com_reg	rw	0x0	
15	wf_pu_adc_vreg_com_dr	rw	0x0	
16	wf_pu_adc_reg	rw	0x0	
17	wf_pu_adc_dr	rw	0x0	
18	wf_rxflt_rstn_reg	rw	0x0	
19	wf_rxflt_rstn_dr	rw	0x0	

20	wf_pu_rxflt_reg	rw	0x0	
21	wf_pu_rxflt_dr	rw	0x0	
22	wf_pu_iref_reg	rw	0x0	
23	wf_pu_iref_dr	rw	0x0	
24	wf_dtmx_cdr_rstn_reg	rw	0x0	
25	wf_dtmx_cdr_rstn_dr	rw	0x0	
26	wf_dtmx_logen_dll_rstn_reg	rw	0x0	
27	wf_dtmx_logen_dll_rstn_dr	rw	0x0	
28	wf_dtmx_bt2wf_tx_en_reg	rw	0x0	
29	wf_dtmx_bt2wf_tx_en_dr	rw	0x0	
30	wf_pu_dtmx_reg	rw	0x0	
31	wf_pu_dtmx_dr	rw	0x0	

44.3 reg_08				0x40344008
Bits	Field Name	Acc	Default	Description
2 : 0	wf_test_bit	rw	0x0	
3	wf_dtmx_pll_test	rw	0x0	
5 : 4	wf_lo_psw_pu_bit	rw	0x3	
7 : 6	wf_rfpll_pu_bit	rw	0x1	
9 : 8	wf_adc_pu_bit	rw	0x2	
11 : 10	wf_rxflt_pu_bit	rw	0x2	
13 : 12	wf_dtmx_logen_pu_bit	rw	0x2	
15 : 14	wf_dtmx_pu_bit_avdd13	rw	0x2	
17 : 16	wf_dtmx_pu_bit_avdd33	rw	0x2	
19 : 18	wf_tia_pu_bit	rw	0x1	
21 : 20	wf_rmx_pu_bit	rw	0x3	
23 : 22	wf_lna_pu_bit	rw	0x1	
25 : 24	wf_lna5g_preamp_pu_bit	rw	0x1	
26	wf_pu_vco_test_buf	rw	0x0	
27	wf_pu_ts_hub	rw	0x0	
28	wf_pu_lo_psw_reg	rw	0x0	
29	wf_pu_lo_psw_dr	rw	0x0	
30	wf_vco_rxlo_en_reg	rw	0x0	
31	wf_vco_rxlo_en_dr	rw	0x0	

44.4 reg_0C				0x4034400C
Bits	Field Name	Acc	Default	Description
1 : 0	wf_pu_tia_td1	rw	0x1	
3 : 2	wf_rmx_logen_dll_rstn_td4	rw	0x1	
5 : 4	wf_logen_phase_mux_rstn_td4	rw	0x2	
7 : 6	wf_pu_rmx_td1	rw	0x1	
10 : 8	wf_pu_rfpll_td3	rw	0x5	
12 : 11	wf_pu_vco_td2	rw	0x1	
14 : 13	wf_pu_lo_psw_td2	rw	0x1	
15	wf_test_enable_vco	rw	0x0	
16	wf_test_enable_rfpll	rw	0x0	
17	wf_vco_test_buf_en	rw	0x0	
18	wf_test_enable_ts_hub	rw	0x0	
19	wf_test_enable_adc	rw	0x0	
21 : 20	wf_test_enable_rxflt	rw	0x0	
22	wf_rxflt_test2gpio_en	rw	0x0	
23	wf_test_enable_iref	rw	0x0	
24	wf_test_enable_dtmx	rw	0x0	
25	wf_test_enable_pa	rw	0x0	
27 : 26	wf_test_enable_tia	rw	0x0	
28	wf_test_enable_rmx	rw	0x0	

29	wf_test_enable_lna	rw	0x0	
30	wf_test_enable_lna5g_preamp	rw	0x0	

44.5 reg_10				0x40344010
Bits	Field Name	Acc	Default	Description
0	wf_rfpll_temp_comp_en	rw	0x1	
1	cm_mdll_auto_reset_en	rw	0x0	
3 : 2	wf_pa24g_bt_mode_en_td1	rw	0x1	
5 : 4	wf_dtmx_bt2wf_tx_en_td1	rw	0x1	
7 : 6	wf_rfpll_vctrl_det_en_td2	rw	0x1	
9 : 8	wf_vco_band_temp_bit_td2	rw	0x1	
11 : 10	wf_pa_pkdet_rstn_td2	rw	0x1	
13 : 12	wf_pa_pkdet_rstn_td1	rw	0x1	
15 : 14	wf_pu_pa_td1	rw	0x1	
17 : 16	wf_dtmx_cdr_rstn_td1	rw	0x1	
19 : 18	wf_logen_phase_mux_rstn_td1	rw	0x1	
21 : 20	wf_dtmx_logen_dll_rstn_td4	rw	0x1	
23 : 22	wf_pu_dtmx_td1	rw	0x1	
25 : 24	wf_pu_lna_td1	rw	0x1	
27 : 26	wf_pu_adc_td1	rw	0x1	
29 : 28	wf_pu_rxflt_td1	rw	0x1	

44.6 reg_14				0x40344014
Bits	Field Name	Acc	Default	Description
3 : 0	wf_lna5g_preamp_cg_vbit	rw	0x6	
6 : 4	wf_lna5g_preamp_reg_vbit	rw	0x4	
7	wf_lna5g_img_cal_en	rw	0x0	
8	wf_lna5g_preamp_bypass_en	ro	0x0	
9	wf_lna5g_preamp_gbit	ro	0x1	
10	wf_lna5g_preamp_main_en	rw	0x1	
12 : 11	wf_pd_td0	rw	0x1	
15 : 13	wfcm_mdll_rstn_td3	rw	0x1	
17 : 16	wfcm_mdll_rstn_td2	rw	0x3	
19 : 18	wfcm_pu_mdll_td2	rw	0x2	
25 : 20	wf_rfpll_vctrl_THL	rw	0xF	
31 : 26	wf_rfpll_vctrl_THH	rw	0x30	

44.7 reg_18				0x40344018
Bits	Field Name	Acc	Default	Description
3 : 0	wf_lna24g_cbit	ro	0x7	
7 : 4	wf_lna5g_cbit	ro	0x7	
8	wf_rmx_1st_en	ro	0x0	
12 : 9	wf_lna_cg_vbit	rw	0x5	
16 : 13	wf_lna_ibias_bit	ro	0xA	
18 : 17	wf_lna_gbit	ro	0x3	
19	wf_lna5g_en	rw	0x0	
20	wf_lna24g_en	rw	0x1	
21	wf_lna5g_preamp_lna_bias_en	rw	0x1	
26 : 22	wf_lna5g_prea-mp_aux_atten_bit	ro	0x0	
27	wf_lna5g_preamp_aux_en	rw	0x0	
31 : 28	wf_lna5g_preamp_ibias_bit	ro	0xA	

44.8 reg_1C	0x4034401C

Bits	Field Name	Acc	Default	Description
2 : 0	wf_rmx_cload_bit	ro	0x2	
5 : 3	wf_rmx_lobias_vbit	rw	0x3	
6	wf_rmx_lobias_en	rw	0x1	
7	wf_rmx_logen_band_sel	rw	0x0	
8	wf_rmx24g_en	rw	0x1	
9	wf_rmx5g_dir_en	ro	0x0	
10	wf_rmx5g_en	rw	0x0	
11	wf_rmx_logen_sf_mode	rw	0x0	
12	wf_rmx_logen_psw_mode	rw	0x1	
15 : 13	wf_rmx_logen_reg_vbit	rw	0x4	
16	wf_rmx_logen_vref_sel_self_bias	rw	0x1	
23 : 17	wf_rmx_logen_sf_avdd_sw_bit	rw	0x13	
26 : 24	wf_lna_reg_vbit	rw	0x3	
27	wf_lna_iref_sel	rw	0x0	
28	wf_lna24g_cap_sel	ro	0x0	

44.9 reg_20 0x40344020

Bits	Field Name	Acc	Default	Description
0	wf_tia_op_hp_mode	ro	0x0	
1	wf_tia_op_lb_mode	ro	0x0	
6 : 2	wf_tia_bw_cbit	ro	0x1C	
9 : 7	wf_tia_bw_rbit	ro	0x4	
14 : 10	wf_tia_ibit	ro	0x19	
17 : 15	wf_tia_gbit	ro	0x7	
18	wf_tia_aux_en	rw	0x0	
19	wf_tia_en	rw	0x1	
23 : 20	wf_rmx_logen_dll_fbc_dly_bit	rw	0x0	
27 : 24	wf_rmx_logen_dll_fref_dly_bit	rw	0x0	
31 : 28	wf_rmx_logen_dll_reg_bit	rw	0x7	

44.10 reg_24 0x40344024

Bits	Field Name	Acc	Default	Description
1 : 0	wf_tia_vcm_ldo_vbit	rw	0x1	
4 : 2	wf_tia_aux_q_rbit	ro	0x3	
7 : 5	wf_tia_aux_mil_rbit	ro	0x4	
10 : 8	wf_tia_aux_mil_cbit	ro	0x4	
15 : 11	wf_tia_aux_bw_bit	ro	0x1D	
20 : 16	wf_tia_aux_ibit	ro	0x19	
23 : 21	wf_tia_aux_gbit	ro	0x3	
26 : 24	wf_tia_aux_input_sel	rw	0x4	
29 : 27	wf_tia_vcm_vbit	rw	0x3	

44.11 reg_28 0x40344028

Bits	Field Name	Acc	Default	Description
5 : 0	wf_pa5g_gbit	ro	0x3F	
6	wf_pa5g_wf_mode_en	rw	0x1	
7	wf_pa_ms	rw	0x0	
15 : 8	wf_tia_q_cal_bit_reg	rw	0x80	
16	wf_tia_q_cal_bit_dr	rw	0x0	
24 : 17	wf_tia_i_cal_bit_reg	rw	0x80	
25	wf_tia_i_cal_bit_dr	rw	0x0	
28 : 26	wf_tia_dc_cal_ibit	ro	0x3	

44.12 reg_2C				0x4034402C
Bits	Field Name	Acc	Default	Description
3 : 0	wf_pa_vm_vbit1	ro	0xB	
6 : 4	wf_pa_vm_ref_vbit	ro	0x4	
7	wf_pa_vm_opam_ibit	rw	0x1	
8	wf_pa_vm_dyn_en	ro	0x0	
9	wf_pa_vm_en	rw	0x1	
10	wf_pa_vh_ref_sel_bg	rw	0x1	
13 : 11	wf_pa_vh_vbit	rw	0x6	
15 : 14	wf_pa_vbg_vbit	rw	0x2	
17 : 16	wf_pa_vbg_tc_bit	rw	0x2	
20 : 18	wf_pa24g_input_cbit	ro	0x0	
26 : 21	wf_pa24g_gbit	ro	0x3F	

44.13 reg_30				0x40344030
Bits	Field Name	Acc	Default	Description
2 : 0	wf_pa5g_vl1_rfdyn_vthbit	rw	0x4	
6 : 3	wf_pa5g_vl1_cgm_ibit	rw	0x8	
9 : 7	wf_pa5g_vl1_vbit	ro	0x4	
12 : 10	wf_pa5g_vl1_ic_sel_bit	rw	0x1	
13	wf_pa5g_vl1_opam_en	rw	0x1	
14	wf_pa5g_vl_sel_vl1	rw	0x1	
16 : 15	wf_pa5g_vl_en_bit	ro	0x3	
19 : 17	wf_pa5g_vm_rfdyn_vthbit	rw	0x4	
20	wf_pa5g_vm_rfdyn_en	ro	0x0	
22 : 21	wf_pa5g_input2_cbit	ro	0x2	
24 : 23	wf_pa5g_input1_cbit	ro	0x2	
30 : 25	wf_pa_vm_vbit2	ro	0x20	

44.14 reg_34				0x40344034
Bits	Field Name	Acc	Default	Description
0	wf_pa24g_vl_rfdyn_en	ro	0x0	
3 : 1	wf_pa24g_vm_rfdyn_vthbit	rw	0x4	
4	wf_pa24g_vm_rfdyn_en	ro	0x0	
6 : 5	wf_pa5g_vl2_resbit	ro	0x3	
8 : 7	wf_pa5g_vl1_resbit	ro	0x3	
11 : 9	wf_pa5g_vl2_rfdyn_gbit	ro	0x4	
14 : 12	wf_pa5g_vl2_rfdyn_vthbit	rw	0x4	
18 : 15	wf_pa5g_vl2_cgm_ibit	rw	0x8	
21 : 19	wf_pa5g_vl2_vbit	ro	0x4	
22	wf_pa5g_vl2_rfdyn_en	ro	0x0	
25 : 23	wf_pa5g_vl2_ic_sel_bit	rw	0x1	
26	wf_pa5g_vl2_opam_en	rw	0x1	
27	wf_pa5g_vl1_rfdyn_en	ro	0x0	
30 : 28	wf_pa5g_vl1_rfdyn_gbit	ro	0x4	

44.15 reg_38				0x40344038
Bits	Field Name	Acc	Default	Description
3 : 0	wf_pa5g_avss_hd1_cbit	rw	0x8	
7 : 4	wf_trx24g_sw_cbit	ro	0x8	
11 : 8	wf_trx5g_sw_cbit	ro	0x8	
14 : 12	wf_pa24g_vl_vbit	ro	0x4	
15	wf_pa24g_vl_opam_en	rw	0x1	
18 : 16	wf_pa24g_vl_ic_sel_bit	rw	0x1	
22 : 19	wf_pa24g_vl_cgm_ibit	ro	0x8	

25 : 23	wf_pa24g_vl_rfdyn_gbit	ro	0x4	
28 : 26	wf_pa24g_vl_rfdyn_vthbit	rw	0x4	

44.16 reg_3C				0x4034403C
Bits	Field Name	Acc	Default	Description
0	wf_pa5g_isense_sel_ic5u	rw	0x0	
1	wf_pa5g_isense_en	rw	0x0	
4 : 2	wf_pa_dpd_cbit	rw	0x4	
5	wf_pa_dpd_en	rw	0x0	
6	wf_pa_pkd_out	ro	0x0	
9 : 7	wf_pa_pkdet_vth_bit	rw	0x4	
10	wf_pa_pkdet_en	rw	0x0	
11	wf_pa_loft_sel_lowpass	rw	0x0	
14 : 12	wf_pa_loft_gbit	rw	0x4	
15	wf_pa_loft_en	rw	0x0	
19 : 16	wf_pa24g_rfop_hd2_cbit_rx	rw	0x8	
23 : 20	wf_pa24g_rfop_hd2_cbit_tx	rw	0x8	

44.17 reg_40				0x40344040
Bits	Field Name	Acc	Default	Description
2 : 0	wf_dtmx_dac_gbit1	ro	0x4	
6 : 3	wf_dtmx_dac_cscd_vbit	rw	0x8	
9 : 7	wf_dtmx_vlo_vbit	rw	0x2	
12 : 10	wf_dtmx_vh_vbit	rw	0x2	
14 : 13	wf_dtmx_path_mode_bit	rw	0x3	
15	wf_dtmx_lo5g_mode	rw	0x1	
16	wf_dtmx_ms	rw	0x0	
19 : 17	wf_dtmx_dvdd_vbit	rw	0x2	
22 : 20	wf_dtmx_btbias_vl_bit	rw	0x4	
23	wf_dtmx_din0_sel_half_lsb	rw	0x1	
25 : 24	wf_trx24g_rx_cbit	ro	0x0	
27 : 26	wf_trx5g_rx_cbit	ro	0x0	
28	wf_pa24g_isense_sel_ic5u	rw	0x0	
29	wf_pa24g_isense_en	rw	0x0	

44.18 reg_44				0x40344044
Bits	Field Name	Acc	Default	Description
0	wf_dtmx_logen_band	rw	0x1	
1	wf_dtmx_logen_vco_div2_en	rw	0x1	
2	wf_dtmx_logen_sf_mode	rw	0x1	
3	wf_dtmx_logen_psw_mode	rw	0x1	
4	wf_dtmx_logen_lo_en	rw	0x1	
5	wf_dtmx_logen_sf_sel_ldo18	rw	0x0	
8 : 6	wf_dtmx_logen_reg_vbit	rw	0x4	
9	wf_dtmx_logen_reg_en	rw	0x1	
11 : 10	wf_dtmx_vbg_b_vbit	rw	0x2	
13 : 12	wf_dtmx_vbg_tc_bit	rw	0x2	
14	wf_dtmx_lb_ext_cap_en	ro	0x0	
18 : 15	wf_dtmx_tuning_cbit	ro	0x8	
22 : 19	wf_dtmx_dac_gbit3	ro	0x7	
26 : 23	wf_dtmx_dac_gbit2	ro	0x7	

44.19 reg_48				0x40344048
Bits	Field Name	Acc	Default	Description

0	wf_dtmx_clk_path0_45_dly_en	rw	0x1	
1	wf_dtmx_clk_iq_dly_en	rw	0x1	
4 : 2	wf_dtmx_clk_phase_sel_bit	rw	0x1	
5	wf_dtmx_clk_edge_sel	rw	0x1	
10 : 6	wf_dtmx_offset_bit	rw	0x0	
14 : 11	wf_dtmx_logen_dll_fbc_dly_bit	rw	0x8	
18 : 15	wf_dtmx_logen_dll_fref_dly_bit	rw	0x8	
22 : 19	wf_dtmx_logen_dll_reg_vbit	rw	0x8	
23	wf_dtmx_clk2cdr_en	rw	0x1	
24	wf_dtmx_logen-_vref_sel_self_bias	rw	0x1	
31 : 25	wf_dtmx_logen_sf_avdd_sw_bit	rw	0x40	

44.20 reg_4C				0x4034404C
Bits	Field Name	Acc	Default	Description
2 : 0	wf_dtmx_dpd_tia_miller_cbit	rw	0x4	
5 : 3	wf_dtmx_dpd_tia_gbit	rw	0x4	
6	wf_dtmx_dpd_en	rw	0x0	
9 : 7	wf_dtmx_cdr_phase_lck_bit	ro	0x0	
11 : 10	wf_dtmx_cdr_dly_lck_bit	ro	0x0	
12	wf_dtmx_cdr_lock	ro	0x0	
13	wf_dtmx_cdr_sel_en	rw	0x1	
16 : 14	wf_dtmx_clk2dac_dly_bit	rw	0x0	
17	wf_dtmx_clk2dac_sel_dig	rw	0x0	
19 : 18	wf_dtmx_cdr_div_freq_sel_bit	rw	0x1	
24 : 20	wf_dtmx_clk2dig_dly_bit	rw	0x0	
27 : 25	wf_dtmx_clk_freq_sel_bit	rw	0x4	

44.21 reg_50				0x40344050
Bits	Field Name	Acc	Default	Description
3 : 0	wf_dtmx_rfdyn_gbit	ro	0x8	
7 : 4	wf_dtmx_idac_padyn_offset_bit	rw	0x0	
8	wf_dtmx_idac_padyn_en	ro	0x0	
9	wf_dtmx_idac_-padyn_clk_sel_pos_edge	rw	0x0	
10	wf_dtmx_idac_ic_sel_dtmx	rw	0x0	
11	wf_dtmx_dpd_img_cal_en	rw	0x0	
14 : 12	wf_dtmx_dpd_lobias_vbit	rw	0x4	
17 : 15	wf_dtmx_dpd_tia_vcm_vbit	rw	0x4	
18	wf_dtmx_dpd_tia_rstn	rw	0x0	
25 : 19	wf_dtmx_dpd_tia_q_rbit	rw	0x3F	
31 : 26	wf_dtmx_dpd_tia_miller_rbit	rw	0x1F	

44.22 reg_54				0x40344054
Bits	Field Name	Acc	Default	Description
2 : 0	wf_rxflt_bw_rbit	ro	0x4	
3	wf_rxflt_iq_swap	rw	0x0	
5 : 4	wf_rxflt_notch_mode	ro	0x0	
6	wf_rxflt_cal_mode	rw	0x0	
8 : 7	wf_rxflt_pwres_bit	rw	0x2	
12 : 9	wf_trx_rsvd_read_bit	ro	0x0	
16 : 13	wf_trx_rsvd_bit	rw	0xC	
17	wf_iref_start_enb	rw	0x1	
18	wf_dtmx_dynbias_en	ro	0x0	
19	wf_dtmx_dynbias_sel_rf	rw	0x0	

20	wf_dtmx_dynbias_sel_dig	rw	0x0	
24 : 21	wf_dtmx_rfdyn_ref_vbit	ro	0x8	
27 : 25	wf_dtmx_rfdyn_vthbit	rw	0x4	

44.23 reg_58				0x40344058
Bits	Field Name	Acc	Default	Description
4 : 0	wf_rxflt_i_calbit	rw	0x10	
7 : 5	wf_rxflt_vcm_bit	rw	0x5	
10 : 8	wf_rxflt_gbit2	ro	0x0	
15 : 11	wf_rxflt_gbit1	ro	0x0	
21 : 16	wf_rxflt_bw_cbit2	ro	0x11	
27 : 22	wf_rxflt_bw_cbit1	ro	0x11	

44.24 reg_5C				0x4034405C
Bits	Field Name	Acc	Default	Description
1 : 0	wf_adc_clk_sel_bit	rw	0x0	
2	wf_adc_aux_en	rw	0x0	
3	wf_adc_core_en	rw	0x0	
7 : 4	wf_rxflt_insel	rw	0x3	
11 : 8	wf_rxflt_op_rq_bit	ro	0xC	
15 : 12	wf_rxflt_op_rm_bit	ro	0xC	
18 : 16	wf_rxflt_op_cm_bit	ro	0x4	
24 : 19	wf_rxflt_op_ibit	ro	0x20	
29 : 25	wf_rxflt_q_calbit	rw	0x10	

44.25 reg_60				0x40344060
Bits	Field Name	Acc	Default	Description
0	wf_adc_lp_mode	rw	0x0	
5 : 1	wf_adc_vref_vbit	rw	0x8	
10 : 6	wf_adc_vref_ibit	rw	0x8	
13 : 11	wf_adc_vcm_vbit	rw	0x3	
17 : 14	wf_adc_reg_ibit	rw	0x3	
20 : 18	wf_adc_reg_vbit	rw	0x5	
24 : 21	wf_adc_dly_bit	rw	0x0	
25	wf_adc_clk_edge_sel	rw	0x0	
26	wf_adc_ti_mode	rw	0x1	
27	wf_adc_slice_sel	rw	0x0	
31 : 28	wf_adc_cmp_ibit	rw	0x4	

44.26 reg_64				0x40344064
Bits	Field Name	Acc	Default	Description
0	wf_rfpll_presc_en	rw	0x1	
1	wf_rfpll_loop_en	rw	0x1	
2	wf_rfpll_sel_avdd12_sf	rw	0x0	
6 : 3	wf_abb_rsvd_read_bit	ro	0x0	
10 : 7	wf_abb_rsvd_bit	rw	0xC	
13 : 11	wf_rx_iq2gpio_en	rw	0x0	
15 : 14	wf_ts_hub_vref_vbit	rw	0x2	
17 : 16	wf_ts_hub_vptat_vbit	rw	0x2	
21 : 18	wf_ts_hub_chn_sel	rw	0x1	
22	wf_ts_hub_chop	rw	0x0	
26 : 23	wf_pdadc_dly_bit	rw	0x2	
27	wf_pdadc_clk_edge_sel	rw	0x0	
31 : 28	wf_adc_clk_cbit	rw	0x8	

44.27 reg_68				0x40344068
Bits	Field Name	Acc	Default	Description
1 : 0	wf_rfpll_lpf_gbit	rw	0x3	
3 : 2	wf_rfpll_lpf_rbit	rw	0x1	
7 : 4	wf_rfpll_cp_ibit	rw	0x8	
9 : 8	wf_rfpll_cp_rbit	rw	0x0	
13 : 10	wf_rfpll_vco_gbit	rw	0xF	
14	wf_rfpll_lpf_notch_bypass	rw	0x0	
16 : 15	wf_rfpll_sinc_mode	rw	0x1	
17	wf_rfpll_loop_mode	rw	0x0	
18	wf_rfpll_frefx2_en	rw	0x0	
19	wf_rfpll_fref_sel	rw	0x1	
23 : 20	wf_rfpll_presc_reg_vbit	rw	0x8	
25 : 24	wf_rfpll_dvdd_ldo_vbit	rw	0x2	
29 : 26	wf_rfpll_pdt_reg_vbit	rw	0x8	

44.28 reg_6C				0x4034406C
Bits	Field Name	Acc	Default	Description
0	wf_rfpll_lock_out	ro	0x0	
17 : 1	wf_rfpll_cnt_out	ro	0x0	
19 : 18	wf_rfpll_vco_cal_sel	rw	0x2	
21 : 20	wf_rfpll_cal_bit	rw	0x2	
22	wf_rfpll_cal_en	ro	0x0	
23	wf_rfpll_cal_clk_edge_sel	rw	0x0	
24	wf_rfpll_fbc_sel	rw	0x0	
25	wf_rfpll_sdm_clk_sel	ro	0x0	
31 : 26	wf_rfpll_pdt_rbit	rw	0x0	

44.29 reg_70				0x40344070
Bits	Field Name	Acc	Default	Description
0	wf_vco_avdd18_reg_en	rw	0x1	
3 : 1	wf_vco_avdd18_reg_bit	rw	0x3	
4	wf_vco_avdd18_reg_bypass	rw	0x1	
5	wf_logen_phase_sync1_clk_sel	rw	0x0	
6	wf_logen_phase_sync_clk_sel	rw	0x0	
7	wf_rfpll_notch_bypass	rw	0x0	
9 : 8	wf_rfpll_sinc_clk_sel	rw	0x0	
11 : 10	wf_rfpll_sinc_clk_div	rw	0x0	
12	wf_rfpll_div_clk_edge_sel	rw	0x0	
21 : 13	wf_rfpll_adc_clk_div	rw	0x18	
22	wf_rfpll_notch_sel_4p2G	rw	0x1	

44.30 reg_74				0x40344074
Bits	Field Name	Acc	Default	Description
3 : 0	wf_vco_buf_reg_bit	rw	0x8	
7 : 4	wf_vco_core_reg_bit	rw	0x8	
12 : 8	wf_vco_band_tail_bit	rw	0x0	
16 : 13	wf_vco_band_temp_bit	rw	0x7	
18 : 17	wf_vco_band_aux_bit	rw	0x1	
31 : 19	wf_vco_band_bit	ro	0xBB8	

44.31 reg_78				0x40344078
Bits	Field Name	Acc	Default	Description
3 : 0	wf_lo_rsvd_bit	rw	0xC	

10 : 4	wf_lo_psw_sf_avdd_sw_bit	rw	0x13	
11	wf_lo_psw_vref_sel_self_bias	rw	0x1	
12	wf_lo_psw_sf_mode	rw	0x0	
13	wf_lo_psw_mode	rw	0x1	
14	wf_vco_mux_sel_4g	rw	0x1	
15	wf_vco_div2_en	rw	0x0	
16	wf_vco_trx_mode	rw	0x1	
17	wf_vco_presclo_en	rw	0x1	
18	wf_vco_lp_mode	rw	0x0	
23 : 19	wf_vco_vctrl_biasn_bit	rw	0xF	
28 : 24	wf_vco_vctrl_biasp_bit	rw	0xF	

44.32 reg_7C				0x4034407C
Bits	Field Name	Acc	Default	Description
7 : 0	wf_apc_rsvd	ro	0xF0	
15 : 8	wf_agc_rsvd	ro	0xF0	
19 : 16	wf_lo_rsvd_read_bit	ro	0x0	

44.33 wf_soft_reset_reg				0x40344080
Bits	Field Name	Acc	Default	Description
0	soft_reset	rw	0x0	

44.34 wf_misc_crtl_reg				0x40344084
Bits	Field Name	Acc	Default	Description
0	pllon_reg	rw	0x0	
1	pllon_trx	rw	0x1	
2	rf_band_sel	rw	0x0	
5 : 3	rxgain_page	rw	0x1	
6	rxgain_mem_sw_req	rw	0x0	
7	rxgain_mem_sw_ack	ro	0x0	
10 : 8	adj_dc_page	rw	0x0	
11	wf_pllon_sleep_en	rw	0x1	

44.35 wf_man_crtl_reg				0x40344088
Bits	Field Name	Acc	Default	Description
5 : 0	tx_gain_reg	rw	0x0	
6	tx_gain_dr	rw	0x0	
7	tx_gain_dr_pulse	rw	0x0	
16 : 8	rx_gain_reg	rw	0x0	
17	rx_gain_dr	rw	0x0	
18	rx_gain_dr_pulse	rw	0x0	
19	rxon_reg	rw	0x0	
20	rxon_dr	rw	0x0	
21	txon_reg	rw	0x0	
22	txon_dr	rw	0x0	
23	bton_reg	rw	0x0	
24	bton_dr	rw	0x1	
25	bttws_reg	rw	0x0	
26	bttws_dr	rw	0x0	

44.36 wf_calclk_reg				0x4034408C
Bits	Field Name	Acc	Default	Description
9 : 0	cal_clk_div_num	rw	0x4F	

10	cal_clk_en	rw	0x0	
16 : 11	cal_clk_div_num_p1	rw	0x6	
22 : 17	cal_clk_div_num_p2	rw	0x13	
23	vco_loen_ctrl	rw	0x0	
25 : 24	debug_mux	rw	0x0	
26	debug_clk_en	rw	0x0	

44.37 wf_ext_trxon_delay_reg				0x40344090
Bits	Field Name	Acc	Default	Description
3 : 0	ext_pa_idle_ctrl_reg	rw	0x0	
7 : 4	ext_pa_tx_ctrl_reg	rw	0x2	
11 : 8	ext_pa_rx_ctrl_reg	rw	0x5	
20 : 12	ext_txon_delay	rw	0x41	
29 : 21	ext_rxon_delay	rw	0x87	
30	ext_pa_ctrl_mode	rw	0x0	

44.38 gain_mem_ctrl_reg				0x40344094
Bits	Field Name	Acc	Default	Description
0	clk_txgain_en	rw	0x0	
1	clk_txgain_sel	rw	0x0	
2	txgain_mem_cfg	rw	0x0	
4	clk_rxgain_en	rw	0x0	
5	clk_rxgain_sel	rw	0x0	
6	rxgain_mem_cfg	rw	0x0	

44.39 gain_mem_fetch_reg				0x40344098
Bits	Field Name	Acc	Default	Description
5 : 0	txgain_mem_fetch_addr	rw	0x0	
15	txgain_mem_fetch_p	rw	0x0	
22 : 16	rxgain_mem_fetch_addr	rw	0x0	
31	rxgain_mem_fetch_p	rw	0x0	

44.40 wf_dpd_txcal_ctrl_reg				0x4034409C
Bits	Field Name	Acc	Default	Description
2 : 0	wf_tia_gbit_reg	rw	0x0	
3	wf_tia_gbit_dr	rw	0x0	
6 : 4	wf_rxflt_gbit1_reg	rw	0x0	
7	wf_rxflt_gbit1_dr	rw	0x0	
9 : 8	wf_rxflt_gbit2_reg	rw	0x0	
10	wf_rxflt_gbit2_dr	rw	0x0	

44.41 pu_done_reg				0x403440A0
Bits	Field Name	Acc	Default	Description
0	txon_co_ant_done	ro	0x0	
1	rxon_co_ant_done	ro	0x0	
2	txon_done	ro	0x0	
3	rxon_done	ro	0x0	
4	pllon_done	ro	0x0	

44.42 dig_reserved_reg				0x403440A4
Bits	Field Name	Acc	Default	Description

44.43 agc_reserved_reg				0x403440A8
Bits	Field Name	Acc	Default	Description
2 : 0	wf_agc_rsvd_reg	rw	0x0	

44.44 agc_table_out_reg0				0x403440AC
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_0	ro	0xA1DD5829	

44.45 agc_table_out_reg1				0x403440B0
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_1	ro	0x3B967267	

44.46 agc_table_out_reg2				0x403440B4
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_2	ro	0x8C0221B9	

44.47 agc_table_out_reg3				0x403440B8
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_3	ro	0x40000022	

44.48 agc_table_out_reg4				0x403440BC
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_4	ro	0x4040732	

44.49 apc_reserved_reg				0x403440C0
Bits	Field Name	Acc	Default	Description
6 : 0	wf_apc_rsvd_reg	rw	0x0	

44.50 apc_table_out_reg0				0x403440C4
Bits	Field Name	Acc	Default	Description
31 : 0	apc_table_out_0	ro	0x505C0FFF	

44.51 apc_table_out_reg1				0x403440C8
Bits	Field Name	Acc	Default	Description
31 : 0	apc_table_out_1	ro	0x443E424D	

44.52 apc_table_out_reg2				0x403440CC
Bits	Field Name	Acc	Default	Description
31 : 0	apc_table_out_2	ro	0x4410EF2	

45 sys_ctrl

45.1 chip_id				0x40500000
Bits	Field Name	Acc	Default	Description
15 : 0	CHIP_ID	ro	0x8820	AIC8820 chip ID
23 : 16	METAL_ID	ro	0x0	Metal ID
31 : 24	BOND_ID	ro	0x0	Bonding ID

45.2 aon_clk_sel				0x40500004
Bits	Field Name	Acc	Default	Description
0	cfg_aon_clk_sel_52m_or_div	rw	0x0	1'b0: 80MHz ; 1'b1: 80MHz divider
1	cfg_aon_clk_80m_hwen	rw	0x0	1'b0: manual mode; 1'b1: auto mode from pwr_ctrl
2	cfg_aon_clk_80m_sel	rw	0x0	1'b0: xtal doubler; 1'b1: pll 480m div6

45.3 aon_clk_mode				0x40500008
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_aon_clk_mode	rw	0xFFFFFFFF	1'b1: manual mode; 1'b0: auto mode

45.4 aon_clk_div				0x4050000C
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_aon_clk_div_denom	rw	0x1	aon clock divider denominator, default div1
8	cfg_aon_clk_div_update	rw	0x0	aon clock divider update

45.5 top_clk_sel				0x40500010
Bits	Field Name	Acc	Default	Description
0	cfg_top_fast_hwen	rw	0x0	1'b0: SW control; 1'b1: power FSM control
1	cfg_top_fast_sel	rw	0x0	1'b0: slow clock; 1'b1: fast clock
2	cfg_top_pll_480m_sel	rw	0x0	1'b0: 480MHz ; 1'b1: 320MHz
3	cfg_top_psram_clk_fast_sel	rw	0x0	1'b0: slow clock; 1'b1: fast clock
4	cfg_top_psram_clk_480m_sel	rw	0x0	1'b0: 480MHz ; 1'b1: 320MHz

45.6 top_hclk_mode				0x40500014
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_top_hclk_mode	rw	0xFF	1'b1: manual mode; 1'b0: auto mode

45.7 top_fclk_mode				0x40500018
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_top_fclk_mode	rw	0xFF	1'b1: manual mode; 1'b0: auto mode

45.8 top_others_clk_mode				0x4050001C
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_top_others_clk_mode	rw	0xFFFF	1'b1: manual mode; 1'b0: auto mode

45.9 hclk_div				0x40500020
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_top_hclk_div_denom	rw	0x1	ahb clock divider denominator, default div1
8	cfg_top_hclk_div_update	rw	0x0	hclk divider update

45.10 fclk_div				0x40500024
Bits	Field Name	Acc	Default	Description
3 : 0	cfg_top_fclk_div_denom	rw	0xD	fast clock divider denominator, default div2
4	cfg_top_fclk_div_update	rw	0x0	fast clock divider update

45.11 psram_clk_div				0x40500028
Bits	Field Name	Acc	Default	Description
3 : 0	cfg_top_psram_clk_div_denom	rw	0xD	psram clock divider denominator, default div2
4	cfg_top_psram_clk_div_update	rw	0x0	psram clk divider update

45.12 cfg_480m_clk_div				0x4050002C
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_top_480m_gen_sel	rw	0x0	clock divider
9 : 8	cfg_top_480m_clk_mode_sel	rw	0x0	mode selector
12	cfg_top_480m_div_update	rw	0x0	real 480M clock divider update

45.13 real_320m_clk_div				0x40500030
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_top_320m_gen_sel	rw	0x0	clock divider
9 : 8	cfg_top_320m_clk_mode_sel	rw	0x0	mode selector
12	cfg_top_320m_div_update	rw	0x0	real 320M clock divider update

45.14 real_480m_clk_div				0x40500034
Bits	Field Name	Acc	Default	Description
3 : 0	cfg_top_real_480m_div_denom	rw	0xF	480M clock divider denominator, default div1
4	cfg_top_real_480m_div_update	rw	0x0	480M clk divider update

45.15 cfg_osc_enable				0x40500038
Bits	Field Name	Acc	Default	Description
0	cfg_top_bt_osc_en	rw	0x0	BT OSC control enable, high active
1	cfg_top_wlan_osc_en	rw	0x0	WLAN OSC control enable, high active
2	cfg_top_bt_osc_manual	rw	0x1	BT OSC manual control, high active
3	cfg_top_wlan_osc_manual	rw	0x1	WLAN OSC manual control, high active
4	cfg_top_wlan_osc_comb_en	rw	0x0	Enable ahb_clk_wifi when comb mode, high active

45.16 reset_cause				0x4050003C
Bits	Field Name	Acc	Default	Description
0	wdg0_rst_happened	ro	0x0	reset happened, high active
1	wdg1_rst_happened	ro	0x0	reset happened, high active
2	wdg2_rst_happened	ro	0x0	reset happened, high active
3	bt_wdg_rst_happened	ro	0x0	reset happened, high active
4	cpu_p_sysrst_req_happened	ro	0x0	reset happened, high active
5	cpu_s0_sysrst_req_happened	ro	0x0	reset happened, high active
6	cpu_s1_sysrst_req_happened	ro	0x0	reset happened, high active
7	global_soft_reset_happened	ro	0x0	reset happened, high active
8	perst_happened	ro	0x0	reset happened, high active

45.17 tports_sel				0x40500040
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_aon_tports_sel	rw	0x0	aon system test ports select
11 : 8	cfg_top_tports_sel_low	rw	0x0	digitial top test ports low 16-bit selector
15 : 12	cfg_top_tports_sel_high	rw	0x0	digitial top test ports high 16-bit selector

45.18 cpu_p_reset_addr				0x40500044
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_cpu_p_reset_addr	rw	0x0	Main CPU boot address

45.19 cpu_s0_reset_addr				0x40500048
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_cpu_s0_reset_addr	rw	0x0	Second CPU0 boot address

45.20 cpu_s1_reset_addr				0x4050004C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_cpu_s1_reset_addr	rw	0x80000	Second CPU1 boot address

45.21 cfg_cpu_sys_ctrl				0x40500050
Bits	Field Name	Acc	Default	Description
3 : 0	cfg_cpu_sys_matrix_remap	rw	0x0	sys matrix remap
4	cfg_cpu_sys_flash_clk_sel	rw	0x0	1'b0: XTAL_2X; 1'b1: fast clock
5	cfg_cpu_sys_uart0_clk_sel	rw	0x0	1'b0: XTAL_2X; 1'b1: fast clock
6	cfg_cpu_sys_uart1_clk_sel	rw	0x0	1'b0: XTAL_2X; 1'b1: fast clock
7	cfg_cpu_sys_uart2_clk_sel	rw	0x0	1'b0: XTAL_2X; 1'b1: fast clock

45.22 cfg_rtc1_pu_dcdc_resetn_bypass				0x40500054
Bits	Field Name	Acc	Default	Description
0	cfg_rtc1_pu_-dcdc_resetn_bypass	rw	0x0	High active.
1	cfg_dcdc_rf_-dcdc_clk_lpo_sel_sw	rw	0x0	
2	cfg_dcdc_rf_dcdc_clk_lpo_sel	rw	0x0	

45.23 cfg_sdio_iomux_sel				0x40500058
Bits	Field Name	Acc	Default	Description
0	cfg_sdio_iomux_bypass_sel	rw	0x1	1'b0: iomux; 1'b1: SDIO function
15 : 1	cfg_gpio_awake_mux_sel	rw	0x20	
17 : 16	cfg_sdio_clk_pulse_bit	rw	0x2	

45.24 cfg_mem_cpu_sys_1pram				0x4050005C
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_cpu_sys_1pram	rw	0x21C8	cpu sys 1p sram configuration

45.25 cfg_mem_bt_sys_1pram				0x40500060
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_bt_sys_1pram	rw	0x21C8	bt sys 1p sram configuration

45.26 bt_hclk_div				0x40500064
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_top_bt_hclk_div_denom	rw	0x2	BT ahb clock divider denominator, default div2
8	cfg_top_bt_hclk_div_update	rw	0x0	BT hclk divider update

45.27 top_tports_bits_sel0				0x40500068
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_top_tports_bit_sel0	rw	0x3020100	top tports sel0

45.28 top_tports_bits_sel1				0x4050006C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_top_tports_bit_sel1	rw	0x7060504	top tports sel1

45.29 top_tports_bits_sel2				0x40500070
Bits	Field Name	Acc	Default	Description

31 : 0	cfg_top_tports_bit_sel2	rw	0xB0A0908	top tports sel2

45.30 top_tports_bits_sel3				0x40500074
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_top_tports_bit_sel3	rw	0xF0E0D0C	top tports sel3

45.31 perst_cfg				0x40500078
Bits	Field Name	Acc	Default	Description
7 : 0	perst_act_dly	rw	0x8	about: (dly+4)*1/128ms
15 : 8	perst_inact_dly	rw	0x0	about: (dly+4)*1/128ms
16	perst_global_req_en	rw	0x0	

45.32 hclk_manual_enable				0x40500100
Bits	Field Name	Acc	Default	Description
0	hclk_cpu_sys_manual_enable	rw	0x1	write 1 enable manual mode
1	hclk_bt_sys_manual_enable	rw	0x1	write 1 enable manual mode
2	hclk_bt_free_manual_enable	rw	0x1	write 1 enable manual mode
3	hclk_wifi_sys_manual_enable	rw	0x1	write 1 enable manual mode
4	hclk_top_vcore_enable	rw	0x1	write 1 enable manual mode

45.33 hclk_manual_disable				0x40500104
Bits	Field Name	Acc	Default	Description
0	hclk_cpu_sys_manual_disable	rw	0x1	write 1 disable manual mode
1	hclk_bt_sys_manual_disable	rw	0x1	write 1 disable manual mode
2	hclk_bt_free_manual_disable	rw	0x1	write 1 disable manual mode
3	hclk_wifi_sys_manual_disable	rw	0x1	write 1 disable manual mode
4	hclk_top_vcore_disable	rw	0x1	write 1 disable manual mode

45.34 fclk_manual_enable				0x40500108
Bits	Field Name	Acc	Default	Description
0	fclk_cpu_sys_manual_enable	rw	0x1	write 1 enable manual mode

45.35 fclk_manual_disable				0x4050010C
Bits	Field Name	Acc	Default	Description
0	fclk_cpu_sys_manual_disable	rw	0x1	write 1 disable manual mode

45.36 others_manual_enable				0x40500110
Bits	Field Name	Acc	Default	Description
0	clk_26m_cpu_-sys_manual_enable	rw	0x1	write 1 enable manual mode
1	clk_52m_cpu_-sys_manual_enable	rw	0x1	write 1 enable manual mode
2	clk_32k_cpu_-sys_manual_enable	rw	0x1	write 1 enable manual mode
3	clk_26m_bt_sys_manual_enable	rw	0x1	write 1 enable manual mode
4	clk_52m_bt_sys_manual_enable	rw	0x1	write 1 enable manual mode
5	clk_32k_bt_sys_manual_enable	rw	0x1	write 1 enable manual mode
6	clk_26m_wifi_-sys_manual_enable	rw	0x1	write 1 enable manual mode
7	clk_52m_wifi_-sys_manual_enable	rw	0x1	write 1 enable manual mode

8	clk_32k_wifi_-sys_manual_enable	rw	0x1	write 1 enable manual mode
9	clk_mem_2x_manual_enable	rw	0x1	write 1 enable manual mode
10	clk_pll_div_-80m_manual_enable	rw	0x1	write 1 enable manual mode

45.37 others_manual_disable				0x40500114
Bits	Field Name	Acc	Default	Description
0	clk_26m_cpu_-sys_manual_disable	rw	0x1	write 1 disable manual mode
1	clk_52m_cpu_-sys_manual_disable	rw	0x1	write 1 disable manual mode
2	clk_32k_cpu_-sys_manual_disable	rw	0x1	write 1 disable manual mode
3	clk_26m_bt_s-ys_manual_disable	rw	0x1	write 1 disable manual mode
4	clk_52m_bt_s-ys_manual_disable	rw	0x1	write 1 disable manual mode
5	clk_32k_bt_sys_manual_disable	rw	0x1	write 1 disable manual mode
6	clk_26m_wifi_-sys_manual_disable	rw	0x1	write 1 disable manual mode
7	clk_52m_wifi_-sys_manual_disable	rw	0x1	write 1 disable manual mode
8	clk_32k_wifi_-sys_manual_disable	rw	0x1	write 1 disable manual mode
9	clk_mem_2x_manual_disable	rw	0x1	write 1 disable manual mode
10	clk_pll_div_-80m_manual_disable	rw	0x1	write 1 enable manual mode

45.38 aon_clk_manual_enable				0x40500118
Bits	Field Name	Acc	Default	Description
0	clk_26m_aon_-sys_manual_enable	rw	0x1	write 1 enable manual mode
1	clk_timer2_manual_enable	rw	0x1	write 1 enable manual mode
2	clk_32k_aon_-sys_manual_enable	rw	0x1	write 1 enable manual mode
3	clk_32k_bt_aon_manual_enable	rw	0x1	write 1 enable manual mode
4	clk_32k_wifi_-aon_manual_enable	rw	0x1	write 1 enable manual mode
5	clk_rtc_wdg2_manual_enable	rw	0x1	write 1 enable manual mode
6	clk_cm_rf_intf_manual_enable	rw	0x1	write 1 enable manual mode
7	clk_pmu_top_manual_enable	rw	0x1	write 1 enable manual mode
8	clk_aon_pwm_manual_enable	rw	0x1	write 1 enable manual mode
9	hclk_wcn_aon_manual_enable	rw	0x1	write 1 enable manual mode
10	hclk_psi_slv_manual_enable	rw	0x1	write 1 enable manual mode
11	pclk_sys_ctrl_manual_enable	rw	0x1	write 1 enable manual mode
12	pclk_timer2_manual_enable	rw	0x1	write 1 enable manual mode
13	pclk_analog_r-eg1_manual_enable	rw	0x1	write 1 enable manual mode
14	pclk_iomux_manual_enable	rw	0x1	write 1 enable manual mode
15	pclk_gpiob_manual_enable	rw	0x1	write 1 enable manual mode
16	pclk_pwr_ctrl_manual_enable	rw	0x1	write 1 enable manual mode
17	pclk_aon_pwm_manual_enable	rw	0x1	write 1 enable manual mode
18	pclk_wdg2_manual_enable	rw	0x1	write 1 enable manual mode
19	pclk_cm_rf_intf_manual_enable	rw	0x1	write 1 enable manual mode
20	pclk_psi_mst_manual_enable	rw	0x1	write 1 enable manual mode
21	pclk_aon_sys_-ctrl_manual_enable	rw	0x1	write 1 enable manual mode

45.39 aon_clk_manual_disable				0x4050011C
Bits	Field Name	Acc	Default	Description
0	clk_26m_aon_-sys_manual_disable	rw	0x1	write 1 disable manual mode
1	clk_timer2_manual_disable	rw	0x1	write 1 disable manual mode
2	clk_32k_aon_-sys_manual_disable	rw	0x1	write 1 disable manual mode
3	clk_32k_bt_a-on_manual_disable	rw	0x1	write 1 disable manual mode
4	clk_32k_wifi_-aon_manual_disable	rw	0x1	write 1 disable manual mode
5	clk_rtc_wdg2_manual_disable	rw	0x1	write 1 disable manual mode
6	clk_cm_rf_intf_manual_disable	rw	0x1	write 1 disable manual mode
7	clk_pmu_top_manual_disable	rw	0x1	write 1 disable manual mode
8	clk_aon_pwm_manual_disable	rw	0x1	write 1 disable manual mode
9	hclk_wcn_aon_manual_disable	rw	0x1	write 1 disable manual mode
10	hclk_psi_slv_manual_disable	rw	0x1	write 1 disable manual mode
11	pclk_sys_ctrl_manual_disable	rw	0x1	write 1 disable manual mode
12	pclk_timer2_manual_disable	rw	0x1	write 1 disable manual mode
13	pclk_analog_r-eg1_manual_disable	rw	0x1	write 1 disable manual mode
14	pclk_iomux_manual_disable	rw	0x1	write 1 disable manual mode
15	pclk_gpiob_manual_disable	rw	0x1	write 1 disable manual mode
16	pclk_pwr_ctrl_manual_disable	rw	0x1	write 1 disable manual mode
17	pclk_aon_pwm_manual_disable	rw	0x1	write 1 disable manual mode
18	pclk_wdg2_manual_disable	rw	0x1	write 1 disable manual mode
19	pclk_cm_rf_i-ntf_manual_disable	rw	0x1	write 1 disable manual mode
20	pclk_psi_mst_manual_disable	rw	0x1	write 1 disable manual mode
21	pclk_aon_sys_-ctrl_manual_disable	rw	0x1	write 1 disable manual mode

45.40 global_soft_resetn_set				0x40500120
Bits	Field Name	Acc	Default	Description
0	global_soft_resetn_set	rw	0x1	write 1 set soft reset

45.41 global_soft_resetn_clr				0x40500124
Bits	Field Name	Acc	Default	Description
0	global_soft_resetn_clear	rw	0x1	write 1 clear soft reset

45.42 hclk_soft_resetn_set				0x40500128
Bits	Field Name	Acc	Default	Description
0	hresetn_cpu_sys_set	rw	0x1	write 1 set soft reset
1	hresetn_bt_sys_set	rw	0x1	write 1 set soft reset
2	hresetn_bt_free_set	rw	0x1	write 1 set soft reset
3	hresetn_wifi_sys_set	rw	0x1	write 1 set soft reset
4	hresetn_vcore_top_set	rw	0x1	write 1 set soft reset
5	hresetn_cpu_p_set	rw	0x1	write 1 set soft reset
6	hresetn_cpu_s0_set	rw	0x0	write 1 set soft reset
7	hresetn_cpu_s1_set	rw	0x0	write 1 set soft reset

45.43 hclk_soft_resetn_clr				0x4050012C
Bits	Field Name	Acc	Default	Description
0	hresetn_cpu_sys_clear	rw	0x1	write 1 clear soft reset
1	hresetn_bt_sys_clear	rw	0x1	write 1 clear soft reset

2	hresetn_bt_free_clear	rw	0x1	write 1 clear soft reset
3	hresetn_wifi_sys_clear	rw	0x1	write 1 clear soft reset
4	hresetn_vcore_top_clear	rw	0x1	write 1 clear soft reset
5	hresetn_cpu_p_clear	rw	0x1	write 1 clear soft reset
6	hresetn_cpu_s0_clear	rw	0x0	write 1 clear soft reset
7	hresetn_cpu_s1_clear	rw	0x0	write 1 clear soft reset

45.44 fclk_soft_resetn_set				0x40500130
Bits	Field Name	Acc	Default	Description
0	fresetn_cpu_sys_set	rw	0x1	write 1 set soft reset

45.45 fclk_soft_resetn_clr				0x40500134
Bits	Field Name	Acc	Default	Description
0	fresetn_cpu_sys_clear	rw	0x1	write 1 clear soft reset

45.46 others_soft_resetn_set				0x40500138
Bits	Field Name	Acc	Default	Description
0	resetn_26m_cpu_sys_set	rw	0x1	write 1 set soft reset
1	resetn_52m_cpu_sys_set	rw	0x1	write 1 set soft reset
2	resetn_32k_cpu_sys_set	rw	0x1	write 1 set soft reset
3	resetn_26m_bt_sys_set	rw	0x1	write 1 set soft reset
4	resetn_52m_bt_sys_set	rw	0x1	write 1 set soft reset
5	resetn_32k_bt_sys_set	rw	0x1	write 1 set soft reset
6	resetn_32k_wifi_sys_set	rw	0x1	write 1 set soft reset
14	resetn_wifi_sys_set	rw	0x1	write 1 set soft reset
15	resetn_bt_sys_set	rw	0x1	write 1 set soft reset

45.47 others_soft_resetn_clr				0x4050013C
Bits	Field Name	Acc	Default	Description
0	resetn_26m_cpu_sys_clear	rw	0x1	write 1 clear soft reset
1	resetn_52m_cpu_sys_clear	rw	0x1	write 1 clear soft reset
2	resetn_32k_cpu_sys_clear	rw	0x1	write 1 clear soft reset
3	resetn_26m_bt_sys_clear	rw	0x1	write 1 clear soft reset
4	resetn_52m_bt_sys_clear	rw	0x1	write 1 clear soft reset
5	resetn_32k_bt_sys_clear	rw	0x1	write 1 clear soft reset
6	resetn_32k_wifi_sys_clear	rw	0x1	write 1 clear soft reset
14	resetn_wifi_sys_clear	rw	0x1	write 1 clear soft reset
15	resetn_bt_sys_clear	rw	0x1	write 1 clear soft reset

45.48 aon_soft_resetn_set				0x40500140
Bits	Field Name	Acc	Default	Description
0	resetn_26m_aon_sys_set	rw	0x1	write 1 set soft reset
1	resetn_timer2_set	rw	0x1	write 1 set soft reset
2	resetn_32k_aon_sys_set	rw	0x1	write 1 set soft reset
3	resetn_32k_bt_aon_set	rw	0x0	write 1 set soft reset
4	resetn_32k_wifi_aon_set	rw	0x0	write 1 set soft reset
5	resetn_rtc_wdg2_set	rw	0x1	write 1 set soft reset
6	resetn_cm_rf_intf_set	rw	0x1	write 1 set soft reset
7	resetn_pmu_top_set	rw	0x1	write 1 set soft reset
8	resetn_psi_slv_set	rw	0x1	write 1 set soft reset
9	resetn_aon_pwm_set	rw	0x1	write 1 set soft reset
10	resetn_pwr_ctrl_set	rw	0x1	write 1 set soft reset
11	hresetn_pmu_psi_slv_set	rw	0x1	write 1 set soft reset

12	hresetn_aon_matrix_set	rw	0x1	write 1 set soft reset
13	hresetn_cm0p_set	rw	0x1	write 1 set soft reset
14	hresetn_aon_sram_set	rw	0x1	write 1 set soft reset
15	hresetn_wcn_aon_set	rw	0x0	write 1 set soft reset
16	hresetn_psi_slv_set	rw	0x1	write 1 set soft reset
17	presetn_sys_ctrl_set	rw	0x1	write 1 set soft reset
18	presetn_timer2_set	rw	0x1	write 1 set soft reset
19	presetn_analog_reg1_set	rw	0x1	write 1 set soft reset
20	presetn_iomux_set	rw	0x1	write 1 set soft reset
21	presetn_gpiob_set	rw	0x1	write 1 set soft reset
22	presetn_pwr_ctrl_set	rw	0x1	write 1 set soft reset
23	presetn_aon_pwm_set	rw	0x1	write 1 set soft reset
24	presetn_wdg2_set	rw	0x1	write 1 set soft reset
25	presetn_cm_rf_intf_set	rw	0x1	write 1 set soft reset
26	presetn_psi_mst_set	rw	0x1	write 1 set soft reset
27	presetn_aon_sys_ctrl_set	rw	0x1	write 1 set soft reset
31	resetn_wcn_aon_set	rw	0x1	write 1 set soft reset

45.49 aon_soft_resetn_clr				0x40500144
Bits	Field Name	Acc	Default	Description
0	resetn_26m_aon_sys_clear	rw	0x1	write 1 clear soft reset
1	resetn_timer2_clear	rw	0x1	write 1 clear soft reset
2	resetn_32k_aon_sys_clear	rw	0x1	write 1 clear soft reset
3	resetn_32k_bt_aon_clear	rw	0x0	write 1 clear soft reset
4	resetn_32k_wifi_aon_clear	rw	0x0	write 1 clear soft reset
5	resetn_rtc_wdg2_clear	rw	0x1	write 1 clear soft reset
6	resetn_cm_rf_intf_clear	rw	0x1	write 1 clear soft reset
7	resetn_pmu_top_clear	rw	0x1	write 1 clear soft reset
8	resetn_psi_slv_clear	rw	0x1	write 1 clear soft reset
9	resetn_aon_pwm_clear	rw	0x1	write 1 clear soft reset
10	resetn_pwr_ctrl_clear	rw	0x1	write 1 clear soft reset
11	resetn_pmu_psi_slv_clear	rw	0x1	write 1 clear soft reset
12	hresetn_aon_matrix_clear	rw	0x1	write 1 clear soft reset
13	hresetn_cm0p_clear	rw	0x1	write 1 clear soft reset
14	hresetn_aon_sram_clear	rw	0x1	write 1 clear soft reset
15	hresetn_wcn_aon_clear	rw	0x0	write 1 clear soft reset
16	hresetn_psi_slv_clear	rw	0x1	write 1 clear soft reset
17	presetn_sys_ctrl_clear	rw	0x1	write 1 clear soft reset
18	presetn_timer2_clear	rw	0x1	write 1 clear soft reset
19	presetn_analog_reg1_clear	rw	0x1	write 1 clear soft reset
20	presetn_iomux_clear	rw	0x1	write 1 clear soft reset
21	presetn_gpiob_clear	rw	0x1	write 1 clear soft reset
22	presetn_pwr_ctrl_clear	rw	0x1	write 1 clear soft reset
23	presetn_aon_pwm_clear	rw	0x1	write 1 clear soft reset
24	presetn_wdg2_clear	rw	0x1	write 1 clear soft reset
25	presetn_cm_rf_intf_clear	rw	0x1	write 1 clear soft reset
26	presetn_psi_mst_clear	rw	0x1	write 1 clear soft reset
27	presetn_aon_sys_ctrl_clear	rw	0x1	write 1 clear soft reset
31	resetn_wcn_aon_clear	rw	0x1	

45.50 boot_mode				0x40500148
Bits	Field Name	Acc	Default	Description
3 : 0	hw_boot_mode	rw	0x0	hardware boot pins
7 : 4	sw_boot_mode	rw	0x0	softeware boot mode

45.51 wdg_reset_en				0x4050014C
Bits	Field Name	Acc	Default	Description
0	wdg0_rst_en	rw	0x0	watchdog0 reset enable, high active
1	wdg1_rst_en	rw	0x0	watchdog1 reset enable, high active
2	wdg2_rst_en	rw	0x0	watchdog2 reset enable, high active
3	bt_wdg_rst_en	rw	0x0	BT watchdog reset enable, high active
4	cpu_p_sysrst_en	rw	0x0	CPU_P system reset enable, high active
5	cpu_s0_sysrst_en	rw	0x0	CPU_S0 system reset enable, high active
6	cpu_s1_sysrst_en	rw	0x0	CPU_S1 system reset enable, high active
8	perst_rst_en	rw	0x0	PCIE perst reset enable, high active

45.52 reserved_reg0				0x40500150
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg0	rw	0x0	SW reserved register 0

45.53 reserved_reg1				0x40500154
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg1	rw	0x0	SW reserved register 1

45.54 reserved_reg2				0x40500158
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg2	rw	0x0	SW reserved register 2

45.55 reserved_reg3				0x4050015C
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg3	rw	0x0	SW reserved register 3

45.56 reserved_reg4				0x40500160
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg4	rw	0x0	SW reserved register 4

45.57 reserved_reg5				0x40500164
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg5	rw	0x0	SW reserved register 5

45.58 reserved_reg6				0x40500168
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg6	rw	0x0	SW reserved register 6

45.59 reserved_reg7				0x4050016C
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg7	rw	0x0	SW reserved register 7

45.60 reserved_reg8				0x40500170
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg8	rw	0x0	SW reserved register 8

45.61 reserved_reg9	0x40500174

Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg9	rw	0x0	SW reserved register 9

45.62 reserved_reg10				0x40500178
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg10	rw	0x0	SW reserved register 10

45.63 reserved_reg11				0x4050017C
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg11	rw	0x0	SW reserved register 11

45.64 reserved_reg12				0x40500180
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg12	rw	0x0	SW reserved register 12

45.65 reserved_reg13				0x40500184
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg13	rw	0x0	SW reserved register 13

45.66 reserved_reg14				0x40500188
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg14	rw	0x0	SW reserved register 14

45.67 reserved_reg15				0x4050018C
Bits	Field Name	Acc	Default	Description
31 : 0	sw_res_reg15	rw	0x0	SW reserved register 15

46 timer2

46.1 TIMER0_CTL				0x40501000
Bits	Field Name	Acc	Default	Description
0	timer0_en	rw	0x0	timer0 enable
1	timer0_reset	rw	0x0	timer0 reset, active high, cleared automatically. When one, the Timer Counter is synchronously reset on the next positive edge of PCLK.
3	timer0_tick_div_bypass	ro	0x0	1, bypass tick divider, TIMER0_TICK_DIV register does not exist. 0, tick divider exists.
7 : 4	timer0_tc_hi_dis	rw	0x0	1, the High 16bit of TC and MR is ignored when generating match/wrap interrupt. 0, this feature is ignored.
9 : 8	timer0_tick_freq	rw	0x1	timer0 tick frequency, used to select TC and adjust counter increment value. 2'b00: tick is 64KHz 2'b01: tick is 64KHz 2'b10: tick is 32KHz 2'b11: tick is 16KHz
12	timer0_match0_en	rw	0x0	1, match0 enable. 0, match0 disable.
13	timer0_match1_en	rw	0x0	1, match1 enable. 0, match1 disable.
14	timer0_match2_en	rw	0x0	1, match2 enable. 0, match2 disable.

15	timer0_match3_en	rw	0x0	1, match3 enable. 0, match3 disable.

46.2 TIMER0_TICK_DIV				0x40501004
Bits	Field Name	Acc	Default	Description
1 : 0	timer0_tick_div	rw	0x1	timer0 tick div value 2'b00: disable tick 2'b01: tick is div2 from rtc clock 2'b10: tick is div4 from rtc clock 2'b11: tick is div8 from rtc clock
4	timer0_tick_div_up	wo	0x0	timer0 tick div value update, cleared automatically

46.3 TIMER0_INT_CTRL				0x40501008
Bits	Field Name	Acc	Default	Description
0	timer0_match0_int_en_set	rw	0x0	Write: timer0 match0 interrupt enable set 1: Set Enable. An interrupt is generated when MR0 matches the value in the TC. 0: Ignored. Read: timer0 match0 interrupt enabled 1: Enalbed. An interrupt is generated when MR0 matches the value in the TC. 0: not Enabled.
1	timer0_match1_int_en_set	rw	0x0	Write: timer0 match1 interrupt enable set 1: Set Enable. An interrupt is generated when MR1 matches the value in the TC. 0: Ignored. Read: timer0 match1 interrupt enabled 1: Enalbed. An interrupt is generated when MR1 matches the value in the TC. 0: not Enabled.
2	timer0_match2_int_en_set	rw	0x0	Write: timer0 match2 interrupt enable set 1: Set Enable. An interrupt is generated when MR2 matches the value in the TC. 0: Ignored. Read: timer0 match2 interrupt enabled 1: Enalbed. An interrupt is generated when MR2 matches the value in the TC. 0: not Enabled.
3	timer0_match3_int_en_set	rw	0x0	Write: timer0 match3 interrupt enable set 1: Set Enable. An interrupt is generated when MR3 matches the value in the TC. 0: Ignored. Read: timer0 match3 interrupt enabled 1: Enalbed. An interrupt is generated when MR3 matches the value in the TC. 0: not Enabled.
4	timer0_wrap0_int_en_set	rw	0x0	Write: timer0 wrap0 interrupt enable set 1: Set Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored. Read: timer0 wrap0 interrupt enabled 1: Enalbed. An interrupt is generated when TC count from all1 to 0x0. 0: not Enabled.
5	timer0_wrap1_int_en_set	rw	0x0	Write: timer0 wrap1 interrupt enable set 1: Set Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored. Read: timer0 wrap1 interrupt enabled 1: Enalbed. An interrupt is generated when TC count from all1 to 0x0. 0: not Enabled.

6	timer0_wrap2_int_en_set	rw	0x0	Write: timer0 wrap2 interrupt enable set 1: Set Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored. Read: timer0 wrap2 interrupt enabled 1: Enalbed. An interrupt is generated when TC count from all1 to 0x0. 0: not Enabled.
7	timer0_wrap3_int_en_set	rw	0x0	Write: timer0 wrap3 interrupt enable set 1: Set Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored. Read: timer0 wrap3 interrupt enabled 1: Enalbed. An interrupt is generated when TC count from all1 to 0x0. 0: not Enabled.
8	timer0_match0_int_en_clr	wo	0x0	Timer0 match0 interrupt enable clear 1: Clear Enable. 0: Ignored.
9	timer0_match1_int_en_clr	wo	0x0	Timer0 match1 interrupt enable clear 1: Clear Enable. 0: Ignored.
10	timer0_match2_int_en_clr	wo	0x0	Timer0 match2 interrupt enable clear 1: Clear Enable. 0: Ignored.
11	timer0_match3_int_en_clr	wo	0x0	Timer0 match3 interrupt enable clear 1: Clear Enable. 0: Ignored.
12	timer0_wrap0_int_en_clr	wo	0x0	Timer0 wrap0 interrupt enable clear 1: Clear Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored.
13	timer0_wrap1_int_en_clr	wo	0x0	Timer0 wrap1 interrupt enable clear 1: Clear Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored.
14	timer0_wrap2_int_en_clr	wo	0x0	Timer0 wrap2 interrupt enable clear 1: Clear Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored.
15	timer0_wrap3_int_en_clr	wo	0x0	Timer0 wrap3 interrupt enable clear 1: Clear Enable. An interrupt is generated when TC count from all1 to 0x0. 0: Ignored.
16	timer0_match0_int_clr	wo	0x0	timer0 match0 interrupt clear, active high
17	timer0_match1_int_clr	wo	0x0	timer0 match1 interrupt clear, active high
18	timer0_match2_int_clr	wo	0x0	timer0 match2 interrupt clear, active high
19	timer0_match3_int_clr	wo	0x0	timer0 match3 interrupt clear, active high
20	timer0_wrap0_int_clr	wo	0x0	timer0 wrap0 interrupt clear
21	timer0_wrap1_int_clr	wo	0x0	timer0 wrap1 interrupt clear
22	timer0_wrap2_int_clr	wo	0x0	timer0 wrap2 interrupt clear
23	timer0_wrap3_int_clr	wo	0x0	timer0 wrap3 interrupt clear

46.4 TIMER0_INT_STATUS				0x4050100C
Bits	Field Name	Acc	Default	Description
0	timer0_match0_int_raw_status	ro	0x0	timer0 match0 interrupt raw status
1	timer0_match1_int_raw_status	ro	0x0	timer0 match1 interrupt raw status
2	timer0_match2_int_raw_status	ro	0x0	timer0 match2 interrupt raw status
3	timer0_match3_int_raw_status	ro	0x0	timer0 match3 interrupt raw status
4	timer0_wrap0_int_raw_status	ro	0x0	timer0 wrap0 interrupt raw status
5	timer0_wrap1_int_raw_status	ro	0x0	timer0 wrap1 interrupt raw status
6	timer0_wrap2_int_raw_status	ro	0x0	timer0 wrap2 interrupt raw status
7	timer0_wrap3_int_raw_status	ro	0x0	timer0 wrap3 interrupt raw status
8	timer0_match0_- int_masked_status	ro	0x0	timer0 match0 interrupt masked status

9	timer0_match1_- int_masked_status	ro	0x0	timer0 match1 interrupt masked status
10	timer0_match2_- int_masked_status	ro	0x0	timer0 match2 interrupt masked status
11	timer0_match3_- int_masked_status	ro	0x0	timer0 match3 interrupt masked status
12	timer0_wrap0_- int_masked_status	ro	0x0	timer0 wrap0 interrupt masked status
13	timer0_wrap1_- int_masked_status	ro	0x0	timer0 wrap1 interrupt masked status
14	timer0_wrap2_- int_masked_status	ro	0x0	timer0 wrap2 interrupt masked status
15	timer0_wrap3_- int_masked_status	ro	0x0	timer0 wrap3 interrupt masked status

46.5 TIMER0_TC_LO 0x40501010

Bits	Field Name	Acc	Default	Description
31 : 0	timer0_tc_lo	rw	0x0	timer0 counter low 32bit The 48-bit TC, unit is 1us.

46.6 TIMER0_TC_HI 0x40501014

Bits	Field Name	Acc	Default	Description
15 : 0	timer0_tc_hi	rw	0x0	timer0 counter high 16bit The 48-bit TC, unit is 1us. Make sure that this high 16bit written before the low 16bit if initialized.

46.7 TIMER0_MR0_LO 0x40501020

Bits	Field Name	Acc	Default	Description
31 : 0	timer0_mr0_lo	rw	0xFFFFFFFF	timer0 match0 register, low 32bit MR0 can be enabled ot generate an interrupt every time MR0 matches the TC

46.8 TIMER0_MR0_HI 0x40501024

Bits	Field Name	Acc	Default	Description
15 : 0	timer0_mr0_hi	rw	0x0	timer0 match0 register, high 16bit MR0 can be enabled to generate an interrupt every time MR0 matches the TC Make sure that this high 16bit configured before the low 16bit.

46.9 TIMER0_MR1_LO 0x40501028

Bits	Field Name	Acc	Default	Description
31 : 0	timer0_mr1_lo	rw	0xFFFFFFFF	timer0 match1 register, low 32bit MR1 can be enabled ot generate an interrupt every time MR1 matches the TC

46.10 TIMER0_MR1_HI 0x4050102C

Bits	Field Name	Acc	Default	Description
15 : 0	timer0_mr1_hi	rw	0x0	timer0 match1 register, high 16bit MR1 can be enabled to generate an interrupt every time MR1 matches the TC Make sure that this high 16bit configured before the low 16bit.

46.11 TIMER0_MR2_LO 0x40501030

Bits	Field Name	Acc	Default	Description
31 : 0	timer0_mr2_lo	rw	0xFFFFFFFF	timer0 match2 register, low 32bit MR2 can be enabled ot generate an interrupt every time MR2 matches the TC

46.12 TIMER0_MR2_HI				0x40501034
Bits	Field Name	Acc	Default	Description
15 : 0	timer0_mr2_hi	rw	0x0	timer0 match2 register, high 16bit MR2 can be enabled to generate an interrupt every time MR2 matches the TC Make sure that this high 16bit configured before the low 16bit.

46.13 TIMER0_MR3_LO				0x40501038
Bits	Field Name	Acc	Default	Description
31 : 0	timer0_mr3_lo	rw	0xFFFFFFFF	timer0 match3 register, low 32bit MR3 can be enabled ot generate an interrupt every time MR3 matches the TC

46.14 TIMER0_MR3_HI				0x4050103C
Bits	Field Name	Acc	Default	Description
15 : 0	timer0_mr3_hi	rw	0x0	timer0 match3 register, high 16bit MR3 can be enabled to generate an interrupt every time MR3 matches the TC Make sure that this high 16bit configured before the low 16bit.

47 ana_reg1

47.1 gpioa_cfg				0x40502000
Bits	Field Name	Acc	Default	Description
2 : 0	cfg_ana_gpioa_ibit	rw	0x7	GPIOA drive strength
5 : 3	cfg_ana_gpioa_pull_up_sel	rw	0x0	
11 : 6	cfg_ana_gpioa_slew_rate	rw	0x3F	GPIOA slew rate selection
12	cfg_ana_gpioa_ie	rw	0x1	GPIOA IE
13	cfg_ana_gpioa_mux_en	rw	0x0	
14	cfg_ana_gpioa_mux_enb	rw	0x1	
31 : 16	cfg_ana_gpioa- _wakeup_polarity	rw	0x0	

47.2 gpiob_cfg1				0x40502004
Bits	Field Name	Acc	Default	Description
1 : 0	cfg_ana_gpiob_ibit	rw	0x3	GPIOB drive strength
3 : 2	cfg_ana_gpiob_pull_up_sel	rw	0x0	
5	cfg_ana_gpiob_mux_en	rw	0x0	
6	cfg_ana_gpiob_mux_enb	rw	0x1	
7	cfg_gpiob_func_mux_sel	rw	0x0	
15 : 8	cfg_ana_gpiob- _wakeup_polarity	rw	0x0	
23 : 16	cfg_gpiob_pull_up_int	rw	0x3	
31 : 24	cfg_gpiob_pull_dn_int	rw	0xFC	

47.3 gpiob_cfg2				0x40502008
Bits	Field Name	Acc	Default	Description

3 : 0	cfg_ana_gpiob_slew_rate	rw	0xF	GPIOB slew rate selection
5 : 4	cfg_ana_gpiob_ie	rw	0x3	GPIOB IE

47.4 agpio_cfg				0x4050200C
Bits	Field Name	Acc	Default	Description
0	cfg_ana_agpio_ibit	rw	0x1	AGPIO drive strength
8	cfg_ana_agpio_ie	rw	0x0	AGPIO IE

47.5 ana_port_ctrl				0x40502018
Bits	Field Name	Acc	Default	Description
0	cfg_ana_test_enable_usb	rw	0x0	
1	cfg_ana_test_enable_pcie	rw	0x0	
2	cfg_ana_test_enable_iref_io	rw	0x0	
5	cfg_ana_pu_iref_io	rw	0x0	
6	cfg_ana_iref_io_start_enb	rw	0x1	
7	cfg_ana_pmic_int_pull_dn	rw	0x1	
8	cfg_ana_pmic_int_pull_up	rw	0x0	
9	cfg_ana_test_enable_pcie_pll	rw	0x0	
10	cfg_ana_gpio_-vio13_sleep_mode	rw	0x0	
14 : 11	cfg_ana_io_test_bit	rw	0x0	

47.6 misc_reg				0x4050201C
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_ana_io_rsvd_bit	rw	0xF0	analog reserved bits
8	cfg_ana_xtal2aud_ibit	rw	0x1	
9	cfg_xtal2aud_en	rw	0x0	
10	cfg_ana_pwrkey2aud_i	rw	0x0	
11	cfg_ana_pwrkey2aud_oen	rw	0x0	
12	pmu_psi_slv_-rdata_wait_timeout_err	ro	0x0	
20 : 13	psi_mst_sig_event	rw	0x0	
24 : 21	cfg_ana_rsvd_bit_core	rw	0xC	
25	cfg_bt_bbpll_auto_pu	rw	0x0	
26	cfg_ana_chip_sleep_i_en	rw	0x0	

47.7 flash_ctrl				0x40502020
Bits	Field Name	Acc	Default	Description
0	cfg_ana_fls_ie	rw	0x1	
1	cfg_ana_fls_ibit	rw	0x1	
2	cfg_ana_fls_pull_up_sel	rw	0x0	
4 : 3	cfg_ana_fls_slew_rate	rw	0x3	
5	cfg_ana_fls_ms	rw	0x0	

47.8 ana_bit_rd_back				0x40502024
Bits	Field Name	Acc	Default	Description
3 : 0	pmu_rsvd_read_bit_rtc1	ro	0x0	
7 : 4	pmu_rsvd_read_bit_core	ro	0x0	
11 : 8	io_rsvd_read_bit	ro	0x0	

48 psi_mst

48.1 ip_revision				0x40503000
Bits	Field Name	Acc	Default	Description
15 : 0	ip_revision	ro	0x200	revision id

48.2 psi_mst_ctrl				0x40503004
Bits	Field Name	Acc	Default	Description
0	write_bit	rw	0x0	write bit flag configure

48.3 psi_mst_pri00				0x40503008
Bits	Field Name	Acc	Default	Description
2 : 0	chnl0_pri	rw	0x0	
5 : 3	chnl1_pri	rw	0x0	
8 : 6	chnl2_pri	rw	0x0	
11 : 9	chnl3_pri	rw	0x0	
14 : 12	chnl4_pri	rw	0x0	
17 : 15	chnl5_pri	rw	0x0	

48.4 psi_mst_pri10				0x4050300C
Bits	Field Name	Acc	Default	Description
2 : 0	chnl10_pri	rw	0x0	
5 : 3	chnl11_pri	rw	0x0	
8 : 6	chnl12_pri	rw	0x0	
11 : 9	chnl13_pri	rw	0x0	

48.5 psi_mst_pri20				0x40503010
Bits	Field Name	Acc	Default	Description
2 : 0	chnl20_pri	rw	0x0	
5 : 3	chnl21_pri	rw	0x0	
8 : 6	chnl22_pri	rw	0x0	
11 : 9	chnl23_pri	rw	0x0	
14 : 12	chnl24_pri	rw	0x0	
17 : 15	chnl25_pri	rw	0x0	
20 : 18	chnl26_pri	rw	0x0	
23 : 21	chnl27_pri	rw	0x0	

48.6 psi_mst_pri30				0x40503014
Bits	Field Name	Acc	Default	Description

48.7 psi_mst_int_en				0x40503018
Bits	Field Name	Acc	Default	Description
0	evt0_fifo_ovf_int_en	rw	0x0	
1	evt1_fifo_ovf_int_en	rw	0x0	
2	evt2_fifo_ovf_int_en	rw	0x0	
3	evt3_fifo_ovf_int_en	rw	0x0	
4	rx_ssc_timeout_err_int_en	rw	0x0	
31	psi_fifo_ovf_int_en	rw	0x0	

48.8 psi_mst_raw_status				0x4050301C
Bits	Field Name	Acc	Default	Description
0	evt0_fifo_ovf_raw_sts	ro	0x0	
1	evt1_fifo_ovf_raw_sts	ro	0x0	

2	evt2_fifo_ovf_raw_sts	ro	0x0	
3	evt3_fifo_ovf_raw_sts	ro	0x0	
4	rx_ssc_timeout_err_raw_sts	ro	0x0	
31	psi_fifo_ovf_raw_sts	ro	0x0	

48.9 psi_mst_mask_status				0x40503020
Bits	Field Name	Acc	Default	Description
0	evt0_fifo_ovf_mask_sts	ro	0x0	
1	evt1_fifo_ovf_mask_sts	ro	0x0	
2	evt2_fifo_ovf_mask_sts	ro	0x0	
3	evt3_fifo_ovf_mask_sts	ro	0x0	
4	rx_ssc_timeout_err_mask_sts	ro	0x0	
31	psi_fifo_ovf_mask_sts	ro	0x0	

48.10 psi_mst_int_clr				0x40503024
Bits	Field Name	Acc	Default	Description
0	evt0_fifo_ovf_int_clr	w1c	0x0	
1	evt1_fifo_ovf_int_clr	w1c	0x0	
2	evt2_fifo_ovf_int_clr	w1c	0x0	
3	evt3_fifo_ovf_int_clr	w1c	0x0	
4	rx_ssc_timeout_err_int_clr	w1c	0x0	
31	psi_fifo_ovf_int_clr	w1c	0x0	

48.11 psi_gssi_cfg0				0x40503028
Bits	Field Name	Acc	Default	Description
6 : 0	rf_gssi_frame_len	rw	0x7	
12 : 7	rf_gssi_cmd_len	rw	0x6	
18 : 13	rf_gssi_data_len	rw	0x6	
24 : 19	rf_gssi_wr_pos	rw	0x6	
25	rf_gssi_wr_pol	rw	0x0	
26	rf_gssi_sck_rev	rw	0x0	
27	rf_gssi_oe_cfg	rw	0x0	
28	rf_gssi_ie_cfg	rw	0x0	
29	rf_gssi_dummy_clk_en	rw	0x1	
30	rf_gssi_fast_mode	rw	0x0	
31	rf_gssi_sck_all_on	rw	0x0	

48.12 psi_gssi_cfg1				0x4050302C
Bits	Field Name	Acc	Default	Description
0	rf_gssi_ng_tx	rw	0x1	
1	rf_gssi_ng_rx	rw	0x1	
9 : 2	rf_gssi_clk_div	rw	0x1	
15 : 10	rf_gssi_wbp_len	rw	0x8	
19 : 16	rf_gssi_dummy_len	rw	0x4	
20	rf_gssi_sample_dly	rw	0x0	
21	rf_gssi_wr_disable	rw	0x0	
22	rf_gssi_rx_ssc_mode	rw	0x1	1: wait rx sync mode , 0: wait fix ssc len mode
25 : 23	rf_gssi_rbp_len	rw	0x0	

48.13 psi_arm_rd_cmd_addr				0x40503030
Bits	Field Name	Acc	Default	Description
0	psi_busy	ro	0x0	
30 : 1	psi_arm_rd_cmd_addr	ro	0x0	

31	psi_arm_rd_req	ro	0x0	

48.14 psi_arm_rd_cmd_data				0x40503034
Bits	Field Name	Acc	Default	Description
31 : 0	psi_arm_rd_cmd_data	ro	0x0	

48.15 psi_arm_debug0				0x40503038
Bits	Field Name	Acc	Default	Description
2 : 0	fifo_wptr	ro	0x0	
6 : 4	fifo_rptr	ro	0x0	
10	fifo_empty_r	ro	0x0	
11	fifo_full_w	ro	0x0	

48.16 psi_arm_debug1				0x4050303C
Bits	Field Name	Acc	Default	Description
5 : 0	psi_sel	ro	0x0	
6	psi_trans_req_pending	ro	0x0	
11 : 9	current_state	ro	0x0	
16 : 13	st_gssi	ro	0x0	

48.17 psi_arm_debug2				0x40503040
Bits	Field Name	Acc	Default	Description
0	evt0_fifo_ovf_req	ro	0x0	
1	evt0_fifo_empty	ro	0x0	
3 : 2	evt0_fifo_rptr	ro	0x0	
5 : 4	evt0_fifo_wptr	ro	0x0	
8	evt1_fifo_ovf_req	ro	0x0	
9	evt1_fifo_empty	ro	0x0	
11 : 10	evt1_fifo_rptr	ro	0x0	
13 : 12	evt1_fifo_wptr	ro	0x0	
16	evt2_fifo_ovf_req	ro	0x0	
17	evt2_fifo_empty	ro	0x0	
19 : 18	evt2_fifo_rptr	ro	0x0	
21 : 20	evt2_fifo_wptr	ro	0x0	
24	evt3_fifo_ovf_req	ro	0x0	
25	evt3_fifo_empty	ro	0x0	
27 : 26	evt3_fifo_rptr	ro	0x0	
29 : 28	evt3_fifo_wptr	ro	0x0	

48.18 psi_cmd_status				0x40503044
Bits	Field Name	Acc	Default	Description
0	arm_wr_req	ro	0x0	
1	arm_rd_req	ro	0x0	
2	chnl2_req	ro	0x0	
3	chnl3_req	ro	0x0	
4	chnl4_req	ro	0x0	
5	chnl5_req	ro	0x0	
10	chnl10_req	ro	0x0	
11	chnl11_req	ro	0x0	
12	chnl12_req	ro	0x0	
13	chnl13_req	ro	0x0	
20	chnl20_req	ro	0x0	
21	chnl21_req	ro	0x0	

22	chnl22_req	ro	0x0	
23	chnl23_req	ro	0x0	
24	chnl24_req	ro	0x0	
25	chnl25_req	ro	0x0	
26	chnl26_req	ro	0x0	
27	chnl27_req	ro	0x0	

48.19 psi_chnl_en				0x40503048
Bits	Field Name	Acc	Default	Description
2	chnl2_trans_en	rw	0x0	
3	chnl3_trans_en	rw	0x0	
4	chnl4_trans_en	rw	0x0	
5	chnl5_trans_en	rw	0x0	
10	chnl10_trans_en	rw	0x0	
11	chnl11_trans_en	rw	0x0	
12	chnl12_trans_en	rw	0x0	
13	chnl13_trans_en	rw	0x0	
20	chnl20_trans_en	rw	0x0	
21	chnl21_trans_en	rw	0x0	
22	chnl22_trans_en	rw	0x0	
23	chnl23_trans_en	rw	0x0	
24	chnl24_trans_en	rw	0x0	
25	chnl25_trans_en	rw	0x0	
26	chnl26_trans_en	rw	0x0	
27	chnl27_trans_en	rw	0x0	

48.20 chnl2_wr_addr				0x4050304C
Bits	Field Name	Acc	Default	Description
31 : 0	chnl2_waddr	rw	0x0	

48.21 chnl3_wr_addr				0x40503050
Bits	Field Name	Acc	Default	Description
31 : 0	chnl3_waddr	rw	0x0	

48.22 chnl4_wr_addr				0x40503054
Bits	Field Name	Acc	Default	Description
31 : 0	chnl4_waddr	rw	0x0	

48.23 chnl5_wr_addr				0x40503058
Bits	Field Name	Acc	Default	Description
31 : 0	chnl5_waddr	rw	0x0	

48.24 chnl10_wr_addr				0x4050306C
Bits	Field Name	Acc	Default	Description
31 : 0	chnl10_waddr	rw	0x0	

48.25 chnl11_wr_addr				0x40503070
Bits	Field Name	Acc	Default	Description
31 : 0	chnl11_waddr	rw	0x0	

48.26 chnl12_wr_addr				0x40503074
Bits	Field Name	Acc	Default	Description
31 : 0	chnl12_waddr	rw	0x0	

48.27 chnl13_wr_addr				0x40503078
Bits	Field Name	Acc	Default	Description
31 : 0	chnl13_waddr	rw	0x0	

48.28 chnl20_wr_addr				0x40503094
Bits	Field Name	Acc	Default	Description
31 : 0	chnl20_waddr	rw	0x0	

48.29 chnl21_wr_addr				0x40503098
Bits	Field Name	Acc	Default	Description
31 : 0	chnl21_waddr	rw	0x0	

48.30 chnl22_wr_addr				0x4050309C
Bits	Field Name	Acc	Default	Description
31 : 0	chnl22_waddr	rw	0x0	

48.31 chnl23_wr_addr				0x405030A0
Bits	Field Name	Acc	Default	Description
31 : 0	chnl23_waddr	rw	0x0	

48.32 chnl24_wr_addr				0x405030A4
Bits	Field Name	Acc	Default	Description
31 : 0	chnl24_waddr	rw	0x0	

48.33 chnl25_wr_addr				0x405030A8
Bits	Field Name	Acc	Default	Description
31 : 0	chnl25_waddr	rw	0x0	

48.34 chnl26_wr_addr				0x405030AC
Bits	Field Name	Acc	Default	Description
31 : 0	chnl26_waddr	rw	0x0	

48.35 chnl27_wr_addr				0x405030B0
Bits	Field Name	Acc	Default	Description
31 : 0	chnl27_waddr	rw	0x0	

48.36 chnl2_wr_low_data				0x40503140
Bits	Field Name	Acc	Default	Description
31 : 0	chnl2_wr_low_data	rw	0x0	

48.37 chnl2_wr_high_data				0x40503144
Bits	Field Name	Acc	Default	Description
31 : 0	chnl2_wr_high_data	rw	0x0	

48.38 chnl3_wr_low_data				0x40503148
Bits	Field Name	Acc	Default	Description
31 : 0	chnl3_wr_low_data	rw	0x0	

48.39 chnl3_wr_high_data				0x4050314C
Bits	Field Name	Acc	Default	Description
31 : 0	chnl3_wr_high_data	rw	0x0	

48.40 chnl4_wr_low_data				0x40503150
Bits	Field Name	Acc	Default	Description
31 : 0	chnl4_wr_low_data	rw	0x0	

48.41 chnl4_wr_high_data				0x40503154
Bits	Field Name	Acc	Default	Description
31 : 0	chnl4_wr_high_data	rw	0x0	

48.42 chnl5_wr_low_data				0x40503158
Bits	Field Name	Acc	Default	Description
31 : 0	chnl5_wr_low_data	rw	0x0	

48.43 chnl5_wr_high_data				0x4050315C
Bits	Field Name	Acc	Default	Description
31 : 0	chnl5_wr_high_data	rw	0x0	

48.44 chnl10_wr_low_data				0x40503180
Bits	Field Name	Acc	Default	Description
31 : 0	chnl10_wr_low_data	rw	0x0	

48.45 chnl10_wr_high_data				0x40503184
Bits	Field Name	Acc	Default	Description
31 : 0	chnl10_wr_high_data	rw	0x0	

48.46 chnl11_wr_low_data				0x40503188
Bits	Field Name	Acc	Default	Description
31 : 0	chnl11_wr_low_data	rw	0x0	

48.47 chnl11_wr_high_data				0x4050318C
Bits	Field Name	Acc	Default	Description
31 : 0	chnl11_wr_high_data	rw	0x0	

48.48 chnl12_wr_low_data				0x40503190
Bits	Field Name	Acc	Default	Description
31 : 0	chnl12_wr_low_data	rw	0x0	

48.49 chnl12_wr_high_data				0x40503194
Bits	Field Name	Acc	Default	Description
31 : 0	chnl12_wr_high_data	rw	0x0	

48.50 chnl13_wr_low_data				0x40503198
Bits	Field Name	Acc	Default	Description
31 : 0	chnl13_wr_low_data	rw	0x0	

48.51 chnl13_wr_high_data				0x4050319C
Bits	Field Name	Acc	Default	Description
31 : 0	chnl13_wr_high_data	rw	0x0	

48.52 evt_trg_cfg00				0x405031C0
Bits	Field Name	Acc	Default	Description
1 : 0	evt0_fifo_trg_cfg	rw	0x0	
3 : 2	evt1_fifo_trg_cfg	rw	0x0	
5 : 4	evt2_fifo_trg_cfg	rw	0x0	
7 : 6	evt3_fifo_trg_cfg	rw	0x0	

48.53 evt_trg_cfg10				0x405031C4
Bits	Field Name	Acc	Default	Description
1 : 0	evt10_nofifo_trg_cfg	rw	0x0	
3 : 2	evt11_nofifo_trg_cfg	rw	0x0	
5 : 4	evt12_nofifo_trg_cfg	rw	0x0	
7 : 6	evt13_nofifo_trg_cfg	rw	0x0	

48.54 psi_rd_cmd_addr				0x40503200
Bits	Field Name	Acc	Default	Description
31 : 0	psi_rd_cmd_addr	rw	0x0	

48.55 psi_wr_cmd_addr				0x40503204
Bits	Field Name	Acc	Default	Description
31 : 0	psi_wr_cmd_addr	rw	0x0	

48.56 psi_wr_cmd_data				0x40503208
Bits	Field Name	Acc	Default	Description
31 : 0	psi_wr_cmd_data	rw	0x0	

48.57 psi_timeout_cfg				0x4050320C
Bits	Field Name	Acc	Default	Description
15 : 0	rx_ssc_timeout_th	rw	0x40	

48.58 pmu_psi_slv_cfg0				0x40503210
Bits	Field Name	Acc	Default	Description
29 : 0	pmu_psi_addr_ll_th	rw	0x1C000000	
30	pmu_psi_auto_det_en	rw	0x0	
31	pmu_psi_force_en	rw	0x1	

48.59 pmu_psi_slv_cfg1				0x40503214
Bits	Field Name	Acc	Default	Description
29 : 0	pmu_psi_addr_hh_th	rw	0x1C003FFF	

49 iomux

49.1 pad_gpioa_0_cfg_reg				0x40504000
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_0_sel	rw	0x0	gpioa_0 select 0 fun_cpu_p_swclk_sel 1 fun_gpioa_0_sel 2 fun_i2cm_scl_sel 3 fun_wf_ext_pa_ctrl_0_sel 4 fun_pcm_fsync_sel 5 fun_pta_ant_sw_0_sel 6 fun_i2s_lrck_0_sel 7 fun_tports0_sel 8 fun_spdif_in_sel 9 fun_spi_lcd_sck_sel 10 fun_pwm_0_sel 11 fun_pcm_dout_sel 12 fun_pcm_clk_sel 13 fun_bt_uart_cts_sel
7 : 4	pad_gpioa_0_ana	rw	0x0	gpioa_0 ANA_Alter_SEL
8	pad_gpioa_0_pull_dn	rw	0x0	gpioa_0 PUll down
9	pad_gpioa_0_pull_up	rw	0x0	gpioa_0 PUll up
16	pad_gpioa_0_pull_frc	rw	0x0	gpioa_0 force enable for pu/pd
17	pad_gpioa_0_oen_reg	rw	0x0	gpioa_0 force outoen value.
19 : 18	pad_gpioa_0_drv_strength	rw	0x0	gpioa_0 driving strength.
20	pad_gpioa_0_out_reg	rw	0x0	gpioa_0 pin output value.
24	pad_gpioa_0_out_frc	rw	0x0	gpioa_0 force output value for output.
28	pad_gpioa_0_oen_frc	rw	0x0	gpioa_0 force enable for outoen.

49.2 pad_gpioa_1_cfg_reg				0x40504004
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_1_sel	rw	0x0	gpioa_1 select 0 fun_cpu_p_swd_sel 1 fun_gpioa_1_sel 2 fun_i2cm_sda_sel 3 fun_wf_ext_pa_ctrl_1_sel 4 fun_pcm_clk_sel 5 fun_pta_ant_sw_1_sel 6 fun_i2s_bck_0_sel 7 fun_tports1_sel 8 fun_spdif_out_sel 9 fun_spi_lcd_csn_0_sel 10 fun_pwm_1_sel 11 fun_pcm_din_sel 12 fun_pcm_dout_sel 13 fun_bt_uart_rts_sel
7 : 4	pad_gpioa_1_ana	rw	0x0	gpioa_1 ANA_Alter_SEL
8	pad_gpioa_1_pull_dn	rw	0x0	gpioa_1 PUll down
9	pad_gpioa_1_pull_up	rw	0x0	gpioa_1 PUll up
16	pad_gpioa_1_pull_frc	rw	0x0	gpioa_1 force enable for pu/pd
17	pad_gpioa_1_oen_reg	rw	0x0	gpioa_1 force outoen value.
19 : 18	pad_gpioa_1_drv_strength	rw	0x0	gpioa_1 driving strength.
20	pad_gpioa_1_out_reg	rw	0x0	gpioa_1 pin output value.
24	pad_gpioa_1_out_frc	rw	0x0	gpioa_1 force output value for output.
28	pad_gpioa_1_oen_frc	rw	0x0	gpioa_1 force enable for outoen.

49.3 pad_gpioa_2_cfg_reg				0x40504008
Bits	Field Name	Acc	Default	Description

3 : 0	pad_gpioa_2_sel	rw	0x0	gpioa_2 select 0 fun_gpioa_2_sel 1 fun_uart0_rx_sel 2 fun_uart1_rx_sel 3 fun_wf_ext_pa_ctrl_2_sel 4 fun_pcm_din_sel 5 fun_debug_clk_sel 6 fun_i2s_dat_in_0_sel 7 fun_tports2_sel 8 fun_pta_ant_sw_0_sel 9 fun_spi_lcd_di_sel 10 fun_pwm_2_sel 11 fun_pcm_fsync_sel 12 fun_pcm_din_sel 13 fun_bt_uart_rx_sel
7 : 4	pad_gpioa_2_ana	rw	0x0	gpioa_2 ANA_Alter_SEL
8	pad_gpioa_2_pull_dn	rw	0x0	gpioa_2 PUll down
9	pad_gpioa_2_pull_up	rw	0x0	gpioa_2 PUll up
16	pad_gpioa_2_pull_frc	rw	0x0	gpioa_2 force enable for pu/pd
17	pad_gpioa_2_oen_reg	rw	0x0	gpioa_2 force outoen value.
19 : 18	pad_gpioa_2_drv_strength	rw	0x0	gpioa_2 driving strength.
20	pad_gpioa_2_out_reg	rw	0x0	gpioa_2 pin output value.
24	pad_gpioa_2_out_frc	rw	0x0	gpioa_2 force output value for output.
28	pad_gpioa_2_oen_frc	rw	0x0	gpioa_2 force enable for outoen.

49.4 pad_gpioa_3_cfg_reg				0x4050400C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_3_sel	rw	0x0	gpioa_3 select 0 fun_gpioa_3_sel 1 fun_uart0_tx_sel 2 fun_uart1_tx_sel 3 fun_wf_ext_pa_ctrl_3_sel 4 fun_pcm_dout_sel 5 fun_aon_uart_tx_sel 6 fun_i2s_dat_out_0_sel 7 fun_tports3_sel 8 fun_pta_ant_sw_1_sel 9 fun_spi_lcd_do_sel 11 fun_pcm_clk_sel 12 fun_pcm_fsync_sel 13 fun_bt_uart_tx_sel 14 fun_mbist_clock_sel
7 : 4	pad_gpioa_3_ana	rw	0x0	gpioa_3 ANA_Alter_SEL
8	pad_gpioa_3_pull_dn	rw	0x0	gpioa_3 PUll down
9	pad_gpioa_3_pull_up	rw	0x0	gpioa_3 PUll up
16	pad_gpioa_3_pull_frc	rw	0x0	gpioa_3 force enable for pu/pd
17	pad_gpioa_3_oen_reg	rw	0x0	gpioa_3 force outoen value.
19 : 18	pad_gpioa_3_drv_strength	rw	0x0	gpioa_3 driving strength.
20	pad_gpioa_3_out_reg	rw	0x0	gpioa_3 pin output value.
24	pad_gpioa_3_out_frc	rw	0x0	gpioa_3 force output value for output.
28	pad_gpioa_3_oen_frc	rw	0x0	gpioa_3 force enable for outoen.

49.5 pad_gpioa_4_cfg_reg				0x40504010
Bits	Field Name	Acc	Default	Description

3 : 0	pad_gpioa_4_sel	rw	0x0	gpioa_4 select 0 fun_gpioa_4_sel 1 fun_uart0_cts_sel 2 fun_uart1_cts_sel 3 fun_uart1_rx_sel 4 fun_bt_uart_rx_sel 5 fun_aud_dac_sync_sel 6 fun_codec_mclk_sel 7 fun_tports4_sel 8 fun_i2s_bck_0_sel 9 fun_spi_lcd_cd_sel 10 fun_pcie_wake_n_sel 13 fun_pcm_fsync_sel 14 fun_dft_jtag_tck_sel
7 : 4	pad_gpioa_4_ana	rw	0x0	gpioa_4 ANA_Alter_SEL
8	pad_gpioa_4_pull_dn	rw	0x0	gpioa_4 PUll down
9	pad_gpioa_4_pull_up	rw	0x0	gpioa_4 PUll up
16	pad_gpioa_4_pull_frc	rw	0x0	gpioa_4 force enable for pu/pd
17	pad_gpioa_4_oen_reg	rw	0x0	gpioa_4 force outoen value.
19 : 18	pad_gpioa_4_drv_strength	rw	0x0	gpioa_4 driving strength.
20	pad_gpioa_4_out_reg	rw	0x0	gpioa_4 pin output value.
24	pad_gpioa_4_out_frc	rw	0x0	gpioa_4 force output value for output.
28	pad_gpioa_4_oen_frc	rw	0x0	gpioa_4 force enable for outoen.

49.6 pad_gpioa_5_cfg_reg				0x40504014
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_5_sel	rw	0x0	gpioa_5 select 0 fun_gpioa_5_sel 1 fun_uart0_rts_sel 2 fun_uart1_rts_sel 3 fun_uart1_tx_sel 4 fun_bt_uart_tx_sel 5 fun_aon_uart_tx_sel 6 fun_debug_clk_sel 7 fun_tports5_sel 8 fun_i2s_lrck_0_sel 9 fun_spi_lcd_fmark_sel 10 fun_pcie_perst_n_sel 13 fun_pcm_din_sel 14 fun_dft_jtag_trst_sel
7 : 4	pad_gpioa_5_ana	rw	0x0	gpioa_5 ANA_Alter_SEL
8	pad_gpioa_5_pull_dn	rw	0x0	gpioa_5 PUll down
9	pad_gpioa_5_pull_up	rw	0x0	gpioa_5 PUll up
16	pad_gpioa_5_pull_frc	rw	0x0	gpioa_5 force enable for pu/pd
17	pad_gpioa_5_oen_reg	rw	0x0	gpioa_5 force outoen value.
19 : 18	pad_gpioa_5_drv_strength	rw	0x0	gpioa_5 driving strength.
20	pad_gpioa_5_out_reg	rw	0x0	gpioa_5 pin output value.
24	pad_gpioa_5_out_frc	rw	0x0	gpioa_5 force output value for output.
28	pad_gpioa_5_oen_frc	rw	0x0	gpioa_5 force enable for outoen.

49.7 pad_gpioa_6_cfg_reg				0x40504018
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_6_sel	rw	0x0	gpioa_6 select 0 fun_gpioa_6_sel 1 fun_i2cm_scl_sel 2 fun_uart2_rx_sel 3 fun_uart1_cts_sel 4 fun_bt_uart_cts_sel 5 fun_psi_s_scl_sel 6 fun_bt_uart_rx_sel 7 fun_tports6_sel 8 fun_tports8_sel 9 fun_spi_lcd_csn_1_sel 10 fun_pcie_clkreq_n_sel 13 fun_pcm_dout_sel

7 : 4	pad_gpioa_6_ana	rw	0x0	gpioa_6 ANA_Alter_SEL
8	pad_gpioa_6_pull_dn	rw	0x0	gpioa_6 PUll down
9	pad_gpioa_6_pull_up	rw	0x0	gpioa_6 PUll up
16	pad_gpioa_6_pull_frc	rw	0x0	gpioa_6 force enable for pu/pd
17	pad_gpioa_6_oen_reg	rw	0x0	gpioa_6 force outoen value.
19 : 18	pad_gpioa_6_drv_strength	rw	0x0	gpioa_6 driving strength.
20	pad_gpioa_6_out_reg	rw	0x0	gpioa_6 pin output value.
24	pad_gpioa_6_out_frc	rw	0x0	gpioa_6 force output value for output.
28	pad_gpioa_6_oen_frc	rw	0x0	gpioa_6 force enable for outoen.

49.8 pad_gpioa_7_cfg_reg				0x4050401C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_7_sel	rw	0x0	gpioa_7 select 0 fun_gpioa_7_sel 1 fun_i2cm_sda_sel 2 fun_uart2_tx_sel 3 fun_uart1_rts_sel 4 fun_bt_uart_rts_sel 5 fun_psi_s_sda_sel 6 fun_bt_uart_tx_sel 7 fun_tports7_sel 8 fun_aon_pwm_0_sel 9 fun_spi_lcd_csn_2_sel 13 fun_pcm_clk_sel
7 : 4	pad_gpioa_7_ana	rw	0x0	gpioa_7 ANA_Alter_SEL
8	pad_gpioa_7_pull_dn	rw	0x0	gpioa_7 PUll down
9	pad_gpioa_7_pull_up	rw	0x0	gpioa_7 PUll up
16	pad_gpioa_7_pull_frc	rw	0x0	gpioa_7 force enable for pu/pd
17	pad_gpioa_7_oen_reg	rw	0x0	gpioa_7 force outoen value.
19 : 18	pad_gpioa_7_drv_strength	rw	0x0	gpioa_7 driving strength.
20	pad_gpioa_7_out_reg	rw	0x0	gpioa_7 pin output value.
24	pad_gpioa_7_out_frc	rw	0x0	gpioa_7 force output value for output.
28	pad_gpioa_7_oen_frc	rw	0x0	gpioa_7 force enable for outoen.

49.9 pad_gpioa_8_cfg_reg				0x40504020
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_8_sel	rw	0x0	gpioa_8 select 0 fun_uart0_rx_sel 1 fun_gpioa_8_sel 2 fun_uart2_cts_sel 3 fun_spdif_in_sel 6 fun_bt_uart_cts_sel
7 : 4	pad_gpioa_8_ana	rw	0x0	gpioa_8 ANA_Alter_SEL
8	pad_gpioa_8_pull_dn	rw	0x0	gpioa_8 PUll down
9	pad_gpioa_8_pull_up	rw	0x0	gpioa_8 PUll up
16	pad_gpioa_8_pull_frc	rw	0x0	gpioa_8 force enable for pu/pd
17	pad_gpioa_8_oen_reg	rw	0x0	gpioa_8 force outoen value.
19 : 18	pad_gpioa_8_drv_strength	rw	0x0	gpioa_8 driving strength.
20	pad_gpioa_8_out_reg	rw	0x0	gpioa_8 pin output value.
24	pad_gpioa_8_out_frc	rw	0x0	gpioa_8 force output value for output.
28	pad_gpioa_8_oen_frc	rw	0x0	gpioa_8 force enable for outoen.

49.10 pad_gpioa_9_cfg_reg				0x40504024
Bits	Field Name	Acc	Default	Description

3 : 0	pad_gpioa_9_sel	rw	0x0	gpioa_9 select 0 fun_uart0_tx_sel 1 fun_gpioa_9_sel 2 fun_uart2_rts_sel 3 fun_spdif_out_sel 4 fun_aon_pwm_1_sel 6 fun_bt_uart_rts_sel 7 fun_tports9_sel 8 fun_i2s_dat_out_0_sel 9 fun_spi_lcd_csn_3_sel
7 : 4	pad_gpioa_9_ana	rw	0x0	gpioa_9 ANA_Alter_SEL
8	pad_gpioa_9_pull_dn	rw	0x0	gpioa_9 PUll down
9	pad_gpioa_9_pull_up	rw	0x0	gpioa_9 PUll up
16	pad_gpioa_9_pull_frc	rw	0x0	gpioa_9 force enable for pu/pd
17	pad_gpioa_9_oen_reg	rw	0x0	gpioa_9 force outoen value.
19 : 18	pad_gpioa_9_drv_strength	rw	0x0	gpioa_9 driving strength.
20	pad_gpioa_9_out_reg	rw	0x0	gpioa_9 pin output value.
24	pad_gpioa_9_out_frc	rw	0x0	gpioa_9 force output value for output.
28	pad_gpioa_9_oen_frc	rw	0x0	gpioa_9 force enable for outoen.

49.11 pad_gpioa_10_cfg_reg				0x40504028
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_10_sel	rw	0x0	gpioa_10 select 0 fun_gpioa_10_sel 1 fun_uart1_rx_sel 2 fun_bt_uart_rx_sel 3 fun_spi_lcd_sck_sel 4 fun_bt_uart_cts_sel 6 fun_sdmmc_data_2_sel 7 fun_tports10_sel 8 fun_i2s_bck_1_sel 9 fun_uart2_rx_sel 10 fun_bt_uart_cts_sel 11 fun_bt_uart_rts_sel 14 fun_dft_jtag_tms_sel
7 : 4	pad_gpioa_10_ana	rw	0x0	gpioa_10 ANA_Alter_SEL
8	pad_gpioa_10_pull_dn	rw	0x0	gpioa_10 PUll down
9	pad_gpioa_10_pull_up	rw	0x0	gpioa_10 PUll up
16	pad_gpioa_10_pull_frc	rw	0x0	gpioa_10 force enable for pu/pd
17	pad_gpioa_10_oen_reg	rw	0x0	gpioa_10 force outoen value.
19 : 18	pad_gpioa_10_drv_strength	rw	0x0	gpioa_10 driving strength.
20	pad_gpioa_10_out_reg	rw	0x0	gpioa_10 pin output value.
24	pad_gpioa_10_out_frc	rw	0x0	gpioa_10 force output value for output.
28	pad_gpioa_10_oen_frc	rw	0x0	gpioa_10 force enable for outoen.

49.12 pad_gpioa_11_cfg_reg				0x4050402C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_11_sel	rw	0x0	gpioa_11 select 0 fun_gpioa_11_sel 1 fun_uart1_tx_sel 2 fun_bt_uart_tx_sel 3 fun_spi_lcd_csn_0_sel 4 fun_bt_uart_rts_sel 5 fun_debug_clk_sel 6 fun_sdmmc_data_3_sel 7 fun_tports11_sel 8 fun_i2s_lrck_1_sel 9 fun_uart2_tx_sel 10 fun_bt_uart_rts_sel 11 fun_bt_uart_rx_sel 14 fun_dft_jtag_tdi_sel
7 : 4	pad_gpioa_11_ana	rw	0x0	gpioa_11 ANA_Alter_SEL
8	pad_gpioa_11_pull_dn	rw	0x0	gpioa_11 PUll down

9	pad_gpioa_11_pull_up	rw	0x0	gpioa_11 PUll up
16	pad_gpioa_11_pull_frc	rw	0x0	gpioa_11 force enable for pu/pd
17	pad_gpioa_11_oen_reg	rw	0x0	gpioa_11 force outoen value.
19 : 18	pad_gpioa_11_drv_strength	rw	0x0	gpioa_11 driving strength.
20	pad_gpioa_11_out_reg	rw	0x0	gpioa_11 pin output value.
24	pad_gpioa_11_out_frc	rw	0x0	gpioa_11 force output value for output.
28	pad_gpioa_11_oen_frc	rw	0x0	gpioa_11 force enable for outoen.

49.13 pad_gpioa_12_cfg_reg 0x40504030

Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_12_sel	rw	0x0	gpioa_12 select 0 fun_gpioa_12_sel 1 fun_uart1_cts_sel 2 fun_bt_uart_cts_sel 3 fun_spi_lcd_di_sel 4 fun_aon_pwm_2_sel 5 fun_pcm_fsync_sel 6 fun_sdmmc_cmd_sel 7 fun_tports12_sel 8 fun_i2s_dat_in_1_sel 9 fun_uart2_cts_sel 10 fun_bt_uart_rx_sel 11 fun_bt_uart_tx_sel
7 : 4	pad_gpioa_12_ana	rw	0x0	gpioa_12 ANA_Alter_SEL
8	pad_gpioa_12_pull_dn	rw	0x0	gpioa_12 PUll down
9	pad_gpioa_12_pull_up	rw	0x0	gpioa_12 PUll up
16	pad_gpioa_12_pull_frc	rw	0x0	gpioa_12 force enable for pu/pd
17	pad_gpioa_12_oen_reg	rw	0x0	gpioa_12 force outoen value.
19 : 18	pad_gpioa_12_drv_strength	rw	0x0	gpioa_12 driving strength.
20	pad_gpioa_12_out_reg	rw	0x0	gpioa_12 pin output value.
24	pad_gpioa_12_out_frc	rw	0x0	gpioa_12 force output value for output.
28	pad_gpioa_12_oen_frc	rw	0x0	gpioa_12 force enable for outoen.

49.14 pad_gpioa_13_cfg_reg 0x40504034

Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_13_sel	rw	0x0	gpioa_13 select 0 fun_gpioa_13_sel 1 fun_uart1_rts_sel 2 fun_bt_uart_rts_sel 3 fun_spi_lcd_do_sel 4 fun_pwm_0_sel 5 fun_pcm_clk_sel 6 fun_sdmmc_clk_sel 7 fun_tports13_sel 8 fun_codec_mclk_sel 9 fun_uart2_rts_sel 10 fun_bt_uart_tx_sel 11 fun_bt_uart_cts_sel
7 : 4	pad_gpioa_13_ana	rw	0x0	gpioa_13 ANA_Alter_SEL
8	pad_gpioa_13_pull_dn	rw	0x0	gpioa_13 PUll down
9	pad_gpioa_13_pull_up	rw	0x0	gpioa_13 PUll up
16	pad_gpioa_13_pull_frc	rw	0x0	gpioa_13 force enable for pu/pd
17	pad_gpioa_13_oen_reg	rw	0x0	gpioa_13 force outoen value.
19 : 18	pad_gpioa_13_drv_strength	rw	0x0	gpioa_13 driving strength.
20	pad_gpioa_13_out_reg	rw	0x0	gpioa_13 pin output value.
24	pad_gpioa_13_out_frc	rw	0x0	gpioa_13 force output value for output.
28	pad_gpioa_13_oen_frc	rw	0x0	gpioa_13 force enable for outoen.

49.15 pad_gpioa_14_cfg_reg 0x40504038

Bits	Field Name	Acc	Default	Description

3 : 0	pad_gpioa_14_sel	rw	0x0	gpioa_14 select 0 fun_gpioa_14_sel 1 fun_i2cm_scl_sel 2 fun_spdif_in_sel 3 fun_spi_lcd_cd_sel 4 fun_pwm_1_sel 5 fun_pcm_din_sel 6 fun_sdmmc_data_0_sel 7 fun_tports14_sel 8 fun_i2s_dat_out_1_sel 9 fun_pta_ant_sw_0_sel 10 fun_uart0_rx_sel 14 fun_dft_jtag_tdo_sel
7 : 4	pad_gpioa_14_ana	rw	0x0	gpioa_14 ANA_Alter_SEL
8	pad_gpioa_14_pull_dn	rw	0x0	gpioa_14 PUll down
9	pad_gpioa_14_pull_up	rw	0x0	gpioa_14 PUll up
16	pad_gpioa_14_pull_frc	rw	0x0	gpioa_14 force enable for pu/pd
17	pad_gpioa_14_oen_reg	rw	0x0	gpioa_14 force outoen value.
19 : 18	pad_gpioa_14_drv_strength	rw	0x0	gpioa_14 driving strength.
20	pad_gpioa_14_out_reg	rw	0x0	gpioa_14 pin output value.
24	pad_gpioa_14_out_frc	rw	0x0	gpioa_14 force output value for output.
28	pad_gpioa_14_oen_frc	rw	0x0	gpioa_14 force enable for outoen.

49.16 pad_gpioa_15_cfg_reg				0x4050403C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpioa_15_sel	rw	0x0	gpioa_15 select 0 fun_gpioa_15_sel 1 fun_i2cm_sda_sel 2 fun_spdif_out_sel 3 fun_spi_lcd_fmark_sel 4 fun_pwm_2_sel 5 fun_pcm_dout_sel 6 fun_sdmmc_data_1_sel 7 fun_tports15_sel 8 fun_i2s_dat_in_0_sel 9 fun_pta_ant_sw_1_sel 10 fun_uart0_tx_sel
7 : 4	pad_gpioa_15_ana	rw	0x0	gpioa_15 ANA_Alter_SEL
8	pad_gpioa_15_pull_dn	rw	0x0	gpioa_15 PUll down
9	pad_gpioa_15_pull_up	rw	0x0	gpioa_15 PUll up
16	pad_gpioa_15_pull_frc	rw	0x0	gpioa_15 force enable for pu/pd
17	pad_gpioa_15_oen_reg	rw	0x0	gpioa_15 force outoen value.
19 : 18	pad_gpioa_15_drv_strength	rw	0x0	gpioa_15 driving strength.
20	pad_gpioa_15_out_reg	rw	0x0	gpioa_15 pin output value.
24	pad_gpioa_15_out_frc	rw	0x0	gpioa_15 force output value for output.
28	pad_gpioa_15_oen_frc	rw	0x0	gpioa_15 force enable for outoen.

49.17 pad_agpio_0_cfg_reg				0x40504040
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_0_sel	rw	0x0	agpio_0 select 0 fun_i2s_dat_out_1_sel 1 fun_gpiob_0_sel 2 fun_tports0_sel 3 fun_tports15_sel
7 : 4	pad_agpio_0_ana	rw	0x0	agpio_0 ANA_Alter_SEL
8	pad_agpio_0_pull_dn	rw	0x0	agpio_0 PUll down
9	pad_agpio_0_pull_up	rw	0x0	agpio_0 PUll up
16	pad_agpio_0_pull_frc	rw	0x0	agpio_0 force enable for pu/pd
17	pad_agpio_0_oen_reg	rw	0x0	agpio_0 force outoen value.
19 : 18	pad_agpio_0_drv_strength	rw	0x0	agpio_0 driving strength.
20	pad_agpio_0_out_reg	rw	0x0	agpio_0 pin output value.
24	pad_agpio_0_out_frc	rw	0x0	agpio_0 force output value for output.

28	pad_agpio_0_oen_frc	rw	0x0	agpio_0 force enable for outoen.

49.18 pad_agpio_1_cfg_reg				0x40504044
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_1_sel	rw	0x0	agpio_1 select 0 fun_i2s_dat_in_0_sel 1 fun_gpiob_1_sel 2 fun_tports1_sel 3 fun_tports14_sel
7 : 4	pad_agpio_1_ana	rw	0x0	agpio_1 ANA_Alter_SEL
8	pad_agpio_1_pull_dn	rw	0x0	agpio_1 PUll down
9	pad_agpio_1_pull_up	rw	0x0	agpio_1 PUll up
16	pad_agpio_1_pull_frc	rw	0x0	agpio_1 force enable for pu/pd
17	pad_agpio_1_oen_reg	rw	0x0	agpio_1 force outoen value.
19 : 18	pad_agpio_1_drv_strength	rw	0x0	agpio_1 driving strength.
20	pad_agpio_1_out_reg	rw	0x0	agpio_1 pin output value.
24	pad_agpio_1_out_frc	rw	0x0	agpio_1 force output value for output.
28	pad_agpio_1_oen_frc	rw	0x0	agpio_1 force enable for outoen.

49.19 pad_agpio_2_cfg_reg				0x40504048
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_2_sel	rw	0x0	agpio_2 select 0 fun_i2s_dat_in_1_sel 1 fun_gpiob_2_sel 2 fun_tports2_sel 3 fun_tports13_sel
7 : 4	pad_agpio_2_ana	rw	0x0	agpio_2 ANA_Alter_SEL
8	pad_agpio_2_pull_dn	rw	0x0	agpio_2 PUll down
9	pad_agpio_2_pull_up	rw	0x0	agpio_2 PUll up
16	pad_agpio_2_pull_frc	rw	0x0	agpio_2 force enable for pu/pd
17	pad_agpio_2_oen_reg	rw	0x0	agpio_2 force outoen value.
19 : 18	pad_agpio_2_drv_strength	rw	0x0	agpio_2 driving strength.
20	pad_agpio_2_out_reg	rw	0x0	agpio_2 pin output value.
24	pad_agpio_2_out_frc	rw	0x0	agpio_2 force output value for output.
28	pad_agpio_2_oen_frc	rw	0x0	agpio_2 force enable for outoen.

49.20 pad_agpio_3_cfg_reg				0x4050404C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_3_sel	rw	0x0	agpio_3 select 0 fun_i2s_bck_1_sel 1 fun_gpiob_3_sel 2 fun_tports3_sel 3 fun_tports12_sel
7 : 4	pad_agpio_3_ana	rw	0x0	agpio_3 ANA_Alter_SEL
8	pad_agpio_3_pull_dn	rw	0x0	agpio_3 PUll down
9	pad_agpio_3_pull_up	rw	0x0	agpio_3 PUll up
16	pad_agpio_3_pull_frc	rw	0x0	agpio_3 force enable for pu/pd
17	pad_agpio_3_oen_reg	rw	0x0	agpio_3 force outoen value.
19 : 18	pad_agpio_3_drv_strength	rw	0x0	agpio_3 driving strength.
20	pad_agpio_3_out_reg	rw	0x0	agpio_3 pin output value.
24	pad_agpio_3_out_frc	rw	0x0	agpio_3 force output value for output.
28	pad_agpio_3_oen_frc	rw	0x0	agpio_3 force enable for outoen.

49.21 pad_agpio_4_cfg_reg				0x40504050
Bits	Field Name	Acc	Default	Description

3 : 0	pad_agpio_4_sel	rw	0x0	agpio_4 select 0 fun_i2s_lrck_1_sel 1 fun_gpiob_4_sel 2 fun_tports4_sel 3 fun_tports11_sel
7 : 4	pad_agpio_4_ana	rw	0x0	agpio_4 ANA_Alter_SEL
8	pad_agpio_4_pull_dn	rw	0x0	agpio_4 PUll down
9	pad_agpio_4_pull_up	rw	0x0	agpio_4 PUll up
16	pad_agpio_4_pull_frc	rw	0x0	agpio_4 force enable for pu/pd
17	pad_agpio_4_oen_reg	rw	0x0	agpio_4 force outoen value.
19 : 18	pad_agpio_4_drv_strength	rw	0x0	agpio_4 driving strength.
20	pad_agpio_4_out_reg	rw	0x0	agpio_4 pin output value.
24	pad_agpio_4_out_frc	rw	0x0	agpio_4 force output value for output.
28	pad_agpio_4_oen_frc	rw	0x0	agpio_4 force enable for outoen.

49.22 pad_agpio_5_cfg_reg				0x40504054
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_5_sel	rw	0x0	agpio_5 select 0 fun_i2s_dat_out_0_sel 1 fun_gpiob_5_sel 2 fun_tports5_sel 3 fun_tports10_sel
7 : 4	pad_agpio_5_ana	rw	0x0	agpio_5 ANA_Alter_SEL
8	pad_agpio_5_pull_dn	rw	0x0	agpio_5 PUll down
9	pad_agpio_5_pull_up	rw	0x0	agpio_5 PUll up
16	pad_agpio_5_pull_frc	rw	0x0	agpio_5 force enable for pu/pd
17	pad_agpio_5_oen_reg	rw	0x0	agpio_5 force outoen value.
19 : 18	pad_agpio_5_drv_strength	rw	0x0	agpio_5 driving strength.
20	pad_agpio_5_out_reg	rw	0x0	agpio_5 pin output value.
24	pad_agpio_5_out_frc	rw	0x0	agpio_5 force output value for output.
28	pad_agpio_5_oen_frc	rw	0x0	agpio_5 force enable for outoen.

49.23 pad_agpio_6_cfg_reg				0x40504058
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_6_sel	rw	0x0	agpio_6 select 0 fun_i2s_bck_0_sel 1 fun_gpiob_6_sel 2 fun_tports6_sel 3 fun_tports9_sel
7 : 4	pad_agpio_6_ana	rw	0x0	agpio_6 ANA_Alter_SEL
8	pad_agpio_6_pull_dn	rw	0x0	agpio_6 PUll down
9	pad_agpio_6_pull_up	rw	0x0	agpio_6 PUll up
16	pad_agpio_6_pull_frc	rw	0x0	agpio_6 force enable for pu/pd
17	pad_agpio_6_oen_reg	rw	0x0	agpio_6 force outoen value.
19 : 18	pad_agpio_6_drv_strength	rw	0x0	agpio_6 driving strength.
20	pad_agpio_6_out_reg	rw	0x0	agpio_6 pin output value.
24	pad_agpio_6_out_frc	rw	0x0	agpio_6 force output value for output.
28	pad_agpio_6_oen_frc	rw	0x0	agpio_6 force enable for outoen.

49.24 pad_agpio_7_cfg_reg				0x4050405C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_7_sel	rw	0x0	agpio_7 select 0 fun_i2s_lrck_0_sel 1 fun_gpiob_7_sel 2 fun_tports7_sel 3 fun_tports8_sel
7 : 4	pad_agpio_7_ana	rw	0x0	agpio_7 ANA_Alter_SEL
8	pad_agpio_7_pull_dn	rw	0x0	agpio_7 PUll down

9	pad_agpio_7_pull_up	rw	0x0	agpio_7 PUll up
16	pad_agpio_7_pull_frc	rw	0x0	agpio_7 force enable for pu/pd
17	pad_agpio_7_oen_reg	rw	0x0	agpio_7 force outoen value.
19 : 18	pad_agpio_7_drv_strength	rw	0x0	agpio_7 driving strength.
20	pad_agpio_7_out_reg	rw	0x0	agpio_7 pin output value.
24	pad_agpio_7_out_frc	rw	0x0	agpio_7 force output value for output.
28	pad_agpio_7_oen_frc	rw	0x0	agpio_7 force enable for outoen.

49.25 pad_agpio_8_cfg_reg 0x40504060

Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_8_sel	rw	0x0	agpio_8 select 0 fun_codec_mclk_sel 1 fun_gpiob_8_sel 2 fun_tports8_sel 3 fun_tports7_sel
7 : 4	pad_agpio_8_ana	rw	0x0	agpio_8 ANA_Alter_SEL
8	pad_agpio_8_pull_dn	rw	0x0	agpio_8 PUll down
9	pad_agpio_8_pull_up	rw	0x0	agpio_8 PUll up
16	pad_agpio_8_pull_frc	rw	0x0	agpio_8 force enable for pu/pd
17	pad_agpio_8_oen_reg	rw	0x0	agpio_8 force outoen value.
19 : 18	pad_agpio_8_drv_strength	rw	0x0	agpio_8 driving strength.
20	pad_agpio_8_out_reg	rw	0x0	agpio_8 pin output value.
24	pad_agpio_8_out_frc	rw	0x0	agpio_8 force output value for output.
28	pad_agpio_8_oen_frc	rw	0x0	agpio_8 force enable for outoen.

49.26 pad_agpio_9_cfg_reg 0x40504064

Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_9_sel	rw	0x0	agpio_9 select 0 fun_psi_m_scl_sel 1 fun_gpiob_9_sel 2 fun_tports9_sel 3 fun_tports6_sel
7 : 4	pad_agpio_9_ana	rw	0x0	agpio_9 ANA_Alter_SEL
8	pad_agpio_9_pull_dn	rw	0x0	agpio_9 PUll down
9	pad_agpio_9_pull_up	rw	0x0	agpio_9 PUll up
16	pad_agpio_9_pull_frc	rw	0x0	agpio_9 force enable for pu/pd
17	pad_agpio_9_oen_reg	rw	0x0	agpio_9 force outoen value.
19 : 18	pad_agpio_9_drv_strength	rw	0x0	agpio_9 driving strength.
20	pad_agpio_9_out_reg	rw	0x0	agpio_9 pin output value.
24	pad_agpio_9_out_frc	rw	0x0	agpio_9 force output value for output.
28	pad_agpio_9_oen_frc	rw	0x0	agpio_9 force enable for outoen.

49.27 pad_agpio_10_cfg_reg 0x40504068

Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_10_sel	rw	0x0	agpio_10 select 0 fun_psi_m_sda_sel 1 fun_gpiob_10_sel 2 fun_tports10_sel 3 fun_tports5_sel
7 : 4	pad_agpio_10_ana	rw	0x0	agpio_10 ANA_Alter_SEL
8	pad_agpio_10_pull_dn	rw	0x0	agpio_10 PUll down
9	pad_agpio_10_pull_up	rw	0x0	agpio_10 PUll up
16	pad_agpio_10_pull_frc	rw	0x0	agpio_10 force enable for pu/pd
17	pad_agpio_10_oen_reg	rw	0x0	agpio_10 force outoen value.
19 : 18	pad_agpio_10_drv_strength	rw	0x0	agpio_10 driving strength.
20	pad_agpio_10_out_reg	rw	0x0	agpio_10 pin output value.
24	pad_agpio_10_out_frc	rw	0x0	agpio_10 force output value for output.
28	pad_agpio_10_oen_frc	rw	0x0	agpio_10 force enable for outoen.

49.28 pad_agpio_11_cfg_reg				0x4050406C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_11_sel	rw	0x0	agpio_11 select 0 fun_aud_dac_sync_sel 1 fun_gpiob_11_sel 2 fun_tports11_sel 3 fun_tports4_sel
7 : 4	pad_agpio_11_ana	rw	0x0	agpio_11 ANA_Alter_SEL
8	pad_agpio_11_pull_dn	rw	0x0	agpio_11 PUll down
9	pad_agpio_11_pull_up	rw	0x0	agpio_11 PUll up
16	pad_agpio_11_pull_frc	rw	0x0	agpio_11 force enable for pu/pd
17	pad_agpio_11_oen_reg	rw	0x0	agpio_11 force outoen value.
19 : 18	pad_agpio_11_drv_strength	rw	0x0	agpio_11 driving strength.
20	pad_agpio_11_out_reg	rw	0x0	agpio_11 pin output value.
24	pad_agpio_11_out_frc	rw	0x0	agpio_11 force output value for output.
28	pad_agpio_11_oen_frc	rw	0x0	agpio_11 force enable for outoen.

49.29 pad_agpio_12_cfg_reg				0x40504070
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_12_sel	rw	0x0	agpio_12 select 0 fun_gpiob_12_sel 2 fun_tports12_sel 3 fun_tports3_sel
7 : 4	pad_agpio_12_ana	rw	0x0	agpio_12 ANA_Alter_SEL
8	pad_agpio_12_pull_dn	rw	0x0	agpio_12 PUll down
9	pad_agpio_12_pull_up	rw	0x0	agpio_12 PUll up
16	pad_agpio_12_pull_frc	rw	0x0	agpio_12 force enable for pu/pd
17	pad_agpio_12_oen_reg	rw	0x0	agpio_12 force outoen value.
19 : 18	pad_agpio_12_drv_strength	rw	0x0	agpio_12 driving strength.
20	pad_agpio_12_out_reg	rw	0x0	agpio_12 pin output value.
24	pad_agpio_12_out_frc	rw	0x0	agpio_12 force output value for output.
28	pad_agpio_12_oen_frc	rw	0x0	agpio_12 force enable for outoen.

49.30 pad_agpio_13_cfg_reg				0x40504074
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_13_sel	rw	0x0	agpio_13 select 0 fun_gpiob_13_sel 2 fun_tports13_sel 3 fun_tports2_sel
7 : 4	pad_agpio_13_ana	rw	0x0	agpio_13 ANA_Alter_SEL
8	pad_agpio_13_pull_dn	rw	0x0	agpio_13 PUll down
9	pad_agpio_13_pull_up	rw	0x0	agpio_13 PUll up
16	pad_agpio_13_pull_frc	rw	0x0	agpio_13 force enable for pu/pd
17	pad_agpio_13_oen_reg	rw	0x0	agpio_13 force outoen value.
19 : 18	pad_agpio_13_drv_strength	rw	0x0	agpio_13 driving strength.
20	pad_agpio_13_out_reg	rw	0x0	agpio_13 pin output value.
24	pad_agpio_13_out_frc	rw	0x0	agpio_13 force output value for output.
28	pad_agpio_13_oen_frc	rw	0x0	agpio_13 force enable for outoen.

49.31 pad_agpio_14_cfg_reg				0x40504078
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_14_sel	rw	0x0	agpio_14 select 0 fun_gpiob_14_sel 1 fun_uart1_rx_sel 2 fun_tports14_sel 3 fun_tports1_sel
7 : 4	pad_agpio_14_ana	rw	0x0	agpio_14 ANA_Alter_SEL
8	pad_agpio_14_pull_dn	rw	0x0	agpio_14 PUll down

9	pad_agpio_14_pull_up	rw	0x0	agpio_14 PUll up
16	pad_agpio_14_pull_frc	rw	0x0	agpio_14 force enable for pu/pd
17	pad_agpio_14_oen_reg	rw	0x0	agpio_14 force outoen value.
19 : 18	pad_agpio_14_drv_strength	rw	0x0	agpio_14 driving strength.
20	pad_agpio_14_out_reg	rw	0x0	agpio_14 pin output value.
24	pad_agpio_14_out_frc	rw	0x0	agpio_14 force output value for output.
28	pad_agpio_14_oen_frc	rw	0x0	agpio_14 force enable for outoen.

49.32 pad_agpio_15_cfg_reg				0x4050407C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_agpio_15_sel	rw	0x0	agpio_15 select 0 fun_gpiob_15_sel 1 fun_uart1_tx_sel 2 fun_tports15_sel 3 fun_tports0_sel
7 : 4	pad_agpio_15_ana	rw	0x0	agpio_15 ANA_Alter_SEL
8	pad_agpio_15_pull_dn	rw	0x0	agpio_15 PUll down
9	pad_agpio_15_pull_up	rw	0x0	agpio_15 PUll up
16	pad_agpio_15_pull_frc	rw	0x0	agpio_15 force enable for pu/pd
17	pad_agpio_15_oen_reg	rw	0x0	agpio_15 force outoen value.
19 : 18	pad_agpio_15_drv_strength	rw	0x0	agpio_15 driving strength.
20	pad_agpio_15_out_reg	rw	0x0	agpio_15 pin output value.
24	pad_agpio_15_out_frc	rw	0x0	agpio_15 force output value for output.
28	pad_agpio_15_oen_frc	rw	0x0	agpio_15 force enable for outoen.

49.33 pad_gpiob_0_cfg_reg				0x40504080
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_0_sel	rw	0x0	gpiob_0 select 0 fun_gpiob_0_sel 1 fun_pcm_fsync_sel 2 fun_i2cm_scl_sel 3 fun_spi_lcd_sck_sel 4 fun_aon_pwm_0_sel 5 fun_wf_ext_pa_ctrl_0_sel 6 fun_tports0_sel 7 fun_tports8_sel
7 : 4	pad_gpiob_0_ana	rw	0x0	gpiob_0 ANA_Alter_SEL
8	pad_gpiob_0_pull_dn	rw	0x0	gpiob_0 PUll down
9	pad_gpiob_0_pull_up	rw	0x0	gpiob_0 PUll up
16	pad_gpiob_0_pull_frc	rw	0x0	gpiob_0 force enable for pu/pd
17	pad_gpiob_0_oen_reg	rw	0x0	gpiob_0 force outoen value.
19 : 18	pad_gpiob_0_drv_strength	rw	0x0	gpiob_0 driving strength.
20	pad_gpiob_0_out_reg	rw	0x0	gpiob_0 pin output value.
24	pad_gpiob_0_out_frc	rw	0x0	gpiob_0 force output value for output.
28	pad_gpiob_0_oen_frc	rw	0x0	gpiob_0 force enable for outoen.

49.34 pad_gpiob_1_cfg_reg				0x40504084
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_1_sel	rw	0x0	gpiob_1 select 0 fun_gpiob_1_sel 1 fun_pcm_clk_sel 2 fun_i2cm_sda_sel 3 fun_spi_lcd_csn_0_sel 4 fun_aon_pwm_1_sel 5 fun_wf_ext_pa_ctrl_1_sel 6 fun_tports1_sel 7 fun_tports9_sel
7 : 4	pad_gpiob_1_ana	rw	0x0	gpiob_1 ANA_Alter_SEL
8	pad_gpiob_1_pull_dn	rw	0x0	gpiob_1 PUll down

9	pad_gpiob_1_pull_up	rw	0x0	gpiob_1 PUll up
16	pad_gpiob_1_pull_frc	rw	0x0	gpiob_1 force enable for pu/pd
17	pad_gpiob_1_oen_reg	rw	0x0	gpiob_1 force outoen value.
19 : 18	pad_gpiob_1_drv_strength	rw	0x0	gpiob_1 driving strength.
20	pad_gpiob_1_out_reg	rw	0x0	gpiob_1 pin output value.
24	pad_gpiob_1_out_frc	rw	0x0	gpiob_1 force output value for output.
28	pad_gpiob_1_oen_frc	rw	0x0	gpiob_1 force enable for outoen.

49.35 pad_gpiob_2_cfg_reg 0x40504088

Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_2_sel	rw	0x0	gpiob_2 select 0 fun_gpiob_2_sel 1 fun_pcm_din_sel 2 fun_psi_m_scl_sel 3 fun_spi_lcd_di_sel 4 fun_aon_pwm_2_sel 5 fun_wf_ext_pa_ctrl_2_sel 6 fun_tports2_sel 7 fun_tports10_sel
7 : 4	pad_gpiob_2_ana	rw	0x0	gpiob_2 ANA_Alter_SEL
8	pad_gpiob_2_pull_dn	rw	0x0	gpiob_2 PUll down
9	pad_gpiob_2_pull_up	rw	0x0	gpiob_2 PUll up
16	pad_gpiob_2_pull_frc	rw	0x0	gpiob_2 force enable for pu/pd
17	pad_gpiob_2_oen_reg	rw	0x0	gpiob_2 force outoen value.
19 : 18	pad_gpiob_2_drv_strength	rw	0x0	gpiob_2 driving strength.
20	pad_gpiob_2_out_reg	rw	0x0	gpiob_2 pin output value.
24	pad_gpiob_2_out_frc	rw	0x0	gpiob_2 force output value for output.
28	pad_gpiob_2_oen_frc	rw	0x0	gpiob_2 force enable for outoen.

49.36 pad_gpiob_3_cfg_reg 0x4050408C

Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_3_sel	rw	0x0	gpiob_3 select 0 fun_gpiob_3_sel 1 fun_pcm_dout_sel 2 fun_psi_m_sda_sel 3 fun_spi_lcd_do_sel 4 fun_spdif_in_sel 5 fun_wf_ext_pa_ctrl_3_sel 6 fun_tports3_sel 7 fun_tports11_sel
7 : 4	pad_gpiob_3_ana	rw	0x0	gpiob_3 ANA_Alter_SEL
8	pad_gpiob_3_pull_dn	rw	0x0	gpiob_3 PUll down
9	pad_gpiob_3_pull_up	rw	0x0	gpiob_3 PUll up
16	pad_gpiob_3_pull_frc	rw	0x0	gpiob_3 force enable for pu/pd
17	pad_gpiob_3_oen_reg	rw	0x0	gpiob_3 force outoen value.
19 : 18	pad_gpiob_3_drv_strength	rw	0x0	gpiob_3 driving strength.
20	pad_gpiob_3_out_reg	rw	0x0	gpiob_3 pin output value.
24	pad_gpiob_3_out_frc	rw	0x0	gpiob_3 force output value for output.
28	pad_gpiob_3_oen_frc	rw	0x0	gpiob_3 force enable for outoen.

49.37 pad_gpiob_4_cfg_reg 0x40504090

Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_4_sel	rw	0x0	gpiob_4 select 0 fun_gpiob_4_sel 1 fun_pwm_0_sel 2 fun_i2s_lrck_0_sel 3 fun_bt_uart_rx_sel 4 fun_spdif_out_sel 6 fun_tports4_sel 7 fun_tports12_sel

7 : 4	pad_gpiob_4_ana	rw	0x0	gpiob_4 ANA_Alter_SEL
8	pad_gpiob_4_pull_dn	rw	0x0	gpiob_4 PUll down
9	pad_gpiob_4_pull_up	rw	0x0	gpiob_4 PUll up
16	pad_gpiob_4_pull_frc	rw	0x0	gpiob_4 force enable for pu/pd
17	pad_gpiob_4_oen_reg	rw	0x0	gpiob_4 force outoen value.
19 : 18	pad_gpiob_4_drv_strength	rw	0x0	gpiob_4 driving strength.
20	pad_gpiob_4_out_reg	rw	0x0	gpiob_4 pin output value.
24	pad_gpiob_4_out_frc	rw	0x0	gpiob_4 force output value for output.
28	pad_gpiob_4_oen_frc	rw	0x0	gpiob_4 force enable for outoen.

49.38 pad_gpiob_5_cfg_reg				0x40504094
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_5_sel	rw	0x0	gpiob_5 select 0 fun_gpiob_5_sel 1 fun_pwm_1_sel 2 fun_i2s_bck_0_sel 3 fun_bt_uart_tx_sel 6 fun_tports5_sel 7 fun_tports13_sel
7 : 4	pad_gpiob_5_ana	rw	0x0	gpiob_5 ANA_Alter_SEL
8	pad_gpiob_5_pull_dn	rw	0x0	gpiob_5 PUll down
9	pad_gpiob_5_pull_up	rw	0x0	gpiob_5 PUll up
16	pad_gpiob_5_pull_frc	rw	0x0	gpiob_5 force enable for pu/pd
17	pad_gpiob_5_oen_reg	rw	0x0	gpiob_5 force outoen value.
19 : 18	pad_gpiob_5_drv_strength	rw	0x0	gpiob_5 driving strength.
20	pad_gpiob_5_out_reg	rw	0x0	gpiob_5 pin output value.
24	pad_gpiob_5_out_frc	rw	0x0	gpiob_5 force output value for output.
28	pad_gpiob_5_oen_frc	rw	0x0	gpiob_5 force enable for outoen.

49.39 pad_gpiob_6_cfg_reg				0x40504098
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_6_sel	rw	0x0	gpiob_6 select 0 fun_gpiob_6_sel 1 fun_pwm_2_sel 2 fun_i2s_dat_in_0_sel 3 fun_bt_uart_cts_sel 6 fun_tports6_sel 7 fun_tports14_sel
7 : 4	pad_gpiob_6_ana	rw	0x0	gpiob_6 ANA_Alter_SEL
8	pad_gpiob_6_pull_dn	rw	0x0	gpiob_6 PUll down
9	pad_gpiob_6_pull_up	rw	0x0	gpiob_6 PUll up
16	pad_gpiob_6_pull_frc	rw	0x0	gpiob_6 force enable for pu/pd
17	pad_gpiob_6_oen_reg	rw	0x0	gpiob_6 force outoen value.
19 : 18	pad_gpiob_6_drv_strength	rw	0x0	gpiob_6 driving strength.
20	pad_gpiob_6_out_reg	rw	0x0	gpiob_6 pin output value.
24	pad_gpiob_6_out_frc	rw	0x0	gpiob_6 force output value for output.
28	pad_gpiob_6_oen_frc	rw	0x0	gpiob_6 force enable for outoen.

49.40 pad_gpiob_7_cfg_reg				0x4050409C
Bits	Field Name	Acc	Default	Description
3 : 0	pad_gpiob_7_sel	rw	0x0	gpiob_7 select 0 fun_gpiob_7_sel 1 fun_aon_pwm_0_sel 2 fun_i2s_dat_out_0_sel 3 fun_bt_uart_rts_sel 6 fun_tports7_sel 7 fun_tports15_sel
7 : 4	pad_gpiob_7_ana	rw	0x0	gpiob_7 ANA_Alter_SEL
8	pad_gpiob_7_pull_dn	rw	0x0	gpiob_7 PUll down

9	pad_gpiob_7_pull_up	rw	0x0	gpiob_7 PUll up
16	pad_gpiob_7_pull_frc	rw	0x0	gpiob_7 force enable for pu/pd
17	pad_gpiob_7_oen_reg	rw	0x0	gpiob_7 force outoen value.
19 : 18	pad_gpiob_7_drv_strength	rw	0x0	gpiob_7 driving strength.
20	pad_gpiob_7_out_reg	rw	0x0	gpiob_7 pin output value.
24	pad_gpiob_7_out_frc	rw	0x0	gpiob_7 force output value for output.
28	pad_gpiob_7_oen_frc	rw	0x0	gpiob_7 force enable for outoen.

50 gpiob

50.1 GPIODATA				0x40505000
Bits	Field Name	Acc	Default	Description
15 : 0	GPIODATA	rw	0x0	GPIO bits data input

50.2 GPIODMSK				0x40505004
Bits	Field Name	Acc	Default	Description
15 : 0	GPIODMSK	rw	0x0	GPIODATA register can be read/write if GPIO DMSK set "1"

50.3 GPIODIR				0x40505008
Bits	Field Name	Acc	Default	Description
15 : 0	GPIODIR	rw	0x0	"1" configure gpio bits to be output "0" configure gpio bits to be input

50.4 GPIOIS				0x4050500C
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIS	rw	0xFFFF	"1" detect signals level "0" detect signals edge

50.5 GPIOIBE				0x40505010
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIBE	rw	0x0	"1" both edges trigger an interrupt, "0" interrupt generation event is controlled by GPIOIEV

50.6 GPIOIEV				0x40505014
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIEV	rw	0xFFFF	GPIO bits interrupt event register: "1" high levels trigger interrupts, "0" low levels trigger interrupts.

50.7 GPIOIE				0x40505018
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIE	rw	0x0	GPIO bits interrupt enable register: "1" corresponding bit interrupt is enabled. "0" corresponding bit interrupt isn't enabled

50.8 GPIORIS				0x4050501C
Bits	Field Name	Acc	Default	Description
15 : 0	GPIORIS	ro	0x0	GPIO bits raw interrupt status register: "1" interrupt condition met "0" condition not met

50.9 GPIOMIS				0x40505020
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOMIS	ro	0x0	GPIO bits masked interrupt status register: "1" Interrupt active "0" interrupt not active

50.10 GPIOIC				0x40505024
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOIC	wo	0x0	GPIO bits interrupt clear register: "1" clears detected interrupt. "0" has no effect.

50.11 GPIOINEN				0x40505028
Bits	Field Name	Acc	Default	Description
15 : 0	GPIOINEN	rw	0x0	GPIO input enable register: "1" input enable. "0" input disable.

50.12 GPIO_FILTER_NUM				0x4050502C
Bits	Field Name	Acc	Default	Description
15 : 0	GPIO_FILTER_NUM	rw	0x3333	

50.13 GPIO_FILTER_DIV_L				0x40505030
Bits	Field Name	Acc	Default	Description
15 : 0	GPIO_FILTER_DIV_L	rw	0x0	

50.14 GPIO_FILTER_DIV_H				0x40505034
Bits	Field Name	Acc	Default	Description
15 : 0	GPIO_FILTER_DIV_H	rw	0x0	

51 pwr_ctrl

51.1 cpu_sys_pwr_ctrl				0x40506000
Bits	Field Name	Acc	Default	Description
9 : 0	cpu_sys_awake_src_en	rw	0x0	
11 : 10	cpu_sys_slp_src_en	rw	0x0	
12	cpu_sys_slp_soft_req	w1c	0x0	
13	cpu_sys_slp_req_vld_clr	w1c	0x0	
15	pon_hwen_cpu_sys	rw	0x1	
16	poff_hwen_cpu_sys	rw	0x1	
17	pon_cpu_sys	rw	0x1	
18	poff_cpu_sys	rw	0x0	
19	timer_poff_pw- r_sys_by_cpu_sys_en	rw	0x0	
20	pwr_sys_pon_by_cpu_sys_en	rw	0x1	
21	pwr_sys_poff_by_cpu_sys_en	rw	0x1	
22	cpu_sys_soft_wakeup_req	w1c	0x0	
23	cpu_sys_awake_req_vld_clr	w1c	0x0	
26 : 24	cfg_cpu_sys_slp_req_mask	rw	0x1	
27	cfg_wifi_core_slep_cpu_sys_en	rw	0x0	
28	cfg_bt_core_slep_cpu_sys_en	rw	0x0	
29	bt_sys_slp_cpu_sys_soft_req	w1c	0x0	
30	wifi_sys_slp_cpu_sys_soft_req	w1c	0x0	

31	cpu_sys_awake_mode	rw	0x1	

51.2 bt_core_pwr_ctrl				0x40506004
Bits	Field Name	Acc	Default	Description
9 : 0	bt_core_awake_src_en	rw	0x0	
11 : 10	bt_core_slp_src_en	rw	0x0	
12	bt_core_slp_soft_req	w1c	0x0	
13	bt_core_slp_req_vld_clr	w1c	0x0	
15	pon_hwen_bt_core	rw	0x1	
16	poff_hwen_bt_core	rw	0x1	
17	pon_bt_core	rw	0x0	
18	poff_bt_core	rw	0x1	
19	timer_poff_pw-r_sys_by_bt_core_en	rw	0x0	
20	pwr_sys_pon_by_bt_core_en	rw	0x1	
21	pwr_sys_poff_by_bt_core_en	rw	0x1	
22	bt_core_soft_wakeup_req	w1c	0x0	
23	bt_core_awake_req_vld_clr	w1c	0x0	
24	bt_core_soft_wakeup_req_to_bt	rw	0x0	
25	bt_core_hw_clr_cpu_sys_slp_en	rw	0x0	
26	bt_sys_awake_mode	rw	0x1	

51.3 wifi_core_pwr_ctrl				0x40506008
Bits	Field Name	Acc	Default	Description
9 : 0	wifi_core_awake_src_en	rw	0x0	
11 : 10	wifi_core_slp_src_en	rw	0x0	
12	wifi_core_slp_soft_req	w1c	0x0	
13	wifi_core_slp_req_vld_clr	w1c	0x0	
15	pon_hwen_wifi_core	rw	0x1	
16	poff_hwen_wifi_core	rw	0x1	
17	pon_wifi_core	rw	0x0	
18	poff_wifi_core	rw	0x1	
19	timer_poff_pw-r_sys_by_wifi_core_en	rw	0x0	
20	pwr_sys_pon_by_wifi_core_en	rw	0x1	
21	pwr_sys_poff_by_wifi_core_en	rw	0x1	
22	wifi_core_soft_wakeup_req	w1c	0x0	
23	wifi_core_awake_req_vld_clr	w1c	0x0	
24	wifi_core_sof-t_wakeup_req_to_wifi	rw	0x0	
25	wifi_core_hw_-clr_cpu_sys_slp_en	rw	0x0	
26	wifi_sys_awake_mode	rw	0x1	
27	wlan_core_tim-er_poff_pwr_sys_sel	rw	0x0	

51.4 pwr_sys_ctrl				0x40506010
Bits	Field Name	Acc	Default	Description
15 : 0	dejitter	rw	0x0	
16	pwr_sys_pon_soft	rw	0x0	
17	pwr_sys_pon_soft_en	rw	0x0	
18	pu_pll_soft	rw	0x0	
19	pu_pll_soft_en	rw	0x0	
20	chip_slp_soft	rw	0x0	
21	chip_slp_soft_en	rw	0x0	
22	cg_pll_soft	rw	0x0	

23	cg_pll_soft_en	rw	0x0	
24	clk_32k_52m_sel_soft	rw	0x1	
25	clk_32k_52m_sel_soft_en	rw	0x0	
26	clk_32k_52m_sw_en	rw	0x1	
27	pwr_sys_poff_hw_en	rw	0x1	
28	xtal_pd_en	rw	0x1	
29	pu_xtal_soft	rw	0x0	
30	pu_xtal_soft_en	rw	0x0	
31	chip_sleep_hw_en	rw	0x1	

51.5 pwr_off_timmer_th1				0x40506014
Bits	Field Name	Acc	Default	Description
31 : 0	bt_core_timer_th	rw	0x80000000	

51.6 pwr_off_timmer_th2				0x40506018
Bits	Field Name	Acc	Default	Description
31 : 0	wifi_core_timer_th	rw	0x80000000	

51.7 pwr_ctrl_delay1				0x4050601C
Bits	Field Name	Acc	Default	Description
15 : 0	pwr_ctrl_pre_delay	rw	0x100	
31 : 16	pwr_ctrl_delay	rw	0x400	

51.8 pwr_ctrl_delay2				0x40506020
Bits	Field Name	Acc	Default	Description
7 : 0	pwr_ctrl_state_delay	rw	0x80	
15 : 8	pwr_sys_xtal_close_delay	rw	0x2	
23 : 16	pwr_sys_xtal_open_delay	rw	0x2	
31 : 24	pwr_sys_pupd_xtal_delay	rw	0x1	

51.9 pwr_sys_delay1				0x40506024
Bits	Field Name	Acc	Default	Description
7 : 0	pwr_sys_dp_open_delay	rw	0xA0	
15 : 8	pwr_sys_dp_close_delay	rw	0x3	
23 : 16	pwr_sys_cg_open_delay	rw	0x1	
31 : 24	pwr_sys_cg_close_delay	rw	0x3	

51.10 pwr_sys_delay2				0x40506028
Bits	Field Name	Acc	Default	Description
7 : 0	pwr_sys_pwr_open_delay	rw	0xC	
15 : 8	pwr_sys_pwr_close_delay	rw	0x1	
23 : 16	pwr_sys_pll_open_delay	rw	0x2	
31 : 24	pwr_sys_pll_close_delay	rw	0x1	

51.11 pwr_ctrl_state1				0x40506030
Bits	Field Name	Acc	Default	Description
1	pon_stab_wifi_core	ro	0x0	
2	pon_stab_bt_core	ro	0x0	
3	pon_stab_cpu_sys	ro	0x0	
5	poff_stab_wifi_core	ro	0x0	
6	poff_stab_bt_core	ro	0x0	

7	poff_stab_cpu_sys	ro	0x0	
11 : 8	state_wifi_core	ro	0x0	
15 : 12	state_bt_core	ro	0x0	
19 : 16	state_cpu_sys	ro	0x0	
21 : 20	wifi_core_cur_state	ro	0x0	
23 : 22	bt_core_cur_state	ro	0x0	
25 : 24	cpu_sys_cur_state	ro	0x0	
29 : 26	pwr_sys_cur_state	ro	0x0	
31 : 30	vcore_sys_cur_state	ro	0x0	

51.12 pwr_ctrl_state2				0x40506034
Bits	Field Name	Acc	Default	Description
0	pon_stab_vcore_sys	ro	0x0	
3	pon_stab_mm_sys	ro	0x0	
4	poff_stab_vcore_sys	ro	0x0	
7	poff_stab_mm_sys	ro	0x0	
19 : 16	state_mm_sys	ro	0x0	
23 : 20	state_vcore_sys	ro	0x0	
24	pu_pll_o	ro	0x0	
25	cg_pll_o	ro	0x0	
26	chip_slp_o	ro	0x0	
27	pwr_ctrl_pu_xtal	ro	0x0	
28	sys_clk_slow_sel	ro	0x0	
29	slow_clk_slow_sel	ro	0x0	
30	pwr_sys_pon	ro	0x0	

51.13 pwr_ctrl_soft_mode1				0x40506038
Bits	Field Name	Acc	Default	Description
5 : 0	cpu_sys_pwr_ctrl_soft	rw	0x0	
11 : 6	bt_core_pwr_ctrl_soft	rw	0x0	
17 : 12	wifi_core_pwr_ctrl_soft	rw	0x0	
31 : 24	pwr_ctrl_clk_32k_div_th	rw	0x98	

51.14 pwr_ctrl_irq_ctrl				0x40506044
Bits	Field Name	Acc	Default	Description
9 : 0	awake_state_clr	w1c	0x0	
13 : 10	slp_state_clr	w1c	0x0	
25 : 16	awake_state_mask	rw	0x0	
29 : 26	slp_state_mask	rw	0x0	
31	pwr_ctrl_irq_en	rw	0x0	

51.15 pwr_ctrl_irq_status				0x40506048
Bits	Field Name	Acc	Default	Description
9 : 0	awake_pos_pulse_raw_state	ro	0x0	
13 : 10	slp_pos_pulse_raw_state	ro	0x0	
14	vcore_sys_awake_raw_status	ro	0x0	
15	cpu_sys_awake_raw_status	ro	0x0	
25 : 16	awake_pos_pulse_state_masked	ro	0x0	
29 : 26	slp_pos_pulse_state_masked	ro	0x0	
30	bt_core_awake_raw_status	ro	0x0	
31	wifi_core_awake_raw_status	ro	0x0	

51.16 mm_sys_pwr_ctrl	0x4050604C

Bits	Field Name	Acc	Default	Description
17	pon_mm_sys	rw	0x1	
18	poff_mm_sys	rw	0x0	

51.17 pwr_ctrl_soft_mode2				0x40506050
Bits	Field Name	Acc	Default	Description
5 : 0	mm_sys_pwr_ctrl_soft	rw	0x0	
11 : 6	vcore_sys_pwr_ctrl_soft	rw	0x36	
14	slp_mode_sel_bt_core	rw	0x0	
15	slp_mode_sel_cpu_sys	rw	0x0	
16	slp_mode_sel_mm_sys	rw	0x0	
17	slp_mode_sel_vcore_sys	rw	0x0	
18	slp_mode_sel_wifi_core	rw	0x0	
21 : 19	cpu_sys_slp_req_mask_d	ro	0x0	
22	cfg_io_mmsys_en_soft	rw	0x0	
23	cfg_io_mmsys_en	rw	0x1	
24	cfg_io_cpusys_en_soft	rw	0x0	
25	cfg_io_cpusys_en	rw	0x1	
26	rtc_slp_out_done_mode_en	rw	0x1	

51.18 vcore_sys_pwr_ctrl				0x40506054
Bits	Field Name	Acc	Default	Description
9 : 0	vcore_sys_awake_src_en	rw	0x0	
11 : 10	vcore_sys_slp_src_en	rw	0x0	
12	vcore_sys_slp_soft_req	w1c	0x0	
13	vcore_sys_slp_req_vld_clr	w1c	0x0	
15	pon_hwen_vcore_sys	rw	0x1	
16	poff_hwen_vcore_sys	rw	0x1	
17	pon_vcore_sys	rw	0x1	
18	poff_vcore_sys	rw	0x0	
19	timer_poff_pw-r_sys_by_vcore_sys_en	rw	0x0	
20	pwr_sys_pon_by_vcore_sys_en	rw	0x1	
21	pwr_sys_poff_by_vcore_sys_en	rw	0x1	
22	vcore_sys_soft_wakeup_req	w1c	0x0	
23	vcore_sys_awake_req_vld_clr	w1c	0x0	
24	vcore_sys_awake_mode	rw	0x1	

51.19 pwr_ctrl_mode				0x4050605C
Bits	Field Name	Acc	Default	Description
30 : 0	pwr_ctrl_mode	rw	0x0	
31	pwr_ctrl_mode_update	rw	0x0	

51.20 ret_ram_ctrl				0x40506060
Bits	Field Name	Acc	Default	Description
3 : 0	cfg_ret_ctrl_cpu_sys_ram	rw	0x0	
7 : 4	cfg_pwr_mux_cpu_sys_lram	rw	0x0	
11 : 8	cfg_pwr_mux_cpu_sys_hram	rw	0x0	
15 : 12	cfg_ret_ctrl_wifi_core_ram	rw	0x0	
19 : 16	cfg_pwr_mux_wifi_core_ram	rw	0x0	
23 : 20	cfg_ret_ctrl_bt_core_ram	rw	0x0	
27 : 24	cfg_pwr_mux_bt_core_ram	rw	0x0	

51.21 pwr_ctrl_monitor				0x40506064
Bits	Field Name	Acc	Default	Description
3 : 0	pwr_ctrl_dbg_mux	rw	0x0	
7 : 4	slp_req_sel	rw	0x0	
11 : 8	awake_req_mask	rw	0x0	
15 : 12	awake_req_raw_clr	w1c	0x0	
19 : 16	awake_vld_state_masked	ro	0x0	
23 : 20	awake_req_sel	rw	0x0	

51.22 pwr_ctrl_soft_mode3				0x40506068
Bits	Field Name	Acc	Default	Description
1 : 0	bt_coreon_soft	rw	0x0	
3 : 2	bt_hold_soft	rw	0x1	
5 : 4	wf_coreon_soft	rw	0x0	
7 : 6	wf_hold_soft	rw	0x1	
9 : 8	cpu_coreon_soft	rw	0x0	
11 : 10	cpu_hold_soft	rw	0x1	
13 : 12	mm_coreon_soft	rw	0x0	
15 : 14	mm_hold_soft	rw	0x1	
17 : 16	top_coreon_soft	rw	0x0	
19 : 18	top_hold_soft	rw	0x1	
22 : 20	cm_coreon_soft	rw	0x0	
26 : 23	cfg_rg_irq_timer2_h_awake_en	rw	0xF	
30 : 27	cfg_rg_irq_others_awake_en	rw	0x0	

51.23 pwr_ctrl_soft_mode4				0x4050606C
Bits	Field Name	Acc	Default	Description
9 : 0	cfg_awake_inv_en	rw	0x0	

51.24 pwr_off_timmer_th3				0x40506070
Bits	Field Name	Acc	Default	Description
31 : 0	cpu_sys_timer_th	rw	0x80000000	

51.25 pwr_off_timmer_th4				0x40506074
Bits	Field Name	Acc	Default	Description
31 : 0	vcore_sys_timer_th	rw	0x80000000	

51.26 bt_core_timer_cnt_ro				0x40506078
Bits	Field Name	Acc	Default	Description
31 : 0	bt_core_timer_cnt_ro	ro	0x0	

51.27 wifi_core_timer_cnt_ro				0x4050607C
Bits	Field Name	Acc	Default	Description
31 : 0	wifi_core_timer_cnt_ro	ro	0x0	

52 aon_pwm

52.1 tmr0_ld_value				0x40507000
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_ld_value	rw	0x0	timer0 load value

52.2 tmr0_cnt_value				0x40507008
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_value	rw	0x0	timer0 counter value

52.3 tmr0_cfg				0x40507010
Bits	Field Name	Acc	Default	Description
0	tmr_mode	rw	0x0	0: free running 1: periodic
1	tmr_run	rw	0x0	timer0 enable

52.4 tmr0_int_ctrl				0x40507014
Bits	Field Name	Acc	Default	Description
0	tmr_int_en	rw	0x0	timer0 interrupt enable
1	tmr_int_raw	ro	0x0	timer0 interrupt raw
2	tmr_int_status	ro	0x0	timer0 interrupt status
3	tmr_int_clr	wo	0x0	timer0 interrupt clear

52.5 tmr0_cnt_read				0x40507018
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_read	rw	0x0	timer0 counter read value

52.6 tmr1_ld_value				0x40507020
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_ld_value	rw	0x0	timer1 load value

52.7 tmr1_cnt_value				0x40507028
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_value	rw	0x0	timer1 counter value

52.8 tmr1_cfg				0x40507030
Bits	Field Name	Acc	Default	Description
0	tmr_mode	rw	0x0	0: free running 1: periodic
1	tmr_run	rw	0x0	timer1 enable

52.9 tmr1_int_ctrl				0x40507034
Bits	Field Name	Acc	Default	Description
0	tmr_int_en	rw	0x0	timer1 interrupt enable
1	tmr_int_raw	ro	0x0	timer1 interrupt raw
2	tmr_int_status	ro	0x0	timer1 interrupt status
3	tmr_int_clr	wo	0x0	timer1 interrupt clear

52.10 tmr1_cnt_read				0x40507038
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_read	rw	0x0	timer1 counter read value

52.11 tmr2_ld_value				0x40507040
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_ld_value	rw	0x0	timer2 load value

52.12 tmr2_cnt_value				0x40507048
Bits	Field Name	Acc	Default	Description
31 : 0	tmr_cnt_value	rw	0x0	timer2 counter value

52.13 tmr2_cfg				0x40507050
Bits	Field Name	Acc	Default	Description
0	tmr_mode	rw	0x0	0: free running 1: periodic
1	tmr_run	rw	0x0	timer2 enable

52.14 tmr2_int_ctrl				0x40507054
Bits	Field Name	Acc	Default	Description
0	tmr_int_en	rw	0x0	timer2 interrupt enable
1	tmr_int_raw	ro	0x0	timer2 interrupt raw
2	tmr_int_status	ro	0x0	timer2 interrupt status
3	tmr_int_clr	wo	0x0	timer2 interrupt clear

52.15 tmr2_cnt_read				0x40507058
Bits	Field Name	Acc	Default	Description
31 : 0	tmr2_cnt_read	rw	0x0	timer2 counter read value

52.16 pwm0_sta_val				0x40507100
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_sta_val	rw	0x0	

52.17 pwm0_end_val				0x40507104
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val	rw	0x0	

52.18 pwm0_step_cfg0				0x40507108
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc	rw	0x0	
31 : 16	pwm_step_num	rw	0x0	

52.19 pwm0_step_cfg1				0x4050710C
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_step_value	rw	0x0	

52.20 pwm0_cfg				0x40507110
Bits	Field Name	Acc	Default	Description
0	pwm_run	rw	0x0	
1	pwm_inc_mode	rw	0x0	
2	pwm_default_val	rw	0x0	
17 : 16	pwm_tmr_sel	rw	0x0	
31	pwm_update	rw	0x0	

52.21 pwm0_int_ctrl				0x40507114
Bits	Field Name	Acc	Default	Description
0	pwm_int_en	rw	0x0	
1	pwm_int_raw	ro	0x0	

2	pwm_int_status	ro	0x0	
3	pwm_int_clr	wo	0x0	

52.22 pwm0_end_val_use				0x40507118
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val_use	ro	0x0	

52.23 pwm0_step_cnt				0x4050711C
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc_cnt	ro	0x0	
31 : 16	pwm_step_num_cnt	ro	0x0	

52.24 pwm1_sta_val				0x40507120
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_sta_val	rw	0x0	

52.25 pwm1_end_val				0x40507124
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val	rw	0x0	

52.26 pwm1_step_cfg0				0x40507128
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc	rw	0x0	
31 : 16	pwm_step_num	rw	0x0	

52.27 pwm1_step_cfg1				0x4050712C
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_step_value	rw	0x0	

52.28 pwm1_cfg				0x40507130
Bits	Field Name	Acc	Default	Description
0	pwm_run	rw	0x0	
1	pwm_inc_mode	rw	0x0	
2	pwm_default_val	rw	0x0	
17 : 16	pwm_tmr_sel	rw	0x0	
31	pwm_update	rw	0x0	

52.29 pwm1_int_ctrl				0x40507134
Bits	Field Name	Acc	Default	Description
0	pwm_int_en	rw	0x0	
1	pwm_int_raw	ro	0x0	
2	pwm_int_status	ro	0x0	
3	pwm_int_clr	wo	0x0	

52.30 pwm1_end_val_use				0x40507138
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val_use	ro	0x0	

52.31 pwm1_step_cnt				0x4050713C
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc_cnt	ro	0x0	
31 : 16	pwm_step_num_cnt	ro	0x0	

52.32 pwm2_sta_val				0x40507140
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_sta_val	rw	0x0	

52.33 pwm2_end_val				0x40507144
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val	rw	0x0	

52.34 pwm2_step_cfg0				0x40507148
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc	rw	0x0	
31 : 16	pwm_step_num	rw	0x0	

52.35 pwm2_step_cfg1				0x4050714C
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_step_value	rw	0x0	

52.36 pwm2_cfg				0x40507150
Bits	Field Name	Acc	Default	Description
0	pwm_run	rw	0x0	
1	pwm_inc_mode	rw	0x0	
2	pwm_default_val	rw	0x0	
17 : 16	pwm_tmr_sel	rw	0x0	
31	pwm_update	rw	0x0	

52.37 pwm2_int_ctrl				0x40507154
Bits	Field Name	Acc	Default	Description
0	pwm_int_en	rw	0x0	
1	pwm_int_raw	ro	0x0	
2	pwm_int_status	ro	0x0	
3	pwm_int_clr	wo	0x0	

52.38 pwm2_end_val_use				0x40507158
Bits	Field Name	Acc	Default	Description
31 : 0	pwm_end_val_use	ro	0x0	

52.39 pwm2_step_cnt				0x4050715C
Bits	Field Name	Acc	Default	Description
15 : 0	pwm_step_cyc_cnt	ro	0x0	
31 : 16	pwm_step_num_cnt	ro	0x0	

53 wdg2

53.1 WDG_LOAD_VAL				0x40508000
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_load_val	rw	0x3FFFF	watchdog timer load value. In reset mode, software should load new value before timer decrease to 0. In interrupt mode, this value is counting start number. The default value is about 8 seconds.

53.2 WDG_CNT				0x40508004
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_cnt	rw	0xFFFFFFFF	watchdog counter value

53.3 WDG_CTRL				0x40508008
Bits	Field Name	Acc	Default	Description
0	wdg_irq_en	rw	0x0	Watchdog interrupt enable bit 0: interrupt is disabled 1: interrupt is enabled For reset mode: wdg_rst_en =1, wdg_irq_en=0. For interrupt mode: wdg_rst_en =0, wdg_irq_en=1. For combined mode: wdg_rst_en =1, wdg_irq_en=1.
1	wdg_run	rw	0x0	Watchdog counter open: 0: counter stops. 1: counter runs.

53.4 WDG_INT_CLR				0x4050800C
Bits	Field Name	Acc	Default	Description
0	wdg_int_clr	wo	0x0	Watchdog interrupt clear Write 1 to this bit to clear interrupt Read this bit always get 0.

53.5 WDG_INT_RAW				0x40508010
Bits	Field Name	Acc	Default	Description
0	wdg_int_raw	ro	0x0	Watchdog interrupt raw status. Watch dog reset cannot clear this raw status. Write wdg_int_clr can clear this raw status.
4	wdg_ld_busy	ro	0x0	Watchdog load busy status 0: Watchdog is ready for new loading 1: Last loading is not completed Software must not load new value when this bit is busy, that is, this bit should be checked before any new loading. This bit is set after a new loading, and lasts two or three RTC clock cycles, about 60us - 92us.

53.6 WDG_INT_MASK				0x40508014
Bits	Field Name	Acc	Default	Description
0	wdg_int_mask	ro	0x0	Watchdog interrupt masked status

53.7 WDG_LOCK				0x40508C00
Bits	Field Name	Acc	Default	Description
15 : 0	wdg_lock	rw	0x0	Watchdog lock control Write 32'h1ACCE551 to this register to unlock watchdog. Write other value to this register to lock watchdog If reading this register, bit-0 is lock status, and other bits are reserved. If watchdog is locked, all control registers cannot be written by software.

54 cm_rf_interface

54.1 reg_00				0x40509000
Bits	Field Name	Acc	Default	Description
1 : 0	bbpll_clk_vco_en_reg	rw	0x2	
3 : 2	bbpll_clk_vco_en_dr	rw	0x0	
4	pu_bbpll_reg	rw	0x0	
5	pu_bbpll_dr	rw	0x0	
6	cm_mdll_rstn_reg	rw	0x0	
7	cm_mdll_rstn_dr	rw	0x0	
11 : 8	cm_mdll_clk26m_en_reg	rw	0x0	
15 : 12	cm_mdll_clk26m_en_dr	rw	0x0	
19 : 16	cm_mdll_outclk_en_reg	rw	0x1	
23 : 20	cm_mdll_outclk_en_dr	rw	0x0	
24	cm_pu_mdll_clk26m_reg	rw	0x0	
25	cm_pu_mdll_clk26m_dr	rw	0x0	
26	cm_pu_mdll_reg	rw	0x0	
27	cm_pu_mdll_dr	rw	0x0	
28	cm_xtal_clk2bbpll_en_reg	rw	0x0	
29	cm_xtal_clk2bbpll_en_dr	rw	0x0	

54.2 reg_04				0x40509004
Bits	Field Name	Acc	Default	Description
3 : 0	cm_xtal_reg_buf_vbit	rw	0xF	
5 : 4	cm_pd_td0	rw	0x1	
7 : 6	bbpll_div1p5_en_td1	rw	0x1	
10 : 8	cm_xtal_clk2bbpll_en_td2	rw	0x5	
11	test_enable_bbpll	rw	0x0	
12	bbpll_vco_high_test_mode	rw	0x0	
13	bbpll_vco_low_test_mode	rw	0x0	
14	cm_test_enable_mdll_clk	rw	0x0	
15	cm_test_enable_mdll	rw	0x0	
16	cm_test_enable_xtal	rw	0x0	
19 : 17	cm_test_bit	rw	0x0	
21 : 20	bbpll_pu_bit	rw	0x3	
23 : 22	cm_mdll_pu_bit	rw	0x3	
25 : 24	cm_xtal_pu_bit	rw	0x1	
26	bbpll_div1p5_en_reg	rw	0x1	
27	bbpll_div1p5_en_dr	rw	0x0	
29 : 28	bbpll_clk_div1p5_en_reg	rw	0x1	
31 : 30	bbpll_clk_div1p5_en_dr	rw	0x0	

54.3 reg_08				0x40509008
Bits	Field Name	Acc	Default	Description
5 : 0	cm_xdblr_count_l_out	ro	0x0	
11 : 6	cm_xdblr_count_h_out	ro	0x0	
14 : 12	cm_xtal_cmp_ibit	rw	0x3	
17 : 15	cm_xtal_clk_delay_bit	rw	0x0	
23 : 18	cm_xtal_cap_fine_bit	rw	0x1F	
26 : 24	cm_xtal_iref_lp_bit	rw	0x5	
30 : 27	cm_xtal_ibias_lp_bit	rw	0x8	
31	cm_xtal_vbcg_en	rw	0x1	

54.4 reg_0C				0x4050900C
Bits	Field Name	Acc	Default	Description

2 : 0	cm_mdll_freq_bit	rw	0x3	
4 : 3	cm_mdll_band_high_bit	rw	0x2	
9 : 5	cm_mdll_div_num	rw	0x7	
10	cm_mdll_frefx2_en	rw	0x1	
11	cm_mdll_colpi-tts_input_sw_notch_en	rw	0x0	
12	cm_mdll_colpi-tts_input_sw_rc_en1	rw	0x0	
13	cm_mdll_colpi-tts_input_sw_rc_en	rw	0x0	
14	cm_mdll_input_sel_colpitts	rw	0x1	
18 : 15	cm_mdll_reg_clk_bit	rw	0x8	
25 : 19	cm_mdll_sf_avdd_sw_bit	rw	0x13	
26	cm_mdll_vref_sel_self_bias	rw	0x1	
27	cm_mdll_sf_sel_ldo18	rw	0x0	
28	cm_mdll_psw_mode	rw	0x1	
29	cm_mdll_sf_mode	rw	0x0	

54.5 reg_10 0x40509010

Bits	Field Name	Acc	Default	Description
0	cm_mdll_lock_det_en	rw	0x1	
1	cm_mdll_auto_reset_en	rw	0x0	
2	cm_mdll_dcc_out_clk	ro	0x0	
3	cm_mdll_dcc_out	ro	0x0	
4	cm_mdll_dcc_en	rw	0x0	
8 : 5	cm_mdll_dcc_vcm_bit	rw	0x3	
9	cm_mdll_dcc_input_swap	rw	0x1	
16 : 10	cm_mdll_dcc_duty_bit	rw	0x3F	
17	cm_mdll_dcc_clk_edge	rw	0x1	
18	cm_mdll_outclk_sel_div2	rw	0x1	
19	cm_mdll_outclk_div2_en	rw	0x1	
20	cm_mdll_ring_reg_lp_mode	rw	0x0	
21	cm_mdll_fref_edge_sel_rise	rw	0x1	
24 : 22	cm_mdll_rstn_vbit	rw	0x7	
28 : 25	cm_mdll_lpf_short_res_bit	rw	0x1	
31 : 29	cm_mdll_cp_ibit	rw	0x4	

54.6 reg_14 0x40509014

Bits	Field Name	Acc	Default	Description
3 : 0	bbpll_pfd_up_dlyf_bit	rw	0x1	
7 : 4	bbpll_pfd_up_dlyc_bit	rw	0x0	
9 : 8	bbpll_clk_80m_duty_bit	rw	0x3	
11 : 10	bbpll_clk_64m_en	rw	0x0	
13 : 12	bbpll_clk_80m_en	rw	0x0	
16 : 14	bbpll_cp_ibit	rw	0x7	
20 : 17	bbpll_reg_vbit	rw	0x8	
21	bbpll_frefx2_en	rw	0x0	
25 : 22	cm_rsvd_read_bit	ro	0x0	
29 : 26	cm_rsvd_bit	rw	0xC	
30	cm_mdll_lock_out	ro	0x0	

54.7 reg_18 0x40509018

Bits	Field Name	Acc	Default	Description
0	bbpll_lock_out	ro	0x0	
1	bbpll_lpf_notch_en	rw	0x0	

2	bbpll_adc_clk160m_en	rw	0x0	
3	bbpll_vco_band_sel	rw	0x1	
4	bbpll_div4_en	rw	0x0	
5	bbpll_pcon_mode	rw	0x1	
6	bbpll_clk_64m_dly_bit	rw	0x0	
8 : 7	bbpll_clk_64m_dly_en	rw	0x0	
9	bbpll_clk_64m_dly_sel	rw	0x0	
13 : 10	bbpll_pfd_dn_dlyf_bit	rw	0x1	
17 : 14	bbpll_pfd_dn_dlyc_bit	rw	0x0	

54.8 cm_1m_reg				0x4050901C
Bits	Field Name	Acc	Default	Description
5 : 0	div_num_1m	rw	0x13	
6	clk_1m_div_en	rw	0x1	
13 : 8	cal_clk_div_num	rw	0x9	
14	cal_clk_en	rw	0x0	
15	debug_cfg	rw	0x0	
17 : 16	xtal_sel	rw	0x3	

54.9 bbpll_sdm_reg1				0x40509020
Bits	Field Name	Acc	Default	Description
0	bbpll_sdm_resetn_reg	rw	0x0	
1	bbpll_sdm_resetn_dr	rw	0x0	
9 : 2	bbpll_sdm_resetn_delay	rw	0x74	
11 : 10	bbpll_sdm_clk_sel_reg	rw	0x1	
14 : 12	bbpll_sdm_intdec_sel	rw	0x3	
15	bbpll_sdm_dither_bypass	rw	0x0	
25 : 24	triangle_wave_cyc_ct	rw	0x0	
27 : 26	triangle_wave_amp_ct	rw	0x0	
28	spread_spectrum_en	rw	0x0	
29	bbpllon_reg	rw	0x0	
30	bbpllon_dr	rw	0x0	
31	bbpllon_done	ro	0x0	

54.10 bbpll_sdm_reg2				0x40509024
Bits	Field Name	Acc	Default	Description
27 : 0	bbpll_sdm_freq_cfg	rw	0x6000000	
28	bbpll_sdm_freq_cfg_pulse	rw	0x0	

54.11 wf_wr_only_reg				0x40509028
Bits	Field Name	Acc	Default	Description
31 : 0	wf_wo_reg	rw	0x0	

54.12 bt_wr_only_reg				0x4050902C
Bits	Field Name	Acc	Default	Description
31 : 0	bt_wo_reg	rw	0x0	

54.13 mdll_fref_reg				0x40509030
Bits	Field Name	Acc	Default	Description
0	mdll_fref_lo	rw	0x0	
1	mdll_fref_md	rw	0x0	

54.14 cm_mdll_lock_ctrl_reg				0x40509034
Bits	Field Name	Acc	Default	Description
9 : 0	mdll_lock_delay	rw	0x34	
10	cm_mdll_lock_chk	rw	0x0	

54.15 cm_mdll_lock_status_reg				0x40509038
Bits	Field Name	Acc	Default	Description
0	cm_mdll_unlock_intr	ro	0x0	
7	cm_mdll_unlock_raw	ro	0x0	

54.16 cm_mdll_unlock_en_reg				0x4050903C
Bits	Field Name	Acc	Default	Description
0	cm_mdll_unlock_en	rw	0x0	

54.17 cm_mdll_unlock_clear_reg				0x40509040
Bits	Field Name	Acc	Default	Description
0	cm_mdll_unlock_clr	rw	0x0	

54.18 dig_reserved_reg				0x40509044
Bits	Field Name	Acc	Default	Description

55 bt_dfe_cfg

55.1 a0x000				0x40620000
Bits	Field Name	Acc	Default	Description
0	rxon_val	rw	0x0	
1	rxon_for	rw	0x0	
2	dm_sync_val	rw	0x0	
3	dm_sync_fix	rw	0x0	
4	fix_rxm	rw	0x0	
5	rx_le_mode	rw	0x0	
7 : 6	rx_rate	rw	0x0	
8	mrx_rstn	rw	0x1	
10	rf_rxon_val	rw	0x0	
11	rf_rxon_for	rw	0x0	
16	txon_val	rw	0x0	
17	txon_for	rw	0x0	
18	tx_tone	rw	0x0	
19	tx_pwr_fix	rw	0x0	
20	fix_txm	rw	0x0	
21	tx_le_mode	rw	0x0	
23 : 22	tx_rate	rw	0x0	
24	mtx_rstn	rw	0x1	
26	rf_txon_val	rw	0x0	
27	rf_txon_for	rw	0x0	
28	btrf_dphi_tone	rw	0x0	
29	btrf_ampq_fix	rw	0x0	
30	btrf_ampi_fix	rw	0x0	

55.2 a0x004				0x40620004
Bits	Field Name	Acc	Default	Description

1 : 0	cfg_fix_rstn	rw	0x0	
27 : 16	cfg_fix_clk	rw	0xF00	

55.3 a0x008				0x40620008
Bits	Field Name	Acc	Default	Description
3 : 0	mslb_idx	rw	0x4	
16 : 8	cfg_clk_mdm	rw	0x0	

55.4 dpll_cgen_00c				0x4062000C
Bits	Field Name	Acc	Default	Description
15 : 0	dpll_cgen_val	rw	0x0	
31 : 16	dpll_cgen_fix	rw	0x0	

55.5 gll_ctrl				0x40620020
Bits	Field Name	Acc	Default	Description
0	gll_rxon_val	rw	0x0	
1	gll_rxon_fix	rw	0x0	
2	gll_txon_val	rw	0x0	
3	gll_txon_fix	rw	0x0	
4	gll_btle_val	rw	0x0	
5	gll_btle_fix	rw	0x0	
6	gll_tedr_val	rw	0x0	
7	gll_tedr_fix	rw	0x0	
9 : 8	gll_rate_val	rw	0x0	
10	gll_sync_fix	rw	0x0	
11	gll_rate_fix	rw	0x0	
12	gll_ifsl_val	rw	0x0	
13	gll_ifsl_fix	rw	0x0	
14	gll_trmp_val	rw	0x0	
15	gll_trmp_fix	rw	0x0	
22 : 16	gll_freq_val	rw	0x0	
23	gll_freq_fix	rw	0x0	
31 : 24	gll_tpwr_val	rw	0x0	

55.6 gll_ctrl2				0x40620024
Bits	Field Name	Acc	Default	Description
0	gll_tpwr_fix	rw	0x0	
1	gll_rfcoef_hld	rw	0x0	

55.7 plriq_26m_080				0x40620080
Bits	Field Name	Acc	Default	Description
4 : 0	g_plriq_dlyp	rw	0x0	
12 : 8	g_plriq_dlya	rw	0x0	
18 : 16	g_plriq_alf2	rw	0x0	
22 : 20	g_plriq_alf1	rw	0x0	
26 : 24	g_plriq_mode	rw	0x3	
31 : 28	g_tx_dly_num_26m	rw	0x0	

55.8 plriq_26m_084				0x40620084
Bits	Field Name	Acc	Default	Description
4 : 0	e_plriq_dlyp	rw	0xE	
12 : 8	e_plriq_dlya	rw	0xE	

18 : 16	e_plriq_alf2	rw	0x4	
22 : 20	e_plriq_alf1	rw	0x4	
26 : 24	e_plriq_mode	rw	0x2	
31 : 28	e_tx_dly_num_26m	rw	0xC	

55.9 a0x090				0x40620090
Bits	Field Name	Acc	Default	Description
0	k_hp_dr0	rw	0x0	
1	k_hp_dr1	rw	0x1	
3	bt_polr_dly_hp	rw	0x0	
4	tx_13to26_cic_edr	rw	0x1	
5	tx_13to26_cic_gfsk	rw	0x1	
6	tx_26to52_div	rw	0x1	
7	tx_52to208_div	rw	0x1	
15 : 12	tx_dly_num_52m	rw	0xA	
19 : 16	tx_dly_num_208m	rw	0xF	
21	fref_52m_fix	rw	0x0	
22	fref_52m_rx	rw	0x1	
23	fref_52m_rx_twsm	rw	0x1	
24	fref_52m_tx_gfsk	rw	0x1	
25	fref_52m_tx_edr	rw	0x0	
27 : 26	rfsel_clk_dpa	rw	0x0	
29 : 28	rfsel_clk_dco_rx	rw	0x0	
31 : 30	rfsel_clk_dco_tx	rw	0x0	

55.10 a0x094				0x40620094
Bits	Field Name	Acc	Default	Description
14 : 0	dac_dc_i	rw	0x0	
30 : 16	dac_dc_q	rw	0x0	

55.11 a0x098				0x40620098
Bits	Field Name	Acc	Default	Description
14 : 0	tx_iqim_phase_err	rw	0x0	
30 : 16	tx_iqim_amp_err	rw	0x0	
31	tx_iqim_bypass	rw	0x1	

55.12 txdfe_rtcv0				0x4062009C
Bits	Field Name	Acc	Default	Description
27 : 0	txdfe_rtcv_val	rw	0x0	
28	txdfe_rtcv_fix	rw	0x0	

55.13 txdfe_rtcv1				0x406200A0
Bits	Field Name	Acc	Default	Description
3 : 0	txdfe_rtcv_thdr	rw	0x4	
7 : 4	txdfe_rtcv_coef_adj	rw	0x0	
11 : 8	txdfe_rtcv_thdr_min	rw	0x3	
15 : 12	txdfe_rtcv_thdr_max	rw	0xC	

55.14 rxdfe_rtcv0				0x406200A4
Bits	Field Name	Acc	Default	Description
27 : 0	rxdfe_rtcv_val	rw	0x0	
28	rxdfe_rtcv_fix	rw	0x0	

55.15 rxdfe_rtcv1				0x406200A8
Bits	Field Name	Acc	Default	Description
3 : 0	rxdfe_rtcv_thdr	rw	0x2	
7 : 4	rxdfe_rtcv_coef_adj	rw	0x0	
11 : 8	rxdfe_rtcv_thdr_min	rw	0x3	
15 : 12	rxdfe_rtcv_thdr_max	rw	0xC	
16	bt_rc_lpf_bypass	rw	0x0	
17	bt_rc_lpf_fir	rw	0x1	

55.16 a0x0f0				0x406200F0
Bits	Field Name	Acc	Default	Description
2 : 0	dbg_sel_top0	rw	0x7	
6 : 4	dbg_sel_top1	rw	0x7	
10 : 8	dbg_sel_top2	rw	0x7	
14 : 12	dbg_sel_top3	rw	0x7	
23 : 16	dbg_sel_mdm	rw	0xFF	
27 : 24	gdb_sel_dpll	rw	0xF	
31 : 28	gdb_sel_fpga	rw	0xF	

55.17 a0x100				0x40620100
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit00	rw	0x0	
12 : 8	gdb_sel_bit01	rw	0x1	
20 : 16	gdb_sel_bit02	rw	0x2	
28 : 24	gdb_sel_bit03	rw	0x3	

55.18 a0x104				0x40620104
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit04	rw	0x4	
12 : 8	gdb_sel_bit05	rw	0x5	
20 : 16	gdb_sel_bit06	rw	0x6	
28 : 24	gdb_sel_bit07	rw	0x7	

55.19 a0x108				0x40620108
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit08	rw	0x8	
12 : 8	gdb_sel_bit09	rw	0x9	
20 : 16	gdb_sel_bit10	rw	0xA	
28 : 24	gdb_sel_bit11	rw	0xB	

55.20 a0x10c				0x4062010C
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit12	rw	0xC	
12 : 8	gdb_sel_bit13	rw	0xD	
20 : 16	gdb_sel_bit14	rw	0xE	
28 : 24	gdb_sel_bit15	rw	0xF	

55.21 a0x110				0x40620110
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit16	rw	0x10	
12 : 8	gdb_sel_bit17	rw	0x11	
20 : 16	gdb_sel_bit18	rw	0x12	

28 : 24	gdb_sel_bit19	rw	0x13	

55.22 a0x114				0x40620114
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit20	rw	0x14	
12 : 8	gdb_sel_bit21	rw	0x15	
20 : 16	gdb_sel_bit22	rw	0x16	
28 : 24	gdb_sel_bit23	rw	0x17	

55.23 a0x118				0x40620118
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit24	rw	0x18	
12 : 8	gdb_sel_bit25	rw	0x19	
20 : 16	gdb_sel_bit26	rw	0x1A	
28 : 24	gdb_sel_bit27	rw	0x1B	

55.24 a0x11c				0x4062011C
Bits	Field Name	Acc	Default	Description
4 : 0	gdb_sel_bit28	rw	0x1C	
12 : 8	gdb_sel_bit29	rw	0x1D	
20 : 16	gdb_sel_bit30	rw	0x1E	
28 : 24	gdb_sel_bit31	rw	0x1F	

55.25 a0x120				0x40620120
Bits	Field Name	Acc	Default	Description
31 : 0	gdb_out_dat	rw	0x0	

55.26 a0x200				0x40620200
Bits	Field Name	Acc	Default	Description
15 : 0	cgen_val0	rw	0x0	
31 : 16	cgen_fix0	rw	0x0	

55.27 a0x204				0x40620204
Bits	Field Name	Acc	Default	Description
15 : 0	cgen_val1	rw	0x0	
31 : 16	cgen_fix1	rw	0x0	

55.28 a0x208				0x40620208
Bits	Field Name	Acc	Default	Description
15 : 0	cgen_val2	rw	0x0	
31 : 16	cgen_fix2	rw	0x0	

55.29 a0x20c				0x4062020C
Bits	Field Name	Acc	Default	Description
7 : 0	cgen_val3	rw	0x0	
23 : 16	cgen_fix3	rw	0x0	
31 : 24	cgen_free	rw	0xFF	

55.30 a0x210	0x40620210

Bits	Field Name	Acc	Default	Description
0	rxmdm_lp	rw	0x0	
1	ghdr_fec_en	rw	0x1	

55.31 cass_ctrl				0x40620280
Bits	Field Name	Acc	Default	Description
0	cass_en	rw	0x0	
1	cass_trx_en	rw	0x0	
2	afh_cass_rsten	rw	0x0	
3	afh_cass_clken	rw	0x0	
4	cass_clr	rw	0x0	
5	cass_ok	ro	0x0	
7 : 6	cass_ct	ro	0x0	
13 : 8	tim_cass_on	rw	0x6	
21 : 16	cass_num	rw	0x20	
29 : 24	cass_cnt	ro	0x0	

55.32 cass_var				0x40620284
Bits	Field Name	Acc	Default	Description
2 : 0	cass_prt_num	rw	0x5	
3	cass_frm_sel	rw	0x0	
6 : 4	tim_norm_dn	rw	0x2	
10 : 8	tim_cass_dn	rw	0x0	
14 : 12	cass_beta	rw	0x3	
18 : 16	cass_idx	rw	0x1	
19	cass_idx_fix	rw	0x1	
20	cass_mem_sel	rw	0x1	
21	cass_mem_stp	rw	0x1	

55.33 cass_rfpwr				0x40620288
Bits	Field Name	Acc	Default	Description
1 : 0	cass_rfpwr_grp0	rw	0x0	
5 : 4	cass_rfpwr_grp1	rw	0x1	
10 : 8	cass_rfpwr_grp2	rw	0x2	
14 : 12	cass_rfpwr_grp3	rw	0x4	
19 : 16	cass_rfpwr_grp4	rw	0x6	
23 : 20	cass_rfpwr_grp5	rw	0x0	
27 : 24	cass_rfpwr_grp6	rw	0x0	

55.34 cass_map0				0x40620290
Bits	Field Name	Acc	Default	Description
31 : 0	cass_map0	rw	0xFFFFFFFF	

55.35 cass_map1				0x40620294
Bits	Field Name	Acc	Default	Description
7 : 0	cass_map1	rw	0xFF	

55.36 cass_rssi0				0x406202A0
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi0	ro	0x0	

55.37 cass_rssi1				0x406202A4
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi1	ro	0x0	

55.38 cass_rssi2				0x406202A8
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi2	ro	0x0	

55.39 cass_rssi3				0x406202AC
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi3	ro	0x0	

55.40 cass_rssi4				0x406202B0
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi4	ro	0x0	

55.41 cass_rssi5				0x406202B4
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi5	ro	0x0	

55.42 cass_rssi6				0x406202B8
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi6	ro	0x0	

55.43 cass_rssi7				0x406202BC
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi7	ro	0x0	

55.44 cass_rssi8				0x406202C0
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi8	ro	0x0	

55.45 cass_rssi9				0x406202C4
Bits	Field Name	Acc	Default	Description
31 : 0	cass_rssi9	ro	0x0	

55.46 a0x500				0x40620500
Bits	Field Name	Acc	Default	Description
0	adc_i_clkedge_sel	rw	0x0	
1	adc_q_clkedge_sel	rw	0x0	
2	adc_iq_format	rw	0x0	
3	adc_iq_swap	rw	0x0	
4	adc_i_negate	rw	0x0	
5	adc_q_negate	rw	0x0	
6	adc_i_dly	rw	0x0	
7	adc_q_dly	rw	0x0	
8	bypass_srrc	rw	0x0	
9	bypass_mixer	rw	0x0	
10	sdm_adc_dith_bypass	rw	0x1	

11	sdm_adc_dith_clkedge_sel	rw	0x0	

55.47 a0x504				0x40620504
Bits	Field Name	Acc	Default	Description
12 : 0	corsrrc_dphi_1m	rw	0x0	
28 : 16	corsrrc_dphi_2m	rw	0x0	

55.48 a0x508				0x40620508
Bits	Field Name	Acc	Default	Description
0	agc_fix_en	rw	0x0	
1	rfpwr_frm_sel	rw	0x1	
2	agc_rx_comb	rw	0x0	
6 : 4	idx_ini	rw	0x3	
10 : 8	idx_min	rw	0x1	
14 : 12	idx_max	rw	0x3	
17 : 16	step_norm	rw	0x1	
19 : 18	step_thlg	rw	0x1	
21 : 20	step_upfl	rw	0x1	
23 : 22	step_dnfl	rw	0x1	
25 : 24	flg_thlg_fix	rw	0x0	
26	en_lg_stp	rw	0x1	
30	agc_ramp_sel	rw	0x1	

55.49 a0x50c				0x4062050C
Bits	Field Name	Acc	Default	Description
6 : 4	time_init	rw	0x1	
14 : 8	time_pcal0	rw	0x20	
22 : 16	time_pcal1	rw	0x20	
30 : 24	time_tran_dig	rw	0x38	

55.50 a0x510				0x40620510
Bits	Field Name	Acc	Default	Description
5 : 0	time_lpfrst	rw	0xA	
15 : 8	dagc_mask	rw	0xFF	
19 : 16	thpup_val	rw	0x6	
23 : 20	thpdn_val	rw	0x6	
29 : 24	time_digtr	rw	0x14	
31 : 30	agc_din_sel	rw	0x2	

55.51 a0x514				0x40620514
Bits	Field Name	Acc	Default	Description
3 : 0	dagc_pth_ovr	rw	0x3	
7 : 4	tim_notch	rw	0x1	
14 : 8	dagc_pin_tar	rw	0x1A	
22 : 16	dagc_pth_max	rw	0xC	
30 : 24	dagc_pth_min	rw	0x28	
31	dagc_fix_en	rw	0x0	

55.52 a0x518				0x40620518
Bits	Field Name	Acc	Default	Description
6 : 0	vitb_pin_tar	rw	0xA	
11 : 8	thpup_val_vitb	rw	0x8	

15 : 12	thpdn_val_vitb	rw	0x8	
18 : 16	agc_din_sht	rw	0x0	
21 : 20	dagc_rssi_ps	rw	0x1	
23 : 22	dagc_rssi_ps_lr	rw	0x2	
24	rsten_srrc	rw	0x0	
25	rsten_notch	rw	0x1	
26	rsten_lpf	rw	0x1	
31 : 27	tim_rcdly	rw	0x6	

55.53 a0x51c				0x4062051C
Bits	Field Name	Acc	Default	Description
0	bypass_dccancel	rw	0x0	
1	dc_calib_en	rw	0x1	
2	dc_cancel_en	rw	0x1	
3	dc_est_restart_dr	rw	0x0	
6 : 4	dc_alpha_rst_ctrl	rw	0x6	
11 : 8	dc_est_wait_time	rw	0x7	
13 : 12	dcc_bw_ct2	rw	0x1	
15 : 14	dc_est_ctrl	rw	0x1	
18 : 16	dc_scale_factor	rw	0x3	
19	dcc_fil_bypass	rw	0x1	
22 : 20	dc_avg_len_coef	rw	0x4	
23	dc_est_mode	rw	0x0	
26 : 24	dcc_mode_ct	rw	0x6	
30 : 28	dc_reg_wr_ctrl	rw	0x1	
31	dc_reg_update_en	rw	0x1	

55.54 a0x520				0x40620520
Bits	Field Name	Acc	Default	Description
15 : 0	dcc_bw_ct	rw	0x0	
23 : 16	dcc_tmr_ct	rw	0x30	
30 : 24	dcc_tmr_ct2	rw	0x10	

55.55 a0x528				0x40620528
Bits	Field Name	Acc	Default	Description
0	notch_en	rw	0x0	
7 : 4	notch_a	rw	0xC	
21 : 8	notch_b	rw	0x3BF3	
26 : 24	dc_reg_index	rw	0x0	
29 : 28	dc_reg_rd_sel	rw	0x0	
30	dc_reg_set_en	rw	0x0	
31	dc_reg_set_dr	rw	0x0	

55.56 a0x52c				0x4062052C
Bits	Field Name	Acc	Default	Description
3 : 0	dagc_pth_ovr_lr	rw	0x3	
14 : 8	dagc_pin_tar_lr	rw	0x14	
22 : 16	dagc_pth_max_lr	rw	0xC	
30 : 24	dagc_pth_min_lr	rw	0x1C	
31	dagc_fix_en_lr	rw	0x0	

55.57 a0x530				0x40620530
Bits	Field Name	Acc	Default	Description

3 : 0	rfdb_idx00	rw	0x0	
7 : 4	rfdb_idx01	rw	0x0	
11 : 8	rfdb_idx02	rw	0x2	
15 : 12	rfdb_idx03	rw	0x1	
19 : 16	rfdb_idx04	rw	0x7	
23 : 20	rfdb_idx05	rw	0x1	
27 : 24	rfdb_idx06	rw	0x3	
31 : 28	rfdb_idx07	rw	0x1	

55.58 a0x534				0x40620534
Bits	Field Name	Acc	Default	Description
6 : 0	rfdb_offset	rw	0x6E	
13 : 8	time_tran_ana	rw	0xD	

55.59 a0x544				0x40620544
Bits	Field Name	Acc	Default	Description
14 : 0	rx_data_i_reg	rw	0x0	
15	rx_data_i_set_dr	rw	0x0	
30 : 16	rx_data_q_reg	rw	0x0	
31	rx_data_q_set_dr	rw	0x0	

55.60 a0x5a0				0x406205A0
Bits	Field Name	Acc	Default	Description
6 : 0	thmax_val0	rw	0x0	
14 : 8	thmax_val1	rw	0x18	
22 : 16	thmax_val2	rw	0x21	
30 : 24	thmax_val3	rw	0x20	

55.61 a0x5a4				0x406205A4
Bits	Field Name	Acc	Default	Description
6 : 0	thmin_val0	rw	0x30	
14 : 8	thmin_val1	rw	0x42	
22 : 16	thmin_val2	rw	0x39	
30 : 24	thmin_val3	rw	0x42	

55.62 a0x5a8				0x406205A8
Bits	Field Name	Acc	Default	Description
2 : 0	agc_alp_a1	rw	0x4	
6 : 4	agc_alp_b1	rw	0x4	
10 : 8	agc_alp_a2	rw	0x2	
14 : 12	agc_alp_b2	rw	0x2	
18 : 16	agc_alp_a3	rw	0x2	
22 : 20	agc_alp_b3	rw	0x2	
26 : 24	agc_alp_c	rw	0x3	
28	agc_alp_alt	rw	0x0	
29	agc_alp_infrm	rw	0x1	

55.63 a0x5ac				0x406205AC
Bits	Field Name	Acc	Default	Description
3 : 0	thmin_dlt	rw	0x6	
7 : 4	thmax_dlt	rw	0x6	
10 : 8	thmin_mrg	rw	0x3	

14 : 12	thmax_mrg	rw	0x3	
22 : 16	thmin_too	rw	0x7F	
30 : 24	thmax_too	rw	0x0	

55.64 a0x5b0				0x406205B0
Bits	Field Name	Acc	Default	Description
2 : 0	prt_num_bt	rw	0x5	
6 : 4	prt_num_le1m	rw	0x5	
10 : 8	prt_num_le2m	rw	0x3	
14 : 12	prt_num_lelr	rw	0x5	

55.65 a0x5b4				0x406205B4
Bits	Field Name	Acc	Default	Description
14 : 0	corlpf_dphi_1m	rw	0x4A4	
15	corlpf_dphi_fix	rw	0x0	
30 : 16	corlpf_dphi_2m	rw	0x3DE	
31	corlpf_bps	rw	0x0	

55.66 a0x5b8				0x406205B8
Bits	Field Name	Acc	Default	Description
6 : 0	thmin_wb	rw	0xC	
14 : 8	thmax_wb	rw	0xA	
18 : 16	agc_alp_wb	rw	0x2	

55.67 a0x5bc				0x406205BC
Bits	Field Name	Acc	Default	Description
6 : 0	thmin_sb	rw	0x9	
14 : 8	thmax_sb	rw	0x0	
21 : 16	dgc_bagc_tar	rw	0x24	
23	dgc_bagc_en	rw	0x1	
27 : 24	dgc_bagc_thd	rw	0x3	

55.68 a0x5c0				0x406205C0
Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a0	rw	0x2000100	

55.69 a0x5c4				0x406205C4
Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a1	rw	0x2000100	

55.70 a0x5c8				0x406205C8
Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a2	rw	0x2000100	

55.71 a0x5cc				0x406205CC
Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a3	rw	0x2000100	

55.72 a0x5d0	0x406205D0

Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a4	rw	0x2000100	

55.73 a0x5d4				0x406205D4
Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a5	rw	0x2000100	

55.74 a0x5d8				0x406205D8
Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a6	rw	0x2000100	

55.75 a0x5dc				0x406205DC
Bits	Field Name	Acc	Default	Description
31 : 0	clkn_th_a7	rw	0x2000100	

55.76 a0x5f0				0x406205F0
Bits	Field Name	Acc	Default	Description
2 : 0	idx_tws	rw	0x7	
6 : 4	idx_nul	rw	0x7	
15 : 8	frm_en_ll	rw	0x0	
23 : 16	frm_en_mrx	ro	0x0	

55.77 wbagc_5f4				0x406205F4
Bits	Field Name	Acc	Default	Description
1 : 0	wbagc_en	rw	0x3	
5 : 4	wbagc_win	rw	0x1	
13 : 8	wbagc_thr	rw	0xA	
21 : 16	wbagc_tran_tim	rw	0x14	
30 : 24	btsar_ifmt	rw	0x0	

55.78 dpll_rtcv_710				0x40620710
Bits	Field Name	Acc	Default	Description
27 : 0	dpll_rtcv_coef_val	rw	0x0	
28	dpll_rtcv_coef_fix	rw	0x0	

55.79 dpll_rtcv_714				0x40620714
Bits	Field Name	Acc	Default	Description
3 : 0	dpll_rtcv_thdr	rw	0x4	
7 : 4	dpll_rtcv_coef_adj	rw	0x2	
11 : 8	dpll_rtcv_thdr_min	rw	0x1	
15 : 12	dpll_rtcv_thdr_max	rw	0xC	
19 : 16	dpll_rtcv_thdr_we0	rw	0xF	
23 : 20	dpll_rtcv_thdr_we1	rw	0x0	

55.80 a0x720				0x40620720
Bits	Field Name	Acc	Default	Description
0	bt_simpl_kvco_cal_en	rw	0x0	
1	bt_simpl_kvco_cal_dr	rw	0x1	
3 : 2	simple_kvco_cal_times	rw	0x0	
5 : 4	simple_kvco_cal_fdlt	rw	0x1	

7	simple_kvco_cal_fneg	rw	0x0	
10 : 8	dco_avg_len	rw	0x4	
14 : 12	simple_kvco_cal_wait_tm	rw	0x7	
18 : 16	dco_tri_fsm	rw	0x4	
19	dco_tri_sel	rw	0x0	
21 : 20	dco_tri_mag	rw	0x0	
24	lpf2_din_div2	rw	0x1	

55.81 a0x724				0x40620724
Bits	Field Name	Acc	Default	Description
0	timer_dpll_sdm_resetn_reg	rw	0x0	
1	timer_dpll_sdm_resetn_dr	rw	0x0	
2	dpll_cal_resetn_reg	rw	0x0	
3	dpll_cal_resetn_dr	rw	0x0	
4	dpll_sdm_resetn_reg	rw	0x0	
5	dpll_sdm_resetn_dr	rw	0x0	
7 : 6	dpll_sdm_clk_sel_reg	rw	0x1	
11 : 8	reset_dpll_sdm_delay	rw	0x1	
12	dpll_cal_opt	rw	0x0	
13	dpll_cal_manual	rw	0x0	
15 : 14	dpll_cal_eachbit_time	rw	0x1	
20 : 16	dpll_cal_lock_delay	rw	0x0	
26 : 24	dpll_cal_init_delay	rw	0x3	
30 : 28	dpll_cal_wait_time	rw	0x0	

55.82 a0x728				0x40620728
Bits	Field Name	Acc	Default	Description
0	dpll_cal_en_reg	rw	0x0	write 1: reg value invert
1	dpll_cal_en_dr	rw	0x0	WT Regs. (normal regs, do not use this addr)
2	dpll_cal_cnt_up	rw	0x0	
3	dpll_cal_cnt_en_sel	rw	0x0	
4	dpll_vco_band_dr	rw	0x0	
5	dpll_vco_band_sel	rw	0x1	
17 : 8	dpll_vco_band_reg	rw	0x100	

55.83 a0x730				0x40620730
Bits	Field Name	Acc	Default	Description
13 : 0	k_hp_reg0	rw	0xE6C	
29 : 16	k_hp_reg1	rw	0xE6C	

55.84 a0x734				0x40620734
Bits	Field Name	Acc	Default	Description
13 : 0	k_hp_reg2	rw	0xE6C	
29 : 16	k_hp_reg3	rw	0xE6C	

55.85 a0x738				0x40620738
Bits	Field Name	Acc	Default	Description
0	dlpf_restart	rw	0x0	
1	dlpf_freq_mode	rw	0x1	
3 : 2	dlpf_mode	rw	0x1	
4	dlpf_coarse_bypass_lpf1	rw	0x1	
5	lpf1_dout_pipe	rw	0x1	
6	gro_sample_clkedge_sel	rw	0x1	

7	bt_dpll_gro_mode_sel	rw	0x1	
9 : 8	dlpf_sinc_order	rw	0x3	
11 : 10	dlpf_sinc_order1	rw	0x3	
21 : 12	dlpf_coarse_code_dr_reg	rw	0x200	
24	dlpf_coarse_code_dr	rw	0x0	
25	dlpf_det_bypass	rw	0x0	
26	rfpll_sdm_dither_bypass	rw	0x1	

55.86 a0x740				0x40620740
Bits	Field Name	Acc	Default	Description
8 : 0	dlpf_dco_code_initial	rw	0xFF	
25 : 16	dlpf_coarse_code_initial	rw	0x1FF	

55.87 a0x744				0x40620744
Bits	Field Name	Acc	Default	Description
1 : 0	dlpf_k1_idx	rw	0x2	
3 : 2	dlpf_k2_idx	rw	0x2	
6 : 4	dlpf_k3_idx_coarse	rw	0x2	
10 : 8	dlpf_k4_idx_coarse	rw	0x2	
13 : 12	dlpf_kp_idx_coarse	rw	0x1	
18 : 16	dlpf_k3_idx_fine1	rw	0x2	
22 : 20	dlpf_k4_idx_fine1	rw	0x2	
25 : 24	dlpf_kp_idx_fine1	rw	0x1	
29 : 28	dlpf_k5_idx	rw	0x2	
31 : 30	dlpf_k6_idx	rw	0x2	

55.88 a0x748				0x40620748
Bits	Field Name	Acc	Default	Description
11 : 0	dlpf_det_fine1_thresh	rw	0xA0	
23 : 12	dlpf_det_fine2_thresh	rw	0xA0	
31 : 24	dly_ini_dco	rw	0x2	

55.89 a0x74c				0x4062074C
Bits	Field Name	Acc	Default	Description
6 : 0	gro_mode2_digital_offset	rw	0x0	
19 : 8	dlpf_det_coarse_thresh	rw	0xA0	
21 : 20	cors_avg_num	rw	0x1	
22	cors_avg_bps	rw	0x0	

55.90 a0x750				0x40620750
Bits	Field Name	Acc	Default	Description
7 : 0	dlpf_cnt_fine1	rw	0xFF	
15 : 8	dlpf_cnt_fine1_52m	rw	0xFF	
23 : 16	dlpf_cnt_fine2	rw	0x7D	
31 : 24	dlpf_cnt_fine2_52m	rw	0x20	

55.91 a0x754				0x40620754
Bits	Field Name	Acc	Default	Description
7 : 0	dlpf_cnt_coarse	rw	0x7D	
15 : 8	dlpf_cnt_coarse_52m	rw	0x20	
18 : 16	dlpf_k3_idx_fine2	rw	0x1	
22 : 20	dlpf_k4_idx_fine2	rw	0x1	

25 : 24	dlpf_kp_idx_fine2	rw	0x1	
28	dlpf_cycle_slip_fine_en	rw	0x1	
29	dlpf_cycle_slip_coarse_en	rw	0x0	

55.92 a0x758				0x40620758
Bits	Field Name	Acc	Default	Description
9 : 0	dlpf_cycle_slip_fine_code	rw	0x40	
21 : 12	dlpf_cycle_slip_coarse_code	rw	0xA	

55.93 phase_off0_75c				0x4062075C
Bits	Field Name	Acc	Default	Description
11 : 0	dlpf_phase_off0	rw	0x15E	
15 : 12	tx_dly_num_plr0	rw	0x6	
27 : 16	dlpf_phase_off1	rw	0xC8	
31 : 28	tx_dly_num_plr1	rw	0x6	

55.94 phase_off1_760				0x40620760
Bits	Field Name	Acc	Default	Description
11 : 0	dlpf_phase_off2	rw	0xB4	
15 : 12	tx_dly_num_plr2	rw	0x6	
27 : 16	dlpf_phase_off3	rw	0x78	
31 : 28	tx_dly_num_plr3	rw	0xC	

55.95 a0x764				0x40620764
Bits	Field Name	Acc	Default	Description
0	dco_clk_swh_val_tx	rw	0x0	
1	dco_clk_swh_fix_tx	rw	0x0	
2	dco_dat_swh_val_tx	rw	0x0	
3	dco_dat_swh_fix_tx	rw	0x0	
6 : 4	dco_clk_fsm_tx	rw	0x3	
13 : 8	dco_dat_tim_tx	rw	0x2	
26 : 16	dco_clk_tim_tx	rw	0x14	
31 : 28	dco_rcc_tim	rw	0x4	

55.96 a0x768				0x40620768
Bits	Field Name	Acc	Default	Description
0	dco_clk_swh_val_rx	rw	0x0	
1	dco_clk_swh_fix_rx	rw	0x0	
2	dco_dat_swh_val_rx	rw	0x0	
3	dco_dat_swh_fix_rx	rw	0x0	
6 : 4	dco_clk_fsm_rx	rw	0x3	
13 : 8	dco_dat_tim_rx	rw	0x2	
26 : 16	dco_clk_tim_rx	rw	0x14	

55.97 a0x76c				0x4062076C
Bits	Field Name	Acc	Default	Description
0	sdm_btdco_sel0	rw	0x0	
1	sdm_btdco_sel1	rw	0x0	
3 : 2	sdm_btdco_sel1_sht	rw	0x0	
23 : 20	num_clkrst_sdm	rw	0x3	
27 : 24	num_clkrst_dco	rw	0x1	
31 : 28	num_clkrst_gro	rw	0x6	

55.98 dacintf_770				0x40620770
Bits	Field Name	Acc	Default	Description
1 : 0	dacintf_delay_0p25_tlo_mode	rw	0x0	
2	dacintf_delay_0p25_tlo_bypass	rw	0x1	
3	dacintf_delay_0p5_tdac_bypass	rw	0x0	
5 : 4	dacintf_scale	rw	0x0	
6	dacintf_rnd_sw	rw	0x0	
7	dacintf_filter_mode	rw	0x0	
8	dacintf_dac_clk_low_mode	rw	0x0	
9	dacintf_lo_mode	rw	0x0	
10	dacintf_dac_iq_swap	rw	0x1	
11	dacintf_dac_iq_format	rw	0x0	
13 : 12	dacintf_dac_q_dly_sel	rw	0x0	
15 : 14	dacintf_dac_i_dly_sel	rw	0x0	

55.99 band_mapd				0x406207A0
Bits	Field Name	Acc	Default	Description
9 : 0	bandm37	rw	0x53	
19 : 10	bandm38	rw	0x54	
29 : 20	bandm39	rw	0x55	

55.100 band_mapc				0x406207A4
Bits	Field Name	Acc	Default	Description
9 : 0	bandm34	rw	0x51	
19 : 10	bandm35	rw	0x52	
29 : 20	bandm36	rw	0x53	

55.101 band_mapb				0x406207A8
Bits	Field Name	Acc	Default	Description
9 : 0	bandm31	rw	0x4E	
19 : 10	bandm32	rw	0x4F	
29 : 20	bandm33	rw	0x50	

55.102 band_mapa				0x406207AC
Bits	Field Name	Acc	Default	Description
9 : 0	bandm28	rw	0x4C	
19 : 10	bandm29	rw	0x4D	
29 : 20	bandm30	rw	0x4E	

55.103 band_map9				0x406207B0
Bits	Field Name	Acc	Default	Description
9 : 0	bandm25	rw	0x4A	
19 : 10	bandm26	rw	0x4A	
29 : 20	bandm27	rw	0x4B	

55.104 band_map8				0x406207B4
Bits	Field Name	Acc	Default	Description
9 : 0	bandm22	rw	0x47	
19 : 10	bandm23	rw	0x48	
29 : 20	bandm24	rw	0x49	

55.105 band_map7				0x406207B8
Bits	Field Name	Acc	Default	Description
9 : 0	bandm19	rw	0x45	
19 : 10	bandm20	rw	0x46	
29 : 20	bandm21	rw	0x47	

55.106 band_map6				0x406207BC
Bits	Field Name	Acc	Default	Description
9 : 0	bandm16	rw	0x43	
19 : 10	bandm17	rw	0x43	
29 : 20	bandm18	rw	0x44	

55.107 band_map5				0x406207C0
Bits	Field Name	Acc	Default	Description
9 : 0	bandm13	rw	0x40	
19 : 10	bandm14	rw	0x41	
29 : 20	bandm15	rw	0x42	

55.108 band_map4				0x406207C4
Bits	Field Name	Acc	Default	Description
9 : 0	bandm10	rw	0x3E	
19 : 10	bandm11	rw	0x3F	
29 : 20	bandm12	rw	0x3F	

55.109 band_map3				0x406207C8
Bits	Field Name	Acc	Default	Description
9 : 0	bandm07	rw	0x3B	
19 : 10	bandm08	rw	0x3C	
29 : 20	bandm09	rw	0x3D	

55.110 band_map2				0x406207CC
Bits	Field Name	Acc	Default	Description
9 : 0	bandm04	rw	0x39	
19 : 10	bandm05	rw	0x3A	
29 : 20	bandm06	rw	0x3B	

55.111 band_map1				0x406207D0
Bits	Field Name	Acc	Default	Description
9 : 0	bandm01	rw	0x37	
19 : 10	bandm02	rw	0x37	
29 : 20	bandm03	rw	0x38	

55.112 band_map0				0x406207D4
Bits	Field Name	Acc	Default	Description
9 : 0	bandm00	rw	0x36	

55.113 band_ctrl				0x406207D8
Bits	Field Name	Acc	Default	Description
0	band_map	rw	0x0	
1	dpll_rstn_sel	rw	0x0	

9 : 4	dpll_rstn_dly	rw	0x0	

55.114 a0x810				0x40620810
Bits	Field Name	Acc	Default	Description
9 : 0	StopDelayReg	rw	0x0	
11 : 10	DumpSize	rw	0x0	
15 : 12	DumpStartWrAddr	rw	0x0	
19 : 16	DumpDataSel	rw	0x0	
23 : 20	StopTrigSel	rw	0x0	

55.115 a0x814				0x40620814
Bits	Field Name	Acc	Default	Description
0	DumpEnable	rw	0x0	
1	bt_dump_clken	rw	0x0	
2	ReStopTrigMode	rw	0x0	
3	ManualStopReg	rw	0x0	
4	DumpWrapModeN	rw	0x0	
5	PhyDumpMuxSel	rw	0x1	
7 : 6	DumpSamplRateSel	rw	0x0	
11 : 8	PhyDumpRamEn	rw	0x0	
16 : 12	bt_dump_clksel	rw	0x0	
19	bt_dump_clkinv	rw	0x0	

55.116 a0x81c				0x4062081C
Bits	Field Name	Acc	Default	Description
0	imbala_cancel_bypass__	rw	0x1	
1	imbala_cal_dr_____	rw	0x0	
2	imbala_calc_rels_____	rw	0x0	
3	imbala_cancel_hold_dr_	rw	0x0	
7 : 4	imbala_cancel_bw_ct_wq	rw	0x4	
11 : 8	imbala_cancel_bw_ct_wa	rw	0x3	
31 : 16	imbala_reg_sel_____	rw	0x0	

55.117 a0x820				0x40620820
Bits	Field Name	Acc	Default	Description
14 : 0	imbala_cancel_wq_reg0_	rw	0x0	
30 : 16	imbala_cancel_wa_reg0_	rw	0x0	

55.118 a0x824				0x40620824
Bits	Field Name	Acc	Default	Description
14 : 0	imbala_cancel_wq_reg1	rw	0x0	
30 : 16	imbala_cancel_wa_reg1_	rw	0x0	

55.119 a0x828				0x40620828
Bits	Field Name	Acc	Default	Description
14 : 0	imbala_cancel_wq_reg2_	rw	0x0	
30 : 16	imbala_cancel_wa_reg2	rw	0x0	

55.120 a0x82c				0x4062082C
Bits	Field Name	Acc	Default	Description
14 : 0	imbala_cancel_wq_reg3_	rw	0x0	

30 : 16	imbala_cancel_wa_reg3	rw	0x0	

55.121 a0x830				0x40620830
Bits	Field Name	Acc	Default	Description
30 : 0	rfpll_sdm_freq_reg	rw	0x0	x*(1-v¿¿(11))
31	rfpll_sdm_freq_dr	rw	0x0	

55.122 a0x834				0x40620834
Bits	Field Name	Acc	Default	Description
3 : 0	fif_tx2m	rw	0x5	
7 : 4	fif_tx1m	rw	0x3	
11 : 8	fif_rx2m	rw	0x5	
15 : 12	fif_rx1m	rw	0x3	
29 : 16	freq_flo_val	rw	0x0	
31	freq_flo_fix	rw	0x0	

55.123 ppm_838				0x40620838
Bits	Field Name	Acc	Default	Description
9 : 0	ppm_val	rw	0x0	
12	ppm_en	rw	0x0	
13	ppm_up	rw	0x0	
27 : 16	flo_shnk	rw	0x0	

55.124 fref_tx_hm				0x4062083C
Bits	Field Name	Acc	Default	Description
14 : 0	fref_num_tx_lo	rw	0xA0	
30 : 16	fref_num_tx_md	rw	0xC8	

55.125 fref_tx				0x40620840
Bits	Field Name	Acc	Default	Description
15 : 8	fref_den_tx	rw	0x1	
30 : 16	fref_num_tx	rw	0xF0	
31	txdfe_div3_en	rw	0x1	

55.126 fref_rx				0x40620844
Bits	Field Name	Acc	Default	Description
7 : 0	dpll_cal_eachbit_cnt	rw	0x28	
15 : 8	fref_den_rx	rw	0x1	
30 : 16	fref_num_rx	rw	0x50	
31	fref_num_tx_fix_hi	rw	0x0	

55.127 coef_rfbt				0x40620848
Bits	Field Name	Acc	Default	Description
9 : 0	rfbt_rx_den	rw	0x244	
25 : 16	rfbt_tx_num	rw	0x100	

55.128 coef_rfwf				0x4062084C
Bits	Field Name	Acc	Default	Description
9 : 0	rfwf_rx_den	rw	0x1D6	
25 : 16	rfwf_tx_num	rw	0x280	

55.129 dco_div_ctrl				0x40620850
Bits	Field Name	Acc	Default	Description
1 : 0	dco_div_rx0	rw	0x3	
3 : 2	dco_div_rx1	rw	0x2	
5 : 4	dco_div_rx2	rw	0x1	
7 : 6	dco_div_rx3	rw	0x0	
9 : 8	dco_div_tx0	rw	0x3	
11 : 10	dco_div_tx1	rw	0x3	
13 : 12	dco_div_tx2	rw	0x1	
15 : 14	dco_div_tx3	rw	0x1	
16	dco_div_fix	rw	0x0	

55.130 a0x86c				0x4062086C
Bits	Field Name	Acc	Default	Description
0	dc_est_restart_reg	rw	0x0	

55.131 rsvd_wc_870				0x40620870
Bits	Field Name	Acc	Default	Description
0	rsvd_wc	w1c	0x0	

55.132 a0x880				0x40620880
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st0	ro	0x0	

55.133 a0x884				0x40620884
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st1	ro	0x0	

55.134 a0x888				0x40620888
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st2	ro	0x0	

55.135 a0x88c				0x4062088C
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st3	ro	0x0	

55.136 a0x890				0x40620890
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st4	ro	0x0	

55.137 a0x894				0x40620894
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st5	ro	0x0	

55.138 a0x898				0x40620898
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st6	ro	0x0	

55.139 a0x89c				0x4062089C
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st7	ro	0x0	

55.140 a0x8a0				0x406208A0
Bits	Field Name	Acc	Default	Description
31 : 0	dpll_cal_status	ro	0x0	

55.141 a0x8a4				0x406208A4
Bits	Field Name	Acc	Default	Description
31 : 0	dpll_cal_hp	ro	0x0	

55.142 a0x8a8				0x406208A8
Bits	Field Name	Acc	Default	Description
31 : 0	dpll_cal_vco	ro	0x0	

55.143 a0x8ac				0x406208AC
Bits	Field Name	Acc	Default	Description
31 : 0	bt_mdm_st8	ro	0x0	

56 bt_mdm_cfg

56.1 a0x010				0x40621010
Bits	Field Name	Acc	Default	Description
9 : 0	gh_bt	rw	0x143	
25 : 16	gh_le	rw	0x200	
29 : 28	edr_coef	rw	0x0	

56.2 a0x014				0x40621014
Bits	Field Name	Acc	Default	Description
3 : 0	dly_spt_tran	rw	0x8	
7 : 4	dly_sym_tran	rw	0x4	
11 : 8	dly_spt_edr	rw	0x0	
15 : 12	dly_sym_edr	rw	0x1	
19 : 16	dly_spt_gfsk	rw	0xA	
23 : 20	dly_sym_gfsk	rw	0x3	
27 : 24	dly_sym_gend	rw	0x4	
30 : 28	edr_s0	rw	0x0	
31	txiq13m_swap	rw	0x0	

56.3 a0x018				0x40621018
Bits	Field Name	Acc	Default	Description
1 : 0	amp_dly_13m	rw	0x2	
3	smooth_sel	rw	0x0	
4	tx_notch_en	rw	0x1	
5	tx_gegap_en	rw	0x1	
11 : 8	tx_notch_step	rw	0x8	
15 : 12	dly_spt_tranp	rw	0x3	
19 : 16	dly_sym_tranp	rw	0x6	
23 : 20	tx_gegap_step	rw	0x0	

56.4 a0x01c				0x4062101C
Bits	Field Name	Acc	Default	Description
14 : 0	dphi_fo	rw	0x762	

56.5 a0x020				0x40621020
Bits	Field Name	Acc	Default	Description
7 : 0	phi_ini	rw	0x20	
11 : 8	dly_spt_gfsk_2m	rw	0xB	
15 : 12	dly_sym_gfsk_2m	rw	0x3	
19 : 16	aup_step_2m	rw	0x7	
23 : 20	adn_step_2m	rw	0x7	
27 : 24	dly_txen_off	rw	0x6	

56.6 a0x024				0x40621024
Bits	Field Name	Acc	Default	Description
3 : 0	dly_sym_txen_2m	rw	0x2	
7 : 4	dly_spt_txen_2m	rw	0x3	
11 : 8	dly_sym_txen	rw	0x2	
15 : 12	dly_spt_txen	rw	0x3	
19 : 16	phi_step	rw	0x2	
23 : 20	aup_step	rw	0x7	
27 : 24	atr_step	rw	0x4	
31 : 28	adn_step	rw	0x7	

56.7 a0x028				0x40621028
Bits	Field Name	Acc	Default	Description
7 : 0	tx_pwr_val	rw	0x6F	
15 : 8	tx_pwr_val_tws	rw	0x6F	
16	tx_pwr_fix_tws	rw	0x1	
29 : 20	gh_tws	rw	0x143	

56.8 dpwr_ctrl				0x40621030
Bits	Field Name	Acc	Default	Description
5 : 0	scl_gfsk	rw	0x14	
13 : 8	scl_gedr	rw	0x14	
21 : 16	scl_edr	rw	0x24	
22	eplr_mem_en	rw	0x1	
23	dpwr_mem_en	rw	0x1	
24	dpwr_att_en	rw	0x0	
25	edly_mem_en	rw	0x1	
27 : 26	dpwr_att_2400	rw	0x0	
29 : 28	dpwr_att_2440	rw	0x0	
31 : 30	dpwr_att_2480	rw	0x0	

56.9 a0x034				0x40621034
Bits	Field Name	Acc	Default	Description
7 : 0	th_dphi	rw	0x80	
15 : 8	th_amp	rw	0x30	
17 : 16	zcp_sel	rw	0x1	
23 : 20	mask_flt_mode	rw	0x3	
31 : 24	dph_prot_th	rw	0x64	

56.10 a0x040				0x40621040
Bits	Field Name	Acc	Default	Description
0	am_bps	rw	0x1	
1	pm_bps	rw	0x1	
6 : 4	pm_shift	rw	0x0	
11 : 8	ampm_thr	rw	0xF	

56.11 a0x044				0x40621044
Bits	Field Name	Acc	Default	Description
9 : 0	am_p0	rw	0x0	
25 : 16	pm_p0	rw	0x0	

56.12 a0x048				0x40621048
Bits	Field Name	Acc	Default	Description
9 : 0	am_p1	rw	0x20	
25 : 16	pm_p1	rw	0x0	

56.13 a0x04c				0x4062104C
Bits	Field Name	Acc	Default	Description
9 : 0	am_p2	rw	0x40	
25 : 16	pm_p2	rw	0x0	

56.14 a0x050				0x40621050
Bits	Field Name	Acc	Default	Description
9 : 0	am_p3	rw	0x60	
25 : 16	pm_p3	rw	0x0	

56.15 a0x054				0x40621054
Bits	Field Name	Acc	Default	Description
9 : 0	am_p4	rw	0x80	
25 : 16	pm_p4	rw	0x0	

56.16 a0x058				0x40621058
Bits	Field Name	Acc	Default	Description
9 : 0	am_p5	rw	0xA0	
25 : 16	pm_p5	rw	0x0	

56.17 a0x05c				0x4062105C
Bits	Field Name	Acc	Default	Description
9 : 0	am_p6	rw	0xC0	
25 : 16	pm_p6	rw	0x0	

56.18 a0x060				0x40621060
Bits	Field Name	Acc	Default	Description
9 : 0	am_p7	rw	0xE0	
25 : 16	pm_p7	rw	0x0	

56.19 a0x064				0x40621064
Bits	Field Name	Acc	Default	Description

9 : 0	am_p8	rw	0x100	
25 : 16	pm_p8	rw	0x0	

56.20 a0x068				0x40621068
Bits	Field Name	Acc	Default	Description
9 : 0	am_p9	rw	0x120	
25 : 16	pm_p9	rw	0x0	

56.21 a0x06c				0x4062106C
Bits	Field Name	Acc	Default	Description
9 : 0	am_p10	rw	0x140	
25 : 16	pm_p10	rw	0x0	

56.22 a0x070				0x40621070
Bits	Field Name	Acc	Default	Description
9 : 0	am_p11	rw	0x160	
25 : 16	pm_p11	rw	0x0	

56.23 a0x074				0x40621074
Bits	Field Name	Acc	Default	Description
9 : 0	am_p12	rw	0x180	
25 : 16	pm_p12	rw	0x0	

56.24 a0x078				0x40621078
Bits	Field Name	Acc	Default	Description
9 : 0	am_p13	rw	0x1A0	
25 : 16	pm_p13	rw	0x0	

56.25 a0x07c				0x4062107C
Bits	Field Name	Acc	Default	Description
9 : 0	am_p14	rw	0x1C0	
25 : 16	pm_p14	rw	0x0	

56.26 a0x080				0x40621080
Bits	Field Name	Acc	Default	Description
9 : 0	am_p15	rw	0x1E0	
25 : 16	pm_p15	rw	0x0	

56.27 a0x084				0x40621084
Bits	Field Name	Acc	Default	Description
9 : 0	am_p16	rw	0x1FF	
25 : 16	pm_p16	rw	0x0	

56.28 a0x400				0x40621400
Bits	Field Name	Acc	Default	Description
23 : 0	demd_g_bt	rw	0xB3B0B	

56.29 a0x404	0x40621404

Bits	Field Name	Acc	Default	Description
23 : 0	demd_g_le	rw	0x135A13	

56.30 a0x408				0x40621408
Bits	Field Name	Acc	Default	Description
23 : 0	sync_g_bt	rw	0xB3B0B	

56.31 a0x40c				0x4062140C
Bits	Field Name	Acc	Default	Description
23 : 0	sync_g_le	rw	0x135A13	

56.32 a0x410				0x40621410
Bits	Field Name	Acc	Default	Description
4 : 0	vit_u_err	rw	0xA	
13 : 8	vit_u_dc	rw	0x8	
21 : 16	vit_angle_th_sel	rw	0x10	
26 : 24	vit_u_sam	rw	0x4	
28	vit_amp_en	rw	0x1	

56.33 a0x414				0x40621414
Bits	Field Name	Acc	Default	Description
23 : 0	sync_g_tws	rw	0xB3B0B	

56.34 a0x418				0x40621418
Bits	Field Name	Acc	Default	Description
23 : 0	demd_g_tws	rw	0xB3B0B	

56.35 a0x420				0x40621420
Bits	Field Name	Acc	Default	Description
4 : 0	gbt_dof_u_err	rw	0x8	
13 : 8	gbt_dof_u_dc	rw	0x1	
20 : 16	gbt_dof_coef_ek	rw	0x8	

56.36 a0x424				0x40621424
Bits	Field Name	Acc	Default	Description
4 : 0	gbt_don_u_err	rw	0xA	
13 : 8	gbt_don_u_dc	rw	0x8	
20 : 16	gbt_don_coef_ek	rw	0x8	

56.37 a0x428				0x40621428
Bits	Field Name	Acc	Default	Description
4 : 0	gbt_cci_u_err	rw	0x8	
13 : 8	gbt_cci_u_dc	rw	0x1	
20 : 16	gbt_cci_coef_ek	rw	0x10	
31 : 24	gbt_phie_max	rw	0x80	

56.38 a0x42c				0x4062142C
Bits	Field Name	Acc	Default	Description
4 : 0	gle_dof_u_err	rw	0x8	

13 : 8	gle_dof_u_dc	rw	0x1	
20 : 16	gle_dof_coef_ek	rw	0x8	

56.39 a0x430				0x40621430
Bits	Field Name	Acc	Default	Description
4 : 0	gle_don_u_err	rw	0xA	
13 : 8	gle_don_u_dc	rw	0x8	
20 : 16	gle_don_coef_ek	rw	0x8	

56.40 a0x434				0x40621434
Bits	Field Name	Acc	Default	Description
4 : 0	gle_cci_u_err	rw	0x8	
13 : 8	gle_cci_u_dc	rw	0x1	
20 : 16	gle_cci_coef_ek	rw	0xA	
31 : 24	gle_phie_max	rw	0x80	

56.41 a0x438				0x40621438
Bits	Field Name	Acc	Default	Description
0	gfsk_mode_ek	rw	0x0	
7 : 4	gfsk_cci_bias	rw	0x8	
13 : 8	gfsk_path_alpha	rw	0x8	
21 : 16	gfsk_angle_th_sel	rw	0x8	
26 : 24	gfsk_u_sam	rw	0x4	

56.42 a0x43c				0x4062143C
Bits	Field Name	Acc	Default	Description
4 : 0	edr_u_step0	rw	0xA	
13 : 8	edr_u_dc0	rw	0x28	
20 : 16	edr_u_step1	rw	0x2	
29 : 24	edr_u_dc1	rw	0x2	

56.43 a0x440				0x40621440
Bits	Field Name	Acc	Default	Description
4 : 0	edr_u_step2	rw	0x8	
13 : 8	edr_u_dc2	rw	0x1	
23 : 16	edr_thdadapt	rw	0x80	
24	edr_enhance	rw	0x1	

56.44 a0x444				0x40621444
Bits	Field Name	Acc	Default	Description
3 : 0	edr_u_sam	rw	0x4	
12 : 8	edr_angle_th_sel	rw	0x8	

56.45 a0x448				0x40621448
Bits	Field Name	Acc	Default	Description
3 : 0	edr_dof_bias	rw	0x8	
7 : 4	edr_cci_bias	rw	0x7	
11 : 8	edr_sym_alpha	rw	0x4	
23 : 16	edr_path_alpha	rw	0x4	
24	edr_dphi_mode	rw	0x0	
25	foini_en	rw	0x0	

56.46 a0x44c				0x4062144C
Bits	Field Name	Acc	Default	Description
8 : 4	edr_sync_len	rw	0xC	
13 : 12	edr_fo_scl	rw	0x3	
17 : 16	edr_ini_md	rw	0x1	
20	edr_err_md	rw	0x0	
21	edr_dc_en	rw	0x0	
30 : 24	sync_ng2e	rw	0x48	

56.47 a0x450				0x40621450
Bits	Field Name	Acc	Default	Description
5 : 0	sync_align_leco	rw	0x4	
13 : 8	sync_align_le1m	rw	0x4	
21 : 16	sync_align_le2m	rw	0x4	
29 : 24	sync_align_bt	rw	0x4	

56.48 a0x454				0x40621454
Bits	Field Name	Acc	Default	Description
5 : 0	sync_align_edr	rw	0x2	

56.49 a0x458				0x40621458
Bits	Field Name	Acc	Default	Description
7 : 0	lr_gmax_p	rw	0x66	
15 : 8	lr_gmax_n	rw	0x66	
23 : 16	bt_gmax_p	rw	0xFF	
31 : 24	bt_gmax_n	rw	0xFF	

56.50 a0x45c				0x4062145C
Bits	Field Name	Acc	Default	Description
7 : 0	lr_gstm_thd	rw	0x1D	
12 : 8	lr_getm_thd	rw	0x4	
23 : 16	lr_gsmd_thd	rw	0x1A	
28 : 24	lr_gemd_thd	rw	0x6	

56.51 a0x460				0x40621460
Bits	Field Name	Acc	Default	Description
7 : 0	lr_gscr_thd	rw	0x16	
12 : 8	lr_gecr_thd	rw	0x4	
17 : 16	lr_cors_thc	rw	0x2	
21 : 20	lr_slid_win	rw	0x1	
29 : 24	foe_mro	rw	0x4	

56.52 a0x464				0x40621464
Bits	Field Name	Acc	Default	Description
4 : 0	bt_g_ratio_thd	rw	0x16	
14 : 8	bt_crg_thd	rw	0x12	
21 : 16	bt_hde_thd	rw	0x3F	
29 : 24	bt_cre_thd	rw	0xD	

56.53 a0x468				0x40621468
Bits	Field Name	Acc	Default	Description

1 : 0	bt_crs_win	rw	0x2	
6 : 4	bt_crs_thd	rw	0x4	
9 : 8	bt_fin_win	rw	0x1	

56.54 a0x46c				0x4062146C
Bits	Field Name	Acc	Default	Description
4 : 0	l1_g_ratio_thd	rw	0x16	
14 : 8	l1_crg_thd	rw	0xF	
21 : 16	l1_hde_thd	rw	0x3F	
29 : 24	l1_cre_thd	rw	0xA	

56.55 a0x470				0x40621470
Bits	Field Name	Acc	Default	Description
1 : 0	l1_crs_win	rw	0x2	
6 : 4	l1_crs_thd	rw	0x4	
9 : 8	l1_fin_win	rw	0x1	

56.56 a0x474				0x40621474
Bits	Field Name	Acc	Default	Description
4 : 0	l2_g_ratio_thd	rw	0x16	
14 : 8	l2_crg_thd	rw	0xF	
21 : 16	l2_hde_thd	rw	0x3F	
29 : 24	l2_cre_thd	rw	0xA	

56.57 a0x478				0x40621478
Bits	Field Name	Acc	Default	Description
1 : 0	l2_crs_win	rw	0x2	
6 : 4	l2_crs_thd	rw	0x4	
9 : 8	l2_fin_win	rw	0x1	

56.58 a0x47c				0x4062147C
Bits	Field Name	Acc	Default	Description
10 : 0	sync_foe_val	rw	0x0	
12	sync_foe_fix	rw	0x0	

56.59 a0x480				0x40621480
Bits	Field Name	Acc	Default	Description
0	edr_sym_len_vld	rw	0x0	
7 : 4	edr_sym_len_adj	rw	0x2	
29 : 16	edr_sym_len	rw	0x0	

56.60 a0x484				0x40621484
Bits	Field Name	Acc	Default	Description
11 : 0	vit_bit_num	rw	0x100	
12	vit_len_fix	rw	0x0	
18 : 16	vit_hdr_tbs	rw	0x2	

56.61 a0x488				0x40621488
Bits	Field Name	Acc	Default	Description
7 : 0	snr_cnte_th	rw	0x0	

15 : 8	snr_eth	rw	0x1E	
23 : 16	snr_avg_th	rw	0x0	
24	snr_elen	rw	0x0	
25	snr_err_mode	rw	0x0	

56.62 a0x48c				0x4062148C
Bits	Field Name	Acc	Default	Description
2 : 0	gfsk_snr_enul	rw	0x4	
6 : 4	gfsk_snr_anul	rw	0x4	
10 : 8	edr_snr_enul	rw	0x5	
14 : 12	edr_snr_anul	rw	0x5	

56.63 a0x550				0x40621550
Bits	Field Name	Acc	Default	Description
3 : 0	aci_bt1m_enter	rw	0xF	
7 : 4	aci_bt1m_dif_low	rw	0x3	
11 : 8	aci_bt1m_dif_hig	rw	0x8	
15 : 12	aci_bt1m_exi_inf	rw	0xF	
19 : 16	aci_bt1m_exi_sig	rw	0x6	
23 : 20	aci_bt1m_frq_stp	rw	0x3	
26 : 24	aci_bt1m_tim_sty	rw	0x0	
30 : 28	aci_bt1m_tim_stp	rw	0x0	

56.64 a0x554				0x40621554
Bits	Field Name	Acc	Default	Description
3 : 0	aci_bt2m_enter	rw	0xF	
7 : 4	aci_bt2m_dif_low	rw	0x3	
11 : 8	aci_bt2m_dif_hig	rw	0x8	
15 : 12	aci_bt2m_exi_inf	rw	0xF	
19 : 16	aci_bt2m_exi_sig	rw	0x6	
23 : 20	aci_bt2m_frq_stp	rw	0x3	
26 : 24	aci_bt2m_tim_sty	rw	0x0	
30 : 28	aci_bt2m_tim_stp	rw	0x0	

56.65 a0x558				0x40621558
Bits	Field Name	Acc	Default	Description
3 : 0	aci_le2m_enter	rw	0xF	
7 : 4	aci_le2m_dif_low	rw	0x3	
11 : 8	aci_le2m_dif_hig	rw	0x8	
15 : 12	aci_le2m_exi_inf	rw	0xF	
19 : 16	aci_le2m_exi_sig	rw	0x6	
23 : 20	aci_le2m_frq_stp	rw	0x3	
26 : 24	aci_le2m_tim_sty	rw	0x0	
30 : 28	aci_le2m_tim_stp	rw	0x0	

56.66 a0x55c				0x4062155C
Bits	Field Name	Acc	Default	Description
3 : 0	aci_lelr_enter	rw	0xF	
7 : 4	aci_lelr_dif_low	rw	0x3	
11 : 8	aci_lelr_dif_hig	rw	0x8	
15 : 12	aci_lelr_exi_inf	rw	0xF	
19 : 16	aci_lelr_exi_sig	rw	0x6	
23 : 20	aci_lelr_frq_stp	rw	0x3	

26 : 24	aci_lelr_tim_sty	rw	0x0	
30 : 28	aci_lelr_tim_stp	rw	0x0	

56.67 a0x560				0x40621560
Bits	Field Name	Acc	Default	Description
1 : 0	aci_bt1m_rssi_ps	rw	0x0	
7 : 4	aci_bt1m_frq_dis	rw	0x0	
11 : 8	aci_bt1m_frq_ini	rw	0x2	
15 : 12	aci_bt1m_frq_sht	rw	0x1	
20 : 16	aci_bt1m_inc	rw	0x1B	
27 : 24	aci_bt1m_dec	rw	0x5	

56.68 a0x564				0x40621564
Bits	Field Name	Acc	Default	Description
1 : 0	aci_bt2m_rssi_ps	rw	0x0	
7 : 4	aci_bt2m_frq_dis	rw	0x0	
11 : 8	aci_bt2m_frq_ini	rw	0x2	
15 : 12	aci_bt2m_frq_sht	rw	0x2	
20 : 16	aci_bt2m_inc	rw	0x1B	
27 : 24	aci_bt2m_dec	rw	0x5	

56.69 a0x568				0x40621568
Bits	Field Name	Acc	Default	Description
1 : 0	aci_le2m_rssi_ps	rw	0x0	
7 : 4	aci_le2m_frq_dis	rw	0x4	
11 : 8	aci_le2m_frq_ini	rw	0x6	
15 : 12	aci_le2m_frq_sht	rw	0x5	
20 : 16	aci_le2m_inc	rw	0x14	
27 : 24	aci_le2m_dec	rw	0x9	
28	aci_le2m_f1m	rw	0x0	

56.70 a0x56c				0x4062156C
Bits	Field Name	Acc	Default	Description
1 : 0	aci_lelr_rssi_ps	rw	0x1	
7 : 4	aci_lelr_frq_dis	rw	0x0	
11 : 8	aci_lelr_frq_ini	rw	0x2	
15 : 12	aci_lelr_frq_sht	rw	0x1	
20 : 16	aci_lelr_inc	rw	0x1B	
27 : 24	aci_lelr_dec	rw	0x5	

56.71 a0x57c				0x4062157C
Bits	Field Name	Acc	Default	Description
0	notchp_bps	rw	0x1	
1	notchp_fix	rw	0x0	
7 : 4	notchp_bw	rw	0x4	
19 : 8	notchp_frqi	rw	0x0	
31 : 20	notchp_frqq	rw	0x0	

56.72 a0x580				0x40621580
Bits	Field Name	Acc	Default	Description
0	notchn_bps	rw	0x1	
1	notchn_fix	rw	0x0	

7 : 4	notchn_bw	rw	0x4	
19 : 8	notchn_frqi	rw	0x0	
31 : 20	notchn_frqq	rw	0x0	

56.73 a0x584				0x40621584
Bits	Field Name	Acc	Default	Description
3 : 0	aci_agcth	rw	0x0	
9 : 4	aci_mask	rw	0x0	
15 : 12	aci_iir_dly	rw	0x2	

56.74 a0x588				0x40621588
Bits	Field Name	Acc	Default	Description
31 : 0	notch_chmap0	rw	0x0	

56.75 a0x58c				0x4062158C
Bits	Field Name	Acc	Default	Description
31 : 0	notch_chmap1	rw	0x0	

56.76 a0x590				0x40621590
Bits	Field Name	Acc	Default	Description
15 : 0	notch_chmap2	rw	0x0	
19 : 16	notch_bw_lelr	rw	0xB	
23 : 20	notch_bw_le2m	rw	0xB	
27 : 24	notch_bw_bt2m	rw	0xB	
31 : 28	notch_bw_bt1m	rw	0xB	

56.77 a0x594				0x40621594
Bits	Field Name	Acc	Default	Description
11 : 0	notch_frqi_1m	rw	0x7EA	
23 : 12	notch_frqq_1m	rw	0x12D	

56.78 a0x598				0x40621598
Bits	Field Name	Acc	Default	Description
11 : 0	notch_frqi_2m	rw	0x7C3	
23 : 12	notch_frqq_2m	rw	0x1F2	

56.79 a0x870				0x40621870
Bits	Field Name	Acc	Default	Description
31 : 0	bt_rsvd_cfg0	rw	0x766C6003	

56.80 a0x874				0x40621874
Bits	Field Name	Acc	Default	Description
31 : 0	bt_rsvd_cfg1	rw	0x7B756F69	

56.81 a0x878				0x40621878
Bits	Field Name	Acc	Default	Description
31 : 0	bt_rsvd_cfg2	rw	0x0	

56.82 a0x87c				0x4062187C
Bits	Field Name	Acc	Default	Description
31 : 0	bt_rsvd_cfg3	rw	0x0	

56.83 a0x8b4				0x406218B4
Bits	Field Name	Acc	Default	Description
23 : 0	phie_crc_st	ro	0x0	

57 bt_rf_interface

57.1 reg_00				0x40622000
Bits	Field Name	Acc	Default	Description
0	bt_dco_rxlo_en_reg	rw	0x0	
1	bt_dco_rxlo_en_dr	rw	0x0	
2	bt_pu_dco_reg	rw	0x0	
3	bt_pu_dco_dr	rw	0x0	
4	bt_dpa_bt2wf_en_reg	rw	0x0	
5	bt_dpa_bt2wf_en_dr	rw	0x0	
6	bt_dpa_cdr_rstn_reg	rw	0x0	
7	bt_dpa_cdr_rstn_dr	rw	0x0	
8	bt_dpa_clk2dac_intf_en_reg	rw	0x0	
9	bt_dpa_clk2dac_intf_en_dr	rw	0x0	
10	bt_pu_dpa_reg	rw	0x0	
11	bt_pu_dpa_dr	rw	0x0	
12	bt_pu_dly_dpa_ldo_reg	rw	0x0	
13	bt_pu_dly_dpa_ldo_dr	rw	0x0	
14	bt_pu_dpa_ldo_reg	rw	0x0	
15	bt_pu_dpa_ldo_dr	rw	0x0	
16	bt_sdm_adc_dig_rstn_reg	rw	0x0	
17	bt_sdm_adc_dig_rstn_dr	rw	0x0	
18	bt_sdm_adc_byp_mode_reg	rw	0x0	
19	bt_sdm_adc_byp_mode_dr	rw	0x0	
20	bt_pu_sdm_adc_reg	rw	0x0	
21	bt_pu_sdm_adc_dr	rw	0x0	
22	bt_tia_rstn_reg	rw	0x0	
23	bt_tia_rstn_dr	rw	0x0	
24	bt_pu_tia_reg	rw	0x0	
25	bt_pu_tia_dr	rw	0x0	
26	bt_pu_rmx_reg	rw	0x0	
27	bt_pu_rmx_dr	rw	0x0	
28	bt_trx24g_wf2bt_en_reg	rw	0x0	
29	bt_trx24g_wf2bt_en_dr	rw	0x0	
30	bt_pu_lna_reg	rw	0x0	
31	bt_pu_lna_dr	rw	0x0	

57.2 reg_04				0x40622004
Bits	Field Name	Acc	Default	Description
2 : 0	bt_test_bit	rw	0x0	
3	bt_dpa_pll_test_mode	rw	0x0	
5 : 4	bt_dpll_pu_bit	rw	0x0	
7 : 6	bt_dco_pu_bit	rw	0x3	
9 : 8	bt_dpa_reg_pu_bit	rw	0x0	
11 : 10	bt_dpa_pu_bit	rw	0x0	
13 : 12	bt_sdm_adc_pu_bit	rw	0x2	
15 : 14	bt_tia_pu_bit	rw	0x1	

17 : 16	bt_rmx_pu_bit	rw	0x1	
19 : 18	bt_lna_pu_bit	rw	0x1	
20	bt_chop_bg_en_reg	rw	0x0	
21	bt_chop_bg_en_dr	rw	0x0	
22	bt_pu_chop_bg_reg	rw	0x0	
23	bt_pu_chop_bg_dr	rw	0x0	
24	bt_dpll_sdadc_clk_en_reg	rw	0x1	
25	bt_dpll_sdadc_clk_en_dr	rw	0x0	
26	bt_dpll_gro_rstn_reg	rw	0x0	
27	bt_dpll_gro_rstn_dr	rw	0x0	
28	bt_pu_dpll_reg	rw	0x0	
29	bt_pu_dpll_dr	rw	0x0	
30	bt_dco_txlo_en_reg	rw	0x1	
31	bt_dco_txlo_en_dr	rw	0x0	

57.3 reg_08				0x40622008
Bits	Field Name	Acc	Default	Description
1 : 0	bt_pu_rmx_td1	rw	0x1	
4 : 2	bt_dpll_gro_rstn_td2	rw	0x3	
6 : 5	bt_dpll_gro_rstn_td1	rw	0x1	
8 : 7	bt_dpa_clk2dac_intf_en_td1	rw	0x1	
10 : 9	bt_dco_txlo_en_td1	rw	0x1	
12 : 11	bt_dco_rxlo_en_td1	rw	0x1	
14 : 13	bt_pu_dco_td1	rw	0x1	
17 : 15	bt_pu_dly_dpa_ldo_td2	rw	0x3	
19 : 18	bt_pu_dpa_ldo_td3	rw	0x1	
21 : 20	bt_chop_bg_en_td1	rw	0x1	
23 : 22	bt_pu_chop_bg_td1	rw	0x3	
24	bt_test_enable_dpa_ldo	rw	0x0	
25	bt_test_enable_chop_bg	rw	0x0	
26	bt_test_enable_dpa	rw	0x0	
27	bt_test_ac_enable_dpll	rw	0x0	
28	bt_test_enable_sdm_adc	rw	0x0	
29	bt_test_enable_tia	rw	0x0	
30	bt_test_enable_rmx	rw	0x0	
31	bt_test_enable_lna	rw	0x0	

57.4 reg_0C				0x4062200C
Bits	Field Name	Acc	Default	Description
1 : 0	bt_lna_gbit	ro	0x3	
2	bt_lna_iref_sel	rw	0x0	
4 : 3	bt_pd_td0	rw	0x1	
7 : 5	btcm_mdll_rstn_td2	rw	0x1	
9 : 8	btcm_mdll_rstn_td1	rw	0x3	
11 : 10	btcm_pu_mdll_td1	rw	0x0	
12	wf_txon_co_ant_reg	rw	0x0	
13	wf_txon_co_ant_dr	rw	0x0	
14	wf_rxon_co_ant_reg	rw	0x0	
15	wf_rxon_co_ant_dr	rw	0x0	
16	co_ant_enable	rw	0x0	
17	cm_mdll_auto_reset_en	rw	0x0	
19 : 18	bt_pu_dpa_off_td1	rw	0x1	
21 : 20	bt_dpa_cdr_rstn_off_td1	rw	0x1	
23 : 22	bt_dpa_cdr_rstn_td1	rw	0x1	
25 : 24	bt_pu_lna_td1	rw	0x1	
27 : 26	bt_pu_sdm_adc_td1	rw	0x1	

29 : 28	bt_tia_rstn_td1	rw	0x1	
31 : 30	bt_pu_tia_td1	rw	0x1	

57.5 reg_10				0x40622010
Bits	Field Name	Acc	Default	Description
4 : 0	bt_tia_if_bit	ro	0x13	
9 : 5	bt_tia_bw_bit	ro	0x6	
10	bt_tia_swap_en	rw	0x0	
11	bt_tia_lpf_mode	rw	0x0	
14 : 12	bt_rmx_lobias_vbit	rw	0x3	
15	bt_lna_cap_sel	ro	0x0	
19 : 16	bt_lna_cbit	ro	0x7	
20	bt_lna_mixer1st_en	ro	0x0	
24 : 21	bt_lna_cg_vbit	rw	0x5	
28 : 25	bt_lna_ibias_bit	ro	0xA	

57.6 reg_14				0x40622014
Bits	Field Name	Acc	Default	Description
2 : 0	bt_tia_rin_bit	ro	0x0	
7 : 3	bt_tia_gain_bit	ro	0xF	
10 : 8	bt_tia_cal_v_range	rw	0x3	
13 : 11	bt_tia_cal_i_range	ro	0x3	
16 : 14	bt_tia_vcm_vbit	rw	0x4	
19 : 17	bt_tia_op_q_rbit	ro	0x3	
22 : 20	bt_tia_op_mil_rbit	ro	0x3	
25 : 23	bt_tia_op_mil_cbit	ro	0x3	
29 : 26	bt_tia_op_ibit	ro	0xC	

57.7 reg_18				0x40622018
Bits	Field Name	Acc	Default	Description
1 : 0	bt_sdm_adc_gbit	ro	0x2	
2	bt_sdm_adc_cal_en	rw	0x1	
5 : 3	bt_sdm_adc_cbit	rw	0x3	
9 : 6	bt_sdm_adc_vcal_bit	rw	0x8	
10	bt_sdm_adc_dith_en	rw	0x1	
11	bt_sdm_adc_clk_sar_div2_en	rw	0x1	
18 : 12	bt_tia_q_cal_bit_reg	rw	0x40	
19	bt_tia_q_cal_bit_dr	rw	0x0	
26 : 20	bt_tia_i_cal_bit_reg	rw	0x40	
27	bt_tia_i_cal_bit_dr	rw	0x0	

57.8 reg_1C				0x4062201C
Bits	Field Name	Acc	Default	Description
2 : 0	bt_dpa_ldo_sink_current_bit	rw	0x4	
4 : 3	bt_sdm_adc_notch_mode	ro	0x0	
6 : 5	bt_sdm_adc_dvdd_pwrsw_bit	rw	0x2	
7	bt_sdm_adc_ana_rstn	rw	0x0	
9 : 8	bt_sdm_adc_sar_en	rw	0x0	
10	bt_sdm_adc_sdm_en	rw	0x1	
12 : 11	bt_sdm_adc_sar_insel	rw	0x0	
16 : 13	bt_sdm_adc_dly_bit	rw	0x2	
17	bt_sdm_adc_dwa_en	rw	0x0	
18	bt_sdm_adc_clk_phsel	rw	0x0	
20 : 19	bt_sdm_adc_nr_mode	ro	0x0	

21	bt_sdm_adc_iq_swap_en	rw	0x0	
23 : 22	bt_sdm_adc_iq_mode	ro	0x0	
29 : 24	bt_sdm_adc_ibit	rw	0xC	

57.9 reg_20				0x40622020
Bits	Field Name	Acc	Default	Description
0	bt_dpa_reg_bypass	rw	0x0	
1	bt_dpa_drv_pwr_sel	rw	0x0	
2	bt_dpa_div_mode	ro	0x0	
3	bt_dpa_ldo_bypass_rx	rw	0x0	
4	bt_dpa_ldo_pull0	rw	0x0	
5	bt_dpa_ldo_vref_sel	rw	0x0	
6	bt_dpa_ldo_vref_cal_en	rw	0x0	
14 : 7	bt_dpa_ldo_rdac_data_in_rx	rw	0x70	
18 : 15	bt_dpa_ldo_rbit1	rw	0x8	
21 : 19	bt_dpa_ldo_current_limit_bit	rw	0x4	
22	bt_dpa_ldo_current_limit_en	rw	0x0	
23	bt_dpa_ldo_sink_current_en	rw	0x1	
25 : 24	bt_dpa_ldo_rbit	rw	0x3	
28 : 26	bt_dpa_ldo_ibit	rw	0x3	

57.10 reg_24				0x40622024
Bits	Field Name	Acc	Default	Description
0	bt_dpa_auto_pm_sync	rw	0x1	
3 : 1	bt_dpa_ant_cbit	rw	0x0	
7 : 4	bt_trx_cap_sw_bit	ro	0x0	
8	bt_dpa_pm_edge_sel	rw	0x0	
11 : 9	bt_dpa_capbank_bit	rw	0x0	
15 : 12	bt_dpa_slice_enn	rw	0xF	
19 : 16	bt_dpa_slice_enp	rw	0xF	
20	bt_dpa_mode_sel	rw	0x0	
22 : 21	bt_dpa_sign_dly_sel	rw	0x1	
23	bt_dpa_sinc_en	rw	0x1	
24	bt_dpa_1p2g_mode	rw	0x0	
25	bt_dpa_retime_sel	rw	0x0	
26	bt_dpa_clk_edge_sel	rw	0x1	
30 : 27	bt_dpa_reg_vbit	rw	0x8	

57.11 reg_28				0x40622028
Bits	Field Name	Acc	Default	Description
3 : 0	bt_dco_reg_var_bit	rw	0x3	
7 : 4	bt_dco_reg_vco_bit	rw	0x3	
8	bt_dco_dvar_clk_edge_sel	rw	0x1	
17 : 9	bt_dco_dvar_din	ro	0xFF	
27 : 18	bt_dco_band	ro	0x1FF	
29 : 28	bt_dpa_cdr_mode	rw	0x0	
31 : 30	bt_dpa_balun_tap_bit	rw	0x3	

57.12 reg_2C				0x4062202C
Bits	Field Name	Acc	Default	Description
1 : 0	bt_dpll_dly_sel4	rw	0x2	
3 : 2	bt_dpll_dly_sel3	rw	0x0	
5 : 4	bt_dpll_dly_sel2	rw	0x1	
7 : 6	bt_dpll_dly_sel1	rw	0x2	

9 : 8	bt_dpll_dly_sel0	rw	0x2	
10	bt_dco_trx_mode	rw	0x1	
14 : 11	bt_dco_band_aux	rw	0x0	
18 : 15	bt_dco_res1_bit	rw	0x0	
22 : 19	bt_dco_res0_bit	rw	0x0	
23	bt_dco_dvdd_en	rw	0x1	
27 : 24	bt_dco_cap_bit	rw	0x0	
28	bt_dco_rot_en	rw	0x0	
30 : 29	bt_dco_dvdd_dec_sel_bit	rw	0x0	
31	bt_dco_reg_var_bypass	rw	0x0	

57.13 reg_30 — 0x40622030

Bits	Field Name	Acc	Default	Description
0	bt_dpll_fref_div2_en	rw	0x0	
2 : 1	bt_dpll_chopper_clk_sel	rw	0x2	
3	bt_dpll_sdm_clk_sel	ro	0x0	
5 : 4	bt_dpll_vco_cal_sel	rw	0x2	
6	bt_dpll_cal_en	ro	0x0	
7	bt_dpll_cal_clk_sel	rw	0x0	
8	bt_dpll_div_mode	ro	0x0	
9	bt_dpll_div_clk_edge_sel	rw	0x0	
10	bt_dpll_clk_am_sel	ro	0x1	
12 : 11	bt_dpll_switch_rbit	rw	0x0	
13	bt_dpll_presc_reg_bypass	rw	0x0	
14	bt_dpll_pfd_ldo_bypass	rw	0x0	
15	bt_dpll_gro_reg_bypass	rw	0x0	
19 : 16	bt_dpll_presc_reg_vbit	rw	0x8	
23 : 20	bt_dpll_gro_reg_vbit	rw	0x8	
25 : 24	bt_dpll_gro_ldo_vbit	rw	0x2	
26	bt_dpll_fbc_sel	rw	0x0	
27	bt_dpll_fref_sel	ro	0x1	
29 : 28	bt_dpll_dly_sel5	rw	0x1	

57.14 reg_34 — 0x40622034

Bits	Field Name	Acc	Default	Description
0	bt_tia2gpio_en	rw	0x0	
2 : 1	bt_chop_bg_osc_bit	rw	0x2	
3	bt_chop_bg_osc_en	rw	0x1	
4	bt_dpll_sdadc_clk_sel	rw	0x0	
5	bt_dpll_chop_clk_en	rw	0x1	
6	bt_dpll2wf_img24g_en	rw	0x0	
7	bt_dpll2wf_img5g_en	rw	0x0	
8	bt_dpll_img_cal_mode	rw	0x0	
25 : 9	bt_dpll_cnt_out	ro	0x0	
31 : 26	bt_dpll_drv_bit	rw	0x0	

57.15 reg_38 — 0x40622038

Bits	Field Name	Acc	Default	Description
3 : 0	bt_rsvd_read_bit	ro	0x0	
11 : 4	bt_rsvd_bit	rw	0xF0	
19 : 12	bt_apc_rsvd	ro	0xF0	
27 : 20	bt_agc_rsvd	ro	0xF0	

57.16 bt_soft_reset_reg — 0x4062203C

Bits	Field Name	Acc	Default	Description
0	soft_reset	rw	0x0	

57.17 bt_man_crtl_reg				0x40622040
Bits	Field Name	Acc	Default	Description
3 : 0	tx_gain_reg	rw	0x0	
4	tx_gain_dr	rw	0x0	
5	tx_gain_dr_pulse	rw	0x0	
9 : 6	rx_gain_reg	rw	0x0	
10	rx_gain_dr	rw	0x0	
11	rx_gain_dr_pulse	rw	0x0	
12	rxon_reg	rw	0x0	
13	rxon_dr	rw	0x0	
14	txon_reg	rw	0x0	
15	txon_dr	rw	0x0	
16	twson_reg	rw	0x0	
17	twson_dr	rw	0x1	
18	pwron	rw	0x0	
19	bt_gro_rstn_sel	rw	0x1	

57.18 bt_calclk_reg				0x40622044
Bits	Field Name	Acc	Default	Description
9 : 0	cal_clk_div_num	rw	0x4F	
10	cal_clk_en	rw	0x0	
12 : 11	debug_mux	rw	0x0	
13	debug_clk_en	rw	0x0	

57.19 gain_mem_ctrl_reg				0x40622048
Bits	Field Name	Acc	Default	Description
0	clk_txgain_en	rw	0x0	
1	clk_txgain_sel	rw	0x0	
2	txgain_mem_cfg	rw	0x0	
4	clk_rxgain_en	rw	0x0	
5	clk_rxgain_sel	rw	0x0	
6	rxgain_mem_cfg	rw	0x0	

57.20 gain_mem_fetch_reg				0x4062204C
Bits	Field Name	Acc	Default	Description
3 : 0	txgain_mem_fetch_addr	rw	0x0	
15	txgain_mem_fetch_p	rw	0x0	
19 : 16	rxgain_mem_fetch_addr	rw	0x0	
31	rxgain_mem_fetch_p	rw	0x0	

57.21 pu_done_reg				0x40622050
Bits	Field Name	Acc	Default	Description
0	txon_done	ro	0x0	
1	rxon_done	ro	0x0	
2	pwron_done	ro	0x0	

57.22 dig_reserved_reg				0x40622054
Bits	Field Name	Acc	Default	Description

57.23 agc_table_out_reg0				0x40622058
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_0	ro	0x6F2663AB	

57.24 agc_table_out_reg1				0x4062205C
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_1	ro	0x83DB	

57.25 agc_table_out_reg2				0x40622060
Bits	Field Name	Acc	Default	Description
31 : 0	agc_table_out_2	ro	0x81000000	

57.26 apc_table_out_reg0				0x40622064
Bits	Field Name	Acc	Default	Description
31 : 0	apc_table_out_0	ro	0x1470	

58 bt_sys_ctrl

58.1 cfg_trap_bypass				0x40680000
Bits	Field Name	Acc	Default	Description
0	cfg_trap_bypass	rw	0x1	1'b1: bypass trap

58.2 clk_sel				0x40680004
Bits	Field Name	Acc	Default	Description
0	cfg_bt_uart_clk_sel	rw	0x0	BT uart clock sel. 1'b0: 52MHz; 1'b1: hclk

58.3 cfg_bt_uart_clk_div				0x40680008
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_clk_bt_uart_div_denom	rw	0x2	uart clock divider denominator, default div2
8	cfg_clk_bt_uart_div_update	rw	0x0	uart clock divider update

58.4 cfg_hclk_mode				0x4068000C
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_hclk_mode	rw	0xFFFF	1'b0: auto mode; 1'b1: manual mode

58.5 cfg_others_clk_mode				0x40680010
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_others_clk_mode	rw	0xFFFF	1'b0: auto mode; 1'b1: manual mode

58.6 cfg_mem_rom				0x40680014
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_rom	rw	0x2006	memory configration

58.7 cfg_mem_1prf				0x40680018
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_1prf	rw	0x2488	memory configration

58.8 cfg_mem_2prf				0x4068001C
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_mem_2prf	rw	0x20D6	memory configration

58.9 cfg_cpu_stclk_div				0x40680020
Bits	Field Name	Acc	Default	Description
7 : 0	cfg_cpu_stclk_div	rw	0x27	CPU stclk divider, default div40; 40MHz/40=1MHz

58.10 cfg_cpu_s1_rxev				0x40680024
Bits	Field Name	Acc	Default	Description
0	cfg_cpu_s1_rxev_bt_dma_en	rw	0x0	
1	cfg_cpu_s1_rxev_bt_uart_en	rw	0x0	
2	cfg_cpu_s1_rxev_pwrctrl_en	rw	0x0	
3	cfg_cpu_s1_rxev_pmu_rtc0_en	rw	0x0	
4	cfg_cpu_s1_rxev_pmu_rtc1_en	rw	0x0	
5	cfg_cpu_s1_rxev_gpiob_en	rw	0x0	
6	cfg_cpu_s1_rxev_aon_pwm_en	rw	0x0	
7	cfg_cpu_s1_rxev_uart0_en	rw	0x0	
8	cfg_cpu_s1_rxev_uart1_en	rw	0x0	
9	cfg_cpu_s1_rxev_uart2_en	rw	0x0	
10	cfg_cpu_s1_rxev_dma_en	rw	0x0	
11	cfg_cpu_s1_rxev_usb_dma_en	rw	0x0	
12	cfg_cpu_s1_rxev_pwm_en	rw	0x0	
13	cfg_cpu_s1_rxev_gpioa_en	rw	0x0	
14	cfg_cpu_s1_rxev_spi_en	rw	0x0	
15	cfg_cpu_s1_rxev_pcie_en	rw	0x0	
16	cfg_cpu_s1_rxev_trng_en	rw	0x0	
17	cfg_cpu_s1_rxev_i2cm_en	rw	0x0	
18	cfg_cpu_s1_rxev_sdmmc_en	rw	0x0	
19	cfg_cpu_s1_rxev_psi_mst_en	rw	0x0	

58.11 hclk_manual_enable				0x40680100
Bits	Field Name	Acc	Default	Description
0	hclk_rom_manual_enable	rw	0x1	write 1'b1 enable manual mode
1	hclk_sram0_manual_enable	rw	0x1	write 1'b1 enable manual mode
2	hclk_cpu_s1_manual_enable	rw	0x1	write 1'b1 enable manual mode
3	hclk_cpu_s1_-gated_manual_enable	rw	0x1	write 1'b1 enable manual mode
4	hclk_trap_manual_enable	rw	0x1	write 1'b1 enable manual mode
5	hclk_irq_manual_enable	rw	0x1	write 1'b1 enable manual mode
6	hclk_bt_apb_manual_enable	rw	0x1	write 1'b1 enable manual mode
7	hclk_bt_dma_manual_enable	rw	0x1	write 1'b1 enable manual mode
8	hclk_bt_matrix_manual_enable	rw	0x1	write 1'b1 enable manual mode
9	hclk_bt_uart_manual_enable	rw	0x1	write 1'b1 enable manual mode
10	pclk_bt_sys_-ctrl_manual_enable	rw	0x1	write 1'b1 enable manual mode
11	pclk_bt_timer_manual_enable	rw	0x1	write 1'b1 enable manual mode
12	pclk_bt_wdg_manual_enable	rw	0x1	write 1'b1 enable manual mode

58.12 hclk_manual_disable				0x40680104
Bits	Field Name	Acc	Default	Description
0	hclk_rom_manual_disable	rw	0x1	write 1'b1 disable manual mode
1	hclk_sram0_manual_disable	rw	0x1	write 1'b1 disable manual mode
2	hclk_cpu_s1_manual_disable	rw	0x1	write 1'b1 disable manual mode

3	hclk_cpu_s1_-gated_manual_disable	rw	0x1	write 1'b1 disable manual mode
4	hclk_trap_manual_disable	rw	0x1	write 1'b1 disable manual mode
5	hclk_irq_manual_disable	rw	0x1	write 1'b1 disable manual mode
6	hclk_bt_apb_manual_disable	rw	0x1	write 1'b1 disable manual mode
7	hclk_bt_dma_manual_disable	rw	0x1	write 1'b1 disable manual mode
8	hclk_bt_matri-x_manual_disable	rw	0x1	write 1'b1 disable manual mode
9	hclk_bt_uart_manual_disable	rw	0x1	write 1'b1 disable manual mode
10	pclk_bt_sys_-ctrl_manual_disable	rw	0x1	write 1'b1 disable manual mode
11	pclk_bt_timer_manual_disable	rw	0x1	write 1'b1 disable manual mode
12	pclk_bt_wdg_manual_disable	rw	0x1	write 1'b1 disable manual mode

58.13 others_manual_enable				0x40680108
Bits	Field Name	Acc	Default	Description
0	others_clk_rt-c_always_manual_enable	rw	0x1	write 1'b1 enable manual mode
1	others_clk_rt-c_wdg_manual_enable	rw	0x1	write 1'b1 enable manual mode
2	others_clk_ti-mer_manual_enable	rw	0x1	write 1'b1 enable manual mode
3	others_clk_bt-_uart_manual_enable	rw	0x1	write 1'b1 enable manual mode

58.14 others_manual_disable				0x4068010C
Bits	Field Name	Acc	Default	Description
0	others_clk_rt-c_always_manual_disable	rw	0x1	write 1'b1 disable manual mode
1	others_clk_rt-c_wdg_manual_disable	rw	0x1	write 1'b1 disable manual mode
2	others_clk_ti-mer_manual_disable	rw	0x1	write 1'b1 disable manual mode
3	others_clk_bt-_uart_manual_disable	rw	0x1	write 1'b1 disable manual mode

58.15 hclk_soft_resetn_set				0x40680110
Bits	Field Name	Acc	Default	Description
0	hresetn_rom_set	rw	0x1	write 1'b1 set soft reset
1	hresetn_sram0_set	rw	0x1	
2	hresetn_trap_set	rw	0x1	write 1'b1 set soft reset
3	hresetn_irq_set	rw	0x1	write 1'b1 set soft reset
4	hresetn_bt_apb_set	rw	0x1	write 1'b1 set soft reset
5	hresetn_bt_dma_set	rw	0x1	write 1'b1 set soft reset
6	hresetn_bt_matrix_set	rw	0x1	write 1'b1 set soft reset
7	hresetn_bt_uart_set	rw	0x1	write 1'b1 set soft reset
8	hresetn_bt_sys_ctrl_set	rw	0x1	write 1'b1 set soft reset
9	hresetn_bt_timer_set	rw	0x1	write 1'b1 set soft reset
10	hresetn_bt_wdg_set	rw	0x1	write 1'b1 set soft reset

58.16 hclk_soft_resetn_clr				0x40680114
Bits	Field Name	Acc	Default	Description
0	hresetn_rom_clear	rw	0x1	write 1'b1 clear soft reset
1	hresetn_sram0_clear	rw	0x1	
2	hresetn_trap_clear	rw	0x1	write 1'b1 clear soft reset
3	hresetn_irq_clear	rw	0x1	write 1'b1 clear soft reset

4	hresetn_bt_apb_clear	rw	0x1	write 1'b1 clear soft reset
5	hresetn_bt_dma_clear	rw	0x1	write 1'b1 clear soft reset
6	hresetn_bt_matrix_clear	rw	0x1	write 1'b1 clear soft reset
7	hresetn_bt_uart_clear	rw	0x1	write 1'b1 clear soft reset
8	hresetn_bt_sys_ctrl_clear	rw	0x1	write 1'b1 clear soft reset
9	hresetn_bt_timer_clear	rw	0x1	write 1'b1 clear soft reset
10	hresetn_bt_wdg_clear	rw	0x1	write 1'b1 clear soft reset

58.17 others_soft_resetn_set				0x40680118
Bits	Field Name	Acc	Default	Description
0	resetn_rtc_wdg_set	rw	0x1	write 1'b1 set soft reset
1	resetn_timer_set	rw	0x1	write 1'b1 set soft reset
2	resetn_bt_uart_set	rw	0x1	write 1'b1 set soft reset

58.18 others_soft_resetn_clr				0x4068011C
Bits	Field Name	Acc	Default	Description
0	resetn_rtc_wdg_clear	rw	0x1	write 1'b1 clear soft reset
1	resetn_timer_clear	rw	0x1	write 1'b1 clear soft reset
2	resetn_bt_uart_clear	rw	0x1	write 1'b1 clear soft reset

59 bt_timer

59.1 tmr_cfg				0x40681000
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_run	rw	0x0	Timer run enable, high active.
1	cfg_tmr_reset	rw	0x0	Timer PC reset, write 1 auto clear.

59.2 tmr_pr				0x40681004
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_pr	rw	0xFFFFFFFF	Prescale register.

59.3 tmr_pc				0x40681008
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_pc	rw	0xFFFFFFFF	Current counter number.

59.4 tmr_mr0				0x4068100C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr0	rw	0xFFFFFFFF	Match point for CPU0.

59.5 tmr_mr1				0x40681010
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr1	rw	0xFFFFFFFF	Match point for CPU1.

59.6 tmr_mr2				0x40681014
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_mr2	rw	0xFFFFFFFF	Match point for CPU2.

59.7 tmr_delta0				0x40681018
Bits	Field Name	Acc	Default	Description

31 : 0	cfg_tmr_delta0	rw	0xFFFFFFFF	Delta counter for CPU0.

59.8 tmr_delta1				0x4068101C
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_delta1	rw	0xFFFFFFFF	Delta counter for CPU1.

59.9 tmr_delta2				0x40681020
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tmr_delta2	rw	0xFFFFFFFF	Delta counter for CPU2.

59.10 tmr_rtc_clk_div				0x40681024
Bits	Field Name	Acc	Default	Description
15 : 0	cfg_tmr_rtc_clk_div	rw	0x19	Timer rtc clock dividier. Default is 26M/26, 1 bit means 1us for PC.
16	cfg_tmr_rtc_clk_div_update	rw	0x0	RTC clock divider update, write auto clear.

59.11 tmr_int_mask_enable				0x40681028
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match-0_int_mask_enable	rw	0x0	Match point0 interrupt mask enable. Write1 to enable.
1	cfg_tmr_match-1_int_mask_enable	rw	0x0	Match point1 interrupt mask enable. Write1 to enable.
2	cfg_tmr_match-2_int_mask_enable	rw	0x0	Match point2 interrupt mask enable. Write1 to enable.

59.12 tmr_int_mask_clear				0x4068102C
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match-0_int_mask_clear	rw	0x0	Match point0 interrupt mask clear. Write1 to clear.
1	cfg_tmr_match-1_int_mask_clear	rw	0x0	Match point1 interrupt mask clear. Write1 to clear.
2	cfg_tmr_match-2_int_mask_clear	rw	0x0	Match point2 interrupt mask clear. Write1 to clear.

59.13 tmr_int_raw				0x40681030
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match0_int_raw	rw	0x0	Match point0 raw interrupt. Write 1 to clear.
1	cfg_tmr_match1_int_raw	rw	0x0	Match point1 raw interrupt. Write 1 to clear.
2	cfg_tmr_match2_int_raw	rw	0x0	Match point2 raw interrupt. Write 1 to clear.

59.14 tmr_int_status				0x40681034
Bits	Field Name	Acc	Default	Description
0	cfg_tmr_match0_int_status	rw	0x0	Match point0 interrupt status. Write 1 to clear.
1	cfg_tmr_match1_int_status	rw	0x0	Match point1 interrupt status. Write 1 to clear.
2	cfg_tmr_match2_int_status	rw	0x0	Match point2 interrupt status. Write 1 to clear.

60 bt_wdg

60.1 WDG_LOAD_VAL				0x40682000
Bits	Field Name	Acc	Default	Description

31 : 0	wdg_load_val	rw	0x3FFFF	watchdog timer load value. In reset mode, software should load new value before timer decrease to 0. In interrupt mode, this value is counting start number. The default value is about 8 seconds.

60.2 WDG_CNT				0x40682004
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_cnt	rw	0xFFFFFFFF	watchdog counter value

60.3 WDG_CTRL				0x40682008
Bits	Field Name	Acc	Default	Description
0	wdg_irq_en	rw	0x0	Watchdog interrupt enable bit 0: interrupt is disabled 1: interrupt is enabled For reset mode: wdg_rst_en =1, wdg_irq_en=0. For interrupt mode: wdg_rst_en =0, wdg_irq_en=1. For combined mode: wdg_rst_en =1, wdg_irq_en=1.
1	wdg_run	rw	0x0	Watchdog counter open: 0: counter stops. 1: counter runs.

60.4 WDG_INT_CLR				0x4068200C
Bits	Field Name	Acc	Default	Description
0	wdg_int_clr	wo	0x0	Watchdog interrupt clear Write 1 to this bit to clear interrupt Read this bit always get 0.

60.5 WDG_INT_RAW				0x40682010
Bits	Field Name	Acc	Default	Description
0	wdg_int_raw	ro	0x0	Watchdog interrupt raw status. Watch dog reset cannot clear this raw status. Write wdg_int_clr can clear this raw status.
4	wdg_ld_busy	ro	0x0	Watchdog load busy status 0: Watchdog is ready for new loading 1: Last loading is not completed Software must not load new value when this bit is busy, that is, this bit should be checked before any new loading. This bit is set after a new loading, and lasts two or three RTC clock cycles, about 60us - 92us.

60.6 WDG_INT_MASK				0x40682014
Bits	Field Name	Acc	Default	Description
0	wdg_int_mask	ro	0x0	Watchdog interrupt masked status

60.7 WDG_LOCK				0x40682C00
Bits	Field Name	Acc	Default	Description
15 : 0	wdg_lock	rw	0x0	Watchdog lock control Write 32'h1ACCE551 to this register to unlock watchdog. Write other value to this register to lock watchdog If reading this register, bit-0 is lock status, and other bits are reserved. If watchdog is locked, all control registers cannot be written by software.

61 cpu_s1_trap

61.1 addr_in_0				0x40690000
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.2 addr_in_1				0x40690004
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.3 addr_in_2				0x40690008
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.4 addr_in_3				0x4069000C
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.5 addr_in_4				0x40690010
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.6 addr_in_5				0x40690014
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.7 addr_in_6				0x40690018
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.8 addr_in_7				0x4069001C
Bits	Field Name	Acc	Default	Description
19 : 0	addr_in	rw	0x0	address to be trapped
31 : 20	addr_in_reserved_0	ro	0x0	

61.9 en				0x40690080
Bits	Field Name	Acc	Default	Description
7 : 0	en	rw	0x0	trap enable for 8 channels
31 : 8	en_reserved_0	ro	0x0	

61.10 trap_out_base				0x40690084
Bits	Field Name	Acc	Default	Description

6 : 0	base_out_reserved	ro	0x0	
31 : 7	base_out	rw	0x0	trap out base address, only high 25 bits are valid.

62 bt_dma

62.1 SAR0				0x406A0000
Bits	Field Name	Acc	Default	Description
31 : 0	SAR	rw	0x0	Current Source Address of DMA transfer. Updated after each source transfer. The SINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every source transfer through the block transfer.

62.2 DAR0				0x406A0008
Bits	Field Name	Acc	Default	Description
31 : 0	DAR	rw	0x0	Current Destination address of DMA transfer. Updated after each destination transfer. The DINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every destination transfer throughout the block transfer.

62.3 LLP0				0x406A0010
Bits	Field Name	Acc	Default	Description
1 : 0	LMS	rw	0x0	List Master Select. Identifies the AHB layer/interface where the memory device that stores the next linked list item resides. This field does not exist if the configuration parameter is not set to NO_HARDCODE. In this case, the read-back value is always the hardcoded value. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (LIST_MASTER_SELECT_1): The memory device stores the next linked list item on AHB master 1 0x1 (LIST_MASTER_SELECT_2): The memory device stores the next linked list item on AHB master 2 0x2 (LIST_MASTER_SELECT_3): The memory device stores the next linked list item on AHB master 3 0x3 (LIST_MASTER_SELECT_4): The memory device
31 : 2	LOC	rw	0x0	Starting Address In Memory of next LLI if block chaining is enabled. Note that the two LSBs of the starting address are not stored because the address is assumed to be aligned to a 32-bit boundary. LLI accesses are always 32-bit accesses (Hsize=2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit.

62.4 CTL0_L				0x406A0018
Bits	Field Name	Acc	Default	Description
0	INT_EN	rw	0x1	Interrupt Enable Bit. If set, then all interrupt-generating sources are enabled. Functions as a global mask bit for all interrupts for the channel; raw* interrupt registers still assert if CTLx.INT_EN=0. Values: 0x0 (INTERRUPT_DISABLE): Interrupt is disabled 0x1 (INTERRUPT_ENABLE): Interrupt is enabled

3 : 1	DST_TR_WIDTH	rw	0x0	Destination Transfer Width. Mapped to AHB bus hsize. For a non-memory peripheral, typically the peripheral (destination) FIFO width. This value must be less than or equal to DMAH_Mk_HDATA_WIDTH where k is the AHB layer 1 to 4 where the destination resides. For the decoding of this field, see the "Setting Up Transfers" section in the DW_ahb_dmac Databook. Values: 0x0 (DST_TR_WIDTH_0): Destination transfer width is 8 bits 0x1 (DST_TR_WIDTH_1): Destination transfer width is 16 bits 0x2 (DST_TR_WIDTH_2): Destination transfer width is 32 bits 0x3 (DST_TR_WIDTH_3): Destination transfer width is 64 bits 0x4 (DST_TR_WIDTH_4): Destination transfer width is 128 bits 0x5 (DST_TR_WIDTH_5): Destination transfer width is 256 bits 0x6 (DST_TR_WIDTH_6): Destination transfer width is 256 bits 0x7 (DST_TR_WIDTH_7): Destination transfer width is 256 bits
6 : 4	SRC_TR_WIDTH	rw	0x0	Source Transfer Width. Mapped to AHB bus hsize. For a non-memory peripheral, typically the peripheral (source) FIFO width. This value must be less than or equl to DMAH_Mk_HDATA_WIDTH, where k is the AHB layer 1 to 4 where the source resides. Values: 0x0 (SRC_TR_WIDTH_0): Source transfer width is 8 bits 0x1 (SRC_TR_WIDTH_1): Source transfer width is 16 bits 0x2 (SRC_TR_WIDTH_2): Source transfer width is 32 bits 0x3 (SRC_TR_WIDTH_3): Source transfer width is 64 bits 0x4 (SRC_TR_WIDTH_4): Source transfer width is 128 bits 0x5 (SRC_TR_WIDTH_5): Source transfer width is 256 bits 0x6 (SRC_TR_WIDTH_6): Source transfer width is 256 bits 0x7 (SRC_TR_WIDTH_7): Source transfer width is 256 bits
8 : 7	DINC	rw	0x0	Destination Address Increment. Indicates whether to increment or decrement the destination address on every destination transfer. If your device is writing data to a destination peripheral FIFO with a fixed address, then set this field to N̈o Change. Values: 0x0 (DINC_0): Increments the destination address 0x1 (DINC_1): Decrements the destination address 0x2 (DINC_2): No change in the destination address 0x3 (DINC_3): No change in the destination address
10 : 9	SINC	rw	0x0	Source Address Increment. Indicates whether to increment or decrement the source address on every source transfer. If the device is fetching data from a source peripheral FIFO with a fixed address, then set this field to "No change". Values: 0x0 (SINC_0): Increments the source address 0x1 (SINC_1): Decrements the source address 0x2 (SINC_2): No change in the source address 0x3 (SINC_3): No change in the source address

13 : 11	DEST_MSIZE	rw	0x1	Destination Burst Transaction Length. Number of data items, each of width CTLx.DST_TR_WIDTH, to be written to the destination every time a destination burst transaction request is made from either the corresponding hardware or software handshaking interface. NOTE: This value is not related to the AHB bus master HBURST bus. Values: 0x0 (DEST_MSIZE_0): Number of data items to be transferred is 1 0x1 (DEST_MSIZE_1): Number of data items to be transferred is 4 0x2 (DEST_MSIZE_2): Number of data items to be transferred is 8 0x3 (DEST_MSIZE_3): Number of data items to be transferred is 16 0x4 (DEST_MSIZE_4): Number of data items to be transferred is 32 0x5 (DEST_MSIZE_5): Number of data items to be transferred is 64 0x6 (DEST_MSIZE_6): Number of data items to be transferred is 128 0x7 (DEST_MSIZE_7): Number of data items to be transferred is 256
16 : 14	SRC_MSIZE	rw	0x1	Source Burst Transaction Length. Number of data items, each of width CTLx.SRC_TR_WIDTH, to be read from the source every time a burst transferred request is made from either the corresponding hardware or software handshaking interface. NOTE: This value is not related to the AHB bus master HBURST bus. For information on the decoding for this field, see the "Setting Up Transfers" section and for more inforamtion about this field, see the "Choosing the Receive Watermark level" section in the DW_ahb_dmac Databook. Values: 0x0 (SRC_MSIZE_0): Number of data items to be transferred is 1 0x1 (SRC_MSIZE_1): Number of data items to be transferred is 4 0x2 (SRC_MSIZE_2): Number of data items to be transferred is 8 0x3 (SRC_MSIZE_3): Number of data items to be transferred is 16 0x4 (SRC_MSIZE_4): Number of data items to be transferred is 32 0x5 (SRC_MSIZE_5): Number of data items to be transferred is 64 0x6 (SRC_MSIZE_6): Number of data items to be transferred is 128 0x7 (SRC_MSIZE_7): Number of data items to be transferred is 256
17	SRC_GATHER_EN	rw	0x0	Source gather enable. Gather on the source side is applicable only when the CTLx.SINC bit indicates an incrementing or decrementing address control. Values: 0x0 (SRC_GATHER_DISABLE): Source gather is disabled 0x1 (SRC_GATHER_ENABLE): Source gather is enabled
18	DST_SCATTER_EN	rw	0x0	Destination scatter enable. Scatter on the destination side is applicable only when the CTLx.DINC bit indicates an incrementing or decrementing address control. Values: 0x0 (DST_SCATTER_DISABLE): Destination Scatter is disabled 0x1 (DST_SCATTER_ENABLE): Destination Scatter is enabled
19	RESERVED	ro	0x0	

22 : 20	TT_FC	rw	0x0	Transfer Type and Flow Control. Flow control can be assigned to the DW_ahb_dmac, the source peripheral, or the destination peripheral. For more information on transfer types and flow control, refer to "Setup/Operation of the DW_ahb_dmac Transfers". Dependencies: If the configuration parameter DMAH_CHx_FC is set to DMA_FC_ONLY, then TT_FC[2] does not exist and TT_FC[2] always reads back 0. If DMAH_CHx_FC is set to SRC_FC_ONLY, then TT_FC[2:1] does not exist and TT_FC[2:1] always reads back 2'b10. If DMAH_CHx_FC is set to DST_FC_ONLY, then TT_FC[2:1] does not exist and TT_FC[2:1] always reads back 2'b11. For multi-block transfers using linked list operation, TT_FC must be constant for all blocks of this multi-block transfer. Values: 0x0 (TT_FC_0): Transfer type is Memory to Memory and Flow Controller is DW_ahb_dmac 0x1 (TT_FC_1): Transfer type is Memory to Peripheral and Flow Controller is DW_ahb_dmac 0x2 (TT_FC_2): Transfer type is Peripheral to Memory and Flow Controller is DW_ahb_dmac 0x3 (TT_FC_3): Transfer type is Peripheral to Peripheral and Flow Controller is DW_ahb_dmac 0x4 (TT_FC_4): Transfer type is Peripheral to Memory and Flow Controller is Peripheral 0x5 (TT_FC_5): Transfer type is Peripheral to Peripheral and Flow Controller is Source Peripheral 0x6 (TT_FC_6): Transfer type is Memory to Peripheral and Flow Controller is Peripheral 0x7 (TT_FC_7): Transfer type is Peripheral to Peripheral and Flow Controller is Destination Peripheral
24 : 23	DMS	rw	0x0	Destination Master Select. Identifies the Master Interface layer where the destination device (peripheral or memory) resides. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (DMS_0): Destination device (peripheral or memory) is accessed from AHB master 1 0x1 (DMS_1): Destination device (peripheral or memory) is accessed from AHB master 2 0x2 (DMS_2): Destination device (peripheral or memory) is accessed from AHB master 3 0x3 (DMS_3): Destination device (peripheral or memory) is accessed from AHB master 4
26 : 25	SMS	rw	0x0	Source Master Select. Identifies the Master Interface layer where the source device (peripheral or memory) resides. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (SMS_0): Source device (peripheral or memory) is accessed from AHB master 1 0x1 (SMS_1): Source device (peripheral or memory) is accessed from AHB master 2 0x2 (SMS_2): Source device (peripheral or memory) is accessed from AHB master 3 0x3 (SMS_3): Source device (peripheral or memory) is accessed from AHB master 4
27	LLP_DST_EN	rw	0x0	Block chaining is enabled on the destination side only if LLP_DST_EN field is high and LLPx.LOC is non-zero. For more information, see "Block Chaining using Linked Lists". Values: 0x0 (LLP_DST_DISABLE): Block chaining using Linked List is disabled on the Destination side 0x1 (LLP_DST_ENABLE): Block chaining using Linked List is enabled on the Destination side

28	LLP_SRC_EN	rw	0x0	Block chaining is enabled on the source side only if the LLP_SRC_EN field is high and LLPx.LOC is non-zero. For more information, see "Block Chaining using Linked Lists". Values: 0x0 (LLP_SRC_DISABLE): Block chaining using Linked List is disabled on the Source side 0x1 (LLP_SRC_ENABLE): Block chaining using Linked List is enabled on the Source side
31 : 29	RESERVED0	ro	0x0	

62.5 CTL0_H				0x406A001C
Bits	Field Name	Acc	Default	Description
10 : 0	BLOCK_TS	rw	0x0	Block Transfer Size. When the DW_ahb_dmac is the flow controller, the user writes this field before the channel is enabled in order to indicate the block size. The number programmed into BLOCK_TS indicates the total number of single transactions to perform for every block transfer; a single transaction is mapped to a single AMBA beat. Width: The width of single transaction is determined by CTLx.SRC_TR_WIDTH. For further information on setting this field, refer to "Transfer Operation". Once the transfer starts, the read-back value is the total number of data items already read from the source peripheral, regardless of what is the flow controller. When the source or destination peripheral is assigned as the flow controller, then the maximum block size that can be read back saturates at DMAH_CHx_MAX_BLK_SIZE, but the actual block size can be greater.
11	RESERVED	ro	0x0	
12	DONE	rw	0x0	Done bit. If status write-back is enabled, the upper word of the control register, CTLx[63:32], is written to the control register location of the Linked List Item (LLI) in system memory at the end of the block transfer with the done bit set. Software can poll the LLI CTLx.DONE bit to see when a block transfer is complete. The LLI CTLx.DONE bit should be cleared when the linked lists are set up in memory prior to enabling the channel. LLI accesses are always 32-bit accesses (Hsize=2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit. For more information, refer to "Multi-Block Transfers". Values: 0x0 (DISABLED): DONE bit is deasserted the enb of block transfer 0x1 (ENABLED): SET the DONE bit at the end of block transfer
31 : 13	RESERVED0	ro	0x0	

62.6 CFG0_L				0x406A0040
Bits	Field Name	Acc	Default	Description
4 : 0	RESERVED	ro	0x0	

7 : 5	CH_PRIOR	rw	0x0	Channel Priority. A priority of 7 is the highest priority, and 0 is the lowest. This field must be programmed within the range 0 to DMAH_NUM_CHANNELS-1. A programmed value outside this range will cause erroneous behavior. Values: 0x0 (CH_PRIOR_0): Channel priority is 0 0x1 (CH_PRIOR_1): Channel priority is 1 0x2 (CH_PRIOR_2): Channel priority is 2 0x3 (CH_PRIOR_3): Channel priority is 3 0x4 (CH_PRIOR_4): Channel priority is 4 0x5 (CH_PRIOR_5): Channel priority is 5 0x6 (CH_PRIOR_6): Channel priority is 6 0x7 (CH_PRIOR_7): Channel priority is 7
8	CH_SUSP	rw	0x0	Channel Suspend. Suspends all DMA data transfers from the source until this bit is cleared. There is no guarantee that the current transaction will complete. Can also be used in conjunction with CFGx.FIFO_EMPTY to cleanly disable a channel without losing any data. For more information, refer to "Disabling a Channel Prior to Transfer Completion". Values: 0x0 (NOT_SUSPENDED): DMA transfer from the source is not suspended 0x1 (SUSPENDED): Suspend DMA transfer from the source
9	FIFO_EMPTY	ro	0x0	Channel FIFO status. Indicates if there is data left in the channel FIFO. Can be used in conjunction with CFGx.CH_SUSP to cleanly disable a channel. For more information, refer to "Disabling a Channel Prior to Transfer Completion". Values: 0x0 (NOT_EMPTY): Channel FIFO is not empty 0x1 (EMPTY): Channel FIFO is empty
10	HS_SEL_DST	rw	0x1	Destination Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces - hardware or software - is active for destination requests on this channel. If the destination peripheral is memory, then this bit is ignored. Values: 0x0 (HARDWARE_HS): Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1 (SOFTWARE_HS): Software handshaking interface. Hardware initiated transaction requests are ignored.
11	HS_SEL_SRC	rw	0x1	Source Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces - hardware or software - is active for source requests on this channel. If the source peripheral is memory, then this bit is ignored. Values: 0x0 (HARDWARE_HS): Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1 (SOFTWARE_HS): Software handshaking interface. Hardware initiated transaction requests are ignored. Value After Reset: 0x1
13 : 12	LOCK_CH_L	rw	0x0	Channel Local Level. Indicates the duration over which CFGx.LOCK_CH applies. This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (LOCK_CH_L_0): Over complete DMA transfer 0x1 (LOCK_CH_L_1): Over complete DMA block transfer 0x2 (LOCK_CH_L_2): Over complete DMA transaction 0x3 (LOCK_CH_L_3): Over complete DMA transaction

15 : 14	LOCK_B_L	rw	0x0	Bus lock level. Indicates the duration over which CFGx.LOCK_B bit applies. This field does not exist if the parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (LOCK_B_L_0): Over complete DMA transfer 0x1 (LOCK_B_L_1): Over complete DMA block transfer 0x2 (LOCK_B_L_2): Over complete DMA transaction 0x3 (LOCK_B_L_3): Over complete DMA transaction
16	LOCK_CH	rw	0x0	Channel Lock Bit. When the channel is granted control of the master bus interface and if the CFGx.LOCK_CH bit is asserted, then no other channels are granted control of the master bus interface for the duration specified in CFGx.LOCK_CH_L. Indicates to the master bus interface arbiter that this channel wants exclusive access to the master bus interface for the duration specified in CFGx.LOCK_CH_L. This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Channel lock bit is not enabled 0x1 (ENABLED): Channel lock bit is enabled
17	LOCK_B	rw	0x0	Bus Lock Bit. When active, the AHB bus master signal hlock is asserted for the duration specified in CFGx.LOCK_B_L. For more information, refer to "Locked DMA Transfers". This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Bus lock bit is not enabled 0x1 (ENABLED): Bus lock bit is enabled
18	DST_HS_POL	rw	0x0	Destination Handshaking Interface Polarity. Values: 0x0 (ACTIVE_HIGH): Destination Handshaking Interface Polarity is Active high 0x1 (ACTIVE_LOW): Destination Handshaking Interface Polarity is Active low
19	SRC_HS_POL	rw	0x0	Source Handshaking Interface Polarity. Values: 0x0 (ACTIVE_HIGH): Source Handshaking Interface Polarity is Active high 0x1 (ACTIVE_LOW): Source Handshaking Interface Polarity is Active low
29 : 20	MAX_ABRST	rw	0x0	Maximum AMBA Burst Length. Maximum AMBA burst length that is used for DMA transfers on this channel. A value of 0 indicates that software is not limiting the maximum AMBA burst length for DMA transfers on this channel. This field does not exist if the configuration parameter DMAH_MABRST is not selected; in this case, the read-back value is always 0, and the maximum AMBA burst length cannot be limited by software.
30	RELOAD_SRC	rw	0x0	Automatic Source Reload. The SARx register can be automatically reloaded from its initial value at the end of every block for multi-block transfers. A new block transfer is then initiated. This field does not exist if the configuration parameter DMAH_CHx_MULTI_BLK_EN is not selected; in this case, the read-back value is always 0. Values: 0x0 (DISABLE): Source Reload Disabled 0x1 (ENABLE): Source Reload Enabled

31	RELOAD_DST	rw	0x0	Automatic Destination Reload. The DARx register can be automatically reloaded from its initial value at the end of every block for multi-block transfers. A new block transfer is then initiated. This register does not exist if the configuration parameter DMAH_CHx_MULTI_BLK_EN is not selected; in this case, the read-back value is always 0. Values: 0x0 (DISABLE): Destination Reload Disabled. 0x1 (ENABLE): Destination Reload Enabled

62.7 CFG0_H				0x406A0044
Bits	Field Name	Acc	Default	Description
0	FCMODE	rw	0x0	Flow Control Mode. Determines when source transaction requests are serviced when the Destination Peripheral is the flow controller. Values: 0x0 (FCMODE_0): Source transaction requests are serviced when they occur. Data pre-fetching is enabled 0x1 (FCMODE_1): Source transaction requests are not serviced until a destination transaction request occurs. In this mode, the amount of data transferred from the source is limited so that it is guaranteed to be transferred to the destination prior to block termination by the destination. Data pre-fetching is disabled.
1	FIFO_MODE	rw	0x0	FIFO Mode Select. Determines how much space or data needs to be available in the FIFO before a burst transaction request is serviced. Values: 0x0 (FIFO_MODE_0): Space/data available for single AHB transfer of the specified transfer width 0x1 (FIFO_MODE_1): Data available is greater than or equal to half the FIFO depth for destination transfers and space available is greater than half the fifo depth for source transfers. The exceptions are at the end of a burst transaction request or at the end of a block transfer.
4 : 2	PROTCTL	rw	0x1	Protection Control bits used to drive the AHB HPROT[3:1] bus. The AMBA Specification recommends that the default of HPROT indicates a non-cached, non-buffered, privileged data access. The reset value is used to indicate such an access. HPROT[0] is tied high because all transfers are data accesses, as there are no opcode fetches. There is a one-to-one mapping of these register bits to the HPROT[3:1] master interface signals. Mapping of HPROT bus is as follows: 1'b1 to HPROT[0] CFGx.PROTCTL[1] to HPROT[1] CFGx.PROTCTL[2] to HPROT[2] CFGx.PROTCTL[3] to HPROT[3]
5	DS_UPD_EN	rw	0x0	Destination Status Update Enable. Destination status information is fetched only from the location pointed to by the DSTATARx register, stored in the DSTATx register and written out to the DSTATx location of the LLI, if DS_UPD_EN is high. This field does not exist if the configuration parameter DMAH_CHx_SATAT_DST is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Destination Status Update is disabled 0x1 (ENABLED): Destination Status Update is enabled

6	SS_UPD_EN	rw	0x0	Source Status Update Enable. Source status information is fetched only from the location pointed to by the SSTATARx register, stored in the SSTATx register and written out to the SSTATx location of the LLI, if SS_UPD_EN is high. NOTE: This enalbe is applicable only if DMAH_CHx_STAT_SRC is set to True. This field does not exist if the configuration parameter DMAH_CHx_STAT_SRC is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Source Status Update is disabled. 0x1 (ENABLED): Source Status Update is enabled.
7	SRC_PER	rw	0x0	Source Hardware Interface.. Assigns a hardware handshaking interface (0 : DMAH_NUM_HS_INT-1) to the source of channel x if the CFGx.HS_SEL_SRC field is 0; otherwise, this field is ignored. The channel can then communicate with the source peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface. NOTE2: This field does not exist if the configuration parameter DMAH_NUM_HS_INT is set to 0.
10 : 8	RESERVED	ro	0x0	
11	DEST_PER	rw	0x0	Destination hardware interface. Assigns a hardware handshaking interface (0 : DMAH_NUM_HS_INT-1) to the destination of channel x if the CFGx.HS_SEL_DST field is 0; otherwise, this field is ignored. The channel can then communicate with the destination peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface. NOTE2: This field does not exist if the configuration parameter DMAH_NUM_HS_INT is set to 0.
31 : 12	RESERVED0	ro	0x0	

62.8 SGR0				0x406A0048
Bits	Field Name	Acc	Default	Description
19 : 0	SGI	rw	0x0	Source Gather Interval.
30 : 20	SGC	rw	0x0	Source Gather Count. Source contiguous tranfer count between successive gather boundaries.
31	RESERVED	ro	0x0	

62.9 DSR0				0x406A0050
Bits	Field Name	Acc	Default	Description
19 : 0	DSI	rw	0x0	Destination Scatter Interval.
30 : 20	DSC	rw	0x0	Destination Scatter Count. Destination contiguous transfer count between successive scatter boundaries.
31	RESERVED	ro	0x0	

62.10 SAR1				0x406A0058
Bits	Field Name	Acc	Default	Description
31 : 0	SAR	rw	0x0	Current Source Address of DMA transfer. Updated after each source transfer. The SINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every source transfer through the block transfer.

62.11 DAR1				0x406A0060
Bits	Field Name	Acc	Default	Description
31 : 0	DAR	rw	0x0	Current Destination address of DMA transfer. Updated after each destination transfer. The DINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every destination transfer throughout the block transfer.

62.12 LLP1				0x406A0068
Bits	Field Name	Acc	Default	Description
1 : 0	LMS	rw	0x0	List Master Select. Identifies the AHB layer/interface where the memory device that stores the next linked list item resides. This field does not exist if the configuration parameter is not set to NO_HARDCODE. In this case, the readback value is always the hardcoded value. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (LIST_MASTER_SELECT_1): The memory device stores the next linked list item on AHB master 1 0x1 (LIST_MASTER_SELECT_2): The memory device stores the next linked list item on AHB master 2 0x2 (LIST_MASTER_SELECT_3): The memory device stores the next linked list item on AHB master 3 0x3 (LIST_MASTER_SELECT_4): The memory device
31 : 2	LOC	rw	0x0	Starting Address In Memory of next LLI if block chaining is enabled. Note that the two LSBs of the starting address are not stored because the address is assumed to be aligned to a 32-bit boundary. LLI accesses are always 32-bit accesses (Hsize=2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit.

62.13 CTL1_L				0x406A0070
Bits	Field Name	Acc	Default	Description
0	INT_EN	rw	0x1	Interrupt Enable Bit. If set, then all interrupt-generating sources are enabled. Functions as a global mask bit for all interrupts for the channel; raw* interrupt registers still assert if CTLx.INT_EN=0. Values: 0x0 (INTERRUPT_DISABLE): Interrupt is disabled 0x1 (INTERRUPT_ENABLE): Interrupt is enabled

3 : 1	DST_TR_WIDTH	rw	0x0	Destination Transfer Width. Mapped to AHB bus hsize. For a non-memory peripheral, typically the peripheral (destination) FIFO width. This value must be less than or equal to DMAH_Mk_HDATA_WIDTH where k is the AHB layer 1 to 4 where the destination resides. For the decoding of this field, see the "Setting Up Transfers" section in the DW_ahb_dmac Databook. Values: 0x0 (DST_TR_WIDTH_0): Destination transfer width is 8 bits 0x1 (DST_TR_WIDTH_1): Destination transfer width is 16 bits 0x2 (DST_TR_WIDTH_2): Destination transfer width is 32 bits 0x3 (DST_TR_WIDTH_3): Destination transfer width is 64 bits 0x4 (DST_TR_WIDTH_4): Destination transfer width is 128 bits 0x5 (DST_TR_WIDTH_5): Destination transfer width is 256 bits 0x6 (DST_TR_WIDTH_6): Destination transfer width is 256 bits 0x7 (DST_TR_WIDTH_7): Destination transfer width is 256 bits
6 : 4	SRC_TR_WIDTH	rw	0x0	Source Transfer Width. Mapped to AHB bus hsize. For a non-memory peripheral, typically the peripheral (source) FIFO width. This value must be less than or equl to DMAH_Mk_HDATA_WIDTH, where k is the AHB layer 1 to 4 where the source resides. Values: 0x0 (SRC_TR_WIDTH_0): Source transfer width is 8 bits 0x1 (SRC_TR_WIDTH_1): Source transfer width is 16 bits 0x2 (SRC_TR_WIDTH_2): Source transfer width is 32 bits 0x3 (SRC_TR_WIDTH_3): Source transfer width is 64 bits 0x4 (SRC_TR_WIDTH_4): Source transfer width is 128 bits 0x5 (SRC_TR_WIDTH_5): Source transfer width is 256 bits 0x6 (SRC_TR_WIDTH_6): Source transfer width is 256 bits 0x7 (SRC_TR_WIDTH_7): Source transfer width is 256 bits
8 : 7	DINC	rw	0x0	Destination Address Increment. Indicates whether to increment or decrement the destination address on every destination transfer. If your device is writing data to a destination peripheral FIFO with a fixed address, then set this field to N̈o Change. Values: 0x0 (DINC_0): Increments the destination address 0x1 (DINC_1): Decrements the destination address 0x2 (DINC_2): No change in the destination address 0x3 (DINC_3): No change in the destination address
10 : 9	SINC	rw	0x0	Source Address Increment. Indicates whether to increment or decrement the source address on every source transfer. If the device is fetching data from a source peripheral FIFO with a fixed address, then set this field to "No change". Values: 0x0 (SINC_0): Increments the source address 0x1 (SINC_1): Decrements the source address 0x2 (SINC_2): No change in the source address 0x3 (SINC_3): No change in the source address

13 : 11	DEST_MSIZE	rw	0x1	Destination Burst Transaction Length. Number of data items, each of width CTLx.DST_TR_WIDTH, to be written to the destination every time a destination burst transaction request is made from either the corresponding hardware or software handshaking interface. NOTE: This value is not related to the AHB bus master HBURST bus. Values: 0x0 (DEST_MSIZE_0): Number of data items to be transferred is 1 0x1 (DEST_MSIZE_1): Number of data items to be transferred is 4 0x2 (DEST_MSIZE_2): Number of data items to be transferred is 8 0x3 (DEST_MSIZE_3): Number of data items to be transferred is 16 0x4 (DEST_MSIZE_4): Number of data items to be transferred is 32 0x5 (DEST_MSIZE_5): Number of data items to be transferred is 64 0x6 (DEST_MSIZE_6): Number of data items to be transferred is 128 0x7 (DEST_MSIZE_7): Number of data items to be transferred is 256
16 : 14	SRC_MSIZE	rw	0x1	Source Burst Transaction Length. Number of data items, each of width CTLx.SRC_TR_WIDTH, to be read from the source every time a burst transferred request is made from either the corresponding hardware or software handshaking interface. NOTE: This value is not related to the AHB bus master HBURST bus. For information on the decoding for this field, see the "Setting Up Transfers" section and for more inforamtion about this field, see the "Choosing the Receive Watermark level" section in the DW_ahb_dmac Databook. Values: 0x0 (SRC_MSIZE_0): Number of data items to be transferred is 1 0x1 (SRC_MSIZE_1): Number of data items to be transferred is 4 0x2 (SRC_MSIZE_2): Number of data items to be transferred is 8 0x3 (SRC_MSIZE_3): Number of data items to be transferred is 16 0x4 (SRC_MSIZE_4): Number of data items to be transferred is 32 0x5 (SRC_MSIZE_5): Number of data items to be transferred is 64 0x6 (SRC_MSIZE_6): Number of data items to be transferred is 128 0x7 (SRC_MSIZE_7): Number of data items to be transferred is 256
17	SRC_GATHER_EN	rw	0x0	Source gather enable. Gather on the source side is applicable only when the CTLx.SINC bit indicates an incrementing or decrementing address control. Values: 0x0 (SRC_GATHER_DISABLE): Source gather is disabled 0x1 (SRC_GATHER_ENABLE): Source gather is enabled
18	DST_SCATTER_EN	rw	0x0	Destination scatter enable. Scatter on the destination side is applicable only when the CTLx.DINC bit indicates an incrementing or decrementing address control. Values: 0x0 (DST_SCATTER_DISABLE): Destination Scatter is disabled 0x1 (DST_SCATTER_ENABLE): Destination Scatter is enabled
19	RESERVED	ro	0x0	

22 : 20	TT_FC	rw	0x0	Transfer Type and Flow Control. Flow control can be assigned to the DW_ahb_dmac, the source peripheral, or the destination peripheral. For more information on transfer types and flow control, refer to "Setup/Operation of the DW_ahb_dmac Transfers". Dependencies: If the configuration parameter DMAH_CHx_FC is set to DMA_FC_ONLY, then TT_FC[2] does not exist and TT_FC[2] always reads back 0. If DMAH_CHx_FC is set to SRC_FC_ONLY, then TT_FC[2:1] does not exist and TT_FC[2:1] always reads back 2'b10. If DMAH_CHx_FC is set to DST_FC_ONLY, then TT_FC[2:1] does not exist and TT_FC[2:1] always reads back 2'b11. For multi-block transfers using linked list operation, TT_FC must be constant for all blocks of this multi-block transfer. Values: 0x0 (TT_FC_0): Transfer type is Memory to Memory and Flow Controller is DW_ahb_dmac 0x1 (TT_FC_1): Transfer type is Memory to Peripheral and Flow Controller is DW_ahb_dmac 0x2 (TT_FC_2): Transfer type is Peripheral to Memory and Flow Controller is DW_ahb_dmac 0x3 (TT_FC_3): Transfer type is Peripheral to Peripheral and Flow Controller is DW_ahb_dmac 0x4 (TT_FC_4): Transfer type is Peripheral to Memory and Flow Controller is Peripheral 0x5 (TT_FC_5): Transfer type is Peripheral to Peripheral and Flow Controller is Source Peripheral 0x6 (TT_FC_6): Transfer type is Memory to Peripheral and Flow Controller is Peripheral 0x7 (TT_FC_7): Transfer type is Peripheral to Peripheral and Flow Controller is Destination Peripheral
24 : 23	DMS	rw	0x0	Destination Master Select. Identifies the Master Interface layer where the destination device (peripheral or memory) resides. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (DMS_0): Destination device (peripheral or memory) is accessed from AHB master 1 0x1 (DMS_1): Destination device (peripheral or memory) is accessed from AHB master 2 0x2 (DMS_2): Destination device (peripheral or memory) is accessed from AHB master 3 0x3 (DMS_3): Destination device (peripheral or memory) is accessed from AHB master 4
26 : 25	SMS	rw	0x0	Source Master Select. Identifies the Master Interface layer where the source device (peripheral or memory) resides. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (SMS_0): Source device (peripheral or memory) is accessed from AHB master 1 0x1 (SMS_1): Source device (peripheral or memory) is accessed from AHB master 2 0x2 (SMS_2): Source device (peripheral or memory) is accessed from AHB master 3 0x3 (SMS_3): Source device (peripheral or memory) is accessed from AHB master 4
27	LLP_DST_EN	rw	0x0	Block chaining is enabled on the destination side only if LLP_DST_EN field is high and LLPx.LOC is non-zero. For more information, see "Block Chaining using Linked Lists". Values: 0x0 (LLP_DST_DISABLE): Block chaining using Linked List is disabled on the Destination side 0x1 (LLP_DST_ENABLE): Block chaining using Linked List is enabled on the Destination side

28	LLP_SRC_EN	rw	0x0	Block chaining is enabled on the source side only if the LLP_SRC_EN field is high and LLPx.LOC is non-zero. For more information, see "Block Chaining using Linked Lists". Values: 0x0 (LLP_SRC_DISABLE): Block chaining using Linked List is disabled on the Source side 0x1 (LLP_SRC_ENABLE): Block chaining using Linked List is enabled on the Source side
31 : 29	RESERVED0	ro	0x0	

62.14 CTL1_H				0x406A0074
Bits	Field Name	Acc	Default	Description
10 : 0	BLOCK_TS	rw	0x0	Block Transfer Size. When the DW_ahb_dmac is the flow controller, the user writes this field before the channel is enabled in order to indicate the block size. The number programmed into BLOCK_TS indicates the total number of single transactions to perform for every block transfer; a single transaction is mapped to a single AMBA beat. Width: The width of single transaction is determined by CTLx.SRC_TR_WIDTH. For further information on setting this field, refer to "Transfer Operation". Once the transfer starts, the read-back value is the total number of data items already read from the source peripheral, regardless of what is the flow controller. When the source or destination peripheral is assigned as the flow controller, then the maximum block size that can be read back saturates at DMAH_CHx_MAX_BLK_SIZE, but the actual block size can be greater.
11	RESERVED	ro	0x0	
12	DONE	rw	0x0	Done bit. If status write-back is enabled, the upper word of the control register, CTLx[63:32], is written to the control register location of the Linked List Item (LLI) in system memory at the end of the block transfer with the done bit set. Software can poll the LLI CTLx.DONE bit to see when a block transfer is complete. The LLI CTLx.DONE bit should be cleared when the linked lists are set up in memory prior to enabling the channel. LLI accesses are always 32-bit accesses (Hsize=2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit. For more information, refer to "Multi-Block Transfers". Values: 0x0 (DISABLED): DONE bit is deasserted the enb of block transfer 0x1 (ENABLED): SET the DONE bit at the end of block transfer
31 : 13	RESERVED0	ro	0x0	

62.15 CFG1_L				0x406A0098
Bits	Field Name	Acc	Default	Description
4 : 0	RESERVED	ro	0x0	

7 : 5	CH_PRIOR	rw	0x0	Channel Priority. A priority of 7 is the highest priority, and 0 is the lowest. This field must be programmed within the range 0 to DMAH_NUM_CHANNELS-1. A programmed value outside this range will cause erroneous behavior. Values: 0x0 (CH_PRIOR_0): Channel priority is 0 0x1 (CH_PRIOR_1): Channel priority is 1 0x2 (CH_PRIOR_2): Channel priority is 2 0x3 (CH_PRIOR_3): Channel priority is 3 0x4 (CH_PRIOR_4): Channel priority is 4 0x5 (CH_PRIOR_5): Channel priority is 5 0x6 (CH_PRIOR_6): Channel priority is 6 0x7 (CH_PRIOR_7): Channel priority is 7
8	CH_SUSP	rw	0x0	Channel Suspend. Suspends all DMA data transfers from the source until this bit is cleared. There is no guarantee that the current transaction will complete. Can also be used in conjunction with CFGx.FIFO_EMPTY to cleanly disable a channel without losing any data. For more information, refer to "Disabling a Channel Prior to Transfer Completion". Values: 0x0 (NOT_SUSPENDED): DMA transfer from the source is not suspended 0x1 (SUSPENDED): Suspend DMA transfer from the source
9	FIFO_EMPTY	ro	0x0	Channel FIFO status. Indicates if there is data left in the channel FIFO. Can be used in conjunction with CFGx.CH_SUSP to cleanly disable a channel. For more information, refer to "Disabling a Channel Prior to Transfer Completion". Values: 0x0 (NOT_EMPTY): Channel FIFO is not empty 0x1 (EMPTY): Channel FIFO is empty
10	HS_SEL_DST	rw	0x1	Destination Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces - hardware or software - is active for destination requests on this channel. If the destination peripheral is memory, then this bit is ignored. Values: 0x0 (HARDWARE_HS): Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1 (SOFTWARE_HS): Software handshaking interface. Hardware initiated transaction requests are ignored.
11	HS_SEL_SRC	rw	0x1	Source Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces - hardware or software - is active for source requests on this channel. If the source peripheral is memory, then this bit is ignored. Values: 0x0 (HARDWARE_HS): Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1 (SOFTWARE_HS): Software handshaking interface. Hardware initiated transaction requests are ignored. Value After Reset: 0x1
13 : 12	LOCK_CH_L	rw	0x0	Channel Local Level. Indicates the duration over which CFGx.LOCK_CH applies. This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (LOCK_CH_L_0): Over complete DMA transfer 0x1 (LOCK_CH_L_1): Over complete DMA block transfer 0x2 (LOCK_CH_L_2): Over complete DMA transaction 0x3 (LOCK_CH_L_3): Over complete DMA transaction

15 : 14	LOCK_B_L	rw	0x0	Bus lock level. Indicates the duration over which CFGx.LOCK_B bit applies. This field does not exist if the parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (LOCK_B_L_0): Over complete DMA transfer 0x1 (LOCK_B_L_1): Over complete DMA block transfer 0x2 (LOCK_B_L_2): Over complete DMA transaction 0x3 (LOCK_B_L_3): Over complete DMA transaction
16	LOCK_CH	rw	0x0	Channel Lock Bit. When the channel is granted control of the master bus interface and if the CFGx.LOCK_CH bit is asserted, then no other channels are granted control of the master bus interface for the duration specified in CFGx.LOCK_CH_L. Indicates to the master bus interface arbiter that this channel wants exclusive access to the master bus interface for the duration specified in CFGx.LOCK_CH_L. This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Channel lock bit is not enabled 0x1 (ENABLED): Channel lock bit is enabled
17	LOCK_B	rw	0x0	Bus Lock Bit. When active, the AHB bus master signal hlock is asserted for the duration specified in CFGx.LOCK_B_L. For more information, refer to "Locked DMA Transfers". This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Bus lock bit is not enabled 0x1 (ENABLED): Bus lock bit is enabled
18	DST_HS_POL	rw	0x0	Destination Handshaking Interface Polarity. Values: 0x0 (ACTIVE_HIGH): Destination Handshaking Interface Polarity is Active high 0x1 (ACTIVE_LOW): Destination Handshaking Interface Polarity is Active low
19	SRC_HS_POL	rw	0x0	Source Handshaking Interface Polarity. Values: 0x0 (ACTIVE_HIGH): Source Handshaking Interface Polarity is Active high 0x1 (ACTIVE_LOW): Source Handshaking Interface Polarity is Active low
29 : 20	MAX_ABRST	rw	0x0	Maximum AMBA Burst Length. Maximum AMBA burst length that is used for DMA transfers on this channel. A value of 0 indicates that software is not limiting the maximum AMBA burst length for DMA transfers on this channel. This field does not exist if the configuration parameter DMAH_MABRST is not selected; in this case, the read-back value is always 0, and the maximum AMBA burst length cannot be limited by software.
30	RELOAD_SRC	rw	0x0	Automatic Source Reload. The SARx register can be automatically reloaded from its initial value at the end of every block for multi-block transfers. A new block transfer is then initiated. This field does not exist if the configuration parameter DMAH_CHx_MULTI_BLK_EN is not selected; in this case, the read-back value is always 0. Values: 0x0 (DISABLE): Source Reload Disabled 0x1 (ENABLE): Source Reload Enabled

31	RELOAD_DST	rw	0x0	Automatic Destination Reload. The DARx register can be automatically reloaded from its initial value at the end of every block for multi-block transfers. A new block transfer is then initiated. This register does not exist if the configuration parameter DMAH_CHx_MULTI_BLK_EN is not selected; in this case, the read-back value is always 0. Values: 0x0 (DISABLE): Destination Reload Disabled. 0x1 (ENABLE): Destination Reload Enabled

62.16 CFG1_H 0x406A009C

Bits	Field Name	Acc	Default	Description
0	FCMODE	rw	0x0	Flow Control Mode. Determines when source transaction requests are serviced when the Destination Peripheral is the flow controller. Values: 0x0 (FCMODE_0): Source transaction requests are serviced when they occur. Data pre-fetching is enabled 0x1 (FCMODE_1): Source transaction requests are not serviced until a destination transaction request occurs. In this mode, the amount of data transferred from the source is limited so that it is guaranteed to be transferred to the destination prior to block termination by the destination. Data pre-fetching is disabled.
1	FIFO_MODE	rw	0x0	FIFO Mode Select. Determines how much space or data needs to be available in the FIFO before a burst transaction request is serviced. Values: 0x0 (FIFO_MODE_0): Space/data available for single AHB transfer of the specified transfer width 0x1 (FIFO_MODE_1): Data available is greater than or equal to half the FIFO depth for destination transfers and space available is greater than half the fifo depth for source transfers. The exceptions are at the end of a burst transaction request or at the end of a block transfer.
4 : 2	PROTCTL	rw	0x1	Protection Control bits used to drive the AHB HPROT[3:1] bus. The AMBA Specification recommends that the default of HPROT indicates a non-cached, non-buffered, privileged data access. The reset value is used to indicate such an access. HPROT[0] is tied high because all transfers are data accesses, as there are no opcode fetches. There is a one-to-one mapping of these register bits to the HPROT[3:1] master interface signals. Mapping of HPROT bus is as follows: 1'b1 to HPROT[0] CFGx.PROTCTL[1] to HPROT[1] CFGx.PROTCTL[2] to HPROT[2] CFGx.PROTCTL[3] to HPROT[3]
5	DS_UPD_EN	rw	0x0	Destination Status Update Enable. Destination status information is fetched only from the location pointed to by the DSTATARx register, stored in the DSTATx register and written out to the DSTATx location of the LLI, if DS_UPD_EN is high. This field does not exist if the configuration parameter DMAH_CHx_SATAT_DST is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Destination Status Update is disabled 0x1 (ENABLED): Destination Status Update is enabled

6	SS_UPD_EN	rw	0x0	Source Status Update Enable. Source status information is fetched only from the location pointed to by the SSTATARx register, stored in the SSTATx register and written out to the SSTATx location of the LLI, if SS_UPD_EN is high. NOTE: This enalbe is applicable only if DMAH_CHx_STAT_SRC is set to True. This field does not exist if the configuration parameter DMAH_CHx_STAT_SRC is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Source Status Update is disabled. 0x1 (ENABLED): Source Status Update is enabled.
7	SRC_PER	rw	0x0	Source Hardware Interface.. Assigns a hardware handshaking interface (0 : DMAH_NUM_HS_INT-1) to the source of channel x if the CFGx.HS_SEL_SRC field is 0; otherwise, this field is ignored. The channel can then communicate with the source peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface. NOTE2: This field does not exist if the configuration parameter DMAH_NUM_HS_INT is set to 0.
10 : 8	RESERVED	ro	0x0	
11	DEST_PER	rw	0x0	Destination hardware interface. Assigns a hardware handshaking interface (0 : DMAH_NUM_HS_INT-1) to the destination of channel x if the CFGx.HS_SEL_DST field is 0; otherwise, this field is ignored. The channel can then communicate with the destination peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface. NOTE2: This field does not exist if the configuration parameter DMAH_NUM_HS_INT is set to 0.
31 : 12	RESERVED0	ro	0x0	

62.17 SGR1				0x406A00A0
Bits	Field Name	Acc	Default	Description
19 : 0	SGI	rw	0x0	Source Gather Interval.
30 : 20	SGC	rw	0x0	Source Gather Count. Source contiguous tranfer count between successive gather boundaries.
31	RESERVED	ro	0x0	

62.18 DSR1				0x406A00A8
Bits	Field Name	Acc	Default	Description
19 : 0	DSI	rw	0x0	Destination Scatter Interval.
30 : 20	DSC	rw	0x0	Destination Scatter Count. Destination contiguous transfer count between successive scatter boundaries.
31	RESERVED	ro	0x0	

62.19 SAR2				0x406A00B0
Bits	Field Name	Acc	Default	Description
31 : 0	SAR	rw	0x0	Current Source Address of DMA transfer. Updated after each source transfer. The SINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every source transfer through the block transfer.

62.20 DAR2				0x406A00B8
Bits	Field Name	Acc	Default	Description
31 : 0	DAR	rw	0x0	Current Destination address of DMA transfer. Updated after each destination transfer. The DINC field in the CTLx register determines whether the address increments, decrements, or is left unchanged on every destination transfer throughout the block transfer.

62.21 LLP2				0x406A00C0
Bits	Field Name	Acc	Default	Description
1 : 0	LMS	rw	0x0	List Master Select. Identifies the AHB layer/interface where the memory device that stores the next linked list item resides. This field does not exist if the configuration parameter is not set to NO_HARDCODE. In this case, the read-back value is always the hardcoded value. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (LIST_MASTER_SELECT_1): The memory device stores the next linked list item on AHB master 1 0x1 (LIST_MASTER_SELECT_2): The memory device stores the next linked list item on AHB master 2 0x2 (LIST_MASTER_SELECT_3): The memory device stores the next linked list item on AHB master 3 0x3 (LIST_MASTER_SELECT_4): The memory device
31 : 2	LOC	rw	0x0	Starting Address In Memory of next LLI if block chaining is enabled. Note that the two LSBs of the starting address are not stored because the address is assumed to be aligned to a 32-bit boundary. LLI accesses are always 32-bit accesses (Hsize=2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit.

62.22 CTL2_L				0x406A00C8
Bits	Field Name	Acc	Default	Description
0	INT_EN	rw	0x1	Interrupt Enable Bit. If set, then all interrupt-generating sources are enabled. Functions as a global mask bit for all interrupts for the channel; raw* interrupt registers still assert if CTLx.INT_EN=0. Values: 0x0 (INTERRUPT_DISABLE): Interrupt is disabled 0x1 (INTERRUPT_ENABLE): Interrupt is enabled

3 : 1	DST_TR_WIDTH	rw	0x0	Destination Transfer Width. Mapped to AHB bus hsize. For a non-memory peripheral, typically the peripheral (destination) FIFO width. This value must be less than or equal to DMAH_Mk_HDATA_WIDTH where k is the AHB layer 1 to 4 where the destination resides. For the decoding of this field, see the "Setting Up Transfers" section in the DW_ahb_dmac Databook. Values: 0x0 (DST_TR_WIDTH_0): Destination transfer width is 8 bits 0x1 (DST_TR_WIDTH_1): Destination transfer width is 16 bits 0x2 (DST_TR_WIDTH_2): Destination transfer width is 32 bits 0x3 (DST_TR_WIDTH_3): Destination transfer width is 64 bits 0x4 (DST_TR_WIDTH_4): Destination transfer width is 128 bits 0x5 (DST_TR_WIDTH_5): Destination transfer width is 256 bits 0x6 (DST_TR_WIDTH_6): Destination transfer width is 256 bits 0x7 (DST_TR_WIDTH_7): Destination transfer width is 256 bits
6 : 4	SRC_TR_WIDTH	rw	0x0	Source Transfer Width. Mapped to AHB bus hsize. For a non-memory peripheral, typically the peripheral (source) FIFO width. This value must be less than or equl to DMAH_Mk_HDATA_WIDTH, where k is the AHB layer 1 to 4 where the source resides. Values: 0x0 (SRC_TR_WIDTH_0): Source transfer width is 8 bits 0x1 (SRC_TR_WIDTH_1): Source transfer width is 16 bits 0x2 (SRC_TR_WIDTH_2): Source transfer width is 32 bits 0x3 (SRC_TR_WIDTH_3): Source transfer width is 64 bits 0x4 (SRC_TR_WIDTH_4): Source transfer width is 128 bits 0x5 (SRC_TR_WIDTH_5): Source transfer width is 256 bits 0x6 (SRC_TR_WIDTH_6): Source transfer width is 256 bits 0x7 (SRC_TR_WIDTH_7): Source transfer width is 256 bits
8 : 7	DINC	rw	0x0	Destination Address Increment. Indicates whether to increment or decrement the destination address on every destination transfer. If your device is writing data to a destination peripheral FIFO with a fixed address, then set this field to N̈o Change. Values: 0x0 (DINC_0): Increments the destination address 0x1 (DINC_1): Decrements the destination address 0x2 (DINC_2): No change in the destination address 0x3 (DINC_3): No change in the destination address
10 : 9	SINC	rw	0x0	Source Address Increment. Indicates whether to increment or decrement the source address on every source transfer. If the device is fetching data from a source peripheral FIFO with a fixed address, then set this field to "No change". Values: 0x0 (SINC_0): Increments the source address 0x1 (SINC_1): Decrements the source address 0x2 (SINC_2): No change in the source address 0x3 (SINC_3): No change in the source address

13 : 11	DEST_MSIZE	rw	0x1	Destination Burst Transaction Length. Number of data items, each of width CTLx.DST_TR_WIDTH, to be written to the destination every time a destination burst transaction request is made from either the corresponding hardware or software handshaking interface. NOTE: This value is not related to the AHB bus master HBURST bus. Values: 0x0 (DEST_MSIZE_0): Number of data items to be transferred is 1 0x1 (DEST_MSIZE_1): Number of data items to be transferred is 4 0x2 (DEST_MSIZE_2): Number of data items to be transferred is 8 0x3 (DEST_MSIZE_3): Number of data items to be transferred is 16 0x4 (DEST_MSIZE_4): Number of data items to be transferred is 32 0x5 (DEST_MSIZE_5): Number of data items to be transferred is 64 0x6 (DEST_MSIZE_6): Number of data items to be transferred is 128 0x7 (DEST_MSIZE_7): Number of data items to be transferred is 256
16 : 14	SRC_MSIZE	rw	0x1	Source Burst Transaction Length. Number of data items, each of width CTLx.SRC_TR_WIDTH, to be read from the source every time a burst transferred request is made from either the corresponding hardware or software handshaking interface. NOTE: This value is not related to the AHB bus master HBURST bus. For information on the decoding for this field, see the "Setting Up Transfers" section and for more inforamtion about this field, see the "Choosing the Receive Watermark level" section in the DW_ahb_dmac Databook. Values: 0x0 (SRC_MSIZE_0): Number of data items to be transferred is 1 0x1 (SRC_MSIZE_1): Number of data items to be transferred is 4 0x2 (SRC_MSIZE_2): Number of data items to be transferred is 8 0x3 (SRC_MSIZE_3): Number of data items to be transferred is 16 0x4 (SRC_MSIZE_4): Number of data items to be transferred is 32 0x5 (SRC_MSIZE_5): Number of data items to be transferred is 64 0x6 (SRC_MSIZE_6): Number of data items to be transferred is 128 0x7 (SRC_MSIZE_7): Number of data items to be transferred is 256
17	SRC_GATHER_EN	rw	0x0	Source gather enable. Gather on the source side is applicable only when the CTLx.SINC bit indicates an incrementing or decrementing address control. Values: 0x0 (SRC_GATHER_DISABLE): Source gather is disabled 0x1 (SRC_GATHER_ENABLE): Source gather is enabled
18	DST_SCATTER_EN	rw	0x0	Destination scatter enable. Scatter on the destination side is applicable only when the CTLx.DINC bit indicates an incrementing or decrementing address control. Values: 0x0 (DST_SCATTER_DISABLE): Destination Scatter is disabled 0x1 (DST_SCATTER_ENABLE): Destination Scatter is enabled
19	RESERVED	ro	0x0	

22 : 20	TT_FC	rw	0x0	Transfer Type and Flow Control. Flow control can be assigned to the DW_ahb_dmac, the source peripheral, or the destination peripheral. For more information on transfer types and flow control, refer to "Setup/Operation of the DW_ahb_dmac Transfers". Dependencies: If the configuration parameter DMAH_CHx_FC is set to DMA_FC_ONLY, then TT_FC[2] does not exist and TT_FC[2] always reads back 0. If DMAH_CHx_FC is set to SRC_FC_ONLY, then TT_FC[2:1] does not exist and TT_FC[2:1] always reads back 2'b10. If DMAH_CHx_FC is set to DST_FC_ONLY, then TT_FC[2:1] does not exist and TT_FC[2:1] always reads back 2'b11. For multi-block transfers using linked list operation, TT_FC must be constant for all blocks of this multi-block transfer. Values: 0x0 (TT_FC_0): Transfer type is Memory to Memory and Flow Controller is DW_ahb_dmac 0x1 (TT_FC_1): Transfer type is Memory to Peripheral and Flow Controller is DW_ahb_dmac 0x2 (TT_FC_2): Transfer type is Peripheral to Memory and Flow Controller is DW_ahb_dmac 0x3 (TT_FC_3): Transfer type is Peripheral to Peripheral and Flow Controller is DW_ahb_dmac 0x4 (TT_FC_4): Transfer type is Peripheral to Memory and Flow Controller is Peripheral 0x5 (TT_FC_5): Transfer type is Peripheral to Peripheral and Flow Controller is Source Peripheral 0x6 (TT_FC_6): Transfer type is Memory to Peripheral and Flow Controller is Peripheral 0x7 (TT_FC_7): Transfer type is Peripheral to Peripheral and Flow Controller is Destination Peripheral
24 : 23	DMS	rw	0x0	Destination Master Select. Identifies the Master Interface layer where the destination device (peripheral or memory) resides. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (DMS_0): Destination device (peripheral or memory) is accessed from AHB master 1 0x1 (DMS_1): Destination device (peripheral or memory) is accessed from AHB master 2 0x2 (DMS_2): Destination device (peripheral or memory) is accessed from AHB master 3 0x3 (DMS_3): Destination device (peripheral or memory) is accessed from AHB master 4
26 : 25	SMS	rw	0x0	Source Master Select. Identifies the Master Interface layer where the source device (peripheral or memory) resides. The maximum value of this field that can be read back is DMAH_NUM_MASTER_INT-1. Values: 0x0 (SMS_0): Source device (peripheral or memory) is accessed from AHB master 1 0x1 (SMS_1): Source device (peripheral or memory) is accessed from AHB master 2 0x2 (SMS_2): Source device (peripheral or memory) is accessed from AHB master 3 0x3 (SMS_3): Source device (peripheral or memory) is accessed from AHB master 4
27	LLP_DST_EN	rw	0x0	Block chaining is enabled on the destination side only if LLP_DST_EN field is high and LLPx.LOC is non-zero. For more information, see "Block Chaining using Linked Lists". Values: 0x0 (LLP_DST_DISABLE): Block chaining using Linked List is disabled on the Destination side 0x1 (LLP_DST_ENABLE): Block chaining using Linked List is enabled on the Destination side

28	LLP_SRC_EN	rw	0x0	Block chaining is enabled on the source side only if the LLP_SRC_EN field is high and LLPx.LOC is non-zero. For more information, see "Block Chaining using Linked Lists". Values: 0x0 (LLP_SRC_DISABLE): Block chaining using Linked List is disabled on the Source side 0x1 (LLP_SRC_ENABLE): Block chaining using Linked List is enabled on the Source side
31 : 29	RESERVED0	ro	0x0	

62.23 CTL2_H				0x406A00CC
Bits	Field Name	Acc	Default	Description
10 : 0	BLOCK_TS	rw	0x0	Block Transfer Size. When the DW_ahb_dmac is the flow controller, the user writes this field before the channel is enabled in order to indicate the block size. The number programmed into BLOCK_TS indicates the total number of single transactions to perform for every block transfer; a single transaction is mapped to a single AMBA beat. Width: The width of single transaction is determined by CTLx.SRC_TR_WIDTH. For further information on setting this field, refer to "Transfer Operation". Once the transfer starts, the read-back value is the total number of data items already read from the source peripheral, regardless of what is the flow controller. When the source or destination peripheral is assigned as the flow controller, then the maximum block size that can be read back saturates at DMAH_CHx_MAX_BLK_SIZE, but the actual block size can be greater.
11	RESERVED	ro	0x0	
12	DONE	rw	0x0	Done bit. If status write-back is enabled, the upper word of the control register, CTLx[63:32], is written to the control register location of the Linked List Item (LLI) in system memory at the end of the block transfer with the done bit set. Software can poll the LLI CTLx.DONE bit to see when a block transfer is complete. The LLI CTLx.DONE bit should be cleared when the linked lists are set up in memory prior to enabling the channel. LLI accesses are always 32-bit accesses (Hsize=2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32-bit. For more information, refer to "Multi-Block Transfers". Values: 0x0 (DISABLED): DONE bit is deasserted the enb of block transfer 0x1 (ENABLED): SET the DONE bit at the end of block transfer
31 : 13	RESERVED0	ro	0x0	

62.24 CFG2_L				0x406A00F0
Bits	Field Name	Acc	Default	Description
4 : 0	RESERVED	ro	0x0	

7 : 5	CH_PRIOR	rw	0x0	Channel Priority. A priority of 7 is the highest priority, and 0 is the lowest. This field must be programmed within the range 0 to DMAH_NUM_CHANNELS-1. A programmed value outside this range will cause erroneous behavior. Values: 0x0 (CH_PRIOR_0): Channel priority is 0 0x1 (CH_PRIOR_1): Channel priority is 1 0x2 (CH_PRIOR_2): Channel priority is 2 0x3 (CH_PRIOR_3): Channel priority is 3 0x4 (CH_PRIOR_4): Channel priority is 4 0x5 (CH_PRIOR_5): Channel priority is 5 0x6 (CH_PRIOR_6): Channel priority is 6 0x7 (CH_PRIOR_7): Channel priority is 7
8	CH_SUSP	rw	0x0	Channel Suspend. Suspends all DMA data transfers from the source until this bit is cleared. There is no guarantee that the current transaction will complete. Can also be used in conjunction with CFGx.FIFO_EMPTY to cleanly disable a channel without losing any data. For more information, refer to "Disabling a Channel Prior to Transfer Completion". Values: 0x0 (NOT_SUSPENDED): DMA transfer from the source is not suspended 0x1 (SUSPENDED): Suspend DMA transfer from the source
9	FIFO_EMPTY	ro	0x0	Channel FIFO status. Indicates if there is data left in the channel FIFO. Can be used in conjunction with CFGx.CH_SUSP to cleanly disable a channel. For more information, refer to "Disabling a Channel Prior to Transfer Completion". Values: 0x0 (NOT_EMPTY): Channel FIFO is not empty 0x1 (EMPTY): Channel FIFO is empty
10	HS_SEL_DST	rw	0x1	Destination Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces - hardware or software - is active for destination requests on this channel. If the destination peripheral is memory, then this bit is ignored. Values: 0x0 (HARDWARE_HS): Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1 (SOFTWARE_HS): Software handshaking interface. Hardware initiated transaction requests are ignored.
11	HS_SEL_SRC	rw	0x1	Source Software or Hardware Handshaking Select. This register selects which of the handshaking interfaces - hardware or software - is active for source requests on this channel. If the source peripheral is memory, then this bit is ignored. Values: 0x0 (HARDWARE_HS): Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1 (SOFTWARE_HS): Software handshaking interface. Hardware initiated transaction requests are ignored. Value After Reset: 0x1
13 : 12	LOCK_CH_L	rw	0x0	Channel Local Level. Indicates the duration over which CFGx.LOCK_CH applies. This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (LOCK_CH_L_0): Over complete DMA transfer 0x1 (LOCK_CH_L_1): Over complete DMA block transfer 0x2 (LOCK_CH_L_2): Over complete DMA transaction 0x3 (LOCK_CH_L_3): Over complete DMA transaction

15 : 14	LOCK_B_L	rw	0x0	Bus lock level. Indicates the duration over which CFGx.LOCK_B bit applies. This field does not exist if the parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (LOCK_B_L_0): Over complete DMA transfer 0x1 (LOCK_B_L_1): Over complete DMA block transfer 0x2 (LOCK_B_L_2): Over complete DMA transaction 0x3 (LOCK_B_L_3): Over complete DMA transaction
16	LOCK_CH	rw	0x0	Channel Lock Bit. When the channel is granted control of the master bus interface and if the CFGx.LOCK_CH bit is asserted, then no other channels are granted control of the master bus interface for the duration specified in CFGx.LOCK_CH_L. Indicates to the master bus interface arbiter that this channel wants exclusive access to the master bus interface for the duration specified in CFGx.LOCK_CH_L. This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Channel lock bit is not enabled 0x1 (ENABLED): Channel lock bit is enabled
17	LOCK_B	rw	0x0	Bus Lock Bit. When active, the AHB bus master signal hlock is asserted for the duration specified in CFGx.LOCK_B_L. For more information, refer to "Locked DMA Transfers". This field does not exist if the configuration parameter DMAH_CHx_LOCK_EN is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Bus lock bit is not enabled 0x1 (ENABLED): Bus lock bit is enabled
18	DST_HS_POL	rw	0x0	Destination Handshaking Interface Polarity. Values: 0x0 (ACTIVE_HIGH): Destination Handshaking Interface Polarity is Active high 0x1 (ACTIVE_LOW): Destination Handshaking Interface Polarity is Active low
19	SRC_HS_POL	rw	0x0	Source Handshaking Interface Polarity. Values: 0x0 (ACTIVE_HIGH): Source Handshaking Interface Polarity is Active high 0x1 (ACTIVE_LOW): Source Handshaking Interface Polarity is Active low
29 : 20	MAX_ABRST	rw	0x0	Maximum AMBA Burst Length. Maximum AMBA burst length that is used for DMA transfers on this channel. A value of 0 indicates that software is not limiting the maximum AMBA burst length for DMA transfers on this channel. This field does not exist if the configuration parameter DMAH_MABRST is not selected; in this case, the read-back value is always 0, and the maximum AMBA burst length cannot be limited by software.
30	RELOAD_SRC	rw	0x0	Automatic Source Reload. The SARx register can be automatically reloaded from its initial value at the end of every block for multi-block transfers. A new block transfer is then initiated. This field does not exist if the configuration parameter DMAH_CHx_MULTI_BLK_EN is not selected; in this case, the read-back value is always 0. Values: 0x0 (DISABLE): Source Reload Disabled 0x1 (ENABLE): Source Reload Enabled

31	RELOAD_DST	rw	0x0	Automatic Destination Reload. The DARx register can be automatically reloaded from its initial value at the end of every block for multi-block transfers. A new block transfer is then initiated. This register does not exist if the configuration parameter DMAH_CHx_MULTI_BLK_EN is not selected; in this case, the read-back value is always 0. Values: 0x0 (DISABLE): Destination Reload Disabled. 0x1 (ENABLE): Destination Reload Enabled

62.25 CFG2_H				0x406A00F4
Bits	Field Name	Acc	Default	Description
0	FCMODE	rw	0x0	Flow Control Mode. Determines when source transaction requests are serviced when the Destination Peripheral is the flow controller. Values: 0x0 (FCMODE_0): Source transaction requests are serviced when they occur. Data pre-fetching is enabled 0x1 (FCMODE_1): Source transaction requests are not serviced until a destination transaction request occurs. In this mode, the amount of data transferred from the source is limited so that it is guaranteed to be transferred to the destination prior to block termination by the destination. Data pre-fetching is disabled.
1	FIFO_MODE	rw	0x0	FIFO Mode Select. Determines how much space or data needs to be available in the FIFO before a burst transaction request is serviced. Values: 0x0 (FIFO_MODE_0): Space/data available for single AHB transfer of the specified transfer width 0x1 (FIFO_MODE_1): Data available is greater than or equal to half the FIFO depth for destination transfers and space available is greater than half the fifo depth for source transfers. The exceptions are at the end of a burst transaction request or at the end of a block transfer.
4 : 2	PROTCTL	rw	0x1	Protection Control bits used to drive the AHB HPROT[3:1] bus. The AMBA Specification recommends that the default of HPROT indicates a non-cached, non-buffered, privileged data access. The reset value is used to indicate such an access. HPROT[0] is tied high because all transfers are data accesses, as there are no opcode fetches. There is a one-to-one mapping of these register bits to the HPROT[3:1] master interface signals. Mapping of HPROT bus is as follows: 1'b1 to HPROT[0] CFGx.PROTCTL[1] to HPROT[1] CFGx.PROTCTL[2] to HPROT[2] CFGx.PROTCTL[3] to HPROT[3]
5	DS_UPD_EN	rw	0x0	Destination Status Update Enable. Destination status information is fetched only from the location pointed to by the DSTATARx register, stored in the DSTATx register and written out to the DSTATx location of the LLI, if DS_UPD_EN is high. This field does not exist if the configuration parameter DMAH_CHx_SATAT_DST is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Destination Status Update is disabled 0x1 (ENABLED): Destination Status Update is enabled

6	SS_UPD_EN	rw	0x0	Source Status Update Enable. Source status information is fetched only from the location pointed to by the SSTATARx register, stored in the SSTATx register and written out to the SSTATx location of the LLI, if SS_UPD_EN is high. NOTE: This enalbe is applicable only if DMAH_CHx_STAT_SRC is set to True. This field does not exist if the configuration parameter DMAH_CHx_STAT_SRC is set to False; in this case, the read-back value is always 0. Values: 0x0 (DISABLED): Source Status Update is disabled. 0x1 (ENABLED): Source Status Update is enabled.
7	SRC_PER	rw	0x0	Source Hardware Interface.. Assigns a hardware handshaking interface (0 : DMAH_NUM_HS_INT-1) to the source of channel x if the CFGx.HS_SEL_SRC field is 0; otherwise, this field is ignored. The channel can then communicate with the source peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface. NOTE2: This field does not exist if the configuration parameter DMAH_NUM_HS_INT is set to 0.
10 : 8	RESERVED	ro	0x0	
11	DEST_PER	rw	0x0	Destination hardware interface. Assigns a hardware handshaking interface (0 : DMAH_NUM_HS_INT-1) to the destination of channel x if the CFGx.HS_SEL_DST field is 0; otherwise, this field is ignored. The channel can then communicate with the destination peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface. NOTE2: This field does not exist if the configuration parameter DMAH_NUM_HS_INT is set to 0.
31 : 12	RESERVED0	ro	0x0	

62.26 SGR2				0x406A00F8
Bits	Field Name	Acc	Default	Description
19 : 0	SGI	rw	0x0	Source Gather Interval.
30 : 20	SGC	rw	0x0	Source Gather Count. Source contiguous tranfer count between successive gather boundaries.
31	RESERVED	ro	0x0	

62.27 DSR2				0x406A0100
Bits	Field Name	Acc	Default	Description
19 : 0	DSI	rw	0x0	Destination Scatter Interval.
30 : 20	DSC	rw	0x0	Destination Scatter Count. Destination contiguous transfer count between successive scatter boundaries.
31	RESERVED	ro	0x0	

62.28 RAW_TFR				0x406A02C0
Bits	Field Name	Acc	Default	Description
2 : 0	RAW	rw	0x0	Raw Status for IntTfr Interrupt Values: 0x0 (INACTIVE): Inactive Raw Interrupt Status 0x1 (ACTIVE): Active Raw Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.29 RAW_BLOCK				0x406A02C8
Bits	Field Name	Acc	Default	Description
2 : 0	RAW	rw	0x0	Raw Status for IntBlock Interrupt Values: 0x0 (INACTIVE): Inactive Raw Interrupt Status 0x1 (ACTIVE): Active Raw Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.30 RAW_SRC_TRAN				0x406A02D0
Bits	Field Name	Acc	Default	Description
2 : 0	RAW	rw	0x0	Raw Status for IntSrcTran Interrupt Values: 0x0 (INACTIVE): Inactive Raw Interrupt Status 0x1 (ACTIVE): Active Raw Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.31 RAW_DST_TRAN				0x406A02D8
Bits	Field Name	Acc	Default	Description
2 : 0	RAW	rw	0x0	Raw Status for IntDstTran Interrupt Values: 0x0 (INACTIVE): Inactive Raw Interrupt Status 0x1 (ACTIVE): Active Raw Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.32 RAW_ERR				0x406A02E0
Bits	Field Name	Acc	Default	Description
2 : 0	RAW	rw	0x0	Raw Status for IntErr Interrupt Values: 0x0 (INACTIVE): Inactive Raw Interrupt Status 0x1 (ACTIVE): Active Raw Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.33 STATUS_TFR				0x406A02E8
Bits	Field Name	Acc	Default	Description
2 : 0	STATUS	rw	0x0	Status for IntTfr Interrupt Values: 0x0 (INACTIVE): Inactive Interrupt Status 0x1 (ACTIVE): Active Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.34 STATUS_BLOCK				0x406A02F0
Bits	Field Name	Acc	Default	Description
2 : 0	STATUS	rw	0x0	Status for IntBlock Interrupt Values: 0x0 (INACTIVE): Inactive Interrupt Status 0x1 (ACTIVE): Active Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.35 STATUS_SRC_TRAN				0x406A02F8
Bits	Field Name	Acc	Default	Description
2 : 0	STATUS	rw	0x0	Status for IntSrcTran Interrupt Values: 0x0 (INACTIVE): Inactive Interrupt Status 0x1 (ACTIVE): Active Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.36 STATUS_DST_TRAN				0x406A0300
Bits	Field Name	Acc	Default	Description
2 : 0	STATUS	rw	0x0	Status for IntDstTran Interrupt Values: 0x0 (INACTIVE): Inactive Interrupt Status 0x1 (ACTIVE): Active Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.37 STATUS_ERR				0x406A0308
Bits	Field Name	Acc	Default	Description
2 : 0	STATUS	rw	0x0	Status for IntErr Interrupt Values: 0x0 (INACTIVE): Inactive Interrupt Status 0x1 (ACTIVE): Active Interrupt Status
31 : 3	RESERVED	ro	0x0	

62.38 MASK_TFR				0x406A0310
Bits	Field Name	Acc	Default	Description
2 : 0	INT_MASK	rw	0x0	Mask for IntTfr Interrupt Values: 0x0 (MASK): Mask the interrupts 0x1 (UNMASK): Unmask the interrupts
7 : 3	RESERVED	ro	0x0	
10 : 8	INT_MASK_WE	rw	0x0	Interrupt Mask Write Enable Values: 0x0 (DISABLED): Interrupt mask write disable 0x1 (ENABLED): Interrupt mask write enable
31 : 11	RESERVED0	ro	0x0	

62.39 MASK_BLOCK				0x406A0318
Bits	Field Name	Acc	Default	Description
2 : 0	INT_MASK	rw	0x0	Mask for IntBlock Interrupt Values: 0x0 (MASK): Mask the interrupts 0x1 (UNMASK): Unmask the interrupts
7 : 3	RESERVED	ro	0x0	
10 : 8	INT_MASK_WE	rw	0x0	Interrupt Mask Write Enable Values: 0x0 (DISABLED): Interrupt mask write disable 0x1 (ENABLED): Interrupt mask write enable
31 : 11	RESERVED0	ro	0x0	

62.40 MASK_SRC_TRAN				0x406A0320
Bits	Field Name	Acc	Default	Description
2 : 0	INT_MASK	rw	0x0	Mask for IntDstTran Interrupt Values: 0x0 (MASK): Mask the interrupts 0x1 (UNMASK): Unmask the interrupts
7 : 3	RESERVED	ro	0x0	
10 : 8	INT_MASK_WE	rw	0x0	Interrupt Mask Write Enable Values: 0x0 (DISABLED): Interrupt mask write disable 0x1 (ENABLED): Interrupt mask write enable
31 : 11	RESERVED0	ro	0x0	

62.41 MASK_DST_TRAN				0x406A0328
Bits	Field Name	Acc	Default	Description

2 : 0	INT_MASK	rw	0x0	Mask for IntDstTran Interrupt Values: 0x0 (MASK): Mask the interrupts 0x1 (UNMASK): Unmask the interrupts
7 : 3	RESERVED	ro	0x0	
10 : 8	INT_MASK_WE	rw	0x0	Interrupt Mask Write Enable Values: 0x0 (DISABLED): Interrupt mask write disable 0x1 (ENABLED): Interrupt mask write enable
31 : 11	RESERVED0	ro	0x0	

62.42 MASK_ERR				0x406A0330
Bits	Field Name	Acc	Default	Description
2 : 0	INT_MASK	rw	0x0	Mask for IntErr Interrupt Values: 0x0 (MASK): Mask the interrupts 0x1 (UNMASK): Unmask the interrupts
7 : 3	RESERVED	ro	0x0	
10 : 8	INT_MASK_WE	rw	0x0	Interrupt Mask Write Enable Values: 0x0 (DISABLED): Interrupt mask write disable 0x1 (ENABLED): Interrupt mask write enable
31 : 11	RESERVED0	ro	0x0	

62.43 CLEAR_TFR				0x406A0338
Bits	Field Name	Acc	Default	Description
2 : 0	CLEAR	rw	0x0	Clear for IntTfr Interrupt Values: 0x0 (NOT_CLEAR): No effect 0x1 (CLEAR): Clears interrupts
31 : 3	RESERVED	ro	0x0	

62.44 CLEAR_BLOCK				0x406A0340
Bits	Field Name	Acc	Default	Description
2 : 0	CLEAR	rw	0x0	Clear for IntBlock Interrupt Values: 0x0 (NOT_CLEAR): No effect 0x1 (CLEAR): Clears interrupts
31 : 3	RESERVED	ro	0x0	

62.45 CLEAR_SRC_TRAN				0x406A0348
Bits	Field Name	Acc	Default	Description
2 : 0	CLEAR	rw	0x0	Clear for IntSrcTran Interrupt Values: 0x0 (NOT_CLEAR): No effect 0x1 (CLEAR): Clears interrupts
31 : 3	RESERVED	ro	0x0	

62.46 CLEAR_DST_TRAN				0x406A0350
Bits	Field Name	Acc	Default	Description
2 : 0	CLEAR	rw	0x0	Clear for IntDstTran Interrupt Values: 0x0 (NOT_CLEAR): No effect 0x1 (CLEAR): Clears interrupts
31 : 3	RESERVED	ro	0x0	

62.47 CLEAR_ERR	0x406A0358

Bits	Field Name	Acc	Default	Description
2 : 0	CLEAR	rw	0x0	Clear for IntErr Interrupt Values: 0x0 (NOT_CLEAR): No effect 0x1 (CLEAR): Clears interrupts
31 : 3	RESERVED	ro	0x0	

62.48 STATUS_INT 0x406A0360

Bits	Field Name	Acc	Default	Description
0	TFR	ro	0x0	OR of the contents of StatusTfr register Values: 0x0 (INACTIVE): OR of the contents of StatusTfr register is 0 0x1 (ACTIVE): OR of the contents of StatusTfr register is 1
1	BLOCK	ro	0x0	OR of the contents of StatusBlock register Values: 0x0 (INACTIVE): OR of the contents of StatusBlock register is 0 0x1 (ACTIVE): OR of the contents of StatusBlock register is 1
2	SRCT	ro	0x0	OR of the contents of StatusSrcTran Values: 0x0 (INACTIVE): OR of the contents of StatusSrcTran register is 0 0x1 (ACTIVE): OR of the contents of StatusSrcTran register is 1
3	DSTT	ro	0x0	OR of the contents of StatusDstTran Values: 0x0 (INACTIVE): OR of the contents of StatusDstTran register is 0 0x1 (ACTIVE): OR of the contents of StatusDstTran register is 1
4	ERR	ro	0x0	OR of the contents of StatusErr Values: 0x0 (INACTIVE): OR of the contents of StatusErr register is 0 0x1 (ACTIVE): OR of the contents of StatusErr register is 1
31 : 5	RESERVED	ro	0x0	

62.49 REQ_SRC 0x406A0368

Bits	Field Name	Acc	Default	Description
2 : 0	SRC_REQ	rw	0x0	Source Software Transaction Request Values: 0x0 (INACTIVE): Source request is not active 0x1 (ACTIVE): Source request is active
7 : 3	RESERVED	ro	0x0	
10 : 8	SRC_REQ_WE	rw	0x0	Source Software Transaction Request write enable
31 : 11	RESERVED0	ro	0x0	

62.50 REQ_DST 0x406A0370

Bits	Field Name	Acc	Default	Description
2 : 0	DST_REQ	rw	0x0	Destination Software Transaction Request Values: 0x0 (INACTIVE): Source request is not active 0x1 (ACTIVE): Source request is active
7 : 3	RESERVED	ro	0x0	
10 : 8	DST_REQ_WE	rw	0x0	Destination Software Transaction Request write enable
31 : 11	RESERVED0	ro	0x0	

62.51 SGL_RQ_SRC 0x406A0378

Bits	Field Name	Acc	Default	Description
2 : 0	SRC_SGLREQ	rw	0x0	Source Single Transaction Request Values: 0x0 (INACTIVE): Source single request is not active 0x1 (ACTIVE): Source single request is active
7 : 3	RESERVED	ro	0x0	
10 : 8	SRC_SGLREQ_WE	rw	0x0	Source Single Transaction Request write enable
31 : 11	RESERVED0	ro	0x0	

62.52 SGL_RQ_DST				0x406A0380
Bits	Field Name	Acc	Default	Description
2 : 0	DST_SGLREQ	rw	0x0	Destination Single Transaction Request Values: 0x0 (INACTIVE): Destination single request is not active 0x1 (ACTIVE): Destination single request is active
7 : 3	RESERVED	ro	0x0	
10 : 8	DST_SGLREQ_WE	rw	0x0	Destination Single Transaction Request write enable
31 : 11	RESERVED0	ro	0x0	

62.53 LST_SRC				0x406A0388
Bits	Field Name	Acc	Default	Description
2 : 0	LSTSRC	rw	0x0	Source Last Transaction Request Values: 0x0 (INACTIVE): Source last request is not active 0x1 (ACTIVE): Source last request is active
7 : 3	RESERVED	ro	0x0	
10 : 8	LSTSRC_WE	rw	0x0	Source Last Transaction Request write enable
31 : 11	RESERVED0	ro	0x0	

62.54 LST_DST				0x406A0390
Bits	Field Name	Acc	Default	Description
2 : 0	LSTDST	rw	0x0	Destination Last Transaction Request Values: 0x0 (INACTIVE): Destination single request is not active 0x1 (ACTIVE): Destination single request is active
7 : 3	RESERVED	ro	0x0	
10 : 8	LSTDST_WE	rw	0x0	Destination Last Transaction Request write enable
31 : 11	RESERVED0	ro	0x0	

62.55 DMA_CFG				0x406A0398
Bits	Field Name	Acc	Default	Description
0	DMA_EN	rw	0x0	DW_ahb_dmac Enable bit. Values: 0x0 (DISABLED): DW_ahb_dmac Disabled 0x1 (ENABLED): DW_ahb_dmac Enabled
31 : 1	RESERVED	ro	0x0	

62.56 CH_EN				0x406A03A0
Bits	Field Name	Acc	Default	Description

2 : 0	CH_EN	rw	0x0	Channel Enable. The ChEnReg.CH_EN bit is automatically cleared by hardware to disable the channel after the last AMBA transfer of the DMA transfer to the destination has completed. Software can therefore poll this bit to determine when this channel is free for a new DMA transfer. Values: 0x0 (DISABLED): Disable the channel 0x1 (ENABLED): Enable the channel
7 : 3	RESERVED	ro	0x0	
10 : 8	CH_EN_WE	rw	0x0	Channel enable register
31 : 11	RESERVED0	ro	0x0	

62.57 DMA_ID				0x406A03A8
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_ID	ro	0x0	Hardcoded DW_ahb_dmac peripheral ID. Value After Reset: DMAH_ID_NUM Exists: Always

62.58 DMA_TEST				0x406A03B0
Bits	Field Name	Acc	Default	Description
0	TEST_SLV_IF	rw	0x0	DMA Test register Values: 0x0 (NORMAL_MODE): Puts the AHB slave interface into Normal mode 0x1 (TEST_MODE): Puts the AHB slave interface into Test mode. In this mode, the readback value of the writable registers always matches the values written
31 : 1	RESERVED	ro	0x0	

62.59 DMA_LP_TIMEOUT				0x406A03B8
Bits	Field Name	Acc	Default	Description
3 : 0	DMA_LP_TIMEOUT	rw	0x0	This field holds timeout value of low power counter.
31 : 4	RESERVED	ro	0x0	

62.60 DMA_COMP_PARAMS_6L				0x406A03C8
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_6L	ro	0x0	

62.61 DMA_COMP_PARAMS_6H				0x406A03CC
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_6H	ro	0x0	

62.62 DMA_COMP_PARAMS_5L				0x406A03D0
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_5L	ro	0x0	

62.63 DMA_COMP_PARAMS_5H				0x406A03D4
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_5H	ro	0x0	

62.64 DMA_COMP_PARAMS_4L				0x406A03D8
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_4L	ro	0x0	

62.65 DMA_COMP_PARAMS_4H				0x406A03DC
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_4H	ro	0x0	

62.66 DMA_COMP_PARAMS_3L				0x406A03E0
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_3L	ro	0x0	

62.67 DMA_COMP_PARAMS_3H				0x406A03E4
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_3H	ro	0x0	

62.68 DMA_COMP_PARAMS_2L				0x406A03E8
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_2L	ro	0x10001800	

62.69 DMA_COMP_PARAMS_2H				0x406A03EC
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_2H	ro	0x0	

62.70 DMA_COMP_PARAMS_1L				0x406A03F0
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_1L	ro	0x0	

62.71 DMA_COMP_PARAMS_1H				0x406A03F4
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_PARAMS_1H	ro	0x0	

62.72 DMA_COMPS_ID_L				0x406A03F8
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_TYPE	ro	0x44571110	DMA Component Type Number = ‘h44571110. This assigned unique hex value is constant and is derived from the two ASCII letters "DW" followed by a 32-bit unsigned number.

62.73 DMA_COMPS_ID_H				0x406A03FC
Bits	Field Name	Acc	Default	Description
31 : 0	DMA_COMP_VERSION	ro	0x3232322A	DMA Component Version - See release notes.

63 bt_uart

63.1 txrx_fifo_divl				0x406B0000
Bits	Field Name	Acc	Default	Description
7 : 0	txrx_data_divl	rw	0x0	Tx data(when write data), Rx data(when read data), Divisor latch(low) config when divisor latch is enable.

63.2 int_en_divh	0x406B0004
Discription: Interrupt enable register; divisor latch high when divisor latch is enable.	

Bits	Field Name	Acc	Default	Description
0	rx_int_en_divh_0	rw	0x0	Enable Received Data Available Interrupt; Divisor latch high bit 0 when divisor latch is enable.
1	tx_int_en_divh_1	rw	0x0	Enable Transmit Holding Register Empty Interrupt; Divisor latch high bit 1 when divisor latch is enable.
2	lstatus_int_en_divh_2	rw	0x0	Enable Receiver Line Status Interrupt; Divisor latch high bit 2 when divisor latch is enable.
3	mstatus_int_en_divh_3	rw	0x0	Enable Modem Status Interrupt; Divisor latch high bit 3 when divisor latch is enable.
4	divh_4	rw	0x0	Divisor latch high bit 4 when divisor latch is enable.
5	timeout_int_en_divh_5	rw	0x0	Enable Timeout Interrupt; Divisor latch high bit 5 when divisor latch is enable.
6	divh_6	rw	0x0	Divisor latch high bit 6 when divisor latch is enable.
7	ptime_divh_7	rw	0x0	Programmable THRE Interrupt Mode Enable; Divisor latch high bit 7 when divisor latch is enable.

63.3 int_id_fifo_ctrl 0x406B0008

Discription: Interrupt identity register or fifo control register

Bits	Field Name	Acc	Default	Description
0	int_id_0_fifo_en	rw	0x1	Read: Interrupt identity bit 0, reset value is '0'; Write: Fifo enable, reset value is '1'
1	int_id_1_rx_reset	rw	0x0	Read: Interrupt identity bit 1; Write: Receiver reset
2	int_id_2_tx_reset	rw	0x0	Read: Interrupt identity bit 2; Write: Transmit reset
3	int_id_3_dma_mode	rw	0x0	Read: Interrupt identity bit 3; Write: Dma mode enable
7 : 6	fifo_enabled	ro	0x3	

63.4 line_ctrl 0x406B000C

Bits	Field Name	Acc	Default	Description
1 : 0	data_length_sel	rw	0x3	0: 5 bit 1: 6 bit 2: 7 bit 3: 8 bit
2	stop	rw	0x0	Number of stop bits
3	parity_en	rw	0x0	Parity enable
4	even_parity_sel	rw	0x0	'1' select even parity
6	break_irq	rw	0x0	
7	div_latch_acc	rw	0x0	Divisor Latch access enable
8	div_mode	rw	0x0	
9	break_en	rw	0x0	

63.5 modem_ctrl 0x406B0010

Bits	Field Name	Acc	Default	Description
0	dtr	rw	0x0	Data terminal ready
1	rts	rw	0x0	Request to send
2	out1	rw	0x0	
3	out2	rw	0x0	
4	loopback	rw	0x0	
5	auto_flow_ctrl_enable	rw	0x0	Auto flow control enable
6	serial_infrared_radiation_en	rw	0x0	Serial infrared radiation enable
7	auto_det	rw	0x0	
8	clk_p	rw	0x0	1: 48M 0: 24M

63.6 line_status 0x406B0014

Bits	Field Name	Acc	Default	Description
0	rx_data_ready	ro	0x0	Rx Data ready(rxfifo not empty)

1	rx_overflow	ro	0x0	Overflow error
5	tx_fifo_empty	ro	0x0	Tx fifo empty or almost empty
6	tx_empty	ro	0x0	Tx fifo empty and shift register empty
7	rx_fifo_error	ro	0x0	Rx fifo error
8	trigger_ready	ro	0x0	Trigger ready(rxfifo not empty)

63.7 modem_status 0x406B0018

Bits	Field Name	Acc	Default	Description
0	delayed_cts	ro	0x0	
4	cts	ro	0x0	

63.8 scratchpad 0x406B001C

Bits	Field Name	Acc	Default	Description
6 : 0	scratchpad	rw	0x0	not used

63.9 fifo_status 0x406B0020

Bits	Field Name	Acc	Default	Description
7 : 0	tx_count	ro	0x0	TX fifo count
16 : 9	rx_count	ro	0x0	RX fifo count
18	tx_fifo_empty	ro	0x1	TX fifo empty
19	tx_fifo_full	ro	0x0	TX fifo full
20	rx_fifo_empty	ro	0x1	RX fifo empty
21	rx_fifo_full	ro	0x0	RX fifo full

63.10 fifo_trigger 0x406B0024

Bits	Field Name	Acc	Default	Description
7 : 0	rx_trigger_th	rw	0x10	
16 : 9	tx_trigger_th	rw	0x10	
18	rx_tout_trig_en	rw	0x0	Enable rx legacy dma req triggered when timeout

63.11 divisor_latch2 0x406B0028

Bits	Field Name	Acc	Default	Description
7 : 0	divisor_adjust_coef	rw	0x0	accuray adjust baudrate

63.12 counter_real 0x406B003C

Bits	Field Name	Acc	Default	Description
19 : 0	counter_real	ro	0x0	

63.13 div_real_slow 0x406B0040

Bits	Field Name	Acc	Default	Description

31 : 0	div_real_slow	rw	0x1000E	when clk_p is 0: slow div_real = div_real_slow[15:1];
				slow div_adjust_real = div_real_slow[31:24];
				when clk_p is 1: slow div_real = div_real_slow[15:0];
				slow div_adjust_real = div_real_slow[23:16];

63.14 div_real_fast 0x406B0044

Bits	Field Name	Acc	Default	Description

31 : 0	div_real_fast	rw	0x660001	when clk_p is 0: fast div_real = div_real_fast[15:1];
				fast div_adjust_real = div_real_fast[31:24];
				when clk_p is 1: fast div_real = div_real_fast[15:0];
				fast div_adjust_real = div_real_fast[23:16];

63.15 tports_sel				0x406B0048
Bits	Field Name	Acc	Default	Description
7 : 0	tports_sel	rw	0x0	Tports Select

64 pcie_ep

64.1 dev_id_vendor_id				0x40700000
Bits	Field Name	Acc	Default	Description
15 : 0	vendor_id	rw	0x16C3	
31 : 16	device_id	rw	0xABCD	

64.2 status_cmd				0x40700004
Bits	Field Name	Acc	Default	Description
0	io_en	rw	0x0	
1	mem_space_en	rw	0x0	
2	bus_master_en	rw	0x0	
3	special_cycle_operation	ro	0x0	
4	mwi_enable	ro	0x0	
5	vga_palette_snoop	ro	0x0	
6	parity_err_en	rw	0x0	
7	idsel_stepping	ro	0x0	
8	serren	rw	0x0	
10	int_en	rw	0x0	
16 : 11	reserv	ro	0x0	
19	int_status	ro	0x0	
20	cap_list	ro	0x1	
21	fast_66mhz_cap	ro	0x0	
23	fast_b2b_cap	ro	0x0	
24	master_dpe	rw	0x0	Write 1 Clear
26 : 25	dev_sel_timing	ro	0x0	
27	signaled_target_abort	rw	0x0	Write 1 Clear
28	rcvd_target_abort	rw	0x0	Write 1 Clear
29	rcvd_master_abort	rw	0x0	Write 1 Clear
30	signaled_sys_err	rw	0x0	Write 1 Clear
31	detected_parity_err	rw	0x0	Write 1 Clear

64.3 class_code_rev_id				0x40700008
Bits	Field Name	Acc	Default	Description
7 : 0	rev_id	rw	0x1	
15 : 8	program_intf	rw	0x0	
23 : 16	subclass_code	rw	0x0	
31 : 24	base_class_code	rw	0x0	

64.4 bist_header_type_latency_cache_line_size				0x4070000C
Bits	Field Name	Acc	Default	Description
7 : 0	cache_line_size	rw	0x0	
15 : 8	latency_master_timer	ro	0x0	
22 : 16	header_type	ro	0x0	
23	multi_func	rw	0x0	
31 : 24	bist	ro	0x0	

64.5 bar0	0x40700010

Bits	Field Name	Acc	Default	Description
0	bar0_mem_io	rw	0x0	
2 : 1	bar0_type	rw	0x0	
3	bar0_prefetch	rw	0x0	
31 : 4	bar0_start	rw	0x0	

64.6 bar1				0x40700014
Bits	Field Name	Acc	Default	Description
0	bar1_mem_io	rw	0x0	
2 : 1	bar1_type	rw	0x0	
3	bar1_prefetch	rw	0x0	
31 : 4	bar1_start	rw	0x0	

64.7 bar2				0x40700018
Bits	Field Name	Acc	Default	Description
0	bar2_mem_io	rw	0x0	
2 : 1	bar2_type	rw	0x0	
3	bar2_prefetch	rw	0x0	
31 : 4	bar2_start	rw	0x0	

64.8 bar3				0x4070001C
Bits	Field Name	Acc	Default	Description
0	bar3_mem_io	rw	0x0	
2 : 1	bar3_type	rw	0x0	
3	bar3_prefetch	rw	0x0	
31 : 4	bar3_start	rw	0x0	

64.9 cardbus_cis_ptr				0x40700028
Bits	Field Name	Acc	Default	Description
31 : 0	cardbus_cis_ptr	rw	0x0	

64.10 subsys_id_subsys_vendor_id				0x4070002C
Bits	Field Name	Acc	Default	Description
15 : 0	subsys_vendor_id	rw	0x0	
31 : 16	subsys_dev_id	rw	0x0	

64.11 exp_rom_base_addr				0x40700030
Bits	Field Name	Acc	Default	Description
0	rom_bar_enable	rw	0x1	
31 : 11	exp_rom_base_addr	rw	0x0	

64.12 pci_cap_ptr				0x40700034
Bits	Field Name	Acc	Default	Description
7 : 0	cap_ptr	rw	0x40	

64.13 intr_line_intr_pin				0x4070003C
Bits	Field Name	Acc	Default	Description
7 : 0	int_line	rw	0xFF	
15 : 8	int_pin	rw	0x1	

64.14 pm_cap				0x40700040
Bits	Field Name	Acc	Default	Description
7 : 0	pm_cap_id	ro	0x1	
15 : 8	pm_next_ptr	rw	0x50	
18 : 16	pm_spec_ver	rw	0x3	
19	pme_clk	ro	0x0	
21	dsi	rw	0x0	
24 : 22	aux_curr	rw	0x7	
25	d1_support	rw	0x1	
26	d2_support	rw	0x0	
31 : 27	pme_support	rw	0x1B	

64.15 pm_ctrl_status				0x40700044
Bits	Field Name	Acc	Default	Description
1 : 0	power_state	rw	0x0	
3	no_soft_rst	rw	0x0	
8	pme_enable	rw	0x0	
12 : 9	data_select	ro	0x0	
14 : 13	data_scale	ro	0x0	
15	pme_status	rw	0x0	Write 1 Clear
22	b2_b3_support	ro	0x0	
23	bus_pwr_clk_con_en	ro	0x0	
31 : 24	data_reg_add_info	ro	0x0	

64.16 msi_cap				0x40700050
Bits	Field Name	Acc	Default	Description
7 : 0	msi_cap_id	ro	0x5	
15 : 8	msi_next_ptr	rw	0x70	
16	msi_enable	rw	0x0	
19 : 17	msi_multi_msg_cap	rw	0x0	
22 : 20	msi_multi_msg_en	rw	0x0	
23	msi_64_bit_addr_cap	rw	0x1	
24	pvm_support	ro	0x0	

64.17 msi_lower_addr				0x40700054
Bits	Field Name	Acc	Default	Description
31 : 2	msi_lower_addr	rw	0x0	

64.18 msi_upper_addr				0x40700058
Bits	Field Name	Acc	Default	Description
31 : 0	msi_upper_addr	rw	0x0	

64.19 msi_data				0x4070005C
Bits	Field Name	Acc	Default	Description
15 : 0	msi_data	rw	0x0	

64.20 pcie_cap				0x40700070
Bits	Field Name	Acc	Default	Description
7 : 0	pcie_cap_id	ro	0x10	
15 : 8	pcie_cap_next_ptr	rw	0x0	
19 : 16	pcie_cap	rw	0x2	
23 : 20	dev_port_type	ro	0x0	

24	slot_imp	rw	0x0	
29 : 25	int_msg_num	rw	0x0	

64.21 dev_cap				0x40700074
Bits	Field Name	Acc	Default	Description
2 : 0	max_payload_size	rw	0x2	
4 : 3	phantom_func_support	rw	0x0	
5	ext_tag_supp	rw	0x0	
8 : 6	ep_l0s_accpt_latency	rw	0x7	
11 : 9	ep_l1_accpt_latency	rw	0x7	
15	role_based_err_report	rw	0x1	
28	flr_cap	rw	0x0	

64.22 dev_ctrl_status				0x40700078
Bits	Field Name	Acc	Default	Description
0	corr_err_report_en	rw	0x0	
1	non_fatal_err_report_en	rw	0x0	
2	fatal_err_report_en	rw	0x0	
3	unsupported_req_rep_en	rw	0x0	
4	en_rel_order	rw	0x1	
7 : 5	max_payload_size_cs	rw	0x0	
8	ext_tag_en	ro	0x0	
9	phantom_func_en	ro	0x0	
10	aux_power_pm_en	rw	0x0	
11	en_no_snoop	ro	0x0	
14 : 12	max_read_req_size	rw	0x2	
15	initiate_flr	rw	0x0	
16	corr_err_detected	rw	0x0	Write 1 clear
17	non_fatal_err_detected	rw	0x0	Write 1 clear
18	fatal_err_detected	rw	0x0	Write 1 clear
19	unsupported_req_detected	rw	0x0	Write 1 clear
20	aux_power_detected	ro	0x1	
21	trans_pending	ro	0x0	

64.23 link_cap				0x4070007C
Bits	Field Name	Acc	Default	Description
3 : 0	max_link_speed	rw	0x1	
9 : 4	max_link_width	rw	0x1	
11 : 10	active_state_link_pm_support	rw	0x3	
14 : 12	l0s_exit_latency	rw	0x3	
17 : 15	l1_exit_latency	rw	0x6	
18	clock_power_man	ro	0x0	
19	surprise_down_err_rep_cap	ro	0x0	
20	dll_active_rep_cap	ro	0x0	
21	link_bw_not_cap	ro	0x0	
22	aspm_opt_compliance	rw	0x1	
31 : 24	port_num	rw	0x0	

64.24 link_ctrl_status				0x40700080
Bits	Field Name	Acc	Default	Description
1 : 0	active_state_link_pm_ctrl	rw	0x0	
3	rcb	rw	0x0	
4	link_disable	rw	0x0	
5	retrain_link	rw	0x0	

6	common_clk_config	rw	0x0	
7	extended_synch	rw	0x0	
8	en_clk_power_man	rw	0x0	
10	link_bw_man_int_en	rw	0x0	
11	link_auto_bw_int_en	ro	0x0	
19 : 16	link_speed	ro	0x1	
24 : 20	nego_link_width	ro	0x1	
27	cap_link_training	ro	0x0	
28	slot_clk_config	rw	0x1	
29	dll_active	ro	0x0	
30	link_bw_man_status	ro	0x0	
31	link_auto_bw_status	ro	0x0	

64.25 dev_cap2 — 0x40700094

Bits	Field Name	Acc	Default	Description
3 : 0	cpl_timeout_range	ro	0xF	
4	cpl_timeout_disable_support	ro	0x1	
5	ari_forward_support	ro	0x0	
6	atomic_routing_supp	ro	0x0	
7	atomic_32_cpl_supp	ro	0x0	
8	atomic_64_cpl_supp	ro	0x0	
9	cas_128_cpl_supp	ro	0x0	
10	no_ro_en_pr2pr_par	ro	0x0	
11	ltr_supp	ro	0x0	
12	tph_cmplt_support_0	ro	0x1	
13	tph_cmplt_support_1	ro	0x0	
19 : 18	obff_support	ro	0x0	
20	cfg_extnd_fmt_support	rw	0x0	
21	cfg_end2end_tlp_prfx_support	rw	0x0	
23 : 22	cfg_max_end2end_tlp_prfxs	rw	0x0	

64.26 dev_ctrl_status2 — 0x40700098

Bits	Field Name	Acc	Default	Description
3 : 0	cpl_timeout_value	rw	0x0	
4	cpl_timeout_disable	rw	0x0	
5	ari_forward_support_cs	ro	0x0	
8	ido_req_en	rw	0x0	
9	ido_cpl_en	rw	0x0	
10	ltr_en	rw	0x0	
14 : 13	obff_en	rw	0x0	
15	cfg_end2end_tlp_pfx_blck	ro	0x0	

64.27 link_cap2 — 0x4070009C

Bits	Field Name	Acc	Default	Description
7 : 1	support_link_speed_vector	ro	0x1	
8	cross_link_support	ro	0x0	

64.28 link_ctrl_status2 — 0x407000A0

Bits	Field Name	Acc	Default	Description
3 : 0	target_link_speed	rw	0x1	
4	enter_compliance	rw	0x0	
5	hw_auto_speed_disable	rw	0x0	
6	sel_deemphasis	ro	0x0	
9 : 7	tx_margin	rw	0x0	

10	enter_modified_compliance	rw	0x0	
11	compliance_sos	rw	0x0	
15 : 12	compliance_preset	rw	0x0	
16	curr_deemphasis	ro	0x1	
17	eq_cpl	ro	0x0	
18	eq_cpl_p1	ro	0x0	
19	eq_cpl_p2	rw	0x0	
20	eq_cpl_p3	rw	0x0	
21	link_eq_req	rw	0x0	write 1 clear

64.29 aer_ext_cap_hdr				0x40700100
Bits	Field Name	Acc	Default	Description
15 : 0	aer_cap_id	rw	0x1	
19 : 16	aer_cap_version	rw	0x2	
31 : 20	aer_cap_next	rw	0x148	

64.30 uncorr_err_status				0x40700104
Bits	Field Name	Acc	Default	Description
4	dl_protocol_err_status	rw	0x0	write 1 clear
5	sur_dwn_err_status	ro	0x0	
12	pois_tlp_err_status	rw	0x0	write 1 clear
13	fc_protocol_err_status	rw	0x0	write 1 clear
14	cmplt_timeout_err_status	rw	0x0	write 1 clear
15	cmplt_abort_err_status	rw	0x0	write 1 clear
16	unexp_cmplt_err_status	rw	0x0	write 1 clear
17	rec_overflow_err_status	rw	0x0	write 1 clear
18	malf_tlp_err_status	rw	0x0	write 1 clear
19	ecrc_err_status	rw	0x0	write 1 clear
20	unsupported_req_err_status	rw	0x0	write 1 clear
22	internal_err_status	rw	0x0	write 1 clear
25	tlp_prfx_blocked_err_status	rw	0x0	write 1 clear

64.31 uncorr_err_mask				0x40700108
Bits	Field Name	Acc	Default	Description
4	dl_protocol_err_mask	rw	0x0	
5	sur_dwn_err_mask	ro	0x0	
12	pois_tlp_err_mask	rw	0x0	
13	fc_protocol_err_mask	rw	0x0	
14	cmplt_timeout_err_mask	rw	0x0	
15	cmplt_abort_err_mask	rw	0x0	
16	unexp_cmplt_err_mask	rw	0x0	
17	rec_overflow_err_mask	rw	0x0	
18	malf_tlp_err_mask	rw	0x0	
19	ecrc_err_mask	rw	0x0	
20	unsupported_req_err_mask	rw	0x0	
22	internal_err_mask	rw	0x1	
24	atomic_egress_- blocked_err_mask	ro	0x0	
25	tlp_prfx_blocked_err_mask	rw	0x0	

64.32 uncorr_err_sev				0x4070010C
Bits	Field Name	Acc	Default	Description
4	dl_protocol_err_severity	rw	0x1	
5	sur_dwn_err_severity	ro	0x1	

12	pois_tlp_err_severity	rw	0x0	
13	fc_protocol_err_severity	rw	0x1	
14	cmplt_timeout_err_severity	rw	0x0	
15	cmplt_abort_err_severity	rw	0x0	
16	unexp_cmplt_err_severity	rw	0x0	
17	rec_overflow_err_severity	rw	0x0	
18	malf_tlp_err_severity	rw	0x1	
19	ecrc_err_severity	rw	0x0	
20	unsupported_req_err_severity	rw	0x0	
22	internal_err_severity	rw	0x1	
24	atomic_egress_-blocked_err_severity	ro	0x0	
25	tlp_prfx_blocked_err_severity	rw	0x0	

64.33 corr_err_status				0x40700110
Bits	Field Name	Acc	Default	Description
0	rx_err_status	rw	0x0	write 1 clear
6	bad_tlp_status	rw	0x0	write 1 clear
7	bad_dllp_status	rw	0x0	write 1 clear
8	replay_no_roleover_status	rw	0x0	write 1 clear
12	rpl_timer_timeout_status	rw	0x0	write 1 clear
13	advisory_non_fatal_err_status	rw	0x0	write 1 clear
14	corrected_int_err_status	rw	0x0	write 1 clear

64.34 corr_err_mask				0x40700114
Bits	Field Name	Acc	Default	Description
0	rx_err_mask	rw	0x0	
6	bad_tlp_mask	rw	0x0	
7	bad_dllp_mask	rw	0x0	
8	replay_no_roleover_mask	rw	0x0	
12	rpl_timer_timeout_mask	rw	0x0	
13	advisory_non_fatal_err_mask	rw	0x1	
14	corrected_int_err_mask	rw	0x1	

64.35 adv_err_cap_ctrl				0x40700118
Bits	Field Name	Acc	Default	Description
4 : 0	first_err_ptr	ro	0x0	
5	ecrc_gen_cap	ro	0x1	
6	ecrc_gen_en	rw	0x0	
7	ecrc_check_cap	ro	0x1	
8	ecrc_check_en	rw	0x0	
11	tlp_prfx_log_present	ro	0x0	

64.36 hdr_log_0				0x4070011C
Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_first_dword	ro	0x0	

64.37 hdr_log_1				0x40700120
Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_second_dword	ro	0x0	

64.38 hdr_log_2	0x40700124

Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_third_dword	ro	0x0	

64.39 hdr_log_3				0x40700128
Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_fourth_dword	ro	0x0	

64.40 tlp_prefix_log				0x40700138
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tlp_pfx_log	ro	0x0	

64.41 tph_ext_cap_hdr				0x40700148
Bits	Field Name	Acc	Default	Description
15 : 0	tph_cap_id	rw	0x17	
19 : 16	tph_req_cap_ver	rw	0x1	
31 : 20	tph_req_next_ptr	rw	0x218	

64.42 tph_req_cap_reg				0x4070014C
Bits	Field Name	Acc	Default	Description
0	tph_req_no_st_mode	ro	0x1	
1	tph_req_cap_int_vec	rw	0x1	
2	tph_req_dev_spec	ro	0x1	
8	tph_req_extended_tph	rw	0x0	
9	tph_req_cap_st_table_loc_0	rw	0x1	
10	tph_req_cap_st_table_loc_1	rw	0x0	
26 : 16	tph_req_cap_st_table_size	rw	0x3	

64.43 tph_req_ctrl_reg				0x40700150
Bits	Field Name	Acc	Default	Description
2 : 0	tph_req_st_mode_select	rw	0x0	
9 : 8	tph_req_ctrl_req_en	rw	0x0	

64.44 tph_st_table_reg_0				0x40700154
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_0	rw	0x0	
15 : 8	tph_req_st_table_higher_0	rw	0x0	
23 : 16	tph_req_st_table_lower_1	rw	0x0	
31 : 24	tph_req_st_table_higher_1	rw	0x0	

64.45 tph_st_table_reg_1				0x40700158
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_2	rw	0x0	
15 : 8	tph_req_st_table_higher_2	rw	0x0	
23 : 16	tph_req_st_table_lower_3	rw	0x0	
31 : 24	tph_req_st_table_higher_3	rw	0x0	

64.46 tph_st_table_reg_2				0x4070015C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_4	rw	0x0	
15 : 8	tph_req_st_table_higher_4	rw	0x0	

23 : 16	tph_req_st_table_lower_5	rw	0x0	
31 : 24	tph_req_st_table_higher_5	rw	0x0	

64.47 tph_st_table_reg_3				0x40700160
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_6	rw	0x0	
15 : 8	tph_req_st_table_higher_6	rw	0x0	
23 : 16	tph_req_st_table_lower_7	rw	0x0	
31 : 24	tph_req_st_table_higher_7	rw	0x0	

64.48 tph_st_table_reg_4				0x40700164
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_8	rw	0x0	
15 : 8	tph_req_st_table_higher_8	rw	0x0	
23 : 16	tph_req_st_table_lower_9	rw	0x0	
31 : 24	tph_req_st_table_higher_9	rw	0x0	

64.49 tph_st_table_reg_5				0x40700168
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_10	rw	0x0	
15 : 8	tph_req_st_table_higher_10	rw	0x0	
23 : 16	tph_req_st_table_lower_11	rw	0x0	
31 : 24	tph_req_st_table_higher_11	rw	0x0	

64.50 tph_st_table_reg_6				0x4070016C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_12	rw	0x0	
15 : 8	tph_req_st_table_higher_12	rw	0x0	
23 : 16	tph_req_st_table_lower_13	rw	0x0	
31 : 24	tph_req_st_table_higher_13	rw	0x0	

64.51 tph_st_table_reg_7				0x40700170
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_14	rw	0x0	
15 : 8	tph_req_st_table_higher_14	rw	0x0	
23 : 16	tph_req_st_table_lower_15	rw	0x0	
31 : 24	tph_req_st_table_higher_15	rw	0x0	

64.52 tph_st_table_reg_8				0x40700174
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_16	rw	0x0	
15 : 8	tph_req_st_table_higher_16	rw	0x0	
23 : 16	tph_req_st_table_lower_17	rw	0x0	
31 : 24	tph_req_st_table_higher_17	rw	0x0	

64.53 tph_st_table_reg_9				0x40700178
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_18	rw	0x0	
15 : 8	tph_req_st_table_higher_18	rw	0x0	
23 : 16	tph_req_st_table_lower_19	rw	0x0	
31 : 24	tph_req_st_table_higher_19	rw	0x0	

64.54 tph_st_table_reg_10				0x4070017C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_20	rw	0x0	
15 : 8	tph_req_st_table_higher_20	rw	0x0	
23 : 16	tph_req_st_table_lower_21	rw	0x0	
31 : 24	tph_req_st_table_higher_21	rw	0x0	

64.55 tph_st_table_reg_11				0x40700180
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_22	rw	0x0	
15 : 8	tph_req_st_table_higher_22	rw	0x0	
23 : 16	tph_req_st_table_lower_23	rw	0x0	
31 : 24	tph_req_st_table_higher_23	rw	0x0	

64.56 tph_st_table_reg_12				0x40700184
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_24	rw	0x0	
15 : 8	tph_req_st_table_higher_24	rw	0x0	
23 : 16	tph_req_st_table_lower_25	rw	0x0	
31 : 24	tph_req_st_table_higher_25	rw	0x0	

64.57 tph_st_table_reg_13				0x40700188
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_26	rw	0x0	
15 : 8	tph_req_st_table_higher_26	rw	0x0	
23 : 16	tph_req_st_table_lower_27	rw	0x0	
31 : 24	tph_req_st_table_higher_27	rw	0x0	

64.58 tph_st_table_reg_14				0x4070018C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_28	rw	0x0	
15 : 8	tph_req_st_table_higher_28	rw	0x0	
23 : 16	tph_req_st_table_lower_29	rw	0x0	
31 : 24	tph_req_st_table_higher_29	rw	0x0	

64.59 tph_st_table_reg_15				0x40700190
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_30	rw	0x0	
15 : 8	tph_req_st_table_higher_30	rw	0x0	
23 : 16	tph_req_st_table_lower_31	rw	0x0	
31 : 24	tph_req_st_table_higher_31	rw	0x0	

64.60 ats_cap_hdr				0x40700218
Bits	Field Name	Acc	Default	Description
15 : 0	ats_cap_id	rw	0xF	
19 : 16	ats_cap_version	rw	0x1	
31 : 20	ats_cap_next_ptr	rw	0x228	

64.61 ats_cap_ctrl				0x4070021C
Bits	Field Name	Acc	Default	Description
4 : 0	ats_invalidate_q_depth	rw	0x0	

5	ats_page_aligned_req	rw	0x1	
6	ats_global_inval_spprtd	rw	0x0	
20 : 16	ats_stu	rw	0x0	
31	ats_enable	rw	0x0	

64.62 l1sub_cap_header				0x40700228
Bits	Field Name	Acc	Default	Description
15 : 0	l1sub_cap_id	rw	0x1E	
19 : 16	l1sub_cap_version	rw	0x1	
31 : 20	l1sub_cap_next_ptr	rw	0x0	

64.63 l1sub_cap				0x4070022C
Bits	Field Name	Acc	Default	Description
0	l1_2_pcipm_support	rw	0x1	
1	l1_1_pcipm_support	rw	0x1	
2	l1_2_aspm_support	rw	0x0	
3	l1_1_aspm_support	rw	0x1	
4	l1_pmsub_support	rw	0x1	
15 : 8	comm_mode_support	rw	0xA	
17 : 16	pwr_on_scale_support	rw	0x0	
23 : 19	pwr_on_value_support	rw	0x5	

64.64 l1sub_ctrl1				0x40700230
Bits	Field Name	Acc	Default	Description
0	l1_2_pcipm_en	rw	0x0	
1	l1_1_pcipm_en	rw	0x0	
2	l1_2_aspm_en	rw	0x0	
3	l1_1_aspm_en	rw	0x0	
15 : 8	t_common_mode	ro	0x0	
25 : 16	l1_2_th_val	rw	0x0	
31 : 29	l1_2_th_sca	rw	0x0	

64.65 l1sub_ctrl2				0x40700234
Bits	Field Name	Acc	Default	Description
1 : 0	t_power_on_scale	rw	0x0	
7 : 3	t_power_on_value	rw	0x5	

64.66 ack_latency_timer				0x40700700
Bits	Field Name	Acc	Default	Description
15 : 0	round_trip_latency_time_limit	rw	0x102F	
31 : 16	replay_time_limit	rw	0x308D	

64.67 vendor_spec_dllp				0x40700704
Bits	Field Name	Acc	Default	Description
31 : 0	vendor_spec_dllp	rw	0xFFFFFFFF	

64.68 port_force				0x40700708
Bits	Field Name	Acc	Default	Description
7 : 0	link_num	rw	0x4	
11 : 8	forced_ltssm	rw	0x0	
15	force_en	rw	0x0	write 1 clear

21 : 16	link_state	rw	0x0	
31 : 24	cpl_sent_count	rw	0x7	

64.69 ack_f_aspm_ctrl				0x4070070C
Bits	Field Name	Acc	Default	Description
7 : 0	ack_freq	rw	0x0	
15 : 8	ack_n_fts	rw	0xF	
23 : 16	common_clk_n_fts	ro	0xF	
26 : 24	l0s_entrance_latency	rw	0x3	
29 : 27	l1_entrance_latency	rw	0x3	
30	enter_aspm	rw	0x0	

64.70 port_link_ctrl				0x40700710
Bits	Field Name	Acc	Default	Description
0	vendor_sprc_dllp_req	rw	0x0	
1	scramble_disable	rw	0x0	
2	loopback_enable	rw	0x0	
3	reset_assert	rw	0x0	
5	dll_link_en	rw	0x1	
6	link_disable	rw	0x0	
7	fast_link_mode	rw	0x0	
11 : 8	link_rate	rw	0x1	
21 : 16	link_capable	rw	0x1	
22	cross_link_en	rw	0x1	
23	cross_link_active	ro	0x0	
24	beacon_enable	rw	0x0	
25	corrupt_lcrc_enable	rw	0x0	
26	extended_synch	rw	0x0	
27	tx_lane_reverse_enable	rw	0x0	

64.71 lane_skew				0x40700714
Bits	Field Name	Acc	Default	Description
23 : 0	insert_lane_skew	rw	0x0	
24	flow_ctrl_disable	rw	0x0	
25	ack_nak_disable	rw	0x0	
31	disable_lane_to_lane_deskew	rw	0x0	

64.72 timer_ctrl_max_func_num				0x40700718
Bits	Field Name	Acc	Default	Description
7 : 0	max_func_num	rw	0x0	
18 : 14	timer_mod_replay_timer	rw	0x1	
23 : 19	timer_mod_ack_nak	rw	0x0	
28 : 24	update_freq_timer	rw	0x0	

64.73 symbol_timer_filter_1				0x4070071C
Bits	Field Name	Acc	Default	Description
10 : 0	skp_int_val	rw	0x500	
14 : 11	eidle_timer	rw	0x0	
15	disable_fc_wd_timer	rw	0x0	
31 : 16	mask_radm_1	rw	0x0	

64.74 filter_mask_2	0x40700720

Bits	Field Name	Acc	Default	Description
31 : 0	mask_radm_2	rw	0x0	

64.75 amba_mul_ob_decomp_np_sub_req_ctrl				0x40700724
Bits	Field Name	Acc	Default	Description
1 : 0	ib_ob_rd_split_burst_en	rw	0x1	

64.76 pl_debug0				0x40700728
Bits	Field Name	Acc	Default	Description
31 : 0	deb_reg_0	ro	0x0	

64.77 pl_debug1				0x4070072C
Bits	Field Name	Acc	Default	Description
31 : 0	deb_reg_1	ro	0x0	

64.78 tx_p_fc_credit_status				0x40700730
Bits	Field Name	Acc	Default	Description
11 : 0	tx_p_data_fc_credit	ro	0x0	
19 : 12	tx_p_header_fc_credit	ro	0x0	

64.79 tx_np_fc_credit_status				0x40700734
Bits	Field Name	Acc	Default	Description
11 : 0	tx_np_data_fc_credit	ro	0x0	
19 : 12	tx_np_header_fc_credit	ro	0x0	

64.80 tx_cpl_fc_credit_status				0x40700738
Bits	Field Name	Acc	Default	Description
11 : 0	tx_cpl_data_fc_credit	ro	0x0	
19 : 12	tx_cpl_header_fc_credit	ro	0x0	

64.81 queue_status				0x4070073C
Bits	Field Name	Acc	Default	Description
0	rx_tlp_fc_credit_non_return	ro	0x0	
1	tx_retry_buffer_ne	ro	0x0	
2	rx_queue_non_empty	ro	0x0	
28 : 16	timer_mod_flow_ctrl	rw	0x0	
31	timer_moid_flow_ctrl_en	ro	0x0	

64.82 vc_tx_arbl_1				0x40700740
Bits	Field Name	Acc	Default	Description
7 : 0	wrr_weight_vc_0	ro	0xF	
15 : 8	wrr_weight_vc_1	ro	0x0	
23 : 16	wrr_weight_vc_2	ro	0x0	
31 : 24	wrr_weight_vc_3	ro	0x0	

64.83 vc_tx_arbl_2				0x40700744
Bits	Field Name	Acc	Default	Description
7 : 0	wrr_weight_vc_4	ro	0x0	
15 : 8	wrr_weight_vc_5	ro	0x0	

23 : 16	wrr_weight_vc_6	ro	0x0	
31 : 24	wrr_weight_vc_7	ro	0x0	

64.84 vc0_p_rx_1_ctrl				0x40700748
Bits	Field Name	Acc	Default	Description
11 : 0	vc0_p_data_credit	ro	0x60	
19 : 12	vc0_p_header_credit	ro	0x1E	
23 : 21	vc0_p_tlp_q_mode	rw	0x1	
30	tlp_type_ordering_vc0	rw	0x1	
31	vc_ordering_rx_q	rw	0x0	

64.85 vc0_np_rx_q_ctrl				0x4070074C
Bits	Field Name	Acc	Default	Description
11 : 0	vc0_np_data_credit	ro	0x7	
19 : 12	vc0_np_header_credit	ro	0x1E	
23 : 21	vc0_np_tlp_q_mode	rw	0x1	

64.86 vc0_cpl_rx_q_ctrl				0x40700750
Bits	Field Name	Acc	Default	Description
11 : 0	vc0_cpl_data_credit	ro	0x0	
19 : 12	vc0_cpl_header_credit	ro	0x0	
23 : 21	vc0_cpl_tlp_q_mode	rw	0x4	

64.87 gen2_ctrl				0x4070080C
Bits	Field Name	Acc	Default	Description
12 : 8	num_of_lanes	rw	0x1	
15 : 13	pre_det_lane	rw	0x0	
16	auto_lane_flip_ctrl_en	rw	0x0	
21	gen1_ei_intf	rw	0x0	

64.88 phy_status				0x40700810
Bits	Field Name	Acc	Default	Description
31 : 0	phy_status	ro	0x0	

64.89 phy_ctrl				0x40700814
Bits	Field Name	Acc	Default	Description
31 : 0	phy_ctrl	rw	0x0	

64.90 pipe_loopback_ctrl				0x407008B8
Bits	Field Name	Acc	Default	Description
15 : 0	lpbk_rxvalid	rw	0x1	
21 : 16	rxstatus_lane	rw	0x0	
26 : 24	rxstatus_value	wo	0x0	
31	pipe_loopback	rw	0x0	

64.91 misc_ctrl_1				0x407008BC
Bits	Field Name	Acc	Default	Description
0	dbi_ro_wr_en	rw	0x0	

64.92 amba_error_resp_default				0x407008D0
Bits	Field Name	Acc	Default	Description
0	amba_err_resp_global	rw	0x0	
2	amba_err_resp_vendorid	rw	0x0	
3	amba_err_resp_flush	rw	0x0	

64.93 amba_ordrmgr_wdog				0x407008DC
Bits	Field Name	Acc	Default	Description
15 : 0	amba_ordrmgr_wdog	rw	0xFFF	
31	amba_ordrmgr_wdog_en	rw	0x1	

64.94 pl_last				0x407008FC
Bits	Field Name	Acc	Default	Description
0	pl_last	ro	0x0	

64.95 iatu_viewport				0x40700900
Bits	Field Name	Acc	Default	Description
2 : 0	iatu_region_index	rw	0x0	
31	iatu_region_dir	rw	0x0	0: outbound, 1: inbound

64.96 iatu_region_ctrl_1				0x40700904
Bits	Field Name	Acc	Default	Description
4 : 0	iatu_type	rw	0x0	
7 : 5	iatu_tc	rw	0x0	
8	iatu_td	rw	0x0	
10 : 9	iatu_attr	rw	0x0	
11	iatu_ido	rw	0x0	
12	iatu_th	rw	0x0	
17 : 16	iatu_at	rw	0x0	
19 : 18	iatu_ph	rw	0x0	
24 : 20	iatu_ctrl_1_func_num	rw	0x0	

64.97 iatu_region_ctrl_2				0x40700908
Bits	Field Name	Acc	Default	Description
7 : 0	iatu_msg_code	rw	0x0	
10 : 8	iatu_inbound_bar_num	rw	0x0	Only in inbound region.
14	iatu_inbound_tc_match_en	rw	0x0	Only in inbound region.
15	iatu_inbound_td_match_en	rw	0x0	Only in inbound region.
16	iatu_inbound_attr_match_en	rw	0x0	Only in inbound region.
17	iatu_inbound_th_match_en	rw	0x0	Only in inbound region.
18	iatu_inbound_at_match_en	rw	0x0	Only in inbound region.
19	iatu_func_bypass_func_num_match_en	rw	0x0	In outbound: func_bypass In inbound: func_num_match_en
20	iatu_inbound_vf_match_en	rw	0x0	Only in inbound region.
21	iatu_inbound_msg_code_match_en	rw	0x0	Only in inbound region.
22	iatu_inbound_ph_match_en	rw	0x0	Only in inbound region.
25 : 24	iatu_inbound_resp_code	rw	0x0	Only in inbound region.
27	iatu_dma_bypass_fuzzy_type_match_mode	rw	0x0	In outbound: dma_bypass In inbound: fuzzy_type_match_mode
28	iatu_cf_shift_mode	rw	0x0	
29	iatu_invert_mode	rw	0x0	
30	iatu_inbound_match_mode	rw	0x0	Only in inbound region. 0: address match mode, 1: bar match mode

31	iatu_region_en	rw	0x0	

64.98 iatu_lwr_base_addr				0x4070090C
Bits	Field Name	Acc	Default	Description
11 : 0	iatu_lwr_base_hw	ro	0x0	
31 : 12	iatu_lwr_base_rw	rw	0x0	

64.99 iatu_upper_base_addr				0x40700910
Bits	Field Name	Acc	Default	Description
31 : 0	iatu_upper_base_rw	rw	0x0	

64.100 iatu_limit_addr				0x40700914
Bits	Field Name	Acc	Default	Description
11 : 0	iatu_limit_addr_hw	ro	0xFFF	
31 : 12	iatu_limit_addr_rw	rw	0x0	

64.101 iatu_lwr_target_addr				0x40700918
Bits	Field Name	Acc	Default	Description
11 : 0	iatu_lwr_target_hw	ro	0x0	
31 : 12	iatu_lwr_target_rw	rw	0x0	

64.102 iatu_upper_target_addr				0x4070091C
Bits	Field Name	Acc	Default	Description
31 : 0	iatu_upper_target_rw	rw	0x0	

64.103 dma_ctrl				0x40700978
Bits	Field Name	Acc	Default	Description
3 : 0	edma_wr_ch_num	rw	0x2	
19 : 16	edma_rd_ch_num	rw	0x2	

64.104 dma_wr_engine_en				0x4070097C
Bits	Field Name	Acc	Default	Description
0	edma_wr_engine_en	rw	0x0	

64.105 dma_wr_doorbell				0x40700980
Bits	Field Name	Acc	Default	Description
2 : 0	edma_wr_doorbell_num	rw	0x0	
31	edma_wr_stop	rw	0x0	

64.106 dma_wr_ch_arb_weight_low				0x40700988
Bits	Field Name	Acc	Default	Description
4 : 0	edma_wr_ch_0_weight	rw	0x1	
9 : 5	edma_wr_ch_1_weight	rw	0x1	
14 : 10	edma_wr_ch_2_weight	rw	0x1	
19 : 15	edma_wr_ch_3_weight	rw	0x1	

64.107 dma_wr_ch_arb_weight_high				0x4070098C
Bits	Field Name	Acc	Default	Description

4 : 0	edma_wr_ch_4_weight	rw	0x1	
9 : 5	edma_wr_ch_5_weight	rw	0x1	
14 : 10	edma_wr_ch_6_weight	rw	0x1	
19 : 15	edma_wr_ch_7_weight	rw	0x1	

64.108 dma_wr_p_req_timer				0x40700998
Bits	Field Name	Acc	Default	Description
11 : 0	edma_wr_p_timer_limit	rw	0xE	
15	edma_wr_p_timer_disable	rw	0x0	

64.109 dma_rd_engine_en				0x4070099C
Bits	Field Name	Acc	Default	Description
0	edma_rd_engine_en	rw	0x0	

64.110 dma_rd_doorbell				0x407009A0
Bits	Field Name	Acc	Default	Description
2 : 0	edma_rd_doorbell_num	rw	0x0	
31	edma_rd_stop	rw	0x0	

64.111 dma_rd_ch_arb_weight_low				0x407009A8
Bits	Field Name	Acc	Default	Description
4 : 0	edma_rd_ch_0_weight	rw	0x1	
9 : 5	edma_rd_ch_1_weight	rw	0x1	
14 : 10	edma_rd_ch_2_weight	rw	0x1	
19 : 15	edma_rd_ch_3_weight	rw	0x1	

64.112 dma_rd_ch_arb_weight_high				0x407009AC
Bits	Field Name	Acc	Default	Description
4 : 0	edma_rd_ch_4_weight	rw	0x1	
9 : 5	edma_rd_ch_5_weight	rw	0x1	
14 : 10	edma_rd_ch_6_weight	rw	0x1	
19 : 15	edma_rd_ch_7_weight	rw	0x1	

64.113 dma_wr_int_status				0x407009BC
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_done_int_status	rw	0x0	
23 : 16	edma_wr_abort_int_status	rw	0x0	

64.114 dma_wr_int_mask				0x407009C4
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_done_int_mask	rw	0xFF	
23 : 16	edma_wr_abort_int_mask	rw	0xFF	

64.115 dma_wr_int_clr				0x407009C8
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_done_int_clr	rw	0x0	write 1 clear
23 : 16	edma_wr_abort_int_clr	rw	0x0	write 1 clear

64.116 dma_wr_err_status	0x407009CC

Bits	Field Name	Acc	Default	Description
7 : 0	edma_app_rd_err_det	ro	0x0	
23 : 16	edma_wr_linkl-ist_elem_fetch_err_det	ro	0x0	

64.117 dma_wr_done_imwr_low				0x407009D0
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_done_low_reg	rw	0x0	

64.118 dma_wr_done_imwr_high				0x407009D4
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_done_high_reg	rw	0x0	

64.119 dma_wr_abort_imwr_low				0x407009D8
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_abort_low_reg	rw	0x0	

64.120 dma_wr_abort_imwr_high				0x407009DC
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_abort_high_reg	rw	0x0	

64.121 dma_wr_ch01_imwr_data				0x407009E0
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_0_data	rw	0x0	
31 : 16	edma_wr_ch_1_data	rw	0x0	

64.122 dma_wr_ch23_imwr_data				0x407009E4
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_2_data	rw	0x0	
31 : 16	edma_wr_ch_3_data	rw	0x0	

64.123 dma_wr_ch45_imwr_data				0x407009E8
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_4_data	rw	0x0	
31 : 16	edma_wr_ch_5_data	rw	0x0	

64.124 dma_wr_ch67_imwr_data				0x407009EC
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_6_data	rw	0x0	
31 : 16	edma_wr_ch_7_data	rw	0x0	

64.125 dma_wr_linked_list_err_en				0x40700A00
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_ch_llraie	rw	0x0	
23 : 16	edma_wr_ch_lllaie	rw	0x0	

64.126 dma_rd_int_status				0x40700A10
Bits	Field Name	Acc	Default	Description

7 : 0	edma_rd_done_int_status	rw	0x0	
23 : 16	edma_rd_abort_int_status	rw	0x0	

64.127 dma_rd_int_mask				0x40700A18
Bits	Field Name	Acc	Default	Description
7 : 0	edma_rd_done_int_mask	rw	0xFF	
23 : 16	edma_rd_abort_int_mask	rw	0xFF	

64.128 dma_rd_int_clr				0x40700A1C
Bits	Field Name	Acc	Default	Description
7 : 0	edma_rd_done_int_clr	rw	0x0	
23 : 16	edma_rd_abort_int_clr	rw	0x0	

64.129 dma_rdd_err_status_low				0x40700A24
Bits	Field Name	Acc	Default	Description
7 : 0	edma_app_wr_err_det	ro	0x0	
23 : 16	edma_rd_linkl-ist_elem_fetch_err_det	ro	0x0	

64.130 dma_rd_err_status_high				0x40700A28
Bits	Field Name	Acc	Default	Description
7 : 0	edma_unsupported_req	ro	0x0	
15 : 8	edma_cpl_abort	ro	0x0	
23 : 16	edma_cpl_timeout	ro	0x0	
31 : 24	edma_data_poisioning	ro	0x0	

64.131 dma_rd_linked_list_err_en				0x40700A34
Bits	Field Name	Acc	Default	Description
7 : 0	edma_rd_ch_llraie	rw	0x0	
23 : 16	edma_rd_ch_lllaie	rw	0x0	

64.132 dma_rd_done_imwr_low				0x40700A3C
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_done_low_reg	rw	0x0	

64.133 dma_rd_done_imwr_high				0x40700A40
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_done_high_reg	rw	0x0	

64.134 dma_rd_abort_imwr_low				0x40700A44
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_abort_low_reg	rw	0x0	

64.135 dma_rd_abort_imwr_high				0x40700A48
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_abort_high_reg	rw	0x0	

64.136 dma_rd_ch01_imwr_data	0x40700A4C

Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_0_data	rw	0x0	
31 : 16	edma_rd_ch_1_data	rw	0x0	

64.137 dma_rd_ch23_imwr_data				0x40700A50
Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_2_data	rw	0x0	
31 : 16	edma_rd_ch_3_data	rw	0x0	

64.138 dma_rd_ch45_imwr_data				0x40700A54
Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_4_data	rw	0x0	
31 : 16	edma_rd_ch_5_data	rw	0x0	

64.139 dma_rd_ch67_imwr_data				0x40700A58
Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_6_data	rw	0x0	
31 : 16	edma_rd_ch_7_data	rw	0x0	

64.140 dma_viewport_sel				0x40700A6C
Bits	Field Name	Acc	Default	Description
2 : 0	edma_ch_num	rw	0x0	
31	edma_ch_dir	rw	0x0	0: write channel 1: read channel

64.141 dma_ch_ctrl1				0x40700A70
Bits	Field Name	Acc	Default	Description
0	edma_cb	rw	0x0	
1	edma_tcb	rw	0x0	
2	edma_llp	rw	0x0	
3	edma_lie	rw	0x0	
4	edma_rie	rw	0x0	
6 : 5	edma_cs	ro	0x0	
8	edma_ccs	rw	0x0	
9	edma_lle	rw	0x0	
16 : 12	edma_func_num	rw	0x0	
24	edma_ns	rw	0x0	
25	edma_ro	rw	0x0	
26	edma_td	rw	0x0	
29 : 27	edma_tc	rw	0x0	
31 : 30	edma_at	rw	0x0	

64.142 dma_ch_ctrl2				0x40700A74
Bits	Field Name	Acc	Default	Description
24 : 17	edma_st	rw	0x0	
30 : 29	edma_ph	rw	0x0	
31	edma_th	rw	0x0	

64.143 dma_trans_size				0x40700A78
Bits	Field Name	Acc	Default	Description
31 : 0	edma_trans_size	rw	0x0	

64.144 dma_sar_low				0x40700A7C
Bits	Field Name	Acc	Default	Description
31 : 0	edma_src_addr_low	rw	0x0	

64.145 dma_sar_high				0x40700A80
Bits	Field Name	Acc	Default	Description
31 : 0	edma_src_addr_high	rw	0x0	

64.146 dma_dar_low				0x40700A84
Bits	Field Name	Acc	Default	Description
31 : 0	edma_dst_addr_low	rw	0x0	

64.147 dma_dar_high				0x40700A88
Bits	Field Name	Acc	Default	Description
31 : 0	edma_dst_addr_high	rw	0x0	

64.148 dma_llp_low				0x40700A8C
Bits	Field Name	Acc	Default	Description
31 : 0	edma_llp_low	rw	0x0	

64.149 dma_llp_high				0x40700A90
Bits	Field Name	Acc	Default	Description
31 : 0	edma_llp_high	rw	0x0	

64.150 aux_clk_freq				0x40700B40
Bits	Field Name	Acc	Default	Description
9 : 0	aux_clk_freq	rw	0xA	

64.151 l1_substates				0x40700B44
Bits	Field Name	Acc	Default	Description
1 : 0	l1sub_t_power_off	rw	0x2	
5 : 2	l1sub_t_l1_2	rw	0x4	
7 : 6	l1sub_t_pclkack	rw	0x3	

65 pcie_ep_dbi2

65.1 dev_id_vendor_id				0x40701000
Bits	Field Name	Acc	Default	Description
15 : 0	vendor_id	rw	0x16C3	
31 : 16	device_id	rw	0xABCD	

65.2 status_cmd				0x40701004
Bits	Field Name	Acc	Default	Description
0	io_en	rw	0x0	
1	mem_space_en	rw	0x0	
2	bus_master_en	rw	0x0	
3	special_cycle_operation	ro	0x0	
4	mwi_enable	ro	0x0	
5	vga_palette_snoop	ro	0x0	

6	parity_err_en	rw	0x0	
7	idsel_stepping	ro	0x0	
8	serren	rw	0x0	
10	int_en	rw	0x0	
16 : 11	reserv	ro	0x0	
19	int_status	ro	0x0	
20	cap_list	ro	0x1	
21	fast_66mhz_cap	ro	0x0	
23	fast_b2b_cap	ro	0x0	
24	master_dpe	rw	0x0	Write 1 Clear
26 : 25	dev_sel_timing	ro	0x0	
27	signaled_target_abort	rw	0x0	Write 1 Clear
28	rcvd_target_abort	rw	0x0	Write 1 Clear
29	rcvd_master_abort	rw	0x0	Write 1 Clear
30	signaled_sys_err	rw	0x0	Write 1 Clear
31	detected_parity_err	rw	0x0	Write 1 Clear

65.3 class_code_rev_id				0x40701008
Bits	Field Name	Acc	Default	Description
7 : 0	rev_id	rw	0x1	
15 : 8	program_intf	rw	0x0	
23 : 16	subclass_code	rw	0x0	
31 : 24	base_class_code	rw	0x0	

65.4 bist_header_type_latency_cache_line_size				0x4070100C
Bits	Field Name	Acc	Default	Description
7 : 0	cache_line_size	rw	0x0	
15 : 8	latency_master_timer	ro	0x0	
22 : 16	header_type	ro	0x0	
23	multi_func	rw	0x0	
31 : 24	bist	ro	0x0	

65.5 bar0				0x40701010
Bits	Field Name	Acc	Default	Description
0	bar0_mem_io	rw	0x0	
2 : 1	bar0_type	rw	0x0	
3	bar0_prefetch	rw	0x0	
31 : 4	bar0_start	rw	0x0	

65.6 bar1				0x40701014
Bits	Field Name	Acc	Default	Description
0	bar1_mem_io	rw	0x0	
2 : 1	bar1_type	rw	0x0	
3	bar1_prefetch	rw	0x0	
31 : 4	bar1_start	rw	0x0	

65.7 bar2				0x40701018
Bits	Field Name	Acc	Default	Description
0	bar2_mem_io	rw	0x0	
2 : 1	bar2_type	rw	0x0	
3	bar2_prefetch	rw	0x0	
31 : 4	bar2_start	rw	0x0	

65.8 bar3				0x4070101C
Bits	Field Name	Acc	Default	Description
0	bar3_mem_io	rw	0x0	
2 : 1	bar3_type	rw	0x0	
3	bar3_prefetch	rw	0x0	
31 : 4	bar3_start	rw	0x0	

65.9 cardbus_cis_ptr				0x40701028
Bits	Field Name	Acc	Default	Description
31 : 0	cardbus_cis_ptr	rw	0x0	

65.10 subsys_id_subsys_vendor_id				0x4070102C
Bits	Field Name	Acc	Default	Description
15 : 0	subsys_vendor_id	rw	0x0	
31 : 16	subsys_dev_id	rw	0x0	

65.11 exp_rom_base_addr				0x40701030
Bits	Field Name	Acc	Default	Description
0	rom_bar_enable	rw	0x1	
31 : 11	exp_rom_base_addr	rw	0x0	

65.12 pci_cap_ptr				0x40701034
Bits	Field Name	Acc	Default	Description
7 : 0	cap_ptr	rw	0x40	

65.13 intr_line_intr_pin				0x4070103C
Bits	Field Name	Acc	Default	Description
7 : 0	int_line	rw	0xFF	
15 : 8	int_pin	rw	0x1	

65.14 pm_cap				0x40701040
Bits	Field Name	Acc	Default	Description
7 : 0	pm_cap_id	ro	0x1	
15 : 8	pm_next_ptr	rw	0x50	
18 : 16	pm_spec_ver	rw	0x3	
19	pme_clk	ro	0x0	
21	dsi	rw	0x0	
24 : 22	aux_curr	rw	0x7	
25	d1_support	rw	0x1	
26	d2_support	rw	0x0	
31 : 27	pme_support	rw	0x1B	

65.15 pm_ctrl_status				0x40701044
Bits	Field Name	Acc	Default	Description
1 : 0	power_state	rw	0x0	
3	no_soft_rst	rw	0x0	
8	pme_enable	rw	0x0	
12 : 9	data_select	ro	0x0	
14 : 13	data_scale	ro	0x0	
15	pme_status	rw	0x0	Write 1 Clear
22	b2_b3_support	ro	0x0	

23	bus_pwr_clk_con_en	ro	0x0	
31 : 24	data_reg_add_info	ro	0x0	

65.16 msi_cap				0x40701050
Bits	Field Name	Acc	Default	Description
7 : 0	msi_cap_id	ro	0x5	
15 : 8	msi_next_ptr	rw	0x70	
16	msi_enable	rw	0x0	
19 : 17	msi_multi_msg_cap	rw	0x0	
22 : 20	msi_multi_msg_en	rw	0x0	
23	msi_64_bit_addr_cap	rw	0x1	
24	pvm_support	ro	0x0	

65.17 msi_lower_addr				0x40701054
Bits	Field Name	Acc	Default	Description
31 : 2	msi_lower_addr	rw	0x0	

65.18 msi_upper_addr				0x40701058
Bits	Field Name	Acc	Default	Description
31 : 0	msi_upper_addr	rw	0x0	

65.19 msi_data				0x4070105C
Bits	Field Name	Acc	Default	Description
15 : 0	msi_data	rw	0x0	

65.20 pcie_cap				0x40701070
Bits	Field Name	Acc	Default	Description
7 : 0	pcie_cap_id	ro	0x10	
15 : 8	pcie_cap_next_ptr	rw	0x0	
19 : 16	pcie_cap	rw	0x2	
23 : 20	dev_port_type	ro	0x0	
24	slot_imp	rw	0x0	
29 : 25	int_msg_num	rw	0x0	

65.21 dev_cap				0x40701074
Bits	Field Name	Acc	Default	Description
2 : 0	max_payload_size	rw	0x2	
4 : 3	phantom_func_support	rw	0x0	
5	ext_tag_supp	rw	0x0	
8 : 6	ep_l0s_accpt_latency	rw	0x7	
11 : 9	ep_l1_accpt_latency	rw	0x7	
15	role_based_err_report	rw	0x1	
28	flr_cap	rw	0x0	

65.22 dev_ctrl_status				0x40701078
Bits	Field Name	Acc	Default	Description
0	corr_err_report_en	rw	0x0	
1	non_fatal_err_report_en	rw	0x0	
2	fatal_err_report_en	rw	0x0	
3	unsupported_req_rep_en	rw	0x0	
4	en_rel_order	rw	0x1	

7 : 5	max_payload_size_cs	rw	0x0	
8	ext_tag_en	ro	0x0	
9	phantom_func_en	ro	0x0	
10	aux_power_pm_en	rw	0x0	
11	en_no_snoop	ro	0x0	
14 : 12	max_read_req_size	rw	0x2	
15	initiate_flr	rw	0x0	
16	corr_err_detected	rw	0x0	Write 1 clear
17	non_fatal_err_detected	rw	0x0	Write 1 clear
18	fatal_err_detected	rw	0x0	Write 1 clear
19	unsupported_req_detected	rw	0x0	Write 1 clear
20	aux_power_detected	ro	0x1	
21	trans_pending	ro	0x0	

65.23 link_cap 0x4070107C

Bits	Field Name	Acc	Default	Description
3 : 0	max_link_speed	rw	0x1	
9 : 4	max_link_width	rw	0x1	
11 : 10	active_state_link_pm_support	rw	0x3	
14 : 12	l0s_exit_latency	rw	0x3	
17 : 15	l1_exit_latency	rw	0x6	
18	clock_power_man	ro	0x0	
19	surprise_down_err_rep_cap	ro	0x0	
20	dll_active_rep_cap	ro	0x0	
21	link_bw_not_cap	ro	0x0	
22	aspm_opt_compliance	rw	0x1	
31 : 24	port_num	rw	0x0	

65.24 link_ctrl_status 0x40701080

Bits	Field Name	Acc	Default	Description
1 : 0	active_state_link_pm_ctrl	rw	0x0	
3	rcb	rw	0x0	
4	link_disable	rw	0x0	
5	retrain_link	rw	0x0	
6	common_clk_config	rw	0x0	
7	extended_synch	rw	0x0	
8	en_clk_power_man	rw	0x0	
10	link_bw_man_int_en	rw	0x0	
11	link_auto_bw_int_en	ro	0x0	
19 : 16	link_speed	ro	0x1	
24 : 20	nego_link_width	ro	0x1	
27	cap_link_training	ro	0x0	
28	slot_clk_config	rw	0x1	
29	dll_active	ro	0x0	
30	link_bw_man_status	ro	0x0	
31	link_auto_bw_status	ro	0x0	

65.25 dev_cap2 0x40701094

Bits	Field Name	Acc	Default	Description
3 : 0	cpl_timeout_range	ro	0xF	
4	cpl_timeout_disable_support	ro	0x1	
5	ari_forward_support	ro	0x0	
6	atomic_routing_supp	ro	0x0	
7	atomic_32_cpl_supp	ro	0x0	
8	atomic_64_cpl_supp	ro	0x0	

9	cas_128_cpl_supp	ro	0x0	
10	no_ro_en_pr2pr_par	ro	0x0	
11	ltr_supp	ro	0x0	
12	tph_cmplt_support_0	ro	0x1	
13	tph_cmplt_support_1	ro	0x0	
19 : 18	obff_support	ro	0x0	
20	cfg_extnd_fmt_support	rw	0x0	
21	cfg_end2end_tlp_prfx_support	rw	0x0	
23 : 22	cfg_max_end2end_tlp_prfxs	rw	0x0	

65.26 dev_ctrl_status2				0x40701098
Bits	Field Name	Acc	Default	Description
3 : 0	cpl_timeout_value	rw	0x0	
4	cpl_timeout_disable	rw	0x0	
5	ari_forward_support_cs	ro	0x0	
8	ido_req_en	rw	0x0	
9	ido_cpl_en	rw	0x0	
10	ltr_en	rw	0x0	
14 : 13	obff_en	rw	0x0	
15	cfg_end2end_tlp_pfx_blck	ro	0x0	

65.27 link_cap2				0x4070109C
Bits	Field Name	Acc	Default	Description
7 : 1	support_link_speed_vector	ro	0x1	
8	cross_link_support	ro	0x0	

65.28 link_ctrl_status2				0x407010A0
Bits	Field Name	Acc	Default	Description
3 : 0	target_link_speed	rw	0x1	
4	enter_compliance	rw	0x0	
5	hw_auto_speed_disable	rw	0x0	
6	sel_deemphasis	ro	0x0	
9 : 7	tx_margin	rw	0x0	
10	enter_modified_compliance	rw	0x0	
11	compliance_sos	rw	0x0	
15 : 12	compliance_preset	rw	0x0	
16	curr_deemphasis	ro	0x1	
17	eq_cpl	ro	0x0	
18	eq_cpl_p1	ro	0x0	
19	eq_cpl_p2	rw	0x0	
20	eq_cpl_p3	rw	0x0	
21	link_eq_req	rw	0x0	write 1 clear

65.29 aer_ext_cap_hdr				0x40701100
Bits	Field Name	Acc	Default	Description
15 : 0	aer_cap_id	rw	0x1	
19 : 16	aer_cap_version	rw	0x2	
31 : 20	aer_cap_next	rw	0x148	

65.30 uncorr_err_status				0x40701104
Bits	Field Name	Acc	Default	Description
4	dl_protocol_err_status	rw	0x0	write 1 clear
5	sur_dwn_err_status	ro	0x0	

12	pois_tlp_err_status	rw	0x0	write 1 clear
13	fc_protocol_err_status	rw	0x0	write 1 clear
14	cmplt_timeout_err_status	rw	0x0	write 1 clear
15	cmplt_abort_err_status	rw	0x0	write 1 clear
16	unexp_cmplt_err_status	rw	0x0	write 1 clear
17	rec_overflow_err_status	rw	0x0	write 1 clear
18	malf_tlp_err_status	rw	0x0	write 1 clear
19	ecrc_err_status	rw	0x0	write 1 clear
20	unsupported_req_err_status	rw	0x0	write 1 clear
22	internal_err_status	rw	0x0	write 1 clear
25	tlp_prfx_blocked_err_status	rw	0x0	write 1 clear

65.31 uncorr_err_mask				0x40701108
Bits	Field Name	Acc	Default	Description
4	dl_protocol_err_mask	rw	0x0	
5	sur_dwn_err_mask	ro	0x0	
12	pois_tlp_err_mask	rw	0x0	
13	fc_protocol_err_mask	rw	0x0	
14	cmplt_timeout_err_mask	rw	0x0	
15	cmplt_abort_err_mask	rw	0x0	
16	unexp_cmplt_err_mask	rw	0x0	
17	rec_overflow_err_mask	rw	0x0	
18	malf_tlp_err_mask	rw	0x0	
19	ecrc_err_mask	rw	0x0	
20	unsupported_req_err_mask	rw	0x0	
22	internal_err_mask	rw	0x1	
24	atomic_egress_-blocked_err_mask	ro	0x0	
25	tlp_prfx_blocked_err_mask	rw	0x0	

65.32 uncorr_err_sev				0x4070110C
Bits	Field Name	Acc	Default	Description
4	dl_protocol_err_severity	rw	0x1	
5	sur_dwn_err_severity	ro	0x1	
12	pois_tlp_err_severity	rw	0x0	
13	fc_protocol_err_severity	rw	0x1	
14	cmplt_timeout_err_severity	rw	0x0	
15	cmplt_abort_err_severity	rw	0x0	
16	unexp_cmplt_err_severity	rw	0x0	
17	rec_overflow_err_severity	rw	0x0	
18	malf_tlp_err_severity	rw	0x1	
19	ecrc_err_severity	rw	0x0	
20	unsupported_req_err_severity	rw	0x0	
22	internal_err_severity	rw	0x1	
24	atomic_egress_-blocked_err_severity	ro	0x0	
25	tlp_prfx_blocked_err_severity	rw	0x0	

65.33 corr_err_status				0x40701110
Bits	Field Name	Acc	Default	Description
0	rx_err_status	rw	0x0	write 1 clear
6	bad_tlp_status	rw	0x0	write 1 clear
7	bad_dllp_status	rw	0x0	write 1 clear
8	replay_no_roleover_status	rw	0x0	write 1 clear
12	rpl_timer_timeout_status	rw	0x0	write 1 clear

13	advisory_non_fatal_err_status	rw	0x0	write 1 clear
14	corrected_int_err_status	rw	0x0	write 1 clear

65.34 corr_err_mask				0x40701114
Bits	Field Name	Acc	Default	Description
0	rx_err_mask	rw	0x0	
6	bad_tlp_mask	rw	0x0	
7	bad_dllp_mask	rw	0x0	
8	replay_no_roleover_mask	rw	0x0	
12	rpl_timer_timeout_mask	rw	0x0	
13	advisory_non_fatal_err_mask	rw	0x1	
14	corrected_int_err_mask	rw	0x1	

65.35 adv_err_cap_ctrl				0x40701118
Bits	Field Name	Acc	Default	Description
4 : 0	first_err_ptr	ro	0x0	
5	ecrc_gen_cap	ro	0x1	
6	ecrc_gen_en	rw	0x0	
7	ecrc_check_cap	ro	0x1	
8	ecrc_check_en	rw	0x0	
11	tlp_prfx_log_present	ro	0x0	

65.36 hdr_log_0				0x4070111C
Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_first_dword	ro	0x0	

65.37 hdr_log_1				0x40701120
Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_second_dword	ro	0x0	

65.38 hdr_log_2				0x40701124
Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_third_dword	ro	0x0	

65.39 hdr_log_3				0x40701128
Bits	Field Name	Acc	Default	Description
31 : 0	hdr_log_fourth_dword	ro	0x0	

65.40 tlp_prefix_log				0x40701138
Bits	Field Name	Acc	Default	Description
31 : 0	cfg_tlp_pfx_log	ro	0x0	

65.41 tph_ext_cap_hdr				0x40701148
Bits	Field Name	Acc	Default	Description
15 : 0	tph_cap_id	rw	0x17	
19 : 16	tph_req_cap_ver	rw	0x1	
31 : 20	tph_req_next_ptr	rw	0x218	

65.42 tph_req_cap_reg	0x4070114C

Bits	Field Name	Acc	Default	Description
0	tph_req_no_st_mode	ro	0x1	
1	tph_req_cap_int_vec	rw	0x1	
2	tph_req_dev_spec	ro	0x1	
8	tph_req_extended_tph	rw	0x0	
9	tph_req_cap_st_table_loc_0	rw	0x1	
10	tph_req_cap_st_table_loc_1	rw	0x0	
26 : 16	tph_req_cap_st_table_size	rw	0x3	

65.43 tph_req_ctrl_reg				0x40701150
Bits	Field Name	Acc	Default	Description
2 : 0	tph_req_st_mode_select	rw	0x0	
9 : 8	tph_req_ctrl_req_en	rw	0x0	

65.44 tph_st_table_reg_0				0x40701154
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_0	rw	0x0	
15 : 8	tph_req_st_table_higher_0	rw	0x0	
23 : 16	tph_req_st_table_lower_1	rw	0x0	
31 : 24	tph_req_st_table_higher_1	rw	0x0	

65.45 tph_st_table_reg_1				0x40701158
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_2	rw	0x0	
15 : 8	tph_req_st_table_higher_2	rw	0x0	
23 : 16	tph_req_st_table_lower_3	rw	0x0	
31 : 24	tph_req_st_table_higher_3	rw	0x0	

65.46 tph_st_table_reg_2				0x4070115C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_4	rw	0x0	
15 : 8	tph_req_st_table_higher_4	rw	0x0	
23 : 16	tph_req_st_table_lower_5	rw	0x0	
31 : 24	tph_req_st_table_higher_5	rw	0x0	

65.47 tph_st_table_reg_3				0x40701160
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_6	rw	0x0	
15 : 8	tph_req_st_table_higher_6	rw	0x0	
23 : 16	tph_req_st_table_lower_7	rw	0x0	
31 : 24	tph_req_st_table_higher_7	rw	0x0	

65.48 tph_st_table_reg_4				0x40701164
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_8	rw	0x0	
15 : 8	tph_req_st_table_higher_8	rw	0x0	
23 : 16	tph_req_st_table_lower_9	rw	0x0	
31 : 24	tph_req_st_table_higher_9	rw	0x0	

65.49 tph_st_table_reg_5				0x40701168
Bits	Field Name	Acc	Default	Description

7 : 0	tph_req_st_table_lower_10	rw	0x0	
15 : 8	tph_req_st_table_higher_10	rw	0x0	
23 : 16	tph_req_st_table_lower_11	rw	0x0	
31 : 24	tph_req_st_table_higher_11	rw	0x0	

65.50 tph_st_table_reg_6				0x4070116C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_12	rw	0x0	
15 : 8	tph_req_st_table_higher_12	rw	0x0	
23 : 16	tph_req_st_table_lower_13	rw	0x0	
31 : 24	tph_req_st_table_higher_13	rw	0x0	

65.51 tph_st_table_reg_7				0x40701170
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_14	rw	0x0	
15 : 8	tph_req_st_table_higher_14	rw	0x0	
23 : 16	tph_req_st_table_lower_15	rw	0x0	
31 : 24	tph_req_st_table_higher_15	rw	0x0	

65.52 tph_st_table_reg_8				0x40701174
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_16	rw	0x0	
15 : 8	tph_req_st_table_higher_16	rw	0x0	
23 : 16	tph_req_st_table_lower_17	rw	0x0	
31 : 24	tph_req_st_table_higher_17	rw	0x0	

65.53 tph_st_table_reg_9				0x40701178
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_18	rw	0x0	
15 : 8	tph_req_st_table_higher_18	rw	0x0	
23 : 16	tph_req_st_table_lower_19	rw	0x0	
31 : 24	tph_req_st_table_higher_19	rw	0x0	

65.54 tph_st_table_reg_10				0x4070117C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_20	rw	0x0	
15 : 8	tph_req_st_table_higher_20	rw	0x0	
23 : 16	tph_req_st_table_lower_21	rw	0x0	
31 : 24	tph_req_st_table_higher_21	rw	0x0	

65.55 tph_st_table_reg_11				0x40701180
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_22	rw	0x0	
15 : 8	tph_req_st_table_higher_22	rw	0x0	
23 : 16	tph_req_st_table_lower_23	rw	0x0	
31 : 24	tph_req_st_table_higher_23	rw	0x0	

65.56 tph_st_table_reg_12				0x40701184
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_24	rw	0x0	
15 : 8	tph_req_st_table_higher_24	rw	0x0	

23 : 16	tph_req_st_table_lower_25	rw	0x0	
31 : 24	tph_req_st_table_higher_25	rw	0x0	

65.57 tph_st_table_reg_13				0x40701188
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_26	rw	0x0	
15 : 8	tph_req_st_table_higher_26	rw	0x0	
23 : 16	tph_req_st_table_lower_27	rw	0x0	
31 : 24	tph_req_st_table_higher_27	rw	0x0	

65.58 tph_st_table_reg_14				0x4070118C
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_28	rw	0x0	
15 : 8	tph_req_st_table_higher_28	rw	0x0	
23 : 16	tph_req_st_table_lower_29	rw	0x0	
31 : 24	tph_req_st_table_higher_29	rw	0x0	

65.59 tph_st_table_reg_15				0x40701190
Bits	Field Name	Acc	Default	Description
7 : 0	tph_req_st_table_lower_30	rw	0x0	
15 : 8	tph_req_st_table_higher_30	rw	0x0	
23 : 16	tph_req_st_table_lower_31	rw	0x0	
31 : 24	tph_req_st_table_higher_31	rw	0x0	

65.60 ats_cap_hdr				0x40701218
Bits	Field Name	Acc	Default	Description
15 : 0	ats_cap_id	rw	0xF	
19 : 16	ats_cap_version	rw	0x1	
31 : 20	ats_cap_next_ptr	rw	0x228	

65.61 ats_cap_ctrl				0x4070121C
Bits	Field Name	Acc	Default	Description
4 : 0	ats_invalidate_q_depth	rw	0x0	
5	ats_page_aligned_req	rw	0x1	
6	ats_global_inval_spprtd	rw	0x0	
20 : 16	ats_stu	rw	0x0	
31	ats_enable	rw	0x0	

65.62 l1sub_cap_header				0x40701228
Bits	Field Name	Acc	Default	Description
15 : 0	l1sub_cap_id	rw	0x1E	
19 : 16	l1sub_cap_version	rw	0x1	
31 : 20	l1sub_cap_next_ptr	rw	0x0	

65.63 l1sub_cap				0x4070122C
Bits	Field Name	Acc	Default	Description
0	l1_2_pcipm_support	rw	0x1	
1	l1_1_pcipm_support	rw	0x1	
2	l1_2_aspm_support	rw	0x0	
3	l1_1_aspm_support	rw	0x1	
4	l1_pmsub_support	rw	0x1	

15 : 8	comm_mode_support	rw	0xA	
17 : 16	pwr_on_scale_support	rw	0x0	
23 : 19	pwr_on_value_support	rw	0x5	

65.64 l1sub_ctrl1				0x40701230
Bits	Field Name	Acc	Default	Description
0	l1_2_pcipm_en	rw	0x0	
1	l1_1_pcipm_en	rw	0x0	
2	l1_2_aspm_en	rw	0x0	
3	l1_1_aspm_en	rw	0x0	
15 : 8	t_common_mode	ro	0x0	
25 : 16	l1_2_th_val	rw	0x0	
31 : 29	l1_2_th_sca	rw	0x0	

65.65 l1sub_ctrl2				0x40701234
Bits	Field Name	Acc	Default	Description
1 : 0	t_power_on_scale	rw	0x0	
7 : 3	t_power_on_value	rw	0x5	

65.66 ack_latency_timer				0x40701700
Bits	Field Name	Acc	Default	Description
15 : 0	round_trip_latency_time_limit	rw	0x102F	
31 : 16	replay_time_limit	rw	0x308D	

65.67 vendor_spec_dllp				0x40701704
Bits	Field Name	Acc	Default	Description
31 : 0	vendor_spec_dllp	rw	0xFFFFFFFF	

65.68 port_force				0x40701708
Bits	Field Name	Acc	Default	Description
7 : 0	link_num	rw	0x4	
11 : 8	forced_ltssm	rw	0x0	
15	force_en	rw	0x0	write 1 clear
21 : 16	link_state	rw	0x0	
31 : 24	cpl_sent_count	rw	0x7	

65.69 ack_f_aspm_ctrl				0x4070170C
Bits	Field Name	Acc	Default	Description
7 : 0	ack_freq	rw	0x0	
15 : 8	ack_n_fts	rw	0xF	
23 : 16	common_clk_n_fts	ro	0xF	
26 : 24	l0s_entrance_latency	rw	0x3	
29 : 27	l1_entrance_latency	rw	0x3	
30	enter_aspm	rw	0x0	

65.70 port_link_ctrl				0x40701710
Bits	Field Name	Acc	Default	Description
0	vendor_sprc_dllp_req	rw	0x0	
1	scramble_disable	rw	0x0	
2	loopback_enable	rw	0x0	
3	reset_assert	rw	0x0	

5	dll_link_en	rw	0x1	
6	link_disable	rw	0x0	
7	fast_link_mode	rw	0x0	
11 : 8	link_rate	rw	0x1	
21 : 16	link_capable	rw	0x1	
22	cross_link_en	rw	0x1	
23	cross_link_active	ro	0x0	
24	beacon_enable	rw	0x0	
25	corrupt_lcrc_enable	rw	0x0	
26	extended_synch	rw	0x0	
27	tx_lane_reverse_enable	rw	0x0	

65.71 lane_skew				0x40701714
Bits	Field Name	Acc	Default	Description
23 : 0	insert_lane_skew	rw	0x0	
24	flow_ctrl_disable	rw	0x0	
25	ack_nak_disable	rw	0x0	
31	disable_lane_to_lane_deskew	rw	0x0	

65.72 timer_ctrl_max_func_num				0x40701718
Bits	Field Name	Acc	Default	Description
7 : 0	max_func_num	rw	0x0	
18 : 14	timer_mod_replay_timer	rw	0x1	
23 : 19	timer_mod_ack_nak	rw	0x0	
28 : 24	update_freq_timer	rw	0x0	

65.73 symbol_timer_filter_1				0x4070171C
Bits	Field Name	Acc	Default	Description
10 : 0	skp_int_val	rw	0x500	
14 : 11	eidle_timer	rw	0x0	
15	disable_fc_wd_timer	rw	0x0	
31 : 16	mask_radm_1	rw	0x0	

65.74 filter_mask_2				0x40701720
Bits	Field Name	Acc	Default	Description
31 : 0	mask_radm_2	rw	0x0	

65.75 amba_mul_ob_decomp_np_sub_req_ctrl				0x40701724
Bits	Field Name	Acc	Default	Description
1 : 0	ib_ob_rd_split_burst_en	rw	0x1	

65.76 pl_debug0				0x40701728
Bits	Field Name	Acc	Default	Description
31 : 0	deb_reg_0	ro	0x0	

65.77 pl_debug1				0x4070172C
Bits	Field Name	Acc	Default	Description
31 : 0	deb_reg_1	ro	0x0	

65.78 tx_p_fc_credit_status	0x40701730

Bits	Field Name	Acc	Default	Description
11 : 0	tx_p_data_fc_credit	ro	0x0	
19 : 12	tx_p_header_fc_credit	ro	0x0	

65.79 tx_np_fc_credit_status				0x40701734
Bits	Field Name	Acc	Default	Description
11 : 0	tx_np_data_fc_credit	ro	0x0	
19 : 12	tx_np_header_fc_credit	ro	0x0	

65.80 tx_cpl_fc_credit_status				0x40701738
Bits	Field Name	Acc	Default	Description
11 : 0	tx_cpl_data_fc_credit	ro	0x0	
19 : 12	tx_cpl_header_fc_credit	ro	0x0	

65.81 queue_status				0x4070173C
Bits	Field Name	Acc	Default	Description
0	rx_tlp_fc_credit_non_return	ro	0x0	
1	tx_retry_buffer_ne	ro	0x0	
2	rx_queue_non_empty	ro	0x0	
28 : 16	timer_mod_flow_ctrl	rw	0x0	
31	timer_moid_flow_ctrl_en	ro	0x0	

65.82 vc_tx_arbl_1				0x40701740
Bits	Field Name	Acc	Default	Description
7 : 0	wrr_weight_vc_0	ro	0xF	
15 : 8	wrr_weight_vc_1	ro	0x0	
23 : 16	wrr_weight_vc_2	ro	0x0	
31 : 24	wrr_weight_vc_3	ro	0x0	

65.83 vc_tx_arbl_2				0x40701744
Bits	Field Name	Acc	Default	Description
7 : 0	wrr_weight_vc_4	ro	0x0	
15 : 8	wrr_weight_vc_5	ro	0x0	
23 : 16	wrr_weight_vc_6	ro	0x0	
31 : 24	wrr_weight_vc_7	ro	0x0	

65.84 vc0_p_rx_1_ctrl				0x40701748
Bits	Field Name	Acc	Default	Description
11 : 0	vc0_p_data_credit	ro	0x60	
19 : 12	vc0_p_header_credit	ro	0x1E	
23 : 21	vc0_p_tlp_q_mode	rw	0x1	
30	tlp_type_ordering_vc0	rw	0x1	
31	vc_ordering_rx_q	rw	0x0	

65.85 vc0_np_rx_q_ctrl				0x4070174C
Bits	Field Name	Acc	Default	Description
11 : 0	vc0_np_data_credit	ro	0x7	
19 : 12	vc0_np_header_credit	ro	0x1E	
23 : 21	vc0_np_tlp_q_mode	rw	0x1	

65.86 vc0_cpl_rx_q_ctrl				0x40701750
Bits	Field Name	Acc	Default	Description
11 : 0	vc0_cpl_data_credit	ro	0x0	
19 : 12	vc0_cpl_header_credit	ro	0x0	
23 : 21	vc0_cpl_tlp_q_mode	rw	0x4	

65.87 gen2_ctrl				0x4070180C
Bits	Field Name	Acc	Default	Description
12 : 8	num_of_lanes	rw	0x1	
15 : 13	pre_det_lane	rw	0x0	
16	auto_lane_flip_ctrl_en	rw	0x0	
21	gen1_ei_intf	rw	0x0	

65.88 phy_status				0x40701810
Bits	Field Name	Acc	Default	Description
31 : 0	phy_status	ro	0x0	

65.89 phy_ctrl				0x40701814
Bits	Field Name	Acc	Default	Description
31 : 0	phy_ctrl	rw	0x0	

65.90 pipe_loopback_ctrl				0x407018B8
Bits	Field Name	Acc	Default	Description
15 : 0	lpbk_rxvalid	rw	0x1	
21 : 16	rxstatus_lane	rw	0x0	
26 : 24	rxstatus_value	wo	0x0	
31	pipe_loopback	rw	0x0	

65.91 misc_ctrl_1				0x407018BC
Bits	Field Name	Acc	Default	Description
0	dbi_ro_wr_en	rw	0x0	

65.92 amba_error_resp_default				0x407018D0
Bits	Field Name	Acc	Default	Description
0	amba_err_resp_global	rw	0x0	
2	amba_err_resp_vendorid	rw	0x0	
3	amba_err_resp_flush	rw	0x0	

65.93 amba_ordrmgr_wdog				0x407018DC
Bits	Field Name	Acc	Default	Description
15 : 0	amba_ordrmgr_wdog	rw	0xFFF	
31	amba_ordrmgr_wdog_en	rw	0x1	

65.94 pl_last				0x407018FC
Bits	Field Name	Acc	Default	Description
0	pl_last	ro	0x0	

65.95 iatu_viewport				0x40701900
Bits	Field Name	Acc	Default	Description

2 : 0	iatu_region_index	rw	0x0	
31	iatu_region_dir	rw	0x0	0: outbound, 1: inbound

65.96 iatu_region_ctrl_1				0x40701904
Bits	Field Name	Acc	Default	Description
4 : 0	iatu_type	rw	0x0	
7 : 5	iatu_tc	rw	0x0	
8	iatu_td	rw	0x0	
10 : 9	iatu_attr	rw	0x0	
11	iatu_ido	rw	0x0	
12	iatu_th	rw	0x0	
17 : 16	iatu_at	rw	0x0	
19 : 18	iatu_ph	rw	0x0	
24 : 20	iatu_ctrl_1_func_num	rw	0x0	

65.97 iatu_region_ctrl_2				0x40701908
Bits	Field Name	Acc	Default	Description
7 : 0	iatu_msg_code	rw	0x0	
10 : 8	iatu_inbound_bar_num	rw	0x0	Only in inbound region.
14	iatu_inbound_tc_match_en	rw	0x0	Only in inbound region.
15	iatu_inbound_td_match_en	rw	0x0	Only in inbound region.
16	iatu_inbound_attr_match_en	rw	0x0	Only in inbound region.
17	iatu_inbound_th_match_en	rw	0x0	Only in inbound region.
18	iatu_inbound_at_match_en	rw	0x0	Only in inbound region.
19	iatu_func_bypass_func_num_match_en	rw	0x0	In outbound: func_bypass In inbound: func_num_match_en
20	iatu_inbound_vf_match_en	rw	0x0	Only in inbound region.
21	iatu_inbound_msg_code_match_en	rw	0x0	Only in inbound region.
22	iatu_inbound_ph_match_en	rw	0x0	Only in inbound region.
25 : 24	iatu_inbound_resp_code	rw	0x0	Only in inbound region.
27	iatu_dma_bypass_fuzzy_type_match_mode	rw	0x0	In outbound: dma_bypass In inbound: fuzzy_type_match_mode
28	iatu_cf_shift_mode	rw	0x0	
29	iatu_invert_mode	rw	0x0	
30	iatu_inbound_match_mode	rw	0x0	Only in inbound region. 0: address match mode, 1: bar match mode
31	iatu_region_en	rw	0x0	

65.98 iatu_lwr_base_addr				0x4070190C
Bits	Field Name	Acc	Default	Description
11 : 0	iatu_lwr_base_hw	ro	0x0	
31 : 12	iatu_lwr_base_rw	rw	0x0	

65.99 iatu_upper_base_addr				0x40701910
Bits	Field Name	Acc	Default	Description
31 : 0	iatu_upper_base_rw	rw	0x0	

65.100 iatu_limit_addr				0x40701914
Bits	Field Name	Acc	Default	Description
11 : 0	iatu_limit_addr_hw	ro	0xFFF	
31 : 12	iatu_limit_addr_rw	rw	0x0	

65.101 iatu_lwr_target_addr				0x40701918
Bits	Field Name	Acc	Default	Description
11 : 0	iatu_lwr_target_hw	ro	0x0	
31 : 12	iatu_lwr_target_rw	rw	0x0	

65.102 iatu_upper_target_addr				0x4070191C
Bits	Field Name	Acc	Default	Description
31 : 0	iatu_upper_target_rw	rw	0x0	

65.103 dma_ctrl				0x40701978
Bits	Field Name	Acc	Default	Description
3 : 0	edma_wr_ch_num	rw	0x2	
19 : 16	edma_rd_ch_num	rw	0x2	

65.104 dma_wr_engine_en				0x4070197C
Bits	Field Name	Acc	Default	Description
0	edma_wr_engine_en	rw	0x0	

65.105 dma_wr_doorbell				0x40701980
Bits	Field Name	Acc	Default	Description
2 : 0	edma_wr_doorbell_num	rw	0x0	
31	edma_wr_stop	rw	0x0	

65.106 dma_wr_ch_arb_weight_low				0x40701988
Bits	Field Name	Acc	Default	Description
4 : 0	edma_wr_ch_0_weight	rw	0x1	
9 : 5	edma_wr_ch_1_weight	rw	0x1	
14 : 10	edma_wr_ch_2_weight	rw	0x1	
19 : 15	edma_wr_ch_3_weight	rw	0x1	

65.107 dma_wr_ch_arb_weight_high				0x4070198C
Bits	Field Name	Acc	Default	Description
4 : 0	edma_wr_ch_4_weight	rw	0x1	
9 : 5	edma_wr_ch_5_weight	rw	0x1	
14 : 10	edma_wr_ch_6_weight	rw	0x1	
19 : 15	edma_wr_ch_7_weight	rw	0x1	

65.108 dma_wr_p_req_timer				0x40701998
Bits	Field Name	Acc	Default	Description
11 : 0	edma_wr_p_timer_limit	rw	0xE	
15	edma_wr_p_timer_disable	rw	0x0	

65.109 dma_rd_engine_en				0x4070199C
Bits	Field Name	Acc	Default	Description
0	edma_rd_engine_en	rw	0x0	

65.110 dma_rd_doorbell				0x407019A0
Bits	Field Name	Acc	Default	Description
2 : 0	edma_rd_doorbell_num	rw	0x0	

31	edma_rd_stop	rw	0x0	

65.111 dma_rd_ch_arb_weight_low				0x407019A8
Bits	Field Name	Acc	Default	Description
4 : 0	edma_rd_ch_0_weight	rw	0x1	
9 : 5	edma_rd_ch_1_weight	rw	0x1	
14 : 10	edma_rd_ch_2_weight	rw	0x1	
19 : 15	edma_rd_ch_3_weight	rw	0x1	

65.112 dma_rd_ch_arb_weight_high				0x407019AC
Bits	Field Name	Acc	Default	Description
4 : 0	edma_rd_ch_4_weight	rw	0x1	
9 : 5	edma_rd_ch_5_weight	rw	0x1	
14 : 10	edma_rd_ch_6_weight	rw	0x1	
19 : 15	edma_rd_ch_7_weight	rw	0x1	

65.113 dma_wr_int_status				0x407019BC
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_done_int_status	rw	0x0	
23 : 16	edma_wr_abort_int_status	rw	0x0	

65.114 dma_wr_int_mask				0x407019C4
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_done_int_mask	rw	0xFF	
23 : 16	edma_wr_abort_int_mask	rw	0xFF	

65.115 dma_wr_int_clr				0x407019C8
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_done_int_clr	rw	0x0	write 1 clear
23 : 16	edma_wr_abort_int_clr	rw	0x0	write 1 clear

65.116 dma_wr_err_status				0x407019CC
Bits	Field Name	Acc	Default	Description
7 : 0	edma_app_rd_err_det	ro	0x0	
23 : 16	edma_wr_linkl-ist_elem_fetch_err_det	ro	0x0	

65.117 dma_wr_done_imwr_low				0x407019D0
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_done_low_reg	rw	0x0	

65.118 dma_wr_done_imwr_high				0x407019D4
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_done_high_reg	rw	0x0	

65.119 dma_wr_abort_imwr_low				0x407019D8
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_abort_low_reg	rw	0x0	

65.120 dma_wr_abort_imwr_high				0x407019DC
Bits	Field Name	Acc	Default	Description
31 : 0	edma_wr_abort_high_reg	rw	0x0	

65.121 dma_wr_ch01_imwr_data				0x407019E0
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_0_data	rw	0x0	
31 : 16	edma_wr_ch_1_data	rw	0x0	

65.122 dma_wr_ch23_imwr_data				0x407019E4
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_2_data	rw	0x0	
31 : 16	edma_wr_ch_3_data	rw	0x0	

65.123 dma_wr_ch45_imwr_data				0x407019E8
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_4_data	rw	0x0	
31 : 16	edma_wr_ch_5_data	rw	0x0	

65.124 dma_wr_ch67_imwr_data				0x407019EC
Bits	Field Name	Acc	Default	Description
15 : 0	edma_wr_ch_6_data	rw	0x0	
31 : 16	edma_wr_ch_7_data	rw	0x0	

65.125 dma_wr_linked_list_err_en				0x40701A00
Bits	Field Name	Acc	Default	Description
7 : 0	edma_wr_ch_llraie	rw	0x0	
23 : 16	edma_wr_ch_lllaie	rw	0x0	

65.126 dma_rd_int_status				0x40701A10
Bits	Field Name	Acc	Default	Description
7 : 0	edma_rd_done_int_status	rw	0x0	
23 : 16	edma_rd_abort_int_status	rw	0x0	

65.127 dma_rd_int_mask				0x40701A18
Bits	Field Name	Acc	Default	Description
7 : 0	edma_rd_done_int_mask	rw	0xFF	
23 : 16	edma_rd_abort_int_mask	rw	0xFF	

65.128 dma_rd_int_clr				0x40701A1C
Bits	Field Name	Acc	Default	Description
7 : 0	edma_rd_done_int_clr	rw	0x0	
23 : 16	edma_rd_abort_int_clr	rw	0x0	

65.129 dma_rdd_err_status_low				0x40701A24
Bits	Field Name	Acc	Default	Description
7 : 0	edma_app_wr_err_det	ro	0x0	
23 : 16	edma_rd_linkl-ist_elem_fetch_err_det	ro	0x0	

65.130 dma_rd_err_status_high				0x40701A28
Bits	Field Name	Acc	Default	Description
7 : 0	edma_unsupported_req	ro	0x0	
15 : 8	edma_cpl_abort	ro	0x0	
23 : 16	edma_cpl_timeout	ro	0x0	
31 : 24	edma_data_poisioning	ro	0x0	

65.131 dma_rd_linked_list_err_en				0x40701A34
Bits	Field Name	Acc	Default	Description
7 : 0	edma_rd_ch_llraie	rw	0x0	
23 : 16	edma_rd_ch_lllaie	rw	0x0	

65.132 dma_rd_done_imwr_low				0x40701A3C
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_done_low_reg	rw	0x0	

65.133 dma_rd_done_imwr_high				0x40701A40
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_done_high_reg	rw	0x0	

65.134 dma_rd_abort_imwr_low				0x40701A44
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_abort_low_reg	rw	0x0	

65.135 dma_rd_abort_imwr_high				0x40701A48
Bits	Field Name	Acc	Default	Description
31 : 0	edma_rd_abort_high_reg	rw	0x0	

65.136 dma_rd_ch01_imwr_data				0x40701A4C
Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_0_data	rw	0x0	
31 : 16	edma_rd_ch_1_data	rw	0x0	

65.137 dma_rd_ch23_imwr_data				0x40701A50
Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_2_data	rw	0x0	
31 : 16	edma_rd_ch_3_data	rw	0x0	

65.138 dma_rd_ch45_imwr_data				0x40701A54
Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_4_data	rw	0x0	
31 : 16	edma_rd_ch_5_data	rw	0x0	

65.139 dma_rd_ch67_imwr_data				0x40701A58
Bits	Field Name	Acc	Default	Description
15 : 0	edma_rd_ch_6_data	rw	0x0	
31 : 16	edma_rd_ch_7_data	rw	0x0	

65.140 dma_viewport_sel				0x40701A6C
Bits	Field Name	Acc	Default	Description
2 : 0	edma_ch_num	rw	0x0	
31	edma_ch_dir	rw	0x0	0: write channel 1: read channel

65.141 dma_ch_ctrl1				0x40701A70
Bits	Field Name	Acc	Default	Description
0	edma_cb	rw	0x0	
1	edma_tcb	rw	0x0	
2	edma_llp	rw	0x0	
3	edma_lie	rw	0x0	
4	edma_rie	rw	0x0	
6 : 5	edma_cs	ro	0x0	
8	edma_ccs	rw	0x0	
9	edma_lle	rw	0x0	
16 : 12	edma_func_num	rw	0x0	
24	edma_ns	rw	0x0	
25	edma_ro	rw	0x0	
26	edma_td	rw	0x0	
29 : 27	edma_tc	rw	0x0	
31 : 30	edma_at	rw	0x0	

65.142 dma_ch_ctrl2				0x40701A74
Bits	Field Name	Acc	Default	Description
24 : 17	edma_st	rw	0x0	
30 : 29	edma_ph	rw	0x0	
31	edma_th	rw	0x0	

65.143 dma_trans_size				0x40701A78
Bits	Field Name	Acc	Default	Description
31 : 0	edma_trans_size	rw	0x0	

65.144 dma_sar_low				0x40701A7C
Bits	Field Name	Acc	Default	Description
31 : 0	edma_src_addr_low	rw	0x0	

65.145 dma_sar_high				0x40701A80
Bits	Field Name	Acc	Default	Description
31 : 0	edma_src_addr_high	rw	0x0	

65.146 dma_dar_low				0x40701A84
Bits	Field Name	Acc	Default	Description
31 : 0	edma_dst_addr_low	rw	0x0	

65.147 dma_dar_high				0x40701A88
Bits	Field Name	Acc	Default	Description
31 : 0	edma_dst_addr_high	rw	0x0	

65.148 dma_llp_low				0x40701A8C
Bits	Field Name	Acc	Default	Description

31 : 0	edma_llp_low	rw	0x0	

65.149 dma_llp_high				0x40701A90
Bits	Field Name	Acc	Default	Description
31 : 0	edma_llp_high	rw	0x0	

65.150 aux_clk_freq				0x40701B40
Bits	Field Name	Acc	Default	Description
9 : 0	aux_clk_freq	rw	0xA	

65.151 l1_substates				0x40701B44
Bits	Field Name	Acc	Default	Description
1 : 0	l1sub_t_power_off	rw	0x2	
5 : 2	l1sub_t_l1_2	rw	0x4	
7 : 6	l1sub_t_pclkack	rw	0x3	

66 pcie_top_ctrl

66.1 pcie_ana_cfg0				0x40780000
Bits	Field Name	Acc	Default	Description
3 : 0	rg_pcie_ana_pll_reg_vbit	rw	0x8	
5 : 4	rg_pcie_ana_pll_pu_bit	rw	0x2	
6	rg_pcie_ana_pll_vco_band	rw	0x1	
9 : 7	rg_pcie_ana_pll_cp_ibit	rw	0x4	
10	rg_pcie_ana_frefx2_en	rw	0x0	
11	rg_pcie_ana_rx_cm_mode	rw	0x1	
12	rg_pcie_ana_dpll_ck_sel	rw	0x1	
13	rg_pcie_ana_clk_edge_sel	rw	0x0	
14	rg_pcie_ana_-vco_low_test_mode	rw	0x0	
15	rg_pcie_ana_-vco_high_test_mode	rw	0x0	
19 : 16	rg_pcie_ana_pfd_dn_dlyc_bit	rw	0x0	
23 : 20	rg_pcie_ana_pfd_dn_dlyf_bit	rw	0x1	
27 : 24	rg_pcie_ana_pfd_up_dlyc_bit	rw	0x0	
31 : 28	rg_pcie_ana_pfd_up_dlyf_bit	rw	0x1	

66.2 pcie_ana_cfg1				0x40780004
Bits	Field Name	Acc	Default	Description
1	rg_pcie_ana_ctle_bypass	rw	0x0	
4 : 2	rg_pcie_ana_ctle_cbit	rw	0x4	
6 : 5	rg_pcie_ana_cdr_gain	rw	0x0	
7	rg_pcie_ana_sel_terr	rw	0x1	
8	rg_pcie_ana_lpf_notch_en	rw	0x0	
10 : 9	rg_pcie_ana_rxdetect_cp	rw	0x2	
12 : 11	rg_pcie_ana_rxdetect_step	rw	0x2	
15 : 13	rg_pcie_ana_iset_sq	rw	0x4	
17 : 16	rg_pcie_ana_idac_mode	rw	0x0	
19	rg_pcie_ana_en_rst_intp	rw	0x0	
23 : 20	rg_pcie_ana_idac_bit	rw	0x8	
27 : 24	rg_pcie_ana_idac_aux_bit	rw	0x8	
28	rg_pcie_ana_test_current	rw	0x0	
29	rg_pcie_ana_gpio_mode	rw	0x0	

30	rg_pcie_ana_tx_cm_mode	rw	0x1	
31	ro_pcie_lock_out	ro	0x0	

66.3 pcie_ana_cfg2				0x40780008
Bits	Field Name	Acc	Default	Description
3 : 0	rg_pcie_ana_rterm_tx	rw	0x8	
7 : 4	rg_pcie_ana_rterm_rx	rw	0x8	
11 : 8	rg_pcie_ana_rterm_tx_lp	rw	0x4	
15 : 12	rg_pcie_ana_idac_bit_beacon	rw	0x0	
19 : 16	rg_pcie_ana_-idac_aux_bit_beacon	rw	0xC	
21 : 20	rg_pcie_ana_ctle_gain	rw	0x2	
22	rg_pcie_ana_div_clk_edge_sel	rw	0x0	
23	rg_pcie_ana_rx_mode_sel	rw	0x1	

66.4 pcie_ana_cfg3				0x4078000C
Bits	Field Name	Acc	Default	Description
0	rg_pcie_ana_plldiv15_en	rw	0x0	
1	rg_pcie_ana_fref_sel_xtal	rw	0x0	
2	rg_pcie_ana_doubler_en	rw	0x0	
3	rg_pcie_ana_clkx2_en	rw	0x0	
4	rg_pcie_ana_clk2dll_en	rw	0x0	
5	rg_pcie_ana_clk2dll_div4_en	rw	0x0	
6	rg_pcie_ana_clk160m2fpga_en	rw	0x0	

66.5 pcie_cfg0				0x40780010
Bits	Field Name	Acc	Default	Description
0	rg_pcie_hw_pon	rw	0x0	
1	rg_sw_power_up_rst_n	rw	0x0	
2	rg_phy_rst_n	rw	0x1	
3	rg_core_rst_n	rw	0x1	
4	rg_non_sticky_rst_n	rw	0x1	
5	rg_sticky_rst_n	rw	0x1	
6	rg_pcs_rst_n	rw	0x1	
11 : 8	rg_bps_perst	rw	0x0	
13 : 12	rg_bps_linkrst	rw	0x2	
14	rg_mpll_en_bypass_pipe_reset	rw	0x1	
15	rg_link_rst_ready	rw	0x0	
23 : 20	rg_auxclk_sel_filt_th	rw	0x7	

66.6 pcie_cfg1				0x40780014
Bits	Field Name	Acc	Default	Description
0	rg_perst_n_dr	rw	0x0	
1	rg_perst_n	rw	0x0	
2	rg_local_ref_clk_req_n_dr	rw	0x0	
3	rg_local_ref_clk_req_n	rw	0x0	
4	rg_pu_pcie_dr	rw	0x0	
5	rg_pu_pcie	rw	0x0	
6	rg_refbuf_en_dr	rw	0x0	
7	rg_refbuf_en	rw	0x0	
8	rg_pcs_en_dr	rw	0x0	
9	rg_pcs_en	rw	0x0	
10	rg_ref_clk_req_n_dr	rw	0x0	
11	rg_ref_clk_req_n	rw	0x0	

12	rg_refclk_en_dr	rw	0x0	
13	rg_refclk_en	rw	0x0	
14	rg_pu_pcie_pll_dr	rw	0x0	
15	rg_pu_pcie_pll	rw	0x0	
16	rg_pcie_clk_en_dr	rw	0x0	
17	rg_pcie_clk_en	rw	0x0	
18	rg_pcie_clk250m2dig_en_dr	rw	0x0	
19	rg_pcie_clk250m2dig_en	rw	0x0	
20	rg_pcie_rstn_dr	rw	0x0	
21	rg_pcie_rstn	rw	0x0	
22	rg_pcie_en_rhs_rx_dr	rw	0x0	
23	rg_pcie_en_rhs_rx	rw	0x0	
24	rg_pcie_en_rhs_tx_dr	rw	0x0	
25	rg_pcie_en_rhs_tx	rw	0x0	
26	rg_pcie_en_idac_dr	rw	0x0	
27	rg_pcie_en_idac	rw	0x0	
28	rg_pcie_ctle_en_dr	rw	0x0	
29	rg_pcie_ctle_en	rw	0x0	
30	rg_pd_pcie_pll_auto_en	rw	0x1	

66.7 pcie_cfg3				0x4078001C
Bits	Field Name	Acc	Default	Description
0	rg_pcs_rx_es_buf_mode	rw	0x0	
1	rg_cdet_bps_rx_elecidle	rw	0x0	
2	rg_pcs_tx_swing	rw	0x0	
5 : 3	rg_pcs_tx_margin	rw	0x0	
6	rg_pcs_tx2rx_loopbk	rw	0x0	
8 : 7	rg_pcs_tx_deemph	rw	0x0	
15 : 9	rg_pcs_tx_swing_low	rw	0x0	
22 : 16	rg_pcs_swing_full_	rw	0x0	
28 : 23	rg_pcs_tx_deemph_gen1	rw	0x0	
29	rg_pcs_common_clocks	rw	0x0	
30	rg_l1_aux_clk_switch_core_clk_gate_en	rw	0x1	
31	rg_l1_mac_powerdown_override_to_p2_en	rw	0x0	

66.8 pcie_cfg4				0x40780020
Bits	Field Name	Acc	Default	Description
5 : 0	rg_pcie_stable_dly	rw	0x19	delay = (1+rg_pcie_stable_dly * 32) / 80 us
14 : 6	rg_pcie_pll_stable_dly	rw	0xFA	delay = (1+rg_pcie_pll_stable_dly* 32) / 80 us
15	rg_wake_dr	rw	0x0	
16	rg_wake	rw	0x0	
17	rg_phy_lane_loopbk_en_dr	rw	0x0	
18	rg_phy_lane_loopbk_en	rw	0x0	
19	rg_phy_tx_invert_mode_dr	rw	0x0	
20	rg_phy_tx_invert_mode	rw	0x0	
21	rg_phy_tx_swizzle_en	rw	0x1	
22	rg_phy_rx_invert_mode_dr	rw	0x0	
23	rg_phy_rx_invert_mode	rw	0x0	
24	rg_phy_rx_swizzle_en	rw	0x0	
25	rg_rx_clk_inv_en	rw	0x0	
26	rg_eios_seen_en	rw	0x1	
29 : 27	rg_pcie_clk_stable_dly	rw	0x2	
30	ro_pcie_pll_stable	ro	0x0	
31	ro_pcie_rstn_release	ro	0x0	

66.9 pcie_cfg5				0x40780024
Bits	Field Name	Acc	Default	Description
0	rg_phy_tx_data_en_dr	rw	0x0	
1	rg_phy_tx_data_en	rw	0x0	
11 : 2	rg_phy_tx_cm_state_dly	rw	0x258	delay=(1+rg_phy_tx_cm_state_dly*32)/80 us
12	rg_phy_tx_cm_en_dr	rw	0x0	
13	rg_phy_tx_cm_en	rw	0x0	
22 : 14	rg_beacon_period	rw	0x53	period=(1+rg_beacon_period * 32)/80 us
23	rg_beacon_en_dr	rw	0x0	
24	rg_beacon_en	rw	0x0	
25	rg_beacon_pattern_dr	rw	0x0	
26	rg_beacon_pattern	rw	0x0	
29 : 27	rg_phy_tx_data_en_dly_sel	rw	0x3	

66.10 pcie_cfg6				0x40780028
Bits	Field Name	Acc	Default	Description
4 : 0	rg_phy_tx_state_dly	rw	0x0	delay=(1+rg_phy_tx_state_dly*32)/80 us
5	rg_phy_tx_data_en_sel	rw	0x0	
6	rg_phy_tx_en_dr	rw	0x0	
7	rg_phy_tx_en	rw	0x0	
8	rg_phy_tx_detect_rx_result_dr	rw	0x0	
9	rg_phy_tx_detect_rx_result	rw	0x0	
17 : 10	rg_phy_tx_detect_rx_ack_dly	rw	0xFA	delay=(1+rg_phy_tx_detect_rx_ack_dly*32)/80 us
25 : 18	rg_phy_tx_detect_rx_req_width	rw	0x53	delay=(1+rg_phy_tx_detect_rx_req_width*32)/80 us
26	rg_phy_tx_detect_rx_req_dr	rw	0x0	
27	rg_phy_tx_detect_rx_req	rw	0x0	
28	rg_phy_tx_detect_rx_result_filt_en	rw	0x0	

66.11 pcie_cfg7				0x4078002C
Bits	Field Name	Acc	Default	Description
0	rg_phy_rx_los_dr	rw	0x0	
1	rg_phy_rx_los	rw	0x0	
2	rg_phy_rx_los_en_dr	rw	0x0	
3	rg_phy_rx_los_en	rw	0x0	
4	rg_phy_rx_en_dr	rw	0x0	
5	rg_phy_rx_en	rw	0x0	
6	rg_phy_rx_en_sel	rw	0x0	
10 : 7	rg_phy_rx_pll_state_dly	rw	0x7	delay=(1+rg_phy_rx_pll_state_dly)/80 us

66.12 pcie_cfg8				0x40780030
Bits	Field Name	Acc	Default	Description
0	rg_app_req_retry_en	rw	0x0	
1	rg_app_req_exit_l1	rw	0x0	
2	rg_app_ready_entr_l23	rw	0x0	
3	rg_app_req_entr_l1	rw	0x0	
4	rg_app_init_rst	rw	0x0	
5	rg_phy_clk_req_n	rw	0x1	
6	rg_app_clk_req_n	rw	0x1	
7	rg_app_ltssm_enable	rw	0x0	
8	rg_apps_pm_xmt_pme	rw	0x0	
10	rg_ahb_reset_ctrl	rw	0x0	
11	ro_perst_n	ro	0x0	
15	ro_link_req_rst	ro	0x0	
16	ro_smlh_link_up	ro	0x0	

17	ro_rdlh_link_up	ro	0x0	
18	ro_cfg_bus_master_en	ro	0x0	
19	ro_cfg_mem_space_en	ro	0x0	
20	ro_pm_xtlh_block_tlp	ro	0x0	
21	ro_pm_linkst_in_l0s	ro	0x0	
22	ro_pm_linkst_in_l1	ro	0x0	
23	ro_pm_linkst_in_l2	ro	0x0	
24	ro_pm_linkst_l2_exit	ro	0x0	
27 : 25	ro_cfg_max_rd_req_size	ro	0x0	
30 : 28	ro_cfg_max_payload_size	ro	0x0	
31	ro_cfg_rcb	ro	0x0	

66.13 pcie_cfg9				0x40780034
Bits	Field Name	Acc	Default	Description
1 : 0	rg_mstr_resp_err_map	rw	0x3	
7 : 2	rg_slv_resp_err_map	rw	0x3F	

66.14 pcie_cfg10				0x40780038
Bits	Field Name	Acc	Default	Description
0	rg_slv_awmisc_info_atu_bypass	rw	0x0	
1	rg_slv_awmisc_info_nw	rw	0x0	
3 : 2	rg_slv_awmisc_info_ats	rw	0x0	
14 : 4	rg_slv_awmisc_info_tph	rw	0x0	

66.15 pcie_cfg11				0x4078003C
Bits	Field Name	Acc	Default	Description
21 : 0	rg_slv_awmisc_info	rw	0x0	

66.16 pcie_cfg12				0x40780040
Bits	Field Name	Acc	Default	Description
0	rg_slv_armisc_info_atu_bypass	rw	0x0	
1	rg_slv_armisc_info_nw	rw	0x0	
3 : 2	rg_slv_armisc_info_ats	rw	0x0	
14 : 4	rg_slv_armisc_info_tph	rw	0x0	

66.17 pcie_cfg13				0x40780044
Bits	Field Name	Acc	Default	Description
21 : 0	rg_slv_armisc_info	rw	0x0	

66.18 pcie_cfg14				0x40780048
Bits	Field Name	Acc	Default	Description
3 : 0	rg_sdm_num_clkrst	rw	0x8	
4	rg_sdm_down_ss_ct	rw	0x1	
6 : 5	rg_sdm_cyc_ct	rw	0x1	
9 : 7	rg_sdm_amp_ct	rw	0x3	
10	rg_sdm_ss_en	rw	0x0	
11	rg_sdm_dither_bypass	rw	0x1	
12	rg_sdm_resetn_dr	rw	0x0	
13	rg_sdm_resetn_reg	rw	0x0	
15 : 14	rg_sdm_clk_sel_reg	rw	0x1	
23 : 16	rg_sdm_resetn_delay	rw	0xE8	

66.19 pcie_cfg15				0x4078004C
Bits	Field Name	Acc	Default	Description
30 : 0	rg_sdm_freq_cfg	rw	0x6400000	
31	rg_sdm_freq_toggle	rw	0x0	

66.20 pcie_cfg16				0x40780050
Bits	Field Name	Acc	Default	Description
31 : 0	rg_msi_req_en	rw	0x0	

66.21 pcie_cfg17				0x40780054
Bits	Field Name	Acc	Default	Description
0	rg_ven_msg_req_p	rw	0x0	
1	rg_ven_msg_en	rw	0x0	
9 : 2	rg_ven_msg_code	rw	0x0	
17 : 10	rg_ven_msg_tag	rw	0x0	
20 : 18	rg_ven_msg_func_num	rw	0x0	
30 : 21	rg_ven_msg_len	rw	0x0	
31	rg_ven_msg_ep	rw	0x0	

66.22 pcie_cfg18				0x40780058
Bits	Field Name	Acc	Default	Description
1 : 0	rg_ven_msg_attr	rw	0x0	
2	rg_ven_msg_td	rw	0x0	
5 : 3	rg_ven_msg_tc	rw	0x0	
10 : 6	rg_ven_msg_type	rw	0x0	
12 : 11	rg_ven_msg_fmt	rw	0x0	

66.23 pcie_cfg19				0x4078005C
Bits	Field Name	Acc	Default	Description
31 : 0	rg_ven_msg_data_low	rw	0x0	

66.24 pcie_cfg20				0x40780060
Bits	Field Name	Acc	Default	Description
31 : 0	rg_ven_msg_data_high	rw	0x0	

66.25 pcie_cfg21				0x40780064
Bits	Field Name	Acc	Default	Description
1 : 0	rg_emb2app_irq_sel	rw	0x3	
2	rg_msi_req_p	rw	0x0	
3	rg_msi_clr	rw	0x0	
6 : 4	rg_msi_func_num	rw	0x0	
7	rg_msi_vector_sw	rw	0x0	
12 : 8	rg_msi_vector	rw	0x0	
18 : 16	rg_msi_tc	rw	0x0	
30	ro_msi_req	ro	0x0	
31	ro_msi_en	ro	0x0	

66.26 pcie_cfg22				0x40780068
Bits	Field Name	Acc	Default	Description
0	rg_cfg_bus_master_en_int_en	rw	0x0	
1	rg_cfg_mem_space_en_int_en	rw	0x0	

2	rg_radm_msg_pending_int_en	rw	0x0	
3	rg_radm_msg_fifo_ovf_int_en	rw	0x0	
4	rg_pm_xtlh_block_tlp_int_en	rw	0x0	
5	rg_pm_linkst_in_l0s_int_en	rw	0x0	
6	rg_pm_linkst_in_l1_int_en	rw	0x0	
7	rg_pm_linkst_in_l2_int_en	rw	0x0	
8	rg_pm_linkst_l2_exit_int_en	rw	0x0	
9	rg_trgt_cpl_timeout_int_en	rw	0x0	
10	rg_radm_cpl_timeout_int_en	rw	0x0	
11	rg_radm_q_not_empty_int_en	rw	0x0	
12	rg_radm_qoverflow_int_en	rw	0x0	
13	rg_radmx_cmpo-ser_lookup_err_int_en	rw	0x0	
14	rg_perst_n_rise_int_en	rw	0x0	
15	rg_linkup_int_en	rw	0x0	
28 : 16	rg_sw_event_int_en	rw	0x0	
29	rg_pm_turnoff_int_en	rw	0x0	
30	rg_perst_n_fall_int_en	rw	0x0	
31	rg_link_req_rst_int_en	rw	0x0	

66.27 pcie_cfg23				0x4078006C
Bits	Field Name	Acc	Default	Description
0	rg_cfg_bus_master_en_int_clr	rw	0x0	
1	rg_cfg_mem_space_en_int_clr	rw	0x0	
2	rg_radm_msg_pending_int_clr	rw	0x0	
3	rg_radm_msg_fifo_ovf_int_clr	rw	0x0	
4	rg_pm_xtlh_block_tlp_int_clr	rw	0x0	
5	rg_pm_linkst_in_l0s_int_clr	rw	0x0	
6	rg_pm_linkst_in_l1_int_clr	rw	0x0	
7	rg_pm_linkst_in_l2_int_clr	rw	0x0	
8	rg_pm_linkst_l2_exit_int_clr	rw	0x0	
9	rg_trgt_cpl_timeout_int_clr	rw	0x0	
10	rg_radm_cpl_timeout_int_clr	rw	0x0	
11	rg_radm_q_not_empty_int_clr	rw	0x0	
12	rg_radm_qoverflow_int_clr	rw	0x0	
13	rg_radmx_cmpo-ser_lookup_err_int_clr	rw	0x0	
14	rg_perst_n_rise_int_clr	rw	0x0	
15	rg_linkup_int_clr	rw	0x0	
16	rg_sw_event0_int_clr	rw	0x0	
17	rg_sw_event1_int_clr	rw	0x0	
18	rg_sw_event2_int_clr	rw	0x0	
19	rg_sw_event3_int_clr	rw	0x0	
20	rg_sw_event4_int_clr	rw	0x0	
21	rg_sw_event5_int_clr	rw	0x0	
22	rg_sw_event6_int_clr	rw	0x0	
23	rg_sw_event7_int_clr	rw	0x0	
24	rg_sw_event8_int_clr	rw	0x0	
25	rg_sw_event9_int_clr	rw	0x0	
26	rg_sw_event10_int_clr	rw	0x0	
27	rg_sw_event11_int_clr	rw	0x0	
28	rg_sw_event12_int_clr	rw	0x0	
29	rg_pm_turnoff_int_clr	rw	0x0	
30	rg_perst_n_fall_int_clr	rw	0x0	
31	rg_link_req_rst_int_clr	rw	0x0	

66.28 pcie_cfg24				0x40780070
Bits	Field Name	Acc	Default	Description
0	ro_cfg_bus_master_en_int_raw	ro	0x0	
1	ro_cfg_mem_space_en_int_raw	ro	0x0	
2	ro_radm_msg_pending_int_raw	ro	0x0	
3	ro_radm_msg_fifo_ovf_int_raw	ro	0x0	
4	ro_pm_xtlh_block_tlp_int_raw	ro	0x0	
5	ro_pm_linkst_in_l0s_int_raw	ro	0x0	
6	ro_pm_linkst_in_l1_int_raw	ro	0x0	
7	ro_pm_linkst_in_l2_int_raw	ro	0x0	
8	ro_pm_linkst_l2_exit_int_raw	ro	0x0	
9	ro_trgt_cpl_timeout_int_raw	ro	0x0	
10	ro_radm_cpl_timeout_int_raw	ro	0x0	
11	ro_radm_q_not_empty_int_raw	ro	0x0	
12	ro_radm_qoverflow_int_raw	ro	0x0	
13	ro_radmx_cmpo-ser_lookup_err_int_raw	ro	0x0	
14	ro_perst_n_rise_int_raw	ro	0x0	
15	ro_linkup_int_raw	ro	0x0	
28 : 16	ro_sw_event_int_raw	ro	0x0	
29	ro_pm_turnoff_int_raw	ro	0x0	
30	ro_perst_n_fall_int_raw	ro	0x0	
31	ro_link_req_rst_int_raw	ro	0x0	

66.29 pcie_cfg25				0x40780074
Bits	Field Name	Acc	Default	Description
0	ro_cfg_bus_m-aster_en_int_status	ro	0x0	
1	ro_cfg_mem_s-pace_en_int_status	ro	0x0	
2	ro_radm_msg_-pending_int_status	ro	0x0	
3	ro_radm_msg_-fifo_ovf_int_status	ro	0x0	
4	ro_pm_xtlh_b-lock_tlp_int_status	ro	0x0	
5	ro_pm_linkst_in_l0s_int_status	ro	0x0	
6	ro_pm_linkst_in_l1_int_status	ro	0x0	
7	ro_pm_linkst_in_l2_int_status	ro	0x0	
8	ro_pm_linkst_l2_exit_int_status	ro	0x0	
9	ro_trgt_cpl_timeout_int_status	ro	0x0	
10	ro_radm_cpl_-timeout_int_status	ro	0x0	
11	ro_radm_q_no-t_empty_int_status	ro	0x0	
12	ro_radm_qoverflow_int_status	ro	0x0	
13	ro_radmx_cmpo-ser_lookup_err_int_status	ro	0x0	
14	ro_perst_n_rise_int_status	ro	0x0	
15	ro_linkup_int_status	ro	0x0	
28 : 16	ro_sw_event_int_status	ro	0x0	
29	ro_pm_turnoff_int_status	ro	0x0	
30	ro_perst_n_fall_int_status	ro	0x0	
31	ro_link_req_rst_int_status	ro	0x0	

66.30 pcie_cfg26				0x40780078
Bits	Field Name	Acc	Default	Description
0	rg_radm_msg_fifo_sw_rst	rw	0x0	

1	rg_radm_msg_fifo_pop	rw	0x0	
2	ro_radm_msg_fifo_empty	ro	0x1	
7 : 3	ro_radm_msg_fifo_used_cnt	ro	0x0	

66.31 pcie_cfg27				0x4078007C
Bits	Field Name	Acc	Default	Description
0	ro_radm_vendor_msg	ro	0x0	
1	ro_radm_msg_unlock	ro	0x0	
2	ro_radm_pm_turnoff	ro	0x0	
19 : 4	ro_radm_msg_req_id	ro	0x0	

66.32 pcie_cfg28				0x40780080
Bits	Field Name	Acc	Default	Description
31 : 0	ro_radm_payload_l	ro	0x0	

66.33 pcie_cfg29				0x40780084
Bits	Field Name	Acc	Default	Description
31 : 0	ro_radm_payload_h	ro	0x0	

66.34 pcie_cfg30				0x40780088
Bits	Field Name	Acc	Default	Description
2 : 0	ro_trgt_timeout_cpl_func_num	ro	0x0	
5 : 3	ro_trgt_timeout_cpl_tc	ro	0x0	
7 : 6	ro_trgt_timeout_cpl_attr	ro	0x0	
19 : 8	ro_trgt_timeout_cpl_len	ro	0x0	
27 : 20	ro_trgt_timeout_lookup_id	ro	0x0	

66.35 pcie_cfg31				0x4078008C
Bits	Field Name	Acc	Default	Description
2 : 0	ro_radm_timeout_func_num	ro	0x0	
5 : 3	ro_radm_timeout_cpl_tc	ro	0x0	
7 : 6	ro_radm_timeout_cpl_attr	ro	0x0	
19 : 8	ro_radm_timeout_cpl_len	ro	0x0	
27 : 20	ro_radm_timeout_cpl_tag	ro	0x0	

66.36 pcie_cfg32				0x40780090
Bits	Field Name	Acc	Default	Description
5 : 0	rg_tport_sel	rw	0x3F	
7 : 6	rg_linkup_int_sel	rw	0x0	
10 : 8	rg_diag_ctrl_bus	rw	0x0	
20 : 12	rg_pcs_rxdetect_delay_low	rw	0x1FF	
29 : 21	rg_pcs_rxdetect_delay_high	rw	0x9F	

66.37 pcie_cfg33				0x40780094
Bits	Field Name	Acc	Default	Description
1 : 0	ro_tph_req_en	ro	0x0	
2	ro_pm_status	ro	0x0	
7 : 3	ro_cfg_pbus_dev_num	ro	0x0	
15 : 8	ro_cfg_pbus_num	ro	0x0	
18 : 16	ro_pm_curnt_state	ro	0x0	
20	ro_radm_q_not_empty	ro	0x0	

21	ro_trgt_lookup_empty	ro	0x1	
31 : 24	ro_trgt_lookup_id	ro	0x0	

66.38 pcie_cfg34				0x40780098
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar0_start_l	ro	0x0	

66.39 pcie_cfg35				0x4078009C
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar0_start_h	ro	0x0	

66.40 pcie_cfg36				0x407800A0
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar0_limit_l	ro	0x0	

66.41 pcie_cfg37				0x407800A4
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar0_limit_h	ro	0x0	

66.42 pcie_cfg38				0x407800A8
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar1_start	ro	0x0	

66.43 pcie_cfg39				0x407800AC
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar1_limit	ro	0x0	

66.44 pcie_cfg40				0x407800B0
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar2_start_l	ro	0x0	

66.45 pcie_cfg41				0x407800B4
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar2_start_h	ro	0x0	

66.46 pcie_cfg42				0x407800B8
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar2_limit_l	ro	0x0	

66.47 pcie_cfg43				0x407800BC
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar2_limit_h	ro	0x0	

66.48 pcie_cfg44				0x407800C0
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar3_start	ro	0x0	

66.49 pcie_cfg45				0x407800C4
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar3_limit	ro	0x0	

66.50 pcie_cfg46				0x407800C8
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar4_start_l	ro	0x0	

66.51 pcie_cfg47				0x407800CC
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar4_start_h	ro	0x0	

66.52 pcie_cfg48				0x407800D0
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar4_limit_l	ro	0x0	

66.53 pcie_cfg49				0x407800D4
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar4_limit_h	ro	0x0	

66.54 pcie_cfg50				0x407800D8
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar5_start	ro	0x0	

66.55 pcie_cfg51				0x407800DC
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_bar5_limit	ro	0x0	

66.56 pcie_cfg52				0x407800E0
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_exp_rom_start	ro	0x0	

66.57 pcie_cfg53				0x407800E4
Bits	Field Name	Acc	Default	Description
31 : 0	ro_cfg_exp_rom_limit	ro	0x0	

66.58 pcie_cfg55				0x407800EC
Bits	Field Name	Acc	Default	Description
0	rg_sw_event0_int_set	rw	0x0	
1	rg_sw_event1_int_set	rw	0x0	
2	rg_sw_event2_int_set	rw	0x0	
3	rg_sw_event3_int_set	rw	0x0	
4	rg_sw_event4_int_set	rw	0x0	
5	rg_sw_event5_int_set	rw	0x0	
6	rg_sw_event6_int_set	rw	0x0	
7	rg_sw_event7_int_set	rw	0x0	
8	rg_sw_event8_int_set	rw	0x0	
9	rg_sw_event9_int_set	rw	0x0	
10	rg_sw_event10_int_set	rw	0x0	

11	rg_sw_event11_int_set	rw	0x0	
12	rg_sw_event12_int_set	rw	0x0	

66.59 pcie_cfg56				0x407800F0
Bits	Field Name	Acc	Default	Description

66.60 pcie_cfg57				0x407800F4
Bits	Field Name	Acc	Default	Description

66.61 pcie_cfg58				0x407800F8
Bits	Field Name	Acc	Default	Description

66.62 pcie_cfg59				0x407800FC
Bits	Field Name	Acc	Default	Description

66.63 pcie_cfg60				0x40780100
Bits	Field Name	Acc	Default	Description
0	rg_tx_prbs_mode	rw	0x0	
1	rg_tx_prbs_load	rw	0x0	
11 : 2	rg_tx_prbs_seed	rw	0x0	

66.64 pcie_cfg61				0x40780104
Bits	Field Name	Acc	Default	Description
0	rg_rx_prbs_mode	rw	0x0	
4 : 1	rg_rx_prbs_time	rw	0x1	
11 : 5	rg_rx_err_tolerance	rw	0x7F	
12	ro_rx_prbs_valid	ro	0x0	
13	ro_rx_prbs_end	ro	0x0	
14	ro_rx_prbs_pass	ro	0x0	
31 : 16	ro_rx_prbs_errcnt	ro	0x0	

66.65 pcie_cfg62				0x40780108
Bits	Field Name	Acc	Default	Description
0	rg_tx_const_en	rw	0x0	
17 : 8	rg_tx_const_data	rw	0x0	

66.66 pcie_cfg63				0x4078010C
Bits	Field Name	Acc	Default	Description
0	rg_mstr_haddr_remap_en	rw	0x0	
1	rg_mstr_haddr_remap_mode	rw	0x0	
11 : 4	rg_mstr_haddr_remap	rw	0x0	

66.67 pcie_cfg64				0x40780110
Bits	Field Name	Acc	Default	Description
0	rg_serdes_en	rw	0x0	
1	rg_serdes_tx_data_en	rw	0x0	
2	rg_serdes_tx_disable_scrambler	rw	0x1	
3	rg_serdes_tx_swizzle_en	rw	0x0	
4	rg_serdes_tx_inv_en	rw	0x0	

5	rg_serdes_tx_prbs_mode	rw	0x0	
6	rg_serdes_tx_prbs7_load	rw	0x0	
13 : 7	rg_serdes_tx_prbs7_seed	rw	0x0	
16	ro_serdes_txfifo_full	ro	0x0	
17	ro_serdes_txfifo_empty	ro	0x1	
21 : 18	ro_serdes_txfifo_rd_cnt	ro	0x0	

66.68 pcie_cfg65				0x40780114
Bits	Field Name	Acc	Default	Description
9 : 0	rg_serdes_tx_const_data	rw	0x0	
10	rg_serdes_tx_const_en	rw	0x0	
11	rg_serdes_tx_test_en	rw	0x0	
19 : 12	rg_serdes_tx_test_data	rw	0x0	
20	rg_serdes_tx_test_charisk	rw	0x0	
21	rg_serdes_tx_test_hack_en	rw	0x0	
31 : 24	rg_serdes_tx_com	rw	0x0	

67 pmu_rtc0

67.1 pmu_rtc0_cfg0				0x70000000
Bits	Field Name	Acc	Default	Description
0	pmu_rtc0_rg_update	w1c	0x0	
1	rtc_rg_pb_in-t_debc_seq_pd_bypass0	rw	0x1	
2	rtc_rg_pb_in-t_debc_seq_pd_bypass1	rw	0x0	
3	rtc_rg_gpio_-debc_seq_pd_bypass0	rw	0x0	
4	rtc_rg_gpio_-debc_seq_pd_bypass1	rw	0x0	
7 : 5	rtc_rg_bg_pd_seq_num	rw	0x7	
8	rtc_rg_force_pbint_neg_pwroff	rw	0x0	
9	rtc_rg_cm_co_xtal_det_out	ro	0x0	
10	rtc_rg_cm_co_xtal_det_clr_en	rw	0x0	
11	rtc_rg_cm_co_xtal_det_hw_en	rw	0x1	
12	rtc_rg_cm_co_xtal_det_sw_en	rw	0x0	
14 : 13	rtc_rg_cm_co_xtal_det_th	rw	0x3	
15	rtc_rg_pmu_ms_vio_rtc0	ro	0x0	
16	rtc_rg_pmu_ms_ldo33_rtc0	ro	0x0	
17	rtc_rg_pmu_m-s_avdd33usb_rtc0	ro	0x0	
31	rtc_rg_pmu_soft_rstn	rw	0x1	

67.2 pmu_rtc0_cfg1				0x70000004
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_ldo_rf_pu_seq_num	rw	0xB	
9 : 5	rtc_rg_ldo_vrtc08_pu_seq_num	rw	0x5	
17 : 10	rtc_rg_vbat_rstn_th	rw	0x10	
18	rtc_rg_pbint_-long_rst_repower_en	rw	0x0	
19	rtc_rg_pbint_long_rst_pd_en	rw	0x0	
20	rtc_rg_bor_repower_en	rw	0x0	
21	rtc_rg_bor_pwroff_en	rw	0x0	
29 : 22	rtc_rg_bor_high_deglitch_th	rw	0xA	
30	rtc_rg_bor_high_deglitch_en	rw	0x0	
31	rtc_rg_bor_en	rw	0x0	

67.3 pmu_rtc0_cfg2				0x70000008
Bits	Field Name	Acc	Default	Description
3 : 0	rtc_rg_ldo_rf_vbit	rw	0xC	
11 : 4	rtc_rg_pbint_-por_trig_cnt_low_th	rw	0x8	
19 : 12	rtc_rg_pbint_-por_trig_cnt_high_th	rw	0x0	
23 : 20	rtc_rg_gpio_cfg	rw	0x0	
24	rtc_rg_charger_on_en	rw	0x1	
25	rtc_rg_charger_on_inv_sel	rw	0x0	
26	rtc_rg_pu_clk_32k_rc_sw	rw	0x0	
27	rtc_rg_pwr_on_rdy_sw	rw	0x0	
31 : 28	rtc_rg_dbg_mux	rw	0x0	

67.4 pmu_rtc0_cfg3				0x7000000C
Bits	Field Name	Acc	Default	Description
7 : 0	rtc_rg_gpio_-por_trig_cnt_low_th	rw	0x8	
15 : 8	rtc_rg_gpio_-por_trig_cnt_high_th	rw	0x0	
16	rtc_rg_wdg_rstn_repower_en	rw	0x0	
17	rtc_rg_wdg_rstn_pd_en	rw	0x1	
18	rtc_rg_pu_dcdc_ldo_mode_sel	rw	0x0	
19	rtc_rg_alm_power_on_en	rw	0x0	
20	rtc_rg_aon_gpio_pwr_on_en	rw	0x0	
21	rtc_rg_aon_gpio_pwr_off_en	rw	0x0	
22	rtc_rg_power_off_pmu_rtc0_en	rw	0x1	
23	rtc_rg_gpio_power_off_rtc0_en	rw	0x0	
24	rtc_rg_wdg_r-stn_power_off_rtc0_en	rw	0x1	
25	rtc_rg_long_pbint_power_off_en	rw	0x1	
26	rtc_rg_pb_in-t_long_rst_power_off_rtc0_en	rw	0x0	
27	rtc_rg_device_set	rw	0x1	
28	rtc_rg_resetn_rtc_cnt	rw	0x0	
29	rtc_rg_presetn_rtc_cnt	rw	0x0	
30	rtc_rg_wdg_rst_rtc_rf_en	rw	0x0	
31	rtc_rg_aon_gpio_neg_clr_en	rw	0x1	

67.5 pmu_rtc0_cfg4				0x70000010
Bits	Field Name	Acc	Default	Description
0	rtc_rg_int_en	rw	0x0	
1	rtc_rg_rtc_cnt_int_en	rw	0x0	
6 : 2	rtc_rg_int_mask	rw	0x0	
7	rtc_rg_rtc_cnt_irq_mask	rw	0x0	
8	rtc_rg_int_clr1	w1c	0x0	
9	rtc_rg_int_clr2	w1c	0x0	
10	rtc_rg_int_clr3	w1c	0x0	
11	rtc_rg_int_clr4	w1c	0x0	
12	rtc_rg_int_clr5	w1c	0x0	
13	rtc_rg_rtc_cnt_irq_clr	w1c	0x0	
18 : 14	rtc_rg_int_raw_status	ro	0x0	
19	rtc_cnt_irq_raw_status	ro	0x0	
24 : 20	rtc_rg_int_mask_status	ro	0x0	
25	rtc_cnt_irq_mask_status	ro	0x0	
28 : 26	rtc_rg_gpioa_hold	rw	0x0	
29	rtc_rg_gpiob_hold	rw	0x0	

30	rtc_rg_agpio_hold	rw	0x0	
31	rtc_rg_fls_hold	rw	0x0	

67.6 pmu_rtc0_cfg5				0x70000014
Bits	Field Name	Acc	Default	Description
31 : 0	rtc_rg_rtc_cnt_ld_value	rw	0x0	

67.7 pmu_rtc0_cfg6				0x70000018
Bits	Field Name	Acc	Default	Description
31 : 0	rtc_cnt_value	ro	0x0	

67.8 pmu_rtc0_cfg7				0x7000001C
Bits	Field Name	Acc	Default	Description
0	rtc_rg_rtc_cnt_enable	rw	0x0	
1	rtc_rg_rtc_cnt_ld_vld	w1c	0x0	
2	rtc_rg_rtc_cnt_mode	rw	0x1	
6 : 3	rtc_rg_bor_low_deglitch_th	rw	0x4	
7	rtc_rg_bor_low_deglitch_en	rw	0x0	
8	rtc_rg_xtal_40m_26m_soft_sel	rw	0x0	
9	rtc_rg_xtal_-40m_26m_hw_detect_high_latch_clr	rw	0x0	
10	xtal_40m_26m_-hw_detect_high_latch	ro	0x0	
11	rtc_rg_pb_int_long_por_set_clr	w1c	0x0	
12	rtc_rg_alm_int_set_clr	w1c	0x0	
13	rtc_rg_wdg_pwr_on_set_clr	w1c	0x0	
14	rtc_rg_bor_pwr_on_set_clr	w1c	0x0	
15	rtc_rg_por_chip_vbat_flag	ro	0x0	
23 : 16	rtc_rg_pwr_on_src_status	ro	0x0	
31 : 24	rtc_rg_pbint_debc_cnt_th	rw	0x0	

67.9 pmu_rtc0_wdg4_cnt_ini				0x70000020
Bits	Field Name	Acc	Default	Description
31 : 0	rtc_rg_wdg4_cnt_ini	rw	0x2625A00	

67.10 rtc_rg_reserved0				0x70000024
Bits	Field Name	Acc	Default	Description

67.11 rtc_rg_reserved1				0x70000028
Bits	Field Name	Acc	Default	Description

68 pmu_rtc1

68.1 pmu_pd_ctrl0				0x70001000
Bits	Field Name	Acc	Default	Description
0	rtc_rg_dcdc_ldo_rf_mode_sel	rw	0x0	
1	rtc_rg_pwr_force_ldo_avdd18	rw	0x0	
2	rtc_rg_pwr_force_ldo_vcore08	rw	0x0	
3	rtc_rg_pwr_force_dcdc_vcore08	rw	0x0	
4	rtc_rg_pwr_force_ldo_pa	rw	0x0	
5	rtc_rg_ldo_avdd18_pd	rw	0x0	

6	rtc_rg_ldo_vcore08_pd	rw	0x0	
7	rtc_rg_dcdc_vcore08_pd	rw	0x1	
8	rtc_rg_ldo_pa_pd	rw	0x0	
9	rtc_rg_bg_pd_set	rw	0x0	
10	rtc_rg_ldo_vrtc08_pd_set	rw	0x0	
11	rtc_rg_dcdc_ldo_rf_pd_set	rw	0x0	
12	rtc_rg_pmu_core_rstn_set	rw	0x0	
13	rtc_rg_pmu_core_iso_set	rw	0x0	
14	rtc_rg_pwr_force_ldo_vflash	rw	0x0	
15	rtc_rg_ldo_vflash_pd	rw	0x0	
16	rtc_rg_ret_n	rw	0x0	
17	rtc_rg_pwr_force_chopp_bg	rw	0x0	
18	rtc_rg_chopp_bg_pd	rw	0x0	
19	rtc_rg_chopp_bg_choppen	rw	0x1	
20	rtc_rg_pmu_c-hopp_bg_pu_lpvref	rw	0x0	
21	rtc_rg_pmu_c-hopp_bg_sw_lpvref_set	rw	0x0	
22	rtc_rg_gpio_sel_ldo18	rw	0x1	
23	rtc_rg_reg18_ulp_pu_set	rw	0x0	
24	rtc_rg_pu_dcdc_core_mode_sel	rw	0x0	
25	rtc_rg_auto_pd_ldo_vcore08_en	rw	0x0	

68.2 pmu_pd_ctrl1				0x70001004
Bits	Field Name	Acc	Default	Description
0	rtc_rg_pwr_off_seq_en	rw	0x0	
1	rtc_rg_ext_soft_rst	w1c	0x0	
2	rtc_rg_core_iso_en	rw	0x0	
3	rtc_rg_global_rstn_en	rw	0x0	
4	rtc_rg_gpio_vio13_lp_en_set	rw	0x0	
5	rtc_rg_dcdc_en_avdd18int	rw	0x1	
6	rtc_rg_dcdc_en_avdd18	rw	0x0	
7	rtc_rg_coreon_set	rw	0x1	
8	rtc_rg_lvl_pullr_enb	rw	0x1	
9	rtc_rg_core_sw_fast_clk_sel_set	rw	0x1	
11 : 10	rtc_rg_avddsense_sel	rw	0x0	
12	rtc_rg_pu_vrtc0sense	rw	0x0	
13	rtc_rg_pu_vbatsense	rw	0x0	

68.3 pmu_seq_num0				0x70001008
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_slp_xtal_pu_seq_num	rw	0x2	
9 : 5	rtc_rg_reset_b_seq_num	rw	0x6	
14 : 10	rtc_rg_ldo_pa_pu_seq_num	rw	0xA	
19 : 15	rtc_rg_ldo_avdd18_pu_seq_num	rw	0x6	
24 : 20	rtc_rg_ldo_d-cdc_vcore08_pu_seq_num	rw	0x1	
31 : 25	rtc_rg_dcdc_dll_start_th	rw	0x30	

68.4 pmu_seq_num1				0x7000100C
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_slp_reset_b_seq_num	rw	0x3	
9 : 5	rtc_rg_slp_xen_en_seq_num	rw	0x2	
14 : 10	rtc_rg_ldo_d-cdc_vcore08_dvfs_seq_num	rw	0xF	

19 : 15	rtc_rg_ldo_v-rtc08_dvfs_seq_num	rw	0xF	
24 : 20	rtc_rg_dcdc_rf_dvfs_seq_num	rw	0xF	
31 : 25	rtc_rg_dcdc_dll_rstn_th	rw	0x18	

68.5 pmu_slp_ctrl0				0x70001010
Bits	Field Name	Acc	Default	Description
0	rtc_rg_slp_ldo_pd_en	rw	0x0	
1	rtc_rg_rtc_slp_pmu_rtc0_pd_en	rw	0x0	
2	rtc_rg_rtc_slp_pmu_rtc1_pd_en	rw	0x0	
3	rtc_rg_ldo_pa_slp_pd_en	rw	0x0	
4	rtc_rg_reg18_ulp_slp_pu_en	rw	0x0	
5	rtc_rg_ldo_d-cdc_vcore08_slp_pd_en	rw	0x0	
6	rtc_rg_ldo_avdd18_slp_pd_en	rw	0x0	
7	rtc_rg_ldo_vrtc08_slp_pd_en	rw	0x0	
8	rtc_rg_core_iso_slp_en	rw	0x0	
9	rtc_rg_coreon_slp_pd_en	rw	0x0	
10	rtc_rg_xtal_slp_pd_en	rw	0x0	
11	rtc_rg_lpo26m_slp_pd_en	rw	0x0	
12	rtc_rg_rc_256k_slp_pd_en	rw	0x0	
13	rtc_rg_ldo_vflash_slp_pd_en	rw	0x0	
14	rtc_rg_gpio_sel_ldo18_slp_en	rw	0x0	
15	rtc_rg_dcdc_-core_adc_slp_pd_en	rw	0x0	
16	rtc_rg_dcdc_rf_adc_slp_pd_en	rw	0x0	
17	rtc_rg_chopp_bg_slp_pd_en	rw	0x0	
18	rtc_rg_cm_xt-al_slp_reg_buf_bypass	rw	0x0	
19	rtc_rg_xdblr_slp_pd_en	rw	0x0	
20	rtc_rg_ldo_v-core08_awake_pu_en	rw	0x0	
21	rtc_rg_aon13_sw2v13_slp_en	rw	0x0	
22	rtc_rg_ldo_a-vdd18_ulpmode_slp_en	rw	0x0	
26 : 23	rtc_rg_deep_sleep_awake_en	rw	0x0	
27	rtc_rg_xtal_slp_hold_en	rw	0x0	
28	rtc_rg_gpio_vio13_lp_slp_en	rw	0x0	

68.6 pmu_slp_ctrl1				0x70001014
Bits	Field Name	Acc	Default	Description
0	rtc_rg_reset_b_slp_en	rw	0x0	
1	rtc_rg_slp_rtc_pclk_gate_en	rw	0x0	
2	rtc_rg_slp_core_clk_fast_sel_en	rw	0x0	
3	rtc_rg_xtal_slp_lp_en	rw	0x0	
4	rtc_rg_gate_to_dbb_slp_en	rw	0x0	
5	rtc_rg_ldo_pa_slp_drop_en	rw	0x0	
6	rtc_rg_ldo_vrtc08_dvfs_slp_en	rw	0x0	
7	rtc_rg_ldo_d-cdc_vcore08_dvfs_slp_en	rw	0x0	
8	rtc_rg_ldo_avdd18_slp_lp_en	rw	0x0	
9	rtc_rg_ldo_avdd18_slp_drop_en	rw	0x0	
10	rtc_rg_dcdc_-core_hys_pfm_mode_slp_en	rw	0x0	
11	rtc_rg_dcdc_-rf_hys_pfm_mode_slp_en	rw	0x0	
12	rtc_rg_dcdc_rf_dvfs_slp_en	rw	0x0	

13	rtc_rg_dcdc_-core_deep_slp_slp_en	rw	0x0	
14	rtc_rg_ldo_pa_slp_lp_en	rw	0x0	
15	rtc_rg_lpo26m_slp_dis_en	rw	0x0	
16	rtc_rg_rc_256k_slp_dis_en	rw	0x0	
17	rtc_rg_clk_26m_slp_dis_en	rw	0x0	
18	rtc_rg_clk_comp_slp_dis_en	rw	0x0	
19	rtc_rg_clk_buf_slp_dis_en	rw	0x0	
20	rtc_rg_clk_6p5m_slp_dis_en	rw	0x0	
21	rtc_rg_dcdc_rf_clk_sel_slp_en	rw	0x0	
22	rtc_rg_dcdc_rf_deep_slp_slp_en	rw	0x0	
23	rtc_rg_clk_52m_slp_dis_en	rw	0x0	
24	rtc_rg_clk_26m_dly_slp_dis_en	rw	0x0	
25	rtc_rg_ldo_vflash_slp_lp_en	rw	0x0	
26	rtc_rg_ldo_vflash_slp_drop_en	rw	0x0	
27	rtc_rg_xtal_avdd_sel_slp_en	rw	0x0	
28	rtc_rg_xtal_clk_div4_sel_slp_en	rw	0x0	
29	rtc_rg_dcdc_core_clk_sel_slp_en	rw	0x0	
30	rtc_rg_ldo_vcore08_slp_lp_en	rw	0x0	
31	rtc_rg_chopp_-bg_sw_lpvref_lp_en	rw	0x0	

68.7 pmu_clk_ctrl0 0x70001018

Bits	Field Name	Acc	Default	Description
0	rtc_rg_clk_d-iv_128k_step_offset_update	rw	0x0	
8 : 1	rtc_rg_clk_d-iv_128k_step_offset_normal	rw	0x0	
16 : 9	rtc_rg_clk_d-iv_128k_step_offset_lp	rw	0x0	
17	rtc_rg_clk_d-iv_128k_lp_mode_h_reg	rw	0x0	
18	rtc_rg_clk_d-iv_128k_lp_mode_h_dr	rw	0x0	
19	rtc_rg_clk_6p5m_div_128k_en	rw	0x1	
20	rtc_rg_xtal_hold_set	rw	0x0	
28 : 21	rtc_rg_pmu_clk_manual_enable	rw	0xFF	

68.8 pmu_clk_ctrl1 0x7000101C

Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_xtal_cap_rf_on_bit	rw	0x10	
10 : 5	rtc_rg_xtal_cap_rf_on_delta	rw	0x38	
15 : 11	rtc_rg_xtal_cap_nor_bit	rw	0x10	
21 : 16	rtc_rg_xtal_cap_nor_delta	rw	0x38	
26 : 22	rtc_rg_xtal_cap_step	rw	0x0	
31 : 27	rtc_rg_xtal_ibias_step	rw	0x0	

68.9 pmu_clk_ctrl2 0x70001020

Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_xtal_cap_lp_bit	rw	0x10	
10 : 5	rtc_rg_xtal_cap_lp_delta	rw	0x38	
11	rtc_rg_xtal_buf_en_set	rw	0x1	
12	rtc_rg_xtal_comp_en_set	rw	0x1	
13	rtc_rg_xtal_clk_26m_en_set	rw	0x1	
14	rtc_rg_xtal_clk_6p5m_en_set	rw	0x1	
15	rtc_rg_rc_256k_en_set	rw	0x1	

16	rtc_rg_rc_128k_sel_set	rw	0x0	
17	rtc_rg_rc_128k_sel	rw	0x1	
18	rtc_rg_pclk_pmu_rtc1_en	rw	0x1	
19	rtc_rg_pu_xtal_set	rw	0x1	
20	rtc_rg_lpo26m_en_set	rw	0x1	
21	rtc_rg_pu_lpo26m_set	rw	0x1	
22	rtc_rg_pu_lpo256k_set	rw	0x1	
23	rtc_rg_clk_128k_out_en_sw_set	rw	0x0	
24	rtc_rg_clk_1-28k_out_en_sw_mode	rw	0x0	
25	rtc_rg_xtal_clk_52m_en_set	rw	0x1	
26	rtc_rg_xtal_clk_26m_dly_en_set	rw	0x1	
27	rtc_rg_xtal_slp_lp_en_set	rw	0x0	
28	rtc_rg_xtal_comp_clk_en_set	rw	0x1	
29	rtc_rg_pu_xdblr_set	rw	0x1	
30	rtc_rg_xdblr_clk_en_set	rw	0x1	
31	rtc_rg_xdblr_clk_sel	rw	0x0	

68.10 pmu_clk_ctrl3				0x70001024
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_xtal_ibias_rf_on_bit	rw	0x19	
9 : 5	rtc_rg_xtal_ibias_nor_bit	rw	0x19	
13 : 10	rtc_rg_lpo256k_fbit	rw	0x7	
14	rtc_rg_dis_clk32k_2power_dr	rw	0x0	
15	rtc_rg_dis_clk32k_2power_reg	rw	0x0	
23 : 16	rtc_rg_clk_26m_en_sw_set	rw	0x0	
31 : 24	rtc_rg_clk_26m_en_sw_mode	rw	0xF8	

68.11 pmu_rstn_ctrl				0x70001028
Bits	Field Name	Acc	Default	Description
0	rtc_rg_presetn_en	rw	0x0	
1	rtc_rg_wdg_soft_rst_en	rw	0x0	
5 : 2	rtc_rg_vbit_uvlo	rw	0x3	
6	rtc_rg_vbit_vio18	rw	0x0	
7	rtc_rg_bor_sel_dly	rw	0x0	
9 : 8	rtc_rg_vbit_ms_vio	rw	0x2	
13 : 10	rtc_rg_bor_vbit	rw	0x6	
14	rtc_rg_bor_sel	rw	0x0	
15	rtc_rg_bor_enable	rw	0x0	
16	rtc_rg_bor_soft_rstn_en	rw	0x0	
17	rtc_rg_chopp_bg_test_en	rw	0x0	
20 : 18	rtc_rg_chopp_bg_test_bit	rw	0x0	
22 : 21	rtc_rg_chopp_bg_fbit	rw	0x2	
23	rtc_rg_chopp_bg_osc_en	rw	0x1	
24	rtc_rg_chopp_bg_clk_en	rw	0x0	
31 : 25	rtc_rg_pmu_r-eset_manual_enable	w1c	0x0	

68.12 pmu_dcdc_rf_ctrl0				0x7000102C
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_dcdc_-rf_vbit_vofb_lpvref_th	rw	0x12	
5	rtc_rg_pu_dc-dc_rf_resetn_bypass	rw	0x0	
11 : 6	rtc_rg_dcdc_rf_dvfs_step_vol	rw	0x0	
12	rtc_rg_dcdc_rf_dvfs_step_en	rw	0x0	

14 : 13	rtc_rg_dcdc_rf_dvfs_step_delay	rw	0x0	
23 : 15	rtc_rg_dcdc_rf_soft_cfg	rw	0x0	
24	rtc_rg_dcdc_- rf_dcdc_clk_lpo_sel_sw	rw	0x0	
25	rtc_rg_dcdc_rf_dcdc_clk_lpo_sel	rw	0x0	
28 : 26	rtc_rg_dcdc_rf_bit_vfb_decap	rw	0x0	
29	rtc_rg_dcdc_rf_force_res_on	rw	0x0	

68.13 pmu_dcdc_rf_ctrl1				0x70001030
Bits	Field Name	Acc	Default	Description
5 : 0	rtc_rg_dcdc_rf_vbit_nor_tune	rw	0x19	
11 : 6	rtc_rg_dcdc_rf_vbit_nor_rf	rw	0x19	
17 : 12	rtc_rg_dcdc_rf_vbit_ds_sw	rw	0x19	
23 : 18	rtc_rg_dcdc_- rf_vbit_vout_valley_delta	rw	0xC	
24	rtc_rg_dcdc_- rf_nor_pfm_mode_en	rw	0x0	
26 : 25	rtc_rg_dcdc_rf_pulse_time	rw	0x1	
27	rtc_rg_dcdc_rf_en_chopp	rw	0x0	
28	rtc_rg_dcdc_rf_en_autozero	rw	0x1	
29	rtc_rg_dcdc_rf_dis_nld	rw	0x1	
30	rtc_rg_dcdc_rf_en_ldo_curlim	rw	0x1	
31	rtc_rg_dcdc_rf_en_nsw	rw	0x0	

68.14 pmu_dcdc_rf_ctrl2				0x70001034
Bits	Field Name	Acc	Default	Description
3 : 0	rtc_rg_dcdc_rf_adc_dly_bit	rw	0x1	
6 : 4	rtc_rg_dcdc_rf_adc_cmp_ibit	rw	0x4	
8 : 7	rtc_rg_dcdc_rf_op_ibit	rw	0x1	
10 : 9	rtc_rg_dcdc_rf_pga_gain_bit	rw	0x1	
11	rtc_rg_aon13_sw2v13_sw_en	rw	0x0	
12	rtc_rg_aon13_sw2v13_sw_set	rw	0x0	
14	rtc_rg_dcdc_rf_pu_pga	rw	0x1	
15	rtc_rg_dcdc_rf_pu_dpwm	rw	0x1	
19 : 16	rtc_rg_dcdc_rf_noverlap_bit	rw	0x8	
20	rtc_rg_dcdc_rf_dll_band	rw	0x1	
25 : 21	rtc_rg_dcdc_rf_start_count	rw	0x10	
28 : 26	rtc_rg_dcdc_rf_lpo_freq_bit	rw	0x4	
31 : 29	rtc_rg_dcdc_rf_dly_bit_clk_zcs	rw	0x4	

68.15 pmu_dcdc_rf_ctrl3				0x70001038
Bits	Field Name	Acc	Default	Description
0	rtc_rg_dcdc_rf_ldo_lpm	rw	0x0	
3 : 1	rtc_rg_dcdc_rf_ibit_ldo	rw	0x7	
4	rtc_rg_dcdc_en_fresp_nm18	rw	0x1	
5	rtc_rg_dcdc_en_fresp_vbp	rw	0x1	
6	rtc_rg_dcdc_sw_vnm_tov13	rw	0x0	
7	rtc_rg_dcdc_- rf_noverlap_ana_hold	rw	0x1	
8	rtc_rg_dcdc_- rf_noverlap_ana_pw	rw	0x0	
9	rtc_rg_dcdc_- rf_pu_res_sample_soft_set	rw	0x0	
10	rtc_rg_dcdc_- rf_pu_res_sample_soft_en	rw	0x0	
23 : 21	rtc_rg_dcdc_rf_en_pmos	rw	0x7	

26 : 24	rtc_rg_dcdc_rf_en_nmos	rw	0x7	
30 : 28	rtc_rg_dcdc_fbit_lpo_1p6m	rw	0x4	

68.16 pmu_ldo_vcore08_ctrl				0x7000103C
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_ldo_v-core08_vosel_nor_tune	rw	0x16	
9 : 5	rtc_rg_ldo_vcore08_vosel_nor_rf	rw	0x16	
14 : 10	rtc_rg_ldo_vcore08_vosel_ds_sw	rw	0x16	
19 : 15	rtc_rg_ldo_v-core08_dvfs_step_vol	rw	0x0	
20	rtc_rg_ldo_v-core08_dvfs_step_en	rw	0x0	
22 : 21	rtc_rg_ldo_v-core08_dvfs_step_delay	rw	0x0	
23	rtc_rg_ldo_vcore08_soft_lim	rw	0x1	
24	rtc_rg_ldo_v-core08_slp_lp_en_set	rw	0x0	
28 : 25	rtc_rg_ldo_vcore08_ibit_lim	rw	0xF	
29	rtc_rg_ldo_vcore08_en_lim	rw	0x1	

68.17 pmu_ldo_vrtc08_ctrl				0x70001040
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_ldo_v-rtc08_vosel_nor_tune	rw	0x19	
9 : 5	rtc_rg_ldo_vrtc08_vosel_ds_sw	rw	0x19	
14 : 10	rtc_rg_ldo_v-rtc08_dvfs_step_vol	rw	0x0	
15	rtc_rg_ldo_vrtc08_dvfs_step_en	rw	0x0	
17 : 16	rtc_rg_ldo_v-rtc08_dvfs_step_delay	rw	0x0	
18	rtc_rg_ldo_vrtc08_disable_clk	rw	0x0	
19	rtc_rg_ldo_vrtc08_more_iss	rw	0x0	

68.18 pmu_other_ldo_ctrl0				0x70001044
Bits	Field Name	Acc	Default	Description
3 : 0	rtc_rg_ldo_pa_vbit_ds_sw	rw	0xD	
7 : 4	rtc_rg_ldo_pa_vbit_nor_tune	rw	0xD	
15 : 11	rtc_rg_vbit_ret	rw	0xC	
19 : 16	rtc_rg_ldo_avdd18_vbit_ds_sw	rw	0x3	
23 : 20	rtc_rg_ldo_a-vdd18_vbit_nor_tune	rw	0x3	
24	rtc_rg_ldo_avdd18_lp_en_set	rw	0x0	
25	rtc_rg_ldo_pa_en_ldo_curlim	rw	0x1	
28 : 26	rtc_rg_ldo_pa_ibit_lim	rw	0x4	
30	rtc_rg_ldo_pa_lp_en_set	rw	0x0	

68.19 pmu_other_ldo_ctrl1				0x70001048
Bits	Field Name	Acc	Default	Description
0	rtc_rg_reg18_ulp_vbit	rw	0x0	
1	rtc_rg_ldo33_bypass	rw	0x0	
2	rtc_rg_ldo_avdd18_lpmode_set	rw	0x0	
3	rtc_rg_reg18_ulp_enclamp_set	rw	0x0	
4	rtc_rg_ldo18_en_curlim	rw	0x1	
7 : 5	rtc_rg_ldo18_ibit	rw	0x4	

8	rtc_rg_ldo_a-vdd18_ulpmode_set	rw	0x0	
10 : 9	rtc_rg_vrtc0_vbit_ms_avdd33	rw	0x2	
11	rtc_rg_vrtc0_low_detclk	rw	0x0	
13 : 12	rtc_rg_vrtc0_vbit_ms_ldo33	rw	0x2	

68.20 pmu_long_pwr_key_cfg				0x7000104C
Bits	Field Name	Acc	Default	Description
0	rtc_rg_pbint_7s_sw_rst_mode	rw	0x0	
4 : 1	rtc_rg_pbint_7s_rst_threshold	rw	0x8	
5	rtc_rg_pbint_7s_rst_mode	rw	0x0	
6	rtc_rg_pbint_7s_rst_disable	rw	0x1	

68.21 pmu_rf_on_cfg				0x70001050
Bits	Field Name	Acc	Default	Description
0	rtc_rg_rf_tx_on_xtal_mode_en	rw	0x0	
1	rtc_rg_rf_tx-_on_ldo_dcdc_vcore08_dvfs_mode_en	rw	0x0	
2	rtc_rg_rf_tx-_on_dcdc_rf_hys_pfm_mode_en	rw	0x0	
3	rtc_rg_rf_tx-_on_dcdc_core_hys_pfm_mode_en	rw	0x0	
4	rtc_rg_rf_tx-_on_dcdc_rf_dvfs_mode_en	rw	0x0	
5	rtc_rg_rf_rx_on_xtal_mode_en	rw	0x0	
6	rtc_rg_rf_rx-_on_ldo_dcdc_vcore08_dvfs_mode_en	rw	0x0	
7	rtc_rg_rf_rx-_on_dcdc_core_hys_pfm_mode_en	rw	0x0	
8	rtc_rg_rf_rx-_on_dcdc_rf_hys_pfm_mode_en	rw	0x0	
9	rtc_rg_rf_rx-_on_dcdc_rf_dvfs_mode_en	rw	0x0	
10	rtc_rg_pu_md-ll_ldo_dcdc_vcore08_dvfs_mode_en	rw	0x0	
11	rtc_rg_pu_md-ll_dcdc_core_hys_pfm_mode_en	rw	0x0	
12	rtc_rg_pu_md-ll_dcdc_rf_hys_pfm_mode_en	rw	0x0	
13	rtc_rg_pu_md-ll_dcdc_rf_dvfs_mode_en	rw	0x0	
14	rtc_rg_pu_mdll_xtal_mode_en	rw	0x0	
15	rtc_rg_wf_rf_rx_en	rw	0x0	
16	rtc_rg_wf_rf_tx_en	rw	0x0	
17	rtc_rg_bt_rf_rx_en	rw	0x0	
18	rtc_rg_bt_rf_tx_en	rw	0x0	

68.22 pmu_psm_cfg0				0x70001054
Bits	Field Name	Acc	Default	Description
0	rtc_rg_psm_reg_update	rw	0x0	
1	rtc_rg_psm_soft_reset	rw	0x0	
2	rtc_rg_psm_26m_cal_cnt_sw_en	rw	0x0	
3	rtc_rg_psm_clk_32k_gate_en	rw	0x1	
4	rtc_rg_psm_sw_cal_en	rw	0x0	
5	rtc_rg_psm_xtal_lp_cal_en	rw	0x0	
6	rtc_rg_psm_deep_sleep_cal_en	rw	0x0	
7	rtc_rg_psm_auto_cal_en	rw	0x0	
8	rtc_rg_psm_cal_en	rw	0x0	

9	rtc_rg_psm_en	rw	0x0	
31 : 13	rtc_rg_psm_div_demon_factor	rw	0x4C4B4	

68.23 pmu_psm_cfg1				0x70001058
Bits	Field Name	Acc	Default	Description
15 : 0	rtc_rg_psm_xtal_lp_cal_offset	rw	0x0	

68.24 pmu_psm_cfg2				0x7000105C
Bits	Field Name	Acc	Default	Description
9 : 0	rtc_rg_psm_d-eep_sleep_cnt_interval_th	rw	0x0	
17 : 10	rtc_rg_psm_cal_pre_cnt_th	rw	0x2	
21 : 18	rtc_rg_psm_128k_cal_cnt_p	rw	0xA	
25 : 22	rtc_rg_psm_128k_cal_cnt_n	rw	0x3	

68.25 pmu_psm_cfg3				0x70001060
Bits	Field Name	Acc	Default	Description
15 : 0	rtc_rg_psm_26m_cal_cnt_up_th	rw	0x1000	
31 : 16	rtc_rg_psm_26m_cal_cnt_dn_th	rw	0x0	

68.26 pmu_psm_cfg4				0x70001064
Bits	Field Name	Acc	Default	Description
15 : 0	rtc_rg_psm_cnt_interval_phase	rw	0x0	
31 : 16	rtc_rg_psm_cnt_interval_th	rw	0x0	

68.27 pmu_psm_cfg5				0x70001068
Bits	Field Name	Acc	Default	Description
27 : 0	rtc_rg_psm_26m_cal_cnt_sw_set	rw	0x0	

68.28 pmu_psm_cfg6				0x7000106C
Bits	Field Name	Acc	Default	Description
27 : 0	psm_cal_cnt_ro	ro	0x0	
28	psm_reg_update_pulse_vld	ro	0x0	

68.29 pmu_psm_cfg7				0x70001070
Bits	Field Name	Acc	Default	Description
3 : 0	psm_cal_cnt_n_lat_ro	ro	0x0	
31 : 4	psm_cal_cnt_lat_ro	ro	0x0	

68.30 pmu_monitor_cfg				0x70001074
Bits	Field Name	Acc	Default	Description
3 : 0	rtc_rg_dbg_mux	rw	0x0	
7 : 4	rtc_rg_pmu_top_monitor_mux	rw	0x0	

68.31 pmu_status				0x70001078
Bits	Field Name	Acc	Default	Description
0	rtc_rg_por_chip_pd_flag_clr	w1c	0x0	
1	rtc_rg_hard_-long_chip_pd_flag_clr	w1c	0x0	

2	rtc_rg_pbint_7s_flag_clr	w1c	0x0	
3	rtc_rg_pb_int_7s_sw_rst_flag_clr	w1c	0x0	
4	rtc_rg_wdg_soft_rst_flag_clr	w1c	0x0	
5	rtc_rg_soft_rst_flag_clr	w1c	0x0	
8	reg_soft_rst_flag	ro	0x0	
9	pb_int_7s_sw_rst_flag	ro	0x0	
10	wdg_soft_rst_flag	ro	0x0	
11	lp_mode_h_flag	ro	0x0	
15 : 12	rtc_slp_out0_cur_state	ro	0x0	
19 : 16	rtc_slp_out1_cur_state	ro	0x0	
20	rtc_xtal_ibias_stab	ro	0x0	
21	rtc_xtal_cap_stab	ro	0x0	
25 : 22	por_seq_cur_state	ro	0x0	
29 : 26	deep_sleep_awake_status	ro	0x0	

68.32 pmu_int_irq 0x7000107C

Bits	Field Name	Acc	Default	Description
0	rtc_psm_cal_done_raw_status	ro	0x0	
1	rtc_psm_cal_-update_raw_status	ro	0x0	
2	bor_rst_raw_status	ro	0x0	
4	rtc_psm_cal_done_mask_status	ro	0x0	
5	rtc_psm_cal_-update_mask_status	ro	0x0	
6	rtc_bor_rst_mask_status	ro	0x0	
8	rtc_rg_psm_c-al_done_status_mask	rw	0x0	
9	rtc_rg_psm_c-al_update_status_mask	rw	0x0	
10	rtc_rg_bor_rst_status_mask	rw	0x0	
12	rtc_rg_psm_cal_done_status_clr	w1c	0x0	
13	rtc_rg_psm_c-al_update_status_clr	w1c	0x0	
14	rtc_rg_bor_rst_status_clr	w1c	0x0	
16	rtc_rg_pmu_rtc1_int_en	rw	0x0	

68.33 pmu_xtal_clk_detect 0x70001080

Bits	Field Name	Acc	Default	Description
11 : 0	rtc_rg_xtal_-clk_lpo256k_detect_th	rw	0x514	
12	rtc_rg_xtal_detect_clk_soft_en	w1c	0x0	
27 : 16	xtal_clk_lpo2-56k_detect_cnt_latch	ro	0x0	
28	xtal_40m_26m_hw_detect	ro	0x0	
29	xtal_detect_busy	ro	0x0	

68.34 pmu_cm_xtal_cfg1 0x70001084

Bits	Field Name	Acc	Default	Description
0	rtc_rg_cm_xt-al_rf_on_reg_buf_bypass	rw	0x0	
5 : 1	rtc_rg_xtal_ibias_lp_bit	rw	0x5	
8 : 6	rtc_rg_cm_xtal_iref_lp_bit	rw	0x5	
12 : 10	rtc_rg_cm_xtal_iref_bit	rw	0x5	
14 : 13	rtc_rg_cm_xdblr_pulse_bit	rw	0x2	
17 : 15	rtc_rg_cm_xtal_buf_rbit_p	rw	0x7	
20 : 18	rtc_rg_cm_xtal_buf_rbit_n	rw	0x0	

22	rtc_rg_cm_xtal_reg_buf_bypass	rw	0x0	
26 : 23	rtc_rg_cm_xtal_reg_vbit	rw	0xF	
27	rtc_rg_cm_xtal_reg_bypass	rw	0x1	
28	rtc_rg_cm_xtal_rf_on_reg_bypass	rw	0x1	
30 : 29	rtc_rg_cm_xtal_pulse_width_bit	rw	0x2	
31	rtc_rg_cm_xtal_avdd_sel	rw	0x0	

68.35 pmu_cm_xtal_cfg2				0x70001088
Bits	Field Name	Acc	Default	Description
6 : 4	rtc_rg_cm_xtal_comp_ibit	rw	0x4	
7	rtc_rg_cm_xtal_mode_sel_colpitts_dr	rw	0x0	
8	rtc_rg_cm_xtal_mode_sel_colpitts_reg	rw	0x0	
9	rtc_rg_cm_xtal_mode_sel_colpitts_ro	ro	0x0	
10	rtc_rg_cm_xtal2dig_en_set	rw	0x1	
18 : 11	rtc_rg_pmic_rsvd_bit	rw	0xF0	
21 : 19	rtc_rg_cm_xtal_rf_on_iref_bit	rw	0x5	
25 : 22	rtc_rg_cm_xtal_rf_on_reg_vbit	rw	0xF	
27	rtc_rg_xtal_clk_div4_sel	rw	0x1	
28	rtc_rg_xtal_coreon_set	rw	0x1	

68.36 pmu_clk_div_128k				0x7000108C
Bits	Field Name	Acc	Default	Description
14 : 0	rtc_rg_clk_div_128k_cnt_frc	rw	0x25D4	
22 : 16	rtc_rg_clk_div_128k_cnt_int	rw	0x31	

68.37 pmu_ldo_vflash_ctrl				0x70001090
Bits	Field Name	Acc	Default	Description
3 : 0	rtc_rg_ldo_vflash_vbit_ds_sw	rw	0x9	
7 : 4	rtc_rg_ldo_vflash_vbit_nor_tune	rw	0x9	
12	rtc_rg_ldo_vflash_ulpmode	rw	0x0	
13	rtc_rg_ldo_vflash_lpmode_set	rw	0x0	
16 : 14	rtc_rg_ldo_vflash_ibit_lim	rw	0x4	
17	rtc_rg_ldo_vflash_en_ldo_curlim	rw	0x1	

68.38 pmu_dcdc_core_ctrl0				0x70001094
Bits	Field Name	Acc	Default	Description
1 : 0	rtc_rg_dcdc_vbit_vbp	rw	0x2	
3 : 2	rtc_rg_dcdc_vbit_nm18	rw	0x2	
11 : 6	rtc_rg_dcdc_core_dvfs_step_vol	rw	0x0	
12	rtc_rg_dcdc_core_dvfs_step_en	rw	0x0	
14 : 13	rtc_rg_dcdc_core_dvfs_step_delay	rw	0x0	
23 : 15	rtc_rg_dcdc_core_soft_cfg	rw	0x0	
24	rtc_rg_dcdc_core_dcdc_clk_lpo_sel_sw	rw	0x0	
25	rtc_rg_dcdc_core_dcdc_clk_lpo_sel	rw	0x0	
28 : 26	rtc_rg_dcdc_core_bit_vfb_decap	rw	0x0	
29	rtc_rg_dcdc_core_force_res_on	rw	0x0	

68.39 pmu_dcdc_core_ctrl1				0x70001098
Bits	Field Name	Acc	Default	Description
4 : 0	rtc_rg_dcdc_-core_vbit_nor_tune	rw	0x19	
9 : 5	rtc_rg_dcdc_core_vbit_nor_rf	rw	0x19	
14 : 10	rtc_rg_dcdc_core_vbit_ds_sw	rw	0x19	
22 : 18	rtc_rg_dcdc_-core_vbit_vout_valley_delta	rw	0x8	
24	rtc_rg_dcdc_-core_nor_pfm_mode_en	rw	0x0	
26 : 25	rtc_rg_dcdc_core_pulse_time	rw	0x1	
27	rtc_rg_dcdc_core_en_chopp	rw	0x0	
28	rtc_rg_dcdc_core_en_autozero	rw	0x1	
29	rtc_rg_dcdc_core_dis_nld	rw	0x1	
31	rtc_rg_pu_dc-dc_core_resetn_bypass	rw	0x0	

68.40 pmu_dcdc_core_ctrl2				0x7000109C
Bits	Field Name	Acc	Default	Description
3 : 0	rtc_rg_dcdc_core_adc_dly_bit	rw	0x1	
6 : 4	rtc_rg_dcdc_core_adc_cmp_ibit	rw	0x4	
8 : 7	rtc_rg_dcdc_core_op_ibit	rw	0x1	
10 : 9	rtc_rg_dcdc_core_pga_gain_bit	rw	0x1	
14	rtc_rg_dcdc_core_pu_pga	rw	0x1	
15	rtc_rg_dcdc_core_pu_dpwm	rw	0x1	
19 : 16	rtc_rg_dcdc_core_noverlap_bit	rw	0x8	
20	rtc_rg_dcdc_core_dll_band	rw	0x1	
25 : 21	rtc_rg_dcdc_core_start_count	rw	0x10	
28 : 26	rtc_rg_dcdc_core_lpo_freq_bit	rw	0x4	
31 : 29	rtc_rg_dcdc_-core_dly_bit_clk_zcs	rw	0x4	

68.41 pmu_dcdc_core_ctrl3				0x700010A0
Bits	Field Name	Acc	Default	Description
0	rtc_rg_dcdc_large_i_nm18	rw	0x0	
1	rtc_rg_dcdc_large_i_vbp	rw	0x0	
6	rtc_rg_dcdc_-core_pu_res_sample_soft_set	rw	0x0	
7	rtc_rg_dcdc_-core_pu_res_sample_soft_en	rw	0x0	
8	rtc_rg_dcdc_core_en_nsw	rw	0x0	
9	rtc_rg_dcdc_-core_noverlap_ana_pw	rw	0x0	
10	rtc_rg_dcdc_-core_noverlap_ana_hold	rw	0x1	
11	rtc_rg_dcdc_core_sw_2_rf	rw	0x0	
14 : 12	rtc_rg_dcdc_core_time_adder	rw	0x5	
23 : 21	rtc_rg_dcdc_core_en_pmos	rw	0x7	
26 : 24	rtc_rg_dcdc_core_en_nmos	rw	0x7	
27	rtc_rg_dcdc_-vcore_ss_end_mode	rw	0x0	
28	rtc_rg_dcdc_vcore_ss_end_set	rw	0x0	

68.42 pmu_dcdc_css_cfg				0x700010A4
Bits	Field Name	Acc	Default	Description
0	rtc_rg_dcdc_css_clkout_en	rw	0x1	
1	rtc_rg_dcdc_css_bypass	rw	0x1	

6 : 2	rtc_rg_dcdc_- css_trngl_wave_peak	rw	0x1F	
11 : 7	rtc_rg_dcdc_- css_trngl_wave_step	rw	0x1	
19 : 12	rtc_rg_dcdc_- css_trngl_wave_phase	rw	0x1	
27 : 20	rtc_rg_dcdc_- css_trngl_wave_period	rw	0x20	

68.43 pmu_clk_ctrl4				0x700010A8
Bits	Field Name	Acc	Default	Description
0	rtc_rg_cm_xdblr_clk_in_sel1	rw	0x0	
1	rtc_rg_cm_xdblr_dll_mode_en	rw	0x0	
2	rtc_rg_cm_xdblr_duty_en	rw	0x0	
4 : 3	rtc_rg_cm_xd- blr_pulse_bit_input	rw	0x2	
6 : 5	rtc_rg_cm_xd- blr_pulse_bit_output	rw	0x2	

69 wdg4

69.1 WDG_LOAD_VAL				0x70001400
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_load_val	rw	0x3FFFF	watchdog timer load value. In reset mode, software should load new value before timer decrease to 0. In interrupt mode, this value is counting start number. The default value is about 8 seconds.

69.2 WDG_CNT				0x70001404
Bits	Field Name	Acc	Default	Description
31 : 0	wdg_cnt	rw	0xFFFFFFFF	watchdog counter value

69.3 WDG_CTRL				0x70001408
Bits	Field Name	Acc	Default	Description
0	wdg_irq_en	rw	0x0	Watchdog interrupt enable bit 0: interrupt is disabled 1: interrupt is enabled For reset mode: wdg_rst_en =1, wdg_irq_en=0. For interrupt mode: wdg_rst_en =0, wdg_irq_en=1. For combined mode: wdg_rst_en =1, wdg_irq_en=1.
1	wdg_run	rw	0x0	Watchdog counter open: 0: counter stops. 1: counter runs.

69.4 WDG_INT_CLR				0x7000140C
Bits	Field Name	Acc	Default	Description
0	wdg_int_clr	wo	0x0	Watchdog interrupt clear Write 1 to this bit to clear interrupt Read this bit always get 0.

69.5 WDG_INT_RAW				0x70001410
Bits	Field Name	Acc	Default	Description
0	wdg_int_raw	ro	0x0	Watchdog interrupt raw status. Watch dog reset cannot clear this raw status. Write wdg_int_clr can clear this raw status.

4	wdg_ld_busy	ro	0x0	Watchdog load busy status 0: Watchdog is ready for new loading 1: Last loading is not completed Software must not load new value when this bit is busy, that is, this bit should be checked before any new loading. This bit is set after a new loading, and lasts two or three RTC clock cycles, about 60us - 92us.

69.6 WDG_INT_MASK				0x70001414
Bits	Field Name	Acc	Default	Description
0	wdg_int_mask	ro	0x0	Watchdog interrupt masked status

69.7 WDG_LOCK				0x70002000
Bits	Field Name	Acc	Default	Description
15 : 0	wdg_lock	rw	0x0	Watchdog lock control Write 32'h1ACCE551 to this register to unlock watchdog. Write other value to this register to lock watchdog If reading this register, bit-0 is lock status, and other bits are reserved. If watchdog is locked, all control registers cannot be written by software.

70 dcdc_common

70.1 dcdc_ref_clk_div				0x70002000
Bits	Field Name	Acc	Default	Description
6 : 0	rg_dcdc_ref_clk_div_denom	rw	0x10	
7	rg_dcdc_ref_clk_div_update	rw	0x0	

70.2 dcdc_clk_ctrl0				0x70002004
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_ref_clk_sel	rw	0x1	

70.3 dcdc_clk_ctrl1				0x70002008
Bits	Field Name	Acc	Default	Description
3 : 0	rg_dcdc_rf_clk_offset	rw	0x0	
7 : 4	rg_dcdc_aud_clk_offset	rw	0x8	

70.4 dcdc_clk_ctrl2				0x7000200C
Bits	Field Name	Acc	Default	Description
6 : 0	rg_dcdc_ref_clk_div_denom_tx	rw	0x10	
7	rg_tx_on_ref_clk_enable	rw	0x0	

70.5 dcdc_clk_ctrl3				0x70002010
Bits	Field Name	Acc	Default	Description
6 : 0	rg_dcdc_ref_clk_div_denom_rx	rw	0x10	
7	rg_rx_on_ref_clk_enable	rw	0x0	

71 dcdc_rf_ctrl

71.1 dcdc_clk_ctrl0	0x70002100

Bits	Field Name	Acc	Default	Description
0	rg_dcdc_clk_lpo_inv_en	rw	0x1	
2	rg_dcdc_clk_en	rw	0x1	
3	rg_dcdc_clk_lpo_sel	rw	0x0	
4	rg_dcdc_clk_-out_dly_deepsleep_en	rw	0x0	
5	rg_dcdc_clk_-updn_deepsleep_en	rw	0x0	

71.2 dcdc_clk_ctrl1				0x70002104
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_pfm_denom	rw	0x2	
6	rg_dcdc_pfm_denom_update	rw	0x0	

71.3 dcdc_clk_ctrl2				0x70002108
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod0_denom	rw	0x1	
6	rg_dcdc_dcm_-clk_mod0_denom_update	rw	0x0	

71.4 dcdc_clk_ctrl3				0x7000210C
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod1_denom	rw	0x4	
6	rg_dcdc_dcm_-clk_mod1_denom_update	rw	0x0	

71.5 dcdc_clk_ctrl4				0x70002110
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod2_denom	rw	0x2	
6	rg_dcdc_dcm_-clk_mod2_denom_update	rw	0x0	

71.6 dcdc_clk_ctrl5				0x70002114
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod3_denom	rw	0x1	
6	rg_dcdc_dcm_-clk_mod3_denom_update	rw	0x0	

71.7 dcdc_clk_ctrl6				0x70002118
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_smpl_mode	rw	0x1	
2	rg_dcdc_clk_dn_sel	rw	0x0	
6 : 4	rg_dcdc_clk_lpo_denom	rw	0x2	
7	rg_dcdc_clk_-lpo_denom_update	rw	0x0	

71.8 dcdc_clk_ctrl7				0x7000211C
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_clk_dn_phase	rw	0x3	
3 : 2	rg_dcdc_clk_up_phase	rw	0x1	

71.9 dcdc_ctrl_cfg0				0x70002130
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_ss_bypass	rw	0x1	soft start bypss
1	rg_dcdc_clk_auto_enable_mode	rw	0x1	
2	rg_dcdc_ns_auto_track_en	rw	0x1	
4	rg_dcdc_clk_sel	rw	0x0	
7 : 5	rg_dcdc_offset_l	rw	0x0	

71.10 dcdc_ctrl_cfg1				0x70002134
Bits	Field Name	Acc	Default	Description
6 : 0	rg_dcdc_offset_h	rw	0x40	
7	rg_dcdc_offset_update	rw	0x0	

71.11 dcdc_ctrl_cfg2				0x70002138
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_cnt_mode_th	rw	0xFF	

71.12 dcdc_ctrl_cfg3				0x7000213C
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_cnt_mode1_th	rw	0xFF	

71.13 dcdc_ctrl_cfg4				0x70002140
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_dcm_clk_int_up_th	rw	0x10	

71.14 dcdc_ctrl_cfg5				0x70002144
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_dcm_clk_int_dn_th	rw	0x10	

71.15 dcdc_ctrl_cfg6				0x70002148
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagp_th0	rw	0x5	
6 : 4	rg_dcdc_flagp_th1	rw	0x6	

71.16 dcdc_ctrl_cfg7				0x7000214C
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagp_th2	rw	0x7	

71.17 dcdc_ctrl_cfg8				0x70002150
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagn_th0	rw	0x2	
6 : 4	rg_dcdc_flagn_th1	rw	0x4	

71.18 dcdc_ctrl_cfg9				0x70002154
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagn_th2	rw	0x5	

71.19 dcdc_ctrl_cfg10	0x70002158

Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_clk_div	rw	0xC	

71.20 dcdc_ctrl_cfg11				0x7000215C
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_low_cnt_vth1	rw	0x4	

71.21 dcdc_ctrl_cfg12				0x70002160
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_low_cnt_vth2	rw	0x7	

71.22 dcdc_ctrl_cfg13				0x70002164
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_low_cnt_vth3	rw	0xD	

71.23 dcdc_ctrl_cfg14				0x70002168
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_pfm_high_cnt_vth	rw	0x19	
6	rg_dcdc_hysco-mp_high_det_bps	rw	0x0	
7	rg_dcdc_pfm_low_det_mode	rw	0x0	

71.24 dcdc_ctrl_cfg15				0x7000216C
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_nlg_scale0	rw	0x0	
3 : 2	rg_dcdc_nlg_scale1	rw	0x0	
5 : 4	rg_dcdc_nlg_scale2	rw	0x1	
7 : 6	rg_dcdc_nlg_scale3	rw	0x2	

71.25 dcdc_ctrl_cfg16				0x70002170
Bits	Field Name	Acc	Default	Description
3 : 0	rg_dcdc_ks	rw	0x2	
6 : 4	rg_dcdc_kp	rw	0x3	

71.26 dcdc_ctrl_cfg17				0x70002174
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_kz	rw	0x2	
7 : 4	rg_dcdc_kw_l	rw	0x8	

71.27 dcdc_ctrl_cfg18				0x70002178
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_kw_h	rw	0xD	
7	rg_dcdc_kw_update	rw	0x0	

71.28 dcdc_ctrl_cfg19				0x7000217C
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_diff_vth_sel	rw	0x2	
6 : 4	rg_dcdc_fp_int	rw	0x5	

71.29 dcdc_ctrl_cfg20				0x70002180
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_fp_gain	rw	0x5	
5 : 4	rg_dcdc_w_step	rw	0x1	

71.30 dcdc_ctrl_cfg21				0x70002184
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_vx_int_vth	rw	0x2	

71.31 dcdc_ctrl_cfg22				0x70002188
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_add2d_init	rw	0x5F	

71.32 dcdc_ctrl_cfg23				0x7000218C
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_add3d_init	rw	0x5F	

71.33 dcdc_ctrl_cfg24				0x70002190
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_hys_pfm_gain3	rw	0x10	
13 : 8	rg_dcdc_hys_-pfm_gain3_deepsleep	rw	0x10	

71.34 dcdc_ctrl_cfg25				0x70002194
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dout_pw_lim	rw	0x1E	

71.35 dcdc_ctrl_cfg26				0x70002198
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dout_offset	rw	0x28	

71.36 dcdc_ctrl_cfg27				0x7000219C
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dout1_offset	rw	0x28	

71.37 dcdc_ctrl_cfg28				0x700021A0
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step0_dout	rw	0x2	
6 : 4	rg_dcdc_ss_step1_dout	rw	0x2	

71.38 dcdc_ctrl_cfg29				0x700021A4
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step2_dout	rw	0x3	
6 : 4	rg_dcdc_ss_step3_dout	rw	0x4	

71.39 dcdc_ctrl_cfg30				0x700021A8
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step4_dout	rw	0x4	

6 : 4	rg_dcdc_ss_step5_dout	rw	0x4	

71.40 dcdc_ctrl_cfg31				0x700021AC
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step6_dout	rw	0x5	
6 : 4	rg_dcdc_ss_step7_dout	rw	0x5	

71.41 dcdc_ctrl_cfg32				0x700021B0
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step8_dout	rw	0x5	
6 : 4	rg_dcdc_ss_step9_dout	rw	0x5	

71.42 dcdc_ctrl_cfg33				0x700021B4
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_dcm_clk_mode_init	rw	0x1	

71.43 dcdc_ctrl_cfg34				0x700021B8
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_dcm_clk_mode	rw	0x0	
2	rg_dcdc_dcm_clk_mode_force	rw	0x0	
3	rg_dcdc_ns_hys_gain	rw	0x0	
4	rg_dcdc_pu_lpo_sw_mode	rw	0x0	
5	rg_dcdc_pu_lpo_sw	rw	0x0	

71.44 dcdc_ctrl_cfg35				0x700021BC
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_vth_scale0	rw	0x0	
3 : 2	rg_dcdc_vth_scale1	rw	0x2	
5 : 4	rg_dcdc_vth_scale2	rw	0x1	
7 : 6	rg_dcdc_vth_scale3	rw	0x0	

71.45 dcdc_ctrl_cfg36				0x700021C0
Bits	Field Name	Acc	Default	Description
3 : 0	rg_tports_sel	rw	0x0	
5 : 4	rg_dcdc_adc_gain_norm	rw	0x2	

71.46 dcdc_ctrl_cfg37				0x700021C4
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_gain3_limit_vth	rw	0xE	

71.47 dcdc_ctrl_cfg38				0x700021C8
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_deepsleep_sw_mode	rw	0x0	
1	rg_dcdc_deepsleep_sw	rw	0x0	
2	rg_dcdc_hys_-pfm_mode_sw_mode	rw	0x0	
3	rg_dcdc_hys_pfm_mode_sw	rw	0x0	
4	rg_dcdc_force_res_on_sw_mode	rw	0x0	
5	rg_dcdc_force_res_on	rw	0x1	

71.48 dcdc_ctrl_cfg39				0x700021CC
Bits	Field Name	Acc	Default	Description
3 : 0	rg_dcdc_en_pmos	rw	0xF	
7 : 4	rg_dcdc_en_nmos	rw	0xF	
8	rg_dcdc_en_n-mos_pmos_sw_mode	rw	0x1	

71.49 dcdc_ctrl_cfg40				0x700021D0
Bits	Field Name	Acc	Default	Description
3 : 0	rg_dcdc_dcm_flag_th	rw	0x5	
4	rg_dcdc_dcm_flag_sel	rw	0x0	
5	rg_dcdc_mode_flag_frc_en	rw	0x0	
6	rg_dcdc_mode_flag_frc_value	rw	0x0	

71.50 dcdc_ctrl_cfg41				0x700021D4
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_tx_on_dcm_enable	rw	0x0	
2 : 1	rg_dcdc_tx_on_dcm_clk_mode	rw	0x1	
4	rg_dcdc_rx_on_dcm_enable	rw	0x0	
6 : 5	rg_dcdc_rx_on_dcm_clk_mode	rw	0x1	

71.51 dcdc_ctrl_cfg42				0x700021D8
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_tx_o-n_mode_flag_enable	rw	0x0	
1	rg_dcdc_tx_on_mode_flag	rw	0x0	
2	rg_dcdc_rx_o-n_mode_flag_enable	rw	0x0	
3	rg_dcdc_rx_on_mode_flag	rw	0x0	
5 : 4	rg_dcdc_clk_dly_mode	rw	0x2	

71.52 dcdc_ctrl_cfg43				0x700021DC
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_dpwm_-frc_on_deepsleep	rw	0x0	
1	rg_dcdc_dpwm_frc_on_norm	rw	0x1	
2	rg_dcdc_dpwm_-frc_on_sw_mode	rw	0x0	
3	rg_dcdc_sdm_quant_bypass	rw	0x1	
4	rg_dcdc_gain3_dither_en	rw	0x0	
7 : 5	rg_dcdc_gain3_dither_gain	rw	0x0	

71.53 dcdc_status				0x700021E0
Bits	Field Name	Acc	Default	Description
0	ro_dcdc_mode_flag	ro	0x0	
1	ro_dcdc_pw_lim	ro	0x0	
3 : 2	ro_dcdc_dcm_clk_mode	ro	0x1	
4	ro_dcdc_hys_pfm_mode	ro	0x0	
5	ro_dcdc_ss_win_end	ro	0x0	
6	ro_dcdc_ss_fail_flag	ro	0x0	
17 : 8	ro_dcdc_in	ro	0x0	

72 dcdc_core_ctrl

72.1 dcdc_clk_ctrl0				0x70002200
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_clk_lpo_inv_en	rw	0x1	
2	rg_dcdc_clk_en	rw	0x1	
3	rg_dcdc_clk_lpo_sel	rw	0x0	
4	rg_dcdc_clk_-out_dly_deepsleep_en	rw	0x0	
5	rg_dcdc_clk_-updn_deepsleep_en	rw	0x0	

72.2 dcdc_clk_ctrl1				0x70002204
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_pfm_denom	rw	0x2	
6	rg_dcdc_pfm_denom_update	rw	0x0	

72.3 dcdc_clk_ctrl2				0x70002208
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod0_denom	rw	0x1	
6	rg_dcdc_dcm_-clk_mod0_denom_update	rw	0x0	

72.4 dcdc_clk_ctrl3				0x7000220C
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod1_denom	rw	0x2	
6	rg_dcdc_dcm_-clk_mod1_denom_update	rw	0x0	

72.5 dcdc_clk_ctrl4				0x70002210
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod2_denom	rw	0x2	
6	rg_dcdc_dcm_-clk_mod2_denom_update	rw	0x0	

72.6 dcdc_clk_ctrl5				0x70002214
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dcm_clk_mod3_denom	rw	0x1	
6	rg_dcdc_dcm_-clk_mod3_denom_update	rw	0x0	

72.7 dcdc_clk_ctrl6				0x70002218
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_smpl_mode	rw	0x1	
2	rg_dcdc_clk_dn_sel	rw	0x0	
6 : 4	rg_dcdc_clk_lpo_denom	rw	0x2	
7	rg_dcdc_clk_-lpo_denom_update	rw	0x0	

72.8 dcdc_clk_ctrl7				0x7000221C
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_clk_dn_phase	rw	0x3	
3 : 2	rg_dcdc_clk_up_phase	rw	0x1	

72.9 dcdc_ctrl_cfg0				0x70002230
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_ss_bypass	rw	0x1	soft start bypss
1	rg_dcdc_clk_auto_enable_mode	rw	0x1	
2	rg_dcdc_ns_auto_track_en	rw	0x1	
4	rg_dcdc_clk_sel	rw	0x0	
7 : 5	rg_dcdc_offset_l	rw	0x0	

72.10 dcdc_ctrl_cfg1				0x70002234
Bits	Field Name	Acc	Default	Description
6 : 0	rg_dcdc_offset_h	rw	0x40	
7	rg_dcdc_offset_update	rw	0x0	

72.11 dcdc_ctrl_cfg2				0x70002238
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_cnt_mode_th	rw	0xFF	

72.12 dcdc_ctrl_cfg3				0x7000223C
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_cnt_mode1_th	rw	0xFF	

72.13 dcdc_ctrl_cfg4				0x70002240
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_dcm_clk_int_up_th	rw	0x10	

72.14 dcdc_ctrl_cfg5				0x70002244
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_dcm_clk_int_dn_th	rw	0x10	

72.15 dcdc_ctrl_cfg6				0x70002248
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagp_th0	rw	0x5	
6 : 4	rg_dcdc_flagp_th1	rw	0x6	

72.16 dcdc_ctrl_cfg7				0x7000224C
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagp_th2	rw	0x7	

72.17 dcdc_ctrl_cfg8				0x70002250
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagn_th0	rw	0x2	
6 : 4	rg_dcdc_flagn_th1	rw	0x4	

72.18 dcdc_ctrl_cfg9				0x70002254
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_flagn_th2	rw	0x5	

72.19 dcdc_ctrl_cfg10	0x70002258

Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_clk_div	rw	0xC	

72.20 dcdc_ctrl_cfg11				0x7000225C
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_low_cnt_vth1	rw	0x4	

72.21 dcdc_ctrl_cfg12				0x70002260
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_low_cnt_vth2	rw	0x7	

72.22 dcdc_ctrl_cfg13				0x70002264
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_pfm_low_cnt_vth3	rw	0xD	

72.23 dcdc_ctrl_cfg14				0x70002268
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_pfm_high_cnt_vth	rw	0x19	
6	rg_dcdc_hysco-mp_high_det_bps	rw	0x0	
7	rg_dcdc_pfm_low_det_mode	rw	0x0	

72.24 dcdc_ctrl_cfg15				0x7000226C
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_nlg_scale0	rw	0x0	
3 : 2	rg_dcdc_nlg_scale1	rw	0x1	
5 : 4	rg_dcdc_nlg_scale2	rw	0x1	
7 : 6	rg_dcdc_nlg_scale3	rw	0x2	

72.25 dcdc_ctrl_cfg16				0x70002270
Bits	Field Name	Acc	Default	Description
3 : 0	rg_dcdc_ks	rw	0x2	
6 : 4	rg_dcdc_kp	rw	0x3	

72.26 dcdc_ctrl_cfg17				0x70002274
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_kz	rw	0x2	
7 : 4	rg_dcdc_kw_l	rw	0x5	

72.27 dcdc_ctrl_cfg18				0x70002278
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_kw_h	rw	0x15	
7	rg_dcdc_kw_update	rw	0x0	

72.28 dcdc_ctrl_cfg19				0x7000227C
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_diff_vth_sel	rw	0x2	
6 : 4	rg_dcdc_fp_int	rw	0x5	

72.29 dcdc_ctrl_cfg20				0x70002280
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_fp_gain	rw	0x5	
5 : 4	rg_dcdc_w_step	rw	0x1	

72.30 dcdc_ctrl_cfg21				0x70002284
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_vx_int_vth	rw	0x2	

72.31 dcdc_ctrl_cfg22				0x70002288
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_add2d_init	rw	0x5F	

72.32 dcdc_ctrl_cfg23				0x7000228C
Bits	Field Name	Acc	Default	Description
7 : 0	rg_dcdc_add3d_init	rw	0x5F	

72.33 dcdc_ctrl_cfg24				0x70002290
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_hys_pfm_gain3	rw	0xE	
13 : 8	rg_dcdc_hys_-pfm_gain3_deepsleep	rw	0xE	

72.34 dcdc_ctrl_cfg25				0x70002294
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dout_pw_lim	rw	0x1E	

72.35 dcdc_ctrl_cfg26				0x70002298
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dout_offset	rw	0x28	

72.36 dcdc_ctrl_cfg27				0x7000229C
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_dout1_offset	rw	0x28	

72.37 dcdc_ctrl_cfg28				0x700022A0
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step0_dout	rw	0x1	
6 : 4	rg_dcdc_ss_step1_dout	rw	0x2	

72.38 dcdc_ctrl_cfg29				0x700022A4
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step2_dout	rw	0x3	
6 : 4	rg_dcdc_ss_step3_dout	rw	0x3	

72.39 dcdc_ctrl_cfg30				0x700022A8
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step4_dout	rw	0x4	

6 : 4	rg_dcdc_ss_step5_dout	rw	0x4	

72.40 dcdc_ctrl_cfg31				0x700022AC
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step6_dout	rw	0x5	
6 : 4	rg_dcdc_ss_step7_dout	rw	0x5	

72.41 dcdc_ctrl_cfg32				0x700022B0
Bits	Field Name	Acc	Default	Description
2 : 0	rg_dcdc_ss_step8_dout	rw	0x5	
6 : 4	rg_dcdc_ss_step9_dout	rw	0x5	

72.42 dcdc_ctrl_cfg33				0x700022B4
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_dcm_clk_mode_init	rw	0x1	

72.43 dcdc_ctrl_cfg34				0x700022B8
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_dcm_clk_mode	rw	0x0	
2	rg_dcdc_dcm_clk_mode_force	rw	0x0	
3	rg_dcdc_ns_hys_gain	rw	0x0	
4	rg_dcdc_pu_lpo_sw_mode	rw	0x0	
5	rg_dcdc_pu_lpo_sw	rw	0x0	

72.44 dcdc_ctrl_cfg35				0x700022BC
Bits	Field Name	Acc	Default	Description
1 : 0	rg_dcdc_vth_scale0	rw	0x0	
3 : 2	rg_dcdc_vth_scale1	rw	0x1	
5 : 4	rg_dcdc_vth_scale2	rw	0x1	
7 : 6	rg_dcdc_vth_scale3	rw	0x0	

72.45 dcdc_ctrl_cfg36				0x700022C0
Bits	Field Name	Acc	Default	Description
3 : 0	rg_tports_sel	rw	0x0	
5 : 4	rg_dcdc_adc_gain_norm	rw	0x2	

72.46 dcdc_ctrl_cfg37				0x700022C4
Bits	Field Name	Acc	Default	Description
5 : 0	rg_dcdc_gain3_limit_vth	rw	0xA	

72.47 dcdc_ctrl_cfg38				0x700022C8
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_deepsleep_sw_mode	rw	0x0	
1	rg_dcdc_deepsleep_sw	rw	0x0	
2	rg_dcdc_hys_-pfm_mode_sw_mode	rw	0x0	
3	rg_dcdc_hys_pfm_mode_sw	rw	0x0	
4	rg_dcdc_force_res_on_sw_mode	rw	0x0	
5	rg_dcdc_force_res_on	rw	0x1	

72.48 dcdc_ctrl_cfg39				0x700022CC
Bits	Field Name	Acc	Default	Description
3 : 0	rg_dcdc_en_pmos	rw	0xF	
7 : 4	rg_dcdc_en_nmos	rw	0xF	
8	rg_dcdc_en_n-mos_pmos_sw_mode	rw	0x0	

72.49 dcdc_ctrl_cfg40				0x700022D0
Bits	Field Name	Acc	Default	Description
3 : 0	rg_dcdc_dcm_flag_th	rw	0x5	
4	rg_dcdc_dcm_flag_sel	rw	0x0	
5	rg_dcdc_mode_flag_frc_en	rw	0x0	
6	rg_dcdc_mode_flag_frc_value	rw	0x0	

72.50 dcdc_ctrl_cfg41				0x700022D4
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_tx_on_dcm_enable	rw	0x0	
2 : 1	rg_dcdc_tx_on_dcm_clk_mode	rw	0x1	
4	rg_dcdc_rx_on_dcm_enable	rw	0x0	
6 : 5	rg_dcdc_rx_on_dcm_clk_mode	rw	0x1	

72.51 dcdc_ctrl_cfg42				0x700022D8
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_tx_o-n_mode_flag_enable	rw	0x0	
1	rg_dcdc_tx_on_mode_flag	rw	0x0	
2	rg_dcdc_rx_o-n_mode_flag_enable	rw	0x0	
3	rg_dcdc_rx_on_mode_flag	rw	0x0	
5 : 4	rg_dcdc_clk_dly_mode	rw	0x2	

72.52 dcdc_ctrl_cfg43				0x700022DC
Bits	Field Name	Acc	Default	Description
0	rg_dcdc_dpwm_-frc_on_deepsleep	rw	0x0	
1	rg_dcdc_dpwm_frc_on_norm	rw	0x1	
2	rg_dcdc_dpwm_-frc_on_sw_mode	rw	0x0	
3	rg_dcdc_sdm_quant_bypass	rw	0x1	
4	rg_dcdc_gain3_dither_en	rw	0x0	
7 : 5	rg_dcdc_gain3_dither_gain	rw	0x0	

72.53 dcdc_status				0x700022E0
Bits	Field Name	Acc	Default	Description
0	ro_dcdc_mode_flag	ro	0x0	
1	ro_dcdc_pw_lim	ro	0x0	
3 : 2	ro_dcdc_dcm_clk_mode	ro	0x1	
4	ro_dcdc_hys_pfm_mode	ro	0x0	
5	ro_dcdc_ss_win_end	ro	0x0	
6	ro_dcdc_ss_fail_flag	ro	0x0	
17 : 8	ro_dcdc_in	ro	0x0	

List of Tables

INT_ROM

1.1 boot rom 1

INT_BTROM

2.1 bt rom 1

INT_SRAM0

3.1 int sram0 1

INT_SRAM1

4.1 int sram1 1

INT_SRAM2

5.1 int sram2 1

INT_SRAM3

6.1 int sram3 1

INT_SRAM4

7.1 int sram4 1

INT_SRAM5

8.1 int sram5 2

INT_SRAM6

9.1 int sram6 2

INT_SHARED_CACHE_SRAM

10.1 int shared_cache_sram 2

INT_BTRAM

11.1 bt ram 2

INT_AON_SRAM

12.1 aon_sram 2

FLASH_MEM

13.1 flash_mem 2

FLASH_CACHE_MEM

14.1 flash_cache_mem 2

cache

15.1 BUS_CFG0 2
15.2 BUS_CFG1 3
15.3 BUS_CFG2 3
15.4 BUS_CFG3 3
15.5 BUS_CFG4 4
15.6 BUS_CFG5 4
15.7 BUS_CFG6 4
15.8 BUS_CFG7 4
15.9 BUS_REMAP0 4
15.10 BUS_REMAP1 4
15.11 BUS_REMAP2 4
15.12 BUS_REMAP3 4
15.13 BUS_REMAP4 4
15.14 BUS_REMAP5 5
15.15 BUS_REMAP6 5
15.16 BUS_REMAP7 5
15.17 CACHE_CFG0 5
15.18 BUS_STS0 6
15.19 CACHE_STS0 6
15.20 CMD_CFG0 6
15.21 CMD_CFG1 6
15.22 CMD_CFG2 6
15.23 INT_EN 6

15.24 INT_RAW . . . 6
15.25 INT_MASK . . . 7
15.26 INT_CLR . . . 7
15.27 CACHE_WRITE_HIT_CNT . . . 7
15.28 CACHE_WRITE_MISS_CNT . . . 7
15.29 CACHE_READ_HIT_CNT . . . 7
15.30 CACHE_READ_MISS_CNT . . . 7
15.31 CACHE_HACT_CNT . . . 7
15.32 CACHE_HRDY_CNT . . . 7
dma_cx
16.1 DMA_PAUSE . . . 8
16.2 DMA_FRAG_WAIT . . . 8
16.3 REQ_PEND0_EN . . . 8
16.4 REQ_PEND1_EN . . . 8
16.5 DMA_INT_RAW_STATUS . . . 8
16.6 DMA_INT_MASK_STATUS . . . 8
16.7 DMA_REQ_STATUS . . . 9
16.8 DMA_EN_STATUS . . . 9
16.9 DMA_DEBUG_STATUS . . . 9
16.10 DMA_ARB_SEL_STATUS . . . 10
16.11 CRC_CFG . . . 10
16.12 CRC_GEN . . . 10
16.13 CRC_OUT_XOR . . . 10
16.14 CRC_INIT . . . 10
16.15 I_CRC . . . 10
16.16 AES_KEY_0_ADDR . . . 10
16.17 AES_KEY_1_ADDR . . . 10
16.18 AES_KEY_2_ADDR . . . 11
16.19 AES_KEY_3_ADDR . . . 11
16.20 AES_KEY_4_ADDR . . . 11
16.21 AES_KEY_5_ADDR . . . 11
16.22 AES_KEY_6_ADDR . . . 11
16.23 AES_KEY_7_ADDR . . . 11
16.24 AES_IV_0_ADDR . . . 11
16.25 AES_IV_1_ADDR . . . 11
16.26 AES_IV_2_ADDR . . . 11
16.27 AES_IV_3_ADDR . . . 11
16.28 AES_PLAIN_LEN_H_ADDR . . . 12
16.29 AES_PLAIN_LEN_L_ADDR . . . 12
16.30 AES_A_LEN_H_ADDR . . . 12
16.31 AES_A_LEN_L_ADDR . . . 12
16.32 CE_CTRL_ADDR . . . 12
16.33 PKA_REG0_ADDR . . . 12
16.34 CE_CLOCK_GATE_CTRL . . . 12
16.35 TRANS_CTRL0_ADDR . . . 13
16.36 TRANS_CTRL1_ADDR . . . 13
16.37 TRANS_CLOCK_GATE_CTRL . . . 13
16.38 DMA_MON_SEL . . . 13
16.39 dma_reserved0 . . . 13
16.40 dma_reserved1 . . . 13
16.41 dma_reserved2 . . . 13
16.42 CHKSUM_CTRL_ADDR . . . 14
16.43 CHKSUM_RES_ADDR . . . 14
16.44 HASH_LEN_BIT_ADDR . . . 14
16.45 CE_CTRL_2_ADDR . . . 14
16.46 FULL_DMA_CHS . . . 14
16.46.46 FULL_CHN_PAUSE . . . 15
16.46.46 FULL_CHN_REQ . . . 15
16.46.46 FULL_CHN_CFG . . . 15

16.46.46 FULL_CHN_INT . . . 16
16.46.46 FULL_CHN_SRC_ADDR . . . 17
16.46.46 FULL_CHN_DEST_ADDR . . . 17
16.46.46 FULL_CHN_FRAG_LEN . . . 17
16.46.46 FULL_CHN_BLK_LEN . . . 18
16.46.46 FULL_CHN_TRSC_LEN . . . 18
16.46.46 FULL_CHN_TRSF_STEP . . . 18
16.46.46 FULL_CHN_WRAP_PTR . . . 18
16.46.46 FULL_CHN_WRAP_TO . . . 19
16.46.46 FULL_CHN_LLIST_PTR . . . 19
16.46.46 FULL_CHN_FRAG_STEP . . . 19
16.46.46 FULL_CHN_SRC_BLK_STEP . . . 19
16.46.46 FULL_CHN_DEST_BLK_STEP . . . 19
16.47 REQ_CID_0 . . . 19
16.48 REQ_CID_1 . . . 19
16.49 REQ_CID_2 . . . 19
16.50 REQ_CID_3 . . . 20
16.51 REQ_CID_4 . . . 20
16.52 REQ_CID_5 . . . 20
16.53 REQ_CID_6 . . . 20
16.54 REQ_CID_7 . . . 20
16.55 REQ_CID_8 . . . 20
16.56 REQ_CID_9 . . . 20
16.57 REQ_CID_10 . . . 20
16.58 REQ_CID_11 . . . 20
16.59 REQ_CID_12 . . . 21
16.60 REQ_CID_13 . . . 21
16.61 REQ_CID_14 . . . 21
16.62 REQ_CID_15 . . . 21
16.63 REQ_CID_16 . . . 21
16.64 REQ_CID_17 . . . 21
16.65 REQ_CID_18 . . . 21
16.66 REQ_CID_19 . . . 21
16.67 REQ_CID_20 . . . 21
16.68 REQ_CID_21 . . . 21
16.69 REQ_CID_22 . . . 22
16.70 REQ_CID_23 . . . 22
16.71 CHN_1_DMA_ADDR_RD . . . 22
16.72 CHN_1_DMA_ADDR_WR . . . 22
16.73 CHN_2_DMA_ADDR_RD . . . 22
16.74 CHN_2_DMA_ADDR_WR . . . 22
16.75 CHN_3_DMA_ADDR_RD . . . 22
16.76 CHN_3_DMA_ADDR_WR . . . 22
16.77 CHN_4_DMA_ADDR_RD . . . 22
16.78 CHN_4_DMA_ADDR_WR . . . 23
16.79 CHN_5_DMA_ADDR_RD . . . 23
16.80 CHN_5_DMA_ADDR_WR . . . 23
16.81 CHN_6_DMA_ADDR_RD . . . 23
16.82 CHN_6_DMA_ADDR_WR . . . 23
16.83 CHN_7_DMA_ADDR_RD . . . 23
16.84 CHN_7_DMA_ADDR_WR . . . 23
16.85 CHN_8_DMA_ADDR_RD . . . 24
16.86 CHN_8_DMA_ADDR_WR . . . 24
16.87 CHN_9_DMA_ADDR_RD . . . 24
16.88 CHN_9_DMA_ADDR_WR . . . 24
16.89 CHN_10_DMA_ADDR_RD . . . 24
16.90 CHN_10_DMA_ADDR_WR . . . 24
16.91 CHN_11_DMA_ADDR_RD . . . 24
16.92 CHN_11_DMA_ADDR_WR . . . 24
16.93 CHN_12_DMA_ADDR_RD . . . 25

16.94 CHN_12_DMA_ADDR_WR . . . 25
16.95 CHN_13_DMA_ADDR_RD . . . 25
16.96 CHN_13_DMA_ADDR_WR . . . 25
16.97 CHN_14_DMA_ADDR_RD . . . 25
16.98 CHN_14_DMA_ADDR_WR . . . 25
16.99 CHN_LLI_NODE_COUNTER_0 . . . 25
16.100 CHN_LLI_NODE_COUNTER_1 . . . 25
16.101 CHN_LLI_NODE_COUNTER_2 . . . 25
16.102 CHN_LLI_NODE_COUNTER_3 . . . 26
16.103 CHN_LLI_NODE_COUNTER_4 . . . 26
16.104 CHN_LLI_NODE_COUNTER_5 . . . 26
16.105 CHN_LLI_NODE_COUNTER_6 . . . 26
16.106 CHN_LLI_NODE_COUNTER_7 . . . 26
16.107 CHN_LLI_NODE_COUNTER_8 . . . 26
16.108 CHN_LLI_NODE_COUNTER_9 . . . 26
16.109 CHN_LLI_NODE_COUNTER_10 . . . 26
16.110 CHN_LLI_NODE_COUNTER_11 . . . 26
16.111 CHN_LLI_NODE_COUNTER_12 . . . 26
16.112 CHN_LLI_NODE_COUNTER_13 . . . 27
cpu_p_trap
17.1 addr_in_0 . . . 27
17.2 addr_in_1 . . . 27
17.3 addr_in_2 . . . 27
17.4 addr_in_3 . . . 27
17.5 addr_in_4 . . . 27
17.6 addr_in_5 . . . 27
17.7 addr_in_6 . . . 27
17.8 addr_in_7 . . . 27
17.9 en . . . 28
17.10 trap_out_base . . . 28
cpu_s0_trap
18.1 addr_in_0 . . . 28
18.2 addr_in_1 . . . 28
18.3 addr_in_2 . . . 28
18.4 addr_in_3 . . . 28
18.5 addr_in_4 . . . 28
18.6 addr_in_5 . . . 28
18.7 addr_in_6 . . . 28
18.8 addr_in_7 . . . 29
18.9 en . . . 29
18.10 trap_out_base . . . 29
uart0
19.1 txrx_fifo_divl . . . 29
19.2 int_en_divh . . . 29
19.3 int_id_fifo_ctrl . . . 29
19.4 line_ctrl . . . 29
19.5 modem_ctrl . . . 30
19.6 line_status . . . 30
19.7 modem_status . . . 30
19.8 scratchpad . . . 30
19.9 fifo_status . . . 30
19.10 fifo_trigger . . . 30
19.11 divisor_latch2 . . . 31
19.12 counter_real . . . 31
19.13 div_real_slow . . . 31
19.14 div_real_fast . . . 31
19.15 tports_sel . . . 31
uart1

20.1 txrx_fifo_divl . . . 31
20.2 int_en_divh . . . 31
20.3 int_id_fifo_ctrl . . . 32
20.4 line_ctrl . . . 32
20.5 modem_ctrl . . . 32
20.6 line_status . . . 32
20.7 modem_status . . . 32
20.8 scratchpad . . . 33
20.9 fifo_status . . . 33
20.10 fifo_trigger . . . 33
20.11 divisor_latch2 . . . 33
20.12 counter_real . . . 33
20.13 div_real_slow . . . 33
20.14 div_real_fast . . . 33
20.15 tports_sel . . . 33
uart2
21.1 txrx_fifo_divl . . . 33
21.2 int_en_divh . . . 34
21.3 int_id_fifo_ctrl . . . 34
21.4 line_ctrl . . . 34
21.5 modem_ctrl . . . 34
21.6 line_status . . . 34
21.7 modem_status . . . 35
21.8 scratchpad . . . 35
21.9 fifo_status . . . 35
21.10 fifo_trigger . . . 35
21.11 divisor_latch2 . . . 35
21.12 counter_real . . . 35
21.13 div_real_slow . . . 35
21.14 div_real_fast . . . 35
21.15 tports_sel . . . 36
mailbox
22.1 app2emb_trigger . . . 36
22.2 emb2app_rawstatus . . . 36
22.3 emb2app_ack . . . 36
22.4 emb2app_unmaskset . . . 36
22.5 emb2app_unmaskclear . . . 36
22.6 emb2app_linesel_low . . . 36
22.7 emb2app_linesel_high . . . 36
22.8 emb2app_status . . . 36
22.9 mutex0 . . . 36
22.10 mutex1 . . . 36
22.11 mutex2 . . . 37
22.12 mutex3 . . . 37
22.13 mutex4 . . . 37
22.14 mutex5 . . . 37
22.15 mutex6 . . . 37
22.16 mutex7 . . . 37
22.17 comreg0 . . . 37
22.18 comreg1 . . . 37
22.19 comreg2 . . . 37
22.20 comreg3 . . . 37
22.21 comreg4 . . . 37
22.22 comreg5 . . . 38
22.23 comreg6 . . . 38
22.24 comreg7 . . . 38
22.25 emb2app_trigger . . . 38
22.26 app2emb_rawstatus . . . 38
22.27 app2emb_ack . . . 38

22.28 app2emb_unmaskset . . . 38
22.29 app2emb_unmaskclear . . . 38
22.30 app2emb_linesel_low . . . 38
22.31 app2emb_linesel_high . . . 38
22.32 app2emb_status . . . 38
22.33 g1_app2emb_trigger . . . 38
22.34 g1_emb2app_rawstatus . . . 39
22.35 g1_emb2app_ack . . . 39
22.36 g1_emb2app_unmaskset . . . 39
22.37 g1_emb2app_unmaskclear . . . 39
22.38 g1_emb2app_linesel_low . . . 39
22.39 g1_emb2app_linesel_high . . . 39
22.40 g1_emb2app_status . . . 39
22.41 g1_mutex0 . . . 39
22.42 g1_mutex1 . . . 39
22.43 g1_mutex2 . . . 39
22.44 g1_mutex3 . . . 39
22.45 g1_mutex4 . . . 39
22.46 g1_mutex5 . . . 40
22.47 g1_mutex6 . . . 40
22.48 g1_mutex7 . . . 40
22.49 g1_comreg0 . . . 40
22.50 g1_comreg1 . . . 40
22.51 g1_comreg2 . . . 40
22.52 g1_comreg3 . . . 40
22.53 g1_comreg4 . . . 40
22.54 g1_comreg5 . . . 40
22.55 g1_comreg6 . . . 40
22.56 g1_comreg7 . . . 40
22.57 g1_emb2app_trigger . . . 40
22.58 g1_app2emb_rawstatus . . . 41
22.59 g1_app2emb_ack . . . 41
22.60 g1_app2emb_unmaskset . . . 41
22.61 g1_app2emb_unmaskclear . . . 41
22.62 g1_app2emb_linesel_low . . . 41
22.63 g1_app2emb_linesel_high . . . 41
22.64 g1_app2emb_status . . . 41
sd_host
23.1 ctrl . . . 41
23.2 txfifo_entry . . . 41
23.3 rxfifo_entry . . . 42
23.4 cfg . . . 42
23.5 status . . . 42
23.6 cmd_index . . . 42
23.7 cmd_arg . . . 42
23.8 rsp_index . . . 42
23.9 rsp_arg3 . . . 42
23.10 rsp_arg2 . . . 42
23.11 rsp_arg1 . . . 43
23.12 rsp_arg0 . . . 43
23.13 dline_width . . . 43
23.14 block_size . . . 43
23.15 block_cnt . . . 43
23.16 int_status . . . 43
23.17 int_enable . . . 43
23.18 int_clear . . . 43
23.19 trans_speed . . . 44
23.20 mclk_ctrl . . . 44
23.21 fifo_status . . . 44

aud_proc

24.1 aud_proc_function_id . . . 44
24.2 aud_proc_cfg . . . 44
24.3 aud_proc_tx_cfg_ch0 . . . 44
24.4 aud_proc_tx_cfg_ch1 . . . 45
24.5 aud_proc_tx_cfg_ch2 . . . 45
24.6 aud_proc_tx_cfg_ch3 . . . 45
24.7 aud_proc_rx_cfg_ch01 . . . 45
24.8 aud_proc_rx_cfg_ch23 . . . 46
24.9 aud_proc_dac_cfg1 . . . 46
24.10 aud_proc_dac_cfg2 . . . 46
24.11 aud_proc_adc_cfg1 . . . 47
24.12 aud_proc_adc_cfg2 . . . 47
24.13 aud_proc_txrx_cfg_ch0 . . . 48
24.14 aud_proc_txrx_cfg_ch1 . . . 49
24.15 aud_proc_rxtx_cfg_ch0 . . . 49
24.16 aud_proc_rxtx_cfg_ch1 . . . 49
24.17 aud_proc_frac_src . . . 50
24.18 aud_proc_dac_trig . . . 50
24.19 aud_proc_dac_strb . . . 50
24.20 aud_proc_irq . . . 50
24.21 bt_clk_lo_cnt . . . 50
24.22 bt_clk_hi_cnt . . . 50
24.23 bt_clk_lo_cnt_latch . . . 50
24.24 bt_clk_hi_cnt_latch . . . 51
24.25 dac_src_strb_lo_cnt . . . 51
24.26 dac_src_strb_hi_cnt . . . 51
24.27 dpll_alfa1_cfg . . . 51
24.28 dpll_k1_k2_cfg . . . 51
24.29 dpll_alfa2_cfg . . . 51
24.30 dpll_other_cfg1 . . . 51
24.31 dpll_trig_cfg . . . 51
24.32 frac_rate_adj_up_th . . . 51
24.33 frac_rate_adj_dn_th . . . 51
24.34 frac_rate_adj_hw . . . 52
24.35 aud_intf_i2s_cfg0 . . . 52
24.36 aud_intf_i2s_cfg1 . . . 53
24.37 pcm_cfg_0 . . . 55
24.38 pcm_cfg_1 . . . 55
24.39 spdif_userdata_0 . . . 56
24.40 spdif_userdata_1 . . . 56
24.41 spdif_userdata_2 . . . 56
24.42 spdif_userdata_3 . . . 56
24.43 spdif_userdata_4 . . . 56
24.44 spdif_userdata_5 . . . 56
24.45 spdif_channel_stat_0 . . . 56
24.46 spdif_channel_stat_1 . . . 56
24.47 spdif_channel_stat_2 . . . 56
24.48 spdif_channel_stat_3 . . . 57
24.49 spdif_channel_stat_4 . . . 57
24.50 spdif_channel_stat_5 . . . 57
24.51 spdif_output_cfg . . . 57
24.52 spdif_input_cfg . . . 57
24.53 spdif_input_stat1 . . . 57
24.54 spdif_input_stat2 . . . 57
24.55 aud_drc_cfg0 . . . 57
24.56 eq_stereo_coef_ram_cfg0 . . . 58
24.57 eq_stereo_coef_ram_cfg1 . . . 58
24.58 eq_mono_coef_ram_cfg0 . . . 58

24.59 eq_mono_coef_ram_cfg1 . . . 58
24.60 aud_drc_ram_cfg0 . . . 58
24.61 aud_drc_ram_cfg1 . . . 58
24.62 aud_drc_ram_cfg2 . . . 58
24.63 aud_src_ctrl0 . . . 58
24.64 aud_src_ctrl1 . . . 59
24.65 aud_src_ctrl2 . . . 59
24.66 aud_src_hpf_coef1_a1 . . . 59
24.67 aud_src_hpf_coef1_a2 . . . 59
24.68 aud_src_hpf_coef1_b1 . . . 60
24.69 aud_src_hpf_coef1_g0 . . . 60
24.70 aud_src_lpf_coef2_a1 . . . 60
24.71 aud_src_lpf_coef2_a2 . . . 60
24.72 aud_src_lpf_coef2_b1 . . . 60
24.73 aud_src_lpf_coef2_g0 . . . 60
24.74 aud_src_lpf_coef3_a1 . . . 60
24.75 aud_src_lpf_coef3_a2 . . . 60
24.76 aud_src_lpf_coef3_b1 . . . 60
24.77 aud_src_lpf_coef3_g0 . . . 60
24.78 aud_adc_hpf_coef1_a1 . . . 60
24.79 aud_adc_hpf_coef1_a2 . . . 61
24.80 aud_adc_hpf_coef1_b1 . . . 61
24.81 aud_adc_hpf_coef1_g0 . . . 61
24.82 aud_adc_lpf_coef2_a1 . . . 61
24.83 aud_adc_lpf_coef2_a2 . . . 61
24.84 aud_adc_lpf_coef2_b1 . . . 61
24.85 aud_adc_lpf_coef2_g0 . . . 61
24.86 aud_adc_lpf_coef3_a1 . . . 61
24.87 aud_adc_lpf_coef3_a2 . . . 61
24.88 aud_adc_lpf_coef3_b1 . . . 61
24.89 aud_adc_lpf_coef3_g0 . . . 62
24.90 eq_coef_ram_cfg3 . . . 62
24.91 aud_adc_drc_cfg0 . . . 62
24.92 aud_adc_drc_ram_cfg0 . . . 62
24.93 aud_adc_drc_ram_cfg1 . . . 62
24.94 aud_adc_drc_ram_cfg2 . . . 62
spiflash
25.1 tx_cmd_addr . . . 62
25.2 tx_block_size . . . 62
25.3 tx_fifo_data . . . 63
25.4 status . . . 63
25.5 rx_fifo_data . . . 63
25.6 flash_config . . . 63
25.7 fifo_ctrl . . . 63
25.8 dual_spi . . . 63
25.9 read_cmd . . . 63
25.10 nand_cfg1 . . . 64
25.11 nand_cfg2 . . . 64
25.12 dummy_cycle_cfg . . . 64
25.13 diff_128m_cmd . . . 64
25.14 aesKey0 . . . 64
25.15 aesKey1 . . . 64
25.16 aesKey2 . . . 64
25.17 aesKey3 . . . 64
25.18 aesKey4 . . . 64
25.19 aesKey5 . . . 64
25.20 aesKey6 . . . 65
25.21 aesKey7 . . . 65
25.22 initCtrNonce0 . . . 65

25.23 initCtrNonce1 . . . 65
25.24 initCtrNonce2 . . . 65
25.25 encStartAddr . . . 65
25.26 encEndAddr . . . 65
25.27 flash_decrypt_ctrl . . . 65
cpu_sysctrl
26.1 hclk_ctrl_mode . . . 65
26.2 hclk1_ctrl_mode . . . 65
26.3 pclk_ctrl_mode . . . 66
26.4 others_clk_ctrl_mode . . . 66
26.5 cfg_mem_rom . . . 66
26.6 cfg_mem_1prf . . . 66
26.7 cfg_mem_2prf . . . 66
26.8 trap_bypass . . . 66
26.9 cache_bypass . . . 66
26.10 pclk_div . . . 66
26.11 codec_mclk_div . . . 66
26.12 bclk_sel . . . 66
26.13 cpu_sys_tports_sel . . . 66
26.14 phy_dump_sel . . . 67
26.15 clk_msadc_div . . . 67
26.16 cfg_cpu_stclk_div . . . 67
26.17 cfg_usb_gpio . . . 67
26.18 cfg_cpu_p_rxev . . . 67
26.19 cfg_cpu_s0_rxev . . . 67
26.20 cfg_cpu_nmi_en . . . 68
26.21 ro_top_tports . . . 68
26.22 hclk_manual_enable . . . 68
26.23 hclk_manual_disable . . . 69
26.24 hclk1_manual_enable . . . 69
26.25 hclk1_manual_disable . . . 70
26.26 pclk_manual_enable . . . 70
26.27 pclk_manual_disable . . . 70
26.28 others_manual_enable . . . 71
26.29 others_manual_disable . . . 71
26.30 hclk_soft_resetn_set . . . 72
26.31 hclk_soft_resetn_clr . . . 72
26.32 hclk1_soft_resetn_set . . . 73
26.33 hclk1_soft_resetn_clr . . . 73
26.34 pclk_soft_resetn_set . . . 73
26.35 pclk_soft_resetn_clr . . . 73
26.36 others_soft_resetn_set . . . 74
26.37 others_soft_resetn_clr . . . 74
pwm
27.1 tmr0_ld_value . . . 74
27.2 tmr0_cnt_value . . . 74
27.3 tmr0_cfg . . . 74
27.4 tmr0_int_ctrl . . . 75
27.5 tmr0_cnt_read . . . 75
27.6 tmr1_ld_value . . . 75
27.7 tmr1_cnt_value . . . 75
27.8 tmr1_cfg . . . 75
27.9 tmr1_int_ctrl . . . 75
27.10 tmr1_cnt_read . . . 75
27.11 tmr2_ld_value . . . 75
27.12 tmr2_cnt_value . . . 75
27.13 tmr2_cfg . . . 75
27.14 tmr2_int_ctrl . . . 76
27.15 tmr2_cnt_read . . . 76

27.16 pwm0_sta_val . . . 76
27.17 pwm0_end_val . . . 76
27.18 pwm0_step_cfg0 . . . 76
27.19 pwm0_step_cfg1 . . . 76
27.20 pwm0_cfg . . . 76
27.21 pwm0_int_ctrl . . . 76
27.22 pwm0_end_val_use . . . 76
27.23 pwm0_step_cnt . . . 77
27.24 pwm1_sta_val . . . 77
27.25 pwm1_end_val . . . 77
27.26 pwm1_step_cfg0 . . . 77
27.27 pwm1_step_cfg1 . . . 77
27.28 pwm1_cfg . . . 77
27.29 pwm1_int_ctrl . . . 77
27.30 pwm1_end_val_use . . . 77
27.31 pwm1_step_cnt . . . 77
27.32 pwm2_sta_val . . . 77
27.33 pwm2_end_val . . . 78
27.34 pwm2_step_cfg0 . . . 78
27.35 pwm2_step_cfg1 . . . 78
27.36 pwm2_cfg . . . 78
27.37 pwm2_int_ctrl . . . 78
27.38 pwm2_end_val_use . . . 78
27.39 pwm2_step_cnt . . . 78

i2cm

28.1 IC_CON . . . 78
28.2 IC_TAR . . . 79
28.3 IC_SAR . . . 79
28.4 IC_HS_MADDR . . . 79
28.5 IC_DATA_CMD . . . 79
28.6 IC_SS_SCL_HCNT . . . 79
28.7 IC_SS_SCL_LCNT . . . 79
28.8 IC_FS_SCL_HCNT . . . 79
28.9 IC_FS_SCL_LCNT . . . 79
28.10 IC_HS_SCl_HCNT . . . 79
28.11 IC_HS_SCL_LCNT . . . 79
28.12 IC_INTR_STAT . . . 79
28.13 IC_INTR_MASK . . . 80
28.14 IC_RAW_INTR_STAT . . . 80
28.15 IC_RX_TL . . . 80
28.16 IC_TX_TL . . . 81
28.17 IC_CLR_INTR . . . 81
28.18 IC_CLR_RX_UNDER . . . 81
28.19 IC_CLR_RX_OVER . . . 81
28.20 IC_CLR_TX_OVER . . . 81
28.21 IC_CLR_RD_REQ . . . 81
28.22 IC_CLR_TX_ABRT . . . 81
28.23 IC_CLR_RX_DONE . . . 81
28.24 IC_CLR_ACTIVITY . . . 81
28.25 IC_CLR_STOP_DET . . . 81
28.26 IC_CLR_START_DET . . . 81
28.27 IC_CLR_GEN_CALL . . . 81
28.28 IC_ENABLE . . . 82
28.29 IC_STATUS . . . 82
28.30 IC_TXFLR . . . 82
28.31 IC_RXFLR . . . 82
28.32 IC_SDA_HOLD . . . 82
28.33 IC_TX_ABRT_SOURCE . . . 82
28.34 IC_DMA_CR . . . 82

28.35 IC_DMA_TDLR . . . 83
28.36 IC_DMA_RDLR . . . 83
28.37 IC_SDA_SETUP . . . 83
28.38 IC_ACK_GENERAL_CALL . . . 83
28.39 IC_ENABLE_STATUS . . . 83
28.40 IC_FS_SPKLEN . . . 83
28.41 IC_HS_SPKLEN . . . 83
28.42 IC_CLR_RESTART_DET . . . 83
28.43 IC_COMP_PARAM_1 . . . 83
28.44 IC_COMP_VERSION . . . 83
28.45 IC_COMP_TYPE . . . 84
wdg0
29.1 WDG_LOAD_VAL . . . 84
29.2 WDG_CNT . . . 84
29.3 WDG_CTRL . . . 84
29.4 WDG_INT_CLR . . . 84
29.5 WDG_INT_RAW . . . 84
29.6 WDG_INT_MASK . . . 84
29.7 WDG_LOCK . . . 85
wdg1
30.1 WDG_LOAD_VAL . . . 85
30.2 WDG_CNT . . . 85
30.3 WDG_CTRL . . . 85
30.4 WDG_INT_CLR . . . 85
30.5 WDG_INT_RAW . . . 85
30.6 WDG_INT_MASK . . . 86
30.7 WDG_LOCK . . . 86
spi
31.1 SPI_TXD . . . 86
31.2 SPI_CLKD . . . 86
31.3 SPI_CTL0 . . . 86
31.4 SPI_CTL1 . . . 86
31.5 SPI_CTL2 . . . 87
31.6 SPI_CTL3 . . . 87
31.7 SPI_CTL4 . . . 87
31.8 SPI_CTL5 . . . 87
31.9 SPI_INT_EN . . . 88
31.10 SPI_INT_CLR . . . 88
31.11 SPI_INT_RAW_STS . . . 88
31.12 SPI_INT_MASK_STS . . . 88
31.13 SPI_STS1 . . . 89
31.14 SPI_STS2 . . . 89
31.15 DSP_WAIT . . . 89
31.16 SPI_STS3 . . . 90
31.17 SPI_CTL6 . . . 90
31.18 SPI_STS4 . . . 90
31.19 SPI_FIFO_RST . . . 90
31.20 SPI_CTL7 . . . 90
31.21 SPI_STS5 . . . 90
31.22 SPI_CTL8 . . . 91
31.23 SPI_CTL9 . . . 91
31.24 SPI_CTL10 . . . 91
31.25 SPI_CTL11 . . . 91
31.26 SPI_CTL12 . . . 91
31.27 SPI_STS6 . . . 91
31.28 SPI_STS7 . . . 91
31.29 SPI_STS8 . . . 92
31.30 SPI_STS9 . . . 92
timer0

32.1 tmr_cfg . . . 92
32.2 tmr_pr . . . 92
32.3 tmr_pc . . . 92
32.4 tmr_mr0 . . . 92
32.5 tmr_mr1 . . . 92
32.6 tmr_mr2 . . . 92
32.7 tmr_delta0 . . . 92
32.8 tmr_delta1 . . . 92
32.9 tmr_delta2 . . . 92
32.10 tmr_rtc_clk_div . . . 93
32.11 tmr_int_mask_enable . . . 93
32.12 tmr_int_mask_clear . . . 93
32.13 tmr_int_raw . . . 93
32.14 tmr_int_status . . . 93
timer1
33.1 tmr_cfg . . . 93
33.2 tmr_pr . . . 93
33.3 tmr_pc . . . 93
33.4 tmr_mr0 . . . 94
33.5 tmr_mr1 . . . 94
33.6 tmr_mr2 . . . 94
33.7 tmr_delta0 . . . 94
33.8 tmr_delta1 . . . 94
33.9 tmr_delta2 . . . 94
33.10 tmr_rtc_clk_div . . . 94
33.11 tmr_int_mask_enable . . . 94
33.12 tmr_int_mask_clear . . . 94
33.13 tmr_int_raw . . . 94
33.14 tmr_int_status . . . 95
trng
34.1 trng_en . . . 95
34.2 trng_status . . . 95
34.3 trng_data . . . 95
34.4 trng_div_dly . . . 95
34.5 trng_sel . . . 95
ana_reg0
35.1 flash_pad_pull . . . 95
proc_mon
36.1 auto_mode . . . 96
36.2 cfg_start . . . 96
36.3 cfg_auto_stop . . . 96
36.4 enable_duration . . . 96
36.5 cfg_auto_interval . . . 96
36.6 ro_pvt_cnt_hvt . . . 96
36.7 cfg_cnt_last . . . 96
gpioa
37.1 GPIODATA . . . 96
37.2 GPIODMSK . . . 96
37.3 GPIODIR . . . 96
37.4 GPIOIS . . . 97
37.5 GPIOIBE . . . 97
37.6 GPIOIEV . . . 97
37.7 GPIOIE . . . 97
37.8 GPIORIS . . . 97
37.9 GPIOMIS . . . 97
37.10 GPIOIC . . . 97
37.11 GPIOINEN . . . 97
37.12 GPIO_FILTER_NUM . . . 97

37.13 GPIO_FILTER_DIV_L . . . 98
37.14 GPIO_FILTER_DIV_H . . . 98
efuse
38.1 EFUSE_ALL0_INDEX . . . 98
38.2 EFUSE_MODE_CTRL . . . 98
38.3 EFUSE_IP_VER_reg . . . 98
38.4 EFUSE_CFG0 . . . 98
38.5 EFUSE_SEC_EN . . . 98
38.6 EFUSE_SEC_ERR_FLAG . . . 99
38.7 EFUSE_SEC_FLAG_CLR . . . 99
38.8 EFUSE_SEC_MAGIC_NUMBER . . . 100
38.9 EFUSE_FW_CFG . . . 100
38.10 EFUSE_PW_SWT . . . 100
38.11 EFUSE_FUNC_EN_0 . . . 100
38.12 EFUSE_FUNC_EN_1 . . . 100
38.13 EFUSE_FUNC_EN_2 . . . 101
38.14 EFUSE_FUNC_EN_3 . . . 101
38.15 EFUSE_FUNC_EN_4 . . . 101
38.16 EFUSE_FUNC_EN_5 . . . 101
38.17 EFUSE_FUNC_EN_6 . . . 101
38.18 EFUSE_FUNC_EN_7 . . . 101
38.19 EFUSE_FUNC_EN_8 . . . 101
38.20 EFUSE_FUNC_EN_9 . . . 101
38.21 EFUSE_FUNC_EN_10 . . . 102
38.22 EFUSE_FUNC_EN_11 . . . 102
38.23 EFUSE_FUNC_EN_12 . . . 102
38.24 EFUSE_FUNC_EN_13 . . . 102
38.25 EFUSE_FUNC_EN_14 . . . 102
38.26 EFUSE_FUNC_EN_15 . . . 102
38.27 efuse_bist_en_reg . . . 102
38.28 efuse_block0_rw_ctrl_reg . . . 102
38.29 efuse_block1_rw_ctrl_reg . . . 103
38.30 efuse_block2_rw_ctrl_reg . . . 103
38.31 efuse_block3_rw_ctrl_reg . . . 103
38.32 efuse_block4_rw_ctrl_reg . . . 103
38.33 efuse_block5_rw_ctrl_reg . . . 103
38.34 efuse_block6_rw_ctrl_reg . . . 103
38.35 efuse_block7_rw_ctrl_reg . . . 103
38.36 efuse_block8_rw_ctrl_reg . . . 104
38.37 efuse_block9_rw_ctrl_reg . . . 104
38.38 efuse_block10_rw_ctrl_reg . . . 104
38.39 efuse_block11_rw_ctrl_reg . . . 104
38.40 efuse_block12_rw_ctrl_reg . . . 104
38.41 efuse_block13_rw_ctrl_reg . . . 104
38.42 efuse_block14_rw_ctrl_reg . . . 104
38.43 efuse_block15_rw_ctrl_reg . . . 104
38.44 efuse_block16_rw_ctrl_reg . . . 105
38.45 efuse_block17_rw_ctrl_reg . . . 105
38.46 efuse_block18_rw_ctrl_reg . . . 105
38.47 efuse_block19_rw_ctrl_reg . . . 105
38.48 efuse_block20_rw_ctrl_reg . . . 105
38.49 efuse_block21_rw_ctrl_reg . . . 105
38.50 efuse_block22_rw_ctrl_reg . . . 105
38.51 efuse_block23_rw_ctrl_reg . . . 105
38.52 efuse_block24_rw_ctrl_reg . . . 106
38.53 efuse_block25_rw_ctrl_reg . . . 106
38.54 efuse_block26_rw_ctrl_reg . . . 106
38.55 efuse_block27_rw_ctrl_reg . . . 106
38.56 efuse_block28_rw_ctrl_reg . . . 106

38.57 efuse_block29_rw_ctrl_reg . . . 106
38.58 efuse_block30_rw_ctrl_reg . . . 106
38.59 efuse_block31_rw_ctrl_reg . . . 106
38.60 por_read_done_reg . . . 107
38.61 efuse_cfg_reg . . . 107
38.62 efuse_block_en_reg . . . 107
38.63 por_read_data0_reg . . . 107
38.64 por_read_data1_reg . . . 107
38.65 por_read_data2_reg . . . 107
38.66 por_read_data3_reg . . . 107
38.67 EFUSE_MEM . . . 107
msadc
39.1 cfg_msadc_mode . . . 107
39.2 cfg_msadc_sw_ctrl0 . . . 107
39.3 cfg_msadc_sw_ctrl1 . . . 108
39.4 cfg_msadc_ana_ctrl0 . . . 108
39.5 cfg_msadc_ro_acc . . . 108
39.6 cfg_msadc_ro_test_acc_0 . . . 108
39.7 cfg_msadc_ro_test_acc_1 . . . 108
39.8 cfg_msadc_int_raw . . . 108
39.9 cfg_msadc_int_status . . . 108
39.10 cfg_msadc_int_mask . . . 108
usbc
40.1 GOTGCTL . . . 109
40.2 GOTGINT . . . 109
40.3 GAHBCFG . . . 109
40.4 GUSBCFG . . . 110
40.5 GRSTCTL . . . 110
40.6 GINTSTS . . . 110
40.7 GINTMSK . . . 111
40.8 GRXSTSR . . . 112
40.9 GRXSTSP . . . 112
40.10 GRXFSIZ . . . 112
40.11 GNPTXFSIZ . . . 112
40.12 GNPTXSTS . . . 112
40.13 GPVNDCTL . . . 112
40.14 GGPIO . . . 113
40.15 GUID . . . 113
40.16 GSNPSID . . . 113
40.17 GHWCFG1 . . . 113
40.18 GHWCFG2 . . . 113
40.19 GHWCFG3 . . . 114
40.20 GHWCFG4 . . . 114
40.21 GPWRDN . . . 114
40.22 GDFIFOCFG . . . 115
40.23 GADPCTL . . . 115
40.24 STREAMSTSFIFO . . . 115
40.25 STREAMFIRSTPTRFIFO . . . 115
40.26 STREAMLASTPTRFIFO . . . 115
40.27 HPTXFSIZ . . . 115
40.28 DIEPTXF1 . . . 116
40.29 DIEPTXF2 . . . 116
40.30 DIEPTXF3 . . . 116
40.31 DIEPTXF4 . . . 116
40.32 DIEPTXF5 . . . 116
40.33 DIEPTXF6 . . . 116
40.34 DIEPTXF7 . . . 116
40.35 DIEPTXF8 . . . 117
40.36 HCFG . . . 117

40.37 HFIR . . . 117
40.38 HFNUM . . . 117
40.39 HPTXSTS . . . 117
40.40 HAINT . . . 117
40.41 HAINTMSK . . . 117
40.42 HFLBAddr . . . 117
40.43 HPRT . . . 118
40.44 HCCHAR0 . . . 118
40.45 HCSPLT0 . . . 118
40.46 HCINT0 . . . 118
40.47 HCINTMSK0 . . . 119
40.48 HCTSIZ0 . . . 119
40.49 HCDMA0 . . . 119
40.50 HCDMAB0 . . . 119
40.51 HCCHAR1 . . . 119
40.52 HCSPLT1 . . . 119
40.53 HCINT1 . . . 120
40.54 HCINTMSK1 . . . 120
40.55 HCTSIZ1 . . . 120
40.56 HCDMA1 . . . 120
40.57 HCDMAB1 . . . 120
40.58 HCCHAR2 . . . 120
40.59 HCSPLT2 . . . 121
40.60 HCINT2 . . . 121
40.61 HCINTMSK2 . . . 121
40.62 HCTSIZ2 . . . 121
40.63 HCDMA2 . . . 122
40.64 HCDMAB2 . . . 122
40.65 HCCHAR3 . . . 122
40.66 HCSPLT3 . . . 122
40.67 HCINT3 . . . 122
40.68 HCINTMSK3 . . . 122
40.69 HCTSIZ3 . . . 123
40.70 HCDMA3 . . . 123
40.71 HCDMAB3 . . . 123
40.72 HCCHAR4 . . . 123
40.73 HCSPLT4 . . . 123
40.74 HCINT4 . . . 123
40.75 HCINTMSK4 . . . 124
40.76 HCTSIZ4 . . . 124
40.77 HCDMA4 . . . 124
40.78 HCDMAB4 . . . 124
40.79 HCCHAR5 . . . 124
40.80 HCSPLT5 . . . 125
40.81 HCINT5 . . . 125
40.82 HCINTMSK5 . . . 125
40.83 HCTSIZ5 . . . 125
40.84 HCDMA5 . . . 125
40.85 HCDMAB5 . . . 125
40.86 HCCHAR6 . . . 126
40.87 HCSPLT6 . . . 126
40.88 HCINT6 . . . 126
40.89 HCINTMSK6 . . . 126
40.90 HCTSIZ6 . . . 127
40.91 HCDMA6 . . . 127
40.92 HCDMAB6 . . . 127
40.93 HCCHAR7 . . . 127
40.94 HCSPLT7 . . . 127
40.95 HCINT7 . . . 127
40.96 HCINTMSK7 . . . 128

40.97 HCTSIZ7 . . . 128
40.98 HCDMA7 . . . 128
40.99 HCDMAB7 . . . 128
40.100 HCCHAR8 . . . 128
40.101 HCSPLT8 . . . 128
40.102 HCINT8 . . . 128
40.103 HCINTMSK8 . . . 129
40.104 HCTSIZ8 . . . 129
40.105 HCDMA8 . . . 129
40.106 HCDMAB8 . . . 129
40.107 HCCHAR9 . . . 129
40.108 HCSPLT9 . . . 130
40.109 HCINT9 . . . 130
40.110 HCINTMSK9 . . . 130
40.111 HCTSIZ9 . . . 130
40.112 HCDMA9 . . . 130
40.113 HCDMAB9 . . . 131
40.114 HCCHAR10 . . . 131
40.115 HCSPLT10 . . . 131
40.116 HCINT10 . . . 131
40.117 HCINTMSK10 . . . 131
40.118 HCTSIZ10 . . . 132
40.119 HCDMA10 . . . 132
40.120 HCDMAB10 . . . 132
40.121 HCCHAR11 . . . 132
40.122 HCSPLT11 . . . 132
40.123 HCINT11 . . . 132
40.124 HCINTMSK11 . . . 133
40.125 HCTSIZ11 . . . 133
40.126 HCDMA11 . . . 133
40.127 HCDMAB11 . . . 133
40.128 HCCHAR12 . . . 133
40.129 HCSPLT12 . . . 133
40.130 HCINT12 . . . 134
40.131 HCINTMSK12 . . . 134
40.132 HCTSIZ12 . . . 134
40.133 HCDMA12 . . . 134
40.134 HCDMAB12 . . . 134
40.135 HCCHAR13 . . . 134
40.136 HCSPLT13 . . . 135
40.137 HCINT13 . . . 135
40.138 HCINTMSK13 . . . 135
40.139 HCTSIZ13 . . . 135
40.140 HCDMA13 . . . 136
40.141 HCDMAB13 . . . 136
40.142 HCCHAR14 . . . 136
40.143 HCSPLT14 . . . 136
40.144 HCINT14 . . . 136
40.145 HCINTMSK14 . . . 136
40.146 HCTSIZ14 . . . 137
40.147 HCDMA14 . . . 137
40.148 HCDMAB14 . . . 137
40.149 HCCHAR15 . . . 137
40.150 HCSPLT15 . . . 137
40.151 HCINT15 . . . 137
40.152 HCINTMSK15 . . . 138
40.153 HCTSIZ15 . . . 138
40.154 HCDMA15 . . . 138
40.155 HCDMAB15 . . . 138
40.156 DCFG . . . 138

40.157 DCTL . . . 138
40.158 DSTS . . . 139
40.159 DIEPMSK . . . 139
40.160 DOEPMSK . . . 139
40.161 DAINT . . . 140
40.162 DAINTMSK . . . 140
40.163 DVBUSDIS . . . 141
40.164 DVBUSPULSE . . . 141
40.165 DTHRCTL . . . 141
40.166 DIEPEMPMSK . . . 141
40.167 DOEPSTREAMTIMEOUTTH . . . 141
40.168 DOEPSTREAMDBGCTRL . . . 142
40.169 DOEPSTREAMDBGSTATUS1 . . . 142
40.170 DOEPSTREAMDBGSTATUS2 . . . 142
40.171 DOEPSTREAMDBGSTATUS3 . . . 142
40.172 DOEPSTREAMDBGSTATUS4 . . . 142
40.173 DIEPCTL0 . . . 142
40.174 DIEPINT0 . . . 142
40.175 DIEPTSIZ0 . . . 143
40.176 DIEPDMA0 . . . 143
40.177 DTXFSTS0 . . . 143
40.178 DIEPDMAB0 . . . 143
40.179 DIEPCTL1 . . . 143
40.180 DIEPINT1 . . . 143
40.181 DIEPTSIZ1 . . . 144
40.182 DIEPDMA1 . . . 144
40.183 DTXFSTS1 . . . 144
40.184 DIEPDMAB1 . . . 144
40.185 DIEPCTL2 . . . 144
40.186 DIEPINT2 . . . 145
40.187 DIEPTSIZ2 . . . 145
40.188 DIEPDMA2 . . . 145
40.189 DTXFSTS2 . . . 145
40.190 DIEPDMAB2 . . . 145
40.191 DIEPCTL3 . . . 145
40.192 DIEPINT3 . . . 146
40.193 DIEPTSIZ3 . . . 146
40.194 DIEPDMA3 . . . 146
40.195 DTXFSTS3 . . . 146
40.196 DIEPDMAB3 . . . 146
40.197 DIEPCTL4 . . . 146
40.198 DIEPINT4 . . . 147
40.199 DIEPTSIZ4 . . . 147
40.200 DIEPDMA4 . . . 147
40.201 DTXFSTS4 . . . 147
40.202 DIEPDMAB4 . . . 147
40.203 DIEPCTL5 . . . 147
40.204 DIEPINT5 . . . 148
40.205 DIEPTSIZ5 . . . 148
40.206 DIEPDMA5 . . . 148
40.207 DTXFSTS5 . . . 148
40.208 DIEPDMAB5 . . . 148
40.209 DIEPCTL6 . . . 149
40.210 DIEPINT6 . . . 149
40.211 DIEPTSIZ6 . . . 149
40.212 DIEPDMA6 . . . 149
40.213 DTXFSTS6 . . . 149
40.214 DIEPDMAB6 . . . 149
40.215 DIEPCTL7 . . . 150
40.216 DIEPINT7 . . . 150

40.217 DIEPTSIZ7 . . . 150
40.218 DIEPDMA7 . . . 150
40.219 DTXFSTS7 . . . 150
40.220 DIEPDMAB7 . . . 151
40.221 DIEPCTL8 . . . 151
40.222 DIEPINT8 . . . 151
40.223 DIEPTSIZ8 . . . 151
40.224 DIEPDMA8 . . . 151
40.225 DTXFSTS8 . . . 152
40.226 DIEPDMAB8 . . . 152
40.227 DIEPCTL9 . . . 152
40.228 DIEPINT9 . . . 152
40.229 DIEPTSIZ9 . . . 152
40.230 DIEPDMA9 . . . 153
40.231 DTXFSTS9 . . . 153
40.232 DIEPDMAB9 . . . 153
40.233 DIEPCTL10 . . . 153
40.234 DIEPINT10 . . . 153
40.235 DIEPTSIZ10 . . . 154
40.236 DIEPDMA10 . . . 154
40.237 DTXFSTS10 . . . 154
40.238 DIEPDMAB10 . . . 154
40.239 DIEPCTL11 . . . 154
40.240 DIEPINT11 . . . 154
40.241 DIEPTSIZ11 . . . 155
40.242 DIEPDMA11 . . . 155
40.243 DTXFSTS11 . . . 155
40.244 DIEPDMAB11 . . . 155
40.245 DIEPCTL12 . . . 155
40.246 DIEPINT12 . . . 155
40.247 DIEPTSIZ12 . . . 156
40.248 DIEPDMA12 . . . 156
40.249 DTXFSTS12 . . . 156
40.250 DIEPDMAB12 . . . 156
40.251 DOEPCTL0 . . . 156
40.252 DOEPINT0 . . . 156
40.253 DOEPTSIZ0 . . . 157
40.254 DOEPDMA0 . . . 157
40.255 DOEPDMAB0 . . . 157
40.256 DOEPCTL1 . . . 157
40.257 DOEPCTL1_1 . . . 158
40.258 DOEPINT1 . . . 158
40.259 DOEPTSIZ1 . . . 158
40.260 DOEPDMA1 . . . 159
40.261 DOEPDMAB1 . . . 159
40.262 DOEPCTL2 . . . 159
40.263 DOEPCTL2_1 . . . 159
40.264 DOEPINT2 . . . 160
40.265 DOEPTSIZ2 . . . 160
40.266 DOEPDMA2 . . . 160
40.267 DOEPDMAB2 . . . 160
40.268 DOEPCTL3 . . . 160
40.269 DOEPCTL3_1 . . . 161
40.270 DOEPINT3 . . . 161
40.271 DOEPTSIZ3 . . . 162
40.272 DOEPDMA3 . . . 162
40.273 DOEPDMAB3 . . . 162
40.274 DOEPCTL4 . . . 162
40.275 DOEPCTL4_1 . . . 162
40.276 DOEPINT4 . . . 163

40.277 DOEPTSIZ4 . . . 163
40.278 DOEPDMA4 . . . 163
40.279 DOEPDMAB4 . . . 163
40.280 DOEPCTL5 . . . 163
40.281 DOEPCTL5_1 . . . 164
40.282 DOEPINT5 . . . 164
40.283 DOEPTSIZ5 . . . 165
40.284 DOEPDMA5 . . . 165
40.285 DOEPDMAB5 . . . 165
40.286 DOEPCTL6 . . . 165
40.287 DOEPCTL6_1 . . . 165
40.288 DOEPINT6 . . . 166
40.289 DOEPTSIZ6 . . . 166
40.290 DOEPDMA6 . . . 166
40.291 DOEPDMAB6 . . . 166
40.292 DOEPCTL7 . . . 167
40.293 DOEPCTL7_1 . . . 167
40.294 DOEPINT7 . . . 167
40.295 DOEPTSIZ7 . . . 168
40.296 DOEPDMA7 . . . 168
40.297 DOEPDMAB7 . . . 168
40.298 DOEPCTL8 . . . 168
40.299 DOEPCTL8_1 . . . 169
40.300 DOEPINT8 . . . 169
40.301 DOEPTSIZ8 . . . 169
40.302 DOEPDMA8 . . . 170
40.303 DOEPDMAB8 . . . 170
40.304 DOEPCTL9 . . . 170
40.305 DOEPCTL9_1 . . . 170
40.306 DOEPINT9 . . . 171
40.307 DOEPTSIZ9 . . . 171
40.308 DOEPDMA9 . . . 171
40.309 DOEPDMAB9 . . . 171
40.310 DOEPCTL10 . . . 171
40.311 DOEPCTL10_1 . . . 172
40.312 DOEPINT10 . . . 172
40.313 DOEPTSIZ10 . . . 173
40.314 DOEPDMA10 . . . 173
40.315 DOEPDMAB10 . . . 173
40.316 DOEPCTL11 . . . 173
40.317 DOEPCTL11_1 . . . 173
40.318 DOEPINT11 . . . 174
40.319 DOEPTSIZ11 . . . 174
40.320 DOEPDMA11 . . . 174
40.321 DOEPDMAB11 . . . 174
40.322 DOEPCTL12 . . . 174
40.323 DOEPCTL12_1 . . . 175
40.324 DOEPINT12 . . . 175
40.325 DOEPTSIZ12 . . . 176
40.326 DOEPDMA12 . . . 176
40.327 DOEPDMAB12 . . . 176
40.328 PCGCCTL . . . 176
mm_sys_ctrl
41.1 mem_cfg . . . 176
41.2 usb_cfg0 . . . 176
41.3 usb_cfg1 . . . 176
41.4 usb_cfg2 . . . 176
41.5 usb_tport_sel . . . 177
41.6 usb_status . . . 177

41.7 usb_digphy_int_enable . . . 177
41.8 usb_digphy_int_clr . . . 177
41.9 cfg_mm_sys_tports_sel . . . 177
41.10 mem_cfg1 . . . 177
41.11 usb_prbs_cfg . . . 177
41.12 usb_prbs_status . . . 178
41.13 cfg_pu_usb . . . 178
41.14 usb_ana_cfg0 . . . 178
41.15 usb_ana_cfg1 . . . 179
41.16 host_dma_cfg . . . 179
host_dma
42.1 CH_LLI_ROOT0 . . . 179
42.2 CH_LLI_ROOT1 . . . 179
42.3 CH_LLI_ROOT2 . . . 179
42.4 CH_LLI_ROOT3 . . . 179
42.5 DMA_STATUS . . . 179
42.6 INT_RAWSTATUS . . . 180
42.7 INT_UNMASK_SET . . . 180
42.8 INT_UNMASK_CLEAR . . . 180
42.9 INT_ACK . . . 180
42.10 INT_STATUS . . . 180
42.11 DMA_REVISION . . . 181
42.12 ARBITRATION . . . 181
42.13 CHANNEL_MUTEX_SET . . . 181
42.14 CHANNEL_MUTEX_CLEAR . . . 181
42.15 CH_LLI_ROOT4 . . . 181
42.16 AXI_BURST_LENGTH_LIMIT . . . 181
42.17 LLI_COUNTER0 . . . 181
42.18 LLI_COUNTER1 . . . 181
42.19 LLI_COUNTER2 . . . 181
42.20 LLI_COUNTER3 . . . 181
42.21 LLI_COUNTER4 . . . 182
42.22 LLI_COUNTER5 . . . 182
42.23 LLI_COUNTER6 . . . 182
42.24 LLI_COUNTER7 . . . 182
42.25 LLI_COUNTER8 . . . 182
42.26 LLI_COUNTER9 . . . 182
42.27 LLI_COUNTER10 . . . 182
42.28 LLI_COUNTER11 . . . 182
42.29 LLI_COUNTER12 . . . 182
42.30 LLI_COUNTER13 . . . 182
42.31 LLI_COUNTER14 . . . 182
42.32 LLI_COUNTER15 . . . 183
42.33 CONVERSION_CONFIG0 . . . 183
42.34 CONVERSION_CONFIG1 . . . 183
42.35 CONVERSION_CONFIG2 . . . 183
42.36 CONVERSION_CONFIG3 . . . 183
42.37 CONVERSION_DATA00 . . . 183
42.38 CONVERSION_DATA01 . . . 183
42.39 CONVERSION_DATA10 . . . 183
42.40 CONVERSION_DATA11 . . . 183
42.41 CONVERSION_DATA20 . . . 183
42.42 CONVERSION_DATA21 . . . 183
42.43 CONVERSION_DATA30 . . . 184
42.44 CONVERSION_DATA31 . . . 184
dfe_reg_bank
43.1 a0x000 . . . 184
43.2 a0x004 . . . 184
43.3 a0x008 . . . 184

43.4 a0x00c . . . 185
43.5 a0x010 . . . 185
43.6 a0x014 . . . 185
43.7 a0x018 . . . 185
43.8 a0x01c . . . 186
43.9 a0x020 . . . 186
43.10 a0x024 . . . 186
43.11 a0x028 . . . 186
43.12 a0x02c . . . 186
43.13 a0x030 . . . 187
43.14 a0x034 . . . 187
43.15 a0x038 . . . 187
43.16 a0x03c . . . 187
43.17 a0x04c . . . 187
43.18 a0x050 . . . 187
43.19 a0x054 . . . 188
43.20 a0x058 . . . 188
43.21 a0x05c . . . 188
43.22 a0x060 . . . 188
43.23 a0x064 . . . 188
43.24 a0x068 . . . 188
43.25 a0x06c . . . 188
43.26 a0x070 . . . 189
43.27 a0x074 . . . 189
43.28 a0x078 . . . 189
43.29 a0x07c . . . 189
43.30 a0x080 . . . 189
43.31 a0x084 . . . 189
43.32 a0x088 . . . 189
43.33 a0x0c8 . . . 190
43.34 a0x0cc . . . 190
43.35 a0x0d0 . . . 190
43.36 a0x0d4 . . . 190
43.37 a0x0d8 . . . 190
43.38 a0x0dc . . . 191
43.39 a0x0e0 . . . 191
43.40 a0x0e4 . . . 191
43.41 a0x0e8 . . . 191
43.42 a0x0ec . . . 191
43.43 a0x0f0 . . . 191
43.44 a0x0f4 . . . 191
43.45 a0x104 . . . 191
43.46 a0x108 . . . 191
43.47 a0x10c . . . 192
43.48 a0x124 . . . 192
43.49 a0x128 . . . 192
43.50 a0x12c . . . 192
43.51 a0x130 . . . 192
43.52 a0x134 . . . 192
43.53 a0x138 . . . 193
43.54 a0x13c . . . 193
43.55 a0x140 . . . 193
43.56 a0x144 . . . 193
43.57 a0x148 . . . 193
43.58 a0x14c . . . 193
43.59 a0x150 . . . 193
43.60 a0x15c . . . 193
43.61 a0x160 . . . 194
43.62 a0x164 . . . 194
43.63 a0x168 . . . 194

43.64 a0x16c . . . 194
43.65 a0x170 . . . 194
43.66 a0x180 . . . 194
43.67 a0x184 . . . 195
43.68 iqim_estg_ctrl . . . 195
43.69 iqim_estg_phi . . . 195
43.70 iqim_estg_idc0 . . . 195
43.71 iqim_estg_odc0 . . . 195
43.72 iqim_estg_oam0 . . . 195
43.73 iqim_estg_idc1 . . . 195
43.74 iqim_estg_odc1 . . . 195
43.75 iqim_estg_oam1 . . . 196
43.76 wdfs_ctrl_1d0 . . . 196
43.77 wdfs_ctrl_1d4 . . . 196
43.78 wdfs_fifo_1d8 . . . 196
43.79 wdfs_stat_1dc . . . 196
43.80 wdfs_ctrl_1e0 . . . 196
43.81 a0x1e4 . . . 196
43.82 wdfs_ctrl_1e8 . . . 196
43.83 a0x200 . . . 197
43.84 a0x204 . . . 197
43.85 a0x208 . . . 197
43.86 a0x20c . . . 197
43.87 a0x210 . . . 197
43.88 a0x214 . . . 197
43.89 a0x218 . . . 197
43.90 a0x21c . . . 197
43.91 a0x220 . . . 197
43.92 a0x224 . . . 197
43.93 a0x228 . . . 197
43.94 a0x22c . . . 198
43.95 a0x230 . . . 198
43.96 a0x234 . . . 198
43.97 a0x238 . . . 198
43.98 a0x23c . . . 198
43.99 a0x240 . . . 198
43.100 a0x244 . . . 198
43.101 a0x248 . . . 198
43.102 a0x24c . . . 198
43.103 a0x26c . . . 199
43.104 a0x270 . . . 199
43.105 a0x274 . . . 199
43.106 a0x278 . . . 199
43.107 a0x27c . . . 199
43.108 a0x280 . . . 199
43.109 a0x284 . . . 199
43.110 a0x2b4 . . . 200
43.111 a0x2c4 . . . 200
43.112 a0x2c8 . . . 200
43.113 a0x2cc . . . 200
43.114 a0x2d0 . . . 200
43.115 a0x2d4 . . . 200
43.116 a0x2d8 . . . 200
43.117 a0x2dc . . . 200
43.118 a0x2e0 . . . 200
43.119 a0x2e4 . . . 200
43.120 a0x2e8 . . . 200
43.121 a0x2ec . . . 201
43.122 a0x2f0 . . . 201
43.123 a0x2f4 . . . 201

43.124 a0x2f8 . . . 201
43.125 a0x2fc . . . 201
43.126 a0x300 . . . 201
43.127 a0x304 . . . 201
43.128 a0x308 . . . 201
wf_rf_interface
44.1 reg_00 . . . 201
44.2 reg_04 . . . 202
44.3 reg_08 . . . 203
44.4 reg_0C . . . 203
44.5 reg_10 . . . 204
44.6 reg_14 . . . 204
44.7 reg_18 . . . 204
44.8 reg_1C . . . 204
44.9 reg_20 . . . 205
44.10 reg_24 . . . 205
44.11 reg_28 . . . 205
44.12 reg_2C . . . 205
44.13 reg_30 . . . 206
44.14 reg_34 . . . 206
44.15 reg_38 . . . 206
44.16 reg_3C . . . 207
44.17 reg_40 . . . 207
44.18 reg_44 . . . 207
44.19 reg_48 . . . 207
44.20 reg_4C . . . 208
44.21 reg_50 . . . 208
44.22 reg_54 . . . 208
44.23 reg_58 . . . 209
44.24 reg_5C . . . 209
44.25 reg_60 . . . 209
44.26 reg_64 . . . 209
44.27 reg_68 . . . 210
44.28 reg_6C . . . 210
44.29 reg_70 . . . 210
44.30 reg_74 . . . 210
44.31 reg_78 . . . 210
44.32 reg_7C . . . 211
44.33 wf_soft_reset_reg . . . 211
44.34 wf_misc_crtl_reg . . . 211
44.35 wf_man_crtl_reg . . . 211
44.36 wf_calclk_reg . . . 211
44.37 wf_ext_trxon_delay_reg . . . 212
44.38 gain_mem_ctrl_reg . . . 212
44.39 gain_mem_fetch_reg . . . 212
44.40 wf_dpd_txcal_ctrl_reg . . . 212
44.41 pu_done_reg . . . 212
44.42 dig_reserved_reg . . . 212
44.43 agc_reserved_reg . . . 212
44.44 agc_table_out_reg0 . . . 213
44.45 agc_table_out_reg1 . . . 213
44.46 agc_table_out_reg2 . . . 213
44.47 agc_table_out_reg3 . . . 213
44.48 agc_table_out_reg4 . . . 213
44.49 apc_reserved_reg . . . 213
44.50 apc_table_out_reg0 . . . 213
44.51 apc_table_out_reg1 . . . 213
44.52 apc_table_out_reg2 . . . 213
sys_ctrl

45.1 chip_id . . . 213
45.2 aon_clk_sel . . . 214
45.3 aon_clk_mode . . . 214
45.4 aon_clk_div . . . 214
45.5 top_clk_sel . . . 214
45.6 top_hclk_mode . . . 214
45.7 top_fclk_mode . . . 214
45.8 top_others_clk_mode . . . 214
45.9 hclk_div . . . 214
45.10 fclk_div . . . 214
45.11 psram_clk_div . . . 214
45.12 cfg_480m_clk_div . . . 215
45.13 real_320m_clk_div . . . 215
45.14 real_480m_clk_div . . . 215
45.15 cfg_osc_enable . . . 215
45.16 reset_cause . . . 215
45.17 tports_sel . . . 215
45.18 cpu_p_reset_addr . . . 215
45.19 cpu_s0_reset_addr . . . 215
45.20 cpu_s1_reset_addr . . . 215
45.21 cfg_cpu_sys_ctrl . . . 216
45.22 cfg_rtc1_pu_dcdc_resetn_bypass . . . 216
45.23 cfg_sdio_iomux_sel . . . 216
45.24 cfg_mem_cpu_sys_1pram . . . 216
45.25 cfg_mem_bt_sys_1pram . . . 216
45.26 bt_hclk_div . . . 216
45.27 top_tports_bits_sel0 . . . 216
45.28 top_tports_bits_sel1 . . . 216
45.29 top_tports_bits_sel2 . . . 216
45.30 top_tports_bits_sel3 . . . 217
45.31 perst_cfg . . . 217
45.32 hclk_manual_enable . . . 217
45.33 hclk_manual_disable . . . 217
45.34 fclk_manual_enable . . . 217
45.35 fclk_manual_disable . . . 217
45.36 others_manual_enable . . . 217
45.37 others_manual_disable . . . 218
45.38 aon_clk_manual_enable . . . 218
45.39 aon_clk_manual_disable . . . 218
45.40 global_soft_resetn_set . . . 219
45.41 global_soft_resetn_clr . . . 219
45.42 hclk_soft_resetn_set . . . 219
45.43 hclk_soft_resetn_clr . . . 219
45.44 fclk_soft_resetn_set . . . 220
45.45 fclk_soft_resetn_clr . . . 220
45.46 others_soft_resetn_set . . . 220
45.47 others_soft_resetn_clr . . . 220
45.48 aon_soft_resetn_set . . . 220
45.49 aon_soft_resetn_clr . . . 221
45.50 boot_mode . . . 221
45.51 wdg_reset_en . . . 221
45.52 reserved_reg0 . . . 222
45.53 reserved_reg1 . . . 222
45.54 reserved_reg2 . . . 222
45.55 reserved_reg3 . . . 222
45.56 reserved_reg4 . . . 222
45.57 reserved_reg5 . . . 222
45.58 reserved_reg6 . . . 222
45.59 reserved_reg7 . . . 222
45.60 reserved_reg8 . . . 222

45.61 reserved_reg9 . . . 222
45.62 reserved_reg10 . . . 223
45.63 reserved_reg11 . . . 223
45.64 reserved_reg12 . . . 223
45.65 reserved_reg13 . . . 223
45.66 reserved_reg14 . . . 223
45.67 reserved_reg15 . . . 223
timer2
46.1 TIMER0_CTL . . . 223
46.2 TIMER0_TICK_DIV . . . 224
46.3 TIMER0_INT_CTRL . . . 224
46.4 TIMER0_INT_STATUS . . . 225
46.5 TIMER0_TC_LO . . . 226
46.6 TIMER0_TC_HI . . . 226
46.7 TIMER0_MR0_LO . . . 226
46.8 TIMER0_MR0_HI . . . 226
46.9 TIMER0_MR1_LO . . . 226
46.10 TIMER0_MR1_HI . . . 226
46.11 TIMER0_MR2_LO . . . 226
46.12 TIMER0_MR2_HI . . . 227
46.13 TIMER0_MR3_LO . . . 227
46.14 TIMER0_MR3_HI . . . 227
ana_reg1
47.1 gpioa_cfg . . . 227
47.2 gpiob_cfg1 . . . 227
47.3 gpiob_cfg2 . . . 227
47.4 agpio_cfg . . . 228
47.5 ana_port_ctrl . . . 228
47.6 misc_reg . . . 228
47.7 flash_ctrl . . . 228
47.8 ana_bit_rd_back . . . 228
psi_mst
48.1 ip_revision . . . 228
48.2 psi_mst_ctrl . . . 229
48.3 psi_mst_pri00 . . . 229
48.4 psi_mst_pri10 . . . 229
48.5 psi_mst_pri20 . . . 229
48.6 psi_mst_pri30 . . . 229
48.7 psi_mst_int_en . . . 229
48.8 psi_mst_raw_status . . . 229
48.9 psi_mst_mask_status . . . 230
48.10 psi_mst_int_clr . . . 230
48.11 psi_gssi_cfg0 . . . 230
48.12 psi_gssi_cfg1 . . . 230
48.13 psi_arm_rd_cmd_addr . . . 230
48.14 psi_arm_rd_cmd_data . . . 231
48.15 psi_arm_debug0 . . . 231
48.16 psi_arm_debug1 . . . 231
48.17 psi_arm_debug2 . . . 231
48.18 psi_cmd_status . . . 231
48.19 psi_chnl_en . . . 232
48.20 chnl2_wr_addr . . . 232
48.21 chnl3_wr_addr . . . 232
48.22 chnl4_wr_addr . . . 232
48.23 chnl5_wr_addr . . . 232
48.24 chnl10_wr_addr . . . 232
48.25 chnl11_wr_addr . . . 232
48.26 chnl12_wr_addr . . . 232
48.27 chnl13_wr_addr . . . 233

48.28 chnl20_wr_addr . . . 233
48.29 chnl21_wr_addr . . . 233
48.30 chnl22_wr_addr . . . 233
48.31 chnl23_wr_addr . . . 233
48.32 chnl24_wr_addr . . . 233
48.33 chnl25_wr_addr . . . 233
48.34 chnl26_wr_addr . . . 233
48.35 chnl27_wr_addr . . . 233
48.36 chnl2_wr_low_data . . . 233
48.37 chnl2_wr_high_data . . . 233
48.38 chnl3_wr_low_data . . . 234
48.39 chnl3_wr_high_data . . . 234
48.40 chnl4_wr_low_data . . . 234
48.41 chnl4_wr_high_data . . . 234
48.42 chnl5_wr_low_data . . . 234
48.43 chnl5_wr_high_data . . . 234
48.44 chnl10_wr_low_data . . . 234
48.45 chnl10_wr_high_data . . . 234
48.46 chnl11_wr_low_data . . . 234
48.47 chnl11_wr_high_data . . . 234
48.48 chnl12_wr_low_data . . . 234
48.49 chnl12_wr_high_data . . . 234
48.50 chnl13_wr_low_data . . . 235
48.51 chnl13_wr_high_data . . . 235
48.52 evt_trg_cfg00 . . . 235
48.53 evt_trg_cfg10 . . . 235
48.54 psi_rd_cmd_addr . . . 235
48.55 psi_wr_cmd_addr . . . 235
48.56 psi_wr_cmd_data . . . 235
48.57 psi_timeout_cfg . . . 235
48.58 pmu_psi_slv_cfg0 . . . 235
48.59 pmu_psi_slv_cfg1 . . . 235

iomux

49.1 pad_gpioa_0_cfg_reg . . . 236
49.2 pad_gpioa_1_cfg_reg . . . 236
49.3 pad_gpioa_2_cfg_reg . . . 236
49.4 pad_gpioa_3_cfg_reg . . . 237
49.5 pad_gpioa_4_cfg_reg . . . 237
49.6 pad_gpioa_5_cfg_reg . . . 238
49.7 pad_gpioa_6_cfg_reg . . . 238
49.8 pad_gpioa_7_cfg_reg . . . 239
49.9 pad_gpioa_8_cfg_reg . . . 239
49.10 pad_gpioa_9_cfg_reg . . . 239
49.11 pad_gpioa_10_cfg_reg . . . 240
49.12 pad_gpioa_11_cfg_reg . . . 240
49.13 pad_gpioa_12_cfg_reg . . . 241
49.14 pad_gpioa_13_cfg_reg . . . 241
49.15 pad_gpioa_14_cfg_reg . . . 241
49.16 pad_gpioa_15_cfg_reg . . . 242
49.17 pad_agpio_0_cfg_reg . . . 242
49.18 pad_agpio_1_cfg_reg . . . 243
49.19 pad_agpio_2_cfg_reg . . . 243
49.20 pad_agpio_3_cfg_reg . . . 243
49.21 pad_agpio_4_cfg_reg . . . 243
49.22 pad_agpio_5_cfg_reg . . . 244
49.23 pad_agpio_6_cfg_reg . . . 244
49.24 pad_agpio_7_cfg_reg . . . 244
49.25 pad_agpio_8_cfg_reg . . . 245
49.26 pad_agpio_9_cfg_reg . . . 245

49.27 pad_agpio_10_cfg_reg . . . 245
49.28 pad_agpio_11_cfg_reg . . . 246
49.29 pad_agpio_12_cfg_reg . . . 246
49.30 pad_agpio_13_cfg_reg . . . 246
49.31 pad_agpio_14_cfg_reg . . . 246
49.32 pad_agpio_15_cfg_reg . . . 247
49.33 pad_gpiob_0_cfg_reg . . . 247
49.34 pad_gpiob_1_cfg_reg . . . 247
49.35 pad_gpiob_2_cfg_reg . . . 248
49.36 pad_gpiob_3_cfg_reg . . . 248
49.37 pad_gpiob_4_cfg_reg . . . 248
49.38 pad_gpiob_5_cfg_reg . . . 249
49.39 pad_gpiob_6_cfg_reg . . . 249
49.40 pad_gpiob_7_cfg_reg . . . 249

gpiob

50.1 GPIODATA . . . 250
50.2 GPIODMSK . . . 250
50.3 GPIODIR . . . 250
50.4 GPIOIS . . . 250
50.5 GPIOIBE . . . 250
50.6 GPIOIEV . . . 250
50.7 GPIOIE . . . 250
50.8 GPIORIS . . . 250
50.9 GPIOMIS . . . 251
50.10 GPIOIC . . . 251
50.11 GPIOINEN . . . 251
50.12 GPIO_FILTER_NUM . . . 251
50.13 GPIO_FILTER_DIV_L . . . 251
50.14 GPIO_FILTER_DIV_H . . . 251

pwr_ctrl

51.1 cpu_sys_pwr_ctrl . . . 251
51.2 bt_core_pwr_ctrl . . . 252
51.3 wifi_core_pwr_ctrl . . . 252
51.4 pwr_sys_ctrl . . . 252
51.5 pwr_off_timmer_th1 . . . 253
51.6 pwr_off_timmer_th2 . . . 253
51.7 pwr_ctrl_delay1 . . . 253
51.8 pwr_ctrl_delay2 . . . 253
51.9 pwr_sys_delay1 . . . 253
51.10 pwr_sys_delay2 . . . 253
51.11 pwr_ctrl_state1 . . . 253
51.12 pwr_ctrl_state2 . . . 254
51.13 pwr_ctrl_soft_mode1 . . . 254
51.14 pwr_ctrl_irq_ctrl . . . 254
51.15 pwr_ctrl_irq_status . . . 254
51.16 mm_sys_pwr_ctrl . . . 254
51.17 pwr_ctrl_soft_mode2 . . . 255
51.18 vcore_sys_pwr_ctrl . . . 255
51.19 pwr_ctrl_mode . . . 255
51.20 ret_ram_ctrl . . . 255
51.21 pwr_ctrl_monitor . . . 255
51.22 pwr_ctrl_soft_mode3 . . . 256
51.23 pwr_ctrl_soft_mode4 . . . 256
51.24 pwr_off_timmer_th3 . . . 256
51.25 pwr_off_timmer_th4 . . . 256
51.26 bt_core_timer_cnt_ro . . . 256
51.27 wifi_core_timer_cnt_ro . . . 256

aon_pwm

52.1 tmr0_ld_value . . . 256

52.2 tmr0_cnt_value . . . 257
52.3 tmr0_cfg . . . 257
52.4 tmr0_int_ctrl . . . 257
52.5 tmr0_cnt_read . . . 257
52.6 tmr1_ld_value . . . 257
52.7 tmr1_cnt_value . . . 257
52.8 tmr1_cfg . . . 257
52.9 tmr1_int_ctrl . . . 257
52.10 tmr1_cnt_read . . . 257
52.11 tmr2_ld_value . . . 257
52.12 tmr2_cnt_value . . . 257
52.13 tmr2_cfg . . . 258
52.14 tmr2_int_ctrl . . . 258
52.15 tmr2_cnt_read . . . 258
52.16 pwm0_sta_val . . . 258
52.17 pwm0_end_val . . . 258
52.18 pwm0_step_cfg0 . . . 258
52.19 pwm0_step_cfg1 . . . 258
52.20 pwm0_cfg . . . 258
52.21 pwm0_int_ctrl . . . 258
52.22 pwm0_end_val_use . . . 259
52.23 pwm0_step_cnt . . . 259
52.24 pwm1_sta_val . . . 259
52.25 pwm1_end_val . . . 259
52.26 pwm1_step_cfg0 . . . 259
52.27 pwm1_step_cfg1 . . . 259
52.28 pwm1_cfg . . . 259
52.29 pwm1_int_ctrl . . . 259
52.30 pwm1_end_val_use . . . 259
52.31 pwm1_step_cnt . . . 259
52.32 pwm2_sta_val . . . 260
52.33 pwm2_end_val . . . 260
52.34 pwm2_step_cfg0 . . . 260
52.35 pwm2_step_cfg1 . . . 260
52.36 pwm2_cfg . . . 260
52.37 pwm2_int_ctrl . . . 260
52.38 pwm2_end_val_use . . . 260
52.39 pwm2_step_cnt . . . 260
wdg2
53.1 WDG_LOAD_VAL . . . 260
53.2 WDG_CNT . . . 261
53.3 WDG_CTRL . . . 261
53.4 WDG_INT_CLR . . . 261
53.5 WDG_INT_RAW . . . 261
53.6 WDG_INT_MASK . . . 261
53.7 WDG_LOCK . . . 261
cm_rf_interface
54.1 reg_00 . . . 262
54.2 reg_04 . . . 262
54.3 reg_08 . . . 262
54.4 reg_0C . . . 262
54.5 reg_10 . . . 263
54.6 reg_14 . . . 263
54.7 reg_18 . . . 263
54.8 cm_1m_reg . . . 264
54.9 bbpll_sdm_reg1 . . . 264
54.10 bbpll_sdm_reg2 . . . 264
54.11 wf_wr_only_reg . . . 264
54.12 bt_wr_only_reg . . . 264

54.13 mdll_fref_reg . . . 264
54.14 cm_mdll_lock_ctrl_reg . . . 264
54.15 cm_mdll_lock_status_reg . . . 265
54.16 cm_mdll_unlock_en_reg . . . 265
54.17 cm_mdll_unlock_clear_reg . . . 265
54.18 dig_reserved_reg . . . 265
bt_dfe_cfg
55.1 a0x000 . . . 265
55.2 a0x004 . . . 265
55.3 a0x008 . . . 266
55.4 dpll_cgen_00c . . . 266
55.5 gll_ctrl . . . 266
55.6 gll_ctrl2 . . . 266
55.7 plriq_26m_080 . . . 266
55.8 plriq_26m_084 . . . 266
55.9 a0x090 . . . 267
55.10 a0x094 . . . 267
55.11 a0x098 . . . 267
55.12 txdfe_rtcv0 . . . 267
55.13 txdfe_rtcv1 . . . 267
55.14 rxdfe_rtcv0 . . . 267
55.15 rxdfe_rtcv1 . . . 268
55.16 a0x0f0 . . . 268
55.17 a0x100 . . . 268
55.18 a0x104 . . . 268
55.19 a0x108 . . . 268
55.20 a0x10c . . . 268
55.21 a0x110 . . . 268
55.22 a0x114 . . . 269
55.23 a0x118 . . . 269
55.24 a0x11c . . . 269
55.25 a0x120 . . . 269
55.26 a0x200 . . . 269
55.27 a0x204 . . . 269
55.28 a0x208 . . . 269
55.29 a0x20c . . . 269
55.30 a0x210 . . . 269
55.31 cass_ctrl . . . 270
55.32 cass_var . . . 270
55.33 cass_rfpwr . . . 270
55.34 cass_map0 . . . 270
55.35 cass_map1 . . . 270
55.36 cass_rssi0 . . . 270
55.37 cass_rssi1 . . . 270
55.38 cass_rssi2 . . . 271
55.39 cass_rssi3 . . . 271
55.40 cass_rssi4 . . . 271
55.41 cass_rssi5 . . . 271
55.42 cass_rssi6 . . . 271
55.43 cass_rssi7 . . . 271
55.44 cass_rssi8 . . . 271
55.45 cass_rssi9 . . . 271
55.46 a0x500 . . . 271
55.47 a0x504 . . . 272
55.48 a0x508 . . . 272
55.49 a0x50c . . . 272
55.50 a0x510 . . . 272
55.51 a0x514 . . . 272
55.52 a0x518 . . . 272

55.53 a0x51c . . . 273
55.54 a0x520 . . . 273
55.55 a0x528 . . . 273
55.56 a0x52c . . . 273
55.57 a0x530 . . . 273
55.58 a0x534 . . . 274
55.59 a0x544 . . . 274
55.60 a0x5a0 . . . 274
55.61 a0x5a4 . . . 274
55.62 a0x5a8 . . . 274
55.63 a0x5ac . . . 274
55.64 a0x5b0 . . . 275
55.65 a0x5b4 . . . 275
55.66 a0x5b8 . . . 275
55.67 a0x5bc . . . 275
55.68 a0x5c0 . . . 275
55.69 a0x5c4 . . . 275
55.70 a0x5c8 . . . 275
55.71 a0x5cc . . . 275
55.72 a0x5d0 . . . 275
55.73 a0x5d4 . . . 276
55.74 a0x5d8 . . . 276
55.75 a0x5dc . . . 276
55.76 a0x5f0 . . . 276
55.77 wbagc_5f4 . . . 276
55.78 dpll_rtcv_710 . . . 276
55.79 dpll_rtcv_714 . . . 276
55.80 a0x720 . . . 276
55.81 a0x724 . . . 277
55.82 a0x728 . . . 277
55.83 a0x730 . . . 277
55.84 a0x734 . . . 277
55.85 a0x738 . . . 277
55.86 a0x740 . . . 278
55.87 a0x744 . . . 278
55.88 a0x748 . . . 278
55.89 a0x74c . . . 278
55.90 a0x750 . . . 278
55.91 a0x754 . . . 278
55.92 a0x758 . . . 279
55.93 phase_off0_75c . . . 279
55.94 phase_off1_760 . . . 279
55.95 a0x764 . . . 279
55.96 a0x768 . . . 279
55.97 a0x76c . . . 279
55.98 dacintf_770 . . . 280
55.99 band_mapd . . . 280
55.100 band_mapc . . . 280
55.101 band_mapb . . . 280
55.102 band_mapa . . . 280
55.103 band_map9 . . . 280
55.104 band_map8 . . . 280
55.105 band_map7 . . . 280
55.106 band_map6 . . . 281
55.107 band_map5 . . . 281
55.108 band_map4 . . . 281
55.109 band_map3 . . . 281
55.110 band_map2 . . . 281
55.111 band_map1 . . . 281
55.112 band_map0 . . . 281

55.113 band_ctrl . . . 281
55.114 a0x810 . . . 282
55.115 a0x814 . . . 282
55.116 a0x81c . . . 282
55.117 a0x820 . . . 282
55.118 a0x824 . . . 282
55.119 a0x828 . . . 282
55.120 a0x82c . . . 282
55.121 a0x830 . . . 283
55.122 a0x834 . . . 283
55.123 ppm_838 . . . 283
55.124 fref_tx_hm . . . 283
55.125 fref_tx . . . 283
55.126 fref_rx . . . 283
55.127 coef_rfbt . . . 283
55.128 coef_rfwf . . . 283
55.129 dco_div_ctrl . . . 284
55.130 a0x86c . . . 284
55.131 rsvd_wc_870 . . . 284
55.132 a0x880 . . . 284
55.133 a0x884 . . . 284
55.134 a0x888 . . . 284
55.135 a0x88c . . . 284
55.136 a0x890 . . . 284
55.137 a0x894 . . . 284
55.138 a0x898 . . . 284
55.139 a0x89c . . . 284
55.140 a0x8a0 . . . 285
55.141 a0x8a4 . . . 285
55.142 a0x8a8 . . . 285
55.143 a0x8ac . . . 285
bt_mdm_cfg
56.1 a0x010 . . . 285
56.2 a0x014 . . . 285
56.3 a0x018 . . . 285
56.4 a0x01c . . . 286
56.5 a0x020 . . . 286
56.6 a0x024 . . . 286
56.7 a0x028 . . . 286
56.8 dpwr_ctrl . . . 286
56.9 a0x034 . . . 286
56.10 a0x040 . . . 286
56.11 a0x044 . . . 287
56.12 a0x048 . . . 287
56.13 a0x04c . . . 287
56.14 a0x050 . . . 287
56.15 a0x054 . . . 287
56.16 a0x058 . . . 287
56.17 a0x05c . . . 287
56.18 a0x060 . . . 287
56.19 a0x064 . . . 287
56.20 a0x068 . . . 288
56.21 a0x06c . . . 288
56.22 a0x070 . . . 288
56.23 a0x074 . . . 288
56.24 a0x078 . . . 288
56.25 a0x07c . . . 288
56.26 a0x080 . . . 288
56.27 a0x084 . . . 288

56.28 a0x400 . . . 288
56.29 a0x404 . . . 288
56.30 a0x408 . . . 289
56.31 a0x40c . . . 289
56.32 a0x410 . . . 289
56.33 a0x414 . . . 289
56.34 a0x418 . . . 289
56.35 a0x420 . . . 289
56.36 a0x424 . . . 289
56.37 a0x428 . . . 289
56.38 a0x42c . . . 289
56.39 a0x430 . . . 290
56.40 a0x434 . . . 290
56.41 a0x438 . . . 290
56.42 a0x43c . . . 290
56.43 a0x440 . . . 290
56.44 a0x444 . . . 290
56.45 a0x448 . . . 290
56.46 a0x44c . . . 291
56.47 a0x450 . . . 291
56.48 a0x454 . . . 291
56.49 a0x458 . . . 291
56.50 a0x45c . . . 291
56.51 a0x460 . . . 291
56.52 a0x464 . . . 291
56.53 a0x468 . . . 291
56.54 a0x46c . . . 292
56.55 a0x470 . . . 292
56.56 a0x474 . . . 292
56.57 a0x478 . . . 292
56.58 a0x47c . . . 292
56.59 a0x480 . . . 292
56.60 a0x484 . . . 292
56.61 a0x488 . . . 292
56.62 a0x48c . . . 293
56.63 a0x550 . . . 293
56.64 a0x554 . . . 293
56.65 a0x558 . . . 293
56.66 a0x55c . . . 293
56.67 a0x560 . . . 294
56.68 a0x564 . . . 294
56.69 a0x568 . . . 294
56.70 a0x56c . . . 294
56.71 a0x57c . . . 294
56.72 a0x580 . . . 294
56.73 a0x584 . . . 295
56.74 a0x588 . . . 295
56.75 a0x58c . . . 295
56.76 a0x590 . . . 295
56.77 a0x594 . . . 295
56.78 a0x598 . . . 295
56.79 a0x870 . . . 295
56.80 a0x874 . . . 295
56.81 a0x878 . . . 295
56.82 a0x87c . . . 295
56.83 a0x8b4 . . . 296
bt_rf_interface
57.1 reg_00 . . . 296
57.2 reg_04 . . . 296

57.3 reg_08 . . . 297
57.4 reg_0C . . . 297
57.5 reg_10 . . . 298
57.6 reg_14 . . . 298
57.7 reg_18 . . . 298
57.8 reg_1C . . . 298
57.9 reg_20 . . . 299
57.10 reg_24 . . . 299
57.11 reg_28 . . . 299
57.12 reg_2C . . . 299
57.13 reg_30 . . . 300
57.14 reg_34 . . . 300
57.15 reg_38 . . . 300
57.16 bt_soft_reset_reg . . . 300
57.17 bt_man_crtl_reg . . . 301
57.18 bt_calclk_reg . . . 301
57.19 gain_mem_ctrl_reg . . . 301
57.20 gain_mem_fetch_reg . . . 301
57.21 pu_done_reg . . . 301
57.22 dig_reserved_reg . . . 301
57.23 agc_table_out_reg0 . . . 301
57.24 agc_table_out_reg1 . . . 302
57.25 agc_table_out_reg2 . . . 302
57.26 apc_table_out_reg0 . . . 302
bt_sys_ctrl
58.1 cfg_trap_bypass . . . 302
58.2 clk_sel . . . 302
58.3 cfg_bt_uart_clk_div . . . 302
58.4 cfg_hclk_mode . . . 302
58.5 cfg_others_clk_mode . . . 302
58.6 cfg_mem_rom . . . 302
58.7 cfg_mem_1prf . . . 302
58.8 cfg_mem_2prf . . . 302
58.9 cfg_cpu_stclk_div . . . 303
58.10 cfg_cpu_s1_rxev . . . 303
58.11 hclk_manual_enable . . . 303
58.12 hclk_manual_disable . . . 303
58.13 others_manual_enable . . . 304
58.14 others_manual_disable . . . 304
58.15 hclk_soft_resetn_set . . . 304
58.16 hclk_soft_resetn_clr . . . 304
58.17 others_soft_resetn_set . . . 305
58.18 others_soft_resetn_clr . . . 305
bt_timer
59.1 tmr_cfg . . . 305
59.2 tmr_pr . . . 305
59.3 tmr_pc . . . 305
59.4 tmr_mr0 . . . 305
59.5 tmr_mr1 . . . 305
59.6 tmr_mr2 . . . 305
59.7 tmr_delta0 . . . 305
59.8 tmr_delta1 . . . 306
59.9 tmr_delta2 . . . 306
59.10 tmr_rtc_clk_div . . . 306
59.11 tmr_int_mask_enable . . . 306
59.12 tmr_int_mask_clear . . . 306
59.13 tmr_int_raw . . . 306
59.14 tmr_int_status . . . 306
bt_wdg

60.1 WDG_LOAD_VAL . . . 306
60.2 WDG_CNT . . . 307
60.3 WDG_CTRL . . . 307
60.4 WDG_INT_CLR . . . 307
60.5 WDG_INT_RAW . . . 307
60.6 WDG_INT_MASK . . . 307
60.7 WDG_LOCK . . . 307
cpu_s1_trap
61.1 addr_in_0 . . . 308
61.2 addr_in_1 . . . 308
61.3 addr_in_2 . . . 308
61.4 addr_in_3 . . . 308
61.5 addr_in_4 . . . 308
61.6 addr_in_5 . . . 308
61.7 addr_in_6 . . . 308
61.8 addr_in_7 . . . 308
61.9 en . . . 308
61.10 trap_out_base . . . 308
bt_dma
62.1 SAR0 . . . 309
62.2 DAR0 . . . 309
62.3 LLP0 . . . 309
62.4 CTL0_L . . . 309
62.5 CTL0_H . . . 313
62.6 CFG0_L . . . 313
62.7 CFG0_H . . . 316
62.8 SGR0 . . . 317
62.9 DSR0 . . . 317
62.10 SAR1 . . . 317
62.11 DAR1 . . . 317
62.12 LLP1 . . . 318
62.13 CTL1_L . . . 318
62.14 CTL1_H . . . 322
62.15 CFG1_L . . . 322
62.16 CFG1_H . . . 325
62.17 SGR1 . . . 326
62.18 DSR1 . . . 326
62.19 SAR2 . . . 326
62.20 DAR2 . . . 326
62.21 LLP2 . . . 327
62.22 CTL2_L . . . 327
62.23 CTL2_H . . . 331
62.24 CFG2_L . . . 331
62.25 CFG2_H . . . 334
62.26 SGR2 . . . 335
62.27 DSR2 . . . 335
62.28 RAW_TFR . . . 335
62.29 RAW_BLOCK . . . 335
62.30 RAW_SRC_TRAN . . . 336
62.31 RAW_DST_TRAN . . . 336
62.32 RAW_ERR . . . 336
62.33 STATUS_TFR . . . 336
62.34 STATUS_BLOCK . . . 336
62.35 STATUS_SRC_TRAN . . . 336
62.36 STATUS_DST_TRAN . . . 336
62.37 STATUS_ERR . . . 337
62.38 MASK_TFR . . . 337
62.39 MASK_BLOCK . . . 337
62.40 MASK_SRC_TRAN . . . 337

62.41 MASK_DST_TRAN . . . 337
62.42 MASK_ERR . . . 338
62.43 CLEAR_TFR . . . 338
62.44 CLEAR_BLOCK . . . 338
62.45 CLEAR_SRC_TRAN . . . 338
62.46 CLEAR_DST_TRAN . . . 338
62.47 CLEAR_ERR . . . 338
62.48 STATUS_INT . . . 339
62.49 REQ_SRC . . . 339
62.50 REQ_DST . . . 339
62.51 SGL_RQ_SRC . . . 339
62.52 SGL_RQ_DST . . . 340
62.53 LST_SRC . . . 340
62.54 LST_DST . . . 340
62.55 DMA_CFG . . . 340
62.56 CH_EN . . . 340
62.57 DMA_ID . . . 341
62.58 DMA_TEST . . . 341
62.59 DMA_LP_TIMEOUT . . . 341
62.60 DMA_COMP_PARAMS_6L . . . 341
62.61 DMA_COMP_PARAMS_6H . . . 341
62.62 DMA_COMP_PARAMS_5L . . . 341
62.63 DMA_COMP_PARAMS_5H . . . 341
62.64 DMA_COMP_PARAMS_4L . . . 341
62.65 DMA_COMP_PARAMS_4H . . . 342
62.66 DMA_COMP_PARAMS_3L . . . 342
62.67 DMA_COMP_PARAMS_3H . . . 342
62.68 DMA_COMP_PARAMS_2L . . . 342
62.69 DMA_COMP_PARAMS_2H . . . 342
62.70 DMA_COMP_PARAMS_1L . . . 342
62.71 DMA_COMP_PARAMS_1H . . . 342
62.72 DMA_COMPS_ID_L . . . 342
62.73 DMA_COMPS_ID_H . . . 342

bt_uart

63.1 txrx_fifo_divl . . . 342
63.2 int_en_divh . . . 342
63.3 int_id_fifo_ctrl . . . 343
63.4 line_ctrl . . . 343
63.5 modem_ctrl . . . 343
63.6 line_status . . . 343
63.7 modem_status . . . 344
63.8 scratchpad . . . 344
63.9 fifo_status . . . 344
63.10 fifo_trigger . . . 344
63.11 divisor_latch2 . . . 344
63.12 counter_real . . . 344
63.13 div_real_slow . . . 344
63.14 div_real_fast . . . 344
63.15 tports_sel . . . 345

pcie_ep

64.1 dev_id_vendor_id . . . 345
64.2 status_cmd . . . 345
64.3 class_code_rev_id . . . 345
64.4 bist_header_type_latency_cache_line_size . . . 345
64.5 bar0 . . . 345
64.6 bar1 . . . 346
64.7 bar2 . . . 346
64.8 bar3 . . . 346
64.9 cardbus_cis_ptr . . . 346

64.10 subsys_id_subsys_vendor_id . . . 346
64.11 exp_rom_base_addr . . . 346
64.12 pci_cap_ptr . . . 346
64.13 intr_line_intr_pin . . . 346
64.14 pm_cap . . . 346
64.15 pm_ctrl_status . . . 347
64.16 msi_cap . . . 347
64.17 msi_lower_addr . . . 347
64.18 msi_upper_addr . . . 347
64.19 msi_data . . . 347
64.20 pcie_cap . . . 347
64.21 dev_cap . . . 348
64.22 dev_ctrl_status . . . 348
64.23 link_cap . . . 348
64.24 link_ctrl_status . . . 348
64.25 dev_cap2 . . . 349
64.26 dev_ctrl_status2 . . . 349
64.27 link_cap2 . . . 349
64.28 link_ctrl_status2 . . . 349
64.29 aer_ext_cap_hdr . . . 350
64.30 uncorr_err_status . . . 350
64.31 uncorr_err_mask . . . 350
64.32 uncorr_err_sev . . . 350
64.33 corr_err_status . . . 351
64.34 corr_err_mask . . . 351
64.35 adv_err_cap_ctrl . . . 351
64.36 hdr_log_0 . . . 351
64.37 hdr_log_1 . . . 351
64.38 hdr_log_2 . . . 351
64.39 hdr_log_3 . . . 352
64.40 tlp_prefix_log . . . 352
64.41 tph_ext_cap_hdr . . . 352
64.42 tph_req_cap_reg . . . 352
64.43 tph_req_ctrl_reg . . . 352
64.44 tph_st_table_reg_0 . . . 352
64.45 tph_st_table_reg_1 . . . 352
64.46 tph_st_table_reg_2 . . . 352
64.47 tph_st_table_reg_3 . . . 353
64.48 tph_st_table_reg_4 . . . 353
64.49 tph_st_table_reg_5 . . . 353
64.50 tph_st_table_reg_6 . . . 353
64.51 tph_st_table_reg_7 . . . 353
64.52 tph_st_table_reg_8 . . . 353
64.53 tph_st_table_reg_9 . . . 353
64.54 tph_st_table_reg_10 . . . 354
64.55 tph_st_table_reg_11 . . . 354
64.56 tph_st_table_reg_12 . . . 354
64.57 tph_st_table_reg_13 . . . 354
64.58 tph_st_table_reg_14 . . . 354
64.59 tph_st_table_reg_15 . . . 354
64.60 ats_cap_hdr . . . 354
64.61 ats_cap_ctrl . . . 354
64.62 l1sub_cap_header . . . 355
64.63 l1sub_cap . . . 355
64.64 l1sub_ctrl1 . . . 355
64.65 l1sub_ctrl2 . . . 355
64.66 ack_latency_timer . . . 355
64.67 vendor_spec_dllp . . . 355
64.68 port_force . . . 355
64.69 ack_f_aspm_ctrl . . . 356

64.70 port_link_ctrl . . . 356
64.71 lane_skew . . . 356
64.72 timer_ctrl_max_func_num . . . 356
64.73 symbol_timer_filter_1 . . . 356
64.74 filter_mask_2 . . . 356
64.75 amba_mul_ob_decomp_np_sub_req_ctrl . . . 357
64.76 pl_debug0 . . . 357
64.77 pl_debug1 . . . 357
64.78 tx_p_fc_credit_status . . . 357
64.79 tx_np_fc_credit_status . . . 357
64.80 tx_cpl_fc_credit_status . . . 357
64.81 queue_status . . . 357
64.82 vc_tx_arbl_1 . . . 357
64.83 vc_tx_arbl_2 . . . 357
64.84 vc0_p_rx_1_ctrl . . . 358
64.85 vc0_np_rx_q_ctrl . . . 358
64.86 vc0_cpl_rx_q_ctrl . . . 358
64.87 gen2_ctrl . . . 358
64.88 phy_status . . . 358
64.89 phy_ctrl . . . 358
64.90 pipe_loopback_ctrl . . . 358
64.91 misc_ctrl_1 . . . 358
64.92 amba_error_resp_default . . . 358
64.93 amba_ordrmgr_wdog . . . 359
64.94 pl_last . . . 359
64.95 iatu_viewport . . . 359
64.96 iatu_region_ctrl_1 . . . 359
64.97 iatu_region_ctrl_2 . . . 359
64.98 iatu_lwr_base_addr . . . 360
64.99 iatu_upper_base_addr . . . 360
64.100 iatu_limit_addr . . . 360
64.101 iatu_lwr_target_addr . . . 360
64.102 iatu_upper_target_addr . . . 360
64.103 dma_ctrl . . . 360
64.104 dma_wr_engine_en . . . 360
64.105 dma_wr_doorbell . . . 360
64.106 dma_wr_ch_arb_weight_low . . . 360
64.107 dma_wr_ch_arb_weight_high . . . 360
64.108 dma_wr_p_req_timer . . . 361
64.109 dma_rd_engine_en . . . 361
64.110 dma_rd_doorbell . . . 361
64.111 dma_rd_ch_arb_weight_low . . . 361
64.112 dma_rd_ch_arb_weight_high . . . 361
64.113 dma_wr_int_status . . . 361
64.114 dma_wr_int_mask . . . 361
64.115 dma_wr_int_clr . . . 361
64.116 dma_wr_err_status . . . 361
64.117 dma_wr_done_imwr_low . . . 362
64.118 dma_wr_done_imwr_high . . . 362
64.119 dma_wr_abort_imwr_low . . . 362
64.120 dma_wr_abort_imwr_high . . . 362
64.121 dma_wr_ch01_imwr_data . . . 362
64.122 dma_wr_ch23_imwr_data . . . 362
64.123 dma_wr_ch45_imwr_data . . . 362
64.124 dma_wr_ch67_imwr_data . . . 362
64.125 dma_wr_linked_list_err_en . . . 362
64.126 dma_rd_int_status . . . 362
64.127 dma_rd_int_mask . . . 363
64.128 dma_rd_int_clr . . . 363
64.129 dma_rdd_err_status_low . . . 363

64.130 dma_rd_err_status_high . . . 363
64.131 dma_rd_linked_list_err_en . . . 363
64.132 dma_rd_done_imwr_low . . . 363
64.133 dma_rd_done_imwr_high . . . 363
64.134 dma_rd_abort_imwr_low . . . 363
64.135 dma_rd_abort_imwr_high . . . 363
64.136 dma_rd_ch01_imwr_data . . . 363
64.137 dma_rd_ch23_imwr_data . . . 364
64.138 dma_rd_ch45_imwr_data . . . 364
64.139 dma_rd_ch67_imwr_data . . . 364
64.140 dma_viewport_sel . . . 364
64.141 dma_ch_ctrl1 . . . 364
64.142 dma_ch_ctrl2 . . . 364
64.143 dma_trans_size . . . 364
64.144 dma_sar_low . . . 364
64.145 dma_sar_high . . . 365
64.146 dma_dar_low . . . 365
64.147 dma_dar_high . . . 365
64.148 dma_llp_low . . . 365
64.149 dma_llp_high . . . 365
64.150 aux_clk_freq . . . 365
64.151 l1_substates . . . 365
pcie_ep_dbi2
65.1 dev_id_vendor_id . . . 365
65.2 status_cmd . . . 365
65.3 class_code_rev_id . . . 366
65.4 bist_header_type_latency_cache_line_size . . . 366
65.5 bar0 . . . 366
65.6 bar1 . . . 366
65.7 bar2 . . . 366
65.8 bar3 . . . 367
65.9 cardbus_cis_ptr . . . 367
65.10 subsys_id_subsys_vendor_id . . . 367
65.11 exp_rom_base_addr . . . 367
65.12 pci_cap_ptr . . . 367
65.13 intr_line_intr_pin . . . 367
65.14 pm_cap . . . 367
65.15 pm_ctrl_status . . . 367
65.16 msi_cap . . . 368
65.17 msi_lower_addr . . . 368
65.18 msi_upper_addr . . . 368
65.19 msi_data . . . 368
65.20 pcie_cap . . . 368
65.21 dev_cap . . . 368
65.22 dev_ctrl_status . . . 368
65.23 link_cap . . . 369
65.24 link_ctrl_status . . . 369
65.25 dev_cap2 . . . 369
65.26 dev_ctrl_status2 . . . 370
65.27 link_cap2 . . . 370
65.28 link_ctrl_status2 . . . 370
65.29 aer_ext_cap_hdr . . . 370
65.30 uncorr_err_status . . . 370
65.31 uncorr_err_mask . . . 371
65.32 uncorr_err_sev . . . 371
65.33 corr_err_status . . . 371
65.34 corr_err_mask . . . 372
65.35 adv_err_cap_ctrl . . . 372
65.36 hdr_log_0 . . . 372

65.37 hdr_log_1 . . . 372
65.38 hdr_log_2 . . . 372
65.39 hdr_log_3 . . . 372
65.40 tlp_prefix_log . . . 372
65.41 tph_ext_cap_hdr . . . 372
65.42 tph_req_cap_reg . . . 372
65.43 tph_req_ctrl_reg . . . 373
65.44 tph_st_table_reg_0 . . . 373
65.45 tph_st_table_reg_1 . . . 373
65.46 tph_st_table_reg_2 . . . 373
65.47 tph_st_table_reg_3 . . . 373
65.48 tph_st_table_reg_4 . . . 373
65.49 tph_st_table_reg_5 . . . 373
65.50 tph_st_table_reg_6 . . . 374
65.51 tph_st_table_reg_7 . . . 374
65.52 tph_st_table_reg_8 . . . 374
65.53 tph_st_table_reg_9 . . . 374
65.54 tph_st_table_reg_10 . . . 374
65.55 tph_st_table_reg_11 . . . 374
65.56 tph_st_table_reg_12 . . . 374
65.57 tph_st_table_reg_13 . . . 375
65.58 tph_st_table_reg_14 . . . 375
65.59 tph_st_table_reg_15 . . . 375
65.60 ats_cap_hdr . . . 375
65.61 ats_cap_ctrl . . . 375
65.62 l1sub_cap_header . . . 375
65.63 l1sub_cap . . . 375
65.64 l1sub_ctrl1 . . . 376
65.65 l1sub_ctrl2 . . . 376
65.66 ack_latency_timer . . . 376
65.67 vendor_spec_dllp . . . 376
65.68 port_force . . . 376
65.69 ack_f_aspm_ctrl . . . 376
65.70 port_link_ctrl . . . 376
65.71 lane_skew . . . 377
65.72 timer_ctrl_max_func_num . . . 377
65.73 symbol_timer_filter_1 . . . 377
65.74 filter_mask_2 . . . 377
65.75 amba_mul_ob_decomp_np_sub_req_ctrl . . . 377
65.76 pl_debug0 . . . 377
65.77 pl_debug1 . . . 377
65.78 tx_p_fc_credit_status . . . 377
65.79 tx_np_fc_credit_status . . . 378
65.80 tx_cpl_fc_credit_status . . . 378
65.81 queue_status . . . 378
65.82 vc_tx_arbl_1 . . . 378
65.83 vc_tx_arbl_2 . . . 378
65.84 vc0_p_rx_1_ctrl . . . 378
65.85 vc0_np_rx_q_ctrl . . . 378
65.86 vc0_cpl_rx_q_ctrl . . . 378
65.87 gen2_ctrl . . . 379
65.88 phy_status . . . 379
65.89 phy_ctrl . . . 379
65.90 pipe_loopback_ctrl . . . 379
65.91 misc_ctrl_1 . . . 379
65.92 amba_error_resp_default . . . 379
65.93 amba_ordrmgr_wdog . . . 379
65.94 pl_last . . . 379
65.95 iatu_viewport . . . 379
65.96 iatu_region_ctrl_1 . . . 380

65.97 iatu_region_ctrl_2 . . . 380
65.98 iatu_lwr_base_addr . . . 380
65.99 iatu_upper_base_addr . . . 380
65.100 iatu_limit_addr . . . 380
65.101 iatu_lwr_target_addr . . . 380
65.102 iatu_upper_target_addr . . . 381
65.103 dma_ctrl . . . 381
65.104 dma_wr_engine_en . . . 381
65.105 dma_wr_doorbell . . . 381
65.106 dma_wr_ch_arb_weight_low . . . 381
65.107 dma_wr_ch_arb_weight_high . . . 381
65.108 dma_wr_p_req_timer . . . 381
65.109 dma_rd_engine_en . . . 381
65.110 dma_rd_doorbell . . . 381
65.111 dma_rd_ch_arb_weight_low . . . 382
65.112 dma_rd_ch_arb_weight_high . . . 382
65.113 dma_wr_int_status . . . 382
65.114 dma_wr_int_mask . . . 382
65.115 dma_wr_int_clr . . . 382
65.116 dma_wr_err_status . . . 382
65.117 dma_wr_done_imwr_low . . . 382
65.118 dma_wr_done_imwr_high . . . 382
65.119 dma_wr_abort_imwr_low . . . 382
65.120 dma_wr_abort_imwr_high . . . 382
65.121 dma_wr_ch01_imwr_data . . . 383
65.122 dma_wr_ch23_imwr_data . . . 383
65.123 dma_wr_ch45_imwr_data . . . 383
65.124 dma_wr_ch67_imwr_data . . . 383
65.125 dma_wr_linked_list_err_en . . . 383
65.126 dma_rd_int_status . . . 383
65.127 dma_rd_int_mask . . . 383
65.128 dma_rd_int_clr . . . 383
65.129 dma_rdd_err_status_low . . . 383
65.130 dma_rd_err_status_high . . . 384
65.131 dma_rd_linked_list_err_en . . . 384
65.132 dma_rd_done_imwr_low . . . 384
65.133 dma_rd_done_imwr_high . . . 384
65.134 dma_rd_abort_imwr_low . . . 384
65.135 dma_rd_abort_imwr_high . . . 384
65.136 dma_rd_ch01_imwr_data . . . 384
65.137 dma_rd_ch23_imwr_data . . . 384
65.138 dma_rd_ch45_imwr_data . . . 384
65.139 dma_rd_ch67_imwr_data . . . 384
65.140 dma_viewport_sel . . . 384
65.141 dma_ch_ctrl1 . . . 385
65.142 dma_ch_ctrl2 . . . 385
65.143 dma_trans_size . . . 385
65.144 dma_sar_low . . . 385
65.145 dma_sar_high . . . 385
65.146 dma_dar_low . . . 385
65.147 dma_dar_high . . . 385
65.148 dma_llp_low . . . 385
65.149 dma_llp_high . . . 386
65.150 aux_clk_freq . . . 386
65.151 l1_substates . . . 386
pcie_top_ctrl
66.1 pcie_ana_cfg0 . . . 386
66.2 pcie_ana_cfg1 . . . 386
66.3 pcie_ana_cfg2 . . . 387

66.4 pcie_ana_cfg3 . . . 387
66.5 pcie_cfg0 . . . 387
66.6 pcie_cfg1 . . . 387
66.7 pcie_cfg3 . . . 388
66.8 pcie_cfg4 . . . 388
66.9 pcie_cfg5 . . . 389
66.10 pcie_cfg6 . . . 389
66.11 pcie_cfg7 . . . 389
66.12 pcie_cfg8 . . . 389
66.13 pcie_cfg9 . . . 390
66.14 pcie_cfg10 . . . 390
66.15 pcie_cfg11 . . . 390
66.16 pcie_cfg12 . . . 390
66.17 pcie_cfg13 . . . 390
66.18 pcie_cfg14 . . . 390
66.19 pcie_cfg15 . . . 390
66.20 pcie_cfg16 . . . 391
66.21 pcie_cfg17 . . . 391
66.22 pcie_cfg18 . . . 391
66.23 pcie_cfg19 . . . 391
66.24 pcie_cfg20 . . . 391
66.25 pcie_cfg21 . . . 391
66.26 pcie_cfg22 . . . 391
66.27 pcie_cfg23 . . . 392
66.28 pcie_cfg24 . . . 392
66.29 pcie_cfg25 . . . 393
66.30 pcie_cfg26 . . . 393
66.31 pcie_cfg27 . . . 394
66.32 pcie_cfg28 . . . 394
66.33 pcie_cfg29 . . . 394
66.34 pcie_cfg30 . . . 394
66.35 pcie_cfg31 . . . 394
66.36 pcie_cfg32 . . . 394
66.37 pcie_cfg33 . . . 394
66.38 pcie_cfg34 . . . 395
66.39 pcie_cfg35 . . . 395
66.40 pcie_cfg36 . . . 395
66.41 pcie_cfg37 . . . 395
66.42 pcie_cfg38 . . . 395
66.43 pcie_cfg39 . . . 395
66.44 pcie_cfg40 . . . 395
66.45 pcie_cfg41 . . . 395
66.46 pcie_cfg42 . . . 395
66.47 pcie_cfg43 . . . 395
66.48 pcie_cfg44 . . . 395
66.49 pcie_cfg45 . . . 395
66.50 pcie_cfg46 . . . 396
66.51 pcie_cfg47 . . . 396
66.52 pcie_cfg48 . . . 396
66.53 pcie_cfg49 . . . 396
66.54 pcie_cfg50 . . . 396
66.55 pcie_cfg51 . . . 396
66.56 pcie_cfg52 . . . 396
66.57 pcie_cfg53 . . . 396
66.58 pcie_cfg55 . . . 396
66.59 pcie_cfg56 . . . 397
66.60 pcie_cfg57 . . . 397
66.61 pcie_cfg58 . . . 397
66.62 pcie_cfg59 . . . 397
66.63 pcie_cfg60 . . . 397

66.64 pcie_cfg61 . . . 397
66.65 pcie_cfg62 . . . 397
66.66 pcie_cfg63 . . . 397
66.67 pcie_cfg64 . . . 397
66.68 pcie_cfg65 . . . 398
pmu_rtc0
67.1 pmu_rtc0_cfg0 . . . 398
67.2 pmu_rtc0_cfg1 . . . 398
67.3 pmu_rtc0_cfg2 . . . 399
67.4 pmu_rtc0_cfg3 . . . 399
67.5 pmu_rtc0_cfg4 . . . 399
67.6 pmu_rtc0_cfg5 . . . 400
67.7 pmu_rtc0_cfg6 . . . 400
67.8 pmu_rtc0_cfg7 . . . 400
67.9 pmu_rtc0_wdg4_cnt_ini . . . 400
67.10 rtc_rg_reserved0 . . . 400
67.11 rtc_rg_reserved1 . . . 400
pmu_rtc1
68.1 pmu_pd_ctrl0 . . . 400
68.2 pmu_pd_ctrl1 . . . 401
68.3 pmu_seq_num0 . . . 401
68.4 pmu_seq_num1 . . . 401
68.5 pmu_slp_ctrl0 . . . 402
68.6 pmu_slp_ctrl1 . . . 402
68.7 pmu_clk_ctrl0 . . . 403
68.8 pmu_clk_ctrl1 . . . 403
68.9 pmu_clk_ctrl2 . . . 403
68.10 pmu_clk_ctrl3 . . . 404
68.11 pmu_rstn_ctrl . . . 404
68.12 pmu_dcdc_rf_ctrl0 . . . 404
68.13 pmu_dcdc_rf_ctrl1 . . . 405
68.14 pmu_dcdc_rf_ctrl2 . . . 405
68.15 pmu_dcdc_rf_ctrl3 . . . 405
68.16 pmu_ldo_vcore08_ctrl . . . 406
68.17 pmu_ldo_vrtc08_ctrl . . . 406
68.18 pmu_other_ldo_ctrl0 . . . 406
68.19 pmu_other_ldo_ctrl1 . . . 406
68.20 pmu_long_pwr_key_cfg . . . 407
68.21 pmu_rf_on_cfg . . . 407
68.22 pmu_psm_cfg0 . . . 407
68.23 pmu_psm_cfg1 . . . 408
68.24 pmu_psm_cfg2 . . . 408
68.25 pmu_psm_cfg3 . . . 408
68.26 pmu_psm_cfg4 . . . 408
68.27 pmu_psm_cfg5 . . . 408
68.28 pmu_psm_cfg6 . . . 408
68.29 pmu_psm_cfg7 . . . 408
68.30 pmu_monitor_cfg . . . 408
68.31 pmu_status . . . 408
68.32 pmu_int_irq . . . 409
68.33 pmu_xtal_clk_detect . . . 409
68.34 pmu_cm_xtal_cfg1 . . . 409
68.35 pmu_cm_xtal_cfg2 . . . 410
68.36 pmu_clk_div_128k . . . 410
68.37 pmu_ldo_vflash_ctrl . . . 410
68.38 pmu_dcdc_core_ctrl0 . . . 410
68.39 pmu_dcdc_core_ctrl1 . . . 411
68.40 pmu_dcdc_core_ctrl2 . . . 411
68.41 pmu_dcdc_core_ctrl3 . . . 411

68.42 pmu_dcdc_css_cfg . . . 411
68.43 pmu_clk_ctrl4 . . . 412
wdg4
69.1 WDG_LOAD_VAL . . . 412
69.2 WDG_CNT . . . 412
69.3 WDG_CTRL . . . 412
69.4 WDG_INT_CLR . . . 412
69.5 WDG_INT_RAW . . . 412
69.6 WDG_INT_MASK . . . 413
69.7 WDG_LOCK . . . 413
dcdc_common
70.1 dcdc_ref_clk_div . . . 413
70.2 dcdc_clk_ctrl0 . . . 413
70.3 dcdc_clk_ctrl1 . . . 413
70.4 dcdc_clk_ctrl2 . . . 413
70.5 dcdc_clk_ctrl3 . . . 413
dcdc_rf_ctrl
71.1 dcdc_clk_ctrl0 . . . 413
71.2 dcdc_clk_ctrl1 . . . 414
71.3 dcdc_clk_ctrl2 . . . 414
71.4 dcdc_clk_ctrl3 . . . 414
71.5 dcdc_clk_ctrl4 . . . 414
71.6 dcdc_clk_ctrl5 . . . 414
71.7 dcdc_clk_ctrl6 . . . 414
71.8 dcdc_clk_ctrl7 . . . 414
71.9 dcdc_ctrl_cfg0 . . . 414
71.10 dcdc_ctrl_cfg1 . . . 415
71.11 dcdc_ctrl_cfg2 . . . 415
71.12 dcdc_ctrl_cfg3 . . . 415
71.13 dcdc_ctrl_cfg4 . . . 415
71.14 dcdc_ctrl_cfg5 . . . 415
71.15 dcdc_ctrl_cfg6 . . . 415
71.16 dcdc_ctrl_cfg7 . . . 415
71.17 dcdc_ctrl_cfg8 . . . 415
71.18 dcdc_ctrl_cfg9 . . . 415
71.19 dcdc_ctrl_cfg10 . . . 415
71.20 dcdc_ctrl_cfg11 . . . 416
71.21 dcdc_ctrl_cfg12 . . . 416
71.22 dcdc_ctrl_cfg13 . . . 416
71.23 dcdc_ctrl_cfg14 . . . 416
71.24 dcdc_ctrl_cfg15 . . . 416
71.25 dcdc_ctrl_cfg16 . . . 416
71.26 dcdc_ctrl_cfg17 . . . 416
71.27 dcdc_ctrl_cfg18 . . . 416
71.28 dcdc_ctrl_cfg19 . . . 416
71.29 dcdc_ctrl_cfg20 . . . 416
71.30 dcdc_ctrl_cfg21 . . . 417
71.31 dcdc_ctrl_cfg22 . . . 417
71.32 dcdc_ctrl_cfg23 . . . 417
71.33 dcdc_ctrl_cfg24 . . . 417
71.34 dcdc_ctrl_cfg25 . . . 417
71.35 dcdc_ctrl_cfg26 . . . 417
71.36 dcdc_ctrl_cfg27 . . . 417
71.37 dcdc_ctrl_cfg28 . . . 417
71.38 dcdc_ctrl_cfg29 . . . 417
71.39 dcdc_ctrl_cfg30 . . . 417
71.40 dcdc_ctrl_cfg31 . . . 418
71.41 dcdc_ctrl_cfg32 . . . 418
71.42 dcdc_ctrl_cfg33 . . . 418

71.43 dcdc_ctrl_cfg34 . . . 418
71.44 dcdc_ctrl_cfg35 . . . 418
71.45 dcdc_ctrl_cfg36 . . . 418
71.46 dcdc_ctrl_cfg37 . . . 418
71.47 dcdc_ctrl_cfg38 . . . 418
71.48 dcdc_ctrl_cfg39 . . . 418
71.49 dcdc_ctrl_cfg40 . . . 419
71.50 dcdc_ctrl_cfg41 . . . 419
71.51 dcdc_ctrl_cfg42 . . . 419
71.52 dcdc_ctrl_cfg43 . . . 419
71.53 dcdc_status . . . 419
dcdc_core_ctrl
72.1 dcdc_clk_ctrl0 . . . 420
72.2 dcdc_clk_ctrl1 . . . 420
72.3 dcdc_clk_ctrl2 . . . 420
72.4 dcdc_clk_ctrl3 . . . 420
72.5 dcdc_clk_ctrl4 . . . 420
72.6 dcdc_clk_ctrl5 . . . 420
72.7 dcdc_clk_ctrl6 . . . 420
72.8 dcdc_clk_ctrl7 . . . 420
72.9 dcdc_ctrl_cfg0 . . . 421
72.10 dcdc_ctrl_cfg1 . . . 421
72.11 dcdc_ctrl_cfg2 . . . 421
72.12 dcdc_ctrl_cfg3 . . . 421
72.13 dcdc_ctrl_cfg4 . . . 421
72.14 dcdc_ctrl_cfg5 . . . 421
72.15 dcdc_ctrl_cfg6 . . . 421
72.16 dcdc_ctrl_cfg7 . . . 421
72.17 dcdc_ctrl_cfg8 . . . 421
72.18 dcdc_ctrl_cfg9 . . . 421
72.19 dcdc_ctrl_cfg10 . . . 421
72.20 dcdc_ctrl_cfg11 . . . 422
72.21 dcdc_ctrl_cfg12 . . . 422
72.22 dcdc_ctrl_cfg13 . . . 422
72.23 dcdc_ctrl_cfg14 . . . 422
72.24 dcdc_ctrl_cfg15 . . . 422
72.25 dcdc_ctrl_cfg16 . . . 422
72.26 dcdc_ctrl_cfg17 . . . 422
72.27 dcdc_ctrl_cfg18 . . . 422
72.28 dcdc_ctrl_cfg19 . . . 422
72.29 dcdc_ctrl_cfg20 . . . 422
72.30 dcdc_ctrl_cfg21 . . . 423
72.31 dcdc_ctrl_cfg22 . . . 423
72.32 dcdc_ctrl_cfg23 . . . 423
72.33 dcdc_ctrl_cfg24 . . . 423
72.34 dcdc_ctrl_cfg25 . . . 423
72.35 dcdc_ctrl_cfg26 . . . 423
72.36 dcdc_ctrl_cfg27 . . . 423
72.37 dcdc_ctrl_cfg28 . . . 423
72.38 dcdc_ctrl_cfg29 . . . 423
72.39 dcdc_ctrl_cfg30 . . . 423
72.40 dcdc_ctrl_cfg31 . . . 424
72.41 dcdc_ctrl_cfg32 . . . 424
72.42 dcdc_ctrl_cfg33 . . . 424
72.43 dcdc_ctrl_cfg34 . . . 424
72.44 dcdc_ctrl_cfg35 . . . 424
72.45 dcdc_ctrl_cfg36 . . . 424
72.46 dcdc_ctrl_cfg37 . . . 424
72.47 dcdc_ctrl_cfg38 . . . 424

72.48 dcdc_ctrl_cfg39 424
72.49 dcdc_ctrl_cfg40 425
72.50 dcdc_ctrl_cfg41 425
72.51 dcdc_ctrl_cfg42 425
72.52 dcdc_ctrl_cfg43 425
72.53 dcdc_status 425